Introduction to Computational Finance and Financial Econometrics with R

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Table of contents

Preface 3

Acknowledgments 7

1 Return Calculations 8
1.1 The Time Value of Money . . . 8
1.1.1 Future value, present value and simple interest . . . 8
1.1.2 Multiple compounding periods . . . 11
1.1.3 Effective annual rate . . . 12
1.2 Asset Return Calculations . . . 14
1.2.1 Assets . . . 14
1.2.2 Simple returns . . . 16
1.2.3 Continuously compounded returns . . . 28
1.3 Portfolios and Portfolio Returns . . . 35
1.3.1 Multiperiod portfolio returns and rebalancing . . . 37
1.3.2 Continuously Compounded Portfolio Returns . . . 46
1.4 Return Calculations with Data in R . . . 47
1.4.1 Representing time series data using `xts` objects . . . 47
1.4.2 Calculating returns . . . 57
1.4.3 Calculating portfolio returns from time series data . . . 61
1.4.4 Downloading financial data from the internet . . . 65
1.5 Further Reading: Return Calculations . . . 65
1.6 Appendix: Properties of Exponentials and Logarithms . . . 67
1.7 Exercises . . . 68
Analytical Problems . . . 68
Return Calculations using R Packages . . . 72

2 Review of Random Variables 74
2.1 Random Variables . . . 74
2.1.1 Discrete random variables . . . 75
2.1.2 Continuous random variables . . . 78
2.1.3 The cumulative distribution function . . . 82
2.1.4 Quantiles of the distribution of a random variable . . . 83
2.1.5 R functions for discrete and continuous distributions . . . 85
2.1.6 Shape characteristics of probability distributions . . . 88
2.1.7 Linear functions of a random variable . . . 101
2.1.8 Value-at-Risk: An introduction . . . 104
2.2 Bivariate Distributions . . . 108
2.2.1 Discrete random variables . . . 108
2.2.2 Bivariate distributions for continuous random variables . . . 114

2.2.3 Independence . . . 116
2.2.4 Covariance and correlation . . . 118
2.2.5 Bivariate normal distributions . . . 123
2.2.6 Expectation and variance of the sum of two random variables . . . 125
2.3 Multivariate Distributions . . . 127
2.3.1 Discrete random variables . . . 127
2.3.2 Continuous random variables . . . 128
2.3.3 Independence . . . 129
2.3.4 Dependence concepts . . . 129
2.3.5 Linear combinations of N random variables . . . 130
2.3.6 Covariance between linear combinations of random variables . . . 131
2.4 Further Reading: Review of Random Variables . . . 132
2.5 Exercises . . . 133

3 Matrix Algebra Review **137**
3.1 Matrices and Vectors . . . 137
3.2 Basic Matrix Operations . . . 142
3.2.1 Addition and subtraction . . . 142
3.2.2 Scalar multiplication . . . 143
3.2.3 Matrix multiplication . . . 144
3.2.4 The identity matrix . . . 146
3.2.5 Diagonal, lower triangular and upper triangular matrices . . . 147
3.3 Representing Summation Using Matrix Notation . . . 149
3.4 Systems of Linear Equations . . . 151
3.4.1 Rank of a matrix . . . 156
3.4.2 Partitioned matrices and partitioned inverses . . . 157
3.5 Positive Definite Matrices . . . 159
3.5.1 Matrix square root . . . 159
3.6 Multivariate Probability Distributions Using Matrix Algebra . . . 161
3.6.1 Random vectors . . . 161
3.6.2 Covariance matrix . . . 161
3.6.3 Correlation matrix . . . 162
3.6.4 Variance of linear combination of random vectors . . . 163
3.6.5 Covariance between linear combination of two random vectors . . . 164
3.6.6 Variance of a vector of linear functions of a random vector . . . 164
3.6.7 Multivariate normal distribution . . . 164
3.7 Portfolio Math Using Matrix Algebra . . . 166
3.8 Derivatives of Simple Matrix Functions . . . 169
3.9 Further Reading: Matrix Algebra Review . . . 171
3.10 Problems: Matrix Algebra Review . . . 172

4 Time Series Concepts **174**
4.1 Stochastic Processes . . . 175
4.1.1 Stationary stochastic processes . . . 175
4.1.2 Non-Stationary processes . . . 183
4.1.3 Ergodicity . . . 186
4.2 Multivariate Time Series . . . 187
4.2.1 Stationary and ergodic multivariate time series . . . 187

4.3 Time Series Models . . . 190
4.3.1 Moving average models . . . 191
4.3.2 Autoregressive Models . . . 198
4.3.3 Autoregressive Moving Average Models . . . 203
4.3.4 Vector Autoregressive Models . . . 204
4.4 Forecasting . . . 205
4.5 Further Reading: Time Series Concepts . . . 205
4.6 Problems: Time Series Concepts . . . 205

5 Descriptive Statistics for Financial Data 208
5.1 Univariate Descriptive Statistics . . . 208
5.1.1 Example data . . . 209
5.1.2 Time plots . . . 210
5.1.3 Descriptive statistics for the distribution of returns . . . 215
5.1.4 QQ-plots . . . 225
5.1.5 Shape characteristics of the empirical distribution . . . 230
5.1.6 Outliers . . . 233
5.1.7 Box plots . . . 236
5.2 Time Series Descriptive Statistics . . . 238
5.2.1 Sample autocovariances and autocorrelations . . . 238
5.2.2 Time dependence in volatility . . . 240
5.3 Bivariate Descriptive Statistics . . . 243
5.3.1 Scatterplots . . . 243
5.3.2 Sample covariance and correlation . . . 245
5.3.3 Sample cross-lag covariances and correlations . . . 249
5.4 Rolling descriptive statistics . . . 251
5.4.1 Practical issues associated with rolling estimates . . . 258
5.5 Stylized facts for daily and monthly asset returns . . . 259
5.6 Further Reading: Descriptive Statistics . . . 259
5.7 Problems: Descriptive Statistics . . . 260

6 The Gaussian White Noise Return Model 265
6.1 GWN Model Assumptions . . . 265
6.1.1 Regression model representation . . . 266
6.1.2 Location-Scale model representation . . . 268
6.1.3 The GWN model in matrix notation . . . 268
6.1.4 The GWN model for continuously compounded returns . . . 269
6.1.5 GWN model for simple returns . . . 273
6.2 Monte Carlo Simulation of the GWN Model . . . 276
6.2.1 Simulating returns on more than one asset . . . 282
6.3 Scenario Analysis . . . 287
6.4 Further Reading: The GWN Return Model . . . 287
6.5 Exercises . . . 289

7 Estimation of The GWN Model 290
7.1 Estimators and Estimates . . . 291
7.2 Finite Sample Properties of Estimators . . . 292
7.2.1 Bias . . . 293

7.2.2 Precision . . . 294
7.2.3 Estimated standard errors . . . 296
7.3 Asymptotic Properties of Estimators . . . 296
7.3.1 Consistency . . . 296
7.3.2 Asymptotic Normality . . . 297
7.3.3 Central Limit Theorem . . . 298
7.3.4 Asymptotic Confidence Intervals . . . 299
7.4 Estimators for the Parameters of the GWN Model . . . 299
7.5 Statistical Properties of the GWN Model Estimates . . . 306
7.5.1 Bias . . . 306
7.5.2 Precision . . . 307
7.5.3 Sampling Distributions and Confidence Intervals . . . 311
7.6 Using Monte Carlo Simulation to Understand the Statistical Properties of Estimators 317
7.6.1 Evaluating the Statistical Properties of $\hat{\mu}$ Using Monte Carlo Simulation . . . 318
7.6.2 Evaluating the Statistical Properties of $\hat{\sigma}_i^2$ and $\hat{\sigma}_i$ Using Monte Carlo simulation 321
7.6.3 Evaluating the Statistical Properties of $\hat{\sigma}_{ij}$ and $\hat{\rho}_{ij}$ by Monte Carlo simulation 324
7.7 Sampling distributions of $\hat{\mu}$ and $\hat{\Sigma}$ . . . 328
7.7.1 Sampling distribution of $\hat{\mu}$ . . . 328
7.7.2 Sampling distribution of the elements of $\hat{\Sigma}$ . . . 329
7.7.3 Joint sampling distribution between $\hat{\mu}$ and vech($\hat{\Sigma}$) . . . 330
7.8 Further Reading: Estimation of The GWN Model . . . 331
7.9 Problems: Estimation of The GWN Model . . . 331

8 Estimating Functions of GWN Model Parameters 332
8.1 Functions of GWN Model Parameters . . . 332
8.2 Estimation of Functions of GWN Model Parameters . . . 334
8.2.1 Bias . . . 335
8.2.2 Consistency . . . 336
8.3 Asymptotic normality and the delta method . . . 337
8.3.1 The numerical delta method . . . 344
8.3.2 The delta method for vector valued functions (advanced) . . . 346
8.3.3 The numerical delta method for vector valued functions (advanced) . . . 348
8.3.4 The delta method explained . . . 348
8.4 Resampling . . . 349
8.5 The Jackknife . . . 350
8.5.1 The jackknife for vector-valued estimates . . . 355
8.5.2 Pros and Cons of the Jackknife . . . 357
8.6 The Nonparametric Bootstrap . . . 358
8.6.1 Bootstrap bias estimate . . . 359
8.6.2 Bootstrap standard error estimate . . . 359
8.6.3 Bootstrap confidence intervals . . . 360
8.6.4 Performing the nonparametric bootstrap in R . . . 361
8.7 Brute force nonparametric bootstrap procedure . . . 362
8.7.1 Pros and cons of the nonparametric bootstrap . . . 375
8.8 Using Monte Carlo Simulation to Evaluate Delta Method, Jackknife and Bootstrap Standard Errors . . . 376
8.9 Further Reading: Delta Method and Resampling . . . 380
8.10 Exercises . . . 380

9 Hypothesis Testing in the GWN Model **382**
9.1 Review of General Principles . . . 383
9.1.1 Steps for hypothesis testing . . . 383
9.1.2 Hypothesis tests and decisions . . . 384
9.1.3 Significance level and power . . . 385
9.2 Hypothesis Testing in the GWN Model . . . 386
9.2.1 Coefficient tests . . . 386
9.2.2 Model specification tests . . . 388
9.2.3 Data for hypothesis testing examples . . . 389
9.3 Tests for Individual Parameters: t-tests and z-scores . . . 390
9.3.1 Exact tests under normality of data . . . 390
9.3.2 Exact tests with σ unknown . . . 396
9.3.3 Z-scores under asymptotic normality of estimators . . . 398
9.3.4 Relationship between hypothesis tests and confidence intervals . . . 402
9.4 Coefficient tests between model coefficients for two assets . . . 404
9.4.1 Test for equal means between two assets . . . 404
9.4.2 Test for equal Sharpe ratios between two assets . . . 406
9.5 Wald tests for general linear hypotheses (advanced) . . . 408
9.6 Model Specification Tests . . . 408
9.6.1 Tests for normality . . . 408
9.6.2 Tests for no autocorrelation . . . 413
9.6.3 Tests for constant parameters . . . 415
9.6.4 Tests for constant parameters based on rolling estimators . . . 416
9.7 Using Monte Carlo Simulation to Understand Hypothesis Testing . . . 418
9.8 Using the Bootstrap for Hypothesis Testing . . . 418
9.9 Further Reading: Hypothesis Testing in the GWN Model . . . 419
9.10 Exercises . . . 419

10 Introduction to Portfolio Theory **420**
10.1 Assumptions About Returns and Investor Risk Preferences . . . 421
10.2 Portfolios of Two Risky Assets . . . 421
10.2.1 The Portfolio Problem . . . 423
10.2.2 Portfolio Value-at-Risk . . . 428
10.2.3 The set of feasible portfolios . . . 431
10.2.4 Computing the global minimum variance portfolio . . . 434
10.2.5 Correlation and the shape of the portfolio frontier . . . 436
10.3 Efficient portfolios with two risky assets . . . 437
10.3.1 Optimal portfolios . . . 439
10.4 Efficient portfolios with a risk-free asset . . . 439
10.4.1 Portfolios with one risky asset and one risk-free asset . . . 439
10.4.2 Portfolios with one risky asset and one risk-free asset with different borrowing and lending rates . . . 443
10.5 Efficient portfolios with two risky assets and a risk-free asset . . . 444
10.5.1 Solving for the Tangency Portfolio . . . 445
10.5.2 Mutual Fund Separation . . . 446
10.5.3 Interpreting Efficient Portfolios . . . 447
10.5.4 Efficient Portfolios and Value-at-Risk . . . 451
10.6 Application to Real World Portfolios . . . 451

10.7 Further Reading: Introduction to Portfolio Theory . . . 456
10.8 Appendix: Review of Optimization and Constrained Optimization . . . 456
10.9 Exercises . . . 459

11 Portfolio Theory with Matrix Algebra 462
11.1 Portfolios with N Risky Assets . . . 462
11.1.1 Portfolio return and risk characteristics using matrix notation . . . 463
11.1.2 Large portfolios and diversification . . . 469
11.2 Determining the Global Minimum Variance Portfolio Using Matrix Algebra . . . 472
11.2.1 Alternative derivation of global minimum variance portfolio . . . 475
11.3 Determining Mean-Variance Efficient Portfolios Using Matrix Algebra . . . 475
11.3.1 Alternative derivation of an efficient portfolio . . . 479
11.4 Computing the Mean-Variance Efficient Frontier . . . 482
11.4.1 Algorithm for computing efficient frontier . . . 486
11.5 Computing Efficient Portfolios of N risky Assets and a Risk-Free Asset Using Matrix Algebra . . . 490
11.5.1 Computing the tangency portfolio using matrix algebra . . . 490
11.6 Remark . . . 491
11.6.1 Alternative derivation of the tangency portfolio . . . 493
11.6.2 Mutual fund separation theorem again . . . 494
11.7 Computational Problems with Very Large Portfolios . . . 497
11.8 Portfolio Analysis Functions in R . . . 499
11.9 Application to Vanguard Mutual Fund . . . 505
11.10Further Reading: Portfolio Theory with Matrix Algebra . . . 510
11.11Exercises . . . 511

12 Portfolio Theory with Short Sales Constraints 512
12.1 Overview of Short Selling . . . 512
12.2 Portfolio Theory with Short Sales Constraints in a Simplified Setting . . . 513
12.2.1 Two Risky Assets . . . 513
12.2.2 One risky asset and a risk-free asset . . . 518
12.2.3 Two risky assets and risk-free asset . . . 520
12.3 Portfolio Theory with Short Sales Constraints in a General Setting . . . 524
12.3.1 Multiple risky assets . . . 524
12.3.2 Portfolio optimization imposing no-short sales constraints . . . 526
12.4 Quadratic Programming Problems . . . 528
12.4.1 No short sales global minimum variance portfolio . . . 529
12.4.2 No short sales minimum variance portfolio with target expected return . . . 532
12.4.3 No short sales tangency portfolio . . . 536
12.5 Application to Vanguard Mutual Funds . . . 542
12.6 Further Reading: Portfolio Theory with Short Sales Constraints . . . 545
12.7 Exercises . . . 545

13 Portfolio Risk Budgeting 546
13.1 Risk Budgeting Using Portfolio Variance and Portfolio Standard Deviation . . . 546
13.1.1 Case 1: uncorrelated assets . . . 547
13.1.2 Case 2: correlated assets . . . 547
13.1.3 The general case of N assets . . . 548

13.2 Euler's Theorem and Risk Decompositions . . . 548
13.2.1 Homogenous functions of degree one . . . 549
13.2.2 Euler's theorem . . . 550
13.2.3 Risk decomposition using Euler's theorem . . . 552
13.2.4 Interpreting marginal contributions to risk . . . 555
13.3 Portfolio risk reports . . . 556
13.4 Understanding Portfolio Volatility Risk Decompositions . . . 560
13.4.1 $\sigma - \rho$ decomposition for MCR_i^σ . . . 560
13.4.2 $\sigma - \beta$ decomposition for MCR_i^σ . . . 563
13.5 Risk Budgeting for Optimized Portfolios . . . 565
13.6 Risk Parity Portfolios . . . 566
13.6.1 Stock and Bond Portfolios . . . 566
13.6.2 Defining Risk Parity Portfoliofolio . . . 567
13.6.3 Example Data . . . 569
13.6.4 Further Reading: Portfolio Risk Budgeting . . . 574
13.7 Exercises . . . 574

14 Statistical Analysis of Portfolios **575**
14.1 Portfolios of Two Risky Assets . . . 575
14.1.1 Estimation of GWN Model Parameters for Portfolios . . . 576
14.1.2 Estimation error in the risk-return diagram . . . 581
14.1.3 Estimation error in the portfolio frontier . . . 585
14.1.4 Statistical properties of the global minimum variance portfolio . . . 590
14.1.5 Statistical properties of the Sharpe ratio . . . 596
14.2 Hypothesis Testing for Sharpe Ratio . . . 605
14.3 Rolling Analysis of Portfolios . . . 606
14.4 Further Reading: Statistical Analysis of Portfolios . . . 606
14.5 Exercises . . . 606

15 Single Index Model **607**
15.1 Introduction . . . 607
15.2 Motivation . . . 607
15.3 William Sharpe's SI Model . . . 611
15.3.1 Interpretation of single index model . . . 612
15.3.2 Statistical properties of returns in the SI model . . . 615
15.3.3 SI model and portfolios . . . 617
15.4 The SI model in matrix notation . . . 619
15.5 Single index model portfolio theory . . . 621
15.6 Monte Carlo Simulation of the SI Model . . . 621
15.7 Further Reading: Single Index Model . . . 624
15.8 Exercicses . . . 624

16 Estimation of the Single Index Model **625**
16.1 Introduction . . . 625
16.2 Data for examples . . . 626
16.3 Estimation of SI Model . . . 626
16.3.1 Plug-in principle estimates . . . 626
16.3.2 Least squares estimates . . . 629

16.3.3 Simple linear regression in R 631
16.4 Statistical Properties of SI Model Estimates 632
16.4.1 Bias 632
16.4.2 Precision 633
16.4.3 Sampling distribution and confidence intervals. 633
16.5 Further Reading: Single Index Model 634
16.6 Exercicses 634

References **635**

Preface

I am a Professor of Economics at the University of Washington (UW) where I teach the class *Introductional to Computational Finance and Financial Econometrics* (ECON 424). This course started as an experiment. In the winter quarter of 1998 I was on leave visiting the economics department at the University of California Los Angeles (UCLA). To help pay for my leave I, I arranged to teach a standard undergraduate econometrics course. My research at the time was moving toward financial econometrics and I thought it would be fun to teach an econometrics course with most of the examples motivated by financial applications. I thought that combining econometrics and statistics topics with financial applications would be a very effective and popular way of getting students interested in learning and applying econometrics. My intuition was right. My students at UCLA loved the version of the course I offered. After returning to the University of Washington, I created the upper division undergraduate course *Introduction to Computational and Financial Econometrics*. The course was an immediate hit. This book evolved from my lecture notes for the course.

Who is the book written for? What background is required?

My ECON 424 course is taken mainly by senior economics majors who are pursuing the more quantitative Bachelor of Science Degree in Economics at UW. These students typically have at least one year of calculus, intermediate microeconomics with some calculus (so they are exposed to optimization under constraints using Lagrange multipliers), and an introductory course in probability and statistics for economics majors. Some students have taken a course in linear algebra, some have taken an introductory econometrics course, and some have taken a financial economics course. I find that the typical introductory probability and statistics course is at a lower level than is needed for much of the material in my course. Ideally, students should have an intermediate probability and statistics course that uses calculus[1]. As a result, I spend a week at the beginning of the course doing a review of random variables and probability that uses basic calculus. A key part of the review is the manipulation of random variables, computing expectations of functions of random variables, etc., so that the students have the necessary tools to work with simple probabilistic models for asset returns. After the review of probability theory I introduce some basic time series concepts (e.g., stochastic processes and stationarity), as these concepts are often not taught in introductory probability and statistics classes, and I also do a quick review of basic matrix algebra that I use throughout the course. The use of matrix algebra is important for two reasons. First,

[1] Insert reference to recent MIT Press book here

most introductory textbook treatments of portfolio theory show calculations for the simple case of two assets and then talk about how the calculations extend to the general case of many assets but do not show how this is done. Hence, students are not capable of implementing the theory for real world portfolios. I emphasize that with knowledge of just a little bit of matrix algebra - essentially matrix addition, multiplication, inversion, and some matrix differential calculus - the general analysis of portfolio theory can be easily learned and implemented. Second, R is a vectorized and matrix programming language and so basic knowledge of matrix algebra allows students to easily understand and implement the R code for portfolio analysis.

Main topics covered in the book

After the review material, I jump into descriptive statistical analysis of historical financial time series data. I emphasize that my approach to building models for asset returns is based on first looking at the historical data and studying its properties. I start by telling students to pretend that observed returns on assets are realizations from some unknown covariance stationary stochastic process. I review common graphical and numerical descriptive statistics (e.g., histograms, QQ-plots, sample mean, variance, correlation, etc.). The goal of descriptive statistical analysis is to compile a collection of "stylized facts" that can be used as building blocks for probabilistic models for asset returns. For example, I show that monthly asset returns are typically uncorrelated over time, have empirical distributions (histograms) that look like normal bell curves, have means close to zero, are positively correlated with other assets, etc. I also emphasize that some "stylized facts" of asset returns appear to stay the same for different investment horizons (daily, weekly, and monthly returns are uncorrelated) and some do not (e.g., daily and weekly returns have fatter tails than the normal and show evidence of nonlinear time dependence in higher moments)

Why us R?

My early versions of *Introduction to Computational and Financial Econometrics* used Microsoft Excel as the main software tool as R was not yet around then. My decision to use Excel was for the students' benefit. Students having experience with Excel was, and still is, in high demand by employers in the finance industry. It is often said, "the financial world runs out of an Excel spreadsheet". However, Excel is not a good tool for doing statistics. Even though you *can* do statistical analysis in Excel, does not mean that you *should* do statistical analysis in Excel. I searched for Excel add-ins to make doing statistical analysis in Excel easier but I did not like what was available. Everything changed when I met Professor Doug Martin from the UW statistics department. Doug founded the company who created S-PLUS, the commercial version of the S statistical programming language developed by John Chambers at Bell Labs (R became the open source version of S in 2000). S is a beautiful language for exploratory data and statistical analysis and S-PLUS made it easily available to students on windows PCs. In the early 2000s I used Excel together with S-PLUS for my course. This hybrid approach worked well but the difficulties of getting student licenses at the beginning of the quarter was always a headache. As R matured in the mid to late 2000s, I made the switch from S-PLUS to R and never looked back. As the

course evolved and R became more commonly used in the financial world, and Rstudio became the integrated developer environment of choice for R, I eventually dropped the use of Excel and focused solely on R for the applications and examples in the book.

Many economics students lack experience with programming languages and some find R a difficult language to learn and use. My approach to dealing with this reality is to provide all of the R code to do calculations in the class lectures, in the textbook examples, and on homework assignments. In this way students do not have to worry about the coding aspect of the class and can more easily concentrate on the concepts of computational finance and financial econometrics. I find that students learn to use R quite well this way – learning by doing – and most students are quite proficient in R by the end of the course.

There are a lot of free resources on the web for learning R. In my Coursera course I partnered with DataCamp to provide a web-based learning environment for R tailored to the course material. I have found that this has substantially lowered the learning curve for many students. Student mostly use either a MAC or PC for their work (a few geeks use UNIX). I found that some things didn't always work well across platforms (usually issues with the file system). In 2014, I required students to use Rstudio as the environment for using R. This makes learning and working with R even easier. Another advantage of using Rstudio is that my lecture notes and homework assignments are be created using Rmarkdown.

Longer print book

(Shorter ebook). This book is based on my University of Washington sponsored Coursera course Introduction to Computational Finance and Financial Econometrics that has been running every quarter on Coursera since 2013. This Coursera course is based on the Summer 2013 offering of my University of Washington advanced undergraduate economics course of the same name. At the time, my UW course was part of a three course summer certificate in Fundamentals of Quantitative Finance offered by the Professional Masters Program in Computational Finance & Risk Management that was video-recorded and available for online students. An edited version of this course became the Coursera course. The popularity of the course encouraged me to convert the class notes for the course into a short book.

The book was typeset using Lyx with all of the R code examples embedded into the document and evaluated using knitr. I am extremely grateful to Yihui Xie for writing his wonderful book Dynamics Documents with R and knitr as this book would not have been possible without it. I am a firm believer in the "reproducible research paradigm".

The book's accompanying R package `introcompfinr` contains all of the financial data used for the examples presented in the book as well as a number of R functions written by me for estimating some simple models of asset returns, and doing portfolio and risk analysis. The package is freely available on GitHub at https://github.com/ezivot/introcompfinr and can be installed using:

```
install.packages("pak")
pak::pak("ezivot/introcompfinr").
```

How is this book different from other introductory book on financial econometrics?

There are now many good books written on the topic of computational/quantitative finance and financial econometrics with examples in R at various levels of sophistication. Books written for advanced U.S. undergraduates include Carmona (2014) and Tsay (2012). Books written for Masters students in computational finance or financial engineering courses include Fan and Yao (20xx), Lai and Xing (2008), Ruppert and Matteson (2015) and Tsay (2010). All of these books assume the readers have a broader mathematical and statistical background than what I assume here. Hence, this book is truly an "introduction" to the methods of computational finance and financial econometrics and is appropriate for undergraduate economics and finance majors at Universities worldwide. Another feature that distinguishes this book from others is the extensive use of R and illustrated R code throughout the book. In this regard, this book is a "hands on" book in which the student can readily execute the R examples on their own computer while reading the book. This allows the student to also learn R as well as the concepts in the book.

Eric Zivot
Seattle, Washington
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Many people have helped with the development of this book. First and foremost, I would like to thank the students at UW who have taken my course over the last fifteen years. I have received countless comments, corrections, and suggestions that have greatly improved the exposition and material in the book.

A lot of how I think about the statistical analysis of financial data has been shaped by many discussions and interactions with my long-term colleague and good friend R. Douglas Martin. Doug introduced me to the S language and I had the pleasure of working with him at Insightful where we developed the S+FinMetrics module for S-PLUS and I wrote the book *Modeling Financial Time Series with S-PLUS* with my former Ph.D. student Jiahui (Jeffrey) Wang.

My very patient wife Nina Sidneva carefully proof-read many early versions of the chapters as they were under development and caught many of my stupid errors. Xiao Yin proof-read most of the chapters after I converted my Latex notes to Lyx and incorporated the R examples with `knitr`. I could not have written the book without the expert technical help from Bethany Yollin and the wonderful book *Dynamics Documents with R and knitr* by Yihui Xie. Bethany helped with the conversion of my Latex notes to Lyx and the incorporation of the R examples in the text using `knitr`. She also helped to develop and maintain an early version the book's R package, `IntroCompfinR`. Any problem I found with Lyx and `knitr` was solved by referring to Yihui's book. In 2020 I decided to migrate the book from Lyx to Rmarkdown so that everything could be managed within Rstudio and the book could be generated using `bookdown`. My UW PhD RAs Yumming (Herbert) Liu, Pushpak Sarkar and Kovid Puria made this happen. In 2026, I moved from `bookdown` to Quarto.

My interactions with the R community, especially those involved with R/Finance, have been invaluable for learning and using R for the analysis of financial data. In particular, Brian Peterson and Peter Carl have been incredibly helpful and supportive of my book projects. I use their wonderful R package `PerformanceAnalytics` throughout the book. Working with time series data is made so much easier with Achiem Zieles' `zoo` package and Jeff Ryan's `xts` package. Many University of Washington (UW) econ PhD students (Brian Donhauser, Ming He, Wan-Jung Hsu, Kara Ng, Anthony Sanford, Galib Sultan, Cindy Huang) acted as TAs for my undergraduate course and gave constructive feedback. I also thank Coursera for helping me convert my undergraduate course into Coursera format.

1 Return Calculations

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Asset returns are the fundamental object of study in this book and in this chapter I review many important definitions and properties of asset returns with an emphasis on equity returns. It is important to understand the different ways in which asset returns are calculated as they impact model choices for asset returns. For example, simple returns are most commonly used for money and portfolio calculations in practice. However, simple returns have some undesirable properties for time aggregation that make mathematical and statistical modeling difficult. Continuously compounded returns, in contrast, have nicer aggregation properties that make mathematical and statistical modeling easier.

This chapter is organized as follows. In Section 1.1 I cover basic time value of money calculations. In Section 1.2 I cover asset return calculations, including both simple and continuously compounded returns. I define portfolios and detail portfolio return calculations in Section 1.3. In Section 1.4 I illustrate asset return calculations using R.

In this chapter I use the following R packages: **ggplot2**, **introcompfinr**, **methods**, **PerformanceAnalytics**, **quantmod**, **xts**, and **zoo**. Make sure these packages are installed and loaded before replicating the chapter examples.

1.1 The Time Value of Money

In this section I review basic time value of money calculations. The concepts of future value, present value and the compounding of interest are defined and discussed.

1.1.1 Future value, present value and simple interest

I start with the definition of the *future value* of an investment.

Definition 1.1 (Future value)**.** Consider an amount $\$V$ invested for n years at a *simple interest rate* of R per annum (where R is expressed as a decimal). If compounding takes place only at the end of the year, the *future value* after n years is defined as:

$$FV_n = \$V(1+R) \times \cdots \times (1+R) = \$V(1+R)^n. \tag{1.1}$$

Over the first year, $\$V$ grows to $\$V(1+R) = \$V + \$V \times R$ which represents the initial principal $\$V$ plus the payment of simple interest $\$V \times R$ for the year. Over the second year, the new principal $\$V(1+R)$ grows to $\$V(1+R)(1+R) = \$V(1+R)^2$, and so on.

Example 1.1 (Future value with simple interest)**.** Consider putting \$1,000 in an interest checking account that pays a simple annual percentage rate of 3%. The future value after $n = 1, 5$ and 10 years is, respectively,

$$\begin{aligned} FV_1 &= \$1,000 \cdot (1.03)^1 = \$,1030, \\ FV_5 &= \$1,000 \cdot (1.03)^5 = \$1,159.27, \\ FV_{10} &= \$1,000 \cdot (1.03)^{10} = \$1,343.92. \end{aligned}$$

Over the first year, \$30 in interest is accrued; over three years, \$159.27 in interest is accrued; over five years, \$343.92 in interest is accrued. These calculations are easy to perform in R:

```
V = 1000
R = 0.03
FV1 = V*(1 + R)
FV5 = V*(1 + R)^5
FV10 = V*(1 + R)^10
c(FV1, FV5, FV10)
#> [1] 1030 1159 1344
```

■

The future value formula Equation 1.1 defines a relationship between four variables: FV_n, V, R and n. Given three variables, the fourth variable can be determined. For example, given FV_n, R and n and solving for V gives the *present value* formula:

$$V = \frac{FV_n}{(1+R)^n}. \tag{1.2}$$

Here, V represents the present value (or current value) of the future value FV_n that is to be received n years from today. The future value is discounted by the factor $(1+R)^{-n}$ to reflect the fact that dollars to be received in n years are worth less than current dollars because current dollars can be invested today returning R per year. The discount factor $(1+R)^{-n}$ is less than 1, and represents the price today of \$1 to be received in n years when the interest rate is R per year.

Taking FV_n, n and V as given, the *compound annual return* on the investment is defined as:

$$R = \left(\frac{FV_n}{V}\right)^{1/n} - 1. \tag{1.3}$$

Here, the simple interest rate R also represents a type of average rate of return on an investment. Specifically, R is the rate of return which compounded annually over n years transforms the present value V into the future value FV_n.

Finally, given FV_n, V and R you can solve for n:

$$n = \frac{ln(FV_n/V)}{ln(1+R)}. \tag{1.4}$$

The expression Equation 1.4 is called the *horizon period* . The horizon period is the number of years it takes for an investment of $\$V$ to grow to a future value of $\$FV_n$ at a simple interest rate of R per year.

The horizon period formula can be used to determine the number of years it takes for an investment of $\$V$ to double. Setting $FV_n = 2V$ in Equation 1.4 gives:

$$n = \frac{\ln(2)}{\ln(1+R)} \approx \frac{0.7}{R},$$

which uses the approximations $\ln(2) = 0.6931 \approx 0.7$ and $\ln(1+R) \approx R$ for R close to zero. The approximation $n \approx 0.7/R$ is called the *rule of 70*. For example, if $R = 0.07$ (7% per year) then $n \approx 0.7/0.07 = 70/7 = 10$ years for the investment to double.

Example 1.2 (Using the rule of 70)**.** The rule of 70 gives a good approximation as long as the interest rate is not too high.

```
R = seq(0.01, 0.10, by=0.01)
nExact = log(2)/log(1 + R)
nRule70 = 0.7/R
cbind(R, nExact, nRule70)
#>           R nExact nRule70
#>  [1,] 0.01  69.66   70.00
#>  [2,] 0.02  35.00   35.00
#>  [3,] 0.03  23.45   23.33
#>  [4,] 0.04  17.67   17.50
#>  [5,] 0.05  14.21   14.00
#>  [6,] 0.06  11.90   11.67
#>  [7,] 0.07  10.24   10.00
#>  [8,] 0.08   9.01    8.75
#>  [9,] 0.09   8.04    7.78
#> [10,] 0.10   7.27    7.00
```

When $R = 0.01$ it takes approximately 70 years for your money to double. When $R = 0.10$ it takes approximately 7 years. For R values between 1% and 10% the approximation error in the rule of 70 is never more than half a year.

■

1.1.2 Multiple compounding periods

Often interest is paid more than once per year. In the case, the future value formula needs to be adjusted.

Definition 1.2 (Future value with multiple compounding periods)**.** Consider an amount $\$V$ invested for n years at a simple interest rate of R per year. If interest is paid m times per year, then the future value after n years is:

$$FV_n^m = \$V \cdot \left(1 + \frac{R}{m}\right)^{m \cdot n}.$$

The value R/m is called the *periodic interest rate* . As $m \to \infty$, the periodic rate becomes *continuously compounded* and the future value becomes:

$$FV_n^c = \lim_{m \to \infty} \$V \cdot \left(1 + \frac{R}{m}\right)^{m \cdot n} = \$V \cdot e^{R \cdot n},$$

where $e^{(\cdot)}$ is the exponential function and $e^1 = 2.71828$.

Example 1.3 (Future value with different compounding frequencies)**.** For a simple annual percentage rate of 10%, the value of \$1000 at the end of one year ($n = 1$) for different values of m is calculated below.

```
V = 1000
R = 0.1
m = c(1, 2, 4, 356, 10000)
FV = V*(1 + R/m)^(m)
print(cbind(m, FV), digits=7)
#>           m       FV
#> [1,]      1 1100.000
#> [2,]      2 1102.500
#> [3,]      4 1103.813
#> [4,]    356 1105.155
#> [5,] 10000 1105.170
```

The result with continuous compounding is

```
print(V*exp(R), digits=7)
#> [1] 1105.171
```

You get more interest as the number of compounding periods increases but the amount of extra interest earned essentially peaks at daily compounding. The difference between daily and continuous compounding is very small for a \$1,000 investment at 10% simple interest rate. ■

The continuously compounded return analogues to the present value, annual return and horizon period formulas Equation 1.2, Equation 1.3 and Equation 1.4 are:

$$V = e^{-Rn} FV_n, \tag{1.5}$$

$$R = \frac{1}{n} \ln\left(\frac{FV_n}{V}\right), \tag{1.6}$$

$$n = \frac{1}{R} \ln\left(\frac{FV_n}{V}\right). \tag{1.7}$$

1.1.3 Effective annual rate

I now consider the relationship between simple interest rates, periodic rates, effective annual rates and continuously compounded rates. Suppose an investment pays a periodic interest rate of 2% each quarter. This gives rise to a simple annual rate of 8% (2% $\times$ 4 quarters). At the end of the year, \$1,000 invested accrues to:

$$\$1000 \cdot \left(1 + \frac{0.08}{4}\right)^{4 \cdot 1} = \$1082.40.$$

Here, compounding the periodic rate 2% for 4 quarters produces \$82.20 of interest on a \$1,000 investment. What simple interest rate R_A, compounded once per year, would produce the same amount of interest? This *effective annual rate*, R_A, is determined by the relationship:

$$\$1,000 \cdot (1 + R_A) = \$1,082.40.$$

Solving for R_A gives:

$$R_A = \frac{\$1,082.40}{\$1,000} - 1 = 0.0824,$$

or $R_A = 8.24\%$. Here, the effective annual rate 8.4% is the simple interest rate with annual compounding that gives the same future value as the simple rate 8% compounded four times per year. The effective annual rate is greater than the simple annual to account for the payment of interest on interest.

The general relationship between the simple annual rate R with interest compounded m times per year and the effective annual rate, R_A, compounded once per year is given in the following

definition.

Definition 1.3 (Effective annual rate)**.** Let R be a simple annual interest rate with interest compounded m times per year and let R_A be the effective annual rate. The relationship between the simple annual rate and the effective annual rate is given by:

$$(1 + R_A) = \left(1 + \frac{R}{m}\right)^m.$$

Given the simple rate R, you can solve for the effective annual rate using:

$$R_A = \left(1 + \frac{R}{m}\right)^m - 1. \tag{1.8}$$

Given the effective annual rate R_A, you can solve for the corresponding simple rate using:

$$R = m\left[(1 + R_A)^{1/m} - 1\right]. \tag{1.9}$$

The relationship between the effective annual rate and the simple rate that is compounded continuously is:

$$(1 + R_A) = e^R.$$

The continuously compounded effective annual rate is then

$$R_A = e^R - 1. \tag{1.10}$$

Given a continuously compounded effective annual rate, the implied simple rate is

$$R = \ln(1 + R_A). \tag{1.11}$$

Example 1.4 (Determine effective annual rates)**.** The effective annual rates associated with the investments in example Example 1.2 are calculated below.

```
RA = FV/V - 1
print(cbind(m, FV, RA), digits=7)
#>           m       FV        RA
#> [1,]      1 1100.000 0.1000000
#> [2,]      2 1102.500 0.1025000
#> [3,]      4 1103.813 0.1038129
#> [4,]    356 1105.155 0.1051554
#> [5,] 10000 1105.170 0.1051704
```

The effective annual rate with continuous compounding is

```r
print(exp(R) - 1, digits=7)
#> [1] 0.1051709
```

■

Example 1.5 (Determine continuously compounded rate from effective annual rate)**.** Suppose an investment pays a periodic interest rate of 5% every six months ($m = 2, R/2 = 0.05$). In the market this would be quoted as having an annual percentage rate, R_A, of 10%. An investment of \$100 yields $\$100 \cdot (1.05)^2 = \$110.25$ after one year. The effective annual rate, R_A, is then 10.25%. To find the continuously compounded simple rate that gives the same future value as investing at the effective annual rate you solve: $R = \ln(1.1025) = 0.09758$. That is, if interest is compounded continuously at a simple annual rate of 9.758% then \$100 invested today would grow to $\$100 \cdot e^{0.09758} = \$110.25$

■

1.2 Asset Return Calculations

In this section, I review asset return calculations given initial and future prices associated with an investment. I first summarize the main type of assets used for investment purposes. Next, I cover simple return calculations, which are typically reported in practice but are often not convenient for statistical modeling purposes. I then describe continuously compounded return calculations, which are more convenient for statistical modeling purposes. I finish the section with a discussion of portfolio return calculations.

1.2.1 Assets

When I talk about assets in this book, I mean the typical investment assets available for purchase to individuals through brokerage and retirement accounts. These include bonds, stocks, mutual funds, exchange traded funds and notes, and derivative securities. Investment assets can be purchased through online brokerage accounts at firms like Ally Invest, Charles Schwab, E*TRADE, Fidelity, Interactive Brokers, Merrill Lynch, Robinhood, TD Ameritrade, and Vanguard. Due to the growth and competition of online investment accounts, the cost of purchasing investment assets has dropped significantly over time.

Bonds

Bonds are assets that represent debt obligations from corporations or the government used to raise capital (money) to finance operations. A typical bond is characterized by (1) a principal amount, representing initial capital borrowed; (2) maturity date, representing when the principal must be repaid, (3) periodic coupon payments, representing interest paid on specific dates. Bonds issued by

the U.S. government are called *Treasury bonds*, and are generally thought to be the safest bonds as it is very unlikely that the U.S. government will default on its debt obligations. Other government agencies as well as states and municipalities issue bonds to raise money. Corporate bonds are bonds issued by private and public corporations. Bonds other than Treasury bonds have the risk, called *default risk*, that the principal or coupon payments many not be made.

Stocks

A *stock* represents an ownership share in corporation. Public stock represents ownership in a publicly traded corporation, where the shares can be purchased or sold (traded) by the public in organized stock exchanges (markets). In the U.S. the main stock exchanges are the New York Stock Exchange (NYSE) and the National Association of Securities Dealers Automated Quotations (NASDAQ) exchange. Private stock represents ownership in a private corporation whose stock is not traded in an exchange. Private corporations become public companies when they decide to raise money (capital) by issuing new stock in a public exchange through an *initial public offering* (IPO). This new stock purchased by individuals can then be sold to others in the exchange. Ownership of stock often provides voting rights which allows stock holders to have a say in the management of the company. Owners of stock have a right to receive dividends, which are cash payments representing corporate profits. Corporations may or many not pay dividends. Typically, young growing companies reinvest their profits instead of paying dividends. The stocks of these companies are called *growth stocks*. More established companies more often pay dividends instead of reinvesting all profits.

Mutual Funds

A *mutual fund* is a professionally managed investment fund that pools money from many investors to purchase a collection of assets called a *portfolio*. Mutual funds typically invest in bonds, stocks, and money market instruments. Mutual funds are commonly used in *individual retirement accounts* (IRAs) and employee sponsored retirement accounts. Because mutual funds are managed portfolios they charge fees for this service. The amount of the fee depends on the activity of the portfolio manager. Actively managed funds, where the portfolio manager actively picks the assets in the portfolio, have the highest fees. Actively managed mutual funds are offered by many firms such as the Capital Group, Fidelity, Janice, Putnam, and T. Rowe Price. Passively managed funds have lower fees because the managers follow a rule to match a particular asset index such as the S&P 500 index. Passive mutual funds were pioneered by Vandguard. Mutual funds are traded on exchanges but are only traded at the end of each day.

Exchange Traded Funds and Notes

An *exchange traded fund* (ETF) is similar in many ways to a mutual fund, except that an ETF can be bought and sold throughout the day on stock exchanges. Most ETFs are index funds, that is, they hold the same securities in the same proportions as a certain stock market index or bond market index. The most popular ETFs in the U.S. replicate the S&P 500 Index, the total market

index, the NASDAQ-100 index, the price of gold, the "growth" stocks in the Russell 1000 Index, or the index of the largest technology companies. ETNs are similar to ETFs. The main companies issuing ETFs and ETNs are Barclays Capital Inc. (iPath), BlackRock (iShares), Credit Swisse, Direxion, StateStreet (SPDR), Invesco (PowerShares), and ProShares. Each of these companies provide online information on their ETFs and ETNs. You can easily find ETFs and ETNs for particular asset categories and investment strategies at www.etf.com.

Derivative Securities

A *derivative security* is an asset whose value is derived from the behavior of other underlying assets such as bonds, stocks, ETFs, etc. Common derivative securities are forwards, futures, options and swaps. I will not consider derivative securities is this book.

Information and historical price data on many publicly traded assets is freely available at https://finance.yahoo.com/.

1.2.2 Simple returns

Consider purchasing one share of an asset (e.g., stock, bond, ETF, mutual fund, option, etc.) at time t_0 for the price P_{t_0}, and then selling that share at time t_1 for the price P_{t_1}. If there are no intermediate cash flows (e.g., dividends) between t_0 and t_1, the rate of return over the period t_0 to t_1 is defined as the percentage change in price:

$$R(t_0, t_1) = \frac{P_{t_1} - P_{t_0}}{P_{t_0}}. \tag{1.12}$$

The time between t_0 and t_1 is called the *holding period* and Equation 1.12 is called the *holding period return*. In principle, the holding period can be any amount of time: one second, five minutes, eight hours, two days six minutes and two seconds, fifteen years, etc.. To simplify matters, in this chapter and throughout the book I will assume that the holding period is some increment of calendar time; e.g., one day, one month or one year. In particular, I will often assume a default holding period of one month.

Definition 1.4 (Simple net return)**.** Let P_t denote the price at the end of month t of an asset that pays no dividends and let P_{t-1} denote the price at the end of month $t-1$. Then the one-month *simple net return* on an investment in the asset between months $t-1$ and t is defined as:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \Delta P_t. \tag{1.13}$$

Writing $(P_t - P_{t-1})/P_{t-1} = P_t/P_{t-1} - 1$, the *simple gross return* is defined as:

$$1 + R_t = \frac{P_t}{P_{t-1}}. \tag{1.14}$$

The one-month gross return has the interpretation of the future value of \$1 invested in the asset for one-month with simple interest rate R_t. Solving Equation 1.14 for P_t gives

$$P_t = P_{t-1}(1 + R_t),$$

which shows P_t can be thought of as the future value of P_{t-1} invested for one month with simple return R_t. Unless otherwise stated, when I refer to returns I mean net returns. Since asset prices must always be non-negative (a long position in an asset is a limited liability investment), the smallest value for R_t is -1 or -100%.

Example 1.6 (Simple return calculation)**.** Consider a one-month investment in Microsoft stock. Suppose you buy the stock in month $t-1$ (e.g., end of February) at $P_{t-1} = \$85$ and sell the stock the next month (e.g., end of March) for $P_t = \$90$. Further assume that Microsoft does not pay a dividend between months $t-1$ and t. The one-month simple net and gross returns (over the month of March) are then:

$$\begin{aligned} R_t &= \frac{\$90 - \$85}{\$85} = \frac{\$90}{\$85} - 1 = 1.0588 - 1 = 0.0588, \\ 1 + R_t &= 1.0588. \end{aligned}$$

The one-month investment in Microsoft yielded a 5.88% per month return. Alternatively, \$1 invested in Microsoft stock in month $t-1$ grew to \$1.0588 in month t. The calculations using R are:

```
P0 = 90
Pm1 = 85
R0 = (P0 - Pm1)/Pm1
c(R0,1 + R0)
#> [1] 0.0588 1.0588
```

■

The return calculation works the same way if multiple shares of an asset are initially purchased. Let $s \geq 1$ denote the number of shares of an asset purchased at time $t-1$. The initial value of the investment is then $V_{t-1} = s \times P_{t-1}$. The value of the investment at time t is $V_t = s \times P_t$. The rate of return on the investment is the percentage change in value over the holding period:

$$R_t = \frac{V_t - V_{t-1}}{V_{t-1}} = \frac{s \times P_t - s \times P_{t-1}}{s \times P_{t-1}} \tag{1.15}$$

$$= \frac{s \times (P_t - P_{t-1})}{s \times P_{t-1}} = \frac{P_t - P_{t-1}}{P_{t-1}}. \tag{1.16}$$

As a result, the future value of the investment is equal to the current value times the gross return:

$$V_t = V_{t-1} \times (1 + R_t)$$

1.2.2.1 Two-period returns

Definition 1.5 (Two-period return)**.** The *simple two-month return* on an investment in an asset between months $t-2$ and t is defined as the percentage change in price over two months:

$$R_t(2) = \frac{P_t - P_{t-2}}{P_{t-2}} = \frac{P_t}{P_{t-2}} - 1. \tag{1.17}$$

The *two-month gross return* is defined as:

$$1 + R_t(2) = \frac{P_t}{P_{t-2}}. \tag{1.18}$$

Writing

$$\frac{P_t}{P_{t-2}} = \frac{P_t}{P_{t-1}} \times \frac{P_{t-1}}{P_{t-2}}$$

the two-month return can be expressed as:

$$\begin{aligned} R_t(2) &= \frac{P_t}{P_{t-1}} \cdot \frac{P_{t-1}}{P_{t-2}} - 1 \\ &= (1 + R_t)(1 + R_{t-1}) - 1. \end{aligned}$$

Then the simple two-month gross return becomes:

$$1 + R_t(2) = (1 + R_t)(1 + R_{t-1}) = 1 + R_{t-1} + R_t + R_{t-1}R_t,$$

which is the *product* of the two simple one-month gross returns and not one plus the sum of the two one-month returns. Hence, the two-month net return is

$$R_t(2) = R_{t-1} + R_t + R_{t-1}R_t,$$

which is not the sum of the two one-month returns unless one of the one-month returns is zero. If, however, either R_{t-1} or R_t is close to zero then $R_{t-1}R_t \approx 0$ and $1 + R_t(2) \approx 1 + R_{t-1} + R_t$ so that $R_t(2) \approx R_{t-1} + R_t$.

Example 1.7 (Computing two-period returns)**.** Continuing with the previous example, suppose that the price of Microsoft stock in month $t-2$ is \$80 and no dividend is paid between months $t-2$ and t. The two-month net return is:

$$R_t(2) = \frac{\$90 - \$80}{\$80} = \frac{\$90}{\$80} - 1 = 1.1250 - 1 = 0.1250,$$

or 12.50% per two months. The two one-month returns are:

$$R_{t-1} = \frac{\$85 - \$80}{\$80} = 1.0625 - 1 = 0.0625,$$
$$R_t = \frac{\$90 - 85}{\$85} = 1.0588 - 1 = 0.0588,$$

and the geometric average of the two one-month gross returns is: $1 + R_t(2) = 1.0625 \times 1.0588 = 1.1250$.

The calculations can be *vectorized* in R as follows:

```
Pm2 = 80
# create vector of prices
P = c(Pm2, Pm1, P0)
# calculate vector of 1-month returns from vector of prices
R = (P[2:3] - P[1:2])/P[1:2]
R
#> [1] 0.0625 0.0588
# calculate 2-month return from 2 1-month returns
(1 + R[1])*(1 + R[2]) - 1
#> [1] 0.125
# same calculation vectorized using R function cumprod()
cumprod(1+R) - 1
#> [1] 0.0625 0.1250
```

In the above, `P[2:3]` represents the 2nd and 3rd elements of the price vector `P`, and `P[1:2]` represents the 1st and 2nd elements of `P`.

■

It is important to recognize that two-period simple returns are not symmetric in the following sense. Suppose $R_{t-1} = -0.5$ and $R_t = 0.5$ and initial wealth is \$1. After two-periods \$1 falls to

$$\$1 \times (1 + R_{t-1})(1 + R_t) = \$1 \times (1 - 0.5)(1 + 0.5) = \$0.75$$

That is, a 50% drop followed by a 50% gain does not return wealth to its initial level. In the second period, the return has to be 100% in order for wealth to return to its initial level.

1.2.2.2 Multi-period returns

Definition 1.6 (Multi-period return)**.** The k-month simple return on an investment in an asset between months $t-k$ and t is defined as the percentage change in price over k months:

$$R_t(k) = \frac{P_t - P_{t-k}}{P_{t-k}} = \frac{P_t}{P_{t-k}} - 1. \tag{1.19}$$

Writing

$$\frac{P_t}{P_{t-k}} = \frac{P_t}{P_{t-1}} \times \frac{P_{t-1}}{P_{t-2}} \times \cdots \times \frac{P_{t-k+1}}{P_{t-k}} = (1+R_t)(1+R_{t-1})\cdots(1+R_{t-k+1}),$$

the k-month gross return is defined as the product of k one-month gross returns:

$$1 + R_t(k) = (1+R_t)(1+R_{t-1})\cdots(1+R_{t-k+1}) = \prod_{j=0}^{k-1}(1+R_{t-j}). \tag{1.20}$$

If all returns are close to zero then

$$R_t(k) \approx \sum_{j=0}^{k-1} R_{t-j},$$

which is a useful approximation when considering returns over short horizons (e.g., daily returns).

As with the one-month return, the multi-period return can also be derived in the same way if multiple shares of an asset are initially purchased. Using $V_{t-j} = s \times P_{t-j}$ you get

$$\frac{V_t}{V_{t-k}} = \frac{V_t}{V_{t-1}} \times \frac{V_{t-1}}{V_{t-2}} \ldots \frac{V_{t-k+1}}{V_{t-k}} \tag{1.21}$$

$$= \frac{s \times P_t}{s \times P_{t-1}} \times \frac{s \times P_{t-1}}{s \times P_{t-2}} \ldots \frac{s \times P_{t-k+1}}{s \times P_{t-k}} \tag{1.22}$$

$$= (1+R_t)(1+R_{t-1})\cdots(1+R_{t-k+1}). \tag{1.23}$$

As a result, V_t can be interpreted as the future value of V_{t-k} based on the returns $R_{t-k+1}, \cdots, R_t$:

$$V_t = V_{t-k} \times (1+R_{t-k+1})(1+R_{t-k+2})\cdots(1+R_t).$$

1.2.2.3 Adjusting for dividends

If an asset pays a *dividend* (or any cash payment), D_t, sometime between months $t-1$ and t, the *total net return* calculation becomes:

$$R_t^{\text{total}} = \frac{P_t + D_t - P_{t-1}}{P_{t-1}} = \frac{P_t - P_{t-1}}{P_{t-1}} + \frac{D_t}{P_{t-1}}, \tag{1.24}$$

where $(P_t - P_{t-1})/P_{t-1}$ is referred as the *capital gain* and D_t/P_{t-1} is referred to as the *dividend yield*. The *total gross return* is:

$$1 + R_t^{\text{total}} = \frac{P_t + D_t}{P_{t-1}}. \tag{1.25}$$

The formula Equation 1.19 for computing multi-period return remains the same except that one-period gross returns are computed using Equation 1.25.

High dividend paying stocks are often called *income stocks* and have characteristics similar to bonds where the regular dividends play a similar role as coupon interest payments.

Example 1.8 (Compute total return when dividends are paid)**.** Consider a one-month investment in Microsoft stock. Suppose you buy the stock in month $t-1$ at $P_{t-1} = \$85$ and sell the stock the next month for $P_t = \$90$. Further assume that Microsoft pays a \$1 dividend between months $t-1$ and t. The capital gain, dividend yield and total return are then:

$$\begin{aligned} R_t &= \frac{\$90 + \$1 - \$85}{\$85} = \frac{\$90 - \$85}{\$85} + \frac{\$1}{\$85} \\ &= 0.0588 + 0.0118 \\ &= 0.0707. \end{aligned}$$

The one-month investment in Microsoft yields a 7.07% per month total return. The capital gain component is 5.88% and the dividend yield component is 1.18%

■

1.2.2.4 Adjusting for inflation

The return calculations considered so far are based on the *nominal* or current prices of assets. Returns computed from nominal prices are nominal returns. The *real return* on an asset over a particular horizon takes into account the growth rate of the general price level over the horizon. If the nominal price of the asset grows faster than the general price level, then the nominal return will be greater than the inflation rate and the real return will be positive. Conversely, if the nominal price of the asset increases less than the general price level, then the nominal return will be less than the inflation rate and the real return will be negative. Real returns reflect the increase or decrease in purchasing power from an investment.

The computation of real returns on an asset is a two-step process:

1. Deflate the nominal price of the asset by the general price level
2. Compute returns in the usual way using the deflated prices

To illustrate, consider computing the real simple one-period return on an asset. Let P_t denote the nominal price of the asset at time t and let CPI_t denote an index of the general price level (e.g. consumer price index) at time t[1]. The deflated or real price at time t is:

$$P_t^{\text{Real}} = \frac{P_t}{CPI_t},$$

and the real one-period return is:

$$\begin{aligned} R_t^{\text{Real}} &= \frac{P_t^{\text{Real}} - P_{t-1}^{\text{Real}}}{P_{t-1}^{\text{Real}}} = \frac{\frac{P_t}{CPI_t} - \frac{P_{t-1}}{CPI_{t-1}}}{\frac{P_{t-1}}{CPI_{t-1}}} \\ &= \frac{P_t}{P_{t-1}} \cdot \frac{CPI_{t-1}}{CPI_t} - 1. \end{aligned} \tag{1.26}$$

The one-period gross real return return is:

$$1 + R_t^{\text{Real}} = \frac{P_t}{P_{t-1}} \cdot \frac{CPI_{t-1}}{CPI_t}. \tag{1.27}$$

If you define inflation between periods $t-1$ and t as:

$$\pi_t = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} = \%\Delta CPI_t, \tag{1.28}$$

then $1 + \pi_t = CPI_t/CPI_{t-1}$ and Equation 1.26 may be re-expressed as:

$$R_t^{\text{Real}} = \frac{1 + R_t}{1 + \pi_t} - 1. \tag{1.29}$$

Example 1.9 (Compute real return)**.** Consider, again, a one-month investment in Microsoft stock. Suppose the CPI in months $t-1$ and t is 1 and 1.01, respectively, representing a 1% monthly growth rate in the overall price level. The real prices of Microsoft stock are:

$$\begin{aligned} P_{t-1}^{\text{Real}} &= \frac{\$85}{1} = \$85, \\ P_t^{\text{Real}} &= \frac{\$90}{1.01} = \$89.1089, \end{aligned}$$

and the real monthly return is:

$$R_t^{\text{Real}} = \frac{\$89.10891 - \$85}{\$85} = 0.0483.$$

[1]The CPI is usually normalized to 1 or 100 in some base year. I assume that the CPI is normalized to 1 in the base year for simplicity.

The nominal return and inflation over the month are:

$$R_t = \frac{\$90 - \$85}{\$85} = 0.0588,$$
$$\pi_t = \frac{1.01 - 1}{1} = 0.01.$$

Then the real return computed using Equation 1.29 is:

$$R_t^{\text{Real}} = \frac{1.0588}{1.01} - 1 = 0.0483.$$

Notice that simple real return is almost, but not quite, equal to the simple nominal return minus the inflation rate: $R_t^{\text{Real}} \approx R_t - \pi_t = 0.0588 - 0.01 = 0.0488$.

■

1.2.2.5 Adjusting for taxes

Any investment that generates a capital gain is potentially subject to taxation by the government. This taxation reduces the investment return. To see how this works, let V_t denote an initial investment amount, let $R_t > 0$ denote the one-period rate of return on the investment, and let τ denote the tax rate on investment income (capital gain). In the U.S. the tax rate on long-term (holding period more than one year) capital gains is either 0%, 15%, or 20% (depending on income) and the tax rate on short-term capital gains (holding period less than one year) is the tax rate on ordinary income (which could be as high as 35%). The future value without taxes is $V_t(1 + R_t)$, and $V_t R_t$ is the capital gain. The amount owed in taxes is $V_t R_t \tau$ and the amount remaining after taxes is $V_t R_t(1 - \tau)$. The after tax return is

$$\begin{aligned} R_t^{Atax} &= \frac{V_t + V_t R_t(1-\tau) - V_t}{V_t} \\ &= \frac{V_t(1 + R_t(1-\tau) - 1)}{V_t} \\ &= R_t(1-\tau). \end{aligned} \tag{1.30}$$

Hence, the after tax return is the before tax return multiplied by one minus the tax rate.

Example 1.10 (Compute after tax return)**.** Let $V_t = \$100$, $R_t = 0.05$, and $\tau = 0.20$. Then the after tax return is

$$R_t^{Atax} = (0.05)(1 - 0.20) = (0.05)(0.80) = 0.04.$$

The amount paid in taxes is $\$1000 \times (0.05) \times (0.2) = \$10$, and the after tax profit is $\$50 - \$10 = \$40$.
■

1.2.2.6 Return on foreign investment

It is common for an investor in one country to buy an asset in another country as an investment. If the two countries have different currencies then the exchange rate between the two currencies influences the return on the asset. To illustrate, consider an investment in a foreign stock (e.g., a stock trading on the London stock exchange) by a U.S. national (domestic investor). The domestic investor takes U.S. dollars, converts them to the foreign currency (e.g., British Pound) via the exchange rate (price of foreign currency in U.S. dollars) and then purchases the foreign stock using the foreign currency. When the stock is sold, the proceeds in the foreign currency must then be converted back to the domestic currency.

To be more precise, consider the information in the table below:

Month	U.S. $ Cost of 1 Pound £, $E_t^{US/UK}$	Value of British Shares, P_t^{UK}	Value in U.S. $, P_t^{US}
t - 1	$ 1.50	£40	$\$1.50 \times £40 = \$60$
t	$ 1.30	£45	$\$1.30 \times £45 = \$58.5$

Let P_t^{US}, P_t^{UK}, and $E_t^{US/UK}$ denote the U.S. dollar price of the asset, British Pound price of the asset, and currency exchange rate between U.S. dollars and British Pounds, at the end of month t, respectively. The U.S. dollar cost of one share of British stock in month $t-1$ is

$$P_{t-1}^{US} = E_{t-1}^{US/UK} \times P_{t-1}^{UK} = \$1.50 \times \ 40 = \$60,$$

and the U.S. dollar cost at time t is

$$P_t^{US} = E_t^{US/UK} \times P_t^{UK} = \$1.30 \times £45 = \$58.5.$$

The U.S. dollar rate of return is then

$$R_t^{US} = \frac{P_t^{US} - P_{t-1}^{US}}{P_{t-1}^{US}} = \frac{\$58.5 - \$60}{\$60} = -0.025.$$

Notice that

$$\begin{aligned} R_t^{US} &= \frac{E_t^{US/UK} \times P_t^{UK} - E_{t-1}^{US/UK} \times P_{t-1}^{UK}}{E_{t-1}^{US/UK} \times P_{t-1}^{UK}} \\ &= \frac{E_t^{US/UK} \times P_t^{UK}}{E_{t-1}^{US/UK} \times P_{t-1}^{UK}} - 1 \\ &= (1 + R_t^{US/UK})(1 + R_t^{UK}) - 1, \end{aligned} \tag{1.31}$$

where

$$\frac{E_t^{US/UK}}{E_{t-1}^{US/UK}} = 1 + R_t^{US/UK}, \ \frac{P_t^{UK}}{P_{t-1}^{UK}} = 1 + R_t^{UK}.$$

Hence,

$$1 + R_t^{US} = (1 + R_t^{US/UK})(1 + R_t^{UK}).$$

Here, $R_t^{US/UK}$ is the simple rate of return on foreign currency and R_t^{UK} is the simple rate of return on the British stock in British Pounds:

$$\begin{aligned} R_t^{US/UK} &= \frac{E_t^{US/UK} - E_{t-1}^{US/UK}}{E_{t-1}^{US/UK}} = \frac{\$1.30 - \$1.50}{\$1.50} = -0.133, \\ R_t^{UK} &= \frac{P_t^{UK} - P_{t-1}^{UK}}{P_{t-1}^{UK}} = \frac{£45 - £40}{£40} = 0.125. \end{aligned}$$

Hence, the gross return, $1 + R_t^{US}$, for the U.S. investor has two parts: (1) the gross return from an investment in foreign currency, $1 + R_t^{US/UK}$; and (2) the gross return of the foreign asset in the foreign currency, $1 + R_t^{UK}$:

$$\begin{aligned} 1 + R_t^{US} &= (1 + R_t^{US/UK})(1 + R_t^{UK}) \\ &= (0.867)(1.125) \\ &= 0.975. \end{aligned}$$

In this example, the U.S. investor loses money even though the British investment has a positive rate of return because of the large drop in the exchange rate. This example illustrates the importance of considering *exchange rate risk* when investing in foreign assets.

1.2.2.7 Annualizing returns

Very often returns over different horizons are annualized, i.e., converted to an annual return, to facilitate comparisons with other investments. The annualization process depends on the holding period of the investment and an implicit assumption about compounding. I illustrate with several examples using monthly returns. The same techniques work with different holding periods.

Example 1.11 (Compute annualized return from 12 one-month returns)**.** Suppose you have 12 one-month returns and you want to report an annual return on the investment. Then the annual gross and net returns are computed directly from the 12 one-month returns:

$$\begin{aligned} 1 + R_A = 1 + R_t(12) &= \frac{P_t}{P_{t-12}} = (1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-11}), \\ R_A &= R_t(12). \end{aligned}$$

In this case, no compounding is required to create an annual return. ■

Example 1.12 (Compute annualized return from a single one-month return)**.** Suppose you have a one-month investment in an asset with return R_t. What is the annualized return on this investment?

To answer this you must make an assumption about the future returns. For example, if you assume that you receive the same return $R = R_t$ every month for the year, then the gross annual return is:

$$1 + R_A = 1 + R_t(12) = (1 + R)^{12}.$$

That is, the *annual gross return* on the one-month investment is defined as the assumed common monthly return compounded for 12 months. Accordingly, the *annual net return* is then:

$$R_A = (1 + R)^{12} - 1.$$

■

In the above example, it is important to recognize that this annualization calculation assumes the same monthly return R for each of the 12 months. This can be highly inaccurate and misleading if the actual future monthly returns are much different than R. The following examples illustrate.

Example 1.13 (Compute annualized return from one-month return)**.** In the first example, the one-month return, R_t, on Microsoft stock was 5.88%. If you assume that you can get this return for 12 months then the annualized return is: $R_A = (1.0588)^{12} - 1 = 1.9850 - 1 = 0.9850$, or 98.50% per year. Pretty good! However, this is likely to be highly inaccurate since getting 5% every month for a year is highly unlikely for a risky stock like Microsoft.

■

Now, consider a two-month investment with return $R_t(2)$. If you assume that you receive the same two-month return $R(2) = R_t(2)$ for the next six two-month periods, then the gross and net annual returns are:

$$\begin{aligned} 1 + R_A &= (1 + R(2))^6, \\ R_A &= (1 + R(2))^6 - 1. \end{aligned}$$

Here the annual gross return is defined as the two-month return compounded for 6 months.

Example 1.14 (Compute annualized return from two-month return)**.** Suppose the two-month return, $R_t(2)$, on Microsoft stock is 12.5%. If you assume that you can get this two-month return for the next 6 two-month periods then the annualized return is: $R_A = (1.1250)^6 - 1 = 2.0273 - 1 = 1.0273$, or 102.73% per year. As with the previous example, this result is likely to be inaccurate as it is highly unlikely to get 12.5% every two months for the rest of the year.

■

Example 1.15 (Compute annualized return from 6 two-month returns)**.** Suppose your investment horizon is two years. That is, you start your investment at month $t-24$ and cash out at month t. The two-year gross return is then $1+R_t(24) = P_t/P_{t-24}$. What is the annual return on this two-year investment? The process is the same as computing the effective annual rate. To determine the annual return you solve the following relationship for R_A:

$$\begin{aligned}(1+R_A)^2 &= 1+R_t(24) \Rightarrow \\ R_A &= (1+R_t(24))^{1/2} - 1.\end{aligned}$$

In this case, the annual return is compounded twice to get the two-year return and the relationship is then solved for the annual return.

■

Example 1.16 (Compute annualized return from two-year return)**.** Suppose that the price of Microsoft stock 24 months ago is $P_{t-24} = \$50$ and the price today is $P_t = \$90$. The two-year gross return is $1+R_t(24) = \$90/\$50 = 1.8000$ which yields a two-year net return of $R_t(24) = 0.80 = 80\%$. The annual return for this investment is defined as:

$$R_A = (1.800)^{1/2} - 1 = 1.3416 - 1 = 0.3416$$

or 34.16% per year.

■

1.2.2.8 Average returns

For investments over a given horizon, it is often of interest to compute a measure of the average rate of return over the horizon. To illustrate, consider a sequence of monthly investments over n months with monthly returns $R_1, R_2, \ldots, R_n$. The $n-$month return is

$$R(n) = (1+R_1)(1+R_2)\cdots(1+R_n) - 1.$$

What is the average monthly return? There are two possibilities. The first is the *arithmetic average return*:

$$\bar{R}^{Arith} = \frac{1}{n}\left(R_1 + R_2 + \cdots + R_n\right). \tag{1.32}$$

The second is the *geometric average return*:

$$\bar{R}^{Geo} = \left[(1+R_1)(1+R_2)\cdots(1+R_n)\right]^{1/n} - 1. \tag{1.33}$$

Notice that the geometric average return is the monthly return which compounded monthly for n months gives the gross $n-$month return

$$(1 + \bar{R}^{Geo})^n = 1 + R(n).$$

As a measure of average investment performance, the geometric average return is preferred to the arithmetic average return. To see why, consider a two-month investment horizon with monthly returns $R_1 = 0.5$ and $R_2 = -0.5$. One dollar invested over these two months grows to

$$(1 + R_1)(1 + R_2) = (1.5)(0.5) = 0.75,$$

for a two-period return of $R(2) = 0.75 - 1 = -0.25$. Hence, the two-month investment loses 25%. The arithmetic average return is

$$\bar{R}^{Arith} = \frac{1}{2}(0.5 + (-0.5)) = 0.$$

As a measure of investment performance, this is misleading because the investment loses money over the two-month horizon. The average return indicates the investment breaks even. The geometric average return is

$$\bar{R}^{Geo} = [(1.5)(0.5)]^{1/2} - 1 = -0.134.$$

This is a more accurate measure of investment performance because it indicates that the investment eventually loses money. In addition, the compound two-month return using $\bar{R}^{Geo} = -0.134$ is equal to $R(2) = -0.25$.

While the arithmetic average return may not be the most appropriate measure of actual average investment performance, you shall see in the following chapters that it is a useful estimate of expected performance.

1.2.3 Continuously compounded returns

In this section I define continuously compounded returns from simple returns, and describe their properties.

1.2.3.1 One-period returns

Definition 1.7 (Continuously compounded return)**.** Let R_t denote the simple monthly return on an investment. The *continuously compounded monthly return*, r_t, is defined as:

$$r_t = \ln(1 + R_t) = \ln\left(\frac{P_t}{P_{t-1}}\right), \tag{1.34}$$

where $\ln(\cdot)$ is the natural log function[2].

To see why r_t is called the continuously compounded return, take the exponential of both sides of Equation 1.34 to give:

$$e^{r_t} = 1 + R_t = \frac{P_t}{P_{t-1}}.$$

Rearranging and solving for P_t gives

$$P_t = P_{t-1}e^{r_t} = P_{t-t}(1 + R_t),$$

so that r_t is the continuously compounded growth rate in prices between months $t-1$ and t. This is to be contrasted with R_t, which is the simple growth rate in prices between months $t-1$ and t without any compounding. As a result, r_t is always smaller than R_t.

Using the result $\ln(x/y) = \ln(x) - \ln(y)$ you can alternatively compute the continuously compounded return as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(P_t) - \ln(P_{t-1}) = p_t - p_{t-1}, \tag{1.35}$$

where $p_t = \ln(P_t)$. Hence, the continuously compounded monthly return, r_t, can be computed simply by taking the first difference of the natural logarithms of monthly prices.

Although r_t is always smaller than R_t, it is close to R_t when R_t is small. This is stated in the following proposition. ::: {#prp-ccReturnApprox name="Continuously compounded return approximation"} Suppose the simple return R_t is close to zero ($R_t \approx 0$). Then

$$r_t = \ln(1 + R_t) \approx R_t$$

. ::: The above result can be derived using a first order Taylor series approximation to the function $f(x) = \ln(1 + x)$. Recall, for a continuous and differentiable function $f(x)$ the *first order Taylor series approximation* of $f(x)$ about the point $x = x_0$ is

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \text{remainder}, \tag{1.36}$$

where $f'(x_0)$ denotes the derivative of $f(x)$ evaluated at x_0. Let $f(x) = \ln(1 + x)$ and set $x_0 = 0$. Then

Using the first order Taylor series approximation Equation 1.36, the remainder term is small when $x \approx 0$ and $\ln(1 + x) \approx x$ when $x \approx 0$.

Example 1.17 (Compute continuously compounded returns)**.** Using the price and return data from Example 1.6, the continuously compounded monthly return on Microsoft stock can be computed

[2]The continuously compounded return is always defined since asset prices, P_t, are always non-negative. Properties of logarithms and exponentials are discussed in the appendix to this chapter.

in two ways:

$$\begin{aligned} r_t &= \ln(1.0588) = 0.0571 \\ r_t &= \ln(90) - \ln(85) = 4.4998 - 4.4427 = 0.0571 \end{aligned}$$

Notice that r_t is slightly smaller than R_t. Using R, the calculations are:

```
c(log(1 + R0), log(P0) - log(Pm1))
#> [1] 0.0572 0.0572
```

Notice that the R function `log()` computes the natural logarithm $\ln(x)$.

■

Given a monthly continuously compounded return r_t, it is straightforward to solve back for the corresponding simple net return R_t:

$$R_t = e^{r_t} - 1. \tag{1.37}$$

Hence, nothing is lost by considering continuously compounded returns instead of simple returns. Continuously compounded returns are very similar to simple returns as long as the return is relatively small, which it generally will be for monthly or daily returns. Since R_t is bounded from below by -1, the smallest value for r_t is $r_t = ln(1-1) = -\infty$. This, however, does not mean that you could lose an infinite amount of money on an investment. The actual amount of money lost is determined by the simple return Equation 1.37.

Example 1.18 (Determine simple returns from continuously compounded returns)**.** In the previous example, the continuously compounded monthly return on Microsoft stock is $r_t = 5.71\%$. The simple net return is then: $R_t = e^{.0571} - 1 = 0.0588$.

■

For asset return modeling and statistical analysis it is often more convenient to use continuously compounded returns than simple returns. One reason is that continuously compounded returns take values on the entire real line, $-\infty < r_t < \infty$, whereas simple returns are only defined for values larger than -1, $-1 \leq R_t < \infty$. Another reason is due to the additivity property of multi-period continuously compounded returns discussed in the next sub-section.

1.2.3.2 Multi-period returns

The relationship between multi-period continuously compounded returns and one-period continuously compounded returns is more simple than the relationship between multi-period simple returns

and one-period simple returns. To illustrate, consider the two-month continuously compounded return defined as:

$$r_t(2) = \ln(1 + R_t(2)) = \ln\left(\frac{P_t}{P_{t-2}}\right) = p_t - p_{t-2}.$$

Taking exponentials of both sides and rearranging gives:

$$P_t = P_{t-2}e^{r_t(2)}$$

so that $r_t(2)$ is the continuously compounded growth rate of prices between months $t-2$ and t. Using $P_t/P_{t-2} = (P_t/P_{t-1}) \cdot (P_{t-1}/P_{t-2})$ and the fact that $\ln(x \cdot y) = \ln(x) + \ln(y)$ it follows that:

$$\begin{aligned} r_t(2) &= \ln\left(\frac{P_t}{P_{t-1}} \cdot \frac{P_{t-1}}{P_{t-2}}\right) \\ &= \ln\left(\frac{P_t}{P_{t-1}}\right) + \ln\left(\frac{P_{t-1}}{P_{t-2}}\right) \\ &= r_t + r_{t-1}. \end{aligned}$$

Hence the continuously compounded two-month return is just the sum of the two continuously compounded one-month returns. Recall, with simple returns the two-month return is a multiplicative (geometric) sum of two one-month returns.

Example 1.19 (Compute two-month continuously compounded returns)**.** Using the data from Example 1.7, the continuously compounded two-month return on Microsoft stock can be computed in two equivalent ways. The first way uses the difference in the logs of P_t and P_{t-2}:

$$r_t(2) = \ln(90) - \ln(80) = 4.4998 - 4.3820 = 0.1178.$$

The second way uses the sum of the two continuously compounded one-month returns. Here $r_t = \ln(90) - \ln(85) = 0.0571$ and $r_{t-1} = \ln(85) - \ln(80) = 0.0607$ so that:

$$r_t(2) = 0.0571 + 0.0607 = 0.1178.$$

Notice that $r_t(2) = 0.1178 < R_t(2) = 0.1250$. The vectorized calculations in R are:

```
p = log(P)
r = log(1 + R)
c(r[1] + r[2], sum(r), p[3] - p[1], diff(p, 2))
#> [1] 0.118 0.118 0.118 0.118
```

■

Previously, you saw that simple two-period returns were not symmetric. Continuously compounded

two-period returns, however, are symmetric. To see this, suppose $r_{t-1} = -0.5$ and $r_t = 0.5$. Notice that the implied simple returns are not symmetric: $R_{t-1} = e^{-0.5} - 1 = -0.3935$ and $R_t = e^{0.5} - 1 = 0.6487$. Then one dollar invested over two periods becomes:

$$\$1 \times (1 - 0.3935)(1 + 0.6487) = \$1$$

The continuously compounded k-month return is defined by:

$$r_t(k) = \ln(1 + R_t(k)) = \ln\left(\frac{P_t}{P_{t-k}}\right) = p_t - p_{t-k}.$$

Using similar manipulations to the ones used for the continuously compounded two-month return, you can express the continuously compounded k-month return as the sum of k continuously compounded monthly returns:

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j}. \tag{1.38}$$

The additivity of continuously compounded returns to form multiperiod returns is an important property for statistical modeling purposes.

1.2.3.3 Adjusting for dividends

The continuously compounded one-period return adjusted for dividends is defined by Equation 1.34, where R_t is computed using Equation 1.24[3].

Example 1.20 (Compute continuously compounded total return)**.** From example Example 1.8, the total simple return using Equation 1.24 is $R_t = 0.0707$. The continuously compounded total return is then:

$$r_t = \ln(1 + R_t) = \ln(1.0707) = 0.0683.$$

■

1.2.3.4 Adjusting for inflation

Adjusting continuously compounded nominal returns for inflation is particularly simple. The continuously compounded one-period real return is defined as:

$$r_t^{\text{Real}} = \ln(1 + R_t^{\text{Real}}). \tag{1.39}$$

[3]Show formula from Campbell, Lo and MacKinlay.

Using Equation 1.27, it follows that:

$$\begin{aligned} r_t^{\text{Real}} &= \ln\left(\frac{P_t}{P_{t-1}} \cdot \frac{CPI_{t-1}}{CPI_t}\right) \\ &= \ln\left(\frac{P_t}{P_{t-1}}\right) + \ln\left(\frac{CPI_{t-1}}{CPI_t}\right) \\ &= \ln(P_t) - \ln(P_{t-1}) + \ln(CPI_{t-1}) - \ln(CPI_t) \\ &= \ln(P_t) - \ln(P_{t-1}) - (\ln(CPI_t) - \ln(CPI_{t-1})) \\ &= r_t - \pi_t^c, \end{aligned} \tag{1.40}$$

where $r_t = \ln(P_t) - \ln(P_{t-1}) = \ln(1 + R_t)$ is the nominal continuously compounded one-period return and $\pi_t^c = \ln(CPI_t) - \ln(CPI_{t-1}) = \ln(1 + \pi_t)$ is the one-period continuously compounded growth rate in the general price level (continuously compounded one-period inflation rate). Hence, the real continuously compounded return is simply the nominal continuously compounded return minus the the continuously compounded inflation rate.

Example 1.21 (Compute continuously compounded real returns)**.** From Example Example 1.9, the nominal simple return is $R_t = 0.0588$, the monthly inflation rate is $\pi_t = 0.01$, and the real simple return is $R_t^{\text{Real}} = 0.0483$. Using Equation 1.39, the real continuously compounded return is:

$$r_t^{\text{Real}} = \ln(1 + R_t\text{Real}) = \ln(1.0483) = 0.047.$$

Equivalently, using (Equation 1.40) the real return may also be computed as:

$$r_t^{\text{Real}} = r_t - \pi_t c = \ln(1.0588) - \ln(1.01) = 0.047.$$

■

1.2.3.5 Adjusting for taxes

Using Equation 1.30, the continuously compounded return adjusted for taxes is

$$r_t^{Atax} = ln\left(R_t(1-\tau)\right) = ln(R_t) + ln(1-\tau) = r_t + ln(1-\tau)$$

1.2.3.6 Return on a foreign investment

Consider again, the US/UK investment example from sub-section Section 1.2.2.6. Using Equation 1.31, the continuously compounded US return is

$$\begin{aligned} r_t^{US} &= ln\left((1 + R_t^{US/UK})(1 + R_t^{UK})\right) \\ &= ln(1 + R_t^{US/UK}) + ln(1 + R_t^{UK}) \\ &= r_t^{US/UK} + r_t^{UK} \end{aligned} \tag{1.41}$$

where $r_t^{US/UK} = ln(1 + R_t^{US/UK})$ is the continuously compounded currency return and $r_t^{UK} = ln(1 + R_t^{UK})$ is the continuously compounded UK return.

1.2.3.7 Annualizing continuously compounded returns

Just as you annualized simple monthly returns, you can also annualize continuously compounded monthly returns. I illustrate using monthly continuously compounded returns. First, if your investment horizon is one year and you observe 12 monthly continuously compounded returns then the annual continuously compounded return is just the sum of the twelve monthly continuously compounded returns:

$$\begin{aligned} r_A = r_t(12) &= r_t + r_{t-1} + \cdots + r_{t-11} \\ &= \sum_{j=0}^{11} r_{t-j}. \end{aligned}$$

The average continuously compounded monthly return is defined as:

$$\overline{r}_m = \frac{1}{12}\sum_{j=0}^{11} r_{t-j}.$$

Notice that:

$$12 \cdot \overline{r}_m = \sum_{j=0}^{11} r_{t-j}$$

so that you may alternatively express r_A as:

$$r_A = 12 \cdot \overline{r}_m.$$

That is, the continuously compounded annual return is twelve times the average of the continuously compounded monthly returns.

Next, consider a one-month investment in an asset with continuously compounded return $r_t = r$. If you assume that you receive the same return r every month for the year, then the annual

continuously compounded return is just 12 times the monthly continuously compounded return:

$$r_A = r_t(12) = r + r + ... + r = 12 \cdot r.$$

If you have a two-month continuously compounded return of $r(2)$, and you assume that you get this same return every two months for the next year, then the annualized continuously compounded return is

$$r_A = 6 \cdot r(2).$$

Finally, suppose you have a 2-year continuously compounded return $r(24)$. Then the continuously compounded annual return can be defined using

$$2 \cdot r_A = r(24) \Longrightarrow r_A = \frac{r(24)}{2}.$$

1.3 Portfolios and Portfolio Returns

Consider an investment in two assets named asset A and asset B (e.g. Amazon and Boeing stock). Let V_A and V_B denote the dollar amounts invested in assets A and B, respectively. you call such an investment in multiple assets a *portfolio*. The total dollar amount invested in the portfolio is $V_p = V_A + V_B$. The shares of dollars invested in assets A and B are:

$$x_A = \frac{V_A}{V_p}, x_B = \frac{V_B}{V_p}.$$

Notice that $x_A + x_B = 1$ by construction (100% of all wealth is invested in the two assets).

In this book, the collection of investment shares (x_A, x_B) and initial wealth invested V_p defines a portfolio. If V_p is not specified then assume that $V_p = \$1$. For example, one portfolio may be $(x_A = 0.5, x_B = 0.5)$ (equally weighted portfolio) and another may be $(x_A = 0, x_B = 1)$ (everything invested in asset B). Negative values for x_A or x_B are possible and represent short sales (to be discussed in later chapters). Given x_A, x_B, and V_p, the dollar amounts invested in assets A and B are:

$$V_A = x_A \times V_p, V_B = x_B \times V_p.$$

The rate of return on a portfolio can be represented as a share weighted average of the rates of returns on the assets in the portfolio. To see this, let R_A and R_B denote the simple one-period returns on assets A and B. You want to determine the simple one-period return on the portfolio defined by (x_A, x_B) and initial wealth V_p. To do this, note that at the end of the holding period, the investments in assets A and B are worth $V_p \times x_A(1 + R_A)$ and $V_p \times x_B(1 + R_B)$, respectively.

Hence, at the end of the holding period the portfolio is worth:

$$V_p \times [x_A(1 + R_A) + x_B(1 + R_B)] = V_p \times (1 + R_p).$$

Hence, $x_A(1 + R_A) + x_B(1 + R_B) = 1 + R_p$ defines the *portfolio gross return.* The portfolio gross return is equal to a weighted average of the gross returns on assets A and B, where the weights are the portfolio shares x_A and x_B.

To determine the portfolio rate of return, re-write the portfolio gross return as:

$$1 + R_p = x_A + x_B + x_A R_A + x_B R_B = 1 + x_A R_A + x_B R_B,$$

since $x_A + x_B = 1$ by construction. Then the *portfolio rate of return* is:

$$R_p = x_A R_A + x_B R_B,$$

which is equal to a weighted average of the simple returns on assets A and B, where the weights are the portfolio shares x_A and x_B.

In the portfolio rate of return, the components $x_A R_A$ and $x_B R_B$ are the *contributions* of assets A and B to the portfolio return, respectively. These contributions add up to the total portfolio return R_p.

Example 1.22 (Compute portfolio return)**.** Consider a portfolio of Microsoft and Starbucks stock in which you initially purchase ten shares of each stock at the end of month $t - 1$ at the prices $P_{\text{msft},t-1} = \$85$ and $P_{\text{sbux},t-1} = \$30$, respectively. The initial value of the portfolio is $V_{t-1} = 10 \times \$85 + 10 \times \$30 = \$1,150$. The portfolio shares are

$$x_{\text{msft}} = \frac{\$850}{\$1,150} = 0.7391,\ x_{\text{sbux}} = \frac{\$300}{\$1,150} = 0.2609.$$

Suppose at the end of month t, $P_{\text{msft},t} = \$90$ and $P_{\text{sbux},t} = \$28$. Assuming that Microsoft and Starbucks do not pay a dividend between periods $t - 1$ and t, the one-period returns on the two stocks are:

$$R_{\text{msft},t} = \frac{\$90 - \$85}{\$85} = 0.0588,\ R_{\text{sbux},t} = \frac{\$28 - \$30}{\$30} = -0.0667.$$

The one-month rate of return on the portfolio is then:

$$R_{p,t} = (0.7391)(0.0588) + (0.2609)(-0.0667) = 0.02609,$$

The contributions of assets A and B to the portfolio return are:

$$x_A R_A = (0.7391)(0.0588) = 0.0435,\ x_B R_B = (0.2609)(-0.0667) = -0.0174.$$

Asset A contributes positively and asset B contributes negatively to the portfolio return. The portfolio value at the end of month t is:

$$V_t = V_{t-1}(1 + R_{p,t}) = \$1,150 \times (1.02609) = \$1,180$$

In R, the portfolio calculations are:

```
P.msft = c(85, 90)
P.sbux = c(30, 28)
V = P.msft[1]*10 + P.sbux[1]*10
x.msft = 10*P.msft[1]/V
x.sbux = 10*P.sbux[1]/V
R.msft = (P.msft[2] - P.msft[1])/P.msft[1]
R.sbux = (P.sbux[2] - P.sbux[1])/P.sbux[1]
R.p = x.msft*R.msft + x.sbux*R.sbux
V1 = V*(1 + R.p)
c(R.p, x.msft*R.msft, x.sbux*R.sbux, V1)
#> [1]    0.0261    0.0435   -0.0174 1180.0000
```

■

In general, for a portfolio of n assets with investment shares x_i such that $x_1 + \cdots + x_n = 1$, the one-period portfolio gross and simple returns are defined as:

$$1 + R_{p,t} = \sum_{i=1}^{n} x_i(1 + R_{i,t}) \tag{1.42}$$

$$R_{p,t} = \sum_{i=1}^{n} x_i R_{i,t}. \tag{1.43}$$

where $R_{i,t}$ is the one-period simple return on asset i. Asset i's contribution to the portfolio return is

$$\text{ctr}_{i,p} = x_i R_{i,t}. \tag{1.44}$$

and

$$\sum_{i=i}^{n} \text{ctr}_{i,p} = \sum_{i=i}^{n} x_i R_{i,t} = R_{p,t}.$$

1.3.1 Multiperiod portfolio returns and rebalancing

How to compute multiperiod portfolio returns depends on the assumptions made about the portfolio weights each period. Several scenarios exist:

1. Portfolio weights from the initial portfolio are allowed to change over time as prices of the underlying assets change over time. In this case no rebalancing of the portfolio is done. This is called a *buy-and-hold* portfolio.
2. Portfolio weights remain constant over time, which implies that the portfolio is rebalanced at every time period (associated with holding period) to maintain constant weights.
3. Portfolio weights from the initial portfolio are rebalanced at specific time intervals that are different than the holding period (e.g. rebalance quarterly for monthly holding periods). This is a hybrid of scenarios 1 and 2.
4. Portfolio weights are actively changed at each time period associated with the holding period. This is called an *actively managed portfolio*.

How to deal with the above scenarios is best illustrated through examples.

1.3.1.1 No rebalancing of portolio weights

Prices and shares framework

I illustrate the buy-and-hold scenario 1 using a simple portfolio consisting of two assets A and B invested for three months starting at time $t-3$ in a framework where I know the asset prices each period and the initial amounts invested in each asset[4]. In particular, I assume that the initial portfolio consists of $s_A = 100$ shares of asset A and $s_B = 50$ shares of asset B. The end-of-month prices and values ($V_{i,t} = s_i \times P_{i,t}, i = A, B$) of the assets as well as the end-of-month values of the portfolio are given by

```
s.A = 100
s.B = 50
P.A = c(5,7,6,7)
V.A = s.A*P.A
P.B = c(10, 11, 12, 8)
V.B = s.B*P.B
V.p = V.A + V.B
table.vals = cbind(P.A, V.A, P.B, V.B, V.p)
rownames(table.vals) = c("t-3", "t-2", "t-1", "t")
table.vals
#>     P.A V.A P.B V.B  V.p
#> t-3   5 500  10 500 1000
#> t-2   7 700  11 550 1250
#> t-1   6 600  12 600 1200
#> t     7 700   8 400 1100
```

[4]This example is inspired by the `return.portfolio` vignette in **PerformanceAnalytics**).

In the above table the time index t represents the end-of-month t. Slightly abusing notation, I can also say that the time index $t-1$ represents the beginning of month t. This is important because for the portfolio calculations the end-of-month values $V_{A,t-1}$, $V_{B,t-1}$ and $V_{p,t-1}$ are used to calculate the portfolio weights to be applied in month t. That is,

$$x_{A,t-1} = \frac{V_{A,t-1}}{V_{p,t-1}}, \; x_{B,t-1} = \frac{V_{B,t-1}}{V_{p,t-1}}$$

represent the portfolio weights for assets A and B at the beginning of month t. Then, the portfolio return over month t is computed as

$$\begin{aligned} R_{p,t} = \%\Delta V_{p,t} &= \frac{V_{p,t} - V_{p,t-1}}{V_{p,t-1}} && (1.45) \\ &= x_{A,t-1} R_{A,t} + x_{B,t-1} R_{B,t}, && (1.46) \\ & && (1.47) \end{aligned}$$

where

$$\begin{aligned} R_{A,t} = \%\Delta V_{A,t} &= \frac{V_{A,t} - V_{A,t-1}}{V_{A,t-1}} && (1.48) \\ &= \%\Delta P_{A,t} = \frac{P_{A,t} - P_{A,t-1}}{P_{A,t-1}} && (1.49) \\ R_{B,t} = \%\Delta V_{B,t} &= \frac{V_{B,t} - V_{B,t-1}}{V_{B,t-1}} && (1.50) \\ &= \%\Delta P_{B,t} = \frac{P_{B,t} - P_{B,t-1}}{P_{B,t-1}}. && (1.51) \end{aligned}$$

The portfolio return, $R_{p,t}$, is still a share weighted average of the individual asset returns but with time varying portfolio weights $x_{A,t-1}$ and $x_{B,t-1}$.

The R calculations for returns are:

```
R.A = (V.A[2:4] - V.A[1:3])/V.A[1:3]
R.B = (V.B[2:4] - V.B[1:3])/V.B[1:3]
R.p = (V.p[2:4] - V.p[1:3])/V.p[1:3]
table.returns = cbind(R.A, R.B, R.p)
rownames(table.returns) =  c("t-2", "t-1", "t")
table.returns
#>         R.A     R.B     R.p
#> t-2   0.400  0.1000  0.2500
#> t-1 -0.143  0.0909 -0.0400
#> t     0.167 -0.3333 -0.0833
```

The R calculations for the weights are:

```
x.A = V.A/V.p
x.B = V.B/V.p
table.weights = cbind(x.A, x.B)
rownames(table.weights) = c("t-3", "t-2", "t-1", "t")
table.weights
#>       x.A   x.B
#> t-3 0.500 0.500
#> t-2 0.560 0.440
#> t-1 0.500 0.500
#> t   0.636 0.364
```

Here, the initial portfolio formed at the end of month $t-3$ is an equally weighted portfolio of assets A and B. However, the portfolio weights at the end of month $t-2$ are not 0.5 and 0.5 as the prices of assets A and B have changed over the month. Asset A's price has gone up more in percentage terms ($R_{A,t-2} = 40\%$) than asset B's price ($R_{B,t-2} = 10\%$) so the portfolio weight in asset A at the beginning of $t-1$ ($x_{A,t-2} = 0.56$), has become bigger than the portfolio weight in asset B ($x_{B,t-2} = 0.44$).

For the calculation of the portfolio returns, the asset weights at time $t-3$ apply to the asset returns at time $t-2$, the weights at time $t-2$ apply to the returns at time $t-1$, and the weights at time $t-1$ apply to the returns at time t:

```
R.p = x.A[1:3]*R.A + x.B[1:3]*R.B
R.p
#> [1]  0.2500 -0.0400 -0.0833
```

The asset contributions to the portfolio return at time t are

$$\text{ctr}_{A,t} = \frac{\Delta V_{A,t}}{V_{p,t-1}} = x_{A,t-1} \times R_{A,t} \tag{1.52}$$

$$\text{ctr}_{B,t} = \frac{\Delta V_{B,t}}{V_{p,t-1}} = x_{B,t-1} \times R_{B,t} \tag{1.53}$$

The R calculations for the contributions derived from values are:

```
ctr.A = (V.A[2:4] - V.A[1:3])/V.p[1:3]
ctr.B = (V.B[2:4] - V.B[1:3])/V.p[1:3]
table.ctr = cbind(ctr.A, ctr.B, ctr.A+ctr.B)
colnames(table.ctr)[3] = "R.p"
rownames(table.ctr) = c("t-2", "t-1", "t")
```

```
table.ctr
#>       ctr.A  ctr.B     R.p
#> t-2  0.2000  0.050  0.2500
#> t-1 -0.0800  0.040 -0.0400
#> t    0.0833 -0.167 -0.0833
```

The R calculations for the contributions derived from returns are:

```
ctr.A = x.A[1:3]*R.A
ctr.B = x.B[1:3]*R.B
table.ctr = cbind(ctr.A, ctr.B, R.p)
rownames(table.ctr) = c("t-2", "t-1", "t")
table.ctr
#>       ctr.A  ctr.B     R.p
#> t-2  0.2000  0.050  0.2500
#> t-1 -0.0800  0.040 -0.0400
#> t    0.0833 -0.167 -0.0833
```

Returns and weights framework

The above calculations are based on knowing asset prices and initial shares purchased for each asset. In many situations, you only have return information and initial portfolio weights on each asset and the initial value of the portfolio. Using the above example, the initial portfolio weights are $x_{A,t-3} = x_{B,t-3} = 0.5$ and the initial value of the portfolio is $V_{p,t-3} = 1000$. You also know the returns, $R_{i,t}$, each period for the two assets:

```
table.returns[, c("R.A", "R.B")]
#>        R.A     R.B
#> t-2  0.400  0.1000
#> t-1 -0.143  0.0909
#> t    0.167 -0.3333
```

You can deduce the initial asset values using

$$V_{A,t-3} = x_{A,t-3} \times V_{p,t-3} = 0.5 \times 1000 = 500, \tag{1.54}$$

$$V_{B,t-3} = x_{B,t-3} \times V_{p,t-3} = 0.5 \times 1000 = 500. \tag{1.55}$$

The end-of-month asset values for $i = A, B$ are computed using

$$\begin{aligned} V_{i,t-2} &= V_{i,t-3} \times (1 + R_{i,t-2}), \\ V_{i,t-1} &= V_{i,t-2} \times (1 + R_{i,t-1}) = V_{i,t-3} \times (1 + R_{i,t-2})(1 + R_{i,t-1}), \\ V_{i,t} &= V_{i,t-1} \times (1 + R_{i,t}) = V_{i,t-3} \times (1 + R_{i,t-2})(1 + R_{i,t-1})(1 + R_{i,t}). \end{aligned}$$

The end-of-month portfolio values are the sum of the end-of-month asset values. The R calculations are:

```
V.A = V.B = V.p = rep(0, 4)
V.p[1] = 1000
V.A[1] = V.B[1] = 0.5*V.p[1]
V.A[2:4] = cumprod(1+R.A)*V.A[1]
V.B[2:4] = cumprod(1+R.B)*V.B[1]
V.p[2:4] = V.A[2:4] + V.B[2:4]
table.vals = cbind(V.A, V.B, V.p)
rownames(table.vals) = c("t-3", "t-2", "t-1", "t")
table.vals
#>     V.A V.B  V.p
#> t-3 500 500 1000
#> t-2 700 550 1250
#> t-1 600 600 1200
#> t   700 400 1100
```

The portfolio weights at the end-of-month t are computed as

$$x_{i,t} = \frac{V_{i,t}}{V_{p,t}},$$

and the R calculations are:

```
x.A = V.A/V.p
x.B = V.B/V.p
table.weights = cbind(x.A, x.B)
rownames(table.weights) = c("t-3", "t-2", "t-1", "t")
table.weights
#>       x.A   x.B
#> t-3 0.500 0.500
#> t-2 0.560 0.440
#> t-1 0.500 0.500
```

```
#> t   0.636 0.364
```

The portfolio return over month t can be computed using

$$R_{p,t} = \%\Delta V_{p,t} = \frac{V_{p,t} - V_{p,t-1}}{V_{p,t-1}} \tag{1.56}$$

$$= x_{A,t-1} \times R_{A,t} + x_{B,t-1} \times R_{B,t} \tag{1.57}$$

The R calculations for returns based on values are:

```
diff(table.vals[, "V.p"])/table.vals[1:3, "V.p"]
#>     t-2     t-1       t
#>  0.2500 -0.0400 -0.0833
```

The calculation for returns based on weights is

```
x.A[1:3]*R.A + x.B[1:3]*R.B
#> [1]  0.2500 -0.0400 -0.0833
```

1.3.1.2 Constant portfolio weights

If the initial portfolio weights are to be held constant over time then the portfolio typically needs to be *rebalanced* at each time period to adjust the portfolio weights back to the initial weights. Given the constant portfolio weights x_A and x_B the return on the portfolio at every time period is given by

$$R_{p,t} = x_A R_{A,t} + x_B R_{B,t}.$$

The rebalancing of portfolio weights can be illustrated with the example from the previous subsection. The initial portfolio is an equally weighted portfolio, $x_{A,t-3} = x_{B,t-3} = 0.5$. To have constant portfolio weights the portfolio will need to be rebalanced each period so that $x_{A,t-2} = x_{A,t-1} = x_{A,t} = 0.5$ and $x_{B,t-2} = x_{B,t-1} = x_{B,t} = 0.5$. Consider the portfolio at the end-of-month $t-2$ where

$$V_{A,t-2} = V_{A,t-3} \times (1 + R_{A,t-2}) = 500 \times (1 + 0.40) = 700 \tag{1.58}$$

$$V_{B,t-2} = V_{B,t-3} \times (1 + R_{B,t-2}) = 500 \times (1 + 0.10) = 550 \tag{1.59}$$

$$V_{p,t-2} = V_{A,t-2} + V_{B,t-2} = 700 + 550 = 1250 \tag{1.60}$$

$$x_{A,t-2} = \frac{V_{A,t-2}}{V_{p,t-2}} = \frac{700}{1250} = 0.56 \tag{1.61}$$

$$x_{B,t-2} = \frac{V_{B,t-2}}{V_{p,t-2}} = \frac{550}{1250} = 0.44 \tag{1.62}$$

The end-of-month portfolio weights are $x_{A,t-2} = 0.56$ and $x_{B,t-2} = 0.44$, which are not equal to 0.5. Since $V_{p,t-2} = 1250$, to rebalance the portfolio so that $x_{A,t-2} = x_{B,t-2} = 0.5$ you need $V_{A,t-2} = V_{B,t-2} = V_{p,t-2}/2 = 625$. This requires selling $700 - 625 = 75$ of asset A and purchasing 75 of asset B at the end-of-month $t-2$. Then, at the beginning of month $t-1$ you have $V_{A,t-2} = V_{B,t-2} = 625$ and $x_{A,t-2} = x_{B,t-2} = 0.5$.

At the end-of-month $t-1$, for the rebalanced portfolio you have

$$V_{A,t-1} = V_{A,t-2} \times (1 + R_{A,t-1}) = 625 \times (1 - 0.1429) = 535.69 \tag{1.63}$$
$$V_{B,t-1} = V_{B,t-2} \times (1 + R_{B,t-1}) = 625 \times (1 + 0.0909) = 681.81 \tag{1.64}$$
$$V_{p,t-1} = V_{A,t-2} + V_{B,t-2} = 535.69 + 681.81 = 1217.50 \tag{1.65}$$
$$x_{A,t-1} = \frac{V_{A,t-1}}{V_{p,t-1}} = \frac{535.69}{1217.50} = 0.44 \tag{1.66}$$
$$x_{B,t-1} = \frac{V_{B,t-1}}{V_{p,t-1}} = \frac{681.81}{1217.50} = 0.56 \tag{1.67}$$

The end-of-month portfolio weights are $x_{A,t-1} = 0.44$ and $x_{B,t-1} = 0.56$. Since $V_{p,t-1} = 1217.50$, to rebalance the portfolio you need $V_{A,t-1} = V_{B,t-1} = 608.75$. This requires selling 73.06 of asset B and purchasing 73.06 of asset A and the end-of-month t-1. Then, at the beginning of month t you have $V_{A,t-1} = V_{B,t-1} = 608.75$ and $x_{A,t-1} = x_{B,t-1} = 0.5$.

Finally, at the end-of-month t, for the rebalanced portfolio you have

$$V_{A,t} = V_{A,t-1} \times (1 + R_{A,t}) = 608.75 \times (1 + 0.1667) = 710.23 \tag{1.68}$$
$$V_{B,t} = V_{B,t-1} \times (1 + R_{B,t}) = 608.75 \times (1 - 0.3333) = 405.85 \tag{1.69}$$
$$V_{p,t} = V_{A,t-1} + V_{B,t-1} = 710.23 + 405.85 = 1116.08 \tag{1.70}$$
$$x_{A,t} = \frac{V_{A,t}}{V_{p,t}} = \frac{710.23}{1116.08} = 0.64 \tag{1.71}$$
$$x_{B,t} = \frac{V_{B,t}}{V_{p,t}} = \frac{405.85}{1116.08} = 0.36 \tag{1.72}$$

The end-of-month portfolio weights are $x_{A,t} = 0.64$ and $x_{B,t} = 0.36$. Since $V_{p,t} = 1116.08$, to rebalance the portfolio you need $V_{A,t} = V_{B,t} = 558.04$. This requires selling 152.19 of asset A and purchasing 152.19 of asset B and the end-of-month t. Then, at the beginning of month $t+1$ you have $V_{A,t} = V_{B,t} = 558.04$ and $x_{A,t} = x_{B,t} = 0.5$.

The above calculations in R are:

```
value.table = matrix(0, 4, 3)
rownames(value.table) = c("t-3", "t-2", "t-1", "t")
colnames(value.table) = c("V.A", "V.B", "V.p")
value.table["t-3", ] = c(V.A[1], V.B[1], V.p[1])
```

```
for (t in 2:4) {
  value.table[t, "V.A"] = value.table[t-1, "V.A"]*(1 + R.A[t-1])
  value.table[t, "V.B"] = value.table[t-1, "V.B"]*(1 + R.B[t-1])
  value.table[t, "V.p"] = value.table[t, "V.A"] + value.table[t, "V.B"]
  RB = max(value.table[t, "V.A"], value.table[t, "V.B"]) - value.table[t, "V.p"]/2
  if (value.table[t, "V.A"] > value.table[t, "V.B"]) {
    value.table[t, "V.A"] = value.table[t, "V.A"] - RB
    value.table[t, "V.B"] = value.table[t, "V.B"] + RB
    }
  else{
    value.table[t, "V.A"] = value.table[t, "V.A"] + RB
    value.table[t, "V.B"] = value.table[t, "V.B"] - RB
  }
  value.table[t, "V.p"] = value.table[t, "V.A"] + value.table[t, "V.B"]
}
value.table
#>     V.A V.B  V.p
#> t-3 500 500 1000
#> t-2 625 625 1250
#> t-1 609 609 1218
#> t   558 558 1116
```

Notice that the rebalanced portfolio slightly outperforms the buy-and-hold portfolio.

The portfolio return each period can be computed using $R_{p,t} = \%\Delta V_{p,t}$:

```
diff(value.table[, "V.p"])/value.table[1:3, "V.p"]
#>     t-2     t-1       t
#>  0.2500 -0.0260 -0.0833
```

It can also be computed using $R_{p,t} = 0.5 \times R_{A,t} + 0.5 \times R_{B,t}$:

```
0.5*R.A + 0.5*R.B
#> [1]  0.2500 -0.0260 -0.0833
```

Here, you see that the rebalanced portfolio loses less money over month $t-1$ than the buy-and-hold portfolio.

In the above example, at each rebalancing date you sell the asset that has the best performance (sell high) and you buy the asset that has the worst performance (buy low). Hence, rebalancing forces the sage investing advice: "buy low and sell high".

1.3.1.3 Rebalance portfolio at specified dates

When the portfolio is rebalanced to a fixed initial portfolio, such as an equally weighted portfolio, on specific dates the portfolio return calculations become a hybrid of those used for scenarios 1 and 2. The portfolio return is computed as

$$R_{p,t} = x_{A,t-1} \times R_{A,t} + x_{B,t-1} \times R_{B,t},$$

where $x_{i,t-1}(i = A, B)$ are equal to the initial portfolio weights on the rebalancing dates and is computed like the buy-and-hold weights in between the rebalancing dates.

1.3.1.4 Actively managed portfolio

In an actively managed portfolio, the portfolio weights are chosen by the portfolio manager at each time period. Many mutual funds are managed in this way. The return on the actively managed portfolio is computed as

$$R_{p,t} = x_{A,t-1} \times R_{A,t} + x_{B,t-1} \times R_{B,t}$$

where $x_{A,t-1}$ and $x_{B,t-1}$ are the portolio weights chosen by the portfolio manager at the beginning of time t.

1.3.2 Continuously Compounded Portfolio Returns

The continuously compounded portfolio return is defined by Equation 1.34, where R_t is computed using the portfolio return Equation 1.43. However, notice that:

$$r_{p,t} = \ln(1 + R_{p,t}) = \ln\left(1 + \sum_{i=1}^{n} x_i R_{i,t}\right) \neq \sum_{i=1}^{n} x_i r_{i,t}, \tag{1.73}$$

where $r_{i,t}$ denotes the continuously compounded one-period return on asset i. Hence, the continuously compounded portfolio return is not a share weighted average of the individual asset continuously compounded returns. However, if the portfolio return $R_{p,t} = \sum_{i=1}^{n} x_i R_{i,t}$ is not too large then $r_{p,t} \approx R_{p,t}$ otherwise, $R_{p,t} > r_{p,t}$.

Example 1.23 (Compute continuously compounded portfolio returns)**.** Consider a portfolio of Microsoft and Starbucks stock with $x_{\text{msft}} = 0.25$, $x_{\text{sbux}} = 0.75$, $R_{\text{msft},t} = 0.0588$, $R_{\text{sbux},t} = -0.0503$ and $R_{p,t} = -0.02302$. Using (Equation 1.73), the continuously compounded portfolio return is: $r_{p,t} = \ln(1 - 0.02302) = \ln(0.977) = -0.02329$. Using $r_{\text{msft},t} = \ln(1 + 0.0588) = 0.0572$ and $r_{\text{sbux},t} = \ln(1 - 0.0503) = -0.05161$, notice that: $x_{\text{msft}} r_{\text{msft}} + x_{\text{sbux}} r_{\text{sbux}} = -0.02442 \neq r_{p,t}$

■

Because the continuously compounded portfolio return is not a share weighted average of the individual asset continuously compounded returns, the analysis of portfolios is typically performed using simple returns and not continuously compounded returns.

1.4 Return Calculations with Data in R

This section discusses representing time series data in R using `xts` objects, the calculation of returns from historical prices in R, as well as the graphical display of prices and returns.

1.4.1 Representing time series data using `xts` objects

The examples in this section are based on the daily adjusted closing price data for Microsoft and Starbucks stock over the period January 4, 1993 through December 31, 2014[5]. These data are available as the `xts` objects `msftDailyPrices` and `sbuxDailyPrices` in the R package **introcompfinr** [6]

```
head(cbind(msftDailyPrices,sbuxDailyPrices), 3)
#>            MSFT SBUX
#> 1993-01-04 1.89 1.08
#> 1993-01-05 1.92 1.10
#> 1993-01-06 1.98 1.12
```

```
class(msftDailyPrices)
#> [1] "xts" "zoo"
```

There are many different ways of representing a time series of data in R. For financial time series `xts` (extensible time series) objects from the **xts** package are especially convenient and useful. An `xts` object consists of two pieces of information: (1) a `matrix` of numeric data with different time series in the columns, (2) an R object representing the common time indexes associated with the rows of the data. `xts` objects extend and enhance the `zoo` class of time series objects from the **zoo** package written by Achim Zeileis (zoo stands for Z's ordered observations).

The `matrix` of time series data can be extracted from the `xts` object using the **xts** function `coredata()`:

[5] These are prices that are adjusted for dividends and stock splits. That is, any dividend payments have been included in the prices and historical prices have been divided by the split ratio associated with any stock splits.

[6] See the Appendix Working with Time Series Data in R for an overview of `xts` objects and Zivot (2016) for a comprehensive coverage.

```
matrix.data = coredata(msftDailyPrices)
head(matrix.data, 3)
#>      MSFT
#> [1,] 1.89
#> [2,] 1.92
#> [3,] 1.98
```

The resulting `matrix` of data does not have any date information. The date index can be extracted using the **xts** function `index()`:

```
date.index = index(msftDailyPrices)
head(date.index, 3)
#> [1] "1993-01-04" "1993-01-05" "1993-01-06"
```

The object `date.index` is of class `Date`:

```
class(date.index)
#> [1] "Date"
```

When a `Date` object is printed, the date is displayed in the format `yyyy-mm-dd`. However, internally each date represents the number of days since January 1, 1970:

```
as.numeric(date.index[1])
#> [1] 8404
```

Here, `1993-01-04` (January 4, 1993) is 8,404 days after January 1, 1970. This allows for simple *date arithmetic* (e.g. adding and subtracting dates, etc). More sophisticated date arithmetic is available using the functions in the **lubridate** package.

There are several advantages of using `xts` objects to represent financial time series data. Most financial time series do not follow an equally spaced regular calendar. The time index of a `xts` object can be any strictly increasing date sequence which can match the times and dates for which assets trade on an exchange. For example, assets traded on the main US stock exchanges (e.g. NYSE and NASDAQ) trade only on weekdays between 9:30AM and 4:00PM Eastern Standard Time and do not trade on a number of holidays.

Another advantage of `xts` objects is that you can easily extract observations at or between specific dates. For example, to extract the daily price on January 3, 2014 use

```
msftDailyPrices["2014-01-03"]
#>            MSFT
#> 2014-01-03 35.7
```

To extract prices between January 3, 2014 and January 7, 2014, use

```
msftDailyPrices["2014-01-03::2014-01-07"]
#>            MSFT
#> 2014-01-03 35.7
#> 2014-01-06 34.9
#> 2014-01-07 35.2
```

Or to extract all of the prices for January, 2014 use:

```
msftDailyPrices["2014-01"]
#>            MSFT
#> 2014-01-02 35.9
#> 2014-01-03 35.7
#> 2014-01-06 34.9
#> 2014-01-07 35.2
#> 2014-01-08 34.6
#> 2014-01-09 34.3
#> 2014-01-10 34.8
#> 2014-01-13 33.8
#> 2014-01-14 34.6
#> 2014-01-15 35.5
#> 2014-01-16 35.6
#> 2014-01-17 35.2
#> 2014-01-21 35.0
#> 2014-01-22 34.7
#> 2014-01-23 34.9
#> 2014-01-24 35.6
#> 2014-01-27 34.8
#> 2014-01-28 35.0
#> 2014-01-29 35.4
#> 2014-01-30 35.6
#> 2014-01-31 36.6
```

Two or more **xts** objects can be merged together and aligned to a common date index using the **xts** function `merge()`:

```
msftSbuxDailyPrices = merge(msftDailyPrices, sbuxDailyPrices)
head(msftSbuxDailyPrices, n=3)
#>            MSFT SBUX
```

```
#> 1993-01-04 1.89 1.08
#> 1993-01-05 1.92 1.10
#> 1993-01-06 1.98 1.12
```

If two or more `xts` objects have a common date index then they can be combined using `cbind()`.

1.4.1.1 Basic calculations on `xts` objects.

Because `xts` objects typically contain a matrix of numeric data, many R functions that operate on matrices also operate on `xts` objects. For example, to transform `msftDailyPrices` to log prices use

```
msftDailyLogPrices = log(msftDailyPrices)
head(msftDailyLogPrices, 3)
#>             MSFT
#> 1993-01-04 0.637
#> 1993-01-05 0.652
#> 1993-01-06 0.683
```

If two or more `xts` objects have the same time index then they can be added, subtracted, multiplied, and divided. For example,

```
msftPlusSbuxDailyPrices = msftDailyPrices + sbuxDailyPrices
colnames(msftPlusSbuxDailyPrices) = "MSFT+SBUX"
head(msftPlusSbuxDailyPrices,3)
#>            MSFT+SBUX
#> 1993-01-04      2.97
#> 1993-01-05      3.02
#> 1993-01-06      3.10
```

To compute the average of the Microsoft daily prices use

```
mean(msftDailyPrices)
#> [1] 19.9
```

If an R function, for some unknown reason, is not operating correctly on an `xts` object first extract the data using `coredata()` and then call the R function. For example,

```
mean(coredata(msftDailyPrices))
#> [1] 19.9
```

1.4.1.2 Changing the frequency of an `xts` object

Data at the daily frequency is the highest frequency of data considered in this book. However, in many of the examples you want to use data at the weekly or monthly frequency. The conversion of data at a daily frequency to data at a monthly frequency is easy to do with `xts` objects. For example, end-of-month prices can be extracted from the daily prices using the **xts** function `to.monthly()`:

```
msftMonthlyPrices = to.monthly(msftDailyPrices, OHLC=FALSE)
sbuxMonthlyPrices = to.monthly(sbuxDailyPrices, OHLC=FALSE)
msftSbuxMonthlyPrices = to.monthly(msftSbuxDailyPrices, OHLC=FALSE)
head(msftMonthlyPrices, 3)
#>          MSFT
#> Jan 1993 1.92
#> Feb 1993 1.85
#> Mar 1993 2.06
```

By default, `to.monthly()` extracts the data for the last day of the month and creates a **zoo** `yearmon` date index. For monthly data, the `yearmon` date index is convenient for printing and plotting as the month and year are nicely printed. To preserve the `Date` class of the time index and show the end-of-month date, use the optional argument `indexAt = "lastof"` in the call to `to.monthly()`:

```
head(to.monthly(msftDailyPrices, OHLC=FALSE, indexAt="lastof"), 3)
#>            MSFT
#> 1993-01-31 1.92
#> 1993-02-28 1.85
#> 1993-03-31 2.06
```

In the above calls to `to.monthly()`, the optional argument `OHLC=FALSE` prevents the creation of open, high, low, and closing prices for the month.

In a similar fashion, you can extract end-of-week prices using the `xts` function `to.weekly()`:

```
msftWeeklyPrices = to.weekly(msftDailyPrices, OHLC=FALSE)
head(msftWeeklyPrices, 3)
#>            MSFT
#> 1993-01-08 1.94
#> 1993-01-15 2.00
#> 1993-01-22 1.99
```

Here, the weekly data are the closing prices on each Friday of the week:

```
head(weekdays(index(msftWeeklyPrices)),3)
#> [1] "Friday" "Friday" "Friday"
```

1.4.1.3 Plotting `xts` objects with `plot.xts()` and `plot.zoo`

Time plots of `xts` objects can be created with the generic `plot()` function as there is a method function in the **xts** package for objects of class `xts`. For example, Figure 1.1 shows a basic time plot of the monthly closing prices of Microsoft and Starbucks created with:

```
plot(msftSbuxMonthlyPrices, main="Monthly Closing Prices",
     legend.loc="topleft")
```

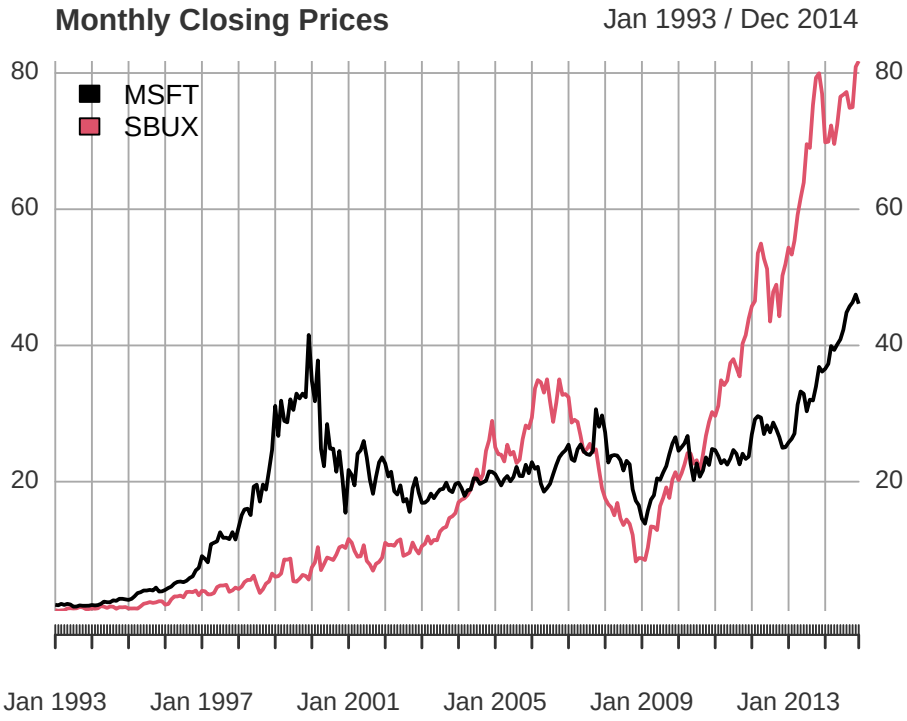


Figure 1.1: Single panel plot (default) with `plot.xts()`

The default plot style in `plot.xts()` is a single-panel plot with multiple series. You can also create multi-panel plots, as in Figure 1.2, by setting the optional argument `multi.panel = TRUE` in the call to `plot.xts()`

```
plot(msftSbuxMonthlyPrices, main="Monthly Closing Prices",
     multi.panel=TRUE)
```

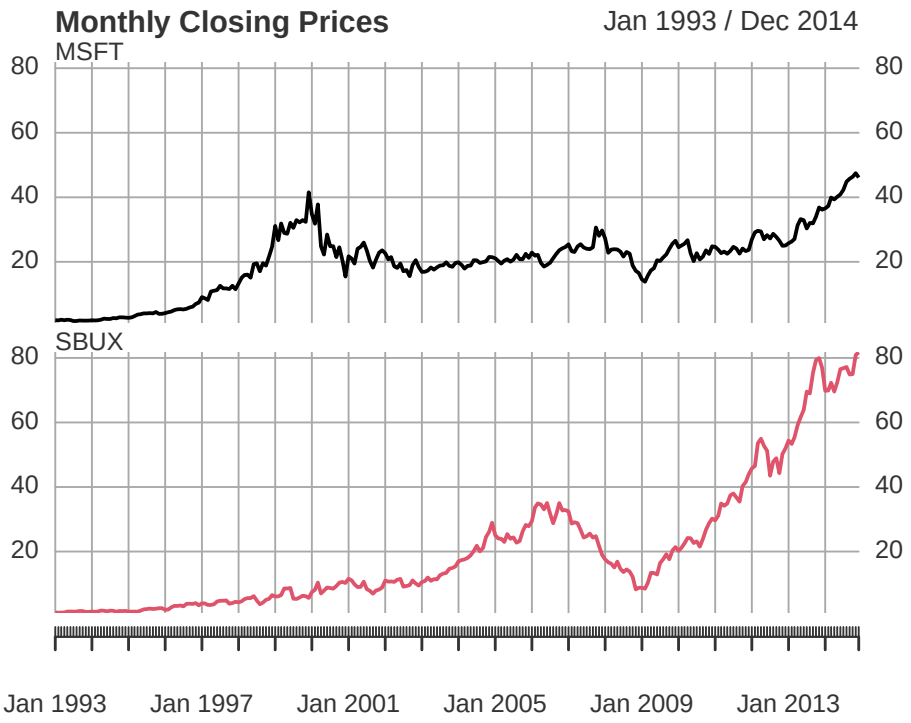


Figure 1.2: Multi-panel time series plot with `plot.xts()`

See the help file for `plot.xts()` for more examples of plotting `xts` objects.

Because the `xts` class inherits from the `zoo` class, the **zoo** method function `plot.zoo()` can also be used to plot `xts` objects. Figure 1.3 is created with

```
plot.zoo(msftSbuxMonthlyPrices, plot.type="single",
         main="Monthly Closing Prices",
         lwd = 2, col=c("black", "red"), ylab="Price")
grid()
legend(x="topleft", legend=colnames(msftSbuxMonthlyPrices),
       lwd = 2, col=c("black", "red"))
```

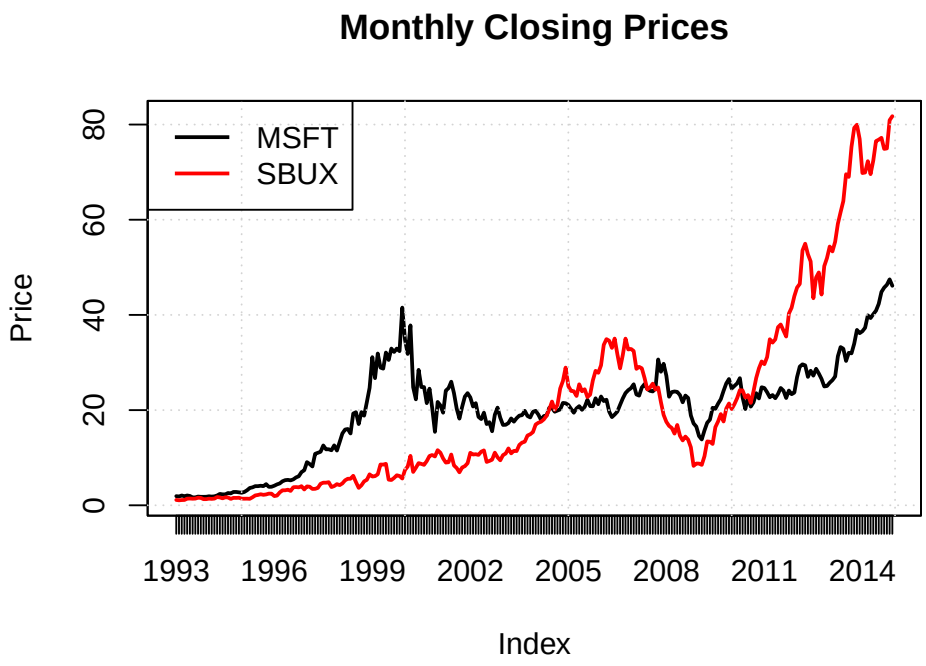


Figure 1.3: Single panel time series plot (default) with `plot.zoo()`

A muti-panel plot, shown in Figure 1.4, can be created using

```
plot.zoo(msftSbuxMonthlyPrices, plot.type="multiple", main="",
         lwd = 2, col=c("black", "red"), cex.axis=0.8)
```

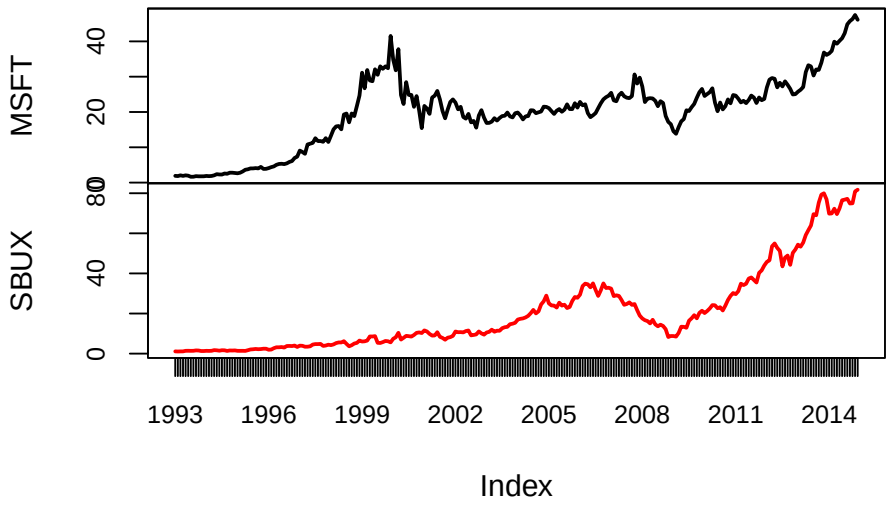


Figure 1.4: Multi-panel time series plot with `plot.zoo()`

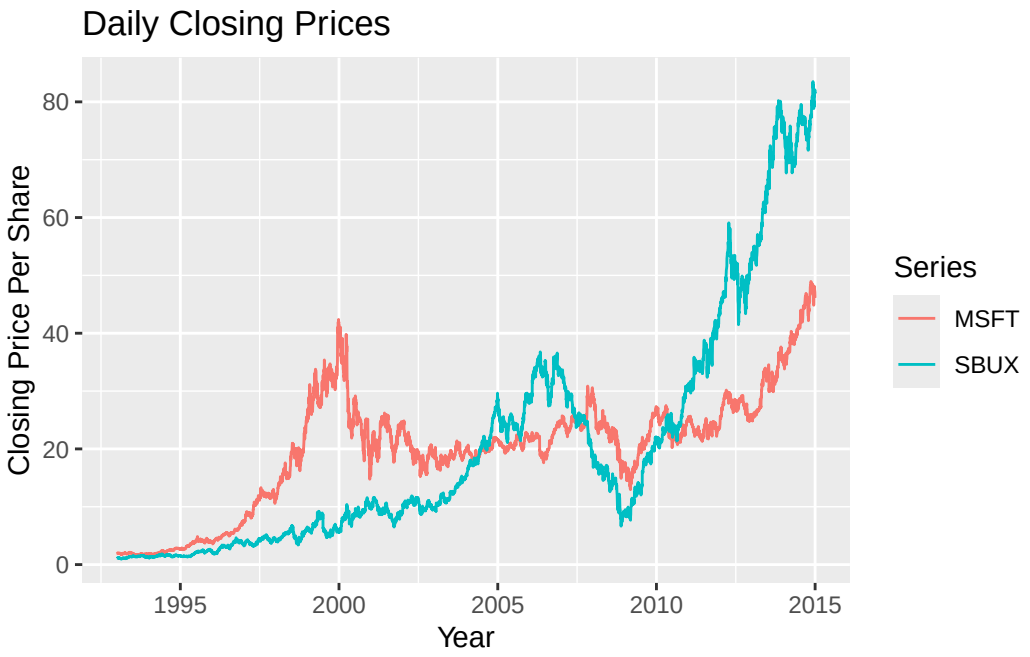
Creating plots with `plot.zoo()` is a bit more involved than with `plot.xts()`, as `plot.xts()` is newer. See the help file for `plot.zoo()` for more examples.

1.4.1.4 Ploting `xts` objects using `autoplot()` from ggplot2

The plots created using `plot.xts()` and `plot.zoo()` are created using the base graphics system in R. This allows for a lot of flexibility but often the resulting plots do not look very exciting or modern. In addition, these functions are not well suited for plotting many time series together in a single plot.

Another very popular graphics system in R is provided by the **ggplot2** package by Hadley Wickham of RStudio. For plotting `xts` objects, especially with multiple columns (data series), the **ggplot2** function `autoplot()` is especially convenient and easy:

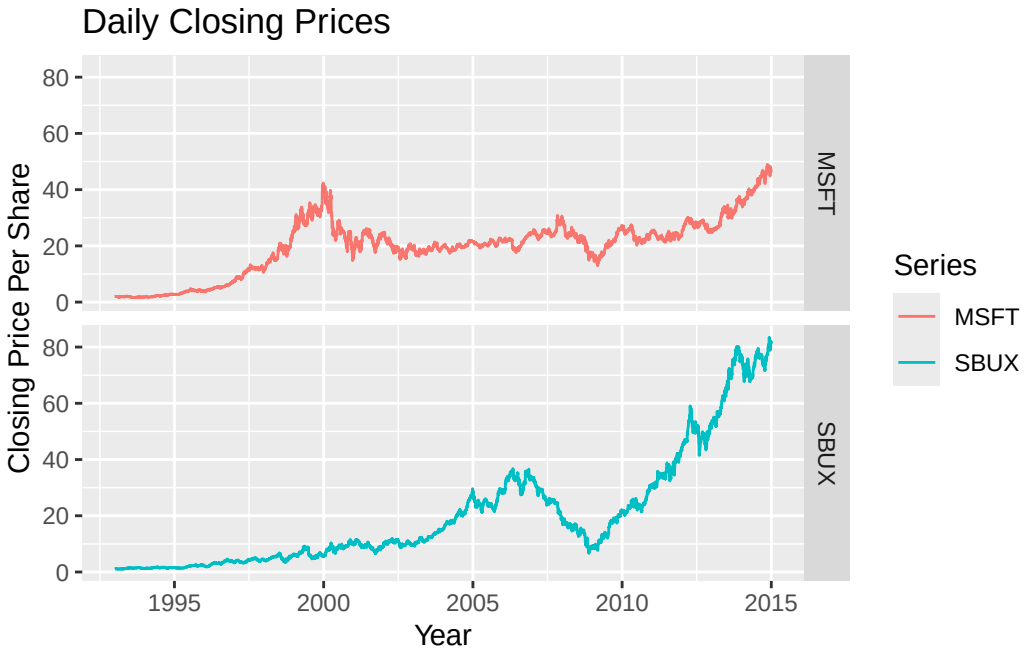
```
autoplot(msftSbuxDailyPrices, facets = NULL) +
  ggtitle("Daily Closing Prices") +
  ylab("Closing Price Per Share") +
  xlab("Year")
```



The syntax for creating graphs with `autoplot()` uses the *grammar of graphics* syntax from the **ggplot2** package. See the R Graphics Cookbook for more details and examples on using the **ggplot2** package. The call to `autoplot()` first invokes the method function `autoplot.zoo()` from the **zoo** package and creates a basic time series plot. Additional layers showing a main title and axes labels are added using `+`. The optional argument `facets = NULL` specifies that all series are plotted together on a single plot with different colors.

To produce a multi-panel plot call `autoplot()` with `facets = Series ~ .`:

```
autoplot(msftSbuxDailyPrices, facets = Series ~ .) +
  ggtitle("Daily Closing Prices") +
  ylab("Closing Price Per Share") +
  xlab("Year")
```



See the help file for `autoplot.zoo()` for more examples.

1.4.2 Calculating returns

In this sub-section, I illustrate how to calculate a time series of returns from a time series of prices.

1.4.2.1 Brute force return calculations

Consider computing simple monthly returns, $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$, from historical prices using the `xts` object `sbuxMonthlyPrices`. The R code for a brute force calculation is:

```
sbuxMonthlyReturns = diff(sbuxMonthlyPrices)/stats::lag(sbuxMonthlyPrices)
head(sbuxMonthlyReturns, n=3)
#>             SBUX
#> Jan 1993      NA
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
```

Here, the `diff()` function computes the first difference in the prices, $P_t - P_{t-1}$, and the `lag()` function computes the lagged price, P_{t-1}.[7] An equivalent calculation is $R_t = \frac{P_t}{P_{t-1}} - 1$:

[7] The method function `lag.xts()` works differently than generic `lag()` function and the method function `lag.zoo()`. In particular, `lag.xts()` with optional argument `k=1` gives the same result as `lag()` and `lag.zoo()` with `k=-1`.

```
head(sbuxMonthlyPrices/stats::lag(sbuxMonthlyPrices) - 1, n=3)
#>             SBUX
#> Jan 1993      NA
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
```

Notice that the return for January, 1993 is `NA` (missing value). To automatically remove this missing value, use the R function `na.omit()` or explicitly exclude the first observation:

```
head(na.omit(sbuxMonthlyReturns), n=3)
#>             SBUX
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
#> Apr 1993  0.0182
```

```
head(sbuxMonthlyReturns[-1], n=3)
#>             SBUX
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
#> Apr 1993  0.0182
```

To compute continuously compounded returns from simple returns, use:

```
sbuxMonthlyReturnsC = na.omit(log(1 + sbuxMonthlyReturns))
head(sbuxMonthlyReturnsC, n=3)
#>             SBUX
#> Feb 1993 -0.0645
#> Mar 1993  0.0465
#> Apr 1993  0.0180
```

Or, equivalently, to compute continuously compounded returns directly from prices, use:

```
sbuxMonthlyReturnsC = na.omit(diff(log(sbuxMonthlyPrices)))
head(sbuxMonthlyReturnsC, n=3)
#>             SBUX
#> Feb 1993 -0.0645
#> Mar 1993  0.0465
#> Apr 1993  0.0180
```

The above calculations used an `xts` object with a single column. If an `xts` object has multiple

columns the same calculations work for each column. For example,

```
msftSbuxMonthlyReturns = diff(msftSbuxMonthlyPrices)/stats::lag(msftSbuxMonthlyPrices)
head(msftSbuxMonthlyReturns, n=3)
#>             MSFT    SBUX
#> Jan 1993      NA      NA
#> Feb 1993 -0.0365 -0.0625
#> Mar 1993  0.1135  0.0476
```

Also,

```
msftSbuxMonthlyReturnsC = diff(log(msftSbuxMonthlyPrices))
head(msftSbuxMonthlyReturnsC, n=3)
#>             MSFT    SBUX
#> Jan 1993      NA      NA
#> Feb 1993 -0.0371 -0.0645
#> Mar 1993  0.1075  0.0465
```

These are examples of *vectorized* calculations. That is, the calculations are performed on all columns simultaneously. This is computationally more efficient than looping the same calculation for each column. In R, you should try to vectorize calculations whenever possible.

1.4.2.2 Calculating returns using `Return.calculate()`

Simple and continuously compounded returns can be computed using the **PerformanceAnalytics** function `Return.calculate()`:

```
# Simple returns
sbuxMonthlyReturns = Return.calculate(sbuxMonthlyPrices)
head(sbuxMonthlyReturns, n=3)
#>             SBUX
#> Jan 1993      NA
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
```

```
# CC returns
sbuxMonthlyReturnsC = Return.calculate(sbuxMonthlyPrices, method="log")
head(sbuxMonthlyReturnsC, n=3)
#>             SBUX
#> Jan 1993      NA
```

```
#> Feb 1993 -0.0645
#> Mar 1993  0.0465
```

`Return.calculate()` also works with `xts` objects with multiple columns:

```
msftSbuxMonthlyReturns = Return.calculate(msftSbuxMonthlyPrices)
head(msftSbuxMonthlyReturns, 3)
#>              MSFT    SBUX
#> Jan 1993       NA      NA
#> Feb 1993 -0.0365 -0.0625
#> Mar 1993  0.1135  0.0476
```

The advantages of using `Return.calculate()` instead of brute force calculations are: (1) clarity of code (you know what is being computed), (2) the same function is used for both simple and continuously compounded returns, (3) built-in error checking.

1.4.2.3 Equity curves

To directly compare the investment performance of two or more assets, plot the simple multi-period cumulative returns of each asset on the same graph. This type of graph, sometimes called an *equity curve*, shows how a one dollar investment amount in each asset grows over time. Better performing assets have higher equity curves. For simple returns, the k-period returns are $R_t(k) = \prod_{j=0}^{k-1}(1+R_{t-j})$ and represent the growth of one dollar invested for k periods. For continuously compounded returns, the k-period returns are $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$. However, this continuously compounded k-period return must be converted to a simple k-period return, using $R_t(k) = \exp(r_t(k)) - 1$, to properly represent the growth of one dollar invested for k periods.

Example 1.24 (Equity curves for Microsoft and Starbucks monthly returns)**.** To create the equity curves for Microsoft and Starbucks based on simple returns use (You can also use the **PerformanceAnalytics** function `chart.CumReturns()`)

```
msftMonthlyReturns = na.omit(Return.calculate(msftMonthlyPrices))
sbuxMonthlyReturns = na.omit(Return.calculate(sbuxMonthlyPrices))
equityCurveMsft = cumprod(1 + msftMonthlyReturns)
equityCurveSbux = cumprod(1 + sbuxMonthlyReturns)
dataToPlot = merge(equityCurveMsft, equityCurveSbux)
plot(dataToPlot, main="Monthly Equity Curves", legend.loc="topleft")
```

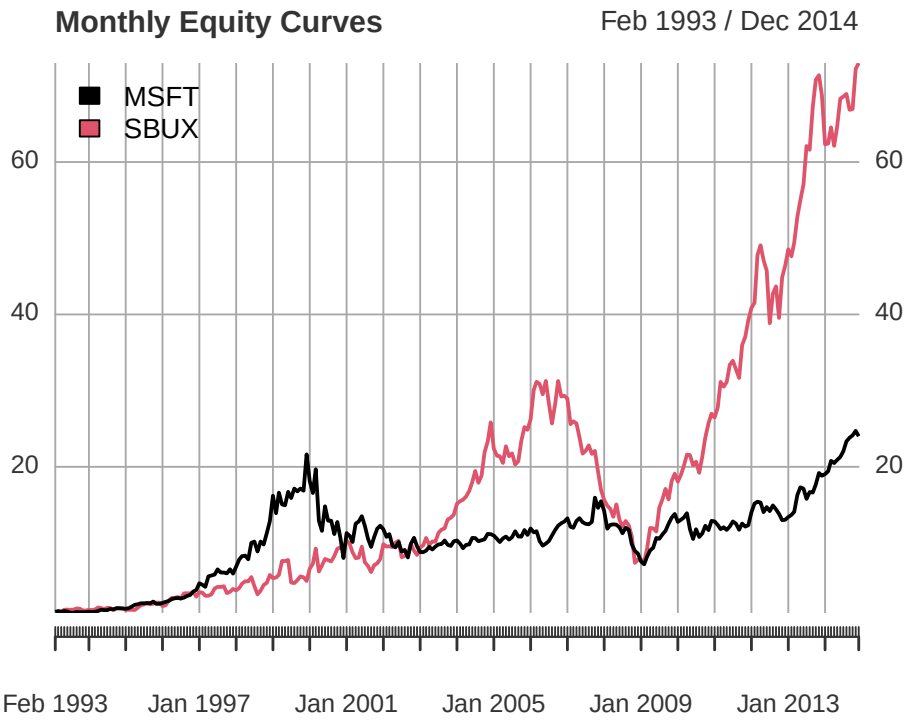


Figure 1.5: Monthly equity curve for Microsoft and Starbucks.

The R function `cumprod()` creates the cumulative products needed for the equity curves. Figure 1.5 shows that a one dollar investment in Starbucks dominated a one dollar investment in Microsoft over the given period. In particular, $1 invested in Microsoft grew to about only $22 (over about 20 years) whereas $1 invested in Starbucks grew over $70. Notice the huge increases and decreases in value of Microsoft during the dot-com bubble and bust over the period 1998 - 2001, and the enormous increase in value of Starbucks from 2009 - 2014.

■

1.4.3 Calculating portfolio returns from time series data

As discussed in sub-section Section 1.3, the calculation of portfolio returns depends on the assumptions made about the portfolio weights over time.

1.4.3.1 Constant portfolio weights

The easiest case assumes that portfolio weights are constant over time. To illustrate, consider an equally weighted portfolio of Microsoft and Starbucks stock. A time series a portfolio monthly returns can be created directly from the monthly returns on Microsoft and Starbucks stock:

```
x.msft = x.sbux = 0.5
msftMonthlyReturns = na.omit(Return.calculate(msftMonthlyPrices))
sbuxMonthlyReturns = na.omit(Return.calculate(sbuxMonthlyPrices))
equalWeightPortMonthlyReturns = x.msft*msftMonthlyReturns +
  x.sbux*sbuxMonthlyReturns
head(equalWeightPortMonthlyReturns, 3)
#>             MSFT
#> Feb 1993 -0.0495
#> Mar 1993  0.0806
#> Apr 1993 -0.0297
```

The same calculation can be performed using the **PerformanceAnalytics** function `Return.portfolio()`:

```
equalWeightPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "months")
head(equalWeightPortMonthlyReturns, 3)
#>          portfolio.returns
#> Feb 1993           -0.0495
#> Mar 1993            0.0806
#> Apr 1993           -0.0297
```

Here, the optional argument `rebalance_on = "months"` specifies that the portfolio is to be rebalanced to the specified weights at the beginning of every month.

If you also want the asset contributions to the portfolio return set the optional argument `contribution = TRUE` (recall, asset contributions sum to the portfolio return):

```
equalWeightPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "months",
                   contribution = TRUE)
head(equalWeightPortMonthlyReturns, 3)
#>          portfolio.returns    MSFT     SBUX
#> Feb 1993           -0.0495 -0.0182 -0.03125
#> Mar 1993            0.0806  0.0568  0.02381
#> Apr 1993           -0.0297 -0.0388  0.00909
```

You can see the rebalancing calculations by setting the optional argument `verbose = TRUE`:

```
equalWeightPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "months",
                   verbose = TRUE)
names(equalWeightPortMonthlyReturns)
#> [1] "returns"      "contribution" "BOP.Weight"   "EOP.Weight"   "BOP.Value"
#> [6] "EOP.Value"
```

In this case `Return.portfolio()` returns a list with the above components. The component `BOP.Weight` shows the beginning-of-period portfolio weights, and the component `EOP.Weight` shows the end-of-period portfolio weights (before any rebalancing).

```
head(equalWeightPortMonthlyReturns$BOP.Weight, 3)
#>          MSFT SBUX
#> Feb 1993  0.5  0.5
#> Mar 1993  0.5  0.5
#> Apr 1993  0.5  0.5
```

With monthly returns and with `rebalance_on = "months"`, the beginning-of-period weights are always rebalanced to the initially supplied weights.

```
head(equalWeightPortMonthlyReturns$EOP.Weight, 3)
#>           MSFT  SBUX
#> Feb 1993 0.507 0.493
#> Mar 1993 0.515 0.485
#> Apr 1993 0.475 0.525
```

Because the returns for Microsoft and Starbux are different every month the end-of-period weights drift away from the beginning-of-period weights.

1.4.3.2 Buy-and-hold portfolio

In the buy-and-hold portfolio, no rebalancing of the initial portfolio is done and the portfolio weights are allowed to evolve over time based on the returns of the assets. Hence, the computation of the portfolio return each month requires knowing the weights on each asset each month. You can use `Return.portfolio()` to compute the buy-and-hold portfolio returns as follows:

```
buyHoldPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux))
head(buyHoldPortMonthlyReturns, 3)
#>          portfolio.returns
#> Feb 1993           -0.0495
#> Mar 1993            0.0810
#> Apr 1993           -0.0319
```

Omitting the optional argument `rebalance_on` tells `Return.portfolio()` to not rebalance the portfolio at all (default behavior of the function).

1.4.3.3 Periodic rebalancing

In practice, a portfolio may be rebalanced at a specific frequency that is different than the frequency of the returns. For example, with monthly returns the portfolio might be rebalanced quarterly or annually. For example, to rebalance the equally weighted portfolio of Microsoft and Starbucks stock each quarter (every three months) use

```
equalWeightQtrPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "quarters")
head(equalWeightQtrPortMonthlyReturns, 3)
#>          portfolio.returns
#> Feb 1993           -0.0495
#> Mar 1993            0.0810
#> Apr 1993           -0.0297
```

In between quarters, the portfolio evolves as a buy-and-hold portfolio. To see the details of the rebalancing set `verbose = TRUE` in the call to `Return.Portfolio()`.

1.4.3.4 Active portfolio weights

If the weights are actively chosen at specific time periods, these weights can be supplied to the function `Return.portfolio()` to compute the portfolio return.

1.4.4 Downloading financial data from the internet

The data for the examples in this book, available in the **introcompFinr** package, were downloaded from finance.yahoo.com using the `getSymbols()` function from the **quantmod** package.

For example, to download daily data on Amazon and Google stock (ticker symbols AMZN and GOOG) from January 3, 2007 through January 3, 2020 use `getSymbols()` as follows:

```
options("getSymbols.warning4.0"=FALSE)
suppressPackageStartupMessages(library(quantmod))
getSymbols(Symbols = c("AMZN", "GOOG"), from="2007-01-03", to="2020-01-03",
           auto.assign=TRUE, warnings=FALSE)
#> [1] "AMZN" "GOOG"
class(AMZN)
#> [1] "xts" "zoo"
```

The `xts` objects `AMZN` and `GOOG` each have six columns containing the daily open price for the day, high price for the day, low price for the day, close price for the day, volume for the day, and (dividend and split) adjusted closing price for the day:

```
head(AMZN, 3)
#>            AMZN.Open AMZN.High AMZN.Low AMZN.Close AMZN.Volume AMZN.Adjusted
#> 2007-01-03      1.93      1.95     1.90       1.93    2.48e+08          1.93
#> 2007-01-04      1.93      1.96     1.91       1.95    1.26e+08          1.95
#> 2007-01-05      1.94      1.94     1.88       1.92    1.32e+08          1.92
```

```
head(GOOG, 3)
#>            GOOG.Open GOOG.High GOOG.Low GOOG.Close GOOG.Volume GOOG.Adjusted
#> 2007-01-03      11.6      11.9     11.5       11.6    3.09e+08          11.5
#> 2007-01-04      11.7      12.1     11.7       12.0    3.17e+08          11.9
#> 2007-01-05      12.0      12.1     11.9       12.1    2.76e+08          12.0
```

1.5 Further Reading: Return Calculations

This chapter describes basic asset return calculations with an emphasis on equity calculations. Similar material is covered in Chapter 2 of (Ruppert and Matteson 2015). Comprehensive coverage of general return calculations is given in (Bacon 2008) and (Christopherson, Carino, and Ferson 2009). The virtues of portfolio rebalancing are discussed in (Ang 2014) and (Qian 2019).

This chapter introduced you to the R packages **PerformanceAnalytics**, **quantmod**, and **xts** for working with financial time series. It is recommended to read the vignettes included in each of these

packages. Jeff Ryan's original quantmod page, while a bit outdated, has many useful examples of working with **quantmod** and **xts**. A good online tutorial for the **xts** package is provided by Joshua Ulrich, one of the creators of **xts**.

There are many useful R packages for the analysis of financial data. An up-to-date list of R packages relevant for finance is given on the Empirical Finance Task View on the comprehensive R archive network (CRAN).

1.6 Appendix: Properties of Exponentials and Logarithms

The computation of continuously compounded returns requires the use of natural logarithms. The natural logarithm function, $\ln(\cdot)$, is the inverse of the exponential function, $e^{(\cdot)} = \exp(\cdot)$, where $e^1 = 2.718$. That is, $\ln(x)$ is defined such that $x = \ln(e^x)$. Figure 1.6 plots e^x and $\ln(x)$. Notice that e^x is always positive and increasing in x. $\ln(x)$ is monotonically increasing in x and is only defined for $x > 0$. Also note that $\ln(1) = 0$ and $\ln(0) = -\infty$.

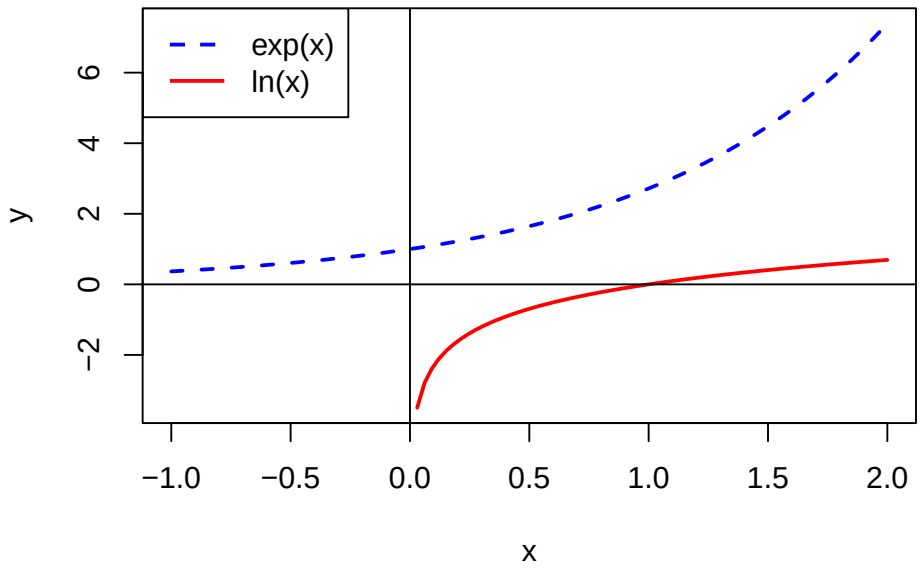


Figure 1.6: Exponential and natural logarithm functions.

The exponential and natural logarithm functions have the following properties:

1. $\ln(x \cdot y) = \ln(x) + \ln(y)$, $x, y > 0$
2. $\ln(x/y) = \ln(x) - \ln(y)$, $x, y > 0$
3. $\ln(x^y) = y \ln(x)$, $x > 0$
4. $\frac{d \ln(x)}{dx} = \frac{1}{x}$, $x > 0$
5. $\frac{d}{dx} \ln(f(x)) = \frac{1}{f(x)} \frac{d}{dx} f(x)$ (chain-rule)
6. $e^x e^y = e^{x+y}$
7. $e^x e^{-y} = e^{x-y}$
8. $(e^x)^y = e^{xy}$
9. $e^{\ln(x)} = x$
10. $\frac{d}{dx} e^x = e^x$
11. $\frac{d}{dx} e^{f(x)} = e^{f(x)} \frac{d}{dx} f(x)$ (chain-rule)

1.7 Exercises

Analytical Problems

Exercise 1.1. In this question you will see how the future value of an investment, FV_n varies with the interest rate, R, and the investment horizon, n. Let the initial amount invested be $V = \$1,000$. For a grid of values $R = 0.01, 0.02, ..., 0.10$ and $n = 1, 2, ... 10$, compute and plot FV_n.

Exercise 1.2. Consider the following credit card offer from a hypothetical bank. The card advertises an annual percentage rate (APR) of 19.34%. However, the bank actually compounds interest charges on a daily basis.

1. Assuming 365 days in the year, what is the daily periodic interest rate being charged?
2. If you carried a $1,000 balance throughout the year, how much would you owe at the end of the year?

Exercise 1.3. Consider "payday loans". Payday loans are short-term loans made to consumers, often for less than two weeks, and are offered by companies such as AmeriCash Advance and National Payday. The loan works like this: You write a check today that is postdated and the company gives you an amount of money less than the check. When the check date arrives, you go to the store and pay cash for the check, or the company cashes the check. For example, suppose AmeriCash Advance allows you to write a postdated check for $125 for 15 days later in return for a $100 loan today.

1. What is the annual percentage rate (APR) on this loan?
2. What is the effective annual rate (EAR)?

Exercise 1.4. Consider the following (actual) monthly adjusted closing price data for Starbucks stock over the period December 2004 through December 2005

Table 1.2: End of Month Price Data for Starbucks Stock

Date	Price
December, 2004	31.2
January, 2005	27.0
February, 2005	25.9
March, 2005	25.8
April, 2005	24.8
May, 2005	27.4
June, 2005	25.8
July, 2005	26.3

Date	Price
August, 2005	24.5
September, 2005	25.1
October, 2005	28.3
November, 2005	30.4
December, 2005	30.5

For the following questions, first do the return calculations by hand. That is, give the appropriate return calculation formula and evaluate the result. Then, check your hand calculations using the corresponding R calculations.

1. Using the data in the table, what is the simple monthly return between the end of December 2004 and the end of January 2005? If you invested $10,000 in Starbucks at the end of December 2004, how much would the investment be worth at the end of January 2005?
2. Using the data in the table, what is the continuously compounded monthly return between December 2004 and January 2005? Convert this continuously compounded return to a simple return (you should get the same answer as in part 1).
3. Assuming that the simple monthly return you computed in part 1 is the same for 12 months, what is the annual return with monthly compounding?
4. Assuming that the continuously compounded monthly return you computed in part 2 is the same for 12 months, what is the continuously compounded annual return?
5. Using the data in the table, compute the actual simple annual return between December 2004 and December 2005. If you invested $10,000 in Starbucks at the end of December 2004, how much would the investment be worth at the end of December 2005? Compare with your result in part 3.
6. Using the data in the table, compute the actual annual continuously compounded return between December 2004 and December 2005. Compare with your result in part 4. Convert this continuously compounded return to a simple return (you should get the same answer as in part 5).

Exercise 1.5. Consider a one month investment in two Northwest stocks: Amazon and Costco. Suppose you buy Amazon and Costco at the end of September at $P_{A,t-1} = \$38.23$, $P_{C,t-1} = \$41.11$ and then sell at the end of the October for $P_{A,t} = \$41.29$ and $P_{C,t} = \$41.74$. (Note: these are actual closing prices for 2004 taken from Yahoo!)

For the following questions, first do the return calculations by hand. That is, give the appropriate return calculation formula and evaluate the result. Then, check your hand calculations using the corresponding R calculations.

1. What are the simple monthly returns for the two stocks?

2. What are the continuously compounded returns for the two stocks?
3. Suppose Costco paid a $0.10 per share cash dividend at the end of October. What is the monthly simple total return on Costco? What is the monthly dividend yield?
4. Suppose the monthly returns on Amazon and Costco from part 1 above are the same every month for 1 year. Compute the simple annual returns as well as the continuously compounded annual returns for the two stocks.
5. At the end of September 2004, suppose you have $10,000 to invest in Amazon and Costco over the next month. If you invest $8000 in Amazon and $2000 in Costco, what are your portfolio shares, x_A and x_C?
6. Continuing with part 5, compute the monthly simple return and the monthly continuously compounded return on the portfolio. Assume that Costco does not pay a dividend.

Exercise 1.6. Consider a 48-month (4 year) investment in two assets: the Vanguard S&P 500 index (VFINX) and Amazon stock (AMZN). Suppose you buy one share of the S&P 500 fund and one share of Amazon stock at the end of January, 2015 for $P_{VFINX,t-48} = 165$, $P_{AMZN,t-48} = 355$, and then sell these shares at the end of January, 2019 for $P_{VFINX,t} = 241$,$P_{AMZN,t} = 1719$ (Note: these are actual adjusted closing prices taken from Yahoo!). In this question, you will see how much money you could have made if you invested in these assets right before the COVID-19 pandemic.

1. What are the simple 48-month (4-year) returns for the two investments?
2. What are the continuously compounded (cc) 48-month (4-year) returns for the two investments? Why are the cc returns smaller?
3. Suppose you invested $10,000 in each asset at the end of January, 2015. How much would each investment be worth at the end of January, 2019?
4. What are the compound annual returns on the two 4-year investments?
5. At the end of January, 2015, suppose you plan to invest in a portfolio of VFINX and AMZN over the next 48 months (4 years). Suppose you purchase 5 shares of the VFINX mutual fund (at $165/share) and 10 shares of AMZN stock (at $355/share). What is the value of your initial investment? What are the portfolio weights in the two assets as of the end of January, 2015?
6. Using the results from part 1, compute the 4-year simple and cc portfolio returns.
7. What is the value of your portfolio at the end of January, 2019? What is the value of your investments in VFINX and AMZN? What is share of wealth invested in VFINX and AMZN (i.e., your portfolio weights)?

Exercise 1.7. Consider a 60-month (5 year) investment in two assets: the Vanguard S&P 500 index (VFINX) and Apple stock (AAPL). Suppose you buy one share of the S&P 500 fund and one share of Apple stock at the end of January, 2010 for $P_{VFINX,t-60} = \$89.91$, $P_{AAPL,t-60} = \$25.88$,

and then sell these shares at the end of January, 2015 for $P_{VFINX,t} = \$184.2$, $P_{AAPL,t} = \$116.7$. (Note: these are actual adjusted closing prices taken from Yahoo!). In this question, you will see how much money you could have made if you invested in these assets right after the 2008 financial crisis.

1. What are the simple 60-month (5-year) returns for the two investments?
2. What are the continuously compounded 60-month (5-year) returns for the two investments?
3. Are the simple and continuously compounded returns close to each other?
4. Suppose you invested \$1,000 in each asset at the end of January,
5. How much would each investment be worth at the end of January, 2015?
6. What is the compound annual return on the two 5 year investments?
7. At the end of January, 2010, suppose you have $\$1,000$ to invest in VFINX and AAPL over the next 60 months (5 years). Suppose you purchase \$400 worth of VFINX and the remainder in AAPL. What are the portfolio weights in the two assets? Using the results from parts 1 and 2, compute the 5-year simple and continuously compounded portfolio returns.

Exercise 1.8. Consider a one month investment in two Northwest stocks: Amazon and Costco. Suppose you buy Amazon and Costco at the end of September at $P_{A,t-1} = \$38.23, P_{C,t-1} = \$41.11$ and then sell at the end of the October for $P_{A,t} = \$41.29, P_{C,t} = \$41.74$. (Note: these are actual closing prices for 2004 taken from Yahoo!)

1. What are the simple monthly returns for the two stocks?
2. What are the continuously compounded returns for the two stocks?
3. Suppose Costco paid a \$0.10 per share cash dividend at the end of October. What is the monthly simple total return on Costco? What is the monthly dividend yield?
4. Suppose the monthly returns on Amazon and Costco from question (a) above are the same every month for 1 year. Compute the simple annual returns as well as the continuously compounded annual returns for the two stocks.
5. At the end of September, 2004, suppose you have $\$10,000$ to invest in Amazon and Costco over the next month. If you invest $\$8,000$ in Amazon and $\$2,000$ in Costco, what are your portfolio shares, x_A and x_C. (Assume partial share purchases are possible)
6. Continuing with the previous question, compute the monthly simple return and the monthly continuously compounded return on the portfolio. Assume that Costco does not pay a dividend.

Exercise 1.9. Consider an investment in a foreign stock (e.g., a stock trading on the London stock exchange) by a U.S. national (domestic investor). The domestic investor takes U.S. dollars, converts them to the foreign currency (e.g. British Pound) via the exchange rate (price of foreign currency in U.S. dollars) and then purchases the foreign stock using the foreign currency. When the stock is sold, the proceeds in the foreign currency must then be converted back to the domestic currency. To be more precise, consider the information in the table below:

Time	Cost of 1 Pound	Value of UK Shares	Value in U.S. $
0	\$1.50	£40	$1.5 \times 40 = 60$
1	\$1.30	£45	$1.3 \times 45 = 58.5$

1. Compute the simple rate of return, R_e, from the prices of foreign currency. This is the return to the domestic investor of investing in the foreign currency.
2. Compute the simple rate of return, R_{UK}, from the UK stock prices.
3. Compute the simple rate of return, R_{US}, from the prices in US dollars.
4. What is the mathematical relationship between R_{US}, R_{UK} and R_e?

Exercise 1.10. Consider an investing in a long-only portfolio of two assets A and B for one period between months $t-1$ and t. The beginning of period investment values in A and B are $V_{A,t-1} > 0$ and $V_{B,t-1} > 0$, and the one-month returns are $R_{A,t-1}$ and $R_{B,t-1}$. The initial value of the portfolio is $V_{p,t-1} = V_{A,t-1} + V_{B,t-1}$. The initial investment weights are $x_{A,t-1} = V_{A,t-1}/V_{p,t-1} > 0$ and $x_{B,t-1} = V_{B,t-1}/V_{p,t-1} > 0$ such that $x_{A,t-1} + x_{B,t-1} = 1$.

1. Show that
$$x_{A,t} = x_{A,t-1} \times \frac{1 + R_{A,t-1}}{1 + R_{p,t-1}},\ x_{B,t} = x_{B,t-1} \times \frac{1 + R_{B,t-1}}{1 + R_{p,t-1}},$$
and that $x_{A,t} + x_{B,t} = 1$.
2. Show that the change in portfolio weights are
$$\Delta x_{A,t} = x_{A,t} - x_{A,t-1} = \frac{x_{A,t-1}(R_{A,t-1} - R_{p,t-1})}{1 + R_{p,t-1}},$$
$$\Delta x_{B,t} = x_{B,t} - x_{B,t-1} = \frac{x_{B,t-1}(R_{B,t-1} - R_{p,t-1})}{1 + R_{p,t-1}}.$$
3. Under what conditions will $\Delta x_{A,t} > 0$, $\Delta x_{A,t} < 0$ and $\Delta x_{A,t} = 0$?
4. Using $R_{p,t-1} = x_{A,t-1}R_{A,t-1} + x_{B,t-1}R_{B,t-1}$ show that
$$\Delta x_{A,t} = \frac{x_{A,t-1}x_{B,t-1}(R_{A,t-1} - R_{B,t-1})}{1 + x_{A,t-1}R_{A,t-1} + x_{B,t-1}R_{B,t-1}}$$
5. Under what conditions will $\Delta x_{A,t} > 0$, $\Delta x_{A,t} < 0$ and $\Delta x_{A,t} = 0$?
6. To rebalance the portfolio at the end-of-month, you need to adjust the portfolio weights such that $x_{A,t} = x_{A,t-1}$ and $x_{B,t} = x_{B,t-1}$. If $R_{A,t-1} > R_{B,t-1}$ do you buy or sell asset A to rebalance the portfolio?

Return Calculations using R Packages

- Give examples of using the **PerformanceAnalytics package**

- Download different asset data from Yahoo!. Extract adjusted closing price data. Compute returns and plot returns and equity curves.

2 Review of Random Variables

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In this chapter I give a review of probability concepts that are necessary for the probability modeling of asset returns and for the statistical analysis of financial data presented in this book. Much of this material is typically covered in an introductory probability and statistics course using calculus. In the course of the review, I will present many examples related to finance and risk management and some important risk concepts, such as value-at-risk, will be introduced. In the examples, I will also show how to use R for probability calculations and for working with probability distributions.

The chapter is outlined as follows. In Section 2.1 I review univariate random variables and properties of univariate probability distributions. I describe in detail important univariate distributions, such as the normal distribution, that are used throughout the book for modeling financial data. In Section 2.2 I cover bivariate probability distributions for two random variables. I discuss and illustrate the important concepts of dependence and independence between two random variables, and focus on the covariance and correlations measures of linear dependence between two random variables. I cover properties of linear combinations of two random variables and illustrate these properties using an example to describe the return and risk properties of a portfolio of two assets. I briefly discuss multivariate distributions and probability calculations of linear combinations of many random variables in Section 2.3.

The R packages I use in this chapter are **mvtnorm** and **sn**. Make sure these packages are downloaded and installed prior to replicating the R examples in this chapter.

2.1 Random Variables

I start with the basic definition of a *random variable*:

Definition 2.1 (Random variable)**.** A *random variable*, X, is a variable that can take on a given set of values, called the *sample space* and denoted S_X, where the likelihood of the values in S_X is determined by X's *probability distribution function* (pdf).

Example 2.1 (Future price of Microsoft stock)**.** Consider the price of Microsoft stock next month, P_{t+1}. Since P_{t+1} is not known with certainty today (t), you can consider it as a random variable.

P_{t+1} must be positive, and realistically it can't get too large. Therefore, the sample space is the set of positive real numbers bounded above by some large number: $S_{P_{t+1}} = \{P_{t+1} : P_{t+1} \in [0, M],\ M > 0\}$. It is an open question as to what is the best characterization of the probability distribution of stock prices. The log-normal distribution is one possibility.[1] ■

Example 2.2 (Return on Microsoft stock)**.** Consider a one-month investment in Microsoft stock. That is, you buy one share of Microsoft stock at the end of month $t-1$ (e.g., end of February) and plan to sell it at the end of month t (e.g., end of March). The simple return over month t on this investment, $R_t = (P_t - P_{t-1})/P_{t-1}$, is a random variable because you do not know what the price will be at the end of the month. In contrast to prices, returns can be positive or negative and are bounded from below by -100%. You can express the sample space as $S_{R_t} = \{R_t : R_t \in [-1, M],\ M > 0\}$. The normal distribution is often a good approximation to the distribution of simple monthly returns, and is a better approximation to the distribution of continuously compounded monthly returns. ■

Example 2.3 (Up-down indicator variable)**.** As a final example, consider a variable X_{t+1} defined to be equal to one if the monthly price change on Microsoft stock, $P_{t+1} - P_t$, is positive, and is equal to zero if the price change is zero or negative. Here, the sample space is the set $S_{X_{t+1}} = \{0, 1\}$. If it is equally likely that the monthly price change is positive or negative (including zero) then the probability that $X_{t+1} = 1$ or $X_{t+1} = 0$ is 0.5. This is an example of a Bernoulli random variable. ■

The next sub-sections define discrete and continuous random variables.

2.1.1 Discrete random variables

Consider a random variable generically denoted X and its set of possible values or sample space denoted S_X.

Definition 2.2 (Discrete random variable)**.** A *discrete random variable* X is one that can take on a finite number of n different values $S_X = \{x_1, x_2, \ldots, x_n\}$ or, at most, a countably infinite number of different values $S_X = \{x_1, x_2, \ldots.\}$.

Definition 2.3 (Probability density function)**.** The pdf of a discrete random variable, denoted $p(x)$, is a function such that $p(x) = \Pr(X = x)$. The pdf must satisfy (i) $p(x) \geq 0$ for all $x \in S_X$; (ii) $p(x) = 0$ for all $x \notin S_X$; and (iii) $\sum_{x \in S_X} p(x) = 1$.

[1] If P_{t+1} is a positive random variable such that $\ln P_{t+1}$ is normally distributed then P_{t+1} has a log-normal distribution.

Example 2.4 (Annual return on Microsoft stock)**.**

Table 2.1: Probability distribution for the annual return on Microsoft.

State of Economy	Sample Space	$\Pr(X = x)$
Depression	-0.30	0.05
Recession	0.0	0.20
Normal	0.10	0.50
Mild Boom	0.20	0.20
Major Boom	0.50	0.05

Let X denote the annual return on Microsoft stock over the next year. You might hypothesize that the annual return will be influenced by the general state of the economy. Consider five possible states of the economy: depression, recession, normal, mild boom and major boom. A stock analyst might forecast different values of the return for each possible state. Hence, X is a discrete random variable that can take on five different values. Table Table 2.1 describes such a probability distribution of the return. A graphical representation of the probability distribution is created with the following R code:

```
r.msft = c(-0.3, 0, 0.1, 0.2, 0.5)
prob.vals = c(0.05, 0.20, 0.50, 0.20, 0.05)
plot(r.msft, prob.vals, type="h", lwd=4, xlab="annual return",
     ylab="probability", xaxt="n")
points(r.msft, prob.vals, pch=16, cex=2, col="blue")
axis(1, at=r.msft)
```

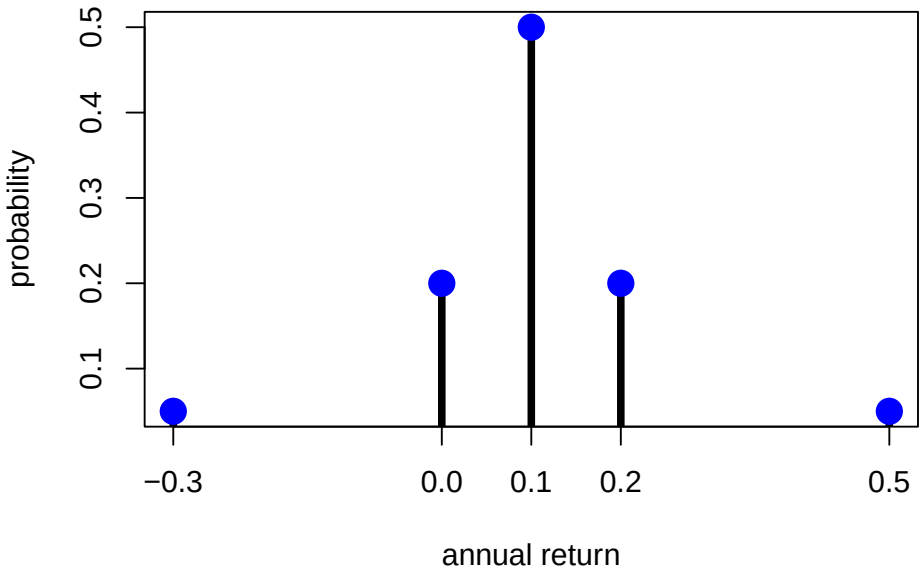


Figure 2.1: Discrete distribution for Microsoft stock.

This distribution is presented in Figure 2.1. The shape of the distribution is revealing. The center and most likely value is at 0.1 and the distribution is symmetric about this value (has the same shape to the left and to the right). Most of the probability is concentrated between 0 and 0.2 and there is very little probability associated with very large or small returns. ■

2.1.1.1 The Bernoulli distribution

Let $X = 1$ if the price next month of Microsoft stock goes up and $X = 0$ if the price goes down (assuming it cannot stay the same). Then X is clearly a discrete random variable with sample space $S_X = \{0, 1\}$. If the probability of the stock price going up or down is the same then $p(0) = p(1) = 1/2$ and $p(0) + p(1) = 1$.

The probability distribution described above can be given an exact mathematical representation known as the *Bernoulli* distribution. Consider two mutually exclusive events generically called "success" and "failure". For example, a success could be a stock price going up or a coin landing heads and a failure could be a stock price going down or a coin landing tails. The process creating the success or failure is called a *Bernoulli trial*. In general, let $X = 1$ if success occurs and let $X = 0$ if failure occurs. Let $\Pr(X = 1) = \pi$, where $0 < \pi < 1$, denote the probability of success. Then $\Pr(X = 0) = 1 - \pi$ is the probability of failure. A mathematical model describing this distribution

is:

$$p(x) = \Pr(X = x) = \pi^x(1-\pi)^{1-x},\ x = 0, 1. \tag{2.1}$$

When $x = 0$, $p(0) = \pi^0(1-\pi)^{1-0} = 1-\pi$ and when $x = 1, p(1) = \pi^1(1-\pi)^{1-1} = \pi$.

2.1.1.2 The binomial distribution

Consider a sequence of independent Bernoulli trials with success probability π generating a sequence of 0-1 variables indicating failures and successes. A *binomial* random variable X counts the number of successes in n Bernoulli trials, and is denoted $X \sim B(n, \pi)$. The sample space is $S_X = \{0, 1, \dots, n\}$ and:

$$\Pr(X = k) = \binom{n}{k}\pi^k(1-\pi)^{n-k}.$$

The term $\binom{n}{k}$ is the binomial coefficient, and counts the number of ways k objects can be chosen from n distinct objects. It is defined by:

$$\binom{n}{k} = \frac{n!}{(n-k)!k!},$$

where $n!$ is the factorial of n, or $n(n-1)\cdots 2 \cdot 1$.

2.1.2 Continuous random variables

Definition 2.4 (Continuous random variable)**.** A *continuous random variable* X is one that can take on any real value. That is, $S_X = \{x : x \in \mathbb{R}\}$.

Definition 2.5 (Probability density function for continuous random variable)**.** The probability density function (pdf) of a continuous random variable X is a nonnegative function f, defined on the real line, such that for any interval A:

$$\Pr(X \in A) = \int_A f(x)dx.$$

That is, $\Pr(X \in A)$ is the area under the probability curve over the interval A. The pdf $f(x)$ must satisfy (i) $f(x) \geq 0$; and (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$.

A typical "bell-shaped" pdf is displayed in Figure 2.2 and the shaded area under the curve between -2 and 1 represents $\Pr(-2 \leq X \leq 1)$. For a continuous random variable, $f(x) \neq \Pr(X = x)$ but rather gives the height of the probability curve at x. In fact, $\Pr(X = x) = 0$ for all values of x. That is, probabilities are not defined over single points. They are only defined over intervals. As a result, for a continuous random variable X:

$$\Pr(a \leq X \leq b) = \Pr(a < X \leq b) = \Pr(a < X < b) = \Pr(a \leq X < b).$$

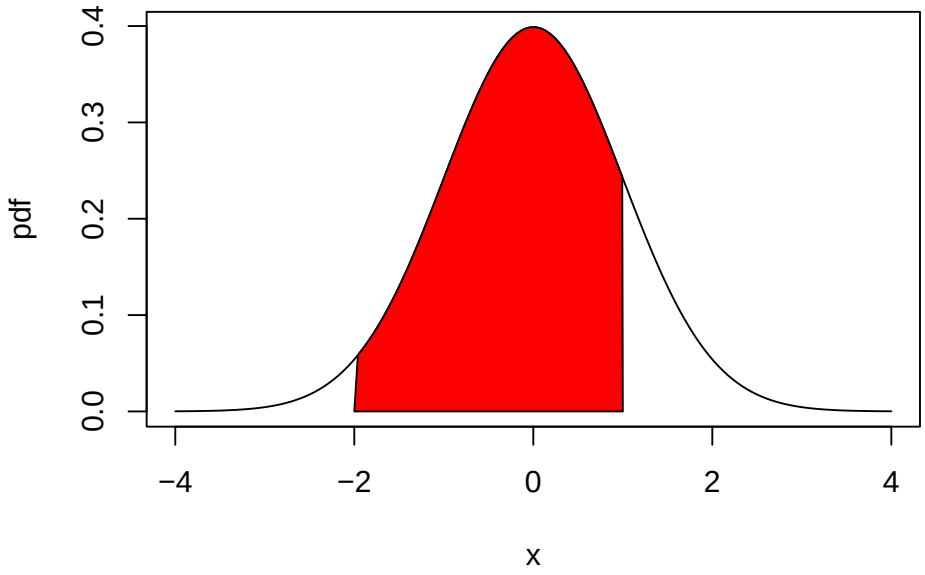


Figure 2.2: $Pr(-2 \leq X \leq 1)$ is represented by the area under the probability curve.

2.1.2.1 The uniform distribution on an interval

Let X denote the annual return on Microsoft stock and let a and b be two real numbers such that $a < b$. Suppose that the annual return on Microsoft stock can take on any value between a and b. That is, the sample space is restricted to the interval $S_X = \{x \in \mathcal{R} : a \leq x \leq b\}$. Further suppose that the probability that X will belong to any subinterval of S_X is proportional to the length of the interval. In this case, it is said that X is *uniformly distributed* on the interval $[a, b]$. The pdf of X has a very simple mathematical form:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{otherwise,} \end{cases}$$

and is presented graphically as a rectangle. It is easy to see that the area under the curve over the interval $[a, b]$ (area of rectangle) integrates to one:

$$\int_a^b \frac{1}{b-a}\, dx = \frac{1}{b-a}\int_a^b dx = \frac{1}{b-a}\,[x]_a^b = \frac{1}{b-a}[b-a] = 1.$$

Example 2.5 (Uniform distribution on [-1 1])**.** Let $a = -1$ and $b = 1$, so that $b - a = 2$. Consider

computing the probability that X will be between -0.5 and 0.5. You must solve:

$$\Pr(-0.5 \leq X \leq 0.5) = \int_{-0.5}^{0.5} \frac{1}{2}\, dx = \frac{1}{2}\left[x\right]_{-0.5}^{0.5} = \frac{1}{2}\left[0.5 - (-0.5)\right] = \frac{1}{2}.$$

Next, consider computing the probability that the return will fall in the interval $[0, \delta]$ where δ is some small number less than $b = 1$:

$$\Pr(0 \leq X \leq \delta) = \frac{1}{2}\int_{0}^{\delta} dx = \frac{1}{2}[x]_0^{\delta} = \frac{1}{2}\delta.$$

As $\delta \to 0$, $\Pr(0 \leq X \leq \delta) \to \Pr(X = 0)$.

Using the above result notice that

$$\lim_{\delta \to 0} \Pr(0 \leq X \leq \delta) = \Pr(X = 0) = \lim_{\delta \to 0} \frac{1}{2}\delta = 0.$$

Hence, for continuous random variables probabilities are defined on intervals but not at distinct points. ■

2.1.2.2 The standard normal distribution

The normal or Gaussian distribution is perhaps the most famous and most useful continuous distribution in all of statistics. The shape of the normal distribution is the familiar "bell curve". As you shall see, it can be used to describe the probabilistic behavior of stock returns although other distributions may be more appropriate.

If a random variable X follows a *standard normal distribution* then you often write $X \sim N(0, 1)$ as short-hand notation. This distribution is centered at zero and has inflection points at ± 1.[2] The pdf of a standard normal random variable is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}x^2},\ -\infty \leq x \leq \infty. \tag{2.2}$$

It can be shown via the change of variables formula in calculus that the area under the standard normal curve is one:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}x^2} dx = 1.$$

The standard normal distribution is illustrated in Figure 2.3. Notice that the distribution is *symmetric* about zero; i.e., the distribution has exactly the same form to the left and right of zero. Because the standard normal pdf formula is used so often it is given its own special symbol $\phi(x)$.

[2] An inflection point is a point where a curve goes from concave up to concave down, or vice versa.

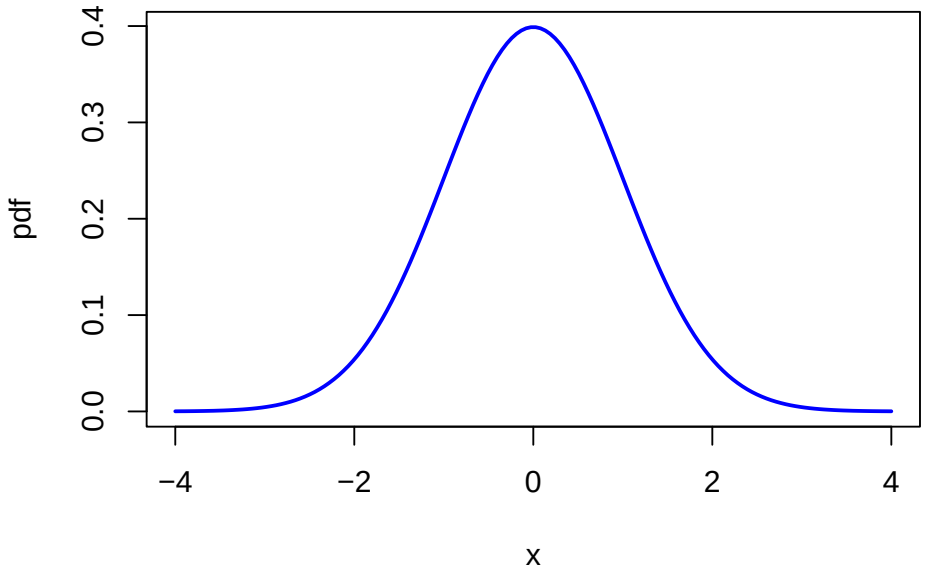


Figure 2.3: Standard normal density.

The normal distribution has the annoying feature that the area under the normal curve cannot be evaluated analytically. That is:

$$\Pr(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}x^2} \, dx,$$

does not have a closed form solution. The above integral must be computed by numerical approximation. Areas under the normal curve, in one form or another, are given in tables in almost every introductory statistics book and standard statistical software can be used to find these areas. Some useful approximate results are:

$$\Pr(-1 \leq X \leq 1) \approx 0.67,$$
$$\Pr(-2 \leq X \leq 2) \approx 0.95,$$
$$\Pr(-3 \leq X \leq 3) \approx 0.99.$$

2.1.3 The cumulative distribution function

Definition 2.6 (Cumulative distribution function)**.** The *cumulative distribution function* (cdf) of a random variable X (discrete or continuous), denoted F_X, is the probability that $X \leq x$:

$$F_X(x) = \Pr(X \leq x), \ -\infty \leq x \leq \infty.$$

The cdf has the following properties:

(i) If $x_1 < x_2$ then $F_X(x_1) \leq F_X(x_2)$
(ii) $F_X(-\infty) = 0$ and $F_X(\infty) = 1$
(iii) $\Pr(X > x) = 1 - F_X(x)$
(iv) $\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$
(v) $F'_X(x) = \frac{d}{dx}F_X(x) = f(x)$ if X is a continuous random variable and $F_X(x)$ is continuous and differentiable.

Example 2.6 (CDF for a discrete random variable)**.** The cdf for the discrete distribution of Microsoft from Table Table 2.1 is given by:

$$F_X(x) = \begin{cases} 0, & x < -0.3 \\ 0.05, & -0.3 \leq x < 0 \\ 0.25, & 0 \leq x < 0.1 \\ 0.75 & 0.1 \leq x < 0.2 \\ 0.95 & 0.2 \leq x < 0.5 \\ 1 & x > 0.5 \end{cases}$$

and is illustrated in Figure 2.4. ■

Example 2.7 (CDF for a uniform random variable)**.** The cdf for the uniform distribution over $[a, b]$ can be determined analytically:

$$\begin{aligned} F_X(x) = \Pr(X < x) &= \int_{-\infty}^{x} f(t)\ dt \\ &= \frac{1}{b-a}\int_a^x dt = \frac{1}{b-a}[t]_a^x = \frac{x-a}{b-a}. \end{aligned}$$

You can determine the pdf of X directly from the cdf via,

$$f(x) = F'_X(x) = \frac{d}{dx}F_X(x) = \frac{1}{b-a}.$$

■

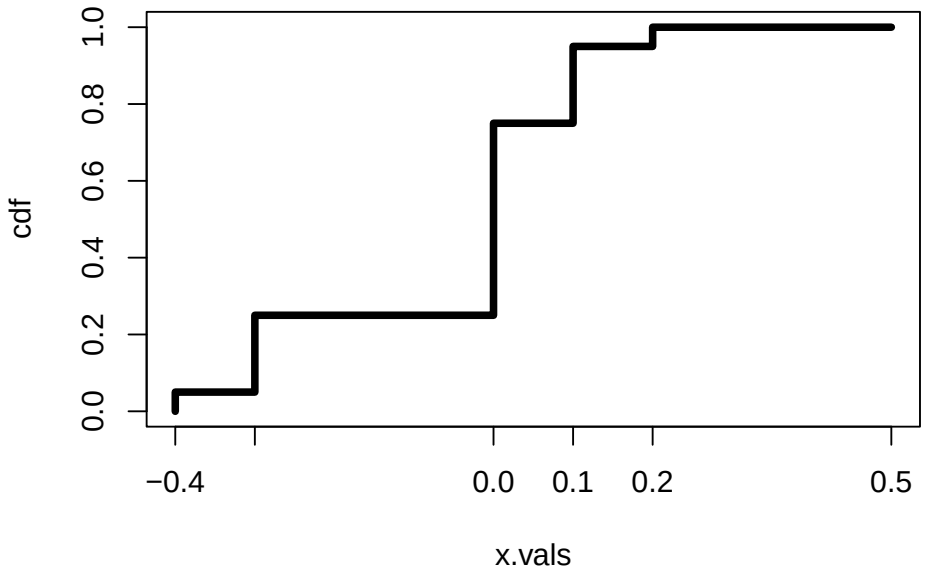


Figure 2.4: CDF of Discrete Distribution for Microsoft Stock Return.

Example 2.8 (CDF for a standard normal random variable)**.** The cdf of standard normal random variable X is used so often in statistics that it is given its own special symbol:

$$\Phi(x) = F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \, dz. \tag{2.3}$$

The cdf $\Phi(x)$, however, does not have an analytic representation like the cdf of the uniform distribution and so the integral in Equation 2.3 must be approximated using numerical techniques. A graphical representation of $\Phi(x)$ is given in Figure 2.5. Because the standard normal pdf, $\phi(x)$, is symmetric about zero and bell-shaped the standard normal CDF, $\Phi(x)$, has an "S" shape where the middle of the "S" is at zero. ■

2.1.4 Quantiles of the distribution of a random variable

Definition 2.7 (Quantiles of the distribution of a random variable)**.** Consider a random variable X with continuous cdf $F_X(x)$. For $0 \leq \alpha \leq 1$, the $100 \cdot \alpha\%$ *quantile* of the distribution for X is the value q_α that satisfies:

$$F_X(q_\alpha) = \Pr(X \leq q_\alpha) = \alpha.$$

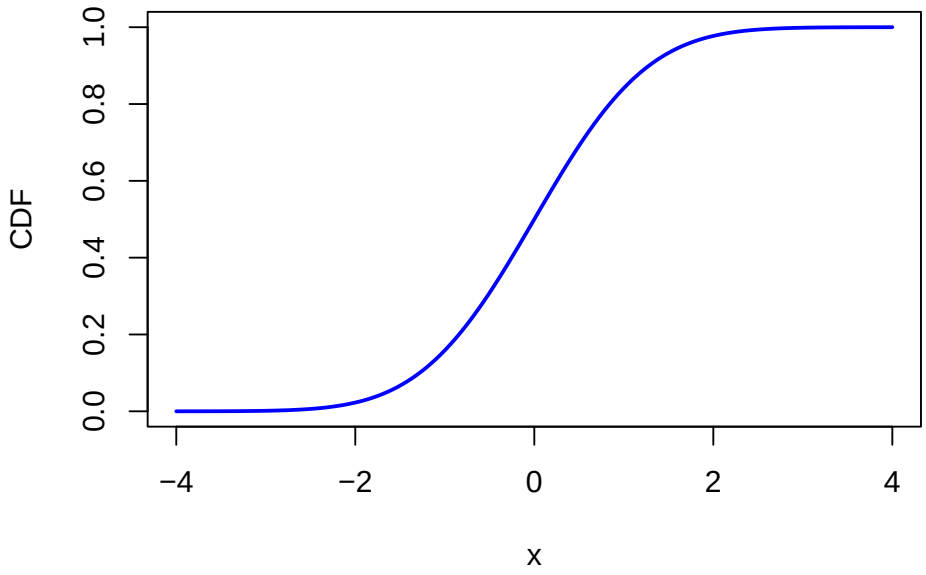


Figure 2.5: Standard normal cdf $\Phi(x)$.

For example, the 5% quantile of X, $q_{0.05}$, satisfies,

$$F_X(q_{0.05}) = \Pr(X \leq q_{0.05}) = 0.05.$$

The *median* of the distribution is the 50% quantile. That is, the median, $q_{0.5}$, satisfies,

$$F_X(q_{0.5}) = \Pr(X \leq q_{0.5}) = 0.5.$$

If F_X is invertible then q_α may be determined analytically as:

$$q_\alpha = F_X^{-1}(\alpha) \tag{2.4}$$

where F_X^{-1} denotes the inverse function of F_X.[3] Hence, the 5% quantile and the median may be determined from

$$q_{0.05} = F_X^{-1}(.05), q_{0.5} = F_X^{-1}(.5).$$

The inverse cdf F_X^{-1} is sometimes called the *quantile function*.

[3]The inverse function of F_X , denoted F_X^{-1}, has the property that $F_X^{-1}(F_X(x)) = x$. F_X^{-1} will exist if F_X is strictly increasing and is continuous.

Example 2.9 (Quantiles from a uniform distribution)**.** Let $X \sim U[a, b]$ where $b > a$. Recall, the cdf of X is given by:

$$F_X(x) = \frac{x-a}{b-a}, \ a \leq x \leq b,$$

which is continuous and strictly increasing. Given $\alpha \in [0, 1]$ such that $F_X(x) = \alpha$, solving for x gives the inverse cdf:

$$x = F_X^{-1}(\alpha) = \alpha(b-a) + a. \tag{2.5}$$

Using Equation 2.5, the 5% quantile and median, for example, are given by:

$$\begin{aligned} q_{0.05} &= F_X^{-1}(.05) = .05(b-a) + a = .05b + .95a, \\ q_{0.5} &= F_X^{-1}(.5) = .5(b-a) + a = .5(a+b). \end{aligned}$$

If $a = 0$ and $b = 1$, then $q_{0.05} = 0.05$ and $q_{0.5} = 0.5$. ■

Example 2.10 (Quantiles from a standard normal distribution)**.** Let $X \sim N(0, 1)$. The quantiles of the standard normal distribution are determined by solving

$$q_\alpha = \Phi^{-1}(\alpha), \tag{2.6}$$

where Φ^{-1} denotes the inverse of the cdf Φ. This inverse function must be approximated numerically and is available in most spreadsheets and statistical software. Using the numerical approximation to the inverse function, the 1%, 2.5%, 5%, 10% quantiles and median are given by:

$$\begin{aligned} q_{0.01} &= \Phi^{-1}(.01) = -2.33, \ q_{0.025} = \Phi^{-1}(.025) = -1.96, \\ q_{0.05} &= \Phi^{-1}(.05) = -1.645, \ q_{0.10} = \Phi^{-1}(.10) = -1.28, \\ q_{0.5} &= \Phi^{-1}(.5) = 0. \end{aligned}$$

Often, the standard normal quantile is denoted z_α. ■

2.1.5 R functions for discrete and continuous distributions

R has built-in functions for a number of discrete and continuous distributions that are commonly used in probability modeling and statistical analysis. These are summarized in Table Table 2.2. For each distribution, there are four functions starting with `d`, `p`, `q`, and `r` that compute density (pdf) values, cumulative probabilities (cdf), quantiles (inverse cdf) and random draws, respectively. Consider, for example, the functions associated with the normal distribution. The functions `dnorm()`, `pnorm()` and `qnorm()` evaluate the standard normal density Equation 2.2, the cdf Equation 2.3, and the inverse cdf or quantile function Equation 2.6, respectively, with the default values `mean = 0` and `sd = 1`. The function `rnorm()` returns a specified number of simulated values from the normal distribution.

Table 2.2: Probability distributions in base R.

Distribution	**Function (root)**	Parameters	Defaults
beta	`beta`	shape1, shape2	_, _
binomial	`binom`	size, prob	_, _
cauchy	`cauchy`	location, scale	0, 1
chi-squared	`chisq`	df, ncp	_, 1
F	`f`	df1, df2	_, _
gamma	`gamma`	shape, rate, scale	_, 1, 1/rate
geometric	`geom`	prob	_
hyper-geometric	`hyper`	m, n, k	_, _, _
log-normal	`lnorm`	meanlog, sdlog	0, 1
logistic	`logis`	location, scale	0, 1
negative binomial	`nbinom`	size, prob, mu	_,_,_
normal	`norm`	mean, sd	0, 1
Poisson	`pois`	Lambda	1
Student's t	`t`	df, ncp	_, 1
uniform	`unif`	min, max	0, 1
Weibull	`weibull`	shape, scale	_, 1
Wilcoxon	`wilcoxon`	m, n	_, _

Commonly used distributions used in finance application are the binomial, chi-squared, log normal, normal, and Student's t.

2.1.5.1 Finding areas under the standard normal curve

Let X denote a standard normal random variable. Given that the total area under the normal curve is one and the distribution is symmetric about zero, the following results hold:

- $\Pr(X \leq z) = 1 - \Pr(X \geq z)$ and $\Pr(X \geq z) = 1 - \Pr(X \leq z)$
- $\Pr(X \geq z) = \Pr(X \leq -z)$
- $\Pr(X \geq 0) = \Pr(X \leq 0) = 0.5$

The following examples show how to do probability calculations with standard normal random variables.

Example 2.11 (Finding areas under the normal curve using R)**.** First, consider finding $\Pr(X \geq 2)$. By the symmetry of the normal distribution, $\Pr(X \geq 2) = \Pr(X \leq -2) = \Phi(-2)$. In R use:

```
pnorm(-2)
#> [1] 0.0228
```

Next, consider finding $\Pr(-1 \leq X \leq 2)$. Using the cdf, you compute $\Pr(-1 \leq X \leq 2) = \Pr(X \leq 2) - \Pr(X \leq -1) = \Phi(2) - \Phi(-1)$. In R use:

```
pnorm(2) - pnorm(-1)
#> [1] 0.819
```

Finally, using R the exact values for $\Pr(-1 \leq X \leq 1)$, $\Pr(-2 \leq X \leq 2)$ and $\Pr(-3 \leq X \leq 3)$ are:

```
pnorm(1) - pnorm(-1)
#> [1] 0.683
pnorm(2) - pnorm(-2)
#> [1] 0.954
pnorm(3) - pnorm(-3)
#> [1] 0.997
```

■

2.1.5.2 Plotting distributions

When working with a probability distribution, it is a good idea to make plots of the pdf or cdf to reveal important characteristics. The following examples illustrate plotting distributions using R.

Example 2.12 (Plotting the standard normal curve)**.** The graphs of the standard normal pdf and cdf in Figures Figure 2.3 and Figure 2.5 were created using the following R code:

```
# plot pdf
x.vals = seq(-4, 4, length=150)
plot(x.vals, dnorm(x.vals), type="l", lwd=2, col="blue",
     xlab="x", ylab="pdf")
# plot cdf
plot(x.vals, pnorm(x.vals), type="l", lwd=2, col="blue",
     xlab="x", ylab="CDF")
```

■

Example 2.13 (Shading a region under the standard normal curve)**.** Figure 2.2 showing $\Pr(-2 \leq X \leq 1)$ as a red shaded area is created with the following code:

```
lb = -2
ub = 1
x.vals = seq(-4, 4, length=150)
d.vals = dnorm(x.vals)
# plot normal density
plot(x.vals, d.vals, type="l", xlab="x", ylab="pdf")
i = x.vals >= lb & x.vals <= ub
# add shaded region between -2 and 1
polygon(c(lb, x.vals[i], ub), c(0, d.vals[i], 0), col="red")
```

■

2.1.6 Shape characteristics of probability distributions

Very often you would like to know certain shape characteristics of a probability distribution. You might want to know where the distribution is centered, and how spread out the distribution is about the central value. You might want to know if the distribution is symmetric about the center or if the distribution has a long left or right tail. For stock returns you might want to know about the likelihood of observing extreme values for returns representing market crashes. This means that you would like to know about the amount of probability in the extreme tails of the distribution. In this section I will discuss four important shape characteristics of a probability distribution:

1. *expected value* (mean): measures the center of mass of a distribution
2. *variance* and *standard deviation* (volatility): measures the spread about the mean
3. *skewness*: measures symmetry about the mean
4. *kurtosis*: measures "tail thickness"

2.1.6.1 Expected Value

The *expected value* of a random variable X, denoted $E[X]$ or μ_X, measures the center of mass of the pdf.

Definition 2.8 (Expected value of discrete random variable)**.** For a discrete random variable X with sample space S_X, the expected value is defined as:

$$\mu_X = E[X] = \sum_{x \in S_X} x \cdot \Pr(X = x). \tag{2.7}$$

Equation 2.7 shows that $E[X]$ is a probability weighted average of the possible values of X.

Example 2.14 (Expected value of discrete random variable)**.** Using the discrete distribution for the return on Microsoft stock in Table Table 2.1, the expected return is computed as:

$$\begin{aligned} E[X] =& (-0.3) \cdot (0.05) + (0.0) \cdot (0.20) + (0.1) \cdot (0.5) \\ &+ (0.2) \cdot (0.2) + (0.5) \cdot (0.05) \\ =& 0.10. \end{aligned}$$

■

Example 2.15 (Expected value of Bernoulli and binomial random variables)**.** Let X be a Bernoulli random variable with success probability π. Then,

$$E[X] = 0 \cdot (1 - \pi) + 1 \cdot \pi = \pi.$$

That is, the expected value of a Bernoulli random variable is its probability of success. Now, let $Y \sim B(n, \pi)$. It can be shown that:

$$E[Y] = \sum_{k=0}^{n} k \binom{n}{k} \pi^k (1 - \pi)^{n-k} = n\pi. \tag{2.8}$$

■

Definition 2.9 (Expected value of a continuous random variable)**.** For a continuous random variable X with pdf $f(x)$, the *expected value* is defined as:

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\ dx. \tag{2.9}$$

Example 2.16 (Expected value of a uniform random variable)**.** Suppose X has a uniform distribution over the interval $[a, b]$. Then,

$$\begin{aligned} E[X] &= \frac{1}{b-a} \int_a^b x\ dx = \frac{1}{b-a} \left[\frac{1}{2} x^2 \right]_a^b \\ &= \frac{1}{2(b-a)} \left[b^2 - a^2 \right] \\ &= \frac{(b-a)(b+a)}{2(b-a)} = \frac{b+a}{2}. \end{aligned}$$

If $b = -1$ and $a = 1$, then $E[X] = 0$. ■

Example 2.17 (Expected value of a standard normal random variable)**.** Let $X \sim N(0,1)$. Then it can be shown that:

$$E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \, dx = 0.$$

Hence, the standard normal distribution is centered at zero. ■

2.1.6.2 Expectation of a function of a random variable

The other shape characteristics of the distribution of a random variable X are based on expectations of certain functions of X.

Definition 2.10 (Expected value of a function of a random variable)**.** Let $g(X)$ denote some function of the random variable X. If X is a discrete random variable with sample space S_X then:

$$E[g(X)] = \sum_{x \in S_X} g(x) \cdot \Pr(X = x), \tag{2.10}$$

and if X is a continuous random variable with pdf f then,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) \, dx. \tag{2.11}$$

2.1.6.3 Variance and standard deviation

The variance of a random variable X, denoted $\text{var}(X)$ or σ_X^2, measures the spread of the distribution about the mean using the function $g(X) = (X - \mu_X)^2$. If most values of X are close to μ_X then on average $(X - \mu_X)^2$ will be small. In contrast, if many values of X are far below and/or far above μ_X then on average $(X - \mu_X)^2$ will be large. Squaring the deviations about μ_X guarantees a positive value.

Definition 2.11 (Variance and standard deviation)**.** For a discrete or continuous random variable X, the variance and standard deviation of X are defined as:

$$\sigma_X^2 = \text{var}(X) = E[(X - \mu_X)^2], \tag{2.12}$$

$$\sigma_X = \text{sd}(X) = \sqrt{\sigma_X^2} = \sigma_X. \tag{2.13}$$

Because σ_X^2 represents an average squared deviation, it is not in the same units as X. The standard deviation of X is the square root of the variance and is in the same units as X. For "bell-shaped" distributions, σ_X measures the typical size of a deviation from the mean value.

The computation of Equation 2.12 can often be simplified by using the result:

$$\text{var}(X) = E[(X - \mu_X)^2] = E[X^2] - \mu_X^2 \tag{2.14}$$

You will learn how to derive this result later in the chapter.

Example 2.18 (Variance and standard deviation for a discrete random variable)**.** Using the discrete distribution for the return on Microsoft stock in Table Table 2.1 and the result that $\mu_X = 0.1$, it follows that:

$$\begin{aligned}\text{var}(X) &= (-0.3 - 0.1)^2 \cdot (0.05) + (0.0 - 0.1)^2 \cdot (0.20) + (0.1 - 0.1)^2 \cdot (0.5) \\ &\quad + (0.2 - 0.1)^2 \cdot (0.2) + (0.5 - 0.1)^2 \cdot (0.05) \\ &= 0.020.\end{aligned}$$

Alternatively, you can compute var(X) using Equation 2.14:

$$\begin{aligned}E[X^2] - \mu_X^2 &= (-0.3)^2 \cdot (0.05) + (0.0)^2 \cdot (0.20) + (0.1)^2 \cdot (0.5) \\ &\quad + (0.2)^2 \cdot (0.2) + (0.5)^2 \cdot (0.05) - (0.1)^2 \\ &= 0.020.\end{aligned}$$

The standard deviation is $\sqrt{0.020} = 0.141$. Given that the distribution is fairly bell-shaped you can say that typical values deviate from the mean value of 10% by about $\pm 14.1\%$. ■

Example 2.19 (Variance and standard deviation of a Bernoulli and binomial random variables)**.** Let X be a Bernoulli random variable with success probability π. Given that $\mu_X = \pi$ it follows that:

$$\begin{aligned}\text{var}(X) &= (0 - \pi)^2 \cdot (1 - \pi) + (1 - \pi)^2 \cdot \pi \\ &= \pi^2(1 - \pi) + (1 - \pi)^2\pi \\ &= \pi(1 - \pi)\left[\pi + (1 - \pi)\right] \\ &= \pi(1 - \pi), \\ \text{sd}(X) &= \sqrt{\pi(1 - \pi)}.\end{aligned}$$

Now, let $Y \sim B(n, \pi)$: It can be shown that:

$$\text{var}(Y) = n\pi(1 - \pi)$$

and so $\text{sd}(Y) = \sqrt{n\pi(1 - \pi)}$. ■

Example 2.20 (Variance and standard deviation of a uniform random variable)**.** Let $X \sim U[a, b]$.

Using Equation 2.14 and $\mu_X = \frac{a+b}{2}$, after some algebra, it can be shown that:

$$\text{var}(X) = E[X^2] - \mu_X^2 = \frac{1}{b-a}\int_a^b x^2\, dx - \left(\frac{a+b}{2}\right)^2 = \frac{1}{12}(b-a)^2,$$

and $\text{sd}(X) = (b-a)/\sqrt{12}$. ■

Example 2.21 (Variance and standard deviation of a standard normal random variable)**.** Let $X \sim N(0,1)$. Here, $\mu_X = 0$ and it can be shown that:

$$\sigma_X^2 = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}\, dx = 1.$$

It follows that $\text{sd}(X) = 1$. ■

2.1.6.4 The general normal distribution

Recall, if X has a standard normal distribution then $E[X] = 0$, $\text{var}(X) = 1$. A *general normal* random variable X has $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$ and is denoted $X \sim N(\mu_X, \sigma_X^2)$. Its pdf is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp\left\{-\frac{1}{2\sigma_X^2}(x-\mu_X)^2\right\},\ -\infty \le x \le \infty. \tag{2.15}$$

Showing that $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$ is a bit of work and is good calculus practice. As with the standard normal distribution, areas under the general normal curve cannot be computed analytically. Using numerical approximations, it can be shown that:

$$\begin{aligned}
\Pr(\mu_X - \sigma_X < X < \mu_X + \sigma_X) &\approx 0.67,\\
\Pr(\mu_X - 2\sigma_X < X < \mu_X + 2\sigma_X) &\approx 0.95,\\
\Pr(\mu_X - 3\sigma_X < X < \mu_X + 3\sigma_X) &\approx 0.99.
\end{aligned}$$

Hence, for a general normal random variable about 95% of the time you expect to see values within $\pm$ 2 standard deviations from its mean. Observations more than three standard deviations from the mean are very unlikely.

Example 2.22 (Normal distribution for monthly returns)**.** Let R denote the monthly return on an investment in Microsoft stock, and assume that it is normally distributed with mean $\mu_R = 0.01$ and standard deviation $\sigma_R = 0.10$. That is, $R \sim N(0.01, (0.10)^2)$. Notice that $\sigma_R^2 = 0.01$ and is not in units of return per month. Figure 2.6 illustrates the distribution and is created using:

```
mu.r = 0.01
sd.r = 0.1
```

```
x.vals = seq(-4, 4, length=150)*sd.r + mu.r
plot(x.vals, dnorm(x.vals, mean=mu.r, sd=sd.r), type="l", lwd=2,
     col="blue", xlab="x", ylab="pdf")
```

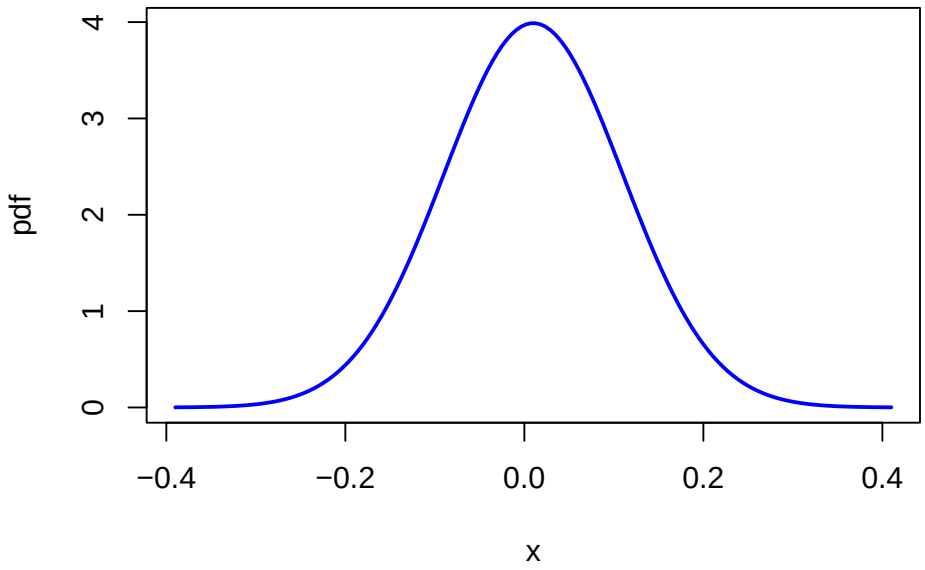


Figure 2.6: Normal distribution for the monthly returns on Microsoft: $R \sim N(0.01, (0.10)^2)$.

Notice that essentially all of the probability lies between -0.3 and 0.3. Using the R function `pnorm()`, you can easily compute the probabilities $\Pr(R < -0.5)$, $\Pr(R < 0)$, $\Pr(R > 0.5)$ and $\Pr(R > 1)$:

```
pnorm(-0.5, mean=0.01, sd=0.1)
#> [1] 1.7e-07
pnorm(0, mean=0.01, sd=0.1)
#> [1] 0.46
1 - pnorm(0.5, mean=0.01, sd=0.1)
#> [1] 4.79e-07
1 - pnorm(1, mean=0.01, sd=0.1)
#> [1] 0
```

Using the R function `qnorm()`, you can find the quantiles $q_{0.01}$, $q_{0.05}$, $q_{0.95}$ and $q_{0.99}$:

```
a.vals = c(0.01, 0.05, 0.95, 0.99)
qnorm(a.vals, mean=0.01, sd=0.10)
#> [1] -0.223 -0.154  0.174  0.243
```

Hence, over the next month, there are 1% and 5% chances of losing more than 22.2% and 15.5%, respectively. In addition, there are 5% and 1% chances of gaining more than 17.5% and 24.3%, respectively. ■

Example 2.23 (Risk-return tradeoff)**.** Consider the following investment problem. You can invest in two non-dividend paying stocks, Amazon and Boeing, over the next month. Let R_A denote the monthly return on Amazon and R_B denote the monthly return on Boeing. Assume that $R_A \sim N(0.02, (0.10)^2)$ and $R_B \sim N(0.01, (0.05)^2)$. Figure 2.7 shows the pdfs for the two returns. Notice that $\mu_A = 0.02 > \mu_B = 0.01$ but also that $\sigma_A = 0.10 > \sigma_B = 0.05$. The return you expect on Amazon is bigger than the return you expect on Boeing but the variability of Amazon's return is also greater than the variability of Boeing's return. The high return variability (volatility) of Amazon reflects the higher risk associated with investing in Amazon compared to investing in Boeing. If you invest in Boeing you get a lower expected return, but you also get less return variability or risk. For example, the probability of losing more than 10% in one month for Amazon is

```
pnorm(-0.1, 0.02, 0.1)
#> [1] 0.115
```

and the probability of losing more than 10% in one month for Boeing is

```
pnorm(-0.1, 0.01, 0.05)
#> [1] 0.0139
```

Of course, there is also a higher probability for upside return for Amazon than for Boeing. This example illustrates the fundamental "no free lunch" principle of economics and finance: you can't get something for nothing. In general, to get a higher expected return you must be prepared to take on higher risk. ■

Example 2.24 (Why the normal distribution may not be appropriate for simple returns)**.** Let R_t denote the simple annual return on an asset, and suppose that $R_t \sim N(0.05, (0.50)^2)$. Because asset prices must be non-negative, R_t must always be larger than -1. However, the normal distribution is defined for $-\infty \leq R_t \leq \infty$ and based on the assumed normal distribution $\Pr(R_t < -1) = 0.018$. That is, there is a 1.8% chance that R_t is smaller than -1. This implies that there is a 1.8% chance that the asset price at the end of the year will be negative! This is one reason why the normal distribution may not be appropriate for simple returns. ■

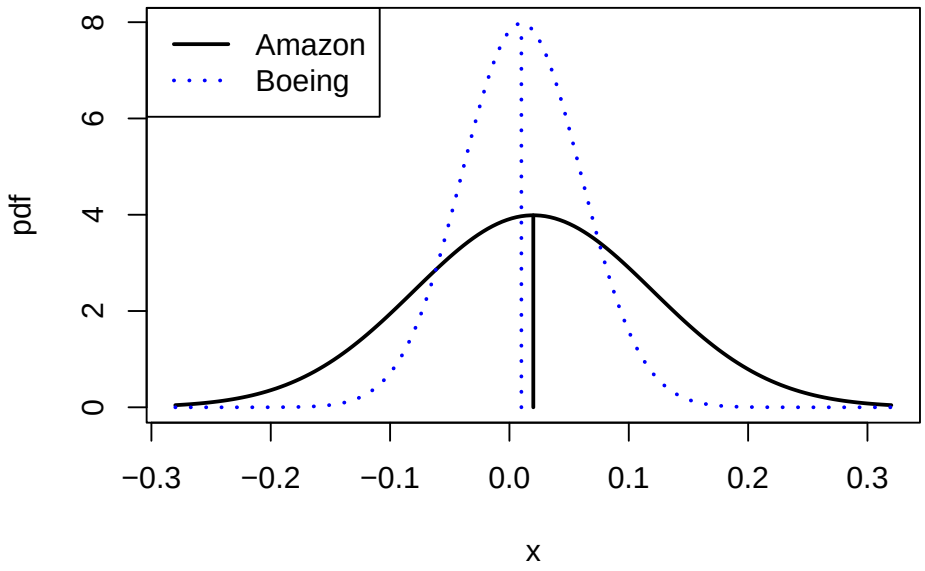


Figure 2.7: Risk-return tradeoff between one-month investments in Amazon and Boeing stock.

Example 2.25 (The normal distribution is more appropriate for continuously compounded returns)**.** Let $r_t = \ln(1 + R_t)$ denote the continuously compounded annual return on an asset, and suppose that $r_t \sim N(0.05, (0.50)^2)$. Unlike the simple return, the continuously compounded return can take on values less than -1. In fact, r_t is defined for $-\infty \leq r_t \leq \infty$. For example, suppose $r_t = -2$. This implies a simple return of $R_t = e^{-2} - 1 = -0.865$.[4] Then $\Pr(r_t \leq -2) = \Pr(R_t \leq -0.865) = 0.00002$. Although the normal distribution allows for values of r_t smaller than -1, the implied simple return R_t will always be greater than -1. ■

2.1.6.5 The Log-Normal Distribution

Let $X \sim N(\mu_X, \sigma_X^2)$, which is defined for $-\infty < X < \infty$. The *log-normally* distributed random variable Y is determined from the normally distributed random variable X using the transformation $Y = e^X$. Here, Y is log-normally distributed and you write:

$$Y \sim \ln N(\mu_X, \sigma_X^2),\ 0 < Y < \infty. \tag{2.16}$$

[4] If $r_t = -\infty$ then $R_t = e^{r_t} - 1 = e^{-\infty} - 1 = 0 - 1 = -1$.

The pdf of the log-normal distribution for Y can be derived from the normal distribution for X using the change-of-variables formula from calculus and is given by:

$$f(y) = \frac{1}{y\sigma_X\sqrt{2\pi}} e^{-\frac{(\ln y - \mu_X)^2}{2\sigma_X^2}}. \tag{2.17}$$

Due to the exponential transformation, Y is only defined for non-negative values. It can be shown that:

$$\mu_Y = E[Y] = e^{\mu_X + \sigma_X^2/2},$$
$$\sigma_Y^2 = \text{var}(Y) = e^{2\mu_X + \sigma_X^2}(e^{\sigma_X^2} - 1).$$

Example 2.26 (Log-normal distribution for simple returns)**.** Let $r_t = \ln(P_t/P_{t-1})$ denote the continuously compounded monthly return on an asset and assume that $r_t \sim N(0.05, (0.50)^2)$. That is, $\mu_r = 0.05$ and $\sigma_r = 0.50$. Let $R_t = (P_t - P_{t-1})/P_t$ denote the simple monthly return. The relationship between r_t and R_t is given by $r_t = \ln(1 + R_t)$ and $1 + R_t = e^{r_t}$. Since r_t is normally distributed, $1 + R_t$ is log-normally distributed. Notice that the distribution of $1 + R_t$ is only defined for positive values of $1 + R_t$. This is appropriate since the smallest value that R_t can take on is -1. Using Equation 2.18, the mean and variance for $1 + R_t$ are given by:

$$\mu_{1+R} = e^{0.05 + (0.5)^2/2} = 1.191$$
$$\sigma_{1+R}^2 = e^{2(0.05) + (0.5)^2}(e^{(0.5)^2} - 1) = 0.563$$

The pdfs for r_t and R_t are shown in Figure 2.8. ■

2.1.6.6 Skewness

Definition 2.12 (Skewness)**.** The *skewness* of a random variable X, denoted skew(X), measures the symmetry of a distribution about its mean value using the function $g(X) = (X - \mu_X)^3/\sigma_X^3$, where σ_X^3 is just sd(X) raised to the third power:

$$\text{skew}(X) = \frac{E[(X - \mu_X)^3]}{\sigma_X^3} \tag{2.18}$$

When X is far below its mean, $(X - \mu_X)^3$ is a big negative number. When X is far above its mean, $(X - \mu_X)^3$ is a big positive number. Hence, if there are more big values of X below μ_X then skew$(X) < 0$. Conversely, if there are more big values of X above μ_X then skew$(X) > 0$. If X has a symmetric distribution, then skew$(X) = 0$ since positive and negative values in Equation 2.18 cancel out. If skew$(X) > 0$, then the distribution of X has a *long right tail* and if skew$(X) < 0$ the distribution of X has a *long left tail*. These cases are illustrated in Figure 2.9.

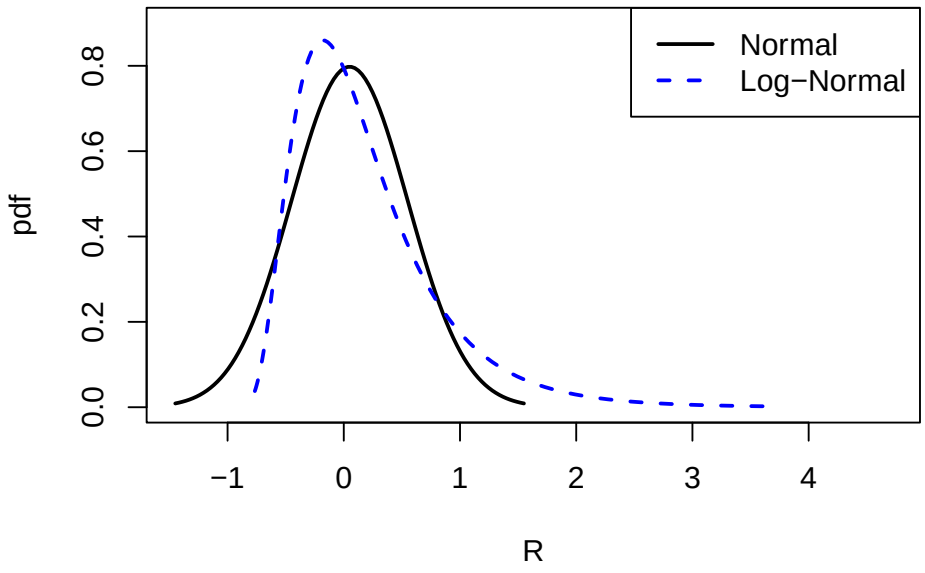


Figure 2.8: Normal distribution for r_t and log-normal distribution for $R_t = e^{r_t} - 1$.

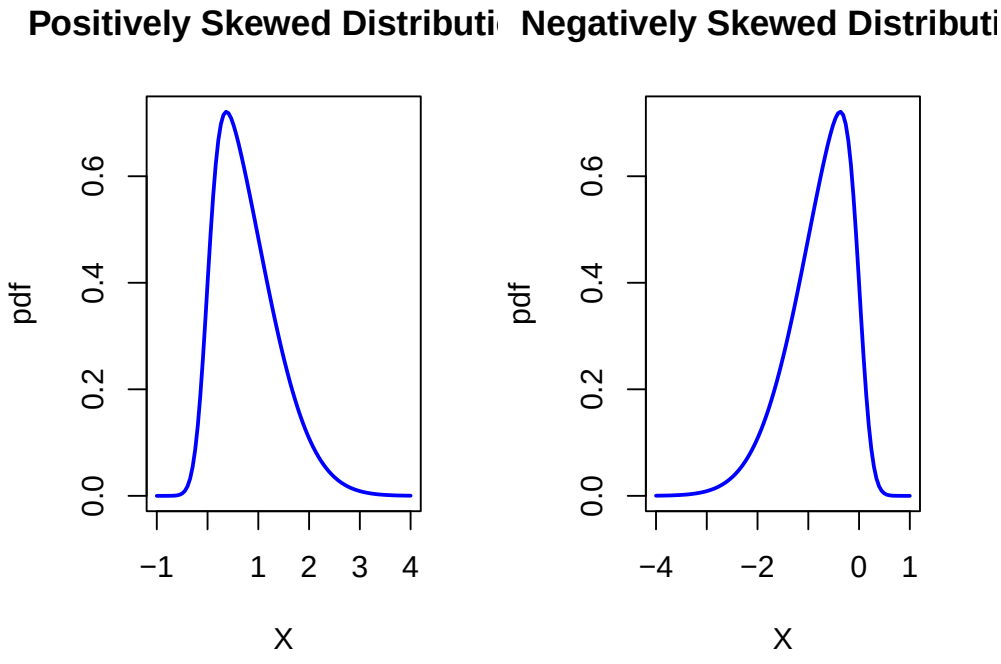


Figure 2.9: Skewed distributions.

Example 2.27 (Skewness for a discrete random variable)**.** From the discrete distribution for the return on Microsoft stock in Table Table 2.1, and the results $\mu_X = 0.1$ and $\sigma_X = 0.141$, it follows that:

$$\begin{aligned}\text{skew}(X) &= [(-0.3-0.1)^3 \cdot (0.05) + (0.0-0.1)^3 \cdot (0.20) + (0.1-0.1)^3 \cdot (0.5) \\ &\quad + (0.2-0.1)^3 \cdot (0.2) + (0.5-0.1)^3 \cdot (0.05)]/(0.141)^3 \\ &= 0.0.\end{aligned}$$

■

Example 2.28 (Skewness for a normal random variable)**.** Suppose X has a general normal distribution with mean μ_X and variance σ_X^2. Then it can be shown that:

$$\text{skew}(X) = \int_{-\infty}^{\infty} \frac{(x-\mu_X)^3}{\sigma_X^3} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma_X^2}(x-\mu_X)^2}\, dx = 0.$$

This result is expected since the normal distribution is symmetric about it's mean value μ_X. ■

Example 2.29 (Skewness for a log-Normal random variable)**.** Let $Y = e^X$, where $X \sim N(\mu_X, \sigma_X^2)$, be a log-normally distributed random variable with parameters μ_X and σ_X^2. Then it can be shown that:

$$\text{skew}(Y) = \left(e^{\sigma_X^2} + 2\right)\sqrt{e^{\sigma_X^2} - 1} > 0.$$

Notice that skew(Y) is always positive, indicating that the distribution of Y has a long right tail, and that it is an increasing function of σ_X^2. This positive skewness is illustrated in Figure 2.8. ■

2.1.6.7 Kurtosis

Definition 2.13 (Kurtosis)**.** The *kurtosis* of a random variable X, denoted kurt(X), measures the thickness in the tails of a distribution and is based on $g(X) = (X-\mu_X)^4/\sigma_X^4$:

$$\text{kurt}(X) = \frac{E[(X-\mu_X)^4]}{\sigma_X^4}, \tag{2.19}$$

where σ_X^4 is just sd(X) raised to the fourth power.

Since kurtosis is based on deviations from the mean raised to the fourth power, large deviations get lots of weight. Hence, distributions with large kurtosis values are ones where there is the possibility of extreme values. In contrast, if the kurtosis is small then most of the observations are tightly clustered around the mean and there is very little probability of observing extreme values.

Example 2.30 (Kurtosis for a discrete random variable)**.** From the discrete distribution for the return on Microsoft stock in Table Table 2.1, and the results that $\mu_X = 0.1$ and $\sigma_X = 0.141$, it follows that:

$$\begin{aligned}\text{kurt}(X) &= [(-0.3-0.1)^4 \cdot (0.05) + (0.0-0.1)^4 \cdot (0.20) + (0.1-0.1)^4 \cdot (0.5) \\ &\quad + (0.2-0.1)^4 \cdot (0.2) + (0.5-0.1)^4 \cdot (0.05)]/(0.141)^4 \\ &= 6.5.\end{aligned}$$

■

Example 2.31 (Kurtosis for a normal random variable)**.** Suppose X has a general normal distribution with mean μ_X and variance σ_X^2. Then it can be shown that:

$$\text{kurt}(X) = \int_{-\infty}^{\infty} \frac{(x-\mu_X)^4}{\sigma_X^4} \cdot \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{1}{2}(\frac{x-\mu_X}{\sigma_X})^2}\, dx = 3.$$

Hence, a kurtosis of 3 is a benchmark value for tail thickness of bell-shaped distributions. If a distribution has a kurtosis greater than 3, then the distribution has thicker tails than the normal distribution. If a distribution has kurtosis less than 3, then the distribution has thinner tails than the normal. ■

Sometimes the kurtosis of a random variable is described relative to the kurtosis of a normal random variable. This relative value of kurtosis is referred to as *excess kurtosis* and is defined as:

$$\text{ekurt}(X) = \text{kurt}(X) - 3 \tag{2.20}$$

If the excess kurtosis of a random variable is equal to zero then the random variable has the same kurtosis as a normal random variable. If excess kurtosis is greater than zero, then kurtosis is larger than that for a normal; if excess kurtosis is less than zero, then kurtosis is less than that for a normal.

2.1.6.8 The Student's t Distribution

The kurtosis of a random variable gives information on the tail thickness of its distribution. The normal distribution, with kurtosis equal to three, gives a benchmark for the tail thickness of symmetric distributions. A distribution similar to the standard normal distribution but with fatter tails, and hence larger kurtosis, is the Student's t distribution. If X has a Student's t distribution with degrees of freedom parameter v, denoted $X \sim t_v$, then its pdf has the form:

$$f(x) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\Gamma\left(\frac{v}{2}\right)} \left(1 + \frac{x^2}{v}\right)^{-\left(\frac{v+1}{2}\right)}, \quad -\infty < x < \infty, \; v > 0. \tag{2.21}$$

where $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$ denotes the gamma function. It can be shown that:

$$\begin{aligned} E[X] &= 0, \ v > 1 \\ \text{var}(X) &= \frac{v}{v-2}, \ v > 2, \\ \text{skew}(X) &= 0, \ v > 3, \\ \text{kurt}(X) &= \frac{6}{v-4} + 3, \ v > 4. \end{aligned}$$

The parameter v controls the scale and tail thickness of the distribution. If v is close to four, then the kurtosis is large and the tails are thick. If $v < 4$, then $\text{kurt}(X) = \infty$. As $v \to \infty$ the Student's t pdf approaches that of a standard normal random variable and $\text{kurt}(X) = 3$. Figure 2.10 shows plots of the Student's t density for various values of v.

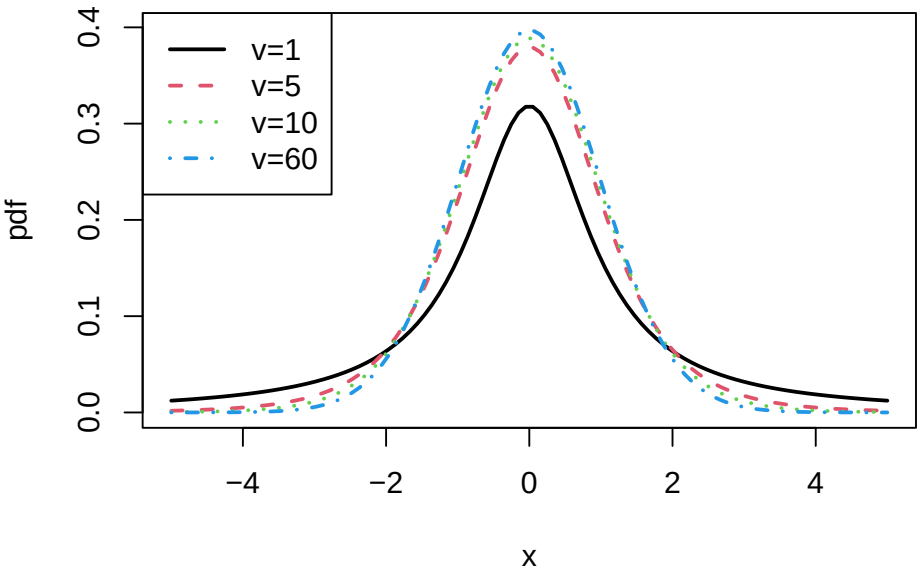


Figure 2.10: Student's t density with $v = 1, 5, 10$ and 60.

Example 2.32 (Computing tail probabilities and quantiles from the Student's t distribution)**.** The R functions `pt()` and `qt()` can be used to compute the cdf and quantiles of a Student's t random variable. For $v = 1, 2, 5, 10, 60, 100$ and ∞ the 1% quantiles can be computed using:

```
v = c(1, 2, 5, 10, 60, 100, Inf)
qt(0.01, df=v)
#> [1] -31.82  -6.96  -3.36  -2.76  -2.39  -2.36  -2.33
```

For $v = 1, 2, 5, 10, 60, 100$ and ∞ the values of $\Pr(X < -3)$ are:

```
pt(-3, df=v)
#> [1] 0.10242 0.04773 0.01505 0.00667 0.00196 0.00170 0.00135
```

■

2.1.7 Linear functions of a random variable

Let X be a random variable either discrete or continuous with $E[X] = \mu_X$, $\text{var}(X) = \sigma_X^2$, and let a and b be known constants. Define a new random variable Y via the linear function of X:

$$Y = g(X) = aX + b.$$

Then the following results hold:

$$\begin{aligned}
\mu_Y &= E[Y] = a \cdot E[X] + b = a \cdot \mu_X + b, \\
\sigma_Y^2 &= \text{var}(Y) = a^2 \text{var}(X) = a^2 \sigma_X^2, \\
\sigma_Y &= \text{sd}(Y) = a \cdot \text{sd}(X) = a \cdot \sigma_X.
\end{aligned}$$

The first result shows that expectation is a linear operation. That is,

$$E[aX + b] = aE[X] + b.$$

The second result shows that adding a constant to X does not affect its variance, and that the effect of multiplying X by the constant a increases the variance of X by the square of a. These results will be used often enough that it is instructive to go through the derivations.

Consider the first result. Let X be a discrete random variable. By the definition of $E[g(X)]$, with $g(X) = b + aX$, you can write:

$$\begin{aligned}
E[Y] &= \sum_{x \in S_X} (ax + b) \cdot \Pr(X = x) \\
&= a \sum_{x \in S_X} x \cdot \Pr(X = x) + b \sum_{x \in S_X} \Pr(X = x) \\
&= a \cdot E[X] + b \cdot 1 \\
&= a \cdot \mu_X + b.
\end{aligned}$$

If X is a continuous random variable then by the linearity of integration:

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} (ax+b) f(x)\ dx = a\int_{-\infty}^{\infty} x f(x)\ dx + b\int_{-\infty}^{\infty} f(x)\ dx \\ &= aE[X] + b. \end{aligned}$$

Next consider the second result. Since $\mu_Y = a\mu_X + b$:

$$\begin{aligned} \text{var}(Y) &= E[(Y-\mu_Y)^2] \\ &= E[(aX+b-(a\mu_X+b))^2] \\ &= E[(a(X-\mu_X)+(b-b))^2] \\ &= E[a^2(X-\mu_X)^2] \\ &= a^2 E[(X-\mu_X)^2] \text{ (by the linearity of } E[\cdot]) \\ &= a^2 \text{var}(X) \end{aligned}$$

Notice that the derivation of the second result works for discrete and continuous random variables.

A normal random variable has the special property that a linear function of it is also a normal random variable. The following proposition states the result.

Proposition 2.1 (Linear function of a normal random variable)**.** *Let $X \sim N(\mu_X, \sigma_X^2)$ and let a and b be constants. Let $Y = aX + b$. Then $Y \sim N(a\mu_X + b, a^2\sigma_X^2)$.*

The above property is special to the normal distribution and may or may not hold for a random variable with a distribution that is not normal.

Example 2.33 (Standardizing a random variable)**.** Let X be a random variable with $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$. Define a new random variable Z as:

$$Z = \frac{X-\mu_X}{\sigma_X} = \frac{1}{\sigma_X}X - \frac{\mu_X}{\sigma_X},$$

which is a linear function $aX + b$ where $a = \frac{1}{\sigma_X}$ and $b = -\frac{\mu_X}{\sigma_X}$. This transformation is called *standardizing* the random variable X since,

$$\begin{aligned} E[Z] &= \frac{1}{\sigma_X}E[X] - \frac{\mu_X}{\sigma_X} = \frac{1}{\sigma_X}\mu_X - \frac{\mu_X}{\sigma_X} = 0, \\ \text{var}(Z) &= \left(\frac{1}{\sigma_X}\right)^2 \text{var}(X) = \frac{\sigma_X^2}{\sigma_X^2} = 1. \end{aligned}$$

Hence, standardization creates a new random variable with mean zero and variance 1. In addition, if X is normally distributed then $Z \sim N(0,1)$. ■

Example 2.34 (Computing probabilities using standardized random variables)**.** Let $X \sim N(2,4)$ and suppose you want to find $\Pr(X > 5)$ but you only know probabilities associated with a standard normal random variable $Z \sim N(0,1)$. You can solve the problem by standardizing X as follows:

$$\Pr(X > 5) = \Pr\left(\frac{X-2}{\sqrt{4}} > \frac{5-2}{\sqrt{4}}\right) = \Pr\left(Z > \frac{3}{2}\right) = 0.06681.$$

■

Standardizing a random variable is often done in the construction of test statistics. For example, the so-called *t-statistic* or *t-ratio* used for testing simple hypotheses on coefficients is constructed by the standardization process.

A non-standard random variable X with mean μ_X and variance σ_X^2 can be created from a standard random variable via the linear transformation:

$$X = \mu_X + \sigma_X \cdot Z. \tag{2.22}$$

This result is useful for modeling purposes as illustrated in the next examples.

Example 2.35 (Quantile of general normal random variable)**.** Let $X \sim N(\mu_X, \sigma_X^2)$. Quantiles of X can be conveniently computed using:

$$q_\alpha = \mu_X + \sigma_X z_\alpha, \tag{2.23}$$

where $\alpha \in (0,1)$ and z_α is the $\alpha \times 100\%$ quantile of a standard normal random variable. This formula is derived as follows. By the definition z_α and Equation 2.22

$$\alpha = \Pr(Z \leq z_\alpha) = \Pr(\mu_X + \sigma_X \cdot Z \leq \mu_X + \sigma_X \cdot z_\alpha) = \Pr(X \leq \mu_X + \sigma_X \cdot z_\alpha),$$

which implies Equation 2.23. ■

Example 2.36 (Generalized Student's t distribution)**.** The Student's t distribution defined in Equation 2.21 has zero mean and variance equal to $v/(v-2)$ where v is the degrees of freedom parameter. You can define a Student's t random variable with non-zero mean μ and variance σ^2 from a Student's t random variable X as follows:

$$Y = \mu + \sigma\left(\frac{v-2}{v}\right)^{1/2} X = \mu + \sigma Z_v,$$

where $Z_v = \left(\frac{v-2}{v}\right)^{1/2} X$. The random variable Z_v is called a *standardized Student's t* random variable with v degrees of freedom since $E[Z_v] = 0$ and $\text{var}(Z_v) = \left(\frac{v-2}{v}\right)\text{var}(X) = \left(\frac{v-2}{v}\right)\left(\frac{v}{v-2}\right) = 1$.

Then

$$\begin{aligned} E[Y] &= \mu + \sigma E[Z] = \mu, \\ \text{var}(Y) &= \sigma^2 \text{var}(Z) = \sigma^2. \end{aligned}$$

I use the notation $Y \sim t_v(\mu, \sigma^2)$ to denote that Y has a *generalized Student's t* distribution with mean μ, variance σ^2, and degrees of freedom v. ■

Example 2.37 (Location scale return model for asset returns)**.** Let r denote the monthly continuously compounded return on an asset with $E[r] = \mu$ and $\text{var}(R) = \sigma^2$. If r has a normal distribution or a general Student's t distribution, then r can be expressed as:

$$r = \mu_r + \sigma_r \cdot \varepsilon,$$

where $\varepsilon \sim N(0, 1)$ or $\varepsilon \sim t_v(0, 1)$. The random variable ε can be interpreted as representing the random news arriving in a given month that makes the observed return differ from its expected value μ. The fact that ε has mean zero means that news, on average, is neutral. The value of σ_r represents the typical size of a news shock. The bigger is σ_r, the larger is the impact of a news shock and vice-versa. The distribution of ε impacts the likelihood of extreme new events (like stock market crashes). If $\varepsilon \sim N(0, 1)$ then extreme news events are not very likely. If $\varepsilon \sim t_v(0, 1)$ with v small (but bigger than 4) then extreme news events may happen from time-to-time. ■

2.1.8 Value-at-Risk: An introduction

As an example of working with linear functions of a normal random variable, and to illustrate the concept of *Value-at-Risk* (VaR), consider an investment of $10,000 in Microsoft stock over the next month. Let R denote the monthly simple return on Microsoft stock and assume that $R \sim N(0.05, (0.10)^2)$. That is, $E[R] = \mu_R = 0.05$ and $\text{var}(R) = \sigma_R^2 = (0.10)^2$. Let W_0 denote the investment value at the beginning of the month and W_1 denote the investment value at the end of the month. In this example, $W_0 = \$10,000$. Consider the following questions:

(i) What is the probability distribution of end-of-month wealth, W_1?
(ii) What is the probability that end-of-month wealth is less than $9,000, and what must the return on Microsoft be for this to happen?
(iii) What is the loss in dollars that would occur if the return on Microsoft stock is equal to its 5% quantile, $q_{.05}$? That is, what is the monthly 5% VaR on the $10,000 investment in Microsoft?

To answer (i), note that end-of-month wealth, W_1, is related to initial wealth W_0 and the return on Microsoft stock R via the linear function:

$$W_1 = W_0(1 + R) = W_0 + W_0 R = \$10,000 + \$10,000 \cdot R.$$

Using the properties of linear functions of a random variable you get:

$$E[W_1] = W_0 + W_0 E[R] = \$10,000 + \$10,000(0.05) = \$10,500,$$

and,

$$\begin{aligned} \text{var}(W_1) &= (W_0)^2 \text{var}(R) = (\$10,000)^2 (0.10)^2, \\ \text{sd}(W_1) &= (\$10,000)(0.10) = \$1,000. \end{aligned}$$

Further, since R is assumed to be normally distributed it follows that W_1 is normally distributed:

$$W_1 \sim N(\$10,500, (\$1,000)^2).$$

To answer (ii), you can use the above normal distribution for W_1 to get:

$$\Pr(W_1 < \$9,000) = 0.067.$$

To find the return that produces end-of-month wealth of \$9,000, or a loss of \$10,000-\$9,000=\$1,000, you solve

$$R = \frac{\$9,000 - \$10,000}{\$10,000} = -0.10.$$

If the monthly return on Microsoft is -10% or less, then end-of-month wealth will be \$9,000 or less. Notice that $R = -0.10$ is the 6.7% quantile of the distribution of R:

$$\Pr(R < -0.10) = 0.067.$$

Question (iii) can be answered in two equivalent ways. First, use $R \sim N(0.05, (0.10)^2)$ and solve for the 5% quantile of Microsoft Stock using Equation 2.23[5]:

$$\begin{aligned} &\Pr(R < q^R_{.05}) = 0.05 \\ &\quad \Rightarrow q^R_{.05} = \mu_R + \sigma_R \cdot z_{.05} = 0.05 + 0.10 \cdot (-1.645) = -0.114. \end{aligned}$$

That is, with 5% probability the return on Microsoft stock is -11.4% or less. Now, if the return on Microsoft stock is -11.4% the loss in investment value that occurs with 5% probability is

$$\text{VaR}_{0.05} = W_0 \cdot q^R_{.05} = \$10,000 \cdot (0.114) = -\$1,144.$$

Since losses are typically reported as positive numbers, $\text{VaR}_{0.05} = \$1,144$ is the 5% VaR over the next month on the \$10,000 investment in Microsoft stock.

For the second method, use $W_1 \sim N(\$10,500, (\$1,000)^2)$ and solve for the 5% quantile of end-of-

[5] Using R, $z_{.05} = \text{qnorm}(0.05) = -1.645$.

month wealth directly:

$$\Pr(W_1 < q_{.05}^{W_1}) = 0.05 \Rightarrow q_{.05}^{W_1} = \$8,856.$$

This corresponds to a loss of investment value of \$10,000-\$8,856=\$1,144. Hence, if W_0 represents the initial wealth and $q_{.05}^{W_1}$ is the 5% quantile of the distribution of W_1 then the 5% VaR is:

$$\text{VaR}_{.05} = W_0 - q_{.05}^{W_1} = \$10,000 - \$8,856 = \$1,144.$$

The following is the usual definition of VaR.

Definition 2.14 (Value-at-Risk)**.** If W_0 represents the initial wealth in dollars and q_α^R is the $\alpha \times 100\%$ quantile of distribution of the simple return R for small values of α, then the $\alpha \times 100\%$ VaR is defined as:

$$\text{VaR}_\alpha = -W_0 \cdot q_\alpha^R. \tag{2.24}$$

In words, VaR_α represents the dollar loss that could occur with probability α. By convention, it is reported as a positive number (hence the multiplication by minus one as $q_\alpha^R < 0$ for small values of α).

2.1.8.1 Value-at-Risk calculations for continuously compounded returns

The above calculations illustrate how to calculate VaR using the normal distribution for simple returns. However, as argued in Example 31, the normal distribution may not be appropriate for characterizing the distribution of simple returns and may be more appropriate for characterizing the distribution of continuously compounded returns. Let R denote the simple monthly return, let $r = \ln(1 + R)$ denote the continuously compounded return and assume that $r \sim N(\mu_r, \sigma_r^2)$. The $\alpha \times 100\%$ monthly VaR on an investment of $\$W_0$ is computed using the following steps:

1. Compute the $\alpha \cdot 100\%$ quantile, q_α^r, from the normal distribution for the continuously compounded return r:
$$q_\alpha^r = \mu_r + \sigma_r z_\alpha,$$
where z_α is the $\alpha \cdot 100\%$ quantile of the standard normal distribution.
2. Convert the continuously compounded return quantile, q_α^r, to a simple return quantile using the transformation:
$$q_\alpha^R = e^{q_\alpha^r} - 1$$
3. Compute VaR using the simple return quantile Equation 2.24.

Step 2 works because quantiles are preserved under monotonically increasing transformations of random variables.

Example 2.38 (Computing VaR from simple and continuously compounded returns using R)**.** Let R denote the simple monthly return on Microsoft stock and assume that $R \sim N(0.05, (0.10)^2)$. Consider an initial investment of $W_0 = \$10,000$. To compute the 1% and 5% VaR values over the next month use:

```
mu.R = 0.05
sd.R = 0.10
w0 = 10000
q.01.R = mu.R + sd.R*qnorm(0.01)
q.05.R = mu.R + sd.R*qnorm(0.05)
VaR.01 = -q.01.R*w0
VaR.05 = -q.05.R*w0
VaR.01
#> [1] 1826
VaR.05
#> [1] 1145
```

Hence with 1% and 5% probability the loss over the next month is at least \$1,826 and \$1,145, respectively.

Let r denote the continuously compounded return on Microsoft stock and assume that $r \sim N(0.05, (0.10)^2)$. To compute the 1% and 5% VaR values over the next month use:

```
mu.r = 0.05
sd.r = 0.10
q.01.R = exp(mu.r + sd.r*qnorm(0.01)) - 1
q.05.R = exp(mu.r + sd.r*qnorm(0.05)) - 1
VaR.01 = -q.01.R*w0
VaR.05 = -q.05.R*w0
VaR.01
#> [1] 1669
VaR.05
#> [1] 1082
```

Notice that when $1 + R = e^r$ has a log-normal distribution, the 1% and 5% VaR values (losses) are slightly smaller than when R is normally distributed. This is due to the positive skewness of the log-normal distribution. ■

2.2 Bivariate Distributions

So far I have only considered probability distributions for a single random variable. In many situations, you want to be able to characterize the probabilistic behavior of two or more random variables simultaneously. For example, you might want to know about the probabilistic behavior of the returns on a two asset portfolio. In this section, I discuss bivariate distributions and important concepts related to the analysis of two random variables.

2.2.1 Discrete random variables

Let X and Y be discrete random variables with sample spaces S_X and S_Y, respectively. The likelihood that X and Y takes values in the joint sample space $S_{XY} = S_X \times S_Y$ (Cartesian product of the individual sample spaces) is determined by the *joint probability distribution* $p(x, y) = \Pr(X = x, Y = y)$. The function $p(x, y)$ satisfies:

- (i) $p(x, y) > 0$ for $x, y \in S_{XY}$;
- (ii) $p(x, y) = 0$ for $x, y \notin S_{XY}$;
- (iii) $\sum_{x,y \in S_{XY}} p(x, y) = \sum_{x \in S_X} \sum_{y \in S_Y} p(x, y) = 1$.

Example 2.39 (Bivariate discrete distribution for stock returns)**.** Let X denote the monthly return (in percent) on Microsoft stock and let Y denote the monthly return on Starbucks stock. For simplicity suppose that the sample spaces for X and Y are $S_X = \{0, 1, 2, 3\}$ and $S_Y = \{0, 1\}$ so that the random variables X and Y are discrete. The joint sample space is the two dimensional grid $S_{XY} = \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1), (3, 0), (3, 1)\}$. Table Table 2.3 illustrates the joint distribution for X and Y. From the table, $p(0, 0) = \Pr(X = 0, Y = 0) = 1/8$. Notice that the sum of all the entries in the table sum to unity. ■

Table 2.3: Discrete bivariate distribution for Microsoft and Starbucks stock prices.

		Y		
	%	**0**	**1**	$\Pr(X)$
	0	1/8	0	1/8
X	1	2/8	1/8	3/8
	2	1/8	2/8	3/8
	3	0	1/8	1/8
	$\Pr(Y)$	**4/8**	**4/8**	**1**

2.2.1.1 Marginal Distributions

The joint probability distribution tells us the probability of X and Y occurring together. What if you only want to know about the probability of X occurring, or the probability of Y occurring?

Example 2.40 (Find $\Pr(X = 0)$ and $\Pr(Y = 1)$ from joint distribution)**.** Consider the joint distribution in Table Table 2.3. What is $\Pr(X = 0)$ regardless of the value of Y? Now X can occur if $Y = 0$ or if $Y = 1$ and since these two events are mutually exclusive you get:

$$\Pr(X = 0) = \Pr(X = 0, Y = 0) + \Pr(X = 0, Y = 1) = 0 + 1/8 = 1/8.$$

Notice that this probability is equal to the horizontal (row) sum of the probabilities in the table at $X = 0$. You can find $\Pr(Y = 1)$ in a similar fashion:

$$\begin{aligned}\Pr(Y = 1) &= \Pr(X = 0, Y = 1) + \Pr(X = 1, Y = 1) + \Pr(X = 2, Y = 1) + \Pr(X = 3, Y = 1) \\ &= 0 + 1/8 + 2/8 + 1/8 = 4/8.\end{aligned}$$

This probability is the vertical (column) sum of the probabilities in the table at $Y = 1$. ■

Definition 2.15 (Marginal distribution)**.** The probability $\Pr(X = x)$ is called the *marginal probability* of X and is given by:

$$\Pr(X = x) = \sum_{y \in S_Y} \Pr(X = x, Y = y). \tag{2.25}$$

Similarly, the marginal probability of $Y = y$ is given by:

$$\Pr(Y = y) = \sum_{x \in S_X} \Pr(X = x, Y = y). \tag{2.26}$$

Example 2.41 (Marginal probabilities of discrete bivariate distribution)**.** The marginal probabilities of $X = x$ are given in the last column of Table Table 2.3, and the marginal probabilities of $Y = y$ are given in the last row of Table Table 2.3. Notice that these probabilities sum to 1. For future reference, note that $E[X] = 3/2, \text{var}(X) = 3/4, E[Y] = 1/2$, and $\text{var}(Y) = 1/4$. ■

2.2.1.2 Conditional Distributions

For random variables in Table Table 2.3, suppose you know that the random variable Y takes on the value $Y = 0$. How does this knowledge affect the likelihood that X takes on the values 0, 1, 2 or 3? For example, what is the probability that $X = 0$ *given that* you know $Y = 0$? To find this

probability, you can use Bayes' law and compute the *conditional probability*:

$$\Pr(X = 0|Y = 0) = \frac{\Pr(X = 0, Y = 0)}{\Pr(Y = 0)} = \frac{1/8}{4/8} = 1/4.$$

The notation $\Pr(X = 0|Y = 0)$ is read as "the probability that $X = 0$ given that $Y = 0$" . Notice that $\Pr(X = 0|Y = 0) = 1/4 > \Pr(X = 0) = 1/8$. Hence, knowledge that $Y = 0$ increases the likelihood that $X = 0$. Clearly, X *depends* on Y.

Now suppose that you know that $X = 0$. How does this knowledge affect the probability that $Y = 0$? To find out compute:

$$\Pr(Y = 0|X = 0) = \frac{\Pr(X = 0, Y = 0)}{\Pr(X = 0)} = \frac{1/8}{1/8} = 1.$$

Notice that $\Pr(Y = 0|X = 0) = 1 > \Pr(Y = 0) = 1/2$. That is, knowledge that $X = 0$ makes it certain that $Y = 0$.

The conditional probability that $X = x$ given that $Y = y$ (provided $\Pr(Y = y) \neq 0$) is,

$$p(x|y) = \Pr(X = x|Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)} = \frac{p(x, y)}{p(y)}, \tag{2.27}$$

and the conditional probability that $Y = y$ given that $X = x$ (provided $\Pr(X = x) \neq 0$) is,

$$p(y|x) = \Pr(Y = y|X = x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} = \frac{p(x, y)}{p(x)}. \tag{2.28}$$

Example 2.42 (Conditional distributions)**.** For the bivariate distribution in Table Table 2.3, the conditional probabilities along with marginal probabilities are summarized in Tables Table 2.4 and Table 2.5. Notice that the marginal distribution of X is centered at $x = 3/2$ whereas the conditional distribution of $X|Y = 0$ is centered at $x = 1$ and the conditional distribution of $X|Y = 1$ is centered at $x = 2$. ■

Table 2.4: Conditional probability distribution of X from bivariate discrete distribution.

x	$\Pr(X = x)$	$\Pr(X\|Y = 0)$	$\Pr(X\|Y = 1)$
0	1/8	2/8	0
1	3/8	4/8	2/8
2	3/8	4/8	4/8
3	1/8	0	2/8

Table 2.5: Conditional distribution of Y from bivariate discrete distribution.

y	$\Pr(Y=y)$	$\Pr(Y\|X=0)$	$\Pr(Y\|X=1)$	$\Pr(Y\|X=2)$	$\Pr(Y\|X=3)$
0	1/2	1	2/3	1/3	0
1	1/2	0	1/3	2/3	1

2.2.1.3 Conditional expectation and conditional variance

Just as I defined shape characteristics of the marginal distributions of X and Y, I can also define shape characteristics of the conditional distributions of $X|Y=y$ and $Y|X=x$. The most important shape characteristics are the *conditional expectation* (*conditional mean*) and the *conditional variance*. The conditional mean of $X|Y=y$ is denoted by $\mu_{X|Y=y} = E[X|Y=y]$, and the conditional mean of $Y|X=x$ is denoted by $\mu_{Y|X=x} = E[Y|X=x]$.

Definition 2.16 (Conditional expectation)**.** For discrete random variables X and Y, the conditional expectations are defined as:

$$\mu_{X|Y=y} = E[X|Y=y] = \sum_{x \in S_X} x \cdot \Pr(X=x|Y=y), \tag{2.29}$$

$$\mu_{Y|X=x} = E[Y|X=x] = \sum_{y \in S_Y} y \cdot \Pr(Y=y|X=x). \tag{2.30}$$

Similarly, the conditional variance of $X|Y=y$ is denoted by $\sigma^2_{X|Y=y} = \text{var}(X|Y=y)$ and the conditional variance of $Y|X=x$ is denoted by $\sigma^2_{Y|X=x} = \text{var}(Y|X=x)$.

Definition 2.17 (Conditional variance)**.** For discrete random variables X and Y, the conditional variances are defined as:

$$\sigma^2_{X|Y=y} = \text{var}(X|Y=y) = \sum_{x \in S_X} (x - \mu_{X|Y=y})^2 \cdot \Pr(X=x|Y=y), \tag{2.31}$$

$$\sigma^2_{Y|X=x} = \text{var}(Y|X=x) = \sum_{y \in S_Y} (y - \mu_{Y|X=x})^2 \cdot \Pr(Y=y|X=x). \tag{2.32}$$

The conditional volatilities are defined as:

$$\sigma_{X|Y=y} = \sqrt{\sigma^2_{X|Y=y}}, \tag{2.33}$$

$$\sigma_{Y|X=x} = \sqrt{\sigma^2_{Y|X=x}}. \tag{2.34}$$

Then conditional mean is the center of mass of the conditional distribution, and the conditional variance and volatility measure the spread of the conditional distribution about the conditional mean.

Example 2.43 (Compute conditional expectation and conditional variance)**.** For the random variables in Table Table 2.3, you can derive the following conditional moments for X:

$$\begin{aligned}
E[X|Y=0] &= 0 \cdot 1/4 + 1 \cdot 1/2 + 2 \cdot 1/4 + 3 \cdot 0 = 1, \\
E[X|Y=1] &= 0 \cdot 0 + 1 \cdot 1/4 + 2 \cdot 1/2 + 3 \cdot 1/4 = 2, \\
\text{var}(X|Y=0) &= (0-1)^2 \cdot 1/4 + (1-1)^2 \cdot 1/2 + (2-1)^2 \cdot 1/2 + (3-1)^2 \cdot 0 = 1/2, \\
\text{var}(X|Y=1) &= (0-2)^2 \cdot 0 + (1-2)^2 \cdot 1/4 + (2-2)^2 \cdot 1/2 + (3-2)^2 \cdot 1/4 = 1/2.
\end{aligned}$$

Compare these values to $E[X] = 3/2$ and $\text{var}(X) = 3/4$. Notice that $\text{var}(X|Y) = 1/2 < \text{var}(X) = 3/4$ so that conditioning on information reduces variability. Also, notice that as y increases, $E[X|Y=y]$ increases.

For Y, similar calculations gives:

$$\begin{aligned}
&E[Y|X=0] = 0, E[Y|X=1] = 1/3, E[Y|X=2] = 2/3, E[Y|X=3] = 1, \\
&\text{var}(Y|X=0) = 0, \text{var}(Y|X=1) = 0.2222, \text{var}(Y|X=2) = 0.2222, \text{var}(Y|X=3) = 0.
\end{aligned}$$

Compare these values to $E[Y] = 1/2$ and $\text{var}(Y) = 1/4$. Notice that as x increases $E[Y|X=x]$ increases. ■

2.2.1.4 Conditional expectation and the regression function

Consider the problem of predicting the value Y given that you know $X = x$. A natural predictor to use is the conditional expectation $E[Y|X=x]$. In this prediction context, the conditional expectation $E[Y|X=x]$ is called the *regression function*. The graph with $E[Y|X=x]$ on the vertical axis and x on the horizontal axis gives the *regression line*. The relationship between Y and the regression function may be expressed using the trivial identity:

$$\begin{aligned}
Y &= E[Y|X=x] + Y - E[Y|X=x] \\
&= E[Y|X=x] + \varepsilon,
\end{aligned}$$

where $\varepsilon = Y - E[Y|X]$ is called the *regression error*.

Example 2.44 (Regression line for bivariate discrete distribution)**.** For the random variables in Table Table 2.3, the regression line is plotted in Figure 2.11. Notice that there is a linear relationship between $E[Y|X=x]$ and x. When such a linear relationship exists you call the regression function a *linear regression*. Linearity of the regression function, however, is not guaranteed. It may be the case that there is a non-linear (e.g., quadratic) relationship between $E[Y|X=x]$ and x. In this case, you call the regression function a *non-linear regression*. ■

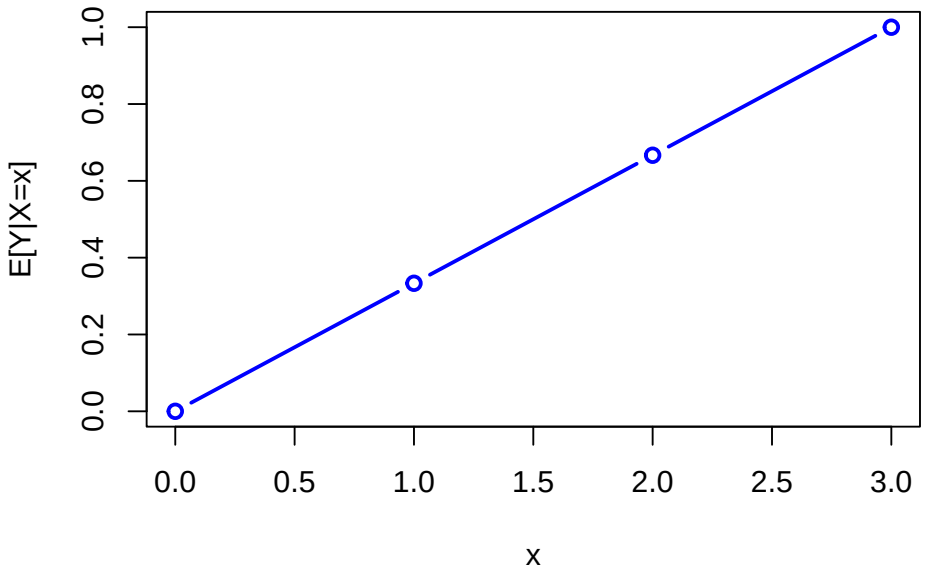


Figure 2.11: Regression function $E[Y|X = x]$ from discrete bivariate distribution.

2.2.1.5 Law of Total Expectations

Before Y is observed, $E[X|Y]$ is a random variable because its value depends on the value of Y which is a random variable. Similarly, $E[Y|X]$ is also a random variable. Since $E[X|Y]$ and $E[Y|X]$ are random variables you can consider their mean values. To illustrate, for the random variables in Table Table 2.3 notice that:

$$\begin{aligned} E[X] &= \sum_{y \in S_Y} E[X|Y = y] \cdot \Pr(Y = y) \\ &= E[X|Y = 0] \cdot \Pr(Y = 0) + E[X|Y = 1] \cdot \Pr(Y = 1) \\ &= 1 \cdot 1/2 + 2 \cdot 1/2 = 3/2, \end{aligned}$$

and,

$$\begin{aligned} E[Y] &= \sum_{x \in S_X} E[Y|X = x] \cdot \Pr(X = x) \\ &= E[Y|X = 0] \cdot \Pr(X = 0) + E[Y|X = 1] \cdot \Pr(X = 1) \\ &+ E[Y|X = 2] \cdot \Pr(X = 2) + E[Y|X = 3] \cdot \Pr(X = 3) = 1/2. \end{aligned}$$

This result is known as the *law of iterated expectations* or the *law of total expectations*.

Proposition 2.2. *For two random variables X and Y (discrete or continuous) it can be shown that:*

$$E[X] = E[E[X|Y]],$$
$$E[Y] = E[E[Y|X]],$$

where the expectation on the first line is taken with respect to the random variable Y, and the the expectation on the second line is taken with respect to the random variable X.

In words, the law of iterated expectations says that the mean of the conditional mean is the unconditional mean.

2.2.2 Bivariate distributions for continuous random variables

Let X and Y be continuous random variables defined over the real line. You can characterize the joint probability distribution of X and Y using the joint probability function (pdf) $f(x, y)$ such that $f(x, y) \geq 0$ and,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\ dx\ dy = 1.$$

The three-dimensional plot of the joint probability distribution gives a probability surface whose total volume is unity. To compute joint probabilities of $x_1 \leq X \leq x_2$ and $y_1 \leq Y \leq y_2$, you need to find the volume under the probability surface over the grid where the intervals $[x_1, x_2]$ and $[y_1, y_2]$ overlap. Finding this volume requries solving the double integral:

$$\Pr(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2) = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f(x, y)\ dx\ dy.$$

Example 2.45 (Standard bivariate normal distribution)**.** A *standard bivariate normal* pdf for X and Y has the form:

$$f(x, y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)},\ -\infty \leq x,\ y \leq \infty \tag{2.35}$$

and has the shape of a symmetric bell (think Liberty Bell) centered at $x = 0$ and $y = 0$. To find $\Pr(-1 < X < 1, -1 < Y < 1)$ you must solve:

$$\int_{-1}^{1} \int_{-1}^{1} \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}\ dx\ dy,$$

which, unfortunately, does not have an analytical solution. Numerical approximation methods are required to evaluate the above integral. The function `pmvnorm()` in the R package **mvtnorm** can be used to evaluate areas under the bivariate standard normal surface. To compute $\Pr(-1 < X < 1, -1 < Y < 1)$ use:

```
pmvnorm(lower=c(-1, -1), upper=c(1, 1))
#> [1] 0.466
#> attr(,"error")
#> [1] 1e-15
#> attr(,"msg")
#> [1] "Normal Completion"
```

Here, $\Pr(-1 < X < 1, -1 < Y < 1) = 0.4661$. The attribute `error` gives the estimated absolute error of the approximation, and the attribute `message` tells the status of the algorithm used for the approximation. See the online help for `pmvnorm` for more details. ■

2.2.2.1 Marginal and conditional distributions

The marginal pdf of X is found by integrating y out of the joint pdf $f(x, y)$ and the marginal pdf of Y is found by integrating x out of the joint pdf $f(x, y)$:

$$f(x) = \int_{-\infty}^{\infty} f(x, y)\ dy, \tag{2.36}$$

$$f(y) = \int_{-\infty}^{\infty} f(x, y)\ dx. \tag{2.37}$$

The conditional pdf of X given that $Y = y$, denoted $f(x|y)$, is computed as

$$f(x|y) = \frac{f(x, y)}{f(y)}, \tag{2.38}$$

and the conditional pdf of Y given that $X = x$ is computed as,

$$f(y|x) = \frac{f(x, y)}{f(x)}. \tag{2.39}$$

The conditional means are computed as

$$\mu_{X|Y=y} = E[X|Y = y] = \int x \cdot p(x|y)\ dx, \tag{2.40}$$

$$\mu_{Y|X=x} = E[Y|X = x] = \int y \cdot p(y|x)\ dy, \tag{2.41}$$

and the conditional variances are computed as,

$$\sigma^2_{X|Y=y} = \text{var}(X|Y=y) = \int (x - \mu_{X|Y=y})^2 p(x|y)\ dx, \tag{2.42}$$

$$\sigma^2_{Y|X=x} = \text{var}(Y|X=x) = \int (y - \mu_{Y|X=x})^2 p(y|x)\ dy. \tag{2.43}$$

Example 2.46 (Conditional and marginal distributions from bivariate standard normal)**.** Suppose X and Y are distributed bivariate standard normal. To find the marginal distribution of X you use Equation 2.36 and solve:

$$f(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}\ dy = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2}\ dy = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

Hence, the marginal distribution of X is standard normal. Similiar calculations show that the marginal distribution of Y is also standard normal. To find the conditional distribution of $X|Y=y$ use Equation 2.38 and solve:

$$\begin{aligned} f(x|y) &= \frac{f(x,y)}{f(x)} = \frac{\frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x^2+y^2)+\frac{1}{2}y^2} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \\ &= f(x) \end{aligned}$$

So, for the standard bivariate normal distribution $f(x|y) = f(x)$ which does not depend on y. Similar calculations show that $f(y|x) = f(y)$. ■

2.2.3 Independence

Let X and Y be two discrete random variables. Intuitively, X is independent of Y if knowledge about Y does not influence the likelihood that $X = x$ for all possible values of $x \in S_X$ and $y \in S_Y$. Similarly, Y is independent of X if knowledge about X does not influence the likelihood that $Y = y$ for all values of $y \in S_Y$. I represent this intuition formally for discrete random variables as follows.

Definition 2.18 (Independence of discrete random variables)**.** Let X and Y be discrete random variables with sample spaces S_X and S_Y, respectively. X and Y are independent random variables if and only if (iff):

$$\begin{aligned} \Pr(X = x|Y = y) &= \Pr(X = x),\ \text{for all } x \in S_X, y \in S_Y \\ \Pr(Y = y|X = x) &= \Pr(Y = y),\ \text{for all } x \in S_X, y \in S_Y \end{aligned}$$

Example 2.47 (Check independence of bivariate discrete random variables)**.** For the data in Table Table 2.3, you know that $\Pr(X = 0|Y = 0) = 1/4 \neq \Pr(X = 0) = 1/8$ so X and Y are not independent. ■

Proposition 2.3 (Independence of discrete random variables)**.** *Let X and Y be discrete random variables with sample spaces S_X and S_Y, respectively. X and Y are independent iff:*

$$\Pr(X = x, Y = y) = \Pr(X = x) \cdot \Pr(Y = y), \text{ for all } x \in S_X, y \in S_Y.$$

Intuition for the above result follows from:

$$\begin{aligned}
\Pr(X = x|Y = y) &= \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)} = \frac{\Pr(X = x) \cdot \Pr(Y = y)}{\Pr(Y = y)} \\
&= \Pr(X = x), \\
\Pr(Y = y|X = x) &= \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} = \frac{\Pr(X = x) \cdot \Pr(Y = y)}{\Pr(X = x)} \\
&= \Pr(Y = y),
\end{aligned}$$

which shows that X and Y are independent.

For continuous random variables, you have the following definition of independence.

Definition 2.19 (Independence of continuous random variables)**.** Let X and Y be continuous random variables with marginal pdfs $f(x)$ and $f(y)$, and conditional pdfs $f(x|y)$ and $f(y|x)$. Then X and Y are independent iff:

$$\begin{aligned}
f(x|y) &= f(x), \text{ for } -\infty < x, y < \infty, \\
f(y|x) &= f(y), \text{ for } -\infty < x, y < \infty.
\end{aligned}$$

As with discrete random variables, you have the following result for continuous random variables.

Proposition 2.4 (Independence of continuous random variables)**.** *Let X and Y be continuous random variables with marginal pdfs $f(x)$ and $f(y)$, and joint pdf $f(x, y)$. Then X and Y are independent iff:*

$$f(x, y) = f(x)f(y)$$

This result is extremely useful in practice because it gives us an easy way to compute the joint pdf for two independent random variables: you simply compute the product of the marginal distributions.

Example 2.48 (Constructing the bivariate standard normal distribution)**.** Let $X \sim N(0,1)$, $Y \sim N(0,1)$ and let X and Y be independent. Then,

$$f(x,y) = f(x)f(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}.$$

This result is a special case of the general bivariate normal distribution to be discussed in the next sub-section. ■

A useful property of the independence between two random variables is the following Proposition.

Proposition 2.5 (Property of independent random variables)**.** *If X and Y are independent then $g(X)$ and $h(Y)$ are independent for any functions $g(\cdot)$ and $h(\cdot)$.*

For example, if X and Y are independent then X^2 and Y^2 are also independent.

2.2.4 Covariance and correlation

Let X and Y be two discrete random variables. Figure 2.12 displays several bivariate probability scatterplots (where equal probabilities are given on the dots). In panel (a) you see no linear relationship between X and Y. In panel (b) you see a perfect positive linear relationship between X and Y and in panel (c) you see a perfect negative linear relationship. In panel (d) you see a positive, but not perfect, linear relationship; in panel (e) you see a negative, but not perfect, linear relationship. Finally, in panel (f) you see no systematic linear relationship but you see a strong nonlinear (parabolic) relationship. The *covariance* between X and Y measures the *direction* of the linear relationship between the two random variables. The *correlation* between X and Y measures the *direction* and the *strength* of the linear relationship between the two random variables.

Let X and Y be two random variables with $E[X] = \mu_X$, $\text{var}(X) = \sigma_X^2$, $E[Y] = \mu_Y$ and $\text{var}(Y) = \sigma_Y^2$.

Definition 2.20 (Covariance)**.** The covariance between two random variables X and Y is given by:

$$\begin{aligned}
\sigma_{XY} &= \text{cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] \\
&= \sum_{x \in S_X} \sum_{y \in S_Y} (x-\mu_X)(y-\mu_Y) \Pr(X=x, Y=y) \text{ for discrete } X, Y, \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y) p(x,y)\, dx\, dy \text{ for continuous } X, Y.
\end{aligned}$$

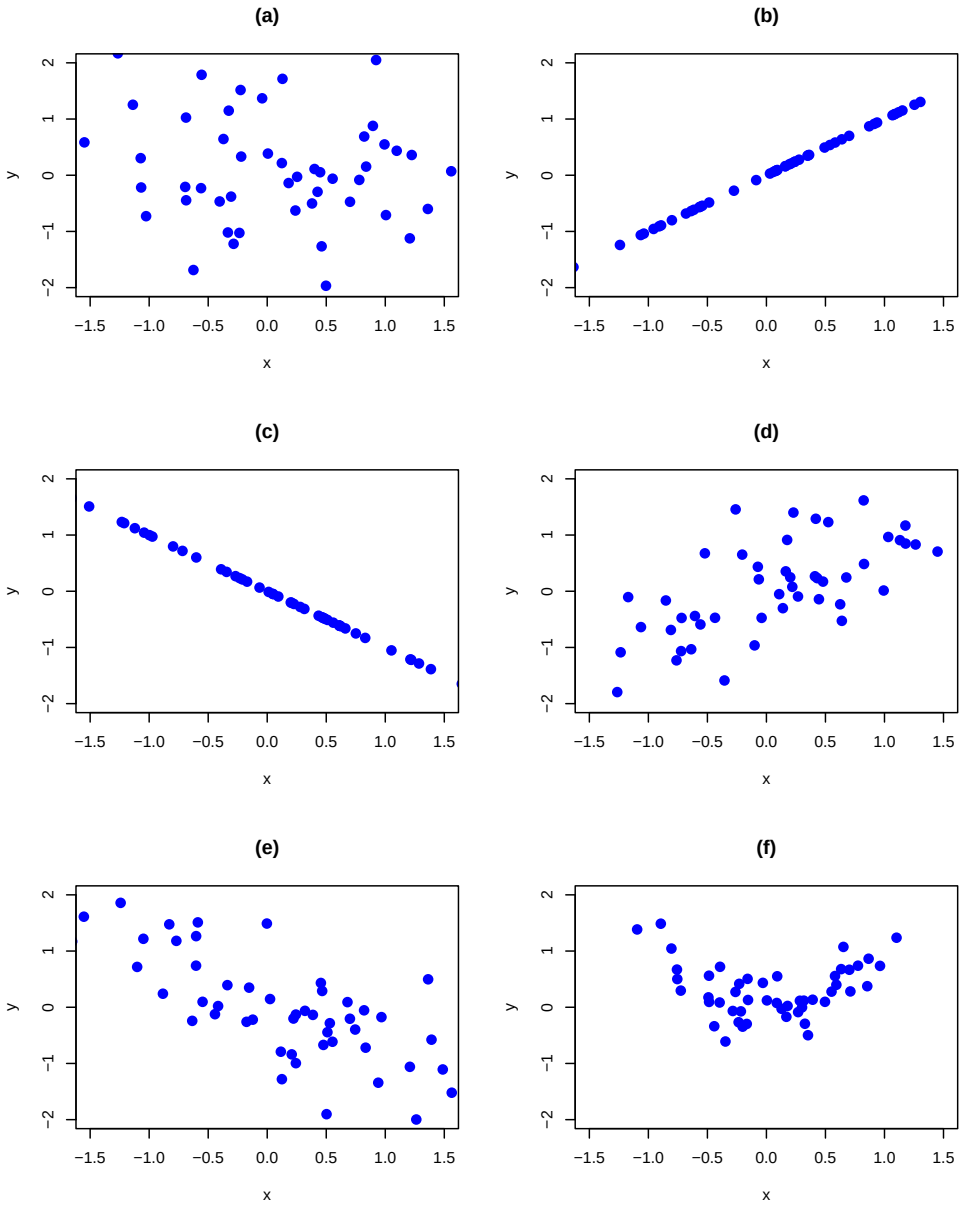


Figure 2.12: Probability scatterplots illustrating dependence between X and Y.

Definition 2.21 (Correlation)**.** The correlation between two random variables X and Y is given by:

$$\rho_{XY} = \text{cor}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

The correlation coefficient, ρ_{XY}, is a scaled version of the covariance.

To see how covariance measures the direction of linear association, consider the probability scatterplot in Figure 2.13. In the figure, each pair of points occurs with equal probability. The plot is separated into quadrants (right to left, top to bottom). In the first quadrant (black circles), the realized values satisfy $x < \mu_X, y > \mu_Y$ so that the product $(x-\mu_X)(y-\mu_Y) < 0$. In the second quadrant (blue squares), the values satisfy $x > \mu_X$ and $y > \mu_Y$ so that the product $(x-\mu_X)(y-\mu_Y) > 0$. In the third quadrant (red triangles), the values satisfy $x < \mu_X$ and $y < \mu_Y$ so that the product $(x - \mu_X)(y - \mu_Y) > 0$. Finally, in the fourth quadrant (green diamonds), $x > \mu_X$ but $y < \mu_Y$ so that the product $(x - \mu_X)(y - \mu_Y) < 0$. Covariance is then a probability weighted average all of the product terms in the four quadrants. For the values in Figure 2.13, this weighted average is positive because most of the values are in the second and third quadrants.

Example 2.49 (Calculate covariance and correlation for discrete random variables)**.** For the data in Table Table 2.3, you get:

$$\begin{aligned}\sigma_{XY} = \text{cov}(X, Y) &= (0 - 3/2)(0 - 1/2) \cdot 1/8 + (0 - 3/2)(1 - 1/2) \cdot 0 \\ &\quad + \cdots + (3 - 3/2)(1 - 1/2) \cdot 1/8 = 1/4 \\ \rho_{XY} = \text{cor}(X, Y) &= \frac{1/4}{\sqrt{(3/4) \cdot (1/2)}} = 0.577\end{aligned}$$

■

2.2.4.1 Properties of covariance and correlation

Let X and Y be random variables and let a and b be constants. Some important properties of $\text{cov}(X, Y)$ are summarized in the following Proposition:

Proposition 2.6 (Properties of covariance)**.**

1. $\text{cov}(X, X) = \text{var}(X)$
2. $\text{cov}(X, Y) = \text{cov}(Y, X)$
3. $\text{cov}(X, Y) = E[XY] - E[X]E[Y] = E[XY] - \mu_X \mu_Y$
4. $\text{cov}(aX, bY) = a \cdot b \cdot \text{cov}(X, Y)$
5. *If X and Y are independent then* $\text{cov}(X, Y) = 0$ *(no association $\implies$ no linear association). However, if* $\text{cov}(X, Y) = 0$ *then X and Y are not necessarily independent (no linear association $\not\Rightarrow$ no association).*

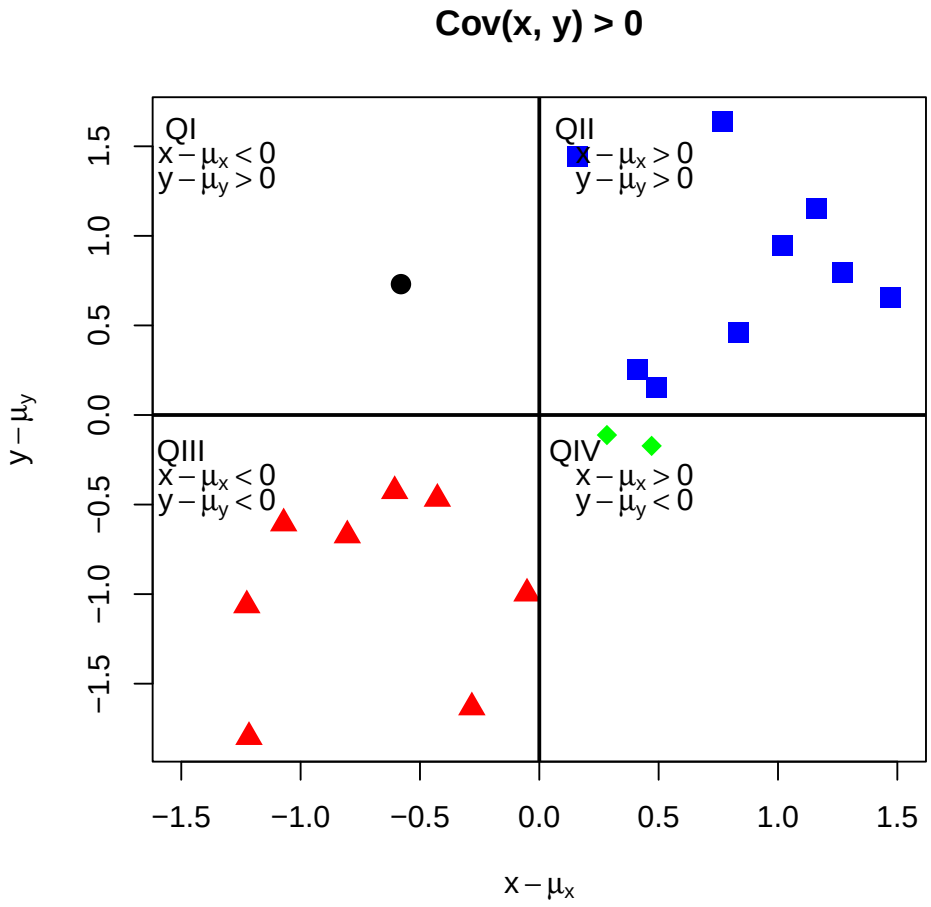


Figure 2.13: Probability scatterplot of discrete distribution with positive covariance. Each pair (X, Y) occurs with equal probability.

6. *If X and Y are jointly normally distributed and $\text{cov}(X,Y) = 0$, then X and Y are independent.*

The first two properties are intuitive. The third property results from expanding the definition of covariance:

$$\begin{aligned}
\text{cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\
&= E\left[XY - X\mu_Y - \mu_X Y + \mu_X\mu_Y\right] \\
&= E[XY] - E[X]\mu_Y - \mu_X E[Y] + \mu_X\mu_Y \\
&= E[XY] - 2\mu_X\mu_Y + \mu_X\mu_Y \\
&= E[XY] - \mu_X\mu_Y
\end{aligned}$$

The fourth property follows from the linearity of expectations:

$$\begin{aligned}
\text{cov}(aX,bY) &= E[(aX-a\mu_X)(bY-b\mu_Y)] \\
&= a\cdot b\cdot E[(X-\mu_X)(Y-\mu_Y)] \\
&= a\cdot b\cdot \text{cov}(X,Y)
\end{aligned}$$

The fourth property shows that the value of $\text{cov}(X,Y)$ depends on the scaling of the random variables X and Y. By simply changing the scale of X or Y you can make $\text{cov}(X,Y)$ equal to any value that you want. Consequently, the numerical value of $\text{cov}(X,Y)$ is not informative about the strength of the linear association between X and Y. However, the sign of $\text{cov}(X,Y)$ is informative about the direction of linear association between X and Y.

The fifth property should be intuitive. Independence between the random variables X and Y means that there is no relationship, linear or nonlinear, between X and Y. However, the lack of a linear relationship between X and Y does not preclude a nonlinear relationship.

The last result illustrates an important property of the normal distribution: lack of covariance implies independence.

Some important properties of $\text{cor}(X,Y)$ are:

Proposition 2.7.

1. $-1 \le \rho_{XY} \le 1$
2. *If $\rho_{XY} = 1$ then X and Y are perfectly positively linearly related. That is, $Y = aX + b$ where $a > 0$.*
3. *If $\rho_{XY} = -1$ then X and Y are perfectly negatively linearly related. That is, $Y = aX + b$ where $a < 0$.*
4. *If $\rho_{XY} = 0$ then X and Y are not linearly related but may be nonlinearly related.*
5. $\text{cor}(aX,bY) = \text{cor}(X,Y)$ *if* $a > 0$ *and* $b > 0$; $\text{cor}(X,Y) = -\text{cor}(X,Y)$ *if* $a > 0, b < 0$ *or* $a < 0, b > 0$.

TBD: discuss the results briefly

2.2.5 Bivariate normal distributions

Let X and Y be distributed bivariate normal. The joint pdf is given by:

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} \times$$
$$\exp\left\{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - \frac{2\rho_{XY}(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right]\right\}$$

where $E[X] = \mu_X$, $E[Y] = \mu_Y$, $\text{sd}(X) = \sigma_X$, $\text{sd}(Y) = \sigma_Y$, and $\rho_{XY} = \text{cor}(X,Y)$. The correlation coefficient ρ_{XY} describes the dependence between X and Y. If $\rho_{XY} = 0$ then the pdf collapses to the pdf of the standard bivariate normal distribution. In this case, X and Y are independent random variables since $f(x,y) = f(x)f(y)$.

It can be shown that the marginal distributions of X and Y are normal: $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$. In addition, it can be shown that the conditional distributions $f(x|y)$ and $f(y|x)$ are also normal with means given by:

$$\mu_{X|Y=y} = \alpha_X + \beta_X \cdot y,$$
$$\mu_{Y|X=x} = \alpha_Y + \beta_Y \cdot x,$$

where,

$$\alpha_X = \mu_X - \beta_X\mu_X,\ \beta_X = \sigma_{XY}/\sigma_Y^2,$$
$$\alpha_Y = \mu_Y - \beta_Y\mu_X,\ \beta_Y = \sigma_{XY}/\sigma_X^2,$$

and variances given by,

$$\sigma_{X|Y=y}^2 = \sigma_X^2 - \sigma_{XY}^2/\sigma_Y^2,$$
$$\sigma_{X|Y=y}^2 = \sigma_Y^2 - \sigma_{XY}^2/\sigma_X^2.$$

Notice that the conditional means (regression functions) are linear functions of x and y, respectively.

Example 2.50 (Expressing the bivariate normal distribution using matrix algebra)**.** The formula for the bivariate normal distribution Equation 2.44 is a bit messy. you can greatly simplify the formula by using matrix algebra. Define the 2×1 vectors $\mathbf{x} = (x,y)'$ and $\mu = (\mu_X, \mu_y)'$, and the 2×2 matrix:

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix}$$

Then the bivariate normal distribution Equation 2.44 may be compactly expressed as:

$$f(\mathbf{x}) = \frac{1}{2\pi \det(\Sigma)^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)} \tag{2.44}$$

where,

$$\det(\Sigma) = \sigma_X^2\sigma_Y^2 - \sigma_{XY}^2 = \sigma_X^2\sigma_Y^2 - \sigma_X^2\sigma_Y^2\rho_{XY}^2 = \sigma_X^2\sigma_Y^2(1-\rho_{XY}^2)$$
$$\Sigma^{-1} = \frac{1}{\det(\Sigma)} \begin{pmatrix} \sigma_Y^2 & -\sigma_{XY} \\ -\sigma_{XY} & \sigma_X^2 \end{pmatrix}$$

■

Example 2.51 (Plotting the bivariate normal distribution)**.** The R package **mvtnorm** contains the functions `dmvnorm()`, `pmvnorm()`, and `qmvnorm()` which can be used to compute the bivariate normal pdf, cdf and quantiles, respectively. Plotting the bivariate normal distribution over a specified grid of x and y values in R can be done with the `persp()` function. First, I specify the parameter values for the joint distribution. Here, I choose $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$ and $\rho = 0.5$. I will use the `dmvnorm()` function to evaluate the joint pdf at these values. To do so, I must specify a covariance matrix of the form:

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}.$$

In R this matrix can be created using:

```
sigma = matrix(c(1, 0.5, 0.5, 1), 2, 2)
```

Next I specify a grid of x and y values between -3 and 3:

```
x = seq(-3, 3, length=25)
y = seq(-3, 3, length=25)
```

To evaluate the joint pdf over the two-dimensional grid I can use the `outer()` function:

```
# function to evaluate bivariate normal pdf on grid
bv.norm <- function(x, y, sigma) {
  z = cbind(x,y)
  return(dmvnorm(z, sigma=sigma))
}
# use outer function to evaluate pdf on 2D grid of x-y values
fxy = outer(x, y, bv.norm, sigma)
```

To create the 3D plot of the joint pdf, I use the `persp()` function:

```
persp(x, y, fxy, theta=60, phi=30, expand=0.5, ticktype="detailed",
      zlab="f(x,y)",col="grey")
```

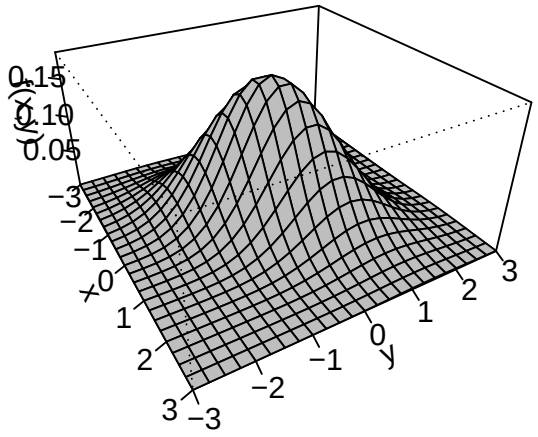


Figure 2.14: Bivariate normal pdf with $\mu_X = \mu_Y = 0$, $\sigma_X\sigma_Y = 1$ and $\rho = 0.5$.

The resulting plot is given in Figure 2.14. ■

2.2.6 Expectation and variance of the sum of two random variables

Proposition 2.8 (Expectation and variance of the sum of two random variables)**.** *Let X and Y be two random variables with well defined means, variances and covariance and let a and b be constants. Then the following results hold:*

$$\begin{aligned} E[aX + bY] &= aE[X] + bE[Y] = a\mu_X + b\mu_Y \\ \text{var}(aX + bY) &= a^2\text{var}(X) + b^2\text{var}(Y) + 2 \cdot a \cdot b \cdot \text{cov}(X, Y) \\ &= a^2\sigma_X^2 + b^2\sigma_Y^2 + 2 \cdot a \cdot b \cdot \sigma_{XY} \end{aligned}$$

The first result states that the expected value of a linear combination of two random variables is equal to a linear combination of the expected values of the random variables. This result indicates

that the expectation operator is a linear operator. In other words, expectation is additive. The second result states that variance of a linear combination of random variables is not a linear combination of the variances of the random variables. In particular, notice that covariance comes up as a term when computing the variance of the sum of two (not independent) random variables. Hence, the variance operator is not, in general, a linear operator. That is, variance, in general, is not additive.

It is instructive to go through the derivation of these results. Let X and Y be discrete random variables. Then,

$$\begin{aligned}
E[aX + bY] &= \sum_{x \in S_X} \sum_{y \in S_y} (ax + by) \Pr(X = x, Y = y) \\
&= \sum_{x \in S_X} \sum_{y \in S_y} ax \Pr(X = x, Y = y) + \sum_{x \in S_X} \sum_{y \in S_y} bx \Pr(X = x, Y = y) \\
&= a \sum_{x \in S_X} x \sum_{y \in S_y} \Pr(X = x, Y = y) + b \sum_{y \in S_y} y \sum_{x \in S_X} \Pr(X = x, Y = y) \\
&= a \sum_{x \in S_X} x \Pr(X = x) + b \sum_{y \in S_y} y \Pr(Y = y) \\
&= aE[X] + bE[Y] = a\mu_X + b\mu_Y.
\end{aligned}$$

The result for continuous random variables is similar. Effectively, the summations are replaced by integrals and the joint probabilities are replaced by the joint pdf.

Next, let X and Y be discrete or continuous random variables. Then, using the definition of variance you get:

$$\begin{aligned}
\text{var}(aX + bY) &= E[(aX + bY - E[aX + bY])^2] \\
&= E[(aX + bY - a\mu_X - b\mu_Y)^2] \\
&= E[(a(X - \mu_X) + b(Y - \mu_Y))^2] \\
&= a^2 E[(X - \mu_X)^2] + b^2 E[(Y - \mu_Y)^2] + 2 \cdot a \cdot b \cdot E[(X - \mu_X)(Y - \mu_Y)] \\
&= a^2 \text{var}(X) + b^2 \text{var}(Y) + 2 \cdot a \cdot b \cdot \text{cov}(X, Y).
\end{aligned}$$

2.2.6.1 Linear combination of two normal random variables

The following proposition gives an important result concerning a linear combination of normal random variables.

Proposition 2.9 (Linear combination of normal random variables)**.** *Let $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$, $\sigma_{XY} = \text{cov}(X, Y)$ and a and b be constants. Define the new random variable Z as:*

$$Z = aX + bY.$$

Then,

$$Z \sim N(\mu_Z, \sigma_Z^2),$$

where $\mu_Z = a\mu_X + b\mu_Y$ *and* $\sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}$.

This important result states that a linear combination of two normally distributed random variables is itself a normally distributed random variable. The proof of the result relies on the change of variables theorem from calculus and is omitted. Not all random variables have the nice property that their distributions are closed under addition.

Example 2.52 (Portfolio of two assets)**.** Consider a portfolio of two stocks A (Amazon) and B (Boeing) with investment shares x_A and x_B with $x_A + x_B = 1$. Let R_A and R_B denote the simple monthly returns on these assets, and assume that $R_A \sim N(\mu_A, \sigma_A^2)$ and $R_B \sim N(\mu_B, \sigma_B^2)$. Furthermore, let $\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B = \text{cov}(R_A, R_B)$. The portfolio return is $R_p = x_A R_A + x_B R_B$, which is a linear function of two random variables. Using the properties of linear functions of random variables, you get:

$$\begin{aligned}
\mu_p &= E[R_p] = x_A E[R_A] + x_B E[R_B] = x_A\mu_A + x_B\mu_B \\
\sigma_p^2 &= \text{var}(R_p) = x_A^2\text{var}(R_A) + x_B^2\text{var}(R_B) + 2x_A x_B\text{cov}(R_A, R_B) \\
&= x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B\sigma_{AB}.
\end{aligned}$$

■

2.3 Multivariate Distributions

Multivariate distributions are used to characterize the joint distribution of a collection of N random variables $X_1, X_2, \ldots, X_N$ for $N > 1$. The mathematical formulation of this joint distribution can be quite complex and typically makes use of matrix algebra. Here, I summarize some basic properties of multivariate distributions without the use of matrix algebra. In chapter Chapter 3, I will show how matrix algebra can greatly simplify the description of multivariate distributions.

2.3.1 Discrete random variables

Let $X_1, X_2, \ldots, X_N$ be N discrete random variables with sample spaces $S_{X_1}, S_{X_2}, \ldots, S_{X_N}$. The likelihood that these random variables take values in the joint sample space $S_{X_1} \times S_{X_2} \times \cdots \times S_{X_N}$ is given by the joint probability function:

$$p(x_1, x_2, \ldots, x_N) = \Pr(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N).$$

For $N > 2$ it is not easy to represent the joint probabilities in a table like Table Table 2.3 or to visualize the distribution.

Marginal distributions for each variable X_i can be derived from the joint distribution as in Equation 2.25 by summing the joint probabilities over the other variables $j \neq i$. For example,

$$p(x_1) = \sum_{x_2 \in S_{X_2}, \ldots, x_N \in S_{X_N}} p(x_1, x_2, \ldots, x_N).$$

With N random variables, there are numerous conditional distributions that can be formed. For example, the distribution of X_1 given $X_2 = x_2, \ldots, X_N = x_N$ is determined using:

$$\Pr(X_1 = x_1 | X_2 = x_2, \ldots, X_N = x_N) = \frac{\Pr(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)}{\Pr(X_2 = x_2, \ldots, X_N = x_N)}.$$

Similarly, the joint distribution of X_1 and X_2 given $X_3 = x_3, \ldots, X_N = x_N$ is given by:

$$\begin{aligned} &\Pr(X_1 = x_1, X_2 = x_2 | X_3 = x_3, \ldots, X_N = x_N) \\ &= \frac{\Pr(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)}{\Pr(X_3 = x_3, \ldots, X_N = x_N)}. \end{aligned}$$

2.3.2 Continuous random variables

Let $X_1, X_2, \ldots, X_N$ be N continuous random variables each taking values on the real line. The joint pdf is a function $f(x_1, x_2, \ldots, x_N) \geq 0$ such that:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \ldots, x_N)\, dx_1\, dx_2 \cdots\, dx_N = 1.$$

Joint probabilities of $x_{11} \leq X_1 \leq x_{12}, x_{21} \leq X_2 \leq x_{22}, \ldots, x_{N1} \leq X_N \leq x_{N2}$ are computed by solving the integral equation:

$$\int_{x_{11}}^{x_{12}} \int_{x_{21}}^{x_{22}} \cdots \int_{x_{N1}}^{x_{N2}} f(x_1, x_2, \ldots, x_N)\, dx_1\, dx_2 \cdots\, dx_N. \tag{2.45}$$

For most multivariate distributions, the integral in Equation 2.45 cannot be solved analytically and must be approximated numerically.

The marginal pdf for x_i is found by integrating the joint pdf with respect to the other variables. For example, the marginal pdf for x_1 is found by solving:

$$f(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \ldots, x_N)\, dx_2 \cdots\, dx_N.$$

Conditional pdf for a single random variable or a collection of random variables are defined in the obvious way.

2.3.3 Independence

A collection of N random variables are independent if their joint distribution factors into the product of all of the marginal distributions:

$$p(x_1, x_2, \ldots, x_N) = p(x_1)p(x_2)\cdots p(x_N) \text{ for } X_i \text{ discrete,}$$
$$f(x_1, x_2, \ldots, x_N) = f(x_1)f(x_2)\cdots f(x_N) \text{ for } X_i \text{ continuous.}$$

In addition, if N random variables are independent then any functions of these random variables are also independent.

2.3.4 Dependence concepts

In general, it is difficult to define dependence concepts for collections of more than two random variables. Dependence is typically only defined between pairwise random variables. Hence, covariance and correlation are also useful concepts when dealing with more than two random variables.

For N random variables $X_1, X_2, \ldots, X_N$, with mean values $\mu_i = E[X_i]$ and variances $\sigma_i^2 = \text{var}(X_i)$, the pairwise covariances and correlations are defined as:

$$\text{cov}(X_i, X_j) = \sigma_{ij} = E[(X_i - \mu_i)(X_j - \mu_i)],$$
$$\text{cov}(X_i, X_j) = \rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j},$$

for $i \neq j$. There are $N(N-1)/2$ pairwise covariances and correlations. Often, these values are summarized using matrix algebra in an $N \times N$ covariance matrix Σ and an $N \times N$ correlation matrix $\mathbf{C}$:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_N^2 \end{pmatrix}, \tag{2.46}$$

$$\mathbf{C} = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1N} \\ \rho_{12} & 1 & \cdots & \rho_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1N} & \rho_{2N} & \cdots & 1 \end{pmatrix}. \tag{2.47}$$

These matrices are formally defined in chapter Chapter 3.

2.3.5 Linear combinations of N random variables

Many of the results for manipulating a collection of random variables generalize in a straightforward way to the case of more than two random variables. The details of the generalizations are not important for our purposes. However, the following results will be used repeatedly throughout the book.

Proposition 2.10 (Linear combination of random variables)**.** *Let $X_1, X_2, \ldots, X_N$ denote a collection of N random variables (discrete or continuous) with means μ_i, variances σ_i^2 and covariances σ_{ij}. Define the new random variable Z as a linear combination:*

$$Z = a_1 X_1 + a_2 X_2 + \cdots + a_N X_N$$

where $a_1, a_2, \ldots, a_N$ are constants. Then the following results hold:

$$\begin{aligned}
\mu_Z = E[Z] &= a_1 E[X_1] + a_2 E[X_2] + \cdots + a_N E[X_N] \\
&= \sum_{i=1}^{N} a_i E[X_i] = \sum_{i=1}^{N} a_i \mu_i.
\end{aligned}$$

$$\begin{aligned}
\sigma_Z^2 = \text{var}(Z) &= a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \cdots + a_N^2\sigma_N^2 \\
&+ 2a_1a_2\sigma_{12} + 2a_1a_3\sigma_{13} + \cdots + a_1a_N\sigma_{1N} \\
&+ 2a_2a_3\sigma_{23} + 2a_2a_4\sigma_{24} + \cdots + a_2a_N\sigma_{2N} \\
&+ \cdots + \\
&+ 2a_{N-1}a_N\sigma_{(N-1)N} \\
&= \sum_{i=1}^{N} a_i^2\sigma_i^2 + 2\sum_{i=1}^{N}\sum_{j\neq i} a_i a_j \sigma_{ij}.
\end{aligned}$$

The derivation of these results is very similar to the bivariate case and so is omitted. In addition, if all of the X_i are normally distributed then Z is also normally distributed with mean μ_Z and variance σ_Z^2 as described above.

The variance of a linear combination of N random variables contains N variance terms and $N(N-1)$ covariance terms. For $N = 2, 5, 10$ and 100 the number of covariance terms in $\text{var}(Z)$ is 2, 20, 90 and 9,900, respectively. Notice that when N is large there are many more covariance terms than variance terms in $\text{var}(Z)$.

The expression for $\text{var}(Z)$ is messy. It can be simplified using matrix algebra notation, as explained in detail in chapter Chapter 3. To preview, define the $N \times 1$ vectors $\mathbf{X} = (X_1, \ldots, X_N)'$ and $\mathbf{a} = (a_1, \ldots, a_N)'$. Then $Z = \mathbf{a}'\mathbf{X}$ and $\text{var}(Z) = \text{var}(\mathbf{a}'\mathbf{X}) = \mathbf{a}'\Sigma\mathbf{a}$, where Σ is the $N \times N$ covariance matrix, which is much more compact than Equation 3.13.

Example 2.53 (Square-root-of-time rule for multi-period continuously compounded returns)**.** Let r_t denote the continuously compounded monthly return on an asset at times $t = 1, \ldots, 12$. Assume that $r_1, \ldots, r_{12}$ are independent and identically distributed (iid) $N(\mu, \sigma^2)$. Recall, the annual continuously compounded return is equal to the sum of twelve monthly continuously compounded returns: $r_A = r(12) = \sum_{t=1}^{12} r_t$. Since each monthly return is normally distributed, the annual return is also normally distributed. The mean of $r(12)$ is:

$$\begin{aligned} E[r(12)] &= E\left[\sum_{t=1}^{12} r_t\right] \\ &= \sum_{t=1}^{12} E[r_t] \text{ (by linearity of expectation)}, \\ &= \sum_{t=1}^{12} \mu \text{ (by identical distributions)}, \\ &= 12 \cdot \mu. \end{aligned}$$

Hence, the expected 12-month (annual) return is equal to 12 times the expected monthly return. The variance of $r(12)$ is:

$$\begin{aligned} \text{var}(r(12)) &= \text{var}\left(\sum_{t=1}^{12} r_t\right) \\ &= \sum_{t=1}^{12} \text{var}(r_t) \text{ (by independence)}, \\ &= \sum_{j=0}^{11} \sigma^2 \text{ (by identical distributions)}, \\ &= 12 \cdot \sigma^2, \end{aligned}$$

so that the annual variance is also equal to 12 times the monthly variance[6]. Hence, the annual standard deviation is $\sqrt{12}$ times the monthly standard deviation: $\text{sd}(r(12)) = \sqrt{12}\sigma$ (this result is known as the *square-root-of-time rule*). Therefore, $r(12) \sim N(12\mu, 12\sigma^2)$. ■

2.3.6 Covariance between linear combinations of random variables

Consider the linear combinations of two random variables:

$$\begin{aligned} Y &= aX_1 + bX_2, \\ Z &= cX_3 + dX_4, \end{aligned}$$

[6]This result often causes some confusion. It is easy to make the mistake and say that the annual variance is $(12)^2 = 144$ time the monthly variance. This result would occur if $r_A = 12r_t$, so that $\text{var}(r_A) = (12)^2\text{var}(r_t) = 144\text{var}(r_t)$.

where a, b, c and d are constants. The covariance between Y and Z is,

$$\begin{aligned}
\text{cov}(Y, Z) &= \text{cov}(aX_1 + bX_2, cX_3 + dX_4) \\
&= E[((aX_1 + bX_2) - (a\mu_1 + b\mu_2))((cX_3 + dX_4) - (c\mu_3 + d\mu_4))] \\
&= E[(a(X_1 - \mu_1) + b(X_2 - \mu_2))(c(X_3 - \mu_3) + d(X_4 - \mu_4))] \\
&= acE[(X_1 - \mu_1)(X_3 - \mu_3)] + adE[(X_1 - \mu_1)(X_4 - \mu_4)] \\
&+ bcE[(X_2 - \mu_2)(X_3 - \mu_3)] + bdE[(X_2 - \mu_2)(X_4 - \mu_4)] \\
&= ac\text{cov}(X_1, X_3) + ad\text{cov}(X_1, X_4) + bc\text{cov}(X_2, X_3) + bd\text{cov}(X_2, X_4).
\end{aligned}$$

Hence, covariance is additive for linear combinations of random variables. The result above extends in an obvious way to arbitrary linear combinations of random variables.

2.4 Further Reading: Review of Random Variables

This material in this chapter is a selected survey of topics, with applications to finance, typically covered in an introductory probability and statistics course using calculus. A good textbook for such a course is (DeGroot and Schervish 2011). Chapter 1 of (Carmona 2014) and the appendix of (Ruppert and Matteson 2015) give a slightly more advanced review than is presented here. Two slightly more advanced books, oriented towards economics students, are (Amemiya 1994) and (Goldberger 1991). Good introductory treatments of probability and statistics using R include (Dalgaard 2002) and (Verzani 2005). (Pfaff 2013) discusses some useful financial asset return distributions and their implementation in R.

2.5 Exercises

Exercise 2.1. Suppose X is a normally distributed random variable with mean 0.05 and variance $(0.10)^2$. That is, $X \sim N(0.05, (0, 10)^2)$. Use the `pnorm()` and `qnorm()` functions in R to compute the following:

1. $\Pr(X \geq 0.10)$, $\Pr(X \leq -0.10)$.
2. $\Pr(-0.05 \leq X \leq 0.15)$.
3. 1% and 5% quantiles, $q^R_{0.01}$ and $q^R_{0.05}$.
4. 95% and 99% quantiles, $q^R_{0.95}$ and $q^R_{0.99}$.

Exercise 2.2. Let X denote the monthly return on Microsoft Stock and let Y denote the monthly return on Starbucks stock. Assume that $X \sim N(0.05, (0.10)^2)$ and $Y \sim N(0.025, (0.05)^2)$.

1. Using a grid of values between -0.25 and 0.35, plot the normal curves for X and Y. Make sure that both normal curves are on the same plot.
2. Plot the points $(\sigma_X, \mu_X) = (0.10, 0.05)$ and $(\sigma_Y, \mu_Y) = (0.05, 0.025)$ in an x-y plot.
3. Comment on the risk-return tradeoffs for the two stocks.

Exercise 2.3. Let X_1, X_2, Y_1, and Y_2 be random variables. Show that

$$\begin{aligned}\text{Cov}(X_1 + X_2, Y_1 + Y_2) =& \text{Cov}(X_1, Y_1) + \text{Cov}(X_1, Y_2) + \\ & \text{Cov}(X_2, Y_1) + \text{Cov}(X_2, Y_2).\end{aligned}$$

Exercise 2.4. Let X be a random variable with $\text{var}(X) < \infty$. Show that $\text{var}(X) = E[(X - \mu_X)^2] = E[X^2] - \mu^2_X$.

Exercise 2.5. Let X and Y be a random variables such that $\text{cov}(X, Y) < \infty$.

1. If $Y = aX + b$ where $a > 0$ show that $\rho_{XY} = \text{cor}(X, Y) = 1$.
2. If $Y = aX + b$ where $a < 0$ show that $\rho_{XY} = \text{cor}(X, Y) = -1$.

Exercise 2.6. Let R denote the simple monthly return on Microsoft stock and let W_0 denote initial wealth to be invested over the month. Assume that $R \sim N(0.04, (0.09)^2)$ and that $W_0 = \$100,000$. Determine the 1% and 5% Value-at-Risk (VaR) over the month on the investment. That is, determine the loss in investment value that may occur over the next month with 1% probability and with 5% probability.

Exercise 2.7. Let R_t denote the simple monthly return and assume $R_t \sim iid\, N(\mu, \sigma^2)$. Consider the 2-period return $R_t(2) = (1 + R_t)(1 + R_{t-1}) - 1$.

1. Assuming that $\text{cov}(R_t, R_{t-1}) = 0$, show that $E[R_t R_{t-1}] = \mu^2$. Hint: use $\text{cov}(R_t, R_{t-1}) = E[R_t R_{t-1}] - E[R_t]E[R_{t-1}]$.
2. Show that $E[R_t(2)] = (1 + \mu^2) - 1$.
3. Is $R_t(2)$ normally distributed? Justify your answer.

Exercise 2.8. Let r denote the continuously compounded monthly return on Microsoft stock and let W_0 denote initial wealth to be invested over the month. Assume that $r \sim iid\, N(0.04, (0.09)^2)$ and that $W_0 = \$100,000$.

1. Determine the 1% and 5% value-at-risk (VaR) over the month on the investment. That is, determine the loss in investment value that may occur over the next month with 1% probability and with 5% probability. (Hint: compute the 1% and 5% quantile from the Normal distribution for r and then convert the continuously compounded return quantile to a simple return quantile using the transformation $R = e^r - 1$.)
2. Let $r(12) = r_1 + r_2 + \cdots + r_{12}$ denote the 12-month (annual) continuously compounded return. Compute $E[r(12)]$, $\text{var}(r(12))$ and $\text{sd}(r(12))$. What is the probability distribution of $r(12)$?
3. Using the probability distribution for $r(12)$, determine the 1% and 5% value-at-risk (VaR) over the year on the investment.

Exercise 2.9. Let R_{VFINX} and R_{AAPL} denote the monthly simple returns on VFINX (Vanguard S&P 500 index) and AAPL (Apple stock). Suppose that $R_{VFINX} \sim i.i.d.\, N(0.013, (0.037)^2)$, $R_{AAPL} \sim i.i.d.\, N(0.028, (0.073)^2)$.

1. Sketch the normal distributions for the two assets on the same graph. Show the mean values and the ranges within 2 standard deviations of the mean. Which asset appears to be more risky?
2. Plot the risk-return tradeoff for the two assets. That is, plot the mean values of each asset on the y-axis and the standard deviations on the x-axis. Comment on the relationship.
3. Let $W_0 = \$1,000$ be the initial wealth invested in each asset. Compute the 1% monthly Value-at-Risk values for each asset. (Hint: $q^Z_{0.01} = -2.326$).
4. Continue with the above question, state in words what the 1% Value-at-Risk numbers represent (i.e., explain what 1% Value-at-Risk for a one month $\$1,000$ investment means).
5. The normal distribution can be used to characterize the probability distribution of monthly simple returns or monthly continuously compounded returns. What are two problems with using the normal distribution for simple returns? Given these two problems, why might it be better to use the normal distribution for continuously compounded returns?

Exercise 2.10. In this question, you will examine the chi-square and Student's t distributions. A chi-square random variable with n degrees of freedom, denoted $X \sim \chi^2_n$, is defined as $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ where $Z_1, Z_2, \ldots, Z_n$ are $iid\, N(0,1)$ random variables. Notice that X only takes positive values. A Student's t random variable with n degrees of freedom, denoted $t \sim t_n$, is

defined as $t = Z/\sqrt{X/n}$, where $Z \sim N(0,1)$, $X \sim \chi^2_n$ and Z is independent of X. The Student's t distribution is similar to the standard normal except that it has fatter tails.

1. On the same graph, plot the probability curves of chi-squared distributed random variables with 1, 2, 5 and 10 degrees of freedom. Use different colors and line styles for each curve. Hint: In R the density of the chi-square distribution is computed using the function `dchisq()`.
2. On the same graph, plot the probability curves of Student's t distributed random variables with 1, 2, 5 and 10 degrees of freedom. Also include the probability curve for the standard normal distribution. Use different colors and line styles for each curve. Hint: In R the density of the chi-square distribution is computed using the function `dt()`.

Exercise 2.11. Consider the following joint distribution of X and Y:

X/Y	1	2	3
1	0.1	0.2	0
2	0.1	0	0.2
3	0	0.1	0.3

1. Find the marginal distributions of X and Y. Using these distributions, compute $E[X]$, $\text{var}(X)$, $\text{sd}(X)$, $E[Y]$, $\text{var}(Y)$ and $\text{sd}(Y)$.
2. Compute $\text{cov}(X,Y)$ and $\text{cor}(X,Y)$.
3. Find the conditional distributions $\Pr(X|Y=y)$ for $y = 1,2,3$ and the conditional distributions $\Pr(Y|X=x)$ for $x = 1,2,3$.
4. Using the conditional distributions $\Pr(X|Y=y)$ for $y = 1,2,3$ and $\Pr(Y|X=x)$ for $x = 1,2,3$ compute $E[X|Y=y]$ and $E[Y|X=x]$.
5. Using the conditional distributions $\Pr(X|Y=y)$ for $y = 1,2,3$ and $\Pr(Y|X=x)$ for $x = 1,2,3$ compute $\text{var}[X|Y=y]$ and $\text{var}[Y|X=x]$.
6. Are X and Y independent? Fully justify your answer.

Exercise 2.12. Let X and Y be distributed bivariate normal with $\mu_X = 0.05$, $\mu_Y = 0.025$, $\sigma_X = 0.10$, and $\sigma_Y = 0.05$.

1. Using R package **mvtnorm** function `rmvnorm()`, simulate 100 observations from the bivariate normal distribution with $\rho_{XY} = 0.9$. Using the `plot()` function create a scatterplot of the observations and comment on the direction and strength of the linear association. Using the function `pmvnorm()`, compute the joint probability $Pr(X \leq, Y \leq 0)$.
2. Using R package **mvtnorm** function `rmvnorm()`, simulate 100 observations from the bivariate normal distribution with $\rho_{XY} = -0.9$. Using the `plot()` function create a scatterplot of the observations and comment on the direction and strength of the linear association. Using the function `pmvnorm()`, compute the joint probability $Pr(X \leq, Y \leq 0)$.

3. Using R package **mvtnorm** function `rmvnorm()`, simulate 100 observations from the bivariate normal distribution with $\rho_{XY} = 0$. Using the `plot()` function create a scatterplot of the observations and comment on the direction and strength of the linear association. Using the function `pmvnorm()`, compute the joint probability $Pr(X \leq, Y \leq 0)$.

3 Matrix Algebra Review

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This chapter reviews some basic matrix algebra concepts that we will use throughout the book. Many of the formulas that involve multiple variables that we derive in the book can be simplified using matrix algebra. This aids in understanding general concepts. For example, portfolio risk and return calculations for portfolios with more than two assets is easy to express using matrix algebra. If returns are jointly normally distributed, then this multivariate normal distribution can be expressed concisely using matrix algebra. The first order conditions involving the portfolio weights for mean-variance optimized portfolios is a system of linear equations that can be set-up and solved using matrix algebra. In addition, R is a matrix programming language. Pretty much anything that you can represent in matrix algebra you can evaluate in R using matrix calculations that have a syntax very much like the matrix algebra representation. Lastly, many R calculations can be efficiently evaluated if they are *vectorized* - that is, if they operate on vectors of elements instead of looping over individual elements.

This chapter is organized as follows. Section 3.1 gives definitions of matrices and vectors and shows how to represent these objects in R. Section 3.2 reviews basic matrix operations (addition, subtraction, multiplication), and Section 3.3 shows how common summation calculations can be represented using matrix notation. Section 3.4 covers systems of linear equations and shows how the inverse of a matrix is related to the solutions to a system of linear equations. Section 3.5 discusses positive definite matrices and Cholesky decomposition. The use of matrix algebra for representing multivariate probability distributions is presented in Section 3.6, and portfolio calculations using matrix algebra is presented in
Section 3.7. Section 3.8 concludes with a discussion of derivatives of some simple matrix functions.

The R packages used in this chapter are **Matrix** and **mvtnorm**. Make sure these packages are downloaded and installed prior to replicating the R examples in this chapter.

3.1 Matrices and Vectors

Definition 3.1 (Matrix)**.** A *matrix* is just an array of numbers. The *dimension* of a matrix is determined by the number of its rows and columns.

For example, a matrix $\mathbf{A}$ with n rows and m columns is illustrated below:

$$\underset{(n\times m)}{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}$$

where a_{ij} denotes the ith row and j^{th} column element of $\mathbf{A}$.

Definition 3.2 (Vector)**.** A *vector* is simply a matrix with 1 column.

For example,

$$\underset{(n\times 1)}{\mathbf{x}} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

is an $n \times 1$ vector with elements $x_1, x_2, \dots, x_n$. Vectors and matrices are often written in bold type (or underlined) to distinguish them from scalars (single elements of vectors or matrices).

Example 3.1 (Matrix creation in R)**.** In R, matrix objects are created using the `matrix()` function. For example, to create the 2×3 matrix:

$$\underset{(2\times 3)}{\mathbf{A}} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

use,

```
matA = matrix(data=c(1,2,3,4,5,6),nrow=2,ncol=3,byrow=TRUE)
matA
#>      [,1] [,2] [,3]
#> [1,]    1    2    3
#> [2,]    4    5    6
class(matA)
#> [1] "matrix" "array"
```

The optional argument `byrow=TRUE` fills the matrix row-by-row.[1] The default is `byrow=FALSE` which fills the matrix column-by-column:

[1] When specifying logical variables in R always spell out `TRUE` and `FALSE` instead of using `T` and `F`. Upon startup R defines the variables `T=TRUE` and `F=FALSE` so that `T` and `F` can be used as substitutes for `TRUE` and `FALSE`, respectively. However, this shortcut is not recommended because the variables `T` and `F` could be reassigned during subsequent programming.

```
matrix(data=c(1,2,3,4,5,6),nrow=2,ncol=3)
#>      [,1] [,2] [,3]
#> [1,]    1    3    5
#> [2,]    2    4    6
```

Matrix objects (i.e., objects of class `"matrix"`) have row and column dimension attributes which can be examined with the `dim()` function:

```
dim(matA)
#> [1] 2 3
```

The rows and columns can be given names using:

```
dimnames(matA) = list(c("row1","row2"),c("col1","col2","col3"))
matA
#>      col1 col2 col3
#> row1    1    2    3
#> row2    4    5    6
```

or,

```
colnames(matA) = c("col1", "col2", "col3")
rownames(matA) = c("row1", "row2")
matA
#>      col1 col2 col3
#> row1    1    2    3
#> row2    4    5    6
```

The elements of a matrix can be extracted or subsetted as follows:

```
matA[1, 2]
#> [1] 2
matA["row1", "col1"]
#> [1] 1
matA[1, ]
#> col1 col2 col3
#>    1    2    3
matA[, 2]
#> row1 row2
#>    2    5
```

To preserve the dimension attributes when subsetting, use the `drop=FALSE` option:

```
matA[1, , drop=FALSE]
#>      col1 col2 col3
#> row1    1    2    3
matA[, 2, drop=FALSE]
#>      col2
#> row1    2
#> row2    5
```

■

Example 3.2 (Creating vectors in R)**.** Vectors can be created in R using a variety of methods. The easiest way is to use the combine function `c()`:

```
xvec = c(1,2,3)
xvec
#> [1] 1 2 3
xvec = 1:3
xvec
#> [1] 1 2 3
xvec = seq(from=1,to=3,by=1)
xvec
#> [1] 1 2 3
```

Vectors of numbers in R are of class `"numeric"` and do not have a dimension attribute:

```
class(xvec)
#> [1] "numeric"
dim(xvec)
#> NULL
```

The elements of a vector can be assigned names using the `names()` function:

```
names(xvec) = c("x1", "x2", "x3")
xvec
#> x1 x2 x3
#>  1  2  3
```

To force a dimension attribute onto a vector, coerce it to a `"matrix"` object using `as.matrix()`:

```
as.matrix(xvec)
#>    [,1]
#> x1    1
#> x2    2
#> x3    3
```

■

Definition 3.3 (Transpose)**.** The *transpose* of an $n \times m$ matrix $\mathbf{A}$ is a new matrix with the rows and columns of $\mathbf{A}$ interchanged, and is denoted $\mathbf{A}'$ or $\mathbf{A}^\top$.

For example,

$$\underset{(2\times 3)}{\mathbf{A}} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad \underset{(3\times 2)}{\mathbf{A}'} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$\underset{(3\times 1)}{\mathbf{x}} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \underset{(1\times 3)}{\mathbf{x}'} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}.$$

Definition 3.4 (Symmetric matrix)**.** A *symmetric* matrix $\mathbf{A}$ is such that $\mathbf{A} = \mathbf{A}'$.

Obviously, symmetry can only occur if $\mathbf{A}$ is a *square* matrix; i.e., the number of rows of $\mathbf{A}$ is equal to the number of columns. For example, consider the 2×2 matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Then,

$$\mathbf{A}' = \mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Example 3.3 (Transpose of a matrix in R)**.** To take the transpose of a matrix or vector, use the `t()` function:

```
matA = matrix(data=c(1,2,3,4,5,6),nrow=2,ncol=3,byrow=TRUE)
matA
#>      [,1] [,2] [,3]
#> [1,]    1    2    3
#> [2,]    4    5    6
t(matA)
#>      [,1] [,2]
```

```
#> [1,]    1    4
#> [2,]    2    5
#> [3,]    3    6
xvec = c(1,2,3)
t(xvec)
#>      [,1] [,2] [,3]
#> [1,]    1    2    3
```

Notice that, when applied to a `"numeric"` vector with n elements, the `t()` function returns a `"matrix"` object with dimension $1 \times n$.

■

3.2 Basic Matrix Operations

In this section we review the basic matrix operations of addition, subtraction, scalar multiplication and multiplication.

3.2.1 Addition and subtraction

Definition 3.5 (Matrix addition)**.** Matrix addition and subtraction are element by element operations and only apply to matrices of the same dimension.

For example, let,

$$\mathbf{A} = \begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix}, \ \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}.$$

Then,

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 4+2 & 9+0 \\ 2+0 & 1+7 \end{bmatrix} = \begin{bmatrix} 6 & 9 \\ 2 & 8 \end{bmatrix},$$

$$\mathbf{A} - \mathbf{B} = \begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 4-2 & 9-0 \\ 2-0 & 1-7 \end{bmatrix} = \begin{bmatrix} 2 & 9 \\ 2 & -6 \end{bmatrix}.$$

Example 3.4 (Matrix addition and subtraction in R)**.** Matrix addition and subtraction is straightforward in R:

```
matA = matrix(c(4,9,2,1),2,2,byrow=TRUE)
matB = matrix(c(2,0,0,7),2,2,byrow=TRUE)
matA
#>      [,1] [,2]
```

```
#> [1,]    4    9
#> [2,]    2    1
matB
#>      [,1] [,2]
#> [1,]    2    0
#> [2,]    0    7
# matrix addition
matC = matA + matB
matC
#>      [,1] [,2]
#> [1,]    6    9
#> [2,]    2    8
# matrix subtraction
matD = matA - matB
matD
#>      [,1] [,2]
#> [1,]    2    9
#> [2,]    2   -6
```

■

3.2.2 Scalar multiplication

Here we refer to the multiplication of a matrix by a scalar number. This is also an element-by-element operation. For example, let $c = 2$ and,

$$\mathbf{A} = \begin{bmatrix} 3 & -1 \\ 0 & 5 \end{bmatrix}.$$

Then,

$$c \cdot \mathbf{A} = \begin{bmatrix} 2 \cdot 3 & 2 \cdot (-1) \\ 2 \cdot (0) & 2 \cdot 5 \end{bmatrix} = \begin{bmatrix} 6 & -2 \\ 0 & 10 \end{bmatrix}.$$

Example 3.5 (Scalar multiplication in R)**.**

```
matA = matrix(c(3,-1,0,5),2,2,byrow=TRUE)
matC = 2*matA
matC
#>      [,1] [,2]
#> [1,]    6   -2
```

```
#> [2,]    0   10
```

■

3.2.3 Matrix multiplication

Matrix multiplication only applies to *conformable* matrices.

Definition 3.6 (Conformable matrices)**.** $\mathbf{A}$ and $\mathbf{B}$ are conformable matrices if the number of columns in $\mathbf{A}$ is equal to the number of rows in $\mathbf{B}$.

For example, if $\mathbf{A}$ is $n \times m$ and $\mathbf{B}$ is $m \times p$ then $\mathbf{A}$ and $\mathbf{B}$ are conformable and the matrix product of $\mathbf{A}$ and $\mathbf{B}$ has dimension $n \times p$. The mechanics of matrix multiplication is best explained by example. Let,

$$\underset{(2\times 2)}{\mathbf{A}} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } \underset{(2\times 3)}{\mathbf{B}} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \end{bmatrix}.$$

Then,

$$\begin{aligned} \underset{(2\times 2)}{\mathbf{A}} \cdot \underset{(2\times 3)}{\mathbf{B}} &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1\cdot 1 + 2\cdot 3 & 1\cdot 2 + 2\cdot 4 & 1\cdot 1 + 2\cdot 2 \\ 3\cdot 1 + 4\cdot 3 & 3\cdot 2 + 4\cdot 4 & 3\cdot 1 + 4\cdot 2 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 10 & 5 \\ 15 & 22 & 11 \end{bmatrix} = \underset{(2\times 3)}{\mathbf{C}} \end{aligned}$$

The resulting matrix $\mathbf{C}$ has 2 rows and 3 columns. The $(1, 1)$ element of $\mathbf{C}$ is the *dot product* of the first row of $\mathbf{A}$ with the first column of $\mathbf{B}$:

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = 1\cdot 1 + 2\cdot 3 = 7.$$

The $(1, 2)$ element of $\mathbf{C}$ is the dot product of the first row of $\mathbf{A}$ with the second column of $\mathbf{B}$:

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 1\cdot 2 + 2\cdot 4 = 10.$$

In general, the (i, j) element of $\mathbf{C}$ is the dot product of the ith row of $\mathbf{A}$ with the jth column of $\mathbf{B}$. If $\mathbf{A}$ is $n \times m$ and $\mathbf{B}$ is $m \times p$ then $\mathbf{C} = \mathbf{A} \cdot \mathbf{B}$ is $n \times p$.

As another example, let,

$$\underset{(2\times 2)}{\mathbf{A}} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } \underset{(2\times 1)}{\mathbf{B}} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}.$$

Then,

$$\underset{(2\times2)}{\mathbf{A}} \cdot \underset{(2\times1)}{\mathbf{B}} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 6 \\ 3 \cdot 2 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 14 \\ 30 \end{bmatrix}.$$

As a final example, let,

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \ \mathbf{y} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

Then

$$\mathbf{x}'\mathbf{y} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 = 32$$

Example 3.6 (Matrix multiplication in R)**.** In R, matrix multiplication is performed with the `%*%` operator. For example:

```
matA = matrix(1:4,2,2,byrow=TRUE)
matB = matrix(c(1,2,1,3,4,2),2,3,byrow=TRUE)
matA
#>      [,1] [,2]
#> [1,]    1    2
#> [2,]    3    4
matB
#>      [,1] [,2] [,3]
#> [1,]    1    2    1
#> [2,]    3    4    2
dim(matA)
#> [1] 2 2
dim(matB)
#> [1] 2 3
matC = matA%*%matB
matC
#>      [,1] [,2] [,3]
#> [1,]    7   10    5
#> [2,]   15   22   11
```

The next example shows that matrix multiplication in R also works on numeric vectors:

```r
matA = matrix(c(1,2,3,4), 2, 2, byrow=TRUE)
vecB = c(2,6)
matA%*%vecB
#>      [,1]
#> [1,]   14
#> [2,]   30
vecX = c(1,2,3)
vecY = c(4,5,6)
t(vecX)%*%vecY
#>      [,1]
#> [1,]   32
crossprod(vecX, vecY)
#>      [,1]
#> [1,]   32
```

■

3.2.3.1 Miscellaneous properties

Matrix multiplication satisfies the distributive property: Let $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ be conformable matrices. Then

$$\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}. \tag{3.1}$$

Also, the transpose of the product of two matrices is the product of the transposes in opposite order:

$$(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'. \tag{3.2}$$

3.2.4 The identity matrix

The *identity matrix* plays a similar role as the number 1. Multiplying any number by 1 gives back that number. In matrix algebra, pre-multiplying or post-multiplying a matrix $\mathbf{A}$ by a conformable identity matrix $\mathbf{I}$ gives back the matrix $\mathbf{A}$. To illustrate, let

$$\mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

denote the 2-dimensional identity matrix and let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

denote an arbitrary 2×2 matrix. Then,

$$\begin{aligned}\mathbf{I}_2 \cdot \mathbf{A} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \mathbf{A},\end{aligned}$$

and,

$$\begin{aligned}\mathbf{A} \cdot \mathbf{I}_2 &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \mathbf{A}.\end{aligned}$$

Example 3.7 (The identity matrix in R)**.** Use the `diag()` function to create an identity matrix:

```
matI = diag(2)
matI
#>      [,1] [,2]
#> [1,]    1    0
#> [2,]    0    1
matA = matrix(c(1,2,3,4), 2, 2, byrow=TRUE)
matI%*%matA
#>      [,1] [,2]
#> [1,]    1    2
#> [2,]    3    4
matA%*%matI
#>      [,1] [,2]
#> [1,]    1    2
#> [2,]    3    4
```

■

3.2.5 Diagonal, lower triangular and upper triangular matrices

Consider an $n \times n$ matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} d_1 & u_{12} & u_{13} & \cdots & u_{1n} \\ l_{21} & d_2 & u_{23} & \cdots & u_{2n} \\ l_{31} & l_{32} & d_3 & \cdots & u_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & l_{n3} & \cdots & d_n \end{bmatrix}.$$

The main diagonal consists of the n elements $\{d_1, d_2 \dots d_n\}$. The lower triangle consists of the $n(n-1)/2$ elements $\{l_{21}, l_{31}, l_{32}, \dots, l_{n(n-1)}\}$ below the main diagonal, and the upper triangle consists of the $n(n-1)/2$ elements $\{u_{12}, u_{13}, u_{23}, \dots, u_{(n-1)n}\}$ above the main diagonal.

An $n-$dimensional *diagonal matrix* $\mathbf{D}$ is an $n \times n$ square matrix with an $n \times 1$ vector $\mathbf{d} = (d_1, \dots, d_n)'$ along the main diagonal and zeros elsewhere:

$$\mathbf{D} = \text{diag}(\mathbf{d}) = \begin{bmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ 0 & 0 & d_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d_n \end{bmatrix}.$$

An $n \times n$ *lower triangular matrix* $\mathbf{L}$ has all values above the main diagonal equal to zero:

$$\mathbf{L} = \begin{bmatrix} d_1 & 0 & 0 & \cdots & 0 \\ l_{21} & d_2 & 0 & \cdots & 0 \\ l_{31} & l_{32} & d_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & l_{n3} & \cdots & d_n \end{bmatrix}.$$

An $n \times n$ *upper triangular matrix* $\mathbf{U}$ has all values below the main diagonal equal to zero:

$$\mathbf{U} = \begin{bmatrix} d_1 & u_{12} & u_{13} & \cdots & u_{1n} \\ 0 & d_2 & u_{23} & \cdots & u_{2n} \\ 0 & 0 & d_3 & \cdots & u_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d_n \end{bmatrix}.$$

Example 3.8 (Creating a diagonal matrix in R)**.** Diagonal matrices can be created with the `diag()` function. To create a 3×3 diagonal matrix with d $= (1, 2, 3)$ along the main diagonal, use

```
matD = diag(1:3)
matD
#>      [,1] [,2] [,3]
#> [1,]    1    0    0
#> [2,]    0    2    0
#> [3,]    0    0    3
```

■

Example 3.9 (Extracting lower and upper triangular elements of a square matrix)**.** The R func-

tions `lower.tri()` and `upper.tri()` can be used to extract the lower and upper triangular parts of a square matrix:

```
matA = matrix(c(1,2,3,4,5,6,7,8,9), 3, 3)
matA
#>      [,1] [,2] [,3]
#> [1,]    1    4    7
#> [2,]    2    5    8
#> [3,]    3    6    9
lower.tri(matA)
#>       [,1]  [,2]  [,3]
#> [1,] FALSE FALSE FALSE
#> [2,]  TRUE FALSE FALSE
#> [3,]  TRUE  TRUE FALSE
upper.tri(matA)
#>       [,1]  [,2]  [,3]
#> [1,] FALSE  TRUE  TRUE
#> [2,] FALSE FALSE  TRUE
#> [3,] FALSE FALSE FALSE
matA[lower.tri(matA)]
#> [1] 2 3 6
matA[upper.tri(matA)]
#> [1] 4 7 8
```

■

3.3 Representing Summation Using Matrix Notation

Consider the sum:

$$\sum_{k=1}^{n} x_k = x_1 + \cdots + x_n.$$

Let $\mathbf{x} = (x_1, \ldots, x_n)'$ be an $n \times 1$ vector and $\mathbf{1} = (1, \ldots, 1)'$ be an $n \times 1$ vector of ones. Then,

$$\mathbf{x}'\mathbf{1} = \begin{bmatrix} x_1 & \ldots & x_n \end{bmatrix} \cdot \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = x_1 + \cdots + x_n = \sum_{k=1}^{n} x_k,$$

and,

$$\mathbf{1}'\mathbf{x} = \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 + \cdots + x_n = \sum_{k=1}^{n} x_k.$$

Next, consider the sum of squared x values,

$$\sum_{k=1}^{n} x_k^2 = x_1^2 + \cdots + x_n^2.$$

This sum can be conveniently represented as,

$$\mathbf{x}'\mathbf{x} = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1^2 + \cdots + x_n^2 = \sum_{k=1}^{n} x_k^2.$$

Last, consider the sum of cross products,

$$\sum_{k=1}^{n} x_k y_k = x_1 y_1 + \cdots x_n y_n.$$

This sum can be compactly represented by,

$$\mathbf{x}'\mathbf{y} = \begin{bmatrix} x_1 & \dots & x_n \vdots \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = x_1 y_1 + \cdots x_n y_n = \sum_{k=1}^{n} x_k y_k.$$

Note that $\mathbf{x}'\mathbf{y} = \mathbf{y}'\mathbf{x}$.

Example 3.10 (Computing sums in R)**.** In R, summing the elements in a vector can be done using matrix algebra.

```
# create vector of 1's and a vector of x
onevec = rep(1,3)
onevec
#> [1] 1 1 1
xvec = c(1,2,3)
# sum elements in x
t(xvec)%*%onevec
#>      [,1]
#> [1,]    6
```

The functions `crossprod()` and `sum()` are generally computationally more efficient:

```
crossprod(xvec,onevec)
#>      [,1]
#> [1,]    6
sum(xvec)
#> [1] 6
```

Sums of squares are best computed using:

```
crossprod(xvec)
#>      [,1]
#> [1,]   14
sum(xvec^2)
#> [1] 14
```

The dot-product two vectors can be conveniently computed using the `crossprod()` function:

```
yvec = 4:6
xvec
#> [1] 1 2 3
yvec
#> [1] 4 5 6
crossprod(xvec,yvec)
#>      [,1]
#> [1,]   32
crossprod(yvec,xvec)
#>      [,1]
#> [1,]   32
```

■

3.4 Systems of Linear Equations

Consider the system of two linear equations:

$$x + y = 1, \tag{3.3}$$

$$2x - y = 1. \tag{3.4}$$

As shown in Figure 3.1, equations Equation 3.3 and Equation 3.4 represent two straight lines which intersect at the point $x = 2/3$ and $y = 1/3$. This point of intersection is determined by solving for

the values of x and y such that $x + y = 2x - y$[2]

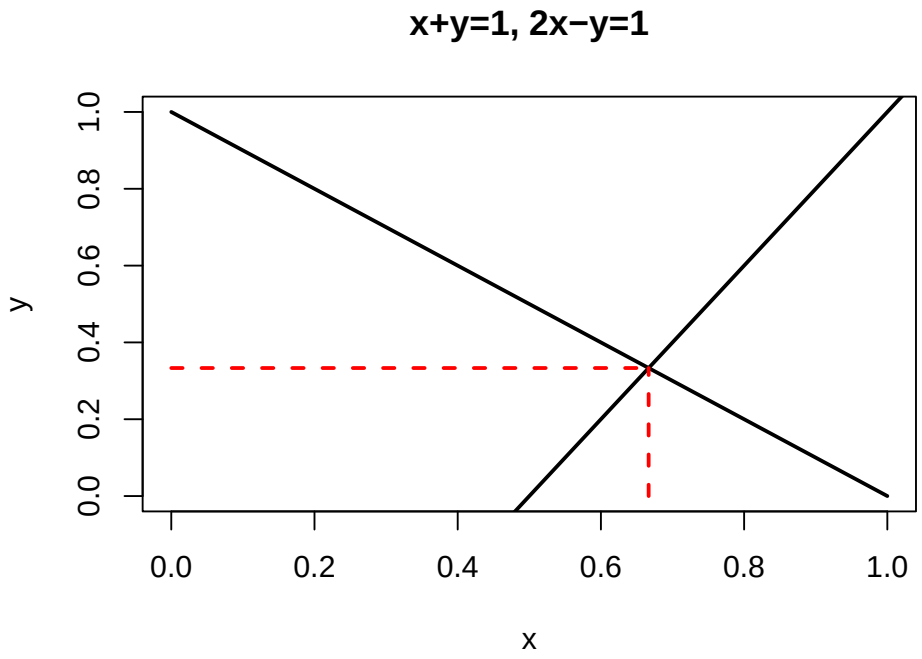


Figure 3.1: System of two linear equations.

The two linear equations can be written in matrix form as:

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

or,

$$\mathbf{A} \cdot \mathbf{z} = \mathbf{b},$$

where,

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}, \ \mathbf{z} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

If there was a (2×2) matrix $\mathbf{B}$, with elements b_{ij}, such that $\mathbf{B} \cdot \mathbf{A} = \mathbf{I}_2$, where $\mathbf{I}_2$ is the (2×2) identity matrix, then we could *solve* for the elements in $\mathbf{z}$ as follows. In the equation $\mathbf{A} \cdot \mathbf{z} = \mathbf{b}$, pre-multiply both sides by $\mathbf{B}$ to give:

$$\begin{aligned} \mathbf{B} \cdot \mathbf{A} \cdot \mathbf{z} &= \mathbf{B} \cdot \mathbf{b} \\ \implies \mathbf{I} \cdot \mathbf{z} &= \mathbf{B} \cdot \mathbf{b} \\ \implies \mathbf{z} &= \mathbf{B} \cdot \mathbf{b}, \end{aligned}$$

[2]Soving for x gives $x = 2y$. Substituting this value into the equation $x + y = 1$ gives $2y + y = 1$ and solving for y gives $y = 1/3$. Solving for x then gives $x = 2/3$.

or

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} b_{11} \cdot 1 + b_{12} \cdot 1 \\ b_{21} \cdot 1 + b_{22} \cdot 1 \end{bmatrix}.$$

If such a matrix $\mathbf{B}$ exists it is called the *inverse* of $\mathbf{A}$ and is denoted $\mathbf{A}^{-1}$. Intuitively, the inverse matrix $\mathbf{A}^{-1}$ plays a similar role as the inverse of a number. Suppose a is a number; e.g., $a = 2$. Then we know that $(1/a) \cdot a = a^{-1}a = 1$. Similarly, in matrix algebra $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_2$ where $\mathbf{I}_2$ is the identity matrix. Now, consider solving the equation $a \cdot x = b$. By simple division we have that $x = (1/a) \cdot b = a^{-1} \cdot b$. Similarly, in matrix algebra, if we want to solve the system of linear equations $\mathbf{Ax} = \mathbf{b}$ we pre-multiply by $\mathbf{A}^{-1}$ and get the solution $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Using $\mathbf{B} = \mathbf{A}^{-1}$, we may express the solution for $\mathbf{z}$ as:

$$\mathbf{z} = \mathbf{A}^{-1}\mathbf{b}.$$

As long as we can determine the elements in $\mathbf{A}^{-1}$ then we can solve for the values of x and y in the vector $\mathbf{z}$. The system of linear equations has a solution as long as the two lines intersect, so we can determine the elements in $\mathbf{A}^{-1}$ provided the two lines are not parallel. If the two lines are parallel, then one of the equations is a multiple of the other. In this case, we say that $\mathbf{A}$ is *not invertible*.

There are general numerical algorithms for finding the elements of $\mathbf{A}^{-1}$ (e.g., Gaussian elimination) and R has these algorithms available. However, if $\mathbf{A}$ is a (2×2) matrix then there is a simple formula for $\mathbf{A}^{-1}$. Let $\mathbf{A}$ be a (2×2) matrix such that:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Then,

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}, \tag{3.5}$$

where $\det(\mathbf{A}) = a_{11}a_{22} - a_{21}a_{12}$ denotes the determinant of $\mathbf{A}$ and is assumed to be not equal to

zero. By brute force matrix multiplication we can verify this formula:

$$\begin{aligned}
\mathbf{A}^{-1}\mathbf{A} &= \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\
&= \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22}a_{11} - a_{12}a_{21} & a_{22}a_{12} - a_{12}a_{22} \\ -a_{21}a_{11} + a_{11}a_{21} & -a_{21}a_{12} + a_{11}a_{22} \end{bmatrix} \\
&= \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22}a_{11} - a_{12}a_{21} & 0 \\ 0 & -a_{21}a_{12} + a_{11}a_{22} \end{bmatrix} \\
&= \begin{bmatrix} \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}a_{22} - a_{21}a_{12}} & 0 \\ 0 & \frac{-a_{21}a_{12} + a_{11}a_{22}}{a_{11}a_{22} - a_{21}a_{12}} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.
\end{aligned}$$

Let's apply the above rule to find the inverse of $\mathbf{A}$ in our example linear system Equation 3.3 and Equation 3.4:

$$\mathbf{A}^{-1} = \frac{1}{-1-2} \begin{bmatrix} -1 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{-1}{3} \end{bmatrix}.$$

Notice that,

$$\mathbf{A}^{-1}\mathbf{A} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{-1}{3} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Our solution for $\mathbf{z}$ is then,

$$\begin{aligned}
\mathbf{z} &= \mathbf{A}^{-1}\mathbf{b} \\
&= \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{-1}{3} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix},
\end{aligned}$$

so that $x = 2/3$ and $y = 1/3$.

Example 3.11 (Solving systems of linear equations in R)**.** In R, the `solve()` function is used to compute the inverse of a matrix and solve a system of linear equations. The linear system $x + y = 1$ and $2x - y = 1$ can be represented using:

```
matA = matrix(c(1,1,2,-1), 2, 2, byrow=TRUE)
vecB = c(1,1)
```

First we solve for $\mathbf{A}^{-1}$.[3]

[3]Notice that the calculations in R do not show $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ exactly. The (1,2) element of $\mathbf{A}^{-1}\mathbf{A}$ is -5.552e-17, which for all practical purposes is zero. However, due to the limitations of machine calculations the result is not exactly

```
matA.inv = solve(matA)
matA.inv
#>       [,1]   [,2]
#> [1,] 0.333  0.333
#> [2,] 0.667 -0.333
matA.inv%*%matA
#>      [,1]      [,2]
#> [1,]    1 -5.55e-17
#> [2,]    0  1.00e+00
matA%*%matA.inv
#>      [,1]     [,2]
#> [1,]    1 5.55e-17
#> [2,]    0 1.00e+00
```

Then we solve the system $\mathbf{z} = \mathbf{A}^{-1}\mathbf{b}$:

```
z = matA.inv%*%vecB
z
#>       [,1]
#> [1,] 0.667
#> [2,] 0.333
```

■

In general, if we have n linear equations in n unknown variables we may write the system of equations as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots &= \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

which we may then express in matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

or

$$\underset{(n\times n)}{\mathbf{A}} \cdot \underset{(n\times 1)}{\mathbf{x}} = \underset{(n\times 1)}{\mathbf{b}} .$$

zero.

The solution to the system of equations is given by:

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b},$$

where $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$ and $\mathbf{I}_n$ is the $(n \times n)$ identity matrix. If the number of equations is greater than two, then we generally use numerical algorithms to find the elements in $\mathbf{A}^{-1}$.

3.4.1 Rank of a matrix

Definition 3.7 (Linear independence of vectors)**.** Consider an $n \times m$ matrix $\mathbf{A}$ with m columns $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m$ where each column is an $n \times 1$ vector and n rows where each row is an $1 \times m$ row vector:

$$\underset{(n\times m)}{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1m} \\ a_{21} & a_{22} & \ldots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nm} \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \ldots & \mathbf{a}_m \end{bmatrix}.$$

Two vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are linearly independent if $c_1\mathbf{a}_1 + c_2\mathbf{a}_2 = 0$ implies that $c_1 = c_2 = 0$. A set of vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m$ are linearly independent if $c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_m\mathbf{a}_m = 0$ implies that $c_1 = c_2 = \cdots = c_m = 0$. That is, no vector $\mathbf{a}_i$ can be expressed as a non-trivial linear combination of the other vectors.

Definition 3.8 (Rank of a matrix)**.** The *column rank* of the $n \times m$ matrix $\mathbf{A}$, denoted rank($\mathbf{A}$), is equal to the maximum number of linearly independent columns. The row rank of a matrix is equal to the maximum number of linearly independent rows, and is given by rank($\mathbf{A}'$).

It turns out that the column rank and row rank of a matrix are the same. Hence, rank($\mathbf{A}$)= rank($\mathbf{A}'$) $\leq \min(m, n)$. If rank($\mathbf{A}$) $= m$ then $\mathbf{A}$ is called *full column rank*, and if rank($\mathbf{A}$) $= n$ then $\mathbf{A}$ is called *full row rank*. If $\mathbf{A}$ is not full rank then it is called *reduced rank*.

Example 3.12 (Determining the rank of a matrix in R)**.** Consider the 2×3 matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

Here, rank($\mathbf{A}$) $\leq min(3, 2) = 2$, the number of rows. Now, rank($\mathbf{A}$) $= 2$ since the rows of $\mathbf{A}$ are linearly independent. In R, the rank of a matrix can be found using the **Matrix** function `rankMatrix()`:

```
library(Matrix)
Amat = matrix(c(1,3,5,2,4,6), 2, 3, byrow=TRUE)
as.numeric(rankMatrix(Amat))
```

```
#> [1] 2
```

■

The rank of an $n \times n$ square matrix $\mathbf{A}$ is directly related to its invertibility. If $\text{rank}(\mathbf{A}) = n$ then $\mathbf{A}^{-1}$ exists. This result makes sense. Since $\mathbf{A}^{-1}$ is used to solve a system of n linear equations in n unknowns, it will have a unique solution as long as no equation in the system can be written as a linear combination of the other equations.

3.4.2 Partitioned matrices and partitioned inverses

Consider a general $n \times m$ matrix $\mathbf{A}$:

$$\underset{(n\times m)}{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}.$$

In some situations we might want to partition $\mathbf{A}$ into sub-matrices containing sub-blocks of the elements of $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix},$$

where the sub-matrices $\mathbf{A}_{11}$, $\mathbf{A}_{12}$, $\mathbf{A}_{21}$, and $\mathbf{A}_{22}$ contain the appropriate sub-elements of $\mathbf{A}$. For example, consider

$$\mathbf{A} = \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ \hline 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{array}\right] = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix},$$

where

$$\begin{aligned} \mathbf{A}_{11} &= \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}, \mathbf{A}_{12} = \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}, \\ \mathbf{A}_{21} &= \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix}, \mathbf{A}_{22} = \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix}. \end{aligned}$$

Example 3.13 (Combining sub-matrices in R)**.** In R sub-matrices can be combined column-wise using `cbind()`, and row-wise using `rbind()`. For example, to create the matrix $\mathbf{A}$ from the sub-matrices use

```
A11mat = matrix(c(1,2,5,6), 2, 2, byrow=TRUE)
A12mat = matrix(c(3,4,7,8), 2, 2, byrow=TRUE)
A21mat = matrix(c(9,10,13,14), 2, 2, byrow=TRUE)
A22mat = matrix(c(11,12,15,16), 2, 2, byrow=TRUE)
Amat = rbind(cbind(A11mat, A12mat), cbind(A21mat, A22mat))
Amat
#>      [,1] [,2] [,3] [,4]
#> [1,]    1    2    3    4
#> [2,]    5    6    7    8
#> [3,]    9   10   11   12
#> [4,]   13   14   15   16
```

■

The basic matrix operations work on sub-matrices in the obvious way. To illustrate, consider another matrix $\mathbf{B}$ that is conformable with $\mathbf{A}$ and partitioned in the same way:

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix}.$$

Then

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} + \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} + \mathbf{B}_{11} & \mathbf{A}_{12} + \mathbf{B}_{12} \\ \mathbf{A}_{21} + \mathbf{B}_{21} & \mathbf{A}_{22} + \mathbf{B}_{22} \end{bmatrix}.$$

If all of the sub-matrices of $\mathbf{A}$ and $\mathbf{B}$ are comformable then

$$\mathbf{AB} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{bmatrix}.$$

The transpose of a partitioned matrix satisfies

$$\mathbf{A}' = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}' = \begin{bmatrix} \mathbf{A}'_{11} & \mathbf{A}'_{21} \\ \mathbf{A}'_{12} & \mathbf{A}'_{22} \end{bmatrix}.$$

Notice the interchange of the two off-diagonal blocks.

A partitioned matrix $\mathbf{A}$ with appropriate invertible sub-matrices $\mathbf{A}_{11}$ and $\mathbf{A}_{22}$ has a partitioned inverse of the form

$$\mathbf{A}^{-1} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{C}^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}\mathbf{A}_{12}\mathbf{C}^{-1} \\ -\mathbf{C}^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{C}^{-1} \end{bmatrix},$$

where $\mathbf{C} = \mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}$. This formula can be verified by direct calculation.

3.5 Positive Definite Matrices

Definition 3.9 (Positive definite matrix)**.** Let $\mathbf{A}$ be an $n \times n$ symmetrix matrix. The matrix $\mathbf{A}$ is *positive definite* if for any $n \times 1$ vector $\mathbf{x} \neq \mathbf{0}$

$$\mathbf{x}'\mathbf{A}\mathbf{x} > 0.$$

This condition implies that there is no non-zero vector $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{0}$, which implies that $\mathbf{A}$ has full rank n.

Definition 3.10 (Positive semi-definite matrix)**.** The matrix $\mathbf{A}$ is *positive semi-definite* if for any $n \times 1$ vector $\mathbf{x} \neq \mathbf{0}$

$$\mathbf{x}'\mathbf{A}\mathbf{x} \geq 0.$$

Hence, if $\mathbf{A}$ is positive semi-definite then there exists some $n \times 1$ vector $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{0}$, which implies that $\mathbf{A}$ does not have full rank n.

3.5.1 Matrix square root

To motivate the idea of the square root of a matrix, consider the square root factorization of a positive number a :

$$a = \sqrt{a} \times \sqrt{a} = a^{1/2} \times a^{1/2}.$$

That is, we can factorize any positive number into the product of its square root. It turns out we can do a similar factorization for a positive definite and symmetric matrix.

Proposition 3.1 (Cholesky factorization)**.** *Let* $\mathbf{A}$ *be an* $n \times n$ *positive definite and symmetric matrix. Then it is possible to factor* $\mathbf{A}$ *as*

$$\mathbf{A} = \mathbf{C}'\mathbf{C},$$

where $\mathbf{C}$ *is an* $n \times n$ *upper triangular matrix with non-negative diagonal elements called the* Cholesky factor *of* $\mathbf{A}$*. This factorization is also called a* matrix square root factorization *by defining the square root matrix* $\mathbf{A}^{1/2} = \mathbf{C}$*. Then we can write* $\mathbf{A} = \mathbf{A}^{1/2\prime}\mathbf{A}^{1/2}$*.* [4] *If all of the diagonal elements of* $\mathbf{C}$ *are positive then* $\mathbf{A}$ *is positive definite. Otherwise,* $\mathbf{A}$ *is positive semi-definite.*

Example 3.14 (Cholesky decomposition in R)**.** The R function `chol()` computes the Cholesky factorization of a square symmetrix matrix. For example, consider the 2×2 correlation matrix:

[4] The Cholesky factorization is one way of defining the matrix square root, where the square root matrix is upper triangular. Other matrix square factorizations exist where the square root matrix is not necessarily upper triangular.

```
Rho = matrix(c(1, 0.75, 0.75, 1), 2, 2)
Rho
#>      [,1] [,2]
#> [1,] 1.00 0.75
#> [2,] 0.75 1.00
```

The Cholesky matrix square root factorization is

```
C = chol(Rho)
C
#>      [,1]  [,2]
#> [1,]    1 0.750
#> [2,]    0 0.661
t(C)%*%C
#>      [,1] [,2]
#> [1,] 1.00 0.75
#> [2,] 0.75 1.00
```

Here, all the diagonal elements of $\mathbf{C}$ are positive so that covariance matrix is positive definite. Next, suppose the correlation matrix is

```
Rho = matrix(c(1, 1, 1, 1), 2, 2)
Rho
#>      [,1] [,2]
#> [1,]    1    1
#> [2,]    1    1
```

so that the two variables are perfectly correlated. Then

```
chol(Rho, pivot=TRUE)
#>      [,1] [,2]
#> [1,]    1    1
#> [2,]    0    0
#> attr(,"pivot")
#> [1] 1 2
#> attr(,"rank")
#> [1] 1
```

Here, we see that one of the diagonal elements of the Cholesky factor is zero and that the correlation matrix is rank deficient.

■

3.6 Multivariate Probability Distributions Using Matrix Algebra

In this section, we show how matrix algebra can be used to simplify many of the messy expressions concerning expectations and covariances between multiple random variables, and we show how certain multivariate probability distributions (e.g., the multivariate normal distribution) can be expressed using matrix algebra.

3.6.1 Random vectors

Let $X_1, \ldots, X_n$ denote n random variables. For $i = 1, \ldots, n$ let $\mu_i = E[X_i]$ and $\sigma_i^2 = \text{var}(X_i)$, and let $\sigma_{ij} = \text{cov}(X_i, X_j)$ for $i \neq j$. Define the $n \times 1$ random vector $\mathbf{X} = (X_1, \ldots, X_n)'$. Associated with $\mathbf{X}$ is the $n \times 1$ vector of expected values:

$$\underset{n\times 1}{\mu} = E[\mathbf{X}] = \begin{pmatrix} E[X_1] \\ \vdots \\ E[X_n] \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix}. \tag{3.6}$$

3.6.2 Covariance matrix

The covariance matrix, Σ, summarizes the variances and covariances of the elements of the random vector $\mathbf{X}$. In general, the covariance matrix of a random vector $\mathbf{X}$ (sometimes called the variance of the vector $\mathbf{X}$) with mean
vector μ is defined as:

$$\text{cov}(\mathbf{X}) = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)'] = \Sigma. \tag{3.7}$$

If $\mathbf{X}$ has n elements then Σ will be the symmetric and positive semi-definite $n \times n$ matrix,

$$\underset{n\times n}{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1n} & \sigma_{2n} & \cdots & \sigma_n^2 \end{pmatrix}. \tag{3.8}$$

For the case $n = 2$, we have by direct calculation

$$\begin{aligned} E[(\mathbf{X}-\mu)(\mathbf{X}-\mu)'] &= E\left[\begin{pmatrix} X_1-\mu_1 \\ X_2-\mu_2 \end{pmatrix} \cdot (X_1-\mu_1, X_2-\mu_2)\right] \\ &= E\left[\begin{pmatrix} (X_1-\mu_1)^2 & (X_1-\mu_1)(X_2-\mu_2) \\ (X_2-\mu_2)(X_1-\mu_1) & (X_2-\mu_2)^2 \end{pmatrix}\right] \\ &= \begin{pmatrix} E[(X_1-\mu_1)^2] & E[(X_1-\mu_1)(X_2-\mu_2)] \\ E[(X_2-\mu_2)(X_1-\mu_1)] & E[(X_2-\mu_2)^2] \end{pmatrix} \\ &= \begin{pmatrix} \text{var}(X_1) & \text{cov}(X_1,X_2) \\ \text{cov}(X_2,X_1) & \text{var}(X_2) \end{pmatrix} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} = \Sigma. \end{aligned}$$

If $\rho_{12} \neq \pm 1$ then $\det(\Sigma) \neq 0$, Σ has rank 2 and is positive definite. However, if $\rho_{12} = \pm 1$ then $\det(\Sigma) = 0$, Σ has rank 1 and is positive semi-definite.

3.6.3 Correlation matrix

The *correlation matrix*, $\mathbf{C}$, summarizes all pairwise correlations between the elements of the $n \times 1$ random vector $\mathbf{X}$ and is given by

$$\mathbf{C} = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{12} & 1 & \cdots & \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1n} & \rho_{2n} & \cdots & 1 \end{pmatrix} \tag{3.9}$$

Let Σ denote the covariance matrix of $\mathbf{X}$. Then the correlation matrix $\mathbf{C}$ can be computed using

$$\mathbf{C} = \mathbf{D}^{-1}\Sigma\mathbf{D}^{-1}, \tag{3.10}$$

where $\mathbf{D}$ is an $n \times n$ diagonal matrix with the standard deviations of the elements of $\mathbf{X}$ along the main diagonal:

$$\mathbf{D} = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \end{pmatrix}.$$

Example 3.15 (Computing correlation matrix from covariance matrix in R)**.** In R the `cov2cor()` function can be used to compute the correlation matrix given the covariance matrix using Equation 3.10:

```r
sigma.1 = 1
sigma.2 = 2
rho.12 = 0.5
sigma.12 = rho.12*sigma.1*sigma.2
Sigma = matrix(c(sigma.1^2, sigma.12, sigma.12, sigma.2^2),
               2, 2, byrow=TRUE)
Sigma
#>      [,1] [,2]
#> [1,]    1    1
#> [2,]    1    4
cov2cor(Sigma)
#>      [,1] [,2]
#> [1,]  1.0  0.5
#> [2,]  0.5  1.0
```

■

3.6.4 Variance of linear combination of random vectors

Consider the $n \times 1$ random vector $\mathbf{X}$ with mean vector μ and covariance matrix Σ. Let $\mathbf{a} = (a_1, \dots, a_n)'$ be an $n \times 1$ vector of constants and consider the random variable $Y = \mathbf{a}'\mathbf{X} = a_1 X_1 + \cdots + a_n X_n$. The expected value of Y is:

$$\mu_Y = E[Y] = E[\mathbf{a}'\mathbf{X}] = \mathbf{a}' E[\mathbf{X}] = \mathbf{a}'\mu. \tag{3.11}$$

The variance of Y is

$$\text{var}(Y) = \text{var}(\mathbf{a}'\mathbf{X}) = E[(\mathbf{a}'\mathbf{X} - \mathbf{a}'\mu)^2] = E[(\mathbf{a}'(\mathbf{X} - \mu))^2] \tag{3.12}$$

since $Y = \mathbf{a}'\mathbf{X}$ is a scalar. Now we use a trick from matrix algebra. If z is a scalar (think of $z = 2$) then $z'z = z \cdot z' = z^2$. Let $z = \mathbf{a}'(\mathbf{X} - \mu)$ and so $z \cdot z' = \mathbf{a}'(\mathbf{X} - \mu)(\mathbf{X} - \mu)'\mathbf{a}$. Then,

$$\begin{aligned} \text{var}(Y) &= E[z^2] = E[z \cdot z'] \\ &= E[\mathbf{a}'(\mathbf{X} - \mu)(\mathbf{X} - \mu)'\mathbf{a}] \\ &= \mathbf{a}' E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)']\mathbf{a} \\ &= \mathbf{a}' \text{cov}(\mathbf{X})\mathbf{a} = \mathbf{a}'\Sigma\mathbf{a}. \end{aligned} \tag{3.13}$$

3.6.5 Covariance between linear combination of two random vectors

Consider the $n \times 1$ random vector $\mathbf{X}$ with mean vector μ and covariance matrix Σ. Let $\mathbf{a} = (a_1, \ldots, a_n)'$ and $\mathbf{b} = (b_1, \ldots, b_n)'$ be $n \times 1$ vectors of constants, and consider the random variable $Y = \mathbf{a}'\mathbf{X} = a_1X_1 + \cdots + a_nX_n$ and $Z = \mathbf{b}'\mathbf{X} = b_1X_1 + \cdots + b_nX_n$. From the definition of covariance we have:

$$\text{cov}(Y, Z) = E[(Y - E[Y])(Z - E[Z])]$$

which may be rewritten in matrix notation as,

$$\begin{aligned} \text{cov}(\mathbf{a}'\mathbf{X}, \mathbf{b}'\mathbf{X}) &= E[(\mathbf{a}'\mathbf{X} - \mathbf{a}'\mu)(\mathbf{b}'\mathbf{X} - \mathbf{b}'\mu)] \\ &= E[\mathbf{a}'(\mathbf{X} - \mu)\mathbf{b}'(\mathbf{X} - \mu)] \\ &= E[\mathbf{a}'(\mathbf{X} - \mu)(\mathbf{X} - \mu)'\mathbf{b}] \\ &= \mathbf{a}'E[(\mathbf{X}\mu)(\mathbf{X} - \mu)']\mathbf{b} \\ &= \mathbf{a}'\Sigma\mathbf{b}. \end{aligned} \tag{3.14}$$

Since $\mathbf{a}'(\mathbf{X} - \mu)$ and $\mathbf{b}'(\mathbf{X} - \mu)$ are scalars, we can use the trick:

$$\mathbf{a}'(\mathbf{X} - \mu)\mathbf{b}'(\mathbf{X} - \mu) = \mathbf{a}'(\mathbf{X} - \mu)(\mathbf{X}\mu)'\mathbf{b}.$$

3.6.6 Variance of a vector of linear functions of a random vector

Consider the $n \times 1$ random vector $\mathbf{X}$ with mean vector μ and covariance matrix Σ. Let $\mathbf{A}$ be an $m \times n$ matrix. Then the $m \times 1$ random vector $\mathbf{Y} = \mathbf{A}\mathbf{X}$ represents m linear functions of $\mathbf{X}$. By the linearity of expectation, we have $E[\mathbf{Y}] = \mathbf{A}E[\mathbf{X}] = \mathbf{A}\mu$. The $m \times m$ covariance matrix of $\mathbf{Y}$ is given by

$$\begin{aligned} \text{var}(\mathbf{Y}) &= E[(\mathbf{Y} - E[\mathbf{Y}])(\mathbf{Y} - E[\mathbf{Y}])'] \\ &= E[(\mathbf{A}\mathbf{X} - \mathbf{A}\mu)(\mathbf{A}\mathbf{X} - \mathbf{A}\mu)'] \\ &= E[\mathbf{A}(\mathbf{X} - \mu)(\mathbf{A}(\mathbf{X} - \mu))'] \\ &= E[\mathbf{A}(\mathbf{X} - \mu)(\mathbf{X} - \mu)'\mathbf{A}'] \\ &= \mathbf{A}E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)']\mathbf{A}' \\ &= \mathbf{A}\Sigma\mathbf{A}'. \end{aligned} \tag{3.15}$$

3.6.7 Multivariate normal distribution

Consider n random variables $X_1, \ldots, X_n$ and assume they are jointly normally distributed. Define the $n \times 1$ vectors $\mathbf{X} = (X_1, \ldots, X_n)'$, $\mathbf{x} = (x_1, \ldots, x_n)'$ and $\mu = (\mu_1, \ldots, \mu_n)'$, and the $n \times n$ covariance

matrix:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1n} & \sigma_{2n} & \cdots & \sigma_n^2 \end{pmatrix}.$$

Then $\mathbf{X} \sim N(\mu, \Sigma)$ means that the random vector $\mathbf{X}$ has a *multivariate normal distribution* with mean vector μ and covariance matrix Σ. The pdf of the multivariate normal distribution can be compactly expressed as:

$$\begin{aligned} f(\mathbf{x}) &= \frac{1}{2\pi^{n/2}\det(\Sigma)^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)} \\ &= (2\pi)^{-n/2}\det(\Sigma)^{-1/2} e^{-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)}. \end{aligned}$$

Example 3.16 (Creating a multivariate normal random vector from a vector of independent standard normal random variables)**.** Let $\mathbf{Z} = (Z_1, \ldots, Z_n)'$ denote a vector of n iid standard normal random variables. Then $\mathbf{Z} \sim N(\mathbf{0}, \mathbf{I}_n)$ where $\mathbf{I}_n$ denotes the $n-$dimensional identity matrix. Given $\mathbf{Z}$, we would like to create a random vector $\mathbf{X}$ such that $\mathbf{X} \sim N(\mu, \Sigma)$. Let $\Sigma^{1/2}$ denote the upper triangular Cholesky factor of Σ such that $\Sigma = \Sigma^{1/2\prime}\Sigma^{1/2}$. Define $\mathbf{X}$ using

$$\mathbf{X} = \mu + \Sigma^{1/2\prime}\mathbf{Z}.$$

Then

$$\begin{aligned} E[\mathbf{X}] &= \mu + \Sigma^{1/2}E[\mathbf{Z}] = \mu, \\ \text{var}(\mathbf{X}) &= \text{var}\left(\Sigma^{1/2}\mathbf{Z}\right) = \Sigma^{1/2\prime}\text{var}(\mathbf{Z})\Sigma^{1/2} = \Sigma^{1/2\prime}\mathbf{I}_n\Sigma^{1/2} = \Sigma. \end{aligned}$$

Hence, $\mathbf{X} \sim N(\mu, \Sigma)$.

■

Example 3.17 (Simulating multivariate normal random vectors in R)**.** Here, we illustrate how to simulate $\mathbf{X} \sim N(\mu, \Sigma)$ where $\mu = (1, 1)'$ and

$$\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix}$$

First, specify the mean vector and covariance matrix for the the bivariate normal distribution:

```
mu.vec = c(1,1)
sigma.1 = 1
sigma.2 = 2
rho.12 = 0.5
```

```
sigma.12 = rho.12*sigma.1*sigma.2
Sigma = matrix(c(sigma.1^2, sigma.12, sigma.12, sigma.2^2),
               2, 2, byrow=TRUE)
mu.vec
#> [1] 1 1
Sigma
#>      [,1] [,2]
#> [1,]    1    1
#> [2,]    1    4
```

Second, compute the Cholesky factorization of Σ:

```
Sigma.5 = chol(Sigma)
Sigma.5
#>      [,1] [,2]
#> [1,]    1 1.00
#> [2,]    0 1.73
```

Third, simulate $\mathbf{Z} \sim N(\mathbf{0}, \mathbf{I}_2)$

```
n = 2
set.seed(123)
Z = rnorm(n)
```

Fourth, compute $\mathbf{X} = \mu + \Sigma^{1/2\prime}\mathbf{Z}$

```
X = mu.vec + t(Sigma.5)%*%Z
X
#>        [,1]
#> [1,] 0.4395
#> [2,] 0.0408
```

■

3.7 Portfolio Math Using Matrix Algebra

Let R_i denote the return on asset $i = A, B, C$ and assume that R_A, R_B and R_C are jointly normally distributed with means, variances and covariances:

$$\mu_i = E[R_i],\ \sigma_i^2 = \text{var}(R_i),\ \text{cov}(R_i, R_j) = \sigma_{ij}.$$

Let x_i denote the share of wealth invested in asset i $(i = A, B, C)$, and assume that all wealth is invested in the three assets so that $x_A + x_B + x_C = 1$. The portfolio return, $R_{p,x}$, is the random variable:

$$R_{p,x} = x_A R_A + x_B R_B + x_C R_C. \tag{3.16}$$

The subscript "x" indicates that the portfolio is constructed using the x-weights x_A, x_B and x_C. The expected return on the portfolio is:

$$\mu_{p,x} = E[R_{p,x}] = x_A \mu_A + x_B \mu_B + x_C \mu_C, \tag{3.17}$$

and the variance of the portfolio return is:

$$\begin{aligned}\sigma^2_{p,x} = \text{var}(R_{p,x}) = x^2_A \sigma^2_A + x^2_B \sigma^2_B + x^2_C \sigma^2_C + 2x_A x_B \sigma_{AB} \\ + 2x_A x_C \sigma_{AC} + 2x_B x_C \sigma_{BC}.\end{aligned}$$

Notice that variance of the portfolio return depends on three variance terms and six covariance terms. Hence, with three assets there are twice as many covariance terms than variance terms contributing to portfolio variance. Even with three assets, the algebra representing the portfolio characteristics Equation 3.16 - Equation 3.18 is cumbersome. We can greatly simplify the portfolio algebra using matrix notation.

Define the following 3×1 column vectors containing the asset returns and portfolio weights:

$$\mathbf{R} = \begin{pmatrix} R_A \\ R_B \\ R_C \end{pmatrix}, \ \mathbf{x} = \begin{pmatrix} x_A \\ x_B \\ x_C \end{pmatrix}.$$

The probability distribution of the random return vector $\mathbf{R}$ is simply the joint distribution of the elements of $\mathbf{R}$. Here all returns are jointly normally distributed and this joint distribution is completely characterized by the means, variances and covariances of the returns. We can easily express these values using matrix notation as follows. The 3×1 vector of portfolio expected values is:

$$E[\mathbf{R}] = E\left[\begin{pmatrix} R_A \\ R_B \\ R_C \end{pmatrix}\right] = \begin{pmatrix} E[R_A] \\ E[R_B] \\ E[R_C] \end{pmatrix} = \begin{pmatrix} \mu_A \\ \mu_B \\ \mu_C \end{pmatrix} = \mu,$$

and the 3×3 covariance matrix of returns is,

$$\begin{aligned}\text{var}(\mathbf{R}) &= \begin{pmatrix} \text{var}(R_A) & \text{cov}(R_A, R_B) & \text{cov}(R_A, R_C) \\ \text{cov}(R_B, R_A) & \text{var}(R_B) & \text{cov}(R_B, R_C) \\ \text{cov}(R_C, R_A) & \text{cov}(R_C, R_B) & \text{var}(R_C) \end{pmatrix} \\ &= \begin{pmatrix} \sigma_A^2 & \sigma_{AB} & \sigma_{AC} \\ \sigma_{AB} & \sigma_B^2 & \sigma_{BC} \\ \sigma_{AC} & \sigma_{BC} & \sigma_C^2 \end{pmatrix} = \Sigma.\end{aligned}$$

Notice that the covariance matrix is symmetric (elements off the diagonal are equal so that $\Sigma = \Sigma'$, where Σ' denotes the transpose of Σ) since $\text{cov}(R_A, R_B) = \text{cov}(R_B, R_A)$, $\text{cov}(R_A, R_C) = \text{cov}(R_C, R_A)$ and $\text{cov}(R_B, R_C) = \text{cov}(R_C, R_B)$.

The return on the portfolio using vector notation is:

$$R_{p,x} = \mathbf{x}'\mathbf{R} = (x_A, x_B, x_C) \cdot \begin{pmatrix} R_A \\ R_B \\ R_C \end{pmatrix} = x_A R_A + x_B R_B + x_C R_C.$$

Similarly, the expected return on the portfolio is:

$$\begin{aligned}\mu_{p,x} &= E[\mathbf{x}'\mathbf{R}] = \mathbf{x}' E[\mathbf{R}] = \mathbf{x}'\mu \\ &= (x_A, x_B, x_C) \cdot \begin{pmatrix} \mu_A \\ \mu_B \\ \mu_C \end{pmatrix} = x_A \mu_A + x_B \mu_B + x_C \mu_C.\end{aligned}$$

The variance of the portfolio is:

$$\begin{aligned}\sigma_{p,x}^2 &= \text{var}(\mathbf{x}'\mathbf{R}) = \mathbf{x}'\Sigma\mathbf{x} = (x_A, x_B, x_C) \cdot \begin{pmatrix} \sigma_A^2 & \sigma_{AB} & \sigma_{AC} \\ \sigma_{AB} & \sigma_B^2 & \sigma_{BC} \\ \sigma_{AC} & \sigma_{BC} & \sigma_C^2 \end{pmatrix} \begin{pmatrix} x_A \\ x_B \\ x_C \end{pmatrix} \\ &= x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + x_C^2 \sigma_C^2 + 2x_A x_B \sigma_{AB} + 2x_A x_C \sigma_{AC} + 2x_B x_C \sigma_{BC}.\end{aligned}$$

Finally, the condition that the portfolio weights sum to one can be expressed as:

$$\mathbf{x}'\mathbf{1} = (x_A, x_B, x_B) \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = x_A + x_B + x_C = 1,$$

where $\mathbf{1}$ is a 3×1 vector with each element equal to 1.

Consider another portfolio with weights $\mathbf{y} = (y_A, y_B, y_C)'$. The return on this portfolio is:

$$R_{p,y} = \mathbf{y}'\mathbf{R} = y_A R_A + y_B R_B + y_C R_C.$$

We often need to compute the covariance between the return on portfolio $\mathbf{x}$ and the return on portfolio $\mathbf{y}$, $\text{cov}(R_{p,x}, R_{p,y})$. This can be easily expressed using matrix algebra:

$$\begin{aligned}\sigma_{xy} &= \text{cov}(R_{p,x}, R_{p,y}) = \text{cov}(\mathbf{x}'\mathbf{R}, \mathbf{y}'\mathbf{R}) \\ &= E[(\mathbf{x}'\mathbf{R} - \mathbf{x}'\mu])(\mathbf{y}'\mathbf{R} - \mathbf{y}'\mu)] = E[\mathbf{x}'(\mathbf{R} - \mu)(\mathbf{R} - \mu)'\mathbf{y}] \\ &= \mathbf{x}'E[(\mathbf{R} - \mu)(\mathbf{R} - \mu)']\mathbf{y} = \mathbf{x}'\Sigma\mathbf{y}.\end{aligned}$$

Notice that,

$$\begin{aligned}\sigma_{xy} = \mathbf{x}'\Sigma\mathbf{y} &= (x_A, x_B, x_C) \cdot \begin{pmatrix} \sigma_A^2 & \sigma_{AB} & \sigma_{AC} \\ \sigma_{AB} & \sigma_B^2 & \sigma_{BC} \\ \sigma_{AC} & \sigma_{BC} & \sigma_C^2 \end{pmatrix} \begin{pmatrix} y_A \\ y_B \\ y_C \end{pmatrix} \\ &= x_A y_A \sigma_A^2 + x_B y_B \sigma_B^2 + x_C y_C \sigma_C^2 \\ &+ (x_A y_B + x_B y_A)\sigma_{AB} + (x_A y_C + x_C y_A)\sigma_{AC} + (x_B y_C + x_C y_B)\sigma_{BC},\end{aligned}$$

which is quite a messy expression!

3.8 Derivatives of Simple Matrix Functions

Result: Let $\mathbf{A}$ be an $n \times n$ symmetric matrix, and let $\mathbf{x}$ and $\mathbf{y}$ be $n \times 1$ vectors. Then,

$$\underset{n\times 1}{\frac{\partial}{\partial \mathbf{x}}} \mathbf{x}'\mathbf{y} = \begin{pmatrix} \frac{\partial}{\partial x_1}\mathbf{x}'\mathbf{y} \\ \vdots \\ \frac{\partial}{\partial x_n}\mathbf{x}'\mathbf{y} \end{pmatrix} = \mathbf{y}, \tag{3.18}$$

$$\underset{n\times 1}{\frac{\partial}{\partial \mathbf{x}}} \mathbf{A}\mathbf{x} = \begin{pmatrix} \frac{\partial}{\partial x_1}(\mathbf{A}\mathbf{x})' \\ \vdots \\ \frac{\partial}{\partial x_n}(\mathbf{A}\mathbf{x})' \end{pmatrix} = \mathbf{A}, \tag{3.19}$$

$$\underset{n\times 1}{\frac{\partial}{\partial \mathbf{x}}} \mathbf{x}'\mathbf{A}\mathbf{x} = \begin{pmatrix} \frac{\partial}{\partial x_1}\mathbf{x}'\mathbf{A}\mathbf{x} \\ \vdots \\ \frac{\partial}{\partial x_n}\mathbf{x}'\mathbf{A}\mathbf{x} \end{pmatrix} = 2\mathbf{A}\mathbf{x}. \tag{3.20}$$

We will demonstrate these results with simple examples. Let,

$$\mathbf{A} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \ \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

First, consider Equation 3.18. Now,

$$\mathbf{x}'\mathbf{y} = x_1y_1 + x_2y_2.$$

Then,

$$\frac{\partial}{\partial \mathbf{x}}\mathbf{x}'\mathbf{y} = \begin{pmatrix} \frac{\partial}{\partial x_1}\mathbf{x}'\mathbf{y} \\ \frac{\partial}{\partial x_2}\mathbf{x}'\mathbf{y} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1}(x_1y_1 + x_2y_2) \\ \frac{\partial}{\partial x_2}(x_1y_1 + x_2y_2) \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \mathbf{y}.$$

Next, consider Equation 3.19. Note that,

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} ax_1 + bx_2 \\ bx_1 + cx_2 \end{pmatrix},$$

and,

$$(\mathbf{A}\mathbf{x})' = (ax_1 + bx_2, bx_1 + cx_2).$$

Then,

$$\frac{\partial}{\partial \mathbf{x}}\mathbf{A}\mathbf{x} = \begin{pmatrix} \frac{\partial}{\partial x_1}(ax_1 + bx_2, bx_1 + cx_2) \\ \frac{\partial}{\partial x_2}(ax_1 + bx_2, bx_1 + cx_2) \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \mathbf{A}.$$

Finally, consider Equation 3.20. We have,

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \begin{pmatrix} x_1 & x_2 \end{pmatrix}\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = ax_1^2 + 2bx_1x_2 + cx_2^2.$$

Then,

$$\begin{aligned}\frac{\partial}{\partial \mathbf{x}}\mathbf{x}'\mathbf{A}\mathbf{x} &= \begin{pmatrix} \frac{\partial}{\partial x_1}(ax_1^2 + 2bx_1x_2 + cx_2^2) \\ \frac{\partial}{\partial x_2}(ax_1^2 + 2bx_1x_2 + cx_2^2) \end{pmatrix} = \begin{pmatrix} 2ax_1 + 2bx_2 \\ 2bx_1 + 2cx_2 \end{pmatrix} \\ &= 2\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2\mathbf{A}\mathbf{x}.\end{aligned}$$

Example 3.18 (Calculating an asset's marginal contribution to portfolio volatility)**.** In portfolio risk budgeting (see chapter Chapter 13), asset i's marginal contribution to portfolio volatility $\sigma_p = (\mathbf{x}'\Sigma\mathbf{x})^{1/2}$ is given by

$$\text{MCR}_\text{i} = \frac{\partial \sigma_p}{\partial x_i} = \frac{\partial (\mathbf{x}'\Sigma\mathbf{x})^{1/2}}{\partial x_i},$$

and approximates how much portfolio volatility changes when the allocation to asset i increases by a small amount. Using the chain rule and matrix derivatives we can compute the entire vector of asset marginal contributions at once:

$$\begin{aligned}\frac{\partial (\mathbf{x}'\Sigma\mathbf{x})^{1/2}}{\partial \mathbf{x}} &= \frac{1}{2}(\mathbf{x}'\Sigma\mathbf{x})^{-1/2}\frac{\partial \mathbf{x}'\Sigma\mathbf{x}}{\partial \mathbf{x}} = \frac{1}{2}(\mathbf{x}'\Sigma\mathbf{x})^{-1/2}2\Sigma\mathbf{x} \\ &= (\mathbf{x}'\Sigma\mathbf{x})^{-1/2}\Sigma\mathbf{x} = \frac{\Sigma\mathbf{x}}{\sigma_p}.\end{aligned}$$

Then asset i's marginal contribution is given by the i-th row of $\frac{\Sigma \mathbf{x}}{\sigma_p}$.

■

Example 3.19 (Finding the global minimum variance portfolio)**.** Let $\mathbf{R}$ denote an $n \times 1$ random vector of asset returns with $E[\mathbf{R}] = \mu$ and $\text{var}(\mathbf{R}) = \Sigma$. The global minimum variance portfolio (see Chapter Chapter 10, Section 10.3) $\mathbf{m}$ solves the constrained minimization problem:

$$\min_{\mathbf{m}} \ \sigma^2_{p,m} = \mathbf{m}'\Sigma\mathbf{m} \text{ s.t. } \mathbf{m}'\mathbf{1} = 1. \tag{3.21}$$

The Lagrangian function is:

$$L(\mathbf{m}, \lambda) = \mathbf{m}'\Sigma\mathbf{m} + \lambda(\mathbf{m}'\mathbf{1} - 1).$$

The first order conditions can be expressed in matrix notation as,

$$\underset{(n\times 1)}{\mathbf{0}} = \frac{\partial L(\mathbf{m}, \lambda)}{\partial \mathbf{m}} = \frac{\partial}{\partial \mathbf{m}}\mathbf{m}'\Sigma\mathbf{m} + \frac{\partial}{\partial \mathbf{m}}\lambda(\mathbf{m}'\mathbf{1} - 1) = 2 \cdot \Sigma\mathbf{m} + \lambda \cdot \mathbf{1} \tag{3.22}$$

$$\underset{(1\times 1)}{0} = \frac{\partial L(\mathbf{m}, \lambda)}{\partial \lambda} = \frac{\partial}{\partial \lambda}\mathbf{m}'\Sigma\mathbf{m} + \frac{\partial}{\partial \lambda}\lambda(\mathbf{m}'\mathbf{1} - 1) = \mathbf{m}'\mathbf{1} - 1 \tag{3.23}$$

These first order conditions represent a system of $n+1$ linear equations in $n+1$ unknowns ($\mathbf{m}$ and λ). These equations can be represented in matrix form as the system

$$\begin{bmatrix} 2\Sigma & \mathbf{1} \\ \mathbf{1}' & 0 \end{bmatrix} \begin{bmatrix} \mathbf{m} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix},$$

which is of the form $\mathbf{Az} = \mathbf{b}$ for

$$\mathbf{A} = \begin{bmatrix} 2\Sigma & \mathbf{1} \\ \mathbf{1}' & 0 \end{bmatrix}, \mathbf{z} = \begin{bmatrix} \mathbf{m} \\ \lambda \end{bmatrix}, \mathbf{b} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}.$$

The portfolio weight vector $\mathbf{m}$ can be found as the first n elements of $\mathbf{z} = \mathbf{A}^{-1}\mathbf{b}$.

■

3.9 Further Reading: Matrix Algebra Review

A classic textbook on linear algebra is (Strang 1980). A review of basic matrix algebra with applications in economics and finance is given in (Chiang 1984). Matrix algebra results that are useful for statistical analysis are presented in (Searle 1982), Gentile (2010) and (Banerjee and Roy 2014). (Lutkepohl 1997) and (Abadir and Magnus 2005) present advanced matrix algebra useful for econometrics and statistics. (Fieller 2015) overviews the basics of matrix algebra for statistics using R.

3.10 Problems: Matrix Algebra Review

Exercise 3.1. Use R to answer the following questions. 1. Create the following matrices and vectors

$$\mathbf{A} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 4 & 8 \\ 6 & 1 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 4 & 4 & 0 \\ 5 & 9 & 1 \\ 2 & 2 & 5 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 5 \\ 2 \\ 7 \end{bmatrix}$$

2. Compute the transposes of the above matrices and vectors. 3. Compute $\mathbf{A} + \mathbf{B}$, $\mathbf{A} - \mathbf{B}$, $2\mathbf{A}$, $\mathbf{Ax}$, $\mathbf{AB}$, $\mathbf{y}'\mathbf{Ax}$

Exercise 3.2. Consider the system of linear equations

$$\begin{aligned} x + y &= 1 \\ 2x + 4y &= 2 \end{aligned}$$

1. Plot the two lines and note the solution to the system of equations. Hint: you can use the R functions `abline()` or `curve()` to plot the lines.
2. Write the system in matrix notation as $\mathbf{Az} = \mathbf{b}$ and solve for $\mathbf{z}$ by computing $\mathbf{z} = \mathbf{A}^{-1}\mathbf{b}$.

Exercise 3.3. Consider creating an equally weighted portfolio of three assets denoted A, B, and C. Assume the following information

$$\mu = \begin{bmatrix} 0.01 \\ 0.04 \\ 0.02 \end{bmatrix}, \Sigma = \begin{bmatrix} 0.10 & 0.30 & 0.10 \\ 0.30 & 0.15 & -0.20 \\ 0.10 & -0.20 & 0.08 \end{bmatrix}.$$

Use R to answer the following questions. 1. Create the vector of portfolio weights. 2. Compute the expected return on the portfolio. 3. Compute the variance and standard deviation of the portfolio.

Exercise 3.4. Let R_i be a random variable denoting the simple return on asset $i\,(i = 1, \dots, N)$ with $E[R_i] = \mu_i$, $\text{var}(R_i) = \sigma_i^2$ and $\text{cov}(R_i, R_j) = \sigma_{ij}$. Define the $N \times 1$ vectors $\mathbf{R} = (R_1, \dots, R_N)'$, $\mu = (\mu_1, \dots, \mu_N)'$, $\mathbf{x} = (x_1, \dots, x_N)'$, $\mathbf{y} = (y_1, \dots, y_N)'$, and $\mathbf{1} = (1, \dots, 1)'$, and the $N \times N$ covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_N^2 \end{pmatrix}.$$

The vectors $\mathbf{x}$ and $\mathbf{y}$ contain portfolio weights (investment shares) that sum to one. Using simple matrix algebra, answer the following questions.

1. For the portfolios defined by the vectors $\mathbf{x}$ and $\mathbf{y}$ give the matrix algebra expressions for the portfolio returns $R_{p,x}$ and $R_{p,y}$ and the portfolio expected returns $\mu_{p,x}$ and $\mu_{p,y}$.

2. For the portfolios defined by the vectors $\mathbf{x}$ and $\mathbf{y}$ give the matrix algebra expressions for the constraint that the portfolio weights sum to one.
3. For the portfolios defined by the vectors $\mathbf{x}$ and $\mathbf{y}$ give the matrix algebra expressions for the portfolio variances $\sigma^2_{p,x}$ and $\sigma^2_{p,y}$, and the covariance between $R_{p,x}$ and $R_{p,y}$.
4. In the expression for the portfolio variance $\sigma^2_{p,x}$, how many variance terms are there? How many unique covariance terms are there?

4 Time Series Concepts

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Financial variables such as asset prices and returns are naturally ordered by time. That is, these variables are *time series* variables. When we construct returns, the time index or data frequency becomes the investment horizon associated with the return. Typical data frequencies are daily, weekly, monthly and annual. In building probability models for time series variables, the time ordering of the data matters because we may think there are important temporal dependencies among the variables. For example, we might have reason to believe that the return on an asset this month is correlated with the return on the same asset in the previous month. This autocorrelation can only be defined if the time ordering of the data is preserved. A major complication in analyzing time series data is that the usual assumption of random sampling from a common population is not appropriate because it does not allow for any kind of time dependence in the time series variables. We would like to retain the notion that the observed data come from some population model, perhaps with time-invariant parameters, but we would like to allow the variables to have time dependencies. Fortunately, we can do this if the time series data come from a *stationary* time series process.

This chapter reviews some basic times series concepts that are essential for describing and modeling financial time series. Section 4.1 defines univariate time series processes and introduces the important concepts of stationarity and ergodicity. Covariance stationary time series processes are defined, which gives meaning to measuring linear time dependence using autocorrelation. The benchmark Gaussian White Noise process and related processes are introduced and illustrated using R. Some common non-stationary time series processes are also discussed including the famous random walk model. Section 4.2 introduces covariance stationary multivariate time series process. Such processes allow for dynamic interactions among groups of time series variables. Section 4.3 discusses time series model building and introduces the class of univariate autoregressive-moving average time series models and multivariate vector autoregression models. The properties of some simple models are derived and it is shown how to simulate observations from these models using R. The chapter concludes with a brief discussion of forecasting from time series models.

The R packages used in this chapter are **mvtnorm** and **vars**. Make sure these packages are installed and loaded before running the R examples in this chapter.

4.1 Stochastic Processes

Definition 4.1 (Stochastic Process)**.** A discrete-time *stochastic process* or *time series process*

$$\{..., Y_1, Y_2, ..., Y_t, Y_{t+1}, ...\} = \{Y_t\}_{t=-\infty}^{\infty}$$

is a sequence of random variables indexed by time t[1].

In most applications, the time index is a regularly spaced index representing calendar time (e.g., days, months, years, etc.) but it can also be irregularly spaced representing event time (e.g., intra-day transaction times). In modeling time series data, the ordering imposed by the time index is important because we often would like to capture the temporal relationships, if any, between the random variables in the stochastic process. In random sampling from a population, the ordering of the random variables representing the sample does not matter because they are independent.

Definition 4.2 (Realization of Stochastic Process)**.** A realization of a stochastic process with T observations is the sequence of observed data

$$\{Y_1 = y_1, Y_2 = y_2, ..., Y_T = y_T\} = \{y_t\}_{t=1}^{T}.$$

The goal of time series modeling is to describe the probabilistic behavior of the underlying stochastic process that is believed to have generated the observed data in a concise way. In addition, we want to be able to use the observed sample to estimate important characteristics of a time series model such as measures of time dependence. In order to do this, we need to make a number of assumptions regarding the joint behavior of the random variables in the stochastic process such that we may treat the stochastic process in much the same way as we treat a random sample from a given population.

4.1.1 Stationary stochastic processes

We often describe random sampling from a population as a sequence of independent, and identically distributed (iid) random variables X_1, X_2 ... such that each X_i is described by the same probability distribution F_X, and write $X_i \sim F_X$. With a time series process, we would like to preserve the identical distribution assumption but we do not want to impose the restriction that each random variable in the sequence is independent of all of the other variables. In many contexts, we would expect some dependence between random variables close together in time (e.g, X_1, and X_2) but little or no dependence between random variables far apart in time (e.g., X_1 and X_{100}). We can allow for this type of behavior using the concepts of *stationarity* and *ergodicity*.

We start with the definition of strict stationarity.

[1]To conserve on notation, we will represent the stochastic process$\{Y_t\}_{t=-\infty}^{\infty}$ simply as $\{Y_t\}$.

Definition 4.3 (Strict stationarity)**.** A stochastic process $\{Y_t\}$ is *strictly stationary* if, for any given finite integer r and for any set of subscripts $t_1, t_2, \dots, t_r$ the joint distribution of $(Y_{t_1}, Y_{t_2}, \dots, Y_{t_r})$ depends only on $t_1 - t, t_2 - t, \dots, t_r - t$ but not on t. In other words, the joint distribution of $(Y_{t_1}, Y_{t_2}, \dots, Y_{t_r})$ is the same as the joint distribution of $(Y_{t_1-t}, Y_{t_2-t}, \dots, Y_{t_r-t})$ for any value of t.

In simple terms, the joint distribution of random variables in a strictly stationary stochastic process is *time invariant*. For example, the joint distribution of (Y_1, Y_5, Y_7) is the same as the distribution of (Y_{12}, Y_{16}, Y_{18}). Just like in an iid sample, in a strictly stationary process all of the individual random variables Y_t $(t = -\infty, \dots, \infty)$ have the same marginal distribution F_Y. This means they all have the same mean, variance etc., assuming these quantities exist. However, assuming strict stationarity does not make any assumption about the correlations between $Y_t, Y_{t_1}, \dots, Y_{t_r}$ other than that the correlation between Y_t and Y_{t_r} only depends on $t - t_r$ (the time between Y_t and Y_{t_r}) and not on t. That is, strict stationarity allows for general temporal dependence between the random variables in the stochastic process.

A useful property of strict stationarity is that it is preserved under general transformations, as summarized in the following proposition.

Proposition 4.1. *Let $\{Y_t\}$ be strictly stationary and let $g(\cdot)$ be any function of the elements in $\{Y_t\}$. Then $\{g(Y_t)\}$, is also strictly stationary.*

For example, if $\{Y_t\}$ is strictly stationary then $\{Y_t^2\}$ and $\{Y_t Y_{t-1}\}$ are also strictly stationary.

The following are some simple examples of strictly stationary processes.

Example 4.1 (iid sequence)**.** If $\{Y_t\}$ is an iid sequence, then it is strictly stationary. ■

Example 4.2 (Non iid sequence)**.** Let $\{Y_t\}$ be an iid sequence and let $X \sim N(0, 1)$ independent of $\{Y_t\}$. Define $Z_t = Y_t + X$. The sequence $\{Z_t\}$ is not an independent sequence (because of the common X) but is an identically distributed sequence and is strictly stationary. ■

Strict stationarity places very strong restrictions on the behavior of a time series. A related concept that imposes weaker restrictions and is convenient for time series model building is *covariance stationarity* (sometimes called *weak stationarity*).

Definition 4.4 (Covariance stationarity)**.** A stochastic process $\{Y_t\}$ is *covariance stationary* if

1. $E[Y_t] = \mu < \infty$ does not depend on t
2. $\text{var}(Y_t) = \sigma^2 < \infty$ does not depend on t
3. $\text{cov}(Y_t, Y_{t-j}) = \gamma_j < \infty$, and depends only on j but not on t for $j = 0, 1, 2, \dots$

The term γ_j is called the jth order *autocovariance.* The jth order *autocorrelation* is defined as:

$$\rho_j = \frac{\text{cov}(Y_t, Y_{t-j})}{\sqrt{\text{var}(Y_t)\text{var}(Y_{t-j})}} = \frac{\gamma_j}{\sigma^2}. \tag{4.1}$$

The autocovariances, γ_j, measure the direction of linear dependence between Y_t and Y_{t-j}. The autocorrelations, ρ_j, measure both the direction and strength of linear dependence between Y_t and Y_{t-j}. With covariance stationarity, instead of assuming the entire joint distribution of a collection of random variables is time invariant we make a weaker assumption that only the mean, variance and autocovariances of the random variables are time invariant. A strictly stationary stochastic process $\{Y_t\}$ such that μ, σ^2, and γ_{ij} exist is a *covariance stationary* stochastic process. However, a covariance stationary process need not be strictly stationary.

The autocovariances and autocorrelations are measures of the linear temporal dependence in a covariance stationary stochastic process. A graphical summary of this temporal dependence is given by the plot of ρ_j against j, and is called the *autocorrelation function* (ACF). Figure 4.1 illustrates an ACF for a hypothetical covariance stationary time series with $\rho_j = (0.9)^j$ for $j = 1, 2, \ldots, 10$ created with

```
rho = 0.9
rhoVec = (rho)^(1:10)
ts.plot(rhoVec, type="h", lwd=2, col="blue", xlab="Lag j",
        ylab=expression(rho[j]))
```

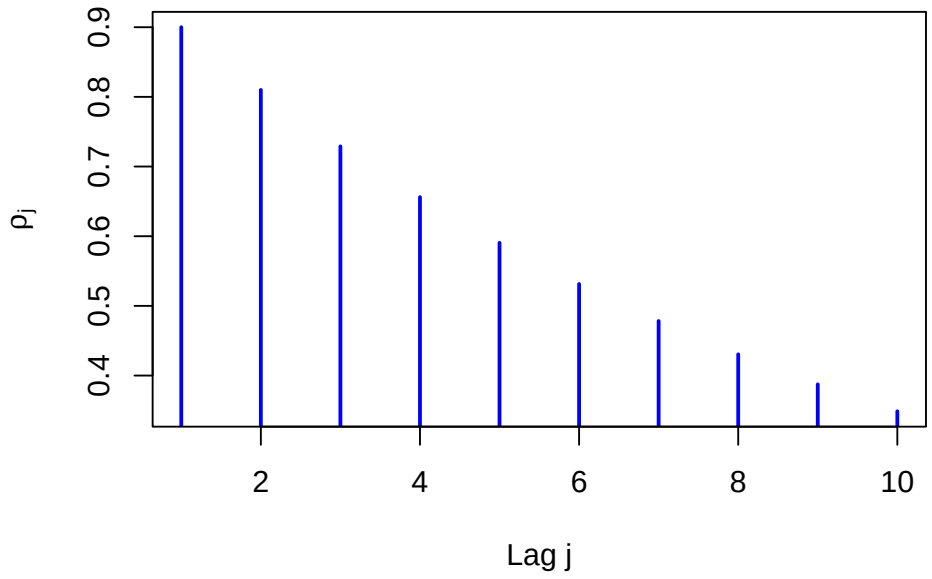


Figure 4.1: ACF for time series with $\rho_j = (0.9)^j$

For this process the strength of linear time dependence decays toward zero geometrically fast as j increases.

The definition of covariance stationarity requires that $E[Y_t] < \infty$ and $\text{var}(Y_t) < \infty$. That is, $E[Y_t]$ and $\text{var}(Y_t)$ must exist and be finite numbers. This is true if Y_t is normally distributed. However, it is not true if, for example, Y_t has a Student's t distribution with one degree of freedom.[2] Hence, a strictly stationary stochastic process $\{Y_t\}$ where the (marginal) pdf of Y_t (for all t) has very fat tails may not be covariance stationary.

Example 4.3 (Gaussian White Noise)**.** Let $Y_t \sim iid\ N(0, \sigma^2)$. Then $\{Y_t\}$ is called a *Gaussian white noise* (GWN) process and is denoted $Y_t \sim \text{GWN}(0, \sigma^2)$. Notice that:

$$\begin{aligned} E[Y_t] &= 0 \text{ independent of } t, \\ \text{var}(Y_t) &= \sigma^2 \text{ independent of } t, \\ \text{cov}(Y_t, Y_{t-j}) &= 0 \text{ (for } j > 0) \text{ independent of } t \text{ for all } j, \end{aligned}$$

so that $\{Y_t\}$ satisfies the properties of a covariance stationary process. The defining characteristic of a GWN process is the lack of any predictable pattern over time in the realized values of the

[2]This is also called a Cauchy distribution. For this distribution $E[Y_t] = \text{var}(Y_t) = \text{cov}(Y_t, Y_{t-j}) = \infty$.

process. The term white noise comes from the electrical engineering literature and represents the absence of any signal.[3]

Simulating observations from a GWN process in R is easy: just simulate iid observations from a normal distribution. For example, to simulate $T = 250$ observations from the GWN(0,1) process use:

```
set.seed(123)
y = rnorm(250)
```

The simulated iid N(0,1) values are generated using the `rnorm()` function. The command `set.seed(123)` initializes R's internal random number generator using the seed value 123. Every time the random number generator seed is set to a particular value, the random number generator produces the same set of random numbers. This allows different people to create the same set of random numbers so that results are reproducible. The simulated data is illustrated in Figure 4.2 created using:

```
ts.plot(y,main="Gaussian White Noise Process",xlab="time",
        ylab="y(t)", col="blue", lwd=2)
abline(h=0)
```

[3] As an example of white noise, think of tuning an AM radio. In between stations there is no signal and all you hear is static. This is white noise.

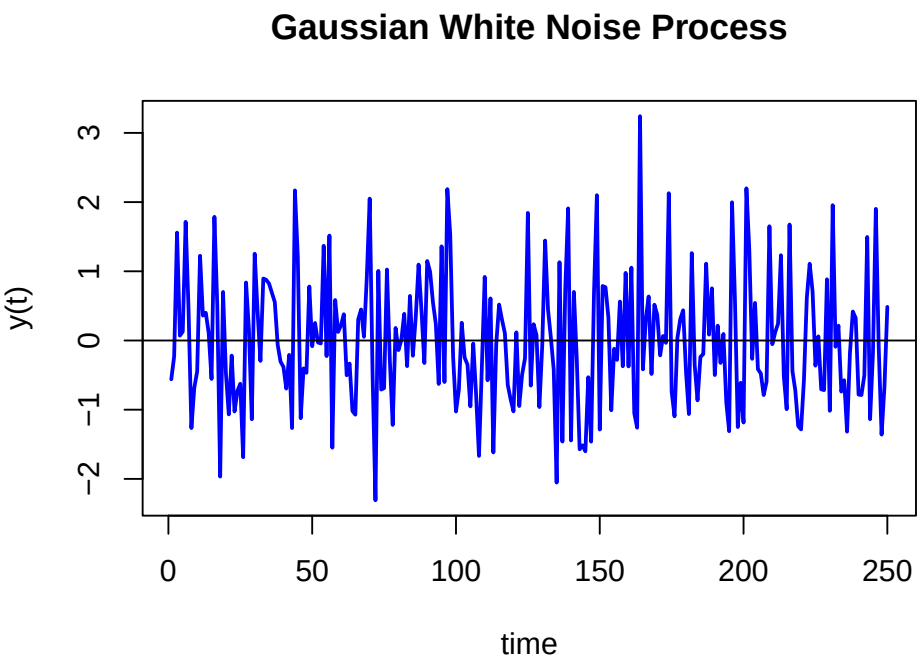


Figure 4.2: Realization of a GWN(0,1) process.

The function `ts.plot()` creates a time series line plot with a dummy time index. An equivalent plot can be created using the generic `plot()` function with optional argument `type="l"`. The data in Figure 4.2 fluctuate randomly about the mean value zero and exhibit a constant volatility of one (typical magnitude of a fluctuation about zero). There is no visual evidence of any predictable pattern in the data. ■

Example 4.4 (GWN model for continuously compounded returns)**.** Let r_t denote the continuously compounded monthly return on Microsoft stock and assume that $r_t \sim \text{iid}\, N(0.01,\, (0.05)^2)$. We can represent this distribution in terms of a GWN process as follows

$$r_t = 0.01 + \varepsilon_t,\ \varepsilon_t \sim \text{GWN}(0,\, (0.05)^2).$$

Here, r_t is constructed as a GWN$(0, \sigma^2)$ process plus a constant. Hence, $\{r_t\}$ is a GWN process with a non-zero mean: $r_t \sim \text{GWN}(0.01,\, (0.05)^2)$. $T = 60$ simulated values of $\{r_t\}$ are computed using:

```
set.seed(123)
y = rnorm(60, mean=0.01, sd=0.05)
ts.plot(y,main="GWN Process for Monthly Continuously Compounded Returns",
        xlab="time",ylab="r(t)", col="blue", lwd=2)
```

```
abline(h=c(0,0.01,-0.04,0.06), lwd=2,
       lty=c("solid","solid","dotted","dotted"),
       col=c("black", "red", "red", "red"))
```

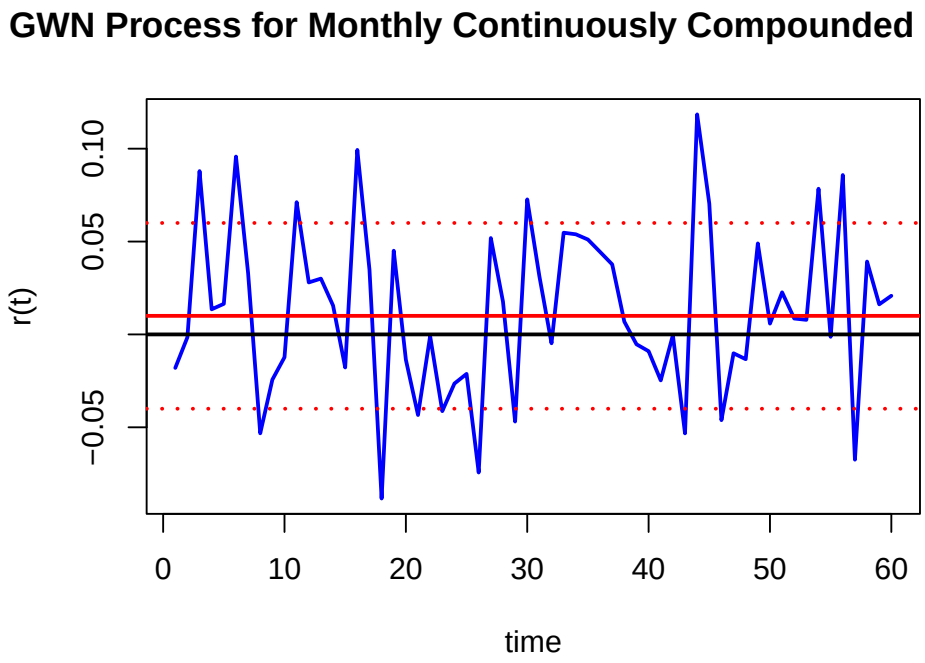


Figure 4.3: Simulated returns from GWN(0.01,$(0.05)^2$).

and are illustrated in Figure 4.3. Notice that the returns fluctuate around the mean value of 0.01 (solid red line) and the size of a typical deviation from the mean is about 0.05. The red dotted lines show the values 0.1 ± 0.05 .

An implication of the GWN assumption for monthly returns is that non-overlapping multi-period returns are also GWN. For example, consider the two-month return $r_t(2) = r_t + r_{t-1}$. The non-overlapping process $\{r_t(2)\} = \{..., r_{t-2}(2), r_t(2), r_{t+2}(2), ...\}$ is GWN with mean $E[r_t(2)] = 2 \cdot \mu = 0.02$, variance $\text{var}(r_t(2)) = 2 \cdot \sigma^2 = 0.005$, and standard deviation $\text{sd}(r_t(2)) = \sqrt{2}\sigma = 0.071$. ■

Example 4.5 (Independent white noise)**.** Let $Y_t \sim$ iid $(0, \sigma^2)$. Then $\{Y_t\}$ is called an *independent white noise* (IWN) process and is denoted $Y_t \sim \text{IWN}(0, \sigma^2)$. The difference between GWN and IWN is that with IWN we don't specify that all random variables are normally distributed. The random variables can have any distribution with mean zero and variance σ^2. To illustrate, suppose $Y_t = \frac{1}{\sqrt{3}} \times t_3$ where t_3 denotes a Student's t distribution with 3 degrees of freedom. This process has $E[Y_t] = 0$ and $\text{var}(Y_t) = 1$.[4] Figure 4.4 shows simulated observations from this process created

[4]Recall, if $Y_t \sim t_v$ then $\text{var}(Y_t) = \frac{v}{v-2}$ for $v > 2$.

using the R commands

```
set.seed(123)
y = (1/sqrt(3))*rt(250, df=3)
ts.plot(y, main="Independent White Noise Process", xlab="time", ylab="y(t)",
        col="blue", lwd=2)
abline(h=0)
```

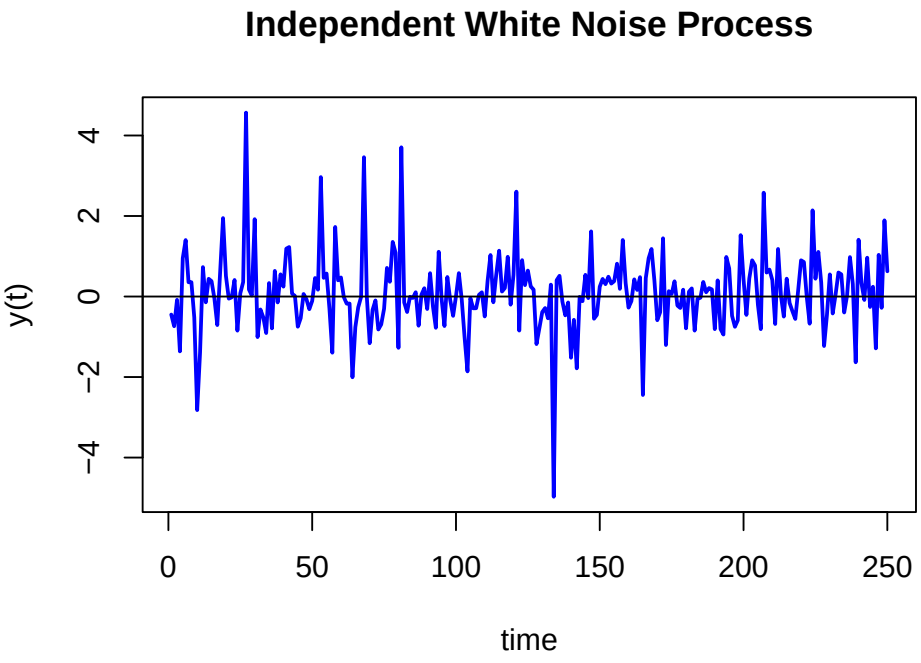


Figure 4.4: Simulation of IWN(0,1) process: $Y_t \sim \frac{1}{\sqrt{3}} \times t_3$

The simulated IWN process resembles the GWN in Figure 4.4 but has more extreme observations. ■

Example 4.6 (Weak white noise)**.** Let $\{Y_t\}$ be a sequence of uncorrelated random variables each with mean zero and variance σ^2. Then $\{Y_t\}$ is called a *weak white noise* (WWN) process and is denoted $Y_t \sim \text{WWN}(0, \sigma^2)$. With a WWN process, the random variables are not independent, only uncorrelated. This allows for potential non-linear temporal dependence between the random variables in the process. ■

4.1.2 Non-Stationary processes

In a covariance stationary stochastic process it is assumed that the means, variances and autocovariances are independent of time. In a *non-stationary* process, one or more of these assumptions is not true. The following examples illustrate some typical non-stationary time series processes.

Example 4.7 (Deterministically trending process)**.** Suppose $\{Y_t\}$ is generated according to the *deterministically trending process*:

$$Y_t = \beta_0 + \beta_1 t + \varepsilon_t,\ \varepsilon_t \sim \text{GWN}(0, \sigma_\varepsilon^2),$$
$$t = 0, 1, 2, ...$$

Then $\{Y_t\}$ is nonstationary because the mean of Y_t depends on t:

$$E[Y_t] = \beta_0 + \beta_1 t.$$

Figure 4.5 shows a realization of this process with $\beta_0 = 0, \beta_1 = 0.1$ and $\sigma_\varepsilon^2 = 1$ created using the R commands:

```
set.seed(123)
e = rnorm(250)
y.dt = 0.1*seq(1,250) + e
ts.plot(y.dt, lwd=2, col="blue", main="Deterministic Trend + Noise")
abline(a=0, b=0.1)
```

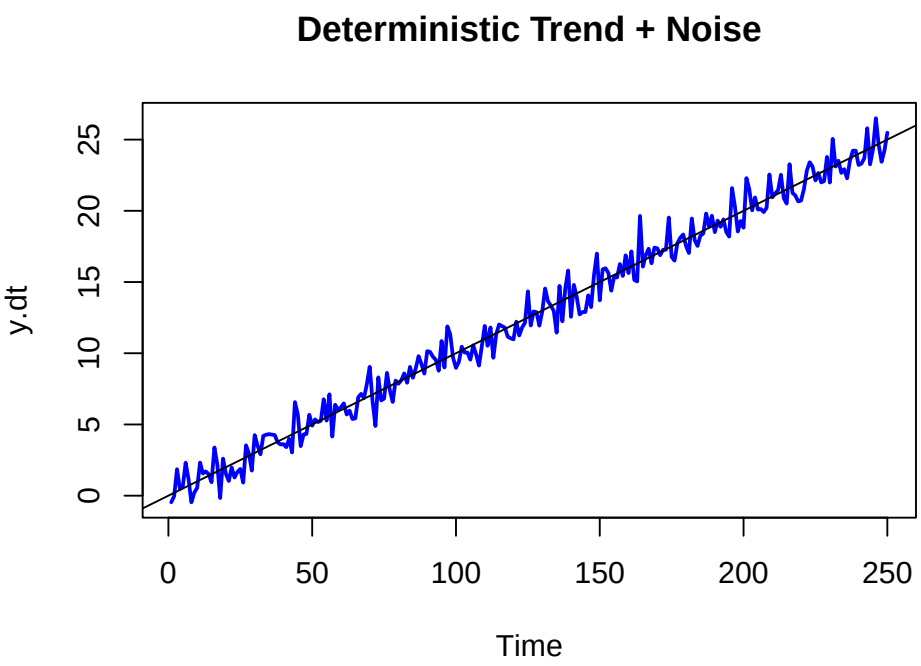


Figure 4.5: Deterministically trending nonstationary process $Y_t = 0.1 \times t + \varepsilon_t, \varepsilon_t \sim \text{GWN}(0, 1)$

Here the non-stationarity is created by the deterministic trend $\beta_0 + \beta_1 t$ in the data. The non-stationary process $\{Y_t\}$ can be transformed into a stationary process by simply subtracting off the trend:

$$X_t = Y_t - \beta_0 - \beta_1 t = \varepsilon_t \sim \text{GWN}(0, \sigma_\varepsilon^2).$$

The detrended process $X_t \sim \text{GWN}(0, \sigma_\varepsilon^2)$ is obviously covariance stationary. ■

Example 4.8 (Random walk)**.** A random walk (RW) process $\{Y_t\}$ is defined as:

$$\begin{aligned} Y_t &= Y_{t-1} + \varepsilon_t, \ \varepsilon_t \sim \text{GWN}(0, \sigma_\varepsilon^2), \\ Y_0 &\text{ is fixed (non-random).} \end{aligned}$$

By recursive substitution starting at $t = 1$, we have:

$$\begin{aligned} Y_1 &= Y_0 + \varepsilon_1, \\ Y_2 &= Y_1 + \varepsilon_2 = Y_0 + \varepsilon_1 + \varepsilon_2, \\ &\vdots \\ Y_t &= Y_{t-1} + \varepsilon_t = Y_0 + \varepsilon_1 + \cdots + \varepsilon_t \\ &= Y_0 + \sum_{j=1}^{t} \varepsilon_j. \end{aligned}$$

Now, $E[Y_t] = Y_0$ which is independent of t. However,

$$\text{var}(Y_t) = \text{var}\left(\sum_{j=1}^{t} \varepsilon_j\right) = \sum_{j=1}^{t} \sigma_\varepsilon^2 = \sigma_\varepsilon^2 \times t,$$

which depends on t, and so $\{Y_t\}$ is not stationary.

Figure 4.6 shows a realization of the RW process with $Y_0 = 0$ and $\sigma_\varepsilon^2 = 1$ created using the R commands:

```
set.seed(321)
e = rnorm(250)
y.rw = cumsum(e)
ts.plot(y.rw, lwd=2, col="blue", main="Random Walk")
abline(h=0)
```

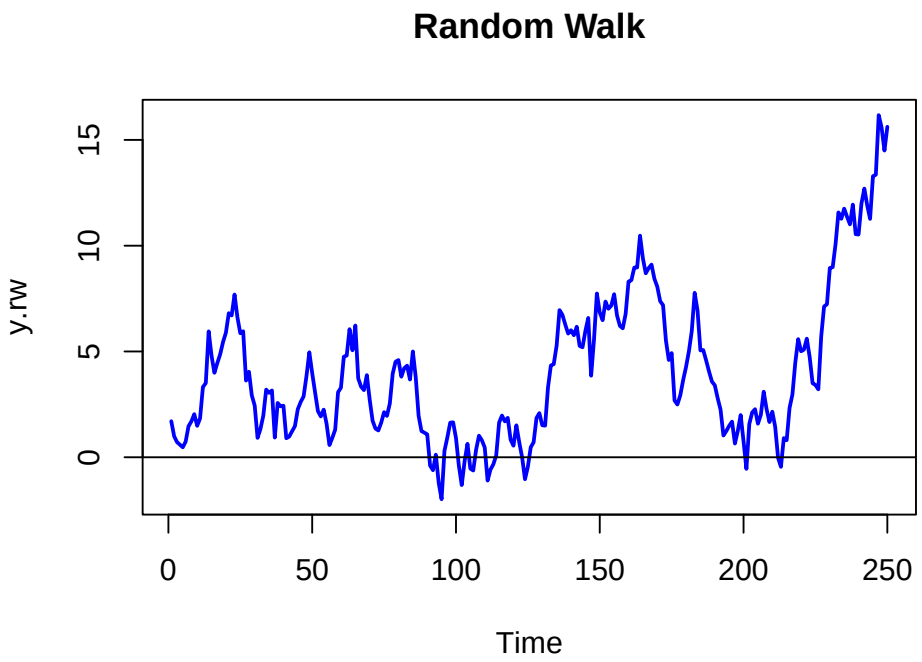


Figure 4.6: Random walk process: $Y_t = Y_{t-1} + \varepsilon_t, \varepsilon_t \sim \text{GWN}(0, 1)$.

The RW process looks much different from the GWN process in Figure 4.2. As the variance of the process increases linearly with time, the uncertainty about where the process will be at a given point in time increases with time.

Although $\{Y_t\}$ is non-stationary, a simple first-differencing transformation, however, yields a co-

variance stationary process:

$$X_t = Y_t - Y_{t-1} = \varepsilon_t \sim \text{GWN}(0, \sigma_\varepsilon^2).$$

■

Example 4.9 (Random walk with drift model for log stock prices)**.** Let r_t denote the continuously compounded monthly return on Microsoft stock and assume that $r_t \sim \text{GWN}(\mu, \sigma^2)$. Since $r_t = \ln(P_t/P_{t-1})$ it follows that $\ln P_t = \ln P_{t-1} + r_t$. Now, re-express r_t as $r_t = \mu + \varepsilon_t$ where $\varepsilon_t \sim \text{GWN}(0, \sigma^2)$. Then $lnP_t = lnP_{t-1} + \mu + \varepsilon_t$. By recursive substitution we have $lnP_t = lnP_0 + \mu t + \sum_{t=1}^{t} \varepsilon_t$ and so $\ln P_t$ follows a random walk process with drift value μ. Here, $E[lnP_t] = \mu t$ and $\text{var}(lnP_t) = \sigma^2 t$ so lnP_t is non-stationary because both the mean and variance depend on t. In this model, prices, however, do not follow a random walk since $P_t = e^{\ln P_t} = P_{t-1}e^{r_t}$. ■

4.1.3 Ergodicity

In a strictly stationary or covariance stationary stochastic process no assumption is made about the strength of dependence between random variables in the sequence. For example, in a covariance stationary stochastic process it is possible that $\rho_1 = \text{cor}(Y_t, Y_{t-1}) = \rho_{100} = \text{cor}(Y_t, Y_{t-100}) = 0.5$, say. However, in many contexts it is reasonable to assume that the strength of dependence between random variables in a stochastic process diminishes the farther apart they become. That is, $\rho_1 > \rho_2 \cdots$ and that eventually $\rho_j = 0$ for j large enough. This diminishing dependence assumption is captured by the concept of *ergodicity*.

Definition 4.5 (Ergodicity (intuitive definition))**.** Intuitively, a stochastic process $\{Y_t\}$ is *ergodic* if any two collections of random variables partitioned far apart in the sequence are essentially independent.

The formal definition of ergodicity is highly technical and requires advanced concepts in probability theory. However, the intuitive definition captures the essence of the concept. The stochastic process $\{Y_t\}$ is ergodic if Y_t and Y_{t-j} are essentially independent if j is large enough.

If a stochastic process $\{Y_t\}$ is covariance stationary and ergodic then strong restrictions are placed on the joint behavior of the elements in the sequence and on the type of temporal dependence allowed.

Example 4.10 (White noise processes)**.** If $\{Y_t\}$ is GWN or IWN then it is both covariance stationary and ergodic. ■

Example 4.11 (Covariance stationary but not ergodic process (White 1984 page 41))**.** Let $Y_t \sim$ GWN$(0,1)$ and let $X \sim N(0,1)$ independent of $\{Y_t\}$. Define $Z_t = Y_t + X$. Then $\{Z_t\}$ is covariance stationary but not ergodic. To see why $\{Z_t\}$ is not ergodic, note that for all $j > 0$:

$$\begin{aligned}
\text{var}(Z_t) &= \text{var}(Y_t + X) = 1 + 1 = 2, \\
\gamma_j &= \text{cov}(Y_t + X, Y_{t-j} + X) = \text{cov}(Y_t, Y_{t-j}) + \text{cov}(Y_t, X) + \text{cov}(Y_{t-j}, X) + \text{cov}(X, X) \\
&= \text{cov}(X, X) = \text{var}(X) = 1, \\
\rho_j &= \frac{1}{2} \text{ for all } j.
\end{aligned}$$

Hence, the correlation between random variables separated far apart does not eventually go to zero and so $\{Z_t\}$ cannot be ergodic. ■

4.2 Multivariate Time Series

Consider n time series variables $\{Y_{1t}\}, \ldots, \{Y_{nt}\}$. A *multivariate time series* is the $(n \times 1)$ vector time series $\{\mathbf{Y}_t\}$ where the ith row of $\{\mathbf{Y}_t\}$ is $\{Y_{it}\}$. That is, for any time t, $\mathbf{Y}_t = (Y_{1t}, \ldots, Y_{nt})'$. Multivariate time series analysis is used when one wants to model and explain the interactions and co-movements among a group of time series variables. In finance, multivariate time series analysis is used to model systems of asset returns, asset prices, exchange rates, the term structure of interest rates, and economic variables, etc.. Many of the time series concepts described previously for univariate time series carry over to multivariate time series in a natural way. Additionally, there are some important time series concepts that are particular to multivariate time series. The following sections give the details of these extensions.

4.2.1 Stationary and ergodic multivariate time series

A multivariate time series $\{\mathbf{Y}_t\}$ is covariance stationary and ergodic if all of its component time series are stationary and ergodic. The mean of $\mathbf{Y}_t$ is defined as the $(n \times 1)$ vector

$$E[\mathbf{Y}_t] = (\mu_1, \ldots, \mu_n)' = \mu,$$

where $\mu_i = E[Y_{it}]$ for $i = 1, \ldots, n$. The variance/covariance matrix of $\mathbf{Y}_t$ is the $(n \times n)$ matrix

$$\begin{aligned}
\text{var}(\mathbf{Y}_t) = \Sigma &= E[(\mathbf{Y}_t - \mu)(\mathbf{Y}_t - \mu)'] \\
&= \begin{pmatrix}
\text{var}(Y_{1t}) & \text{cov}(Y_{1t}, Y_{2t}) & \cdots & \text{cov}(Y_{1t}, Y_{nt}) \\
\text{cov}(Y_{2t}, Y_{1t}) & \text{var}(Y_{2t}) & \cdots & \text{cov}(Y_{2t}, Y_{nt}) \\
\vdots & \vdots & \ddots & \vdots \\
\text{cov}(Y_{nt}, Y_{1t}) & \text{cov}(Y_{nt}, Y_{2t}) & \cdots & \text{var}(Y_{nt})
\end{pmatrix}.
\end{aligned}$$

The matrix Σ has elements $\sigma_{ij} = \text{cov}(Y_{it}, Y_{jt})$ which measure the contemporaneous linear dependence between Y_{it} and Y_{jt} that is time invariant. The correlation matrix of $\mathbf{Y}_t$ is the $(n \times n)$ matrix

$$\text{cor}(\mathbf{Y}_t) = \mathbf{C} = \mathbf{D}^{-1}\Sigma\mathbf{D}^{-1},$$

where $\mathbf{D}$ is an $(n \times n)$ diagonal matrix with j^{th} diagonal element $\sigma_j = \text{sd}(Y_{jt})$.

4.2.1.1 Cross covariance and correlation matrices

For a univariate time series $\{Y_t\}$, the autocovariances, γ_k, and autocorrelations, ρ_k, summarize the linear time dependence in the data. With a multivariate time series $\{\mathbf{Y}_t\}$ each component has autocovariances and autocorrelations but there are also cross lead-lag covariances and correlations between all possible pairs of components. The lag k autocovariances and autocorrelations of Y_{jt}, for $j = 1, \ldots, n$, are defined as

$$\begin{aligned}\gamma_{jj}^k &= \text{cov}(Y_{jt}, Y_{jt-k}),\\ \rho_{jj}^k &= \text{corr}(Y_{jt}, Y_{jt-k}) = \frac{\gamma_{jj}^k}{\sigma_j^2},\end{aligned}$$

and these are symmetric in k: $\gamma_{jj}^k = \gamma_{jj}^{-k}$, $\rho_{jj}^k = \rho_{jj}^{-k}$. The *cross lag-k covariances* and *cross lag-k correlations* between Y_{it} and Y_{jt} are defined as

$$\begin{aligned}\gamma_{ij}^k &= \text{cov}(Y_{it}, Y_{jt-k}),\\ \rho_{ij}^k &= \text{corr}(Y_{jt}, Y_{jt-k}) = \frac{\gamma_{ij}^k}{\sqrt{\sigma_i^2\sigma_j^2}},\end{aligned}$$

and they are not necessarily symmetric in k. In general,

$$\gamma_{ij}^k = \text{cov}(Y_{it}, Y_{jt-k}) \neq \text{cov}(Y_{jt}, Y_{it-k}) = \gamma_{ji}^k.$$

If $\gamma_{ij}^k \neq 0$ for some $k > 0$ then Y_{jt} is said to *lead* Y_{it}. This implies that past values of Y_{jt} are useful for predicting future values of Y_{it}. Similarly, if $\gamma_{ji}^k \neq 0$ for some $k > 0$ then Y_{it} is said to *lead* Y_{jt}. It is possible that Y_{it} leads Y_{jt} and vice-versa. In this case, there is said to be *dynamic feedback* between the two series.

All of the lag k cross covariances and correlations are summarized in the $(n \times n)$ lag k cross

covariance and lag k cross correlation matrices

$$\begin{aligned}\Gamma_k &= E[(\mathbf{Y}_t - \mu)(\mathbf{Y}_{t-k} - \mu)'] \\ &= \begin{pmatrix} \text{cov}(Y_{1t}, Y_{1t-k}) & \text{cov}(Y_{1t}, Y_{2t-k}) & \cdots & \text{cov}(Y_{1t}, Y_{nt-k}) \\ \text{cov}(Y_{2t}, Y_{1t-k}) & \text{cov}(Y_{2t}, Y_{2t-k}) & \cdots & \text{cov}(Y_{2t}, Y_{nt-k}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(Y_{nt}, Y_{1t-k}) & \text{cov}(Y_{nt}, Y_{2t-k}) & \cdots & \text{cov}(Y_{nt}, Y_{nt-k}) \end{pmatrix}, \\ \mathbf{C}_k &= \mathbf{D}^{-1}\Gamma_k\mathbf{D}^{-1}.\end{aligned}$$

The matrices Γ_k and $\mathbf{C}_k$ are not symmetric in k but it is easy to show that $\Gamma_{-k} = \Gamma_k'$ and $\mathbf{C}_{-k} = \mathbf{C}_k'$.

Example 4.12 (Multivariate Gaussian white noise processes)**.** Let $\{\mathbf{Y}_t\}$ be an $n \times 1$ vector time series process. If $\mathbf{Y}_t \sim iid\, N(\mathbf{0}, \Sigma)$ then $\{\mathbf{Y}_t\}$ is called *multivariate Gaussian white noise* and is denoted $\mathbf{Y}_t \sim \text{GWN}(\mathbf{0}, \Sigma)$. Notice that

$$\begin{aligned} E[\mathbf{Y}_t] &= \mathbf{0}, \\ \text{var}(\mathbf{Y}_t) &= \Sigma, \\ \text{cov}(Y_{jt}, Y_{jt-k}) &= \gamma_{jj}^k = 0 \,(\text{for } k > 0) \\ \text{cov}(Y_{it}, Y_{jt-k}) &= \gamma_{ij}^k = 0 \,(\text{for } k > 0) \end{aligned}$$

Hence, the elements of $\{\mathbf{Y}_t\}$ are contemporaneously correlated but exhibit no time dependence. That is, each element of $\mathbf{Y}_t$ exhibits no time dependence and there is no dynamic feedback between any two elements. Simulating observations from $\text{GWN}(\mathbf{0}, \Sigma)$ requires simulating from a multivariate normal distribution, which can be done using the **mvtnorm** function `rmvnorm()`. For example, to simulate and plot $T = 250$ observation from a bivariate $\text{GWN}(\mathbf{0}, \Sigma)$ process with

$$\Sigma = \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow \mathbf{C} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

use:

```
library(mvtnorm)
Sigma = matrix(c(4, 1, 1, 1), 2, 2)
set.seed(123)
Y = rmvnorm(250, sigma=Sigma)
colnames(Y) = c("Y1", "Y2")
ts.plot(Y, lwd=2, col=c("black", "blue"))
abline(h=0)
legend("topleft", legend=c("Y1", "Y2"),
```

```
       lwd=2, col=c("black", "blue"), lty=1)
```

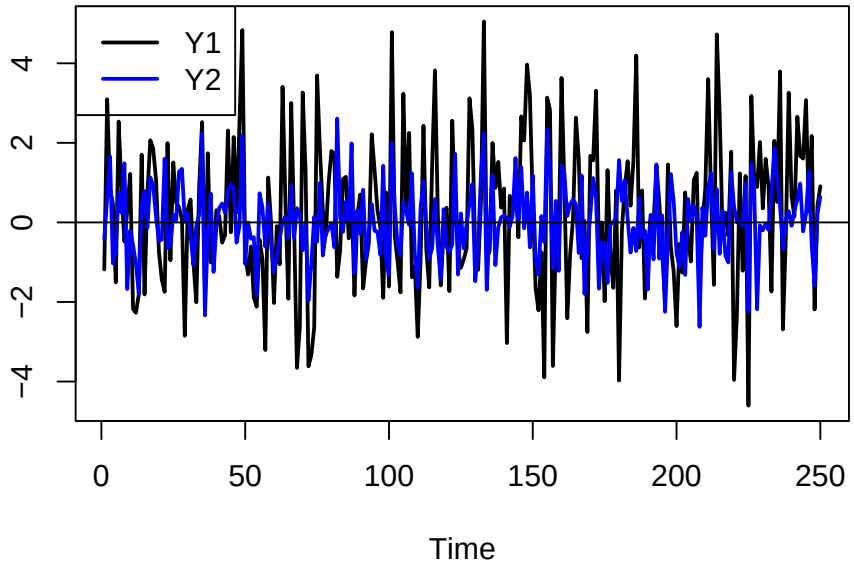


Figure 4.7: Simulated bivariate GWN process.

The simulated values are shown on the same plot in Figure 4.7. Both series fluctuate randomly about zero, and the first series (black line) has larger fluctuations (volatility) than the second series (blue line). The two series are contemporaneously correlated ($\rho_{12} = 0.5$) but are both uncorrelated over time ($\rho_{11}^k = \rho_{22}^k = 0$, $k > 0$) and are not cross-lag correlated ($\rho_{12}^k = \rho_{21}^k = 0$, $k > 0$). ■

4.3 Time Series Models

Time series models are probability models that are used to describe the behavior of a stochastic process. In many cases of interest, it is assumed that the stochastic process to be modeled is covariance stationary and ergodic. Then, the main feature of the process to be modeled is the time dependence between the random variables. In this section, we illustrate some simple models for covariance stationary and ergodic time series that exhibit particular types of linear time dependence captured by autocorrelations. The univariate models, made popular originally by (Box and Jenkins 1976), are called *autoregressive moving average* (ARMA) models. The multivariate model, made popular by (Sims 1980), is called the *vector autoregressive* (VAR) model. These models are used extensively in economics and finance for modeling univariate and multivariate time series.

4.3.1 Moving average models

Moving average models are simple covariance stationary and ergodic time series models built from linear functions of GWN that can capture time dependence between random variables that lasts only for a finite number of lags.

4.3.1.1 MA(1) Model

Suppose you want to create a covariance stationary and ergodic stochastic process $\{Y_t\}$ in which Y_t and Y_{t-1} are correlated but Y_t and Y_{t-j} are not correlated for $j > 1$. That is, the time dependence in the process only lasts for one period. Such a process can be created using the *first order moving average* (MA(1)) model:

$$Y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1},\ -1 < \theta < 1,\ \varepsilon_t \sim \text{GWN}(0, \sigma_\varepsilon^2). \tag{4.2}$$

The MA(1) model is a simple linear function of the GWN random variables ε_t and ε_{t-1}. This linear structure allows for easy analysis of the model. The moving average parameter θ determines the sign and magnitude of the correlation between Y_t and Y_{t-1}. Clearly, if $\theta = 0$ then $Y_t = \mu + \varepsilon_t$ so that $\{Y_t\}$ is GWN with non-zero mean μ and exhibits no time dependence. As will be shown below, the MA(1) model produces a covariance stationary and ergodic process for any (finite) value of θ. The restriction $-1 < \theta < 1$ is called the *invertibility* restriction and will be explained below.

To verify that Equation 4.2 process is a covariance stationary process we must show that the mean, variance and autocovariances are time invariant. For the mean, we have:

$$E[Y_t] = \mu + E[\varepsilon_t] + \theta E[\varepsilon_{t-1}] = \mu,$$

because $E[\varepsilon_t] = E[\varepsilon_{t-1}] = 0$.

For the variance, we have

$$\begin{aligned}
\text{var}(Y_t) = \sigma^2 &= E[(Y_t - \mu)^2] = E[(\varepsilon_t + \theta\varepsilon_{t-1})^2] \\
&= E[\varepsilon_t^2] + 2\theta E[\varepsilon_t\varepsilon_{t-1}] + \theta^2 E[\varepsilon_{t-1}^2] \\
&= \sigma_\varepsilon^2 + 0 + \theta^2\sigma_\varepsilon^2 = \sigma_\varepsilon^2(1 + \theta^2).
\end{aligned}$$

The term $E[\varepsilon_t\varepsilon_{t-1}] = \text{cov}(\varepsilon_t, \varepsilon_{t-1}) = 0$ because $\{\varepsilon_t\}$ is an independent process.

For $\gamma_1 = \text{cov}(Y_t, Y_{t-1})$, we have:

$$\begin{aligned}
\text{cov}(Y_t, Y_{t-1}) &= E[(Y_t - \mu)(Y_{t-1} - \mu)] \\
&= E[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-1} + \theta\varepsilon_{t-2})] \\
&= E[\varepsilon_t\varepsilon_{t-1}] + \theta E[\varepsilon_t\varepsilon_{t-2}] \\
&+ \theta E[\varepsilon_{t-1}^2] + \theta^2 E[\varepsilon_{t-1}\varepsilon_{t-2}] \\
&= 0 + 0 + \theta\sigma_\varepsilon^2 + 0 = \theta\sigma_\varepsilon^2.
\end{aligned}$$

Here, the sign of γ_1 is the same as the sign of θ.

For $\rho_1 = \text{cor}(Y_t, Y_{t-1})$ we have:

$$\rho_1 = \frac{\gamma_1}{\sigma^2} = \frac{\theta\sigma_\varepsilon^2}{\sigma_\varepsilon^2(1+\theta^2)} = \frac{\theta}{(1+\theta^2)}.$$

Clearly, $\rho_1 = 0$ if $\theta = 0$; $\rho_1 > 0$ if $\theta > 0$; $\rho_1 < 0$ if $\theta < 0$. Also, the largest value for $|\rho_1|$ is 0.5 which occurs when $|\theta| = 1$. Hence, a MA(1) model cannot describe a stochastic process that has $|\rho_1| > 0.5$. Also, note that there is more than one value of θ that produces the same value of ρ_1. For example, θ and $1/\theta$ give the same value for ρ_1. The *invertibility restriction* $-1 < \theta < 1$ provides a unique mapping between θ and ρ_1.

For $\gamma_2 = \text{cov}(Y_t, Y_{t-2})$, we have:

$$\begin{aligned}
\text{cov}(Y_t, Y_{t-2}) &= E[(Y_t - \mu)(Y_{t-2} - \mu)] \\
&= E[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-2} + \theta\varepsilon_{t-3})] \\
&= E[\varepsilon_t\varepsilon_{t-2}] + \theta E[\varepsilon_t\varepsilon_{t-3}] \\
&+ \theta E[\varepsilon_{t-1}\varepsilon_{t-2}] + \theta^2 E[\varepsilon_{t-1}\varepsilon_{t-3}] \\
&= 0 + 0 + 0 + 0 = 0.
\end{aligned}$$

Similar calculations can be used to show that $\text{cov}(Y_t, Y_{t-j}) = \gamma_j = 0$ for $j > 1$. Hence, for $j > 1$ we have $\rho_j = 0$ and there is only time dependence between Y_t and Y_{t-1} but no time dependence between Y_t and Y_{t-j} for $j > 1$. Because $\rho_j = 0$ for $j > 1$ the MA(1) process is trivially ergodic.

Example 4.13 (Simulating values from MA(1) process)**.** Consider simulating $T = 250$ observations from Equation 4.2 with $\mu = 1$, $\theta = 0.9$ and $\sigma_\varepsilon = 1$. When simulating an MA(1) process, you need to decide how to start the simulation. The value of Y_t at $t = 0$, y_0, is called the *initial value* and is the starting value for the simulation. Now, the first two observations from Equation 4.2 starting

at $t = 0$ are

$$\begin{aligned} y_0 &= \mu + \varepsilon_0 + \theta\varepsilon_{-1}, \\ y_1 &= \mu + \varepsilon_1 + \theta\varepsilon_0. \end{aligned}$$

It is common practice is to set $\varepsilon_{-1} = \varepsilon_0 = 0$ so that $y_0 = \mu$, $y_1 = \mu + \varepsilon_1 = y_0 + \varepsilon_1$ and ε_1 is the first simulated error term. The remaining observations for $t = 2, \ldots, T$ are created from Equation 4.2. We can implement the simulation in a number of ways in R. The most straightforward way is to use a simple loop:

```
n.obs = 250
mu = 1
theta = 0.9
sigma.e = 1
set.seed(123)
e = rnorm(n.obs, sd = sigma.e)
y = rep(0, n.obs)
y[1] = mu + e[1]
for (i in 2:n.obs) {
    y[i] = mu + e[i] + theta*e[i-1]
}
head(y, n=3)
#> [1] 0.440 0.265 2.352
```

The "for loop" in R can be slow, however, especially for a very large number of simulations. The simulation can be more efficiently implemented using *vectorized* calculations as illustrated below:

```
set.seed(123)
e = rnorm(n.obs, sd = sigma.e)
em1 = c(0, e[1:(n.obs-1)])
y = mu + e + theta*em1
head(y, n=3)
#> [1] 0.440 0.265 2.352
```

The vectorized calculation avoids looping all together and computes all of the simulated values at the same time. This can be considerably faster than the "for loop" calculation.

The MA(1) model is a special case of the more general *autoregressive integrated moving average* (ARIMA) model. R has many built-in functions and several packages for working with ARIMA models. In particular, the R function `arima.sim()` can be used to simulate observations from a MA(1) process. It essentially implements the simulation loop described above. The arguments of

`arima.sim()` are:

```
args(arima.sim)
#> function (model, n, rand.gen = rnorm, innov = rand.gen(n, ...),
#>     n.start = NA, start.innov = rand.gen(n.start, ...), ...)
#> NULL
```

where `model` is a list with named components describing the ARIMA model parameters (excluding the mean value), `n` is the number of simulated observations, `rand.gen` specifies the pdf for ε_t, `innov` is a vector ε_t values of length `n`, `n.start` is the number of pre-simulation (burn-in) values for ε_t, `start.innov` is a vector of `n.start` pre-simulation values for ε_t, and `...` specify any additional arguments for `rand.gen`. For example, to perform the same simulations as above use:

```
ma1.model = list(ma=0.9)
set.seed(123)
y = mu + arima.sim(model=ma1.model, n=250,
                   n.start=1, start.innov=0)
head(y, n=3)
#> Time Series:
#> Start = 1
#> End = 3
#> Frequency = 1
#> [1] 0.440 0.265 2.352
```

The `ma` component of the `"list"` object `ma1.model` specifies the value of θ for the MA(1) model, and is used as an input to the function `arima.sim()`. The options `n.start = 1` and `start.innov = 0` sets the start-up initial value $\varepsilon_0 = 0$. By default, `arima.sim()` sets $\mu = 0$, specifies $\varepsilon_t \sim \text{GWN}(0, 1)$, and returns $\varepsilon_t + \theta\varepsilon_{t-1}$ for $t = 1, \dots, T$. The simulated value for Y_t is constructed by adding the value of `mu` ($\mu = 1$) to the output of `arima.sim()`.

The function `ARMAacf()` can be used to compute the theoretical autocorrelations, ρ_j, from the MA(1) model (recall, $\rho_1 = \theta/(1 + \theta^2)$ and $\rho_j = 0$ for $j > 1$). For example, to compute ρ_j for $j = 1, \dots, 10$ use:

```
ma1.acf = ARMAacf(ar=0, ma=0.9, lag.max=10)
ma1.acf
#>     0     1     2     3     4     5     6     7     8     9    10
#> 1.000 0.497 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
```

Figure 4.8 shows the simulated data and the theoretical ACF created using:

```
par(mfrow=c(2,1))
ts.plot(y,main="MA(1) Process: mu=1, theta=0.9",
xlab="time",ylab="y(t)", col="blue", lwd=2)
abline(h=c(0,1))
plot(0:10, ma1.acf,type="h", col="blue", lwd=2,
main="ACF for MA(1): theta=0.9",xlab="lag",ylab="rho(j)")
abline(h=0)
par(mfrow=c(1,1))
```

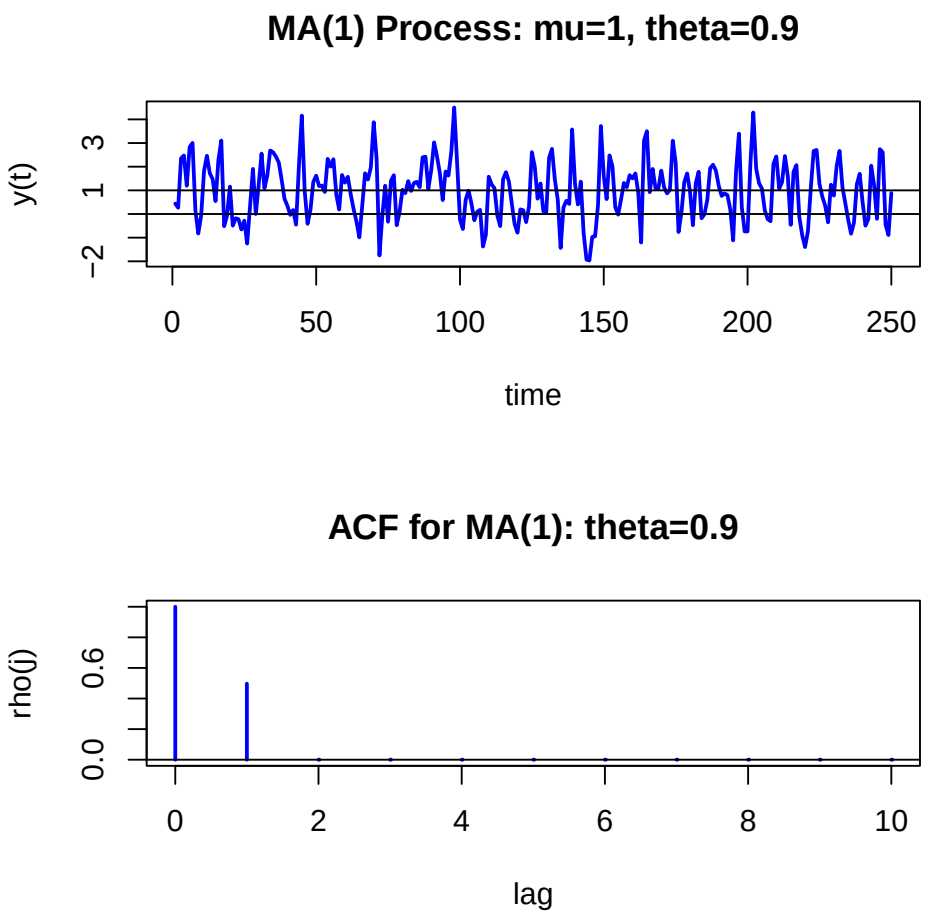


Figure 4.8: Simulated values and theoretical ACF from MA(1) process with $\mu = 1$, $\theta = 0.9$ and $\sigma_\varepsilon^2 = 1$.

Compared to the GWN process in Figure 4.2, the MA(1) process is a bit smoother in its appearance. This is due to the positive one-period time dependence captured by $\rho_1 = 0.4972$. ■

Example 4.14 (MA(1) model for overlapping continuously compounded returns)**.** Let R_t denote

the one-month continuously compounded return and assume that:

$$R_t \sim \text{ GWN}(\mu, \sigma^2).$$

Consider creating a monthly time series of two-month continuously compounded returns using:

$$R_t(2) = R_t + R_{t-1}.$$

The time series of these two-month returns, observed monthly, overlap by one month:

$$\begin{aligned} R_t(2) &= R_t + R_{t-1}, \\ R_{t-1}(2) &= R_{t-1} + R_{t-2}, \\ R_{t-2}(2) &= R_{t-2} + R_{t-3}, \\ &\vdots \end{aligned}$$

The one-month overlap in the two-month returns implies that $\{R_t(2)\}$ follows an MA(1) process. To show this, we need to show that the autocovariances of $\{R_t(2)\}$ behave like the autocovariances of an MA(1) process.

To verify that $\{R_t(2)\}$ follows an MA(1) process, first we have:

$$\begin{aligned} E[R_t(2)] &= E[R_t] + E[R_{t-1}] = 2\mu, \\ \text{var}(R_t(2)) &= \text{var}(R_t + R_{t-1}) = 2\sigma^2. \end{aligned}$$

Next, we have:

$$\text{cov}(R_t(2), R_{t-1}(2)) = \text{cov}(R_t + R_{t-1}, R_{t-1} + R_{t-2}) = \text{cov}(R_{t-1}, R_{t-1}) = \text{var}(R_{t-1}) = \sigma^2,$$

and,

$$\begin{aligned} &\text{cov}(R_t(2), R_{t-2}(2)) = \text{cov}(R_t + R_{t-1}, R_{t-2} + R_{t-3}) = 0, \\ &\text{cov}(R_t(2), R_{t-j}(2)) = 0 \text{ for } j > 1. \end{aligned}$$

Hence, the autocovariances of $\{R_t(2)\}$ are those of an MA(1) process.

Notice that

$$\rho_1 = \frac{\sigma^2}{2\sigma^2} = \frac{1}{2}.$$

What MA(1) process describes $\{R_t(2)\}$? Because $\rho_1 = \frac{\theta}{1+\theta^2} = 0.5$ it follows that $\theta = 1$. Hence, the

MA(1) process has mean 2μ and $\theta = 1$ and can be expressed as the MA(1) model:

$$\begin{aligned} R_t(2) &= 2\mu + \varepsilon_t + \varepsilon_{t-1}, \\ \varepsilon_t &\sim \text{GWN}(0, \sigma^2). \end{aligned}$$

Notice that this is a *non-invertible* MA(1) model. ■

4.3.1.2 MA(q) Model

The MA(q) model has the form

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \tag{4.3}$$

where

$$\varepsilon_t \sim \text{GWN}(0, \sigma_\varepsilon^2).$$

The MA(q) model is stationary and ergodic provided $\theta_1, \ldots, \theta_q$ are finite. The moments of the MA(q) (see end-of-chapter exercises) are

$$\begin{aligned} E[Y_t] &= \mu, \\ \gamma_0 &= \sigma^2(1 + \theta_1^2 + \cdots + \theta_q^2), \\ \gamma_j &= \begin{cases} \left(\theta_j + \theta_{j+1}\theta_1 + \theta_{j+2}\theta_2 + \cdots + \theta_q\theta_{q-j}\right)\sigma^2 \text{ for } j = 1, 2, \ldots, q \\ 0 \text{ for } j > q \end{cases}. \end{aligned}$$

Hence, the ACF of an MA(q) is non-zero up to lag q and is zero afterward.

Example 4.15 (Overlapping returns and MA(q) models)**.** MA(q) models often arise in finance through data aggregation transformations. For example, let $R_t = \ln(P_t/P_{t-1})$ denote the monthly continuously compounded return on an asset with price P_t. Define the annual return at time t using monthly returns as $R_t(12) = \ln(P_t/P_{t-12}) = \sum_{j=0}^{11} R_{t-j}$. Suppose $R_t \sim \text{GWN}(\mu, \sigma^2)$ and consider a sample of monthly returns of size T, $\{R_1, R_2, \ldots, R_T\}$. A sample of annual returns may be created using *overlapping* or *non-overlapping* returns. Let $\{R_{12}(12), R_{13}(12), \ldots, R_T(12)\}$ denote a sample of $T^* = T - 11$ monthly overlapping annual returns and $\{R_{12}(12), R_{24}(12), \ldots, R_T(12)\}$ denote a sample of $T/12$ non-overlapping annual returns. Researchers often use overlapping returns in analysis due to the apparent larger sample size. One must be careful using overlapping returns because the monthly annual return sequence $\{R_t(12)\}$ is not a Gaussian white noise process even

if the monthly return sequence $\{R_t\}$ is. To see this, straightforward calculations give:

$$\begin{aligned} E[R_t(12)] &= 12\mu, \\ \gamma_0 &= \text{var}(R_t(12)) = 12\sigma^2, \\ \gamma_j &= \text{cov}(R_t(12), R_{t-j}(12)) = (12-j)\sigma^2 \text{ for } j < 12, \\ \gamma_j &= 0 \text{ for } j \geq 12. \end{aligned}$$

Since $\gamma_j = 0$ for $j \geq 12$ notice that $\{R_t(12)\}$ behaves like an MA(11) process:

$$\begin{aligned} R_t(12) &= 12\mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_{11}\varepsilon_{t-11}, \\ \varepsilon_t &\sim \text{GWN}(0, \sigma^2). \end{aligned}$$

■

4.3.2 Autoregressive Models

Moving average models can capture almost any kind of autocorrelation structure. However, this may require many moving average terms in Equation 4.3. Another type of simple time series model is the *autoregressive model*. This model can capture complex autocorrelation patterns with a small number of parameters and is used more often in practice than the moving average model.

4.3.2.1 AR(1) Model

Suppose you want to create a covariance stationary and ergodic stochastic process $\{Y_t\}$ in which Y_t and Y_{t-1} are correlated, Y_t and Y_{t-2} are slightly less correlated, Y_t and Y_{t-3} are even less correlated and eventually Y_t and Y_{t-j} are uncorrelated for j large enough. That is, the time dependence in the process decays to zero as the random variables in the process get farther and farther apart. Such a process can be created using the *first order autoregressive* (AR(1)) model:

$$Y_t - \mu = \phi(Y_{t-1} - \mu) + \varepsilon_t, \ -1 < \phi < 1, \ \varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2). \tag{4.4}$$

It can be shown that the AR(1) model is covariance stationary and ergodic provided $-1 < \phi < 1$. We will show that the AR(1) process has the following properties:

$$E[Y_t] = \mu, \tag{4.5}$$

$$\text{var}(Y_t) = \sigma^2 = \sigma_\varepsilon^2/(1-\phi^2), \tag{4.6}$$

$$\text{cov}(Y_t, Y_{t-1}) = \gamma_1 = \sigma^2\phi, \tag{4.7}$$

$$\text{cor}(Y_t, Y_{t-1}) = \rho_1 = \gamma_1/\sigma^2 = \phi, \tag{4.8}$$

$$\text{cov}(Y_t, Y_{t-j}) = \gamma_j = \sigma^2\phi^j, \tag{4.9}$$

$$\text{cor}(Y_t, Y_{t-j}) = \rho_j = \gamma_j/\sigma^2 = \phi^j. \tag{4.10}$$

Notice that the restriction $|\phi| < 1$ implies that:

$$\lim_{j\to\infty} \rho_j = \phi^j = 0,$$

so that Y_t is essentially independent of Y_{t-j} for large j and so $\{Y_t\}$ is ergodic. For example, if $\phi = 0.5$ then $\rho_{10} = (0.5)^{10} = 0.001$; if $\phi = 0.9$ then $\rho_{10} = (0.9)^{10} = 0.349$. Hence, the closer ϕ is to unity the stronger is the time dependence in the process. If $\phi = 1$, then Equation 4.4 becomes the random walk model $Y_t = Y_{t-1} + \varepsilon_t$ and is a non-stationary process.

Verifying covariance stationarity for the AR(1) model is more involved than for the MA(1) model, and establishing the properties Equation 4.5 - Equation 4.10 involves some tricks. In what follows, we will assume that $\{Y_t\}$ is a covariance stationary process and that $|\phi| < 1$. First, consider the derivation for Equation 4.5. We have:

$$\begin{aligned} E[Y_t] &= \mu + \phi(E[Y_{t-1}] - \mu) + E[\varepsilon_t] \\ &= \mu + \phi E[Y_{t-1}] - \phi\mu. \end{aligned}$$

Here, we use the first trick. Given that $\{Y_t\}$ is covariance stationary it follows that $E[Y_t] = E[Y_{t-1}]$. Substituting $E[Y_t] = E[Y_{t-1}]$ into the above and solving for $E[Y_t]$ gives Equation 4.5.

A similar trick can be used to derive Equation 4.6:

$$\text{var}(Y_t) = \phi^2(\text{var}(Y_{t-1})) + \text{var}(\varepsilon_t) = \phi^2(\text{var}(Y_t)) + \sigma_\varepsilon^2,$$

which uses the fact that Y_{t-1} is independent of ε_t (because Y_{t-1} only depends on $t-1$ values) and $\text{var}(Y_t) = \text{var}(Y_{t-1})$ given that $\{Y_t\}$ is covariance stationary. Solving for $\sigma^2 = \text{var}(Y_t)$ gives Equation 4.6.

To determine Equation 4.7, we use another trick. Multiply both sides of Equation 4.4 by $Y_{t-1} - \mu$ and take expectations to give:

$$\gamma_1 = E\left[(Y_t - \mu)(Y_{t-1} - \mu)\right] = \phi E\left[(Y_{t-1} - \mu)^2\right] + E\left[\varepsilon_t (Y_{t-1} - \mu)\right] = \phi\sigma^2,$$

which uses the fact that Y_{t-1} is independent of ε_t, and $\text{var}(Y_t) = \text{var}(Y_{t-1}) = \sigma^2$. Finally, to determine Equation 4.9, multiply both sides of Equation 4.4 by $Y_{t-j} - \mu$ and take expectations to

give:

$$\begin{aligned}\gamma_j &= E\left[(Y_t-\mu)\left(Y_{t-j}-\mu\right)\right] = \phi E\left[(Y_{t-1}-\mu)(Y_{t-j}-\mu)\right] + E\left[\varepsilon_t\left(Y_{t-j}-\mu\right)\right] \\ &= \phi\gamma_{j-1},\end{aligned}$$

which uses the fact that Y_{t-j} is independent of ε_t, and $E\left[(Y_{t-1}-\mu)(Y_{t-j}-\mu)\right] = \gamma_{j-1}$ provided $\{Y_t\}$ is covariance stationary. Using recursive substitution and $\gamma_0 = \sigma^2$ gives Equation 4.9.

The AR(1) model Equation 4.4 is written in what is called the *mean-adjusted form*. The mean-adjusted form can be re-expressed in the form of a linear regression model as follows:

$$\begin{aligned}Y_t - \mu &= \phi(Y_{t-1}-\mu) + \varepsilon_t \Rightarrow \\ Y_t &= \mu - \phi\mu + \phi Y_{t-1} + \varepsilon_t \\ &= c + \phi Y_{t-1} + \varepsilon_t,\end{aligned} \tag{4.11}$$

where $c = (1-\phi)\mu \Rightarrow \mu = c/(1-\phi)$ is the intercept of the linear regression. This regression model form is convenient for estimation by ordinary least squares.

Example 4.16 (Simulating values from AR(1) process)**.** Consider simulating 250 observations from Equation 4.4 with $\mu = 1$, $\phi = 0.9$ and $\sigma_\varepsilon = 1$. To start the simulation, an initial value or start-up value for Y_0 is required. A commonly used initial value is the mean value μ so that $Y_1 = \mu + \varepsilon_1$. As with the MA(1) model, this can be performed using a simple "for loop" in R:

```
phi = 0.9
mu = 1
sigma.e = 1
n.obs = 250
y = rep(0, n.obs)
set.seed(123)
e = rnorm(n.obs, sd=sigma.e)
y[1] = mu + e[1]
for (i in 2:n.obs) {
     y[i] = mu + phi*(y[i-1] - mu) + e[i]
}
head(y, 3)
#> [1] 0.440 0.265 1.898
```

Unfortunately, there is no easy way to vectorize the loop calculation. However, the R function `filter()`, with optional argument `method = "recursive"`, implements the AR(1) recursion efficiently in C code and so is more efficient than the for loop code in R above:

```
y = mu + stats::filter(e, phi, method="recursive")
head(y, 3)
#> Time Series:
#> Start = 1
#> End = 3
#> Frequency = 1
#> [1] 0.440 0.265 1.898
```

The R function `arima.sim()`, which internally uses the `filter()` function, can also be used to simulate observations from an AR(1) process. For the AR(1) model, the function `arima.sim()` simulates the *components form* of the AR(1) model

$$\begin{aligned} Y_t &= \mu + u_{t,} \\ u_t &= \phi u_{t-1} + \epsilon_t. \end{aligned}$$

Hence, to replicate the "for loop" simulation with `arima.sim()` use:

```
ar1.model = list(ar=0.9)
mu = 1
set.seed(123)
y = mu + arima.sim(model=ar1.model,n=250,n.start=1, start.innov=0)
head(y, 3)
#> Time Series:
#> Start = 1
#> End = 3
#> Frequency = 1
#> [1] 0.440 0.265 1.898
```

The R function `ARMAacf()` can be used to compute the theoretical ACF for an AR(1) model as follows

```
ar1.acf = ARMAacf(ar = 0.9, ma = 0, lag.max = 10)
```

The simulated AR(1) values and the ACF are shown in Figure 4.9. Compared to the MA(1) process in Figure 4.8, the realizations from the AR(1) process are much smoother. That is, when Y_t wanders high above its mean it tends to stay above the mean for a while and when it wanders low below the mean it tends to stay below for a while. ■

Example 4.17 (AR(1) models and the speed of mean reversion)**.**

- insert example showing AR(1) models with different values of ρ

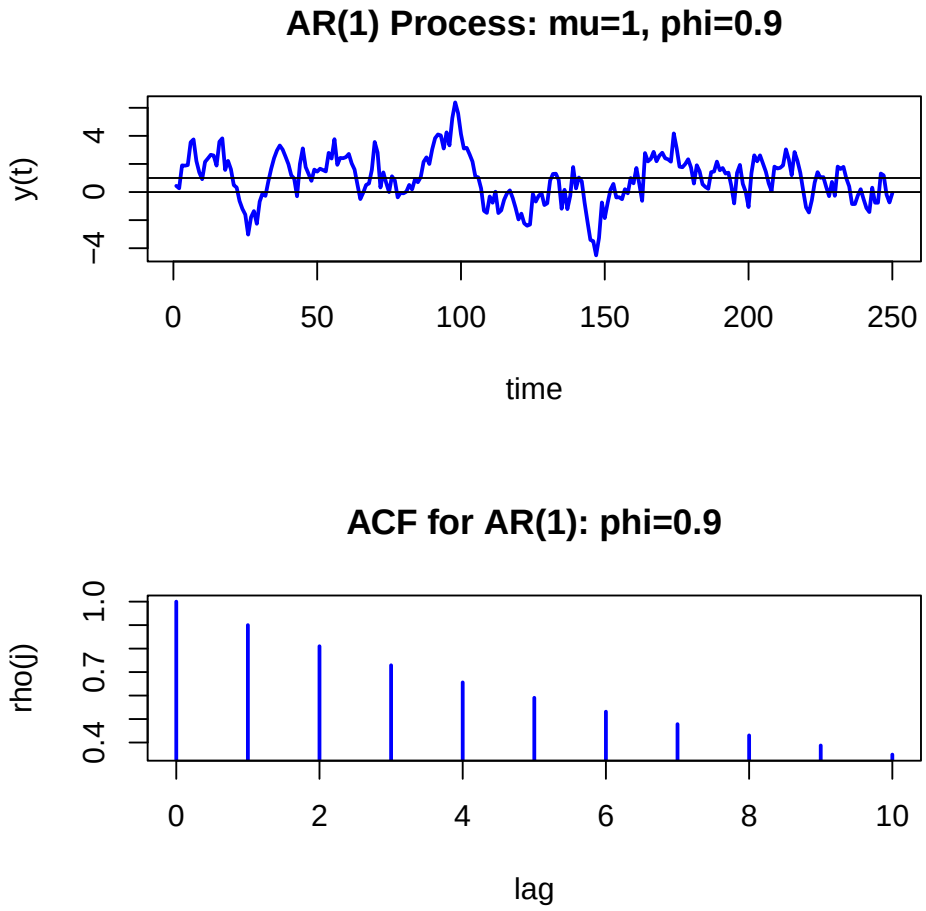


Figure 4.9: Simulated values and ACF from AR(1) model with $\mu = 1, \phi = 0.9$ and $\sigma_\varepsilon^2 = 1$

- discuss the concept of mean reversion
- discuss RW model as special case of AR(1)

■

4.3.2.2 AR(p) Model

The covariance stationary AR(p) model in mean-adjusted form is

$$\begin{aligned} Y_t - \mu &= \phi_1(Y_{t-1} - \mu) + \cdots + \phi_p(Y_{t-p} - \mu) + \varepsilon_t, \\ \varepsilon_t &\sim \text{GWN}(0, \sigma^2_\varepsilon), \end{aligned} \tag{4.12}$$

where $\mu = E[Y_t]$. Like the AR(1), restrictions on the autoregressive parameters $\phi_1, \ldots, \phi_p$ are required for $\{Y_t\}$ to be covariance stationary and ergodic. A detailed treatment of these restrictions is beyond the scope of this book. However, one simple necessary condition for $\{Y_t\}$ to be covariance stationary is $|\phi| < 1$ where $\phi = \phi_1 + \cdots + \phi_p$. Hence, in the AR(p) model the sum of the autoregressive components ϕ has a similar interpretation as the single autoregressive coefficient in the AR(1) model.

The regression form of the AR(p) is

$$Y_t = c + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \varepsilon_t,$$

where $c = \mu/(1 - \phi_1 - \cdots - \phi_p) = \mu/(1 - \phi)$. This form is convenient for estimation purposes because it is in the form of a linear regression.

The regression form of the AR(p) model is used very often in practice because of its simple linear structure and because it can capture a wide variety of autocorrelation patterns such as exponential decay, damped cyclical patterns, and oscillating damped cyclical patterns. Unfortunately, the mathematical derivation of the autocorrelations in the AR(p) model is complicated and tedious (and beyond the scope of this book). The exercises at the end of the chapter illustrate some of the calculations for the AR(2) model.

4.3.3 Autoregressive Moving Average Models

Autoregressive and moving average models can be combined into a general model called the *autoregressive moving average* (ARMA) model. The ARMA model with p autoregressive components and q moving average components, denoted ARMA(p,q) is given by

$$\begin{aligned} Y_t - \mu &= \phi_1(Y_{t-1} - \mu) + \cdots + \phi_p(Y_{t-p} - \mu) \\ &\quad + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} \\ \varepsilon_t &\sim \text{GWN}(0, \sigma^2) \end{aligned} \tag{4.13}$$

The regression formulation is

$$Y_t = c + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \varepsilon_t + \theta\varepsilon_{t-1} + \cdots + \theta\varepsilon_{t-q}$$

where $c = \mu/(1 - \phi_1 - \cdots - \phi_p) = \mu/(1 - \phi)$ and $\phi = \phi_1 + \cdots + \phi_p$. This model combines aspects of the pure moving average models and the pure autoregressive models and can capture many types of autocorrelation patterns. For modeling typical non-seasonal economic and financial data, it is seldom necessary to consider models in which $p > 2$ and $q > 2$. For example, the simple ARMA(1,1) model

$$Y_t - \mu = \phi_1(Y_{t-1} - \mu) + \varepsilon_t + \theta_1\varepsilon_{t-1} \tag{4.14}$$

can capture many realistic autocorrelations observed in data.

4.3.4 Vector Autoregressive Models

The most popular multivariate time series model is the *vector autoregressive* (VAR) model. The VAR model is a multivariate extension of the univariate autoregressive model Equation 4.12. For example, a bivariate VAR(1) model for $\mathbf{Y}_t = (Y_{1t}, Y_{2t})'$ has the form

$$\begin{pmatrix} Y_{1t} \\ Y_{2t} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} a_{11}^1 & a_{12}^1 \\ a_{21}^1 & a_{22}^1 \end{pmatrix} \begin{pmatrix} Y_{1t-1} \\ Y_{2t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

or

$$\begin{aligned} Y_{1t} &= c_1 + a_{11}^1 Y_{1t-1} + a_{12}^1 Y_{2t-1} + \varepsilon_{1t}, \\ Y_{2t} &= c_2 + a_{21}^1 Y_{1t-1} + a_{22}^1 Y_{2t-1} + \varepsilon_{2t}, \end{aligned}$$

where

$$\begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix} \sim \text{iid } N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix} \right).$$

In the equations for Y_1 and Y_2, the lagged values of both Y_1 and Y_2 are present. Hence, the VAR(1) model allows for dynamic feedback between Y_1 and Y_2 and can capture cross-lag correlations between the variables. In matrix notation, the model is

$$\begin{aligned} \mathbf{Y}_t &= \mathbf{A}\mathbf{Y}_{t-1} + \varepsilon_t, \\ \varepsilon_t &\sim N(\mathbf{0}, \Sigma), \end{aligned}$$

where

$$\mathbf{A} = \begin{pmatrix} a_{11}^1 & a_{12}^1 \\ a_{21}^1 & a_{22}^1 \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}.$$

The general VAR(p) model for $\mathbf{Y}_t = (Y_{1t}, Y_{2t}, \ldots, Y_{nt})'$ has the form

$$\mathbf{Y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{Y}_{t-1} + \mathbf{A}_2\mathbf{Y}_{t-2} + \cdots + \mathbf{A}_p\mathbf{Y}_{t-p} + \varepsilon_t,$$

where $\mathbf{A}_i$ are $(n \times n)$ coefficient matrices and $_t$ is an $(n \times 1)$ unobservable zero mean white noise vector process with covariance matrix Σ. VAR models are capable of capturing much of the complicated dynamics observed in stationary multivariate time series.

Example 4.18 (Simulated values from a bivariate VAR(1) process)**.** To be completed

4.4 Forecasting

- One of the main practical uses of time series models is for forecasting future observations
- The existence of time dependence in a covariance stationary time series means that we can exploit this time dependence to obtain forecasts of future observations that are superior to the unconditional mean.
- Describe basic forecasting problem
- Show result that conditional mean is optimal mse forecast
- Show chain-rule of forecasting for AR(1) processes

4.5 Further Reading: Time Series Concepts

This chapter gives a very brief introduction to time series modeling. More thorough treatments of time series analysis with an orientation towards economics and finance with examples in R are given in (Ruppert and Matteson 2015), (Tsay 2010), and (Zivot 2016). The CRAN task view for Time Series is an excellent summary of R packages used for time series analysis.

4.6 Problems: Time Series Concepts

Exercise 4.1. Suppose the time series $\{X_t\}_{t=-\infty}^{\infty}$ is independent white noise. That is $X_t \sim i.i.d.\,(0, \sigma^2)$. Define two new time series $\{Y_t\}_{t=-\infty}^{\infty}$ and $\{Z_t\}_{t=-\infty}^{\infty}$, where $Y_t = X_t^2$ and $Z_t = |X_t|$. Are $\{Y_t\}_{t=-\infty}^{\infty}$ and $\{Z_t\}_{t=-\infty}^{\infty}$ also independent white noise processes? Why or why not?

Exercise 4.2. Realizations from four stochastic processes are given in Figure . Which processes appear to be covariance stationary and which processes appear to be non-stationary? Briefly justify your answers.

Exercise 4.3. Consider the MA(1) model

$$\begin{aligned} Y_t &= 0.05 + \varepsilon_t + \theta\varepsilon_{t-1}, -1 < \theta < 1 \\ \varepsilon_t &\sim iid\, N(0, (0.10)^2). \end{aligned}$$

This process has mean $E[Y_t] = 0.05$.

1. Calculate $\text{var}(Y_t)$ and $\rho_1 = \text{cor}(Y_t, Y_{t-1})$ for $\theta = 0.5$ and $\theta = 0.9$.
2. Using the R function `arima.sim()`, simulate and plot 250 observations of the MA(1) process with $\theta = 0.5$ and $\theta = 0.9$. Briefly comment on the behavior of the simulated data series. Does it look covariance stationary? Does it show evidence of time dependence?

Exercise 4.4. Consider the AR(1) model

$$\begin{aligned} Y_t - 0.05 &= \phi(Y_{t-1} - 0.05) + \varepsilon_t, -1 < \phi < 1 \\ \varepsilon_t &\sim iid\, N(0, (0.10)^2). \end{aligned}$$

This process has mean $E[Y_t] = 0.05$.

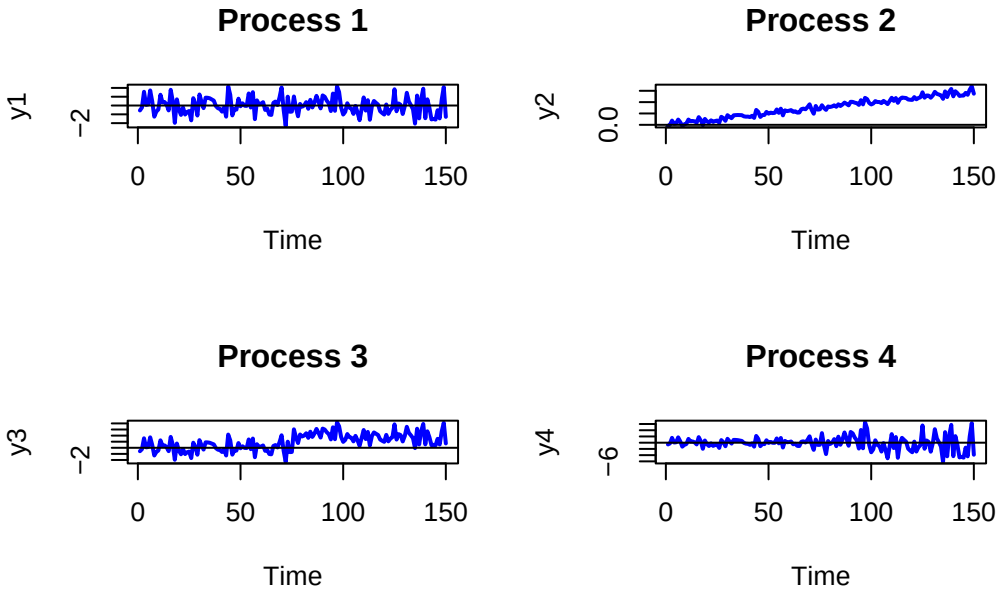
1. Calculate $\text{var}(Y_t)$ for $\phi = 0.5$ and $\phi = 0.9$.
2. Calculate $\rho_j = \text{cor}(Y_t, Y_{t-j})$ for $\phi = 0.5$ and $\phi = 0.9$ and for $j = 1, \ldots, 5$.
3. Using the R function `arima.sim()`, simulate and plot 250 observations of the AR(1) process with $\theta = 0.5$ and $\theta = 0.9$. Briefly comment on the behavior of the simulated data series. Does it look covariance stationary? Does it show evidence of time dependence? How is it different from the MA(1) process

Exercise 4.5. Figure shows a realization of a stochastic process representing a monthly time series of overlapping 2-month continuously compounded returns $r_t(2) = r_t + r_{t-1}$, where the 1-month continuously compounded returns r_t follow a Gaussian White noise process with variance 1.

1. Based on the sample autocorrelations, which time series process is most appropriate for describing the series: MA(1) or AR(1)? Justify your answer.
2. If you think the process is an AR(1) process, what do you think is the value of the autoregressive parameter? If you think the process is a MA(1) process, what do you think is the value of the moving average parameter?

Exercise 4.6. Let Y_t represent a stochastic process. Under what conditions is Y_t covariance stationary?

Exercise 4.7. Realizations from four stochastic processes are given in the Figures below:



Which processes appear to be covariance stationary and which processes appear to be non-stationary? For those processes that you think are non-stationary, explain why the process is non-stationary.

Exercise 4.8. Consider the following model:

$$Y_t = 10 - 0.67Y_{t-1} + \epsilon_t, \quad \epsilon \sim N(0, 1)$$

1. Is it stationary? Why or why not?
2. Find the mean and the variance of this process.

5 Descriptive Statistics for Financial Data

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In this chapter I use graphical and numerical descriptive statistics to study the distribution and dependence properties of daily and monthly asset returns on two representative assets representing a single stock and a large diversified portfolio of stocks. The purpose of this chapter is to introduce the techniques of exploratory data analysis for financial time series and to document a set of stylized facts for monthly and daily asset returns that will be used in later chapters to motivate probability models for asset returns.

This chapter is organized as follows. In Section 5.1 I introduce the example data and reviews graphical and numerical descriptive statistics for univariate data. Graphical descriptive statistics include time plots, histograms, QQ-plots and boxplots. Numerical statistics are the sample statistics associated with the analagous characteristics of univariate distributions reviewed in Chapter 2. In Section 5.2 I cover univariate time series descriptive statistics including sample autocovariances, autocorrelations and the sample autocorrelation function. In Section 5.3 I present bivariate descriptive statistics including scatterplots, sample covariance and correlation, and sample cross-lag covariances and correlations. In Section 5.5 I conclude with a summary of some stylized facts for monthly and daily returns.

The R packages I use in this chapter are **car**, **corrplot**, **introcompfinR**, **PerformanceAnalytics**, **sn**, **tseries**, **zoo** and **xts**. Make sure these packages are installed and loaded before running the R examples in the chapter.

5.1 Univariate Descriptive Statistics

Let $\{R_t\}$ denote a univariate time series of asset returns (simple or continuously compounded). Throughout this chapter I will assume that $\{R_t\}$ is a covariance stationary and ergodic stochastic

process such that:

$$\begin{aligned} E[R_t] &= \mu \text{ independent of } t, \\ \text{var}(R_t) &= \sigma^2 \text{ independent of } t, \\ \text{cov}(R_t, R_{t-j}) &= \gamma_j \text{ independent of } t, \\ \text{corr}(R_t, R_{t-j}) &= \gamma_j/\sigma^2 = \rho_j \text{ independent of } t. \end{aligned}$$

In addition, I will assume that each R_t is identically distributed with unknown pdf $f_R(r)$.

An observed sample of size T of historical asset returns $\{r_t\}_{t=1}^T$ is assumed to be a realization from the stochastic process $\{R_t\}$ for $t = 1, \dots, T$. That is,

$$\{r_t\}_{t=1}^T = \{R_1 = r_1, \dots, R_T = r_T\}$$

The goal of *exploratory data analysis* (eda) is to use the observed sample $\{r_t\}_{t=1}^T$ to learn about the unknown pdf $f_R(r)$ as well as the time dependence properties of $\{R_t\}$.

5.1.1 Example data

I illustrate the descriptive statistical analysis of financial data using daily and monthly adjusted closing prices on Microsoft stock (ticker symbol `msft`) and the S&P 500 index (ticker symbol `^gspc`) over the period January 2, 1998 and May 31, 2012.[1] These data are obtained from (finance.yahoo.com) and are available in the R package **introcompfinr.** I first use the daily and monthly data to illustrate descriptive statistical analysis and to establish a number of stylized facts about the distribution and time dependence in daily and monthly returns.

Example 5.1 (Constructing example data)**.** Daily adjusted closing price data on Microsoft and the S&P 500 index over the period January 4, 1993 through December 31, 2014 are available in the **introcompfinr** package as the `xts` objects `msftDailyPrices` and `sp500DailyPrices`, respectively.

I restrict the sample period to January 2, 1998 and May 31, 2012 using:

```
smpl = "1998-01::2012-05"
msftDailyPrices = msftDailyPrices[smpl]
sp500DailyPrices = sp500DailyPrices[smpl]
```

End-of-month prices are extracted from the daily prices using the **xts** function `to.monthly()`:

[1] An adjusted closing price is adjusted for dividend payments and stock splits. Any dividend payment received between closing dates are added to the close price. If a stock split occurs between the closing dates then the all past prices are divided by the split ratio. The ticker symbol `^gspc` refers to the actual S&P 500 index, which is not a tradeable security. There are several mutual funds (e.g., Vanguard's S&P 500 fund with ticker VFINF) and exchange traded funds (e.g., State Street's SPDR S&P 500 ETF with ticker SPY) which track the S&P 500 index that are investable.

```
msftMonthlyPrices = to.monthly(msftDailyPrices, OHLC=FALSE)
sp500MonthlyPrices = to.monthly(sp500DailyPrices, OHLC=FALSE)
```

It will also be convenient to create a merged `xts` object containing both the Microsoft and S&P 500 index prices:

```
msftSp500DailyPrices = merge(msftDailyPrices, sp500DailyPrices)
msftSp500MonthlyPrices = merge(msftMonthlyPrices, sp500MonthlyPrices)
```

I create `xts` objects containing simple returns using the **PerformanceAnalytics** function `Return.calculate()` (and remove the first `NA` value with `na.omit()`):

```
msftMonthlyRetS = na.omit(Return.calculate(msftMonthlyPrices,
                                           method="simple"))
msftDailyRetS = na.omit(Return.calculate(msftDailyPrices,
                                         method="simple"))
sp500MonthlyRetS = na.omit(Return.calculate(sp500MonthlyPrices,
                                            method="simple"))
sp500DailyRetS = na.omit(Return.calculate(sp500DailyPrices,
                                          method="simple"))
msftSp500MonthlyRetS = na.omit(Return.calculate(msftSp500MonthlyPrices,
                                                method="simple"))
msftSp500DailyRetS = na.omit(Return.calculate(msftSp500DailyPrices,
                                              method="simple"))
```

I also create `xts` objects containing monthly continuously compounded (cc) returns:

```
msftMonthlyRetC = log(1 + msftMonthlyRetS)
sp500MonthlyRetC = log(1 + sp500MonthlyRetS)
msftSp500MonthlyRetC = merge(msftMonthlyRetC, sp500MonthlyRetC)
```

■

5.1.2 Time plots

A natural graphical descriptive statistic for time series data is a *time plot*.

Definition 5.1 (Time Plot)**.** A time plot is simply a line plot with the time series data on the y-axis and the time index on the x-axis.

Time plots are useful for quickly visualizing many features of the time series data.

Example 5.2 (Time plots of monthly prices and returns)**.** A two-panel plot showing the monthly prices is given in Figure 5.1, and is created using the `plot` method for `xts` objects:

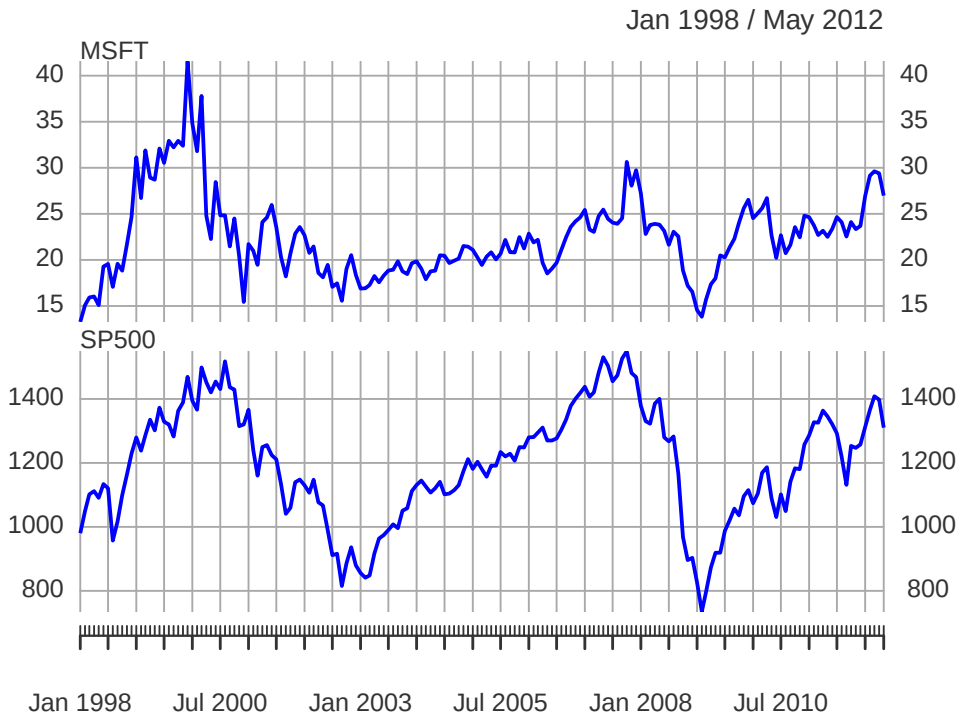


Figure 5.1: End-of-month closing prices on Microsoft stock and the S&P 500 index.

The prices exhibit random walk like behavior (no tendency to revert to a time independent mean) and appear to be non-stationary. Both prices show two large boom-bust periods associated with the dot-com period of the late 1990s and the run-up to the financial crisis of 2008 reflecting some common trend behavior of the two price series.

A time plot for the monthly simple returns is created using:

```
plot(msftSp500MonthlyRetS, multi.panel=TRUE, yaxis.same=FALSE,
     main="", lwd=2, col="blue")
```

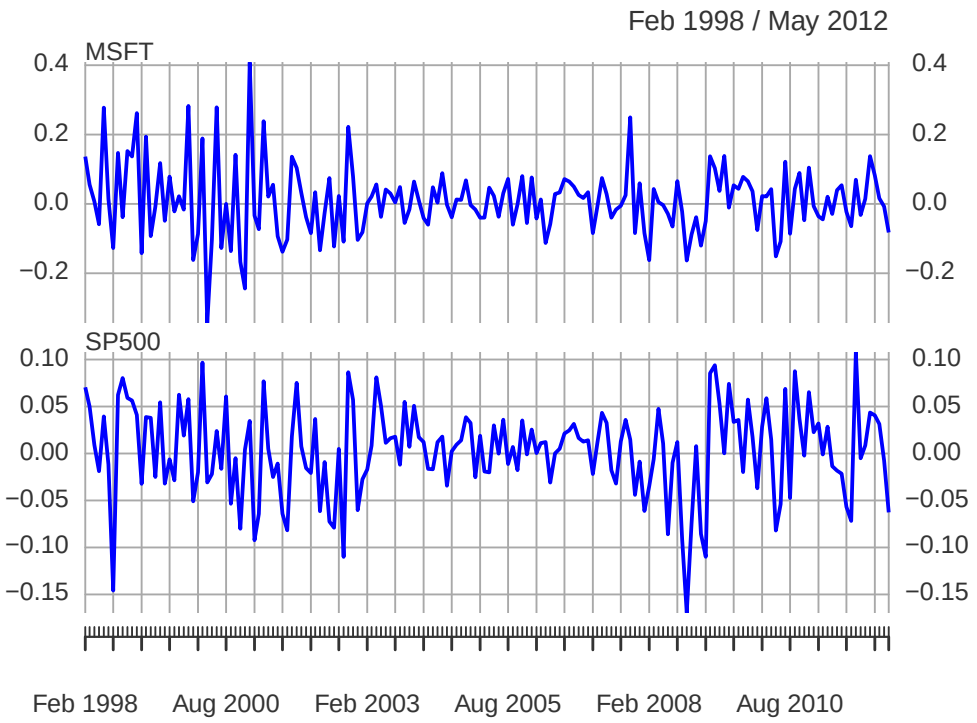


Figure 5.2: Monthly simple returns on Microsoft stock and the S&P 500 index.

and is given in Figure 5.2. In contrast to prices, returns show clear mean-reverting behavior and the common monthly mean values look to be very close to zero. Hence, the common mean value assumption of covariance stationarity looks to be satisfied. However, the volatility (i.e., fluctuation of returns about the mean) of both series appears to change over time. Both series show higher volatility over the periods 1998 - 2003 and 2008 - 2012 than over the period 2003 - 2008. This is an indication of possible non-stationarity in volatility.[2] Also, the coincidence of high and low volatility periods across assets suggests a common driver to the time varying behavior of volatility. There does not appear to be any visual evidence of systematic time dependence in the returns. Later on you will see that the estimated autocorrelations are very close to zero. The returns for Microsoft and the S&P 500 tend to go up and down together suggesting a positive correlation. ■

Example 5.3 (Plotting returns on the same graph)**.** In Figure 5.2, the volatility of the returns on Microsoft and the S&P 500 looks to be similar but this is illusory. The y-axis scale for Microsoft is much larger than the scale for the S&P 500 index and so the volatility of Microsoft returns is actually much larger than the volatility of the S&P 500 returns. Figure 5.3 shows both returns series on the same time plot created using:

[2]The returns can still be covariance stationary and exhibit time varying *conditional* volatility. This is explored in chapter 10.

```
plot(msftSp500MonthlyRetS, main="", multi.panel=FALSE, lwd=2,
     col=c("red", "blue"), lty=c("dashed", "solid"),
     legend.loc = "topright")
```

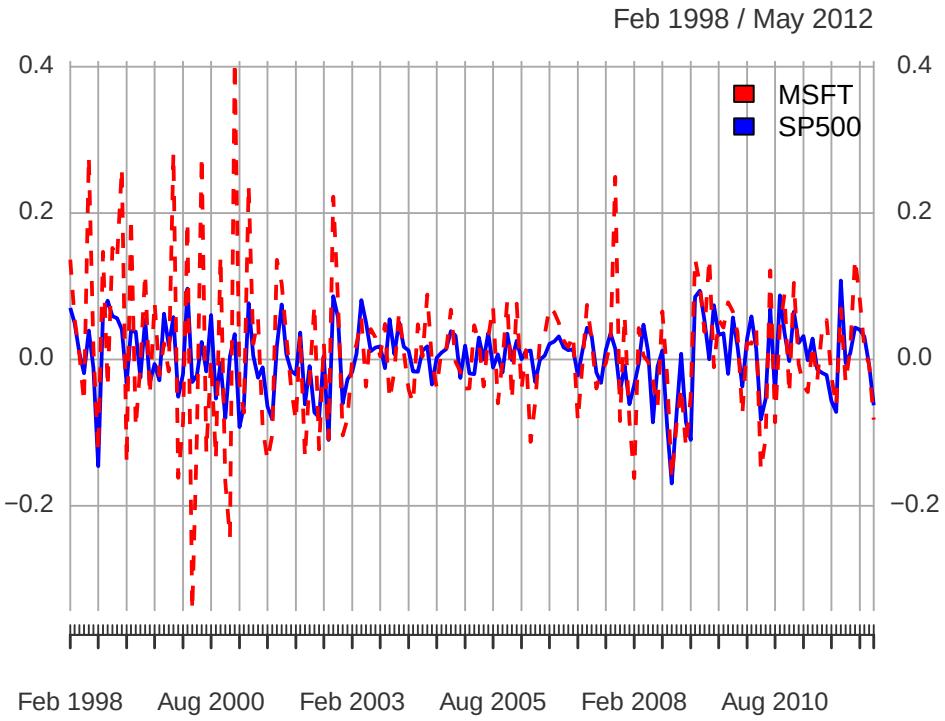


Figure 5.3: Monthly simple returns for Microsoft and S&P 500 index on the same graph.

Now the higher volatility of Microsoft returns, especially before 2003, is clearly visible. However, after 2008 the volatilities of the two series look quite similar. In general, the lower volatility of the S&P 500 index represents risk reduction due to holding a large diversified portfolio. ■

Example 5.4 (Comparing simple and continuously compounded returns)**.** Figure 5.4 compares the simple and continuously compounded (cc) monthly returns for Microsoft created using:

```
retDiff = msftMonthlyRetS - msftMonthlyRetC
dataToPlot = merge(msftMonthlyRetS, msftMonthlyRetC, retDiff)
colnames(dataToPlot) = c("Simple", "cc", "Diff")
plot(dataToPlot, multi.panel=TRUE, yaxis.same=FALSE, main="",
         lwd=2, col=c("black", "blue", "red"))
```

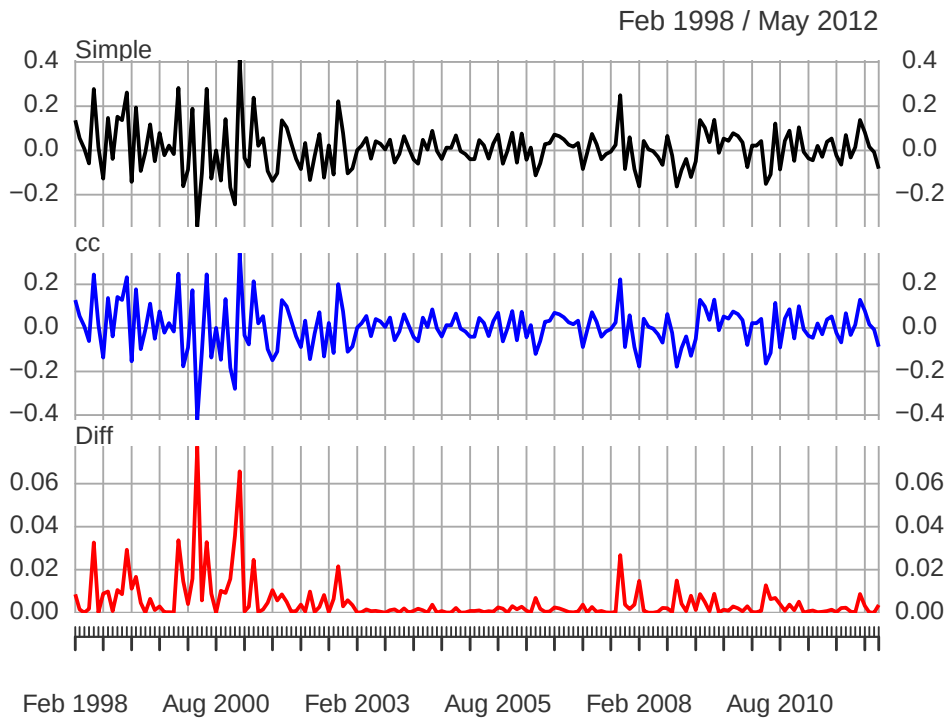


Figure 5.4: Monthly simple and cc returns on Microsoft. Top panel: simple returns; middle panel: cc returns; bottom panel: difference between simple and cc returns.

The top panel shows the simple returns, the middle panel shows the cc returns, and the bottom panel shows the difference between the simple and cc returns. Qualitatively, the simple and cc returns look almost identical. The main differences occur when the simple returns are large in absolute value (e.g., mid 2000, early 2001 and early 2008). In this case the difference between the simple and cc returns can be fairly large (e.g., as large as 0.08 in mid 2000). When the simple return is large and positive, the cc return is not quite as large; when the simple return is large and negative, the cc return is a larger negative number. ■

Example 5.5 (Plotting daily returns)**.** Figure 5.5 shows the daily simple returns on Microsoft and the S&P 500 index created with:

```
plot(msftSp500DailyRetS, multi.panel=TRUE, yaxis.same=FALSE,
     main="", col="blue")
```

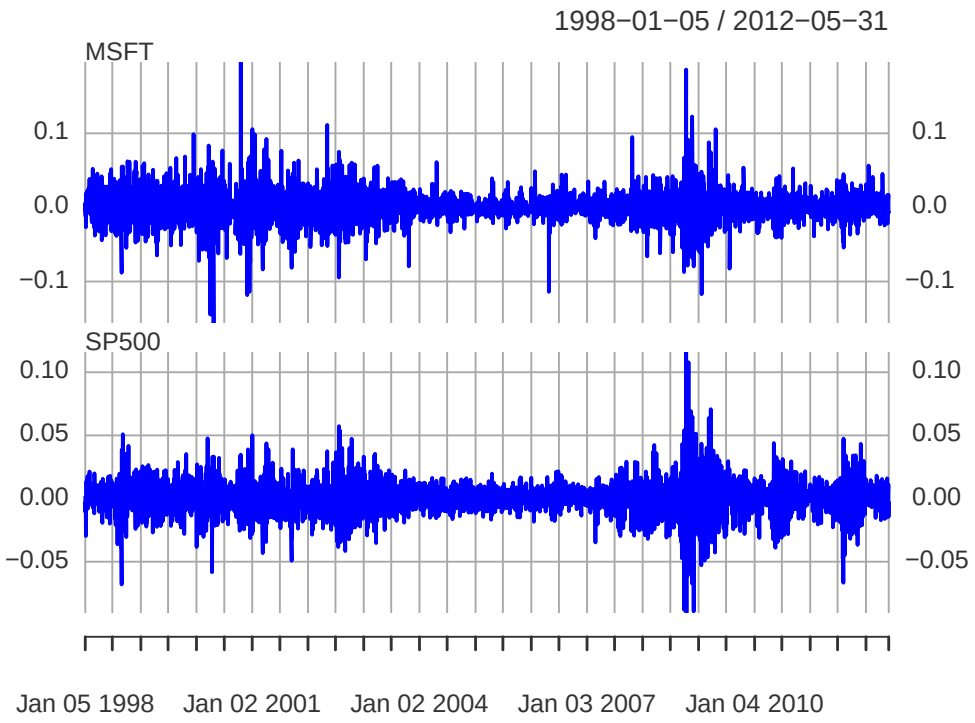


Figure 5.5: Daily returns on Microsoft and the S&P 500 index.

Compared with the monthly returns, the daily returns are closer to zero and the magnitude of the fluctuations (volatility) is smaller. However, the clustering of periods of high and low volatility is more pronounced in the daily returns. As with the monthly returns, the volatility of the daily returns on Microsoft is larger than the volatility of the S&P 500 returns. Also, the daily returns show some large positive and negative "spikes" that represent unusually large (inabsolute value) daily movements (e.g., Microsoft up 20% in one day and down 15% on another day). The monthly returns do not exhibit such extreme movements relative to typical volatility. ■

5.1.3 Descriptive statistics for the distribution of returns

In this section, I consider graphical and numerical descriptive statistics for the unknown marginal pdf, $f_R(r)$, of returns. Recall, I assume that the observed sample $\{r_t\}_{t=1}^T$ is a realization from a covariance stationary and ergodic time series $\{R_t\}$ where each R_t is a continuous random variable with common pdf $f_R(r)$. The goal is to use $\{r_t\}_{t=1}^T$ to describe properties of $f_R(r)$.

I study returns and not prices because prices are random walk non-stationary. Sample descriptive statistics are only meaningful for covariance stationary and ergodic time series.

5.1.3.1 Histograms

Definition 5.2 (Histogram)**.** A *histogram* of returns is a graphical summary used to describe the general shape of the unknown pdf $f_R(r)$. It is constructed as follows. Order returns from smallest to largest. Divide the range of observed values into N equally sized bins. Show the number or fraction of observations in each bin using a bar chart.

Example 5.6 (Histograms for the daily and monthly returns on Microsoft and the S&P 500 index)**.** Figure 5.6 shows the histograms of the daily and monthly returns on Microsoft stock and the S&P 500 index created using the R function `hist()`:

```
par(mfrow=c(2,2))
hist(msftMonthlyRetS, main="", col="cornflowerblue")
hist(msftDailyRetS, main="", col="cornflowerblue")
hist(sp500MonthlyRetS, main="", col="cornflowerblue")
hist(sp500DailyRetS, main="", col="cornflowerblue")
```

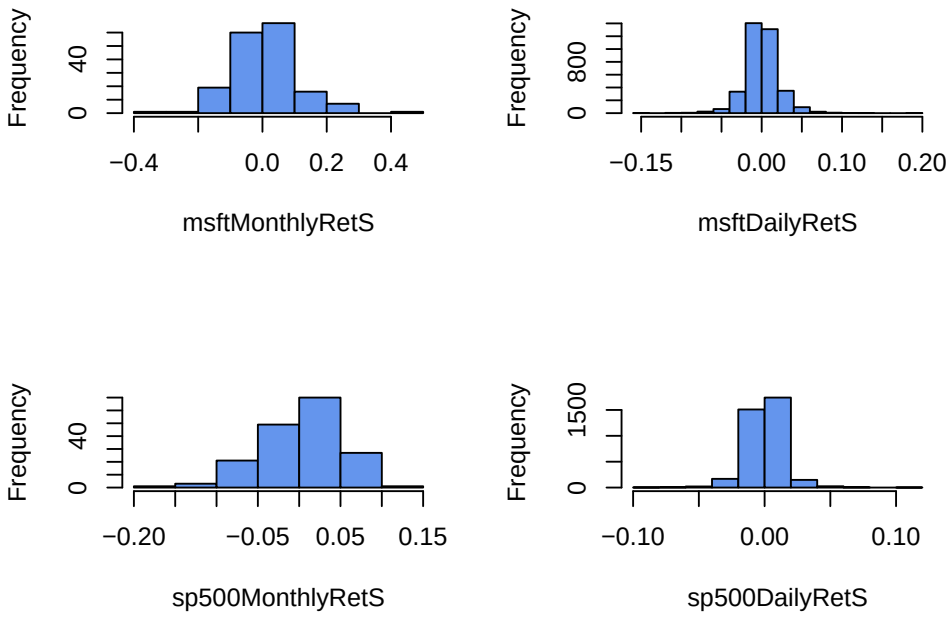


Figure 5.6: Histograms of monthly continuously compounded returns on Microsoft stock and S&P 500 index.

All histograms have a bell-shape like the normal distribution. The histograms of daily returns are centered around zero and those for the monthly return are centered around values slightly larger

than zero. The bulk of the daily (monthly) returns for Microsoft and the S&P 500 are between -5% and 5% (-20% and 20%) and -3% and 3% (-10% and 10%), respectively. The histogram for the S&P 500 monthly returns is slightly skewed left (long left tail) due to more large negative returns than large positive returns whereas the histograms for the other returns are roughly symmetric.

When comparing two or more return distributions, it is useful to use the same bins for each histogram. Figure 5.7 shows the histograms for Microsoft and S&P 500 returns using the same 15 bins, created with the R code:

```
msftHist = hist(msftMonthlyRetS, plot=FALSE, breaks=15)
par(mfrow=c(2,2))
hist(msftMonthlyRetS, main="", col="cornflowerblue")
hist(msftDailyRetS, main="", col="cornflowerblue",
     breaks=msftHist$breaks)
hist(sp500MonthlyRetS, main="", col="cornflowerblue",
     breaks=msftHist$breaks)
hist(sp500DailyRetS, main="", col="cornflowerblue",
     breaks=msftHist$breaks)
```

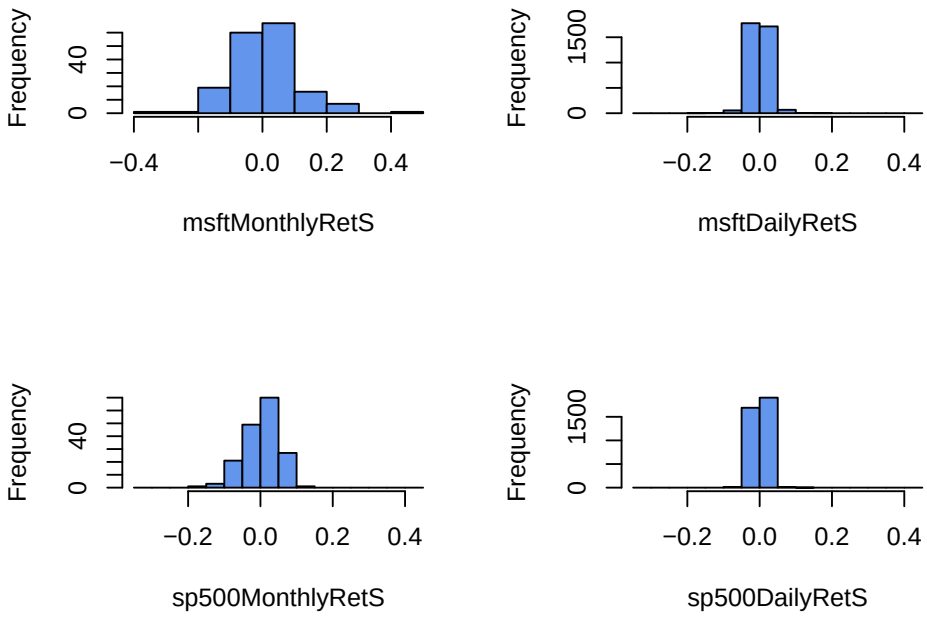


Figure 5.7: Histograms for Microsoft and S&P 500 returns using the same bins.

The common bins for each histogram are specified using the optional argument `breaks=msftHist$breaks`

in the call to `hist()`. Using the same bins for all histograms allows us to see more clearly that the distribution of monthly returns is more spread out than the distributions for daily returns, and that the distribution of S&P 500 returns is more tightly concentrated around zero than the distribution of Microsoft returns. ■

Example 5.7 (Are Microsoft returns normally distributed? A first look)**.** The shape of the histogram for Microsoft returns suggests that a normal distribution might be a good candidate for the unknown distribution of Microsoft returns. To investigate this conjecture, I simulate random returns from a normal distribution with mean and standard deviation calibrated to the Microsoft daily and monthly returns using:

```
set.seed(123)
gwnDaily = rnorm(length(msftDailyRetS),
                 mean=mean(msftDailyRetS), sd=sd(msftDailyRetS))
gwnDaily = xts(gwnDaily, index(msftDailyRetS))
gwnMonthly = rnorm(length(msftMonthlyRetS),
                   mean=mean(msftMonthlyRetS), sd=sd(msftMonthlyRetS))
gwnMonthly = xts(gwnMonthly, index(msftMonthlyRetS))
```

Figure 5.8 shows the Microsoft monthly returns together with the simulated normal returns.

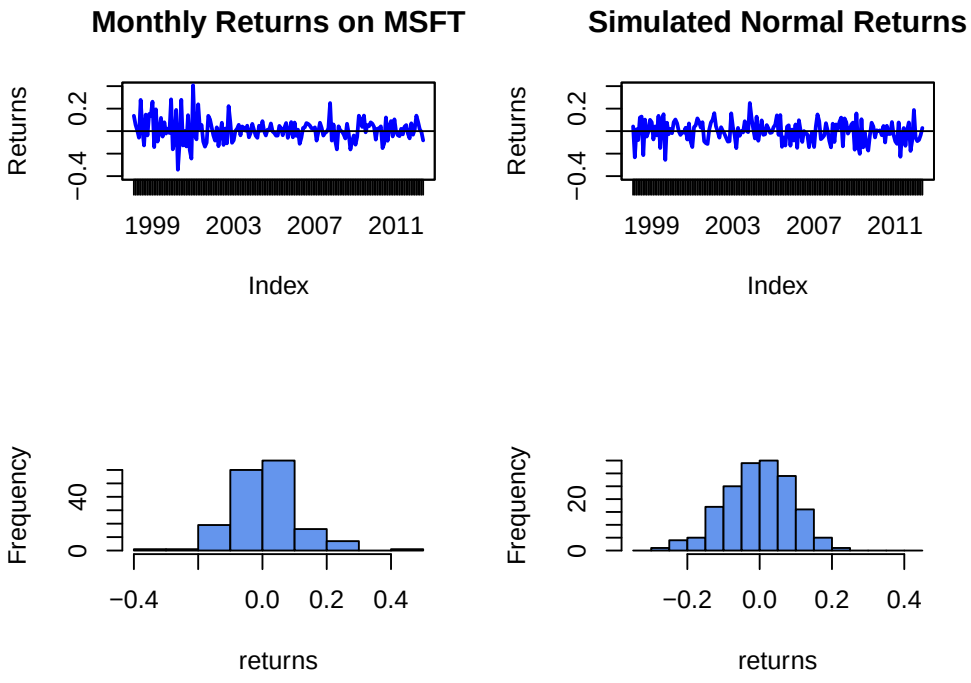


Figure 5.8: Comparison of Microsoft monthly returns with simulated normal returns with the same mean and standard deviation as the Microsoft returns.

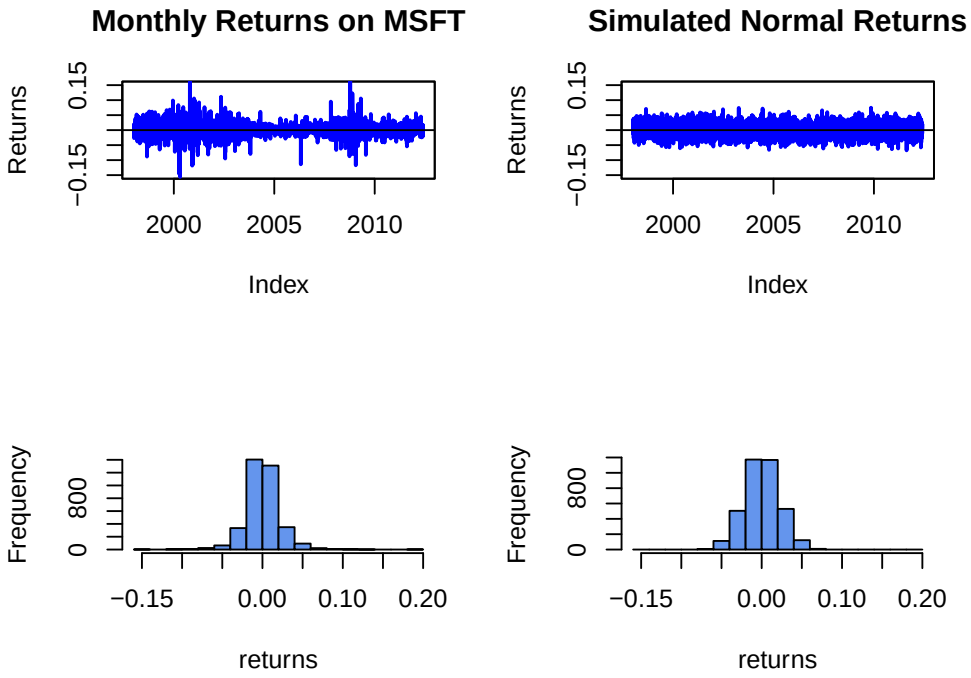


Figure 5.9: Comparison of Microsoft daily returns with simulated normal returns with the same mean and standard deviation as the Microsoft returns.

The simulated normal returns shares many of the same features as the Microsoft returns: both fluctuate randomly about zero. However, there are some important differences. In particular, the volatility of Microsoft returns appears to change over time (large before 2003, small between 2003 and 2008, and large again after 2008) whereas the simulated returns has constant volatility. Additionally, the distribution of Microsoft returns has fatter tails (more extreme large and small returns) than the simulated normal returns. Apart from these features, the simulated normal returns look remarkably like the Microsoft monthly returns.

Figure 5.9 shows the Microsoft daily returns together with the simulated normal returns. The daily returns look much less like GWN than the monthly returns. Here the constant volatility of simulated GWN does not match the volatility patterns of the Microsoft daily returns, and the tails of the histogram for the Microsoft returns are "fatter" than the tails of the GWN histogram. ■

5.1.3.2 Smoothed histogram

Histograms give a good visual representation of the data distribution. The shape of the histogram, however, depends on the number of bins used. With a small number of bins, the histogram often appears blocky and fine details of the distribution are not revealed. With a large number of bins,

the histogram might have many bins with very few observations. The `hist()` function in R smartly chooses the number of bins so that the resulting histogram typically looks good.

The main drawback of the histogram as a descriptive statistic for the underlying pdf of the data is that it is discontinuous. If it is believed that the underlying pdf is continuous, it is desirable to have a continuous graphical summary of the pdf. The *smoothed histogram* achieves this goal. Given a sample of data $\{x_t\}_{t=1}^T$ the R function `density()` computes a smoothed estimate of the underlying pdf $f_X(x)$ at each point x in the bins of the histogram using the formula:

$$\hat{f}_X(x) = \frac{1}{Tb} \sum_{t=1}^{T} k\left(\frac{x - x_t}{b}\right),$$

where $k(\cdot)$ is a continuous smoothing function (typically a standard normal distribution) and b is a bandwidth (or bin-width) parameter that determines the width of the bin around x in which the smoothing takes place. The resulting pdf estimate $\hat{f}_X(x)$ is a two-sided weighted average of the histogram values around x.[3]

Example 5.8 (Smoothed histogram for Microsoft monthly returns)**.** Figure 5.10 shows the histogram of Microsoft returns overlaid with the smoothed histogram created using:

```
par(mfrow=c(1,1))
MSFT.density = density(msftMonthlyRetS)
hist(msftMonthlyRetS, main="", xlab="Microsoft Monthly Returns",
     col="cornflowerblue", probability=TRUE, ylim=c(0,5.5))
points(MSFT.density,type="l", col="orange", lwd=2)
```

[3] In this book, a "hat" on top of a symbol denotes that it is an estimate the underlying symbol.

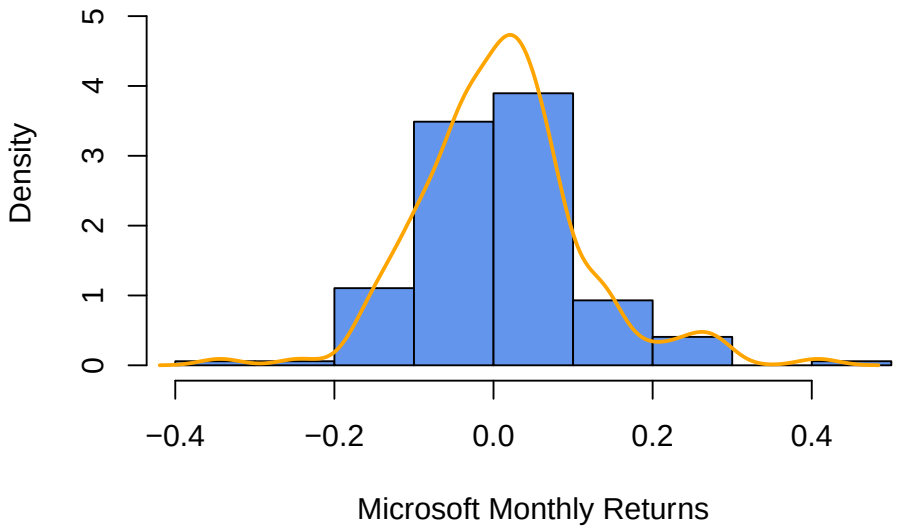


Figure 5.10: Histogram and smoothed density estimate for the monthly returns on Microsoft.

In Figure 5.10, the histogram is normalized (using the argument `probability=TRUE`), so that its total area is equal to one. The smoothed density estimate transforms the blocky shape of the histogram into a smooth continuous graph. ■

5.1.3.3 Empirical CDF

Recall, the CDF of a random variable X is the function $F_X(x) = \Pr(X \leq x)$. The empirical CDF $\hat{F}_X(x)$ is the sample analogue of $F_X(x)$.

Definition 5.3 (Empirical CDF)**.** The *empirical CDF* of a data sample $\{x_t\}_{t=1}^T$ is the function that counts the fraction of observations less than or equal to x:

$$\begin{aligned} \hat{F}_X(x) &= \frac{1}{T}(\#x_i \leq x) \\ &= \frac{\text{number of } x_i \text{ values } \leq x}{\text{sample size}} \end{aligned} \tag{5.1}$$

Example 5.9 (Empirical CDF for monthly returns on Microsoft)**.** Computing the empirical CDF of Microsoft monthly returns at a given point, say $r = 0$, is a straightforward calculation in R:

```
Fhat.0 = sum(msftMonthlyRetS <= 0)/nrow(msftMonthlyRetS)
Fhat.0
#> [1] 0.471
```

Here, the expression `msftMonthlyRetS <= 0` creates a logical vector the same length as `msftMonthlyRetS` that is equal to `TRUE` when returns are less than or equal to 0 and `FALSE` when returns are greater than zero. Then `Fhat.0` is equal to the fraction of `TRUE` values (returns less than or equal to zero) in the data, which gives Equation 5.1 evaluated at zero. The R function `ecdf()` can be used to compute and plot Equation 5.1 for all of the observed returns. For example, Figure 5.11 shows the empirical CDF for Microsoft monthly returns computed using:

```
plot(ecdf(coredata(msftMonthlyRetS)), main="", col="blue")
```

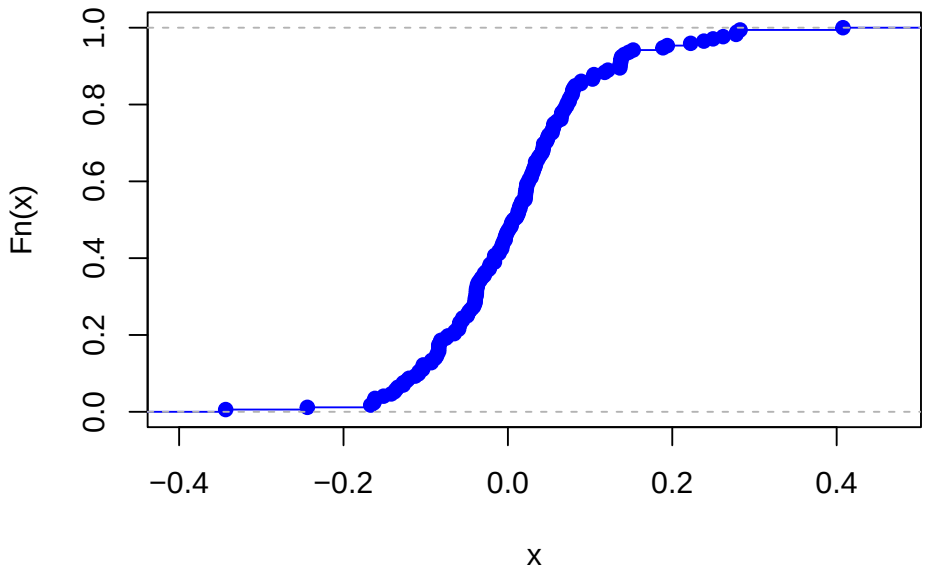


Figure 5.11: Empirical CDF of monthly returns on Microsoft

The "S" shape of the empirical CDF of Microsoft monthly returns is due to the symmetric shape of the histogram of monthly returns and resembles the CDF of a normal distribution. ■

5.1.3.4 Empirical quantiles/percentiles

Recall, for $\alpha \in (0,1)$ the $\alpha \times 100\%$ quantile of the distribution of a continuous random variable X with CDF F_X is the point q_α^X such that $F_X(q_\alpha^X) = \Pr(X \leq q_\alpha^X) = \alpha$. The *empirical quantile* is defined similarly using the empirical CDF of the data.

Definition 5.4 (Empirical Quantile)**.** The $\alpha \times 100\%$ *empirical quantile* (or $100 \times \alpha^{\text{th}}$ *percentile*) of a data sample $\{x_t\}_{t=1}^T$ is the data value $\hat{q}_\alpha$ such that $\alpha \cdot 100\%$ of the data are less than or equal to $\hat{q}_\alpha$.

Empirical quantiles can be easily determined by ordering the data from smallest to largest giving the ordered sample (also known as *order statistics*):

$$x_{(1)} < x_{(2)} < \cdots < x_{(T)}.$$

The empirical quantile $\hat{q}_\alpha$ is the order statistic closest to $\alpha \times T$.[4]

The *empirical quartiles* are the empirical quantiles for $\alpha = 0.25, 0.5$ and 0.75, respectively. The second empirical quartile $\hat{q}_{.50}$ is called the *sample median* and is the data point such that half of the data is less than or equal to its value. The *interquartile range* (IQR) is the difference between the 3rd and 1st quartile:

$$\text{IQR} = \hat{q}_{.75} - \hat{q}_{.25},$$

and shows the size of the middle of the data distribution.

Example 5.10 (Empirical quantiles of the Microsoft and S&P 500 monthly returns)**.** The R function `quantile()` computes empirical quantiles for a single data series. By default, `quantile()` returns the empirical quartiles as well as the minimum and maximum values:

```
apply(msftSp500MonthlyRetS, 2, quantile)
#>          MSFT    SP500
#> 0%   -0.34339 -0.16942
#> 25%  -0.04883 -0.02094
#> 50%   0.00854  0.00809
#> 75%   0.05698  0.03532
#> 100%  0.40765  0.10772
```

In the above code, the R function `apply()` is used to compute the empirical quantiles on each column of the `xts` object `msftSp500MonthlyRetS`. The syntax for `apply()` is:

[4] There is no unique way to determine the empirical quantile from a sample of size N for all values of α. The R function `quantile()` can compute empirical quantile using one of seven different definitions.

```
args(apply)
#> function (X, MARGIN, FUN, ..., simplify = TRUE)
#> NULL
```

where `X` is a multi-column data object, `MARGIN` specifies looping over rows (`MARGIN=1`) or columns (`MARGIN=2`), `FUN` specifies the R function to be applied, and `...` represents any optional arguments to be passed to `FUN`.

Here you see that the lower (upper) quartiles of the Microsoft monthly returns are smaller (larger) than the respective quartiles for the S&P 500 index.

To compute quantiles for a specified α use the `probs` argument. For example, to compute the 1% and 5% quantiles of the monthly returns use:

```
apply(msftSp500MonthlyRetS, 2, quantile, probs=c(0.01,0.05))
#>      MSFT   SP500
#> 1% -0.189 -0.1204
#> 5% -0.137 -0.0818
```

Historically, there is a 1% chance of losing more than 19% (12%) on Microsoft (S&P 500) in a given month, and a 5% chance of losing more than 14% (8%) on Microsoft (S&P 500) in a given month.

To compute the median and IQR values for monthly returns use the R functions `median()` and `IQR()`, respectively:

```
med = apply(msftSp500MonthlyRetS, 2, median)
iqr = apply(msftSp500MonthlyRetS, 2, IQR)
rbind(med, iqr)
#>        MSFT   SP500
#> med 0.00854 0.00809
#> iqr 0.10581 0.05626
```

The median returns are similar (about 0.8% per month) but the IQR for Microsoft is about twice as large as the IQR for the S&P 500 index. ■

5.1.3.5 Empirical/Historical VaR

Recall, the $\alpha \times 100\%$ value-at-risk (VaR) of an investment of $\$W$ is $\text{VaR}_\alpha = -\$W \times q_\alpha^R$, where q_α^R is the $\alpha \times 100\%$ quantile of the probability distribution of the investment simple rate of return R (recall, you multiply by -1 to represent loss as a positive number). The *empirical VaR* or *historical VaR* is defined similarly using the *empirical quantile*.

Definition 5.5 (Empirical/Historical VaR)**.** The $\alpha \times 100\%$ *empirical VaR* (sometimes called *historical VaR* or *historical simulation VaR*) of an investment of $\$W$ is defined as:

$$\text{VaR}_\alpha^{HS} = -\$W \times \hat{q}_\alpha^R, \tag{5.2}$$

where $\hat{q}_\alpha^R$ is the empirical α quantile of a sample of simple returns $\{R_t\}_{t=1}^T$. For a sample of continuously compounded returns $\{r_t\}_{t=1}^T$ with empirical α quantile $\hat{q}_\alpha^r$,

$$\text{VaR}_\alpha^{HS} = -\$W \times (\exp(\hat{q}_\alpha^r) - 1)\,. \tag{5.3}$$

Historical VaR is based on the distribution of the observed returns and not on any assumed distribution for returns (e.g., the normal distribution or the log-normal distribution).

Example 5.11 (Using empirical quantiles to compute historical VaR)**.** Consider investing $W = \$100,000$ in Microsoft and the S&P 500 over a month. The 1% and 5% historical VaR values for these investments based on the historical samples of monthly returns are:

```
W = 100000
msftQuantiles = quantile(msftMonthlyRetS, probs=c(0.01, 0.05))
sp500Quantiles = quantile(sp500MonthlyRetS, probs=c(0.01, 0.05))
msftVaR = -W*msftQuantiles
sp500VaR = -W*sp500Quantiles
cbind(msftVaR, sp500VaR)
#>    msftVaR sp500VaR
#> 1%   18929    12040
#> 5%   13694     8184
```

Based on the empirical distribution of the monthly returns, a $100,000 monthly investment in Microsoft will lose $13,694 or more with 5% probability and will lose $18,929 or more with 1% probability. The corresponding values for the S&P 500 are $8,183 and $12,039, respectively. The historical VaR values for the S&P 500 are considerably smaller than those for Microsoft. In this sense, investing in Microsoft is riskier than investing in the S&P 500 index. ■

5.1.4 QQ-plots

Often it is of interest to see if a given data sample could be viewed as a random sample from a specified probability distribution. One easy and effective way to do this is to compare the empirical quantiles of a data sample to the quantiles from a reference probability distribution. If the quantiles match up, then this provides strong evidence that the reference distribution is appropriate for describing the distribution of the observed data. If the quantiles do no,t match

up, then the observed differences between the empirical quantiles and the reference quantiles can be used to determine a more appropriate reference distribution. It is common to use the normal distribution as the reference distribution, but any distribution can in principle be used.

Definition 5.6 (Quantile-Quantile Plot)**.** The *quantile-quantile plot* (QQ-plot) gives a graphical comparison of the empirical quantiles of a data sample to those from a specified reference distribution. The QQ-plot is an xy-plot with the reference distribution quantiles on the x-axis and the empirical quantiles on the y-axis. If the quantiles exactly match up then the QQ-plot is a straight line. If the quantiles do not match up, then the shape of the QQ-plot indicates which features of the data are not captured by the reference distribution.

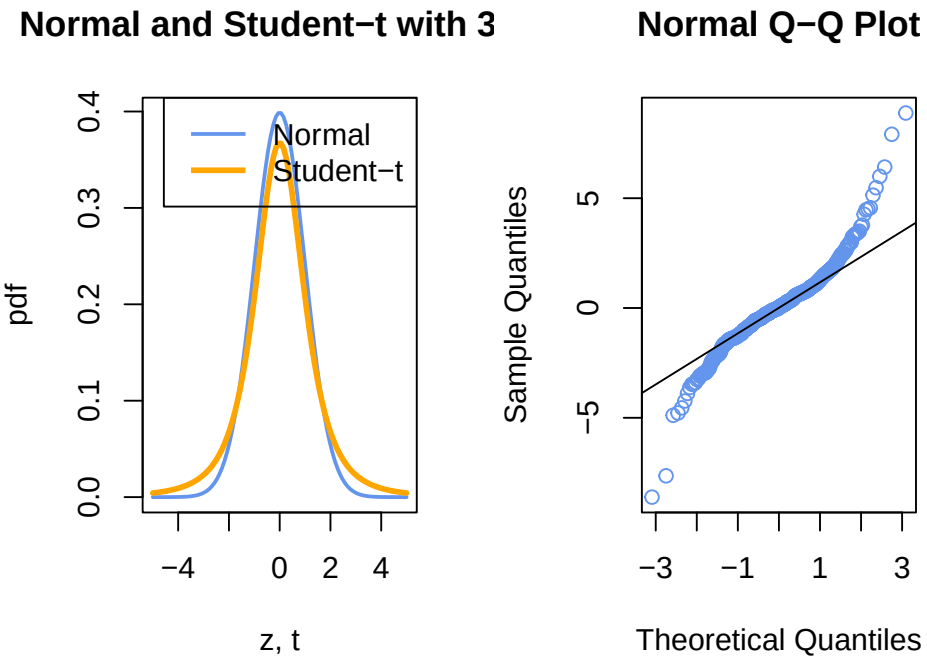


Figure 5.12: Normal QQ-Plot for Fat-tailed Data

Example 5.12 (QQ-plots for non-normal data)**.** Figure 5.12 shows a normal QQ-plot for data simulated from a Student's t distribution with 3 degrees-of-freedom which has fatter tails than the normal distribution. Notice how the normal QQ-plot deviates from the straight line for both large and small quantiles of the normal distribution. This S-shape tells us that both extremely small and extremely large empirical quantiles (on the vertical axis) are larger (in absolute value) than the corresponding theoretical quantiles of the normal distribution.

Figure 5.13 shows a normal QQ-plot for data simulated from a distribution with positive skewness which has more density in the right tail than in the left tail. Here the normal QQ-plot has a convex

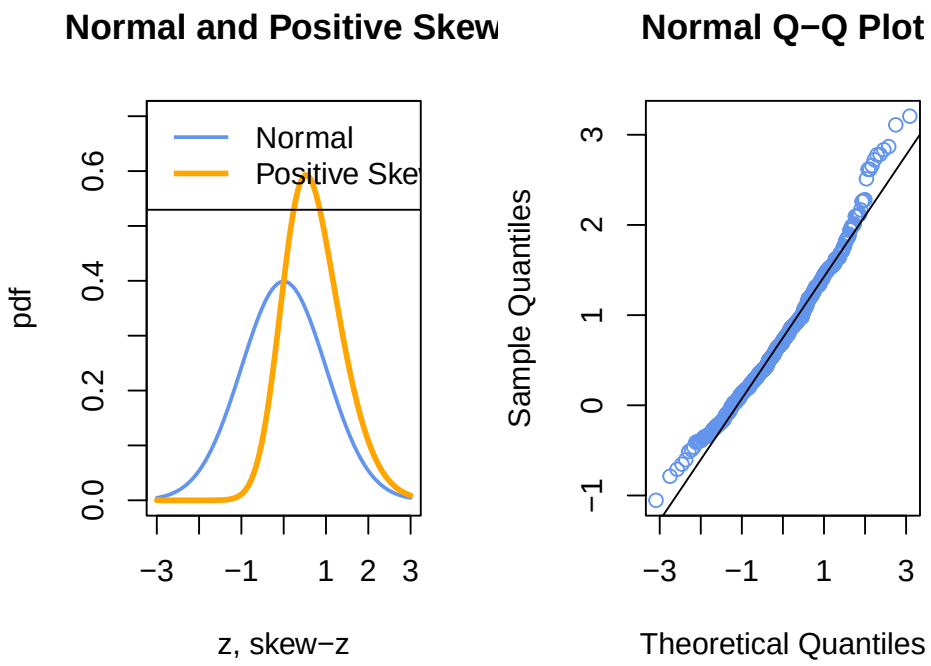


Figure 5.13: Normal QQ-Plot for Positively Skewed Data

shape relative to the straight line indicating that the lower (upper) empirical quantiles are closer to (farther from) zero than the corresponding theoretical quantiles of the normal distribution.

Figure 5.14 shows a normal QQ-plot for data simulated from a distribution with negative skewness which has more density in the left tail than in the right tail. Here the normal QQ-plot has a concave shape relative to the straight line indicating that the lower (upper) empirical quantiles are farther from (closer to) zero than the corresponding theoretical quantiles of the normal distribution. ■

:::

Example 5.13 (Normal QQ-plots for GWN Microsoft and S&P 500 returns)**.** The R function `qqnorm()` creates a QQ-plot for a data sample using the normal distribution as the reference distribution. Figure 5.15 shows normal QQ-plots for the simulated GWN data, Microsoft returns and S&P 500 returns.

The normal QQ-plot for the simulated GWN data is very close to a straight line, as it should be since the data are simulated from a normal distribution.[5] The `qqline()` function draws a straight line through the points to help determine if the quantiles match up. The normal QQ-plots for the Microsoft and S&P 500 monthly returns are linear in the middle of the distribution but

[5]Even with data simulated from a normal distribution, the empirical quantiles in the extreme left and right tails can deviate a bit from the straight line due to estimation error. For this reason, small deviations from linearity in the tails should be ignored.

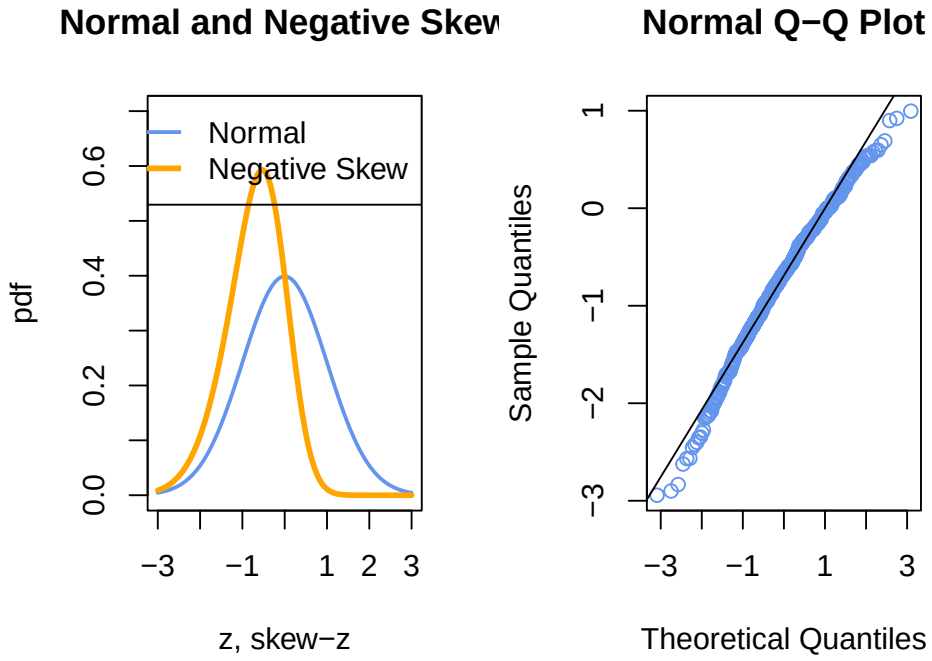


Figure 5.14: Normal QQ-Plot for Negatively Skewed Data

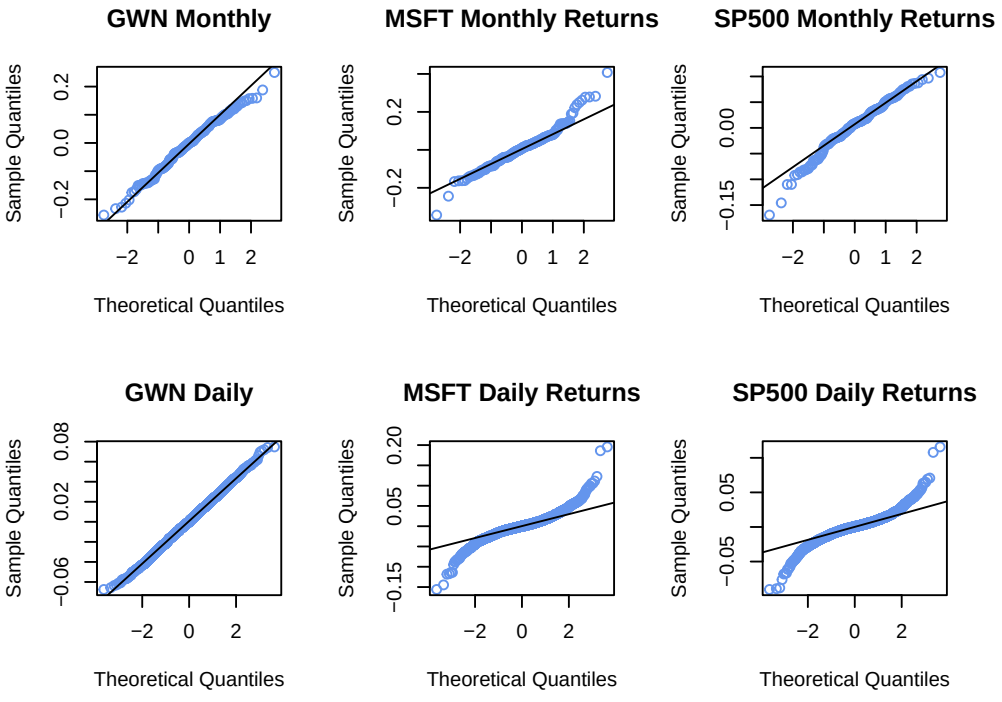


Figure 5.15: Normal QQ-plots for GWN, Microsoft returns and S&P 500 returns.

deviate from linearity in the tails of the distribution. In the normal QQ-plot for Microsoft monthly returns, the theoretical normal quantiles on the x-axis are too small in both the left and right tails because the points fall below the straight line in the left tail and fall above the straight line in the right tail. Hence, the normal distribution does not match the empirical distribution of Microsoft monthly returns in the extreme tails of the distribution. In other words, the Microsoft returns have *fatter tails* than the normal distribution. For the S&P 500 monthly returns, the theoretical normal quantiles are too small only for the left tail of the empirical distribution of returns (points fall below the straight line in the left tail only). This reflects the long left tail (negative skewness) of the empirical distribution of S&P 500 returns. The normal QQ-plots for the daily returns on Microsoft and the S&P 500 index both exhibit a pronounced tilted S-shape with extreme departures from linearity in the left and right tails of the distributions. The daily returns clearly have much fatter tails than the normal distribution. ■

Example 5.14 (Student's t QQ-plot for Microsoft returns)**.** The function `qqPlot()` from the **car** package or the function `chart.QQPlot()` from the **PerformanceAnalytics** package can be used to create a QQ-plot against any reference distribution that has a corresponding quantile function implemented in R. For example, a QQ-plot for the Microsoft returns using a Student's t reference distribution with 5 degrees of freedom can be created using:

```
qqPlot(coredata(msftMonthlyRetS), distribution="t",
       df=5, ylab="MSFT quantiles", envelope=FALSE, id=FALSE)
```

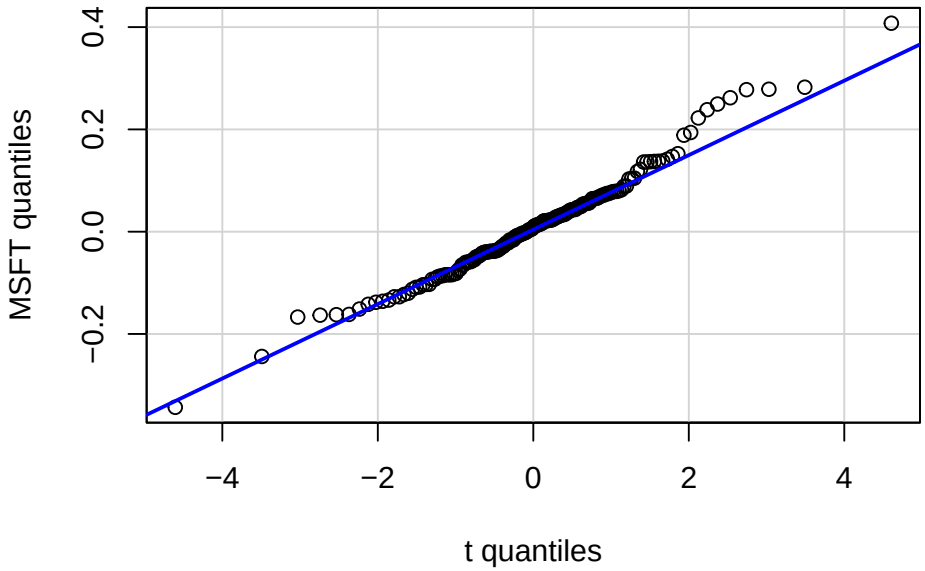


Figure 5.16: QQ-plot of Microsoft returns using Student's t distribution with 5 degrees of freedom as the reference distribution.

In the function `qqPlot()`, the argument `distribution= "t"` specifies that the quantiles are to be computed using the R function `qt()`.[6] Figure 5.16 shows the resulting graph. Here, with a reference distribution with fatter tails than the normal distribution the QQ-plot for Microsoft returns is closer to a straight line. This indicates that the Student's t distribution with 5 degrees of freedom is a better reference distribution for Microsoft returns than the normal distribution. ■

5.1.5 Shape characteristics of the empirical distribution

Recall, for a random variable X the shape measures of center, spread, asymmetry and tail thickness of the pdf are:

$$\begin{aligned}
\text{center} &: \mu_X = E[X], \\
\text{spread} &: \sigma_X^2 = \text{var}(X) = E[(X - \mu_X)^2], \\
\text{spread} &: \sigma_X = \sqrt{\text{var}(X)} \\
\text{asymmetry} &: \text{skew}_X = E[(X - \mu_X)^3]/\sigma^3, \\
\text{tail thickness} &: \text{kurt}_X = E[(X - \mu_X)^4]/\sigma^4.
\end{aligned}$$

[6] The `coredata()` function is used to extract the data as a `"numeric"` object because `qqPlot()` does not work correctly with `"zoo"` or `xts` objects.

Definition 5.7 (Sample Statistics)**.** The shape measures for the empirical distribution (e.g., as measured by the histogram) of a data sample $\{x_t\}_{t=1}^T$ are the *sample statistics*:

$$\hat{\mu}_x = \bar{x} = \frac{1}{T}\sum_{t=1}^{T} x_t, \tag{5.4}$$

$$\hat{\sigma}_x^2 = s_x^2 = \frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})^2, \tag{5.5}$$

$$\hat{\sigma}_x = \sqrt{\hat{\sigma}_x^2}, \tag{5.6}$$

$$\widehat{\text{skew}}_x = \frac{\frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})^3}{s_x^3}, \tag{5.7}$$

$$\widehat{\text{kurt}}_x = \frac{\frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})^4}{s_x^4}. \tag{5.8}$$

The sample mean, $\hat{\mu}_X$, measures the center of the histogram; the sample standard deviation, $\hat{\sigma}_x$, measures the spread of the data about the mean in the same units as the data; the sample skewness, $\widehat{\text{skew}}_x$, measures the asymmetry of the histogram; the sample kurtosis, $\widehat{\text{kurt}}_x$, measures the tail-thickness of the histogram. The sample excess kurtosis, defined as the sample kurtosis minus 3:

$$\widehat{\text{ekurt}}_x = \widehat{\text{kurt}}_x - 3, \tag{5.9}$$

measures the tail thickness of the data sample relative to that of a normal distribution.[7]

Notice that the divisor in Equation 5.5 - Equation 5.8 is $T-1$ and not T. This is called a *degrees-of-freedom* correction. In computing the sample variance, skewness and kurtosis, one degree-of-freedom in the sample is used up in the computation of the sample mean so that there are effectively only $T-1$ observations available to compute the statistics.[8]

Example 5.15 (Sample shape statistics for the returns on Microsoft and S&P 500)**.** The R functions for computing Equation 5.4 - Equation 5.8 are `mean()`, `var()` and `sd()`, respectively. There are no functions for computing Equation 5.7 and Equation 5.8 in base R. The functions `skewness()` and `kurtosis()` in the **PerformanceAnalytics** package compute the sample skewness Equation 5.7 and the sample excess kurtosis Equation 5.9, respectively.[9] The sample statistics for the Microsoft and S&P 500 monthly returns are computed using:

[7] Values with hats, `"^"`, denote sample estimates of the corresponding population quantity. For example, the sample mean $\hat{\mu}_x$ is the sample estimate of the population expected value μ_X.

[8] If there is only one observation in the sample then it is impossible to create a measure of spread in the sample. You need at least two observations to measure deviations from the sample average. Hence the effective sample size for computing the sample variance is $T-1$.

[9] Similar functions are available in the **moments** package.

```
statsMonthly = rbind(apply(msftSp500MonthlyRetS, 2, mean),
                     apply(msftSp500MonthlyRetS, 2, var),
                     apply(msftSp500MonthlyRetS, 2, sd),
                     apply(msftSp500MonthlyRetS, 2, skewness),
                     apply(msftSp500MonthlyRetS, 2, kurtosis))
rownames(statsMonthly) = c("Mean", "Variance", "Std Dev",
                           "Skewness",  "Excess Kurtosis")
round(statsMonthly, digits=4)
#>                    MSFT   SP500
#> Mean             0.0092  0.0028
#> Variance         0.0103  0.0023
#> Std Dev          0.1015  0.0478
#> Skewness         0.4853 -0.5631
#> Excess Kurtosis 1.9148  0.6545
```

The mean and standard deviation for Microsoft monthly returns are 0.9% and 10%, respectively. Annualized, these values are approximately 10.8% (.009 $\times$12) and 34.6% (.10 $\times \sqrt{12}$), respectively. The corresponding monthly and annualized values for S&P 500 returns are .3% and 4.7%, and 4.8% and 16.6%, respectively. Microsoft has a higher mean and volatility than S&P 500. The lower volatility for the S&P 500 reflects risk reduction due to *diversification* (holding many assets in a portfolio). The sample skewness for Microsoft, 0.485, is slightly positive and reflects the approximate symmetry in the histogram in Figure 5.6. The skewness for S&P 500, however, is moderately negative at -0.563 which reflects the somewhat long left tail of the histogram in Figure 5.6. The sample excess kurtosis values for Microsoft and S&P 500 are 1.915 and 0.654, respectively, and indicate that the tails of the histograms are slightly fatter than the tails of a normal distribution.

The sample statistics for the daily returns are:

```
statsDaily = rbind(apply(msftSp500DailyRetS, 2, mean),
                   apply(msftSp500DailyRetS, 2, var),
                   apply(msftSp500DailyRetS, 2, sd),
                   apply(msftSp500DailyRetS, 2, skewness),
                   apply(msftSp500DailyRetS, 2, kurtosis))
rownames(statsDaily) = rownames(statsMonthly)
round(statsDaily, digits=4)
#>                    MSFT  SP500
#> Mean             0.0005 0.0002
#> Variance         0.0005 0.0002
#> Std Dev          0.0217 0.0135
#> Skewness         0.2506 0.0012
```

```
#> Excess Kurtosis 7.3915 6.9971
```

For a daily horizon, the sample mean for both series is zero (to three decimals). As with the monthly returns, the sample standard deviation for Microsoft, 2.2%, is about twice as big as the sample standard deviation for the S&P 500 index, 1.4%. Neither daily return series exhibits much skewness, but the excess kurtosis values are much larger than the corresponding monthly values indicating that the daily returns have much fatter tails and are more non-normally distributed than the monthly return series.

The daily simple returns are very close to the daily cc returns so the square-root-of-time rule (with 20 trading days within the month) can be applied to the sample mean and standard deviation of the simple returns:

```
20*statsDaily["Mean", ]
#>    MSFT   SP500
#> 0.00933 0.00345
sqrt(20)*statsDaily["Std Dev", ]
#>   MSFT  SP500
#> 0.0971 0.0603
```

Comparing these values to the corresponding monthly sample statistics shows that the square-root-of-time-rule applied to the daily mean and standard deviation gives results that are fairly close to the actual monthly mean and standard deviation. ■

5.1.6 Outliers

Figure 5.17 nicely illustrates the concept of an outlier in a data sample. All of the points are following a nice systematic relationship except one - the red dot "outlier". Outliers can be thought of in two ways. First, an outlier can be the result of a data entry error. In this view, the outlier is not a valid observation and should be removed from the data sample. Second, an outlier can be a valid data point whose behavior is seemingly unlike the other data points. In this view, the outlier provides important information and should not be removed from the data sample. For financial market data, outliers are typically extremely large or small values that could be the result of a data entry error (e.g. price entered as 10 instead of 100) or a valid outcome associated with some unexpected bad or good news. Outliers are problematic for data analysis because they can greatly influence the value of certain sample statistics.

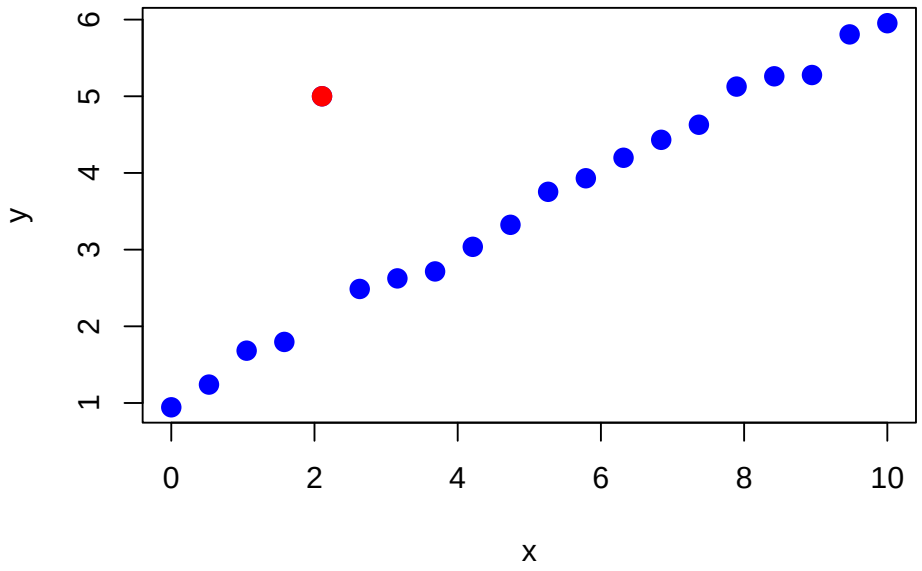


Figure 5.17: Illustration of an "outlier" in a data sample.

Table 5.1: Sample statistics for GWN with and without outlier.

Statistic	**GWN**	**GWN with Outlier**	%Δ
Mean	-0.0054	-0.0072	32%
Variance	0.0086	0.0101	16%
Std. Deviation	0.0929	0.1003	8%
Skewness	-0.2101	-1.0034	378%
Kurtosis	2.7635	7.1530	159%
Median	-0.0002	-0.0002	0%
IQR	0.1390	0.1390	0%

Example 5.16 (Effect of outliers on sample statistics)**.** To illustrate the impact of outliers on sample statistics, the simulated monthly GWN data is polluted by a single large negative outlier:

```
gwnMonthlyOutlier = gwnMonthly
gwnMonthlyOutlier[20] = -sd(gwnMonthly)*6
```

Figure 5.18 shows the resulting data. Visually, the outlier is much smaller than a typical negative observation and creates a pronounced asymmetry in the histogram. Table 5.1 compares the sample statistics Equation 5.4 - Equation 5.8 of the unpolluted and polluted data. All of the sample

statistics are influenced by the single outlier with the skewness and kurtosis being influenced the most. The mean decreases by 32% and the variance and standard deviation increase by 16% and 8%, respectively. The skewness changes by 378% and the kurtosis by 159%. Table 5.1 also shows the median and the IQR, which are quantile-based statistics for the center and spread, respectively. Notice that these statistics are unaffected by the outlier.

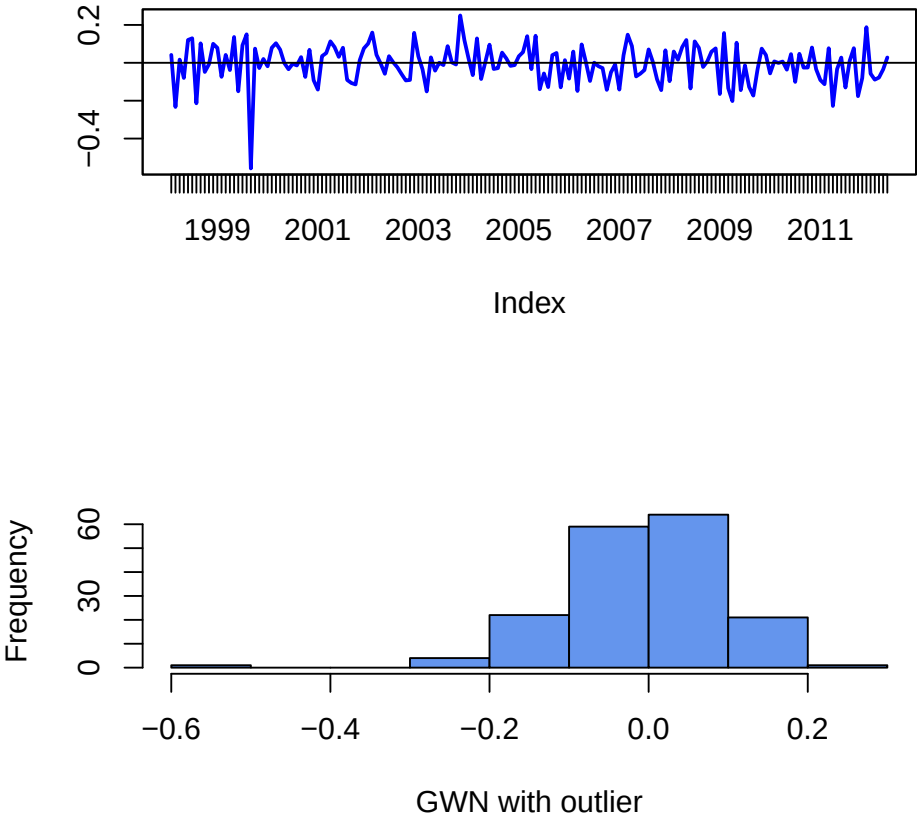


Figure 5.18: Monthly GWN data polluted by a single outlier.

■

The previous example shows that the common sample statistics Equation 5.4 - Equation 5.8 based on the sample average and deviations from the sample average can be greatly influenced by a single outlier, whereas quantile-based sample statistics are not. Sample statistics that are not greatly influenced by a single outlier are called (outlier) *robust statistics*.

5.1.6.1 Defining outliers

A commonly used rule-of-thumb defines an outlier as an observation that is beyond the sample mean plus or minus three times the sample standard deviation. The intuition for this rule comes from the fact that if $X \sim N(\mu, \sigma^2)$ then $\Pr(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.99$. While the intuition for this definition is clear, the practical implementation of the rule using the sample mean, $\hat{\mu}$, and sample standard deviation, $\hat{\sigma}$, is problematic because these statistics can be heavily influenced by the presence of one or more outliers (as illustrated in the previous sub-section). In particular, both $\hat{\mu}$ and $\hat{\sigma}$ become larger (in absolute value) when there are outliers and this may cause the rule-of-thumb rule to miss identifying an outlier or to identify a non-outlier as an outlier. That is, the rule-of-thumb outlier detection rule is not robust to outliers!

A natural fix to the above outlier detection rule-of-thumb is to use outlier robust statistics, such as the median and the IQR, instead of the mean and standard deviation. To this end, a commonly used definition of a moderate outlier in the right tail of the distribution (value above the median) is a data point x such that:

$$\hat{q}_{.75} + 1.5 \cdot \text{IQR} < x < \hat{q}_{.75} + 3 \cdot \text{IQR}. \tag{5.10}$$

To understand this rule, recall that the IQR is where the middle 50% of the data lies. If the data were normally distributed then $\hat{q}_{.75} \approx \hat{\mu} + 0.67 \times \hat{\sigma}$ and $\text{IQR} \approx 1.34 \times \hat{\sigma}$. Then Equation 5.10 is approximately:

$$\hat{\mu} + 2.67 \times \hat{\sigma} < x < \hat{\mu} + 4.67 \times \hat{\sigma}$$

Similarly, a moderate outlier in the left tail of the distribution (value below the median) is a data point such that,

$$\hat{q}_{.25} - 3 \cdot IQR < x < \hat{q}_{.25} - 1.5 \cdot \text{IQR}.$$

Extreme outliers are those observations x even further out in the tails of the distribution and are defined using:

$$x > \hat{q}_{.75} + 3 \cdot \text{IQR},$$
$$x < \hat{q}_{.25} - 3 \cdot \text{IQR}.$$

For example, with normally distributed data an extreme outlier would be an observation that is more than 4.67 standard deviations above the mean.

5.1.7 Box plots

A boxplot of a data series describes the distribution using outlier robust statistics. The basic boxplot is illustrated in Figure 5.19. The middle of the distribution is shown by a rectangular box whose top and bottom sides are approximately the first and third quartiles, $\hat{q}_{.25}$ and $\hat{q}_{.75}$,

respectively. The line roughly in the middle of the box is the sample median, $\hat{q}_{.5}$. The top whisker is a horizontal line that represents either the largest data value or $\hat{q}_{.75} + 1.5 \cdot IQR$, whichever is smaller. Similarly, the bottom whisker is located at either the smallest data value or $\hat{q}_{.25} - 1.5 \cdot IQR$, whichever is closer to $\hat{q}_{.25}$. Data values above the top whisker and below the bottom whisker are displayed as circles and represent extreme observations. If a data distribution is symmetric then the median is in the center of the box and the top and bottom whiskers are approximately the same distance from the box. Fat tailed distributions will have observations above and below the whiskers.

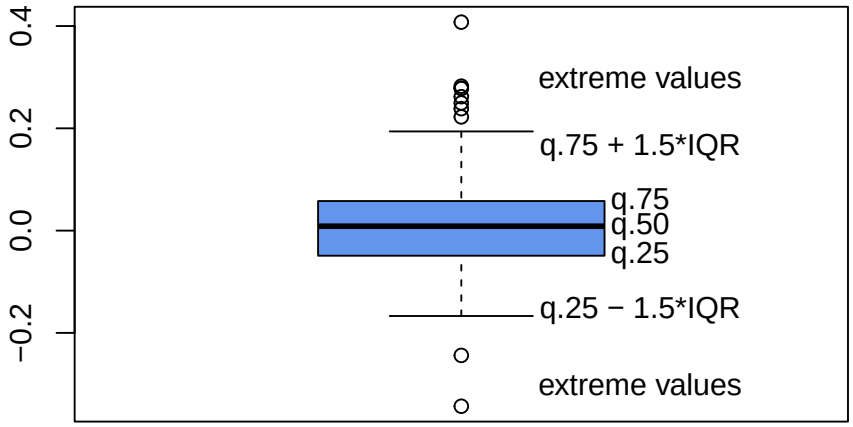


Figure 5.19: Boxplot of return distribution.

Example 5.17 (Boxplots of return distributions)**.** Boxplots are a great tool for visualizing the return distribution of one or more assets. Figure 5.20 shows the boxplots of the monthly continuously compounded and simple returns on Microsoft and the S&P 500 index computed using:[10]

```
dataToPlot = merge(msftMonthlyRetS, msftMonthlyRetC,
                   sp500MonthlyRetS, sp500MonthlyRetC)
colnames(dataToPlot) = c("msftRetS", "msftRetC",
                         "sp500RetS", "sp500RetC")
boxplot(coredata(dataToPlot), main="Simple vs. CC Monthly Returns",
```

[10]The `coredata()` function is used to extract the data from the zoo object because `boxplot()` doesn't like `"xts"` or `"zoo"` objects.

```
        col="cornflowerblue")
```

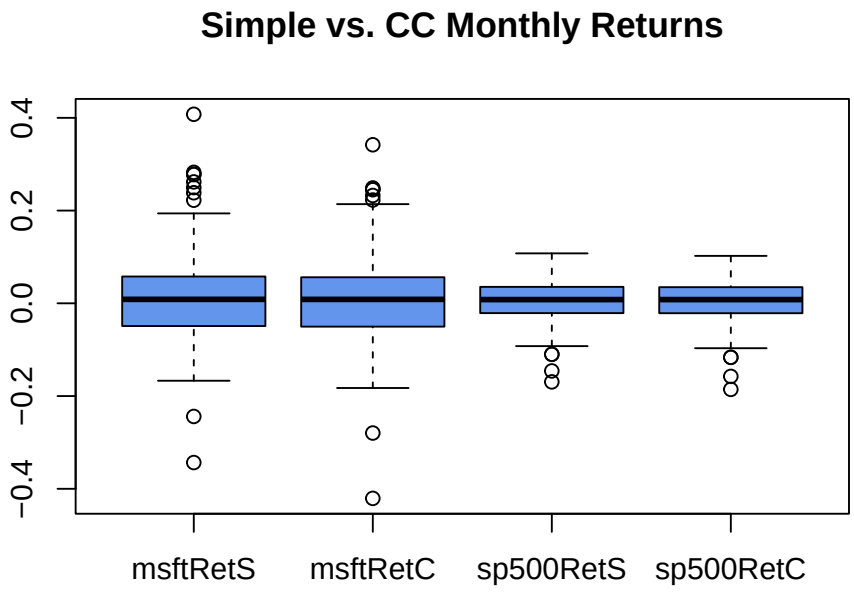


Figure 5.20: Boxplots of the continuously compounded and simple monthly returns on Microsoft stock and the S&P 500 index.

The Microsoft returns have both positive and negative outliers whereas the S&P 500 index returns only have negative outliers. The boxplots also clearly show that the distributions of the continuously compounded and simple returns are very similar except for the extreme tails of the distributions. ■

5.2 Time Series Descriptive Statistics

In this section, I consider graphical and numerical descriptive statistics for summarizing the linear time dependence in a time series.

5.2.1 Sample autocovariances and autocorrelations

For a covariance stationary time series process $\{X_t\}$, the autocovariances $\gamma_j = \text{cov}(X_t, X_{t-j})$ and autocorrelations $\rho_j = \text{cov}(X_t, X_{t-j})/\sqrt{\text{var}(X_t)\text{var}(X_{t-j})} = \gamma_j/\sigma^2$ describe the linear time dependencies in the process. For a sample of data $\{x_t\}_{t=1}^T$, the linear time dependencies are captured by

the *sample autocovariances* and *sample autocorrelations*:

$$\hat{\gamma}_j = \frac{1}{T-1}\sum_{t=j+1}^{T}(x_t - \bar{x})(x_{t-j} - \bar{x}),\ j = 1, 2, \ldots, T - J + 1,$$
$$\hat{\rho}_j = \frac{\hat{\gamma}_j}{\hat{\sigma}^2},\ j = 1, 2, \ldots, T - J + 1,$$

where $\hat{\sigma}^2$ is the sample variance Equation 5.5. The *sample autocorrelation function* (SACF) is a plot of $\hat{\rho}_j$ vs. j, and gives a graphical view of the linear time dependencies in the observed data.

Example 5.18 (SACF for the Microsoft and S&P 500 returns)**.** The R function `acf()` computes and plots the sample autocorrelations $\hat{\rho}_j$ of a time series for a specified number of lags. For example, to compute $\hat{\rho}_j$ for $j = 1, \ldots, 5$ for the monthly and daily returns on Microsoft use:

```
acf(coredata(msftMonthlyRetS), lag.max=5, plot=FALSE)
#>
#> Autocorrelations of series 'coredata(msftMonthlyRetS)', by lag
#>
#>      0      1      2      3      4      5
#>  1.000 -0.193 -0.114  0.193 -0.139 -0.023
acf(coredata(msftDailyRetS), lag.max=5, plot=FALSE)
#>
#> Autocorrelations of series 'coredata(msftDailyRetS)', by lag
#>
#>      0      1      2      3      4      5
#>  1.000 -0.044 -0.018 -0.010 -0.042  0.018
```

None of these sample autocorrelations are very big and indicate lack of linear time dependence in the monthly and daily returns.

Sample autocorrelations can also be visualized using the function `acf()`. Figure 5.21 shows the SACF for the monthly and daily returns on Microsoft and the S&P 500 created using:

```
par(mfrow=c(2,2))
acf(coredata(msftMonthlyRetS), main="msftMonthlyRetS", lwd=2)
acf(coredata(sp500MonthlyRetS), main="sp500MonthlyRetS", lwd=2)
acf(coredata(msftDailyRetS), main="msftDailyRetS", lwd=2)
acf(coredata(sp500DailyRetS), main="sp500DailyRetS", lwd=2)
```

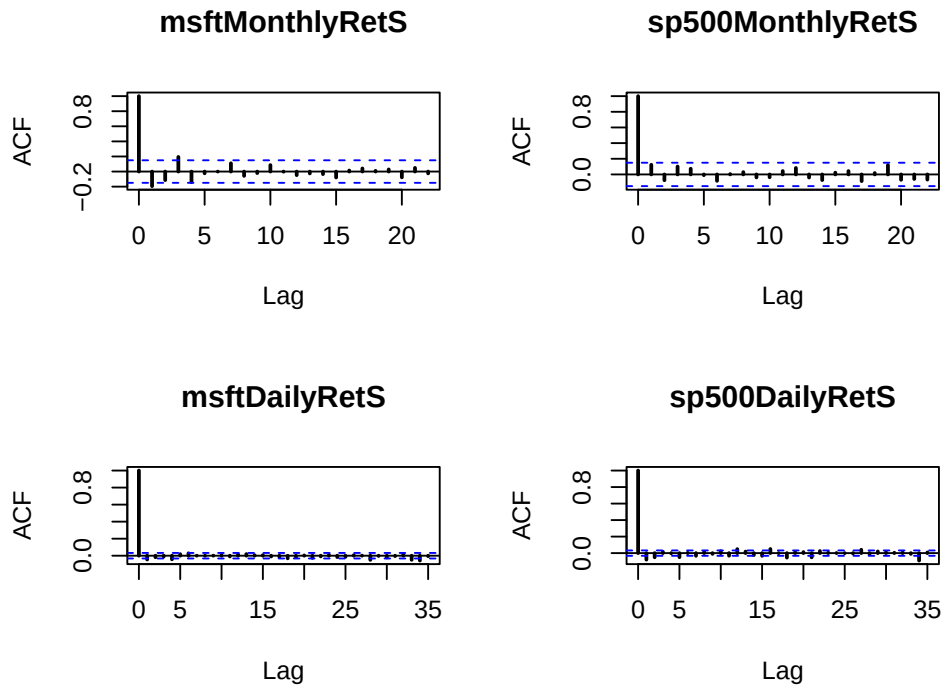


Figure 5.21: SACFs for the monthly and daily returns on Microsoft and the S&P 500 index.

None of the series show strong evidence for linear time dependence. The monthly returns have slightly larger sample autocorrelations at low lags than the daily returns. The dotted lines on the SACF plots are 5% critical values for testing the null hypothesis that the true autocorrelations ρ_j are equal to zero (see Chapter 9). If $\hat{\rho}_j$ crosses the dotted line then the null hypothesis that $\rho_j = 0$ can be rejected at the 5% significance level. ■

5.2.2 Time dependence in volatility

The daily and monthly returns of Microsoft and the S&P 500 index do not exhibit much evidence for linear time dependence. Their returns are essentially uncorrelated over time. However, the lack of autocorrelation does not mean that returns are independent over time. There can be nonlinear time dependence. Indeed, there is strong evidence for a particular type of nonlinear time dependence in the daily returns of Microsoft and the S&P 500 index that is related to time varying volatility. This nonlinear time dependence, however, is less pronounced in the monthly returns.

Example 5.19 (GWN and lack of time independence)**.** Let $y_t \sim \text{GWN}(0, \sigma^2)$. Then y_t is, by construction, independent over time. This implies that any function of y_t is also independent over time. To illustrate, Figure 5.22 shows y_t, y_t^2 and $|y_t|$ together with their SACFs where $y_t \sim \text{GWN}(0, 1)$ for $t = 1, \dots, 500$. Notice that all of the sample autocorrelations are essentially zero which confirms the lack of linear time dependence in y_t, y_t^2 and $|y_t|$.

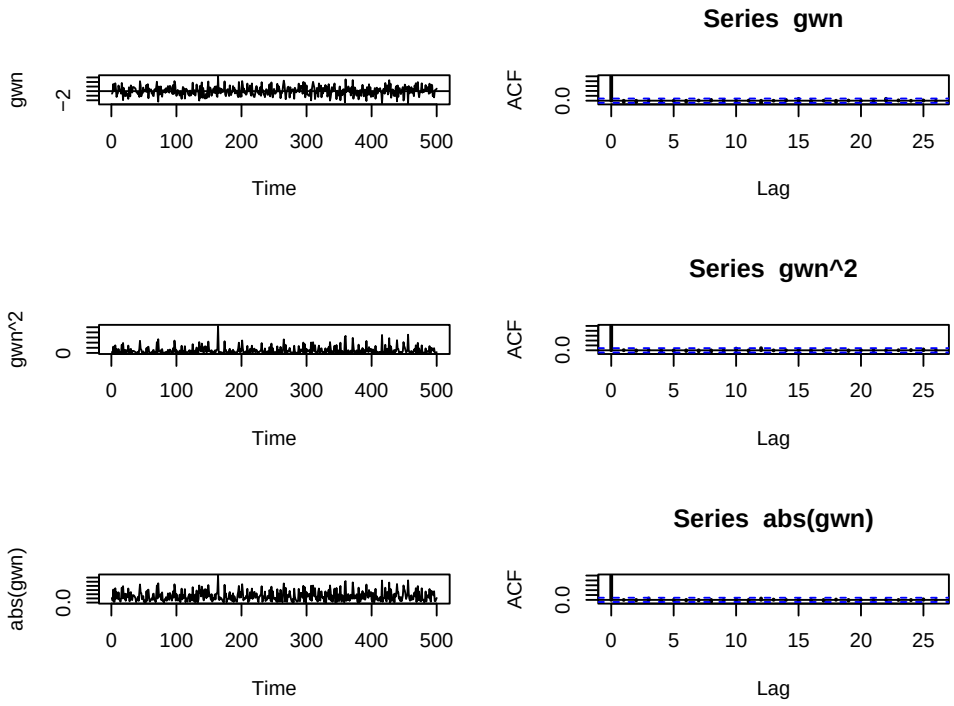


Figure 5.22: The left panel (top to bottom) shows y_t, y_t^2 and $|y_t|$ where y_t is similar to GWN(0,1) for $t = 1, ..., 500$. The right panel shows the corresponding SACFs.

■

5.2.2.1 Nonlinear time dependence in asset returns

Figure 5.5 and Figure 5.2 show that the daily and monthly returns on Microsoft and the S&P 500 index appear to exhibit periods of volatility clustering. That is, high periods of volatility tend to be followed by periods of high volatility and periods of low volatility appear to be followed by periods of low volatility. In other words, the volatility of the returns appears to exhibit some time dependence. This time dependence in volatility causes linear time dependence in the absolute value and squared value of returns.

Example 5.20 (Nonlinear time dependence in the S&P 500 returns)**.** To illustrate nonlinear time dependence in asset returns, Figure 5.23 shows the daily returns, absolute returns and squared returns for the S&P 500 index. Notice how the absolute and squared returns are large (small) when the daily returns are more (less) volatile. Figure 5.24 shows the SACFs for the absolute and squared daily and monthly returns. There is clear evidence of time dependence in the daily absolute and squared returns while there is some evidence of time dependence in the monthly absolute and squared returns. Since the absolute and squared daily returns reflect the volatility of the daily returns, the SACFs show a strong positive linear time dependence in daily return volatility.

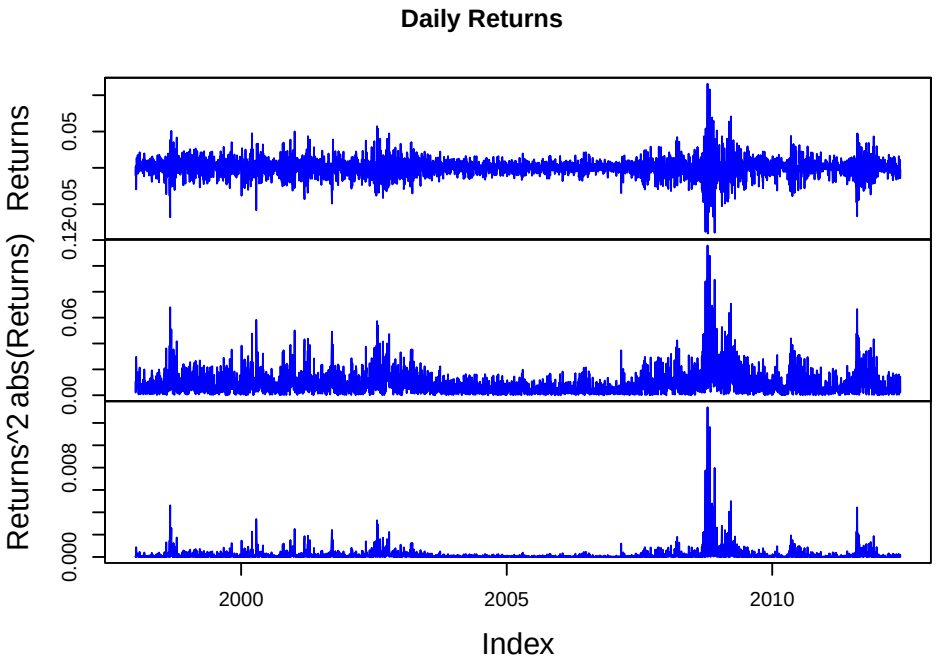


Figure 5.23: Daily returns, absolute returns, and squared returns on the S&P 500 index.

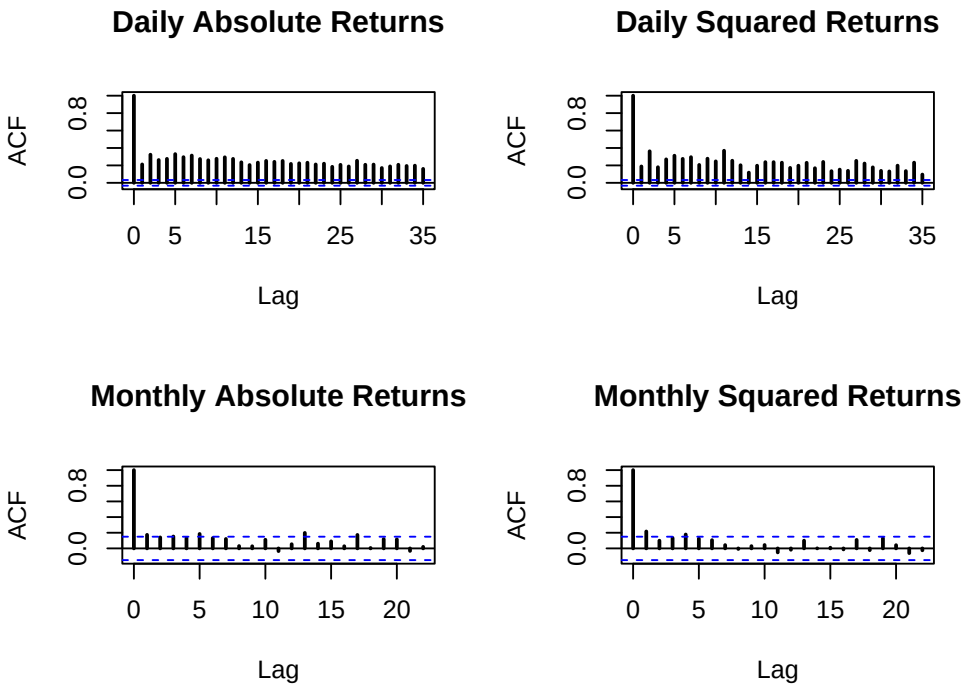


Figure 5.24: SACFs of daily and monthly absolute and squared returns on the S&P 500 index.

■

5.3 Bivariate Descriptive Statistics

In this section, I consider graphical and numerical descriptive statistics for summarizing two or more data series.

5.3.1 Scatterplots

The contemporaneous dependence properties between two data series $\{x_t\}_{t=1}^T$ and $\{y_t\}_{t=1}^T$ can be displayed graphically in a *scatterplot*, which is simply an xy-plot of the bivariate data.

Example 5.21 (Scatterplot of Microsoft and S&P 500 returns)**.** Figure 5.25 shows the scatterplots between the Microsoft and S&P 500 monthly and daily returns created using:

```
par(mfrow=c(1,2))
plot(coredata(sp500MonthlyRetS),coredata(msftMonthlyRetS),
     main="Monthly returns", xlab="S&P500", ylab="MSFT", lwd=2,
     pch=16, cex=1.25, col="blue")
abline(v=mean(sp500MonthlyRetS))
abline(h=mean(msftMonthlyRetS))
plot(coredata(sp500DailyRetS),coredata(msftDailyRetS),
     main="Daily returns", xlab="S&P500", ylab="MSFT", lwd=2,
     pch=16, cex=1.25, col="blue")
abline(v=mean(sp500DailyRetS))
abline(h=mean(msftDailyRetS))
```

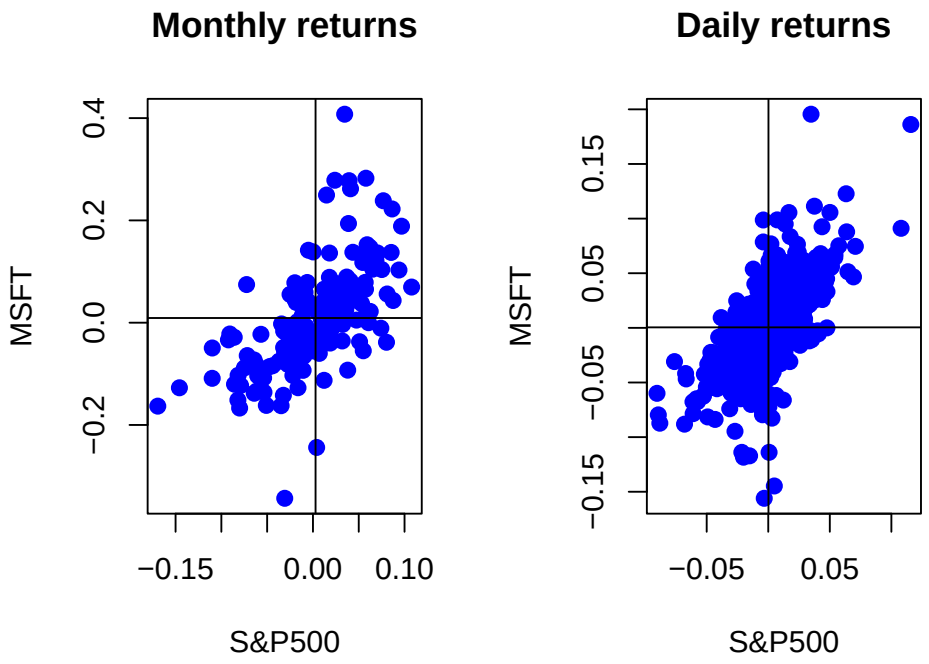


Figure 5.25: Scatterplot of Monthly returns on Microsoft and the S&P 500 index.

The S&P 500 returns are put on the x-axis and the Microsoft returns on the y-axis because the "market", as proxied by the S&P 500, is often thought as an independent variable driving individual asset returns. The upward sloping orientation of the scatterplots indicate a positive linear dependence between Microsoft and S&P 500 returns at both the monthly and daily frequencies. ■

Example 5.22 (Pair-wise scatterplots for multiple series)**.** For more than two data series, the R function `pairs()` plots all pair-wise scatterplots in a single plot. For example, to plot all pair-wise scatterplots for the GWN, Microsoft returns and S&P 500 returns use:

```
dataToPlot = merge(gwnMonthly,msftMonthlyRetS,sp500MonthlyRetS)
pairs(coredata(dataToPlot), col="blue", pch=16, cex=1.25, cex.axis=1.25)
```

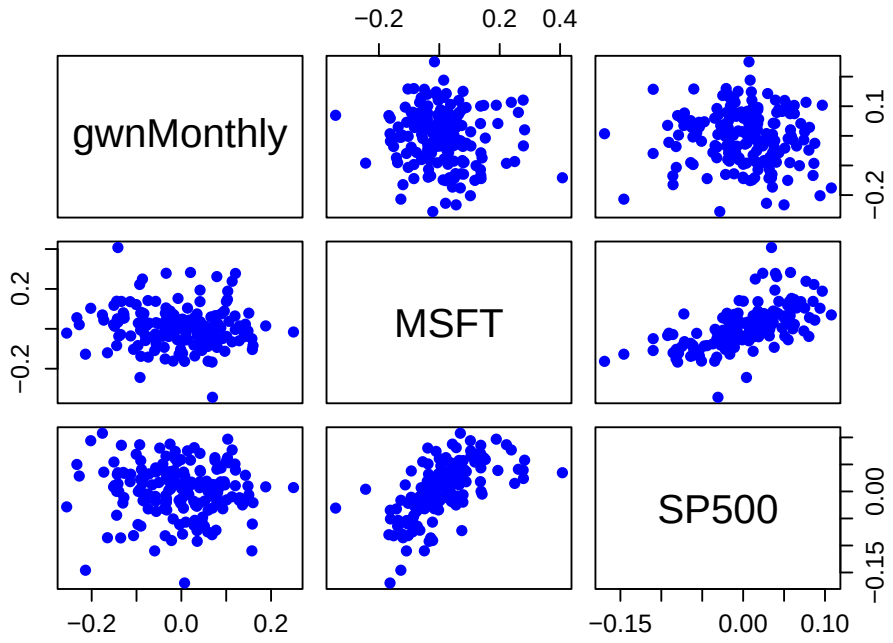


Figure 5.26: Pair-wise scatterplots between simulated GWN, Microsoft returns and S&P 500 returns.

The top row of Figure 5.26 shows the scatterplots between the pairs (MSFT, GWN) and (SP500, GWN), the second row shows the scatterplots between the pairs (GWN, MSFT) and (SP500, MSFT), the third row shows the scatterplots between the pairs (GWN, SP500) and (MSFT, SP500). The plots in the lower triangle are the same as the plots in the upper triangle except the axes are reversed. ■

5.3.2 Sample covariance and correlation

For two random variables X and Y, the direction of linear dependence is captured by the covariance, $\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)]$, and the direction and strength of linear dependence is captured by the correlation, $\rho_{XY} = \sigma_{XY}/\sigma_X\sigma_Y$.

Definition 5.8 (Sample Covariance and Correlation)**.** For two data series $\{x_t\}_{t=1}^T$ and $\{y_t\}_{t=1}^T$, the *sample covariance*,

$$\hat{\sigma}_{xy} = \frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})(y_t - \bar{y}), \tag{5.11}$$

measures the direction of linear dependence, and the *sample correlation*,

$$\hat{\rho}_{xy} = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x \hat{\sigma}_x}, \tag{5.12}$$

measures the direction and strength of linear dependence. In Equation 5.12, $\hat{\sigma}_x$ and $\hat{\sigma}_y$ are the sample standard deviations of $\{x_t\}_{t=1}^T$ and $\{y_t\}_{t=1}^T$, respectively, defined by Equation 5.6.

When more than two data series are being analyzed, it is often convenient to compute all pairwise covariances and correlations at once using matrix algebra. Recall, for a vector of N random variables $\mathbf{X} = (X_1, \ldots, X_N)'$ with mean vector $\mu = (\mu_1, \ldots, \mu_N)'$ the $N \times N$ variance-covariance matrix is defined as:

$$\Sigma = \text{var}(\mathbf{X}) = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)'] = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_N^2 \end{pmatrix}.$$

Definition 5.9 (Sample Covariance and Correlation Matrices)**.** For N data series $\{\mathbf{x}_t\}_{t=1}^T$, where $x_t = (x_{1t}, \ldots, x_{Nt})'$, the *sample covariance matrix* is computed using:

$$\hat{\Sigma} = \frac{1}{T-1} \sum_{t=1}^{T} (\mathbf{x}_t - \hat{\mu})(\mathbf{x}_t - \hat{\mu})' = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \cdots & \hat{\sigma}_{1N} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 & \cdots & \hat{\sigma}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\sigma}_{1N} & \hat{\sigma}_{2N} & \cdots & \hat{\sigma}_N^2 \end{pmatrix}, \tag{5.13}$$

where $\hat{\mu}$ is the $N \times 1$ sample mean vector. Define the $N \times N$ diagonal matrix:

$$\hat{\mathbf{D}} = \begin{pmatrix} \hat{\sigma}_1 & 0 & \cdots & 0 \\ 0 & \hat{\sigma}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{\sigma}_N \end{pmatrix}. \tag{5.14}$$

Then the $N \times N$ *sample correlation matrix* $\hat{\mathbf{C}}$ is computed as:

$$\hat{\mathbf{C}} = \hat{\mathbf{D}}^{-1} \hat{\Sigma} \hat{\mathbf{D}}^{-1} = \begin{pmatrix} 1 & \hat{\rho}_{12} & \cdots & \hat{\rho}_{1N} \\ \hat{\rho}_{12} & 1 & \cdots & \hat{\rho}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\rho}_{1N} & \hat{\rho}_{2N} & \cdots & 1 \end{pmatrix}. \tag{5.15}$$

Example 5.23 (Sample covariance and correlation between Microsoft and S&P 500 returns)**.** The scatterplots of Microsoft and S&P 500 returns in Figure 5.25 suggest positive linear relationships

in the data. You can confirm this by computing the sample covariance and correlation using the R functions `cov()` and `cor()`. For the monthly returns, you get:

```
cov(sp500MonthlyRetS, msftMonthlyRetS)
#>          MSFT
#> SP500 0.00298
cor(sp500MonthlyRetS, msftMonthlyRetS)
#>        MSFT
#> SP500 0.614
```

Indeed, the sample covariance is positive and the sample correlation shows a moderately strong linear relationship. For the daily returns you get:

```
cov(sp500DailyRetS, msftDailyRetS)
#>           MSFT
#> SP500 0.000196
cor(sp500DailyRetS, msftDailyRetS)
#>        MSFT
#> SP500 0.671
```

Here, the daily sample covariance is about twenty times smaller than the monthly covariance (recall the square-root-of time rule), but the daily sample correlation is similar to the monthly sample correlation.

When passed a matrix of data, the `cov()` and `cor()` functions can also be used to compute the sample covariance and correlation matrices $\hat{\Sigma}$ and $\hat{\mathbf{C}}$, respectively. For example,

```
cov(msftSp500MonthlyRetS)
#>          MSFT   SP500
#> MSFT  0.01030 0.00298
#> SP500 0.00298 0.00228
cor(msftSp500MonthlyRetS)
#>        MSFT SP500
#> MSFT  1.000 0.614
#> SP500 0.614 1.000
```

The function `cov2cor()` transforms a sample covariance matrix to a sample correlation matrix using Equation 5.15:

```
cov2cor(cov(msftSp500MonthlyRetS))
#>        MSFT SP500
```

```
#> MSFT   1.000 0.614
#> SP500 0.614 1.000
```

■

Example 5.24 (Visualizing correlation matrices)**.** The R package **corrplot** contains functions for visualizing correlation matrices. This is particularly useful for summarizing the linear dependencies among many data series. For example, Figure 5.27 shows the correlation plot from the **corrplot** function `corrplot.mixed()` created using:

```
dataToPlot = merge(gwnMonthly, msftMonthlyRetS, sp500MonthlyRetS)
cor.mat = cor(dataToPlot)
corrplot.mixed(cor.mat, lower="number", upper="ellipse")
```

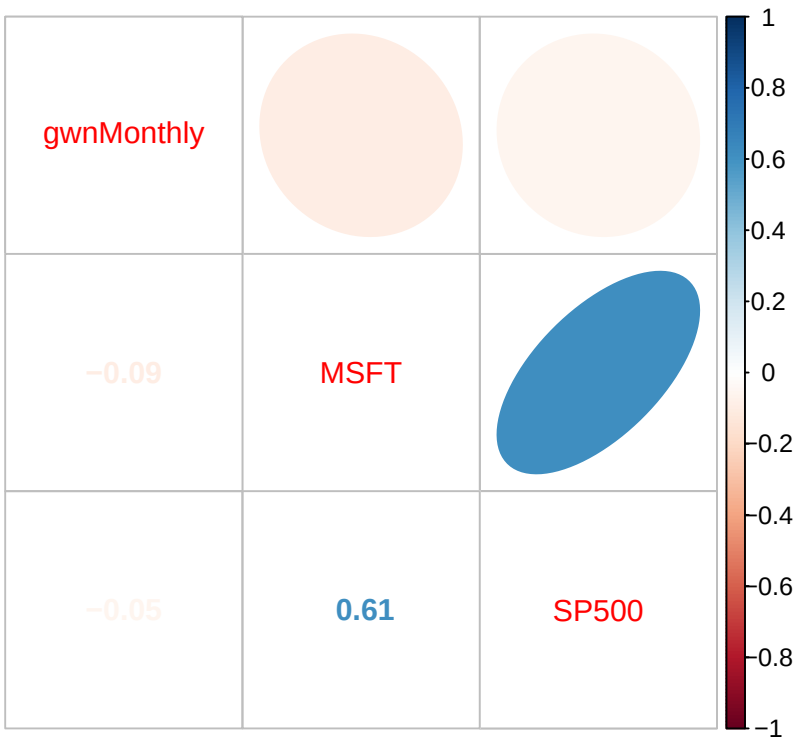


Figure 5.27: Correlation plot created with `corrplot()`.

The color scheme shows the magnitudes of the correlations (blue for positive and red for negative) and the orientation of the ellipses show the magnitude and direction of the linear associations. ■

5.3.3 Sample cross-lag covariances and correlations

The dynamic interactions between two observed time series $\{x_t\}_{t=1}^T$and $\{y_t\}_{t=1}^T$ can be measured using the *sample cross-lag covariances* and *correlations* defined by:

$$\hat{\gamma}_{xy}^k = \widehat{\text{cov}}(X_t, Y_{t-k}),$$
$$\hat{\rho}_{xy}^k = \widehat{\text{corr}}(X_t, Y_{t-k}) = \frac{\hat{\gamma}_{xy}^k}{\sqrt{\hat{\sigma}_x^2 \hat{\sigma}_y^2}}.$$

When more than two data series are being analyzed, all pairwise cross-lag sample covariances and sample correlations can be computed at once using matrix algebra. For a time series of N data series $\{\mathbf{x}_t\}_{t=1}^T$, where $\mathbf{x}_t = (x_{1t}, \dots, x_{Nt})'$, the sample lag k cross-lag covariance and correlation matrices are computed using:

$$\hat{\Gamma}_k = \frac{1}{T-1} \sum_{t=k+1}^{T} (\mathbf{x}_t - \hat{\mu})(\mathbf{x}_{t-k} - \hat{\mu})', \tag{5.16}$$

$$\hat{\mathbf{C}}_k = \hat{\mathbf{D}}^{-1} \hat{\Gamma}_k \hat{\mathbf{D}}^{-1}, \tag{5.17}$$

where $\hat{\mathbf{D}}$ is defined in Equation 5.14.

Example 5.25 (Sample cross-lag covariances and correlations between Microsoft and S&P 500 returns)**.** Consider computing the cross-lag covariance and correlation matrices Equation 5.16 and Equation 5.17 for $k = 0, 1, \dots, 5$ between Microsoft and S&P 500 monthly returns. These matrices may be computed using the R function `acf()` as follows:

```
Ghat = acf(coredata(msftSp500MonthlyRetS), type="covariance",
           lag.max=5, plot=FALSE)
Chat = acf(coredata(msftSp500MonthlyRetS), type="correlation",
           lag.max=5, plot=FALSE)
names(Ghat)
#> [1] "acf"    "type"   "n.used" "lag"    "series" "snames"
```

Here, `Ghat` and `Chat` are objects of class `"acf"` for which there are print and plot methods. The `acf` components of `Ghat` and `Chat` are 3-dimensional arrays containing the cross lag matrices Equation 5.16 and Equation 5.17, respectively. For example, to extract $\hat{\mathbf{C}}_0$ and $\hat{\mathbf{C}}_1$ use:

```
# Chat0
Chat$acf[1,,]
#>       [,1]  [,2]
#> [1,] 1.000 0.614
```

```
#> [2,] 0.614 1.000
# Chat1
Chat$acf[2,,]
#>         [,1]    [,2]
#> [1,] -0.1931 -0.0135
#> [2,] -0.0307  0.1232
```

The print method shows the sample autocorrelations of each variable as well as the pairwise cross-lag correlations:

```
Chat
#>
#> Autocorrelations of series 'coredata(msftSp500MonthlyRetS)', by lag
#>
#> , , MSFT
#>
#>  MSFT        SP500
#>   1.000 ( 0)  0.614 ( 0)
#>  -0.193 ( 1) -0.031 (-1)
#>  -0.114 ( 2) -0.070 (-2)
#>   0.193 ( 3)  0.137 (-3)
#>  -0.139 ( 4) -0.030 (-4)
#>  -0.023 ( 5)  0.032 (-5)
#>
#> , , SP500
#>
#>  MSFT        SP500
#>   0.614 ( 0)  1.000 ( 0)
#>  -0.014 ( 1)  0.123 ( 1)
#>  -0.043 ( 2) -0.074 ( 2)
#>   0.112 ( 3)  0.101 ( 3)
#>  -0.105 ( 4)  0.073 ( 4)
#>  -0.055 ( 5) -0.010 ( 5)
```

These values can also be visualized using the plot method:

```
plot(Chat, lwd=2)
```

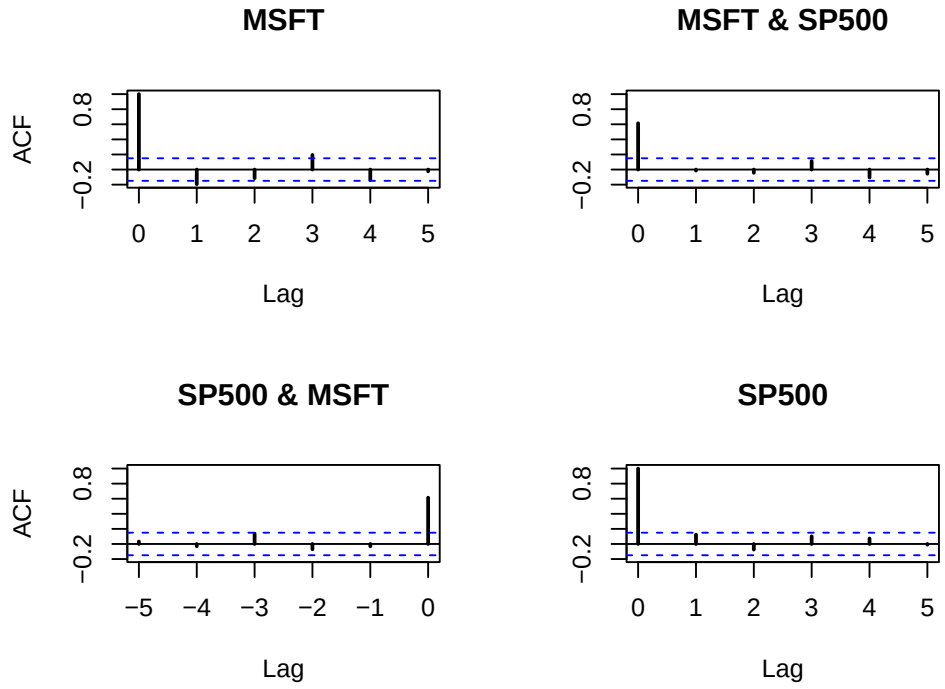


Figure 5.28: Sample cross-lag correlations between Microsoft and S&P 500 returns.

Figure 5.28 shows the resulting four-panel plot. The top-left and bottom-right (diagonal) panels give the SACFs for Microsoft and S&P 500 returns. The top-right plot gives the cross-lag correlations $\hat{\rho}^k_{msft,sp500}$ for $k = 0, 1, \ldots, 5$ and the bottom-left panel gives the cross-lag correlations $\hat{\rho}^k_{sp500,msft}$ for $k = 0, 1, \ldots, 5$. The plots show no evidence of any dynamic feedback between the return series. ■

5.4 Rolling descriptive statistics

The assumption that asset returns are covariance stationary implies that the asset return means (μ_i), variances (σ_i^2), covariances (σ_{ij}), and correlations (ρ_{ij}) are constant over time for all assets i and j. Essentially, this means that the economic environment remains constant over time. Clearly, this is a strong assumption and you have seen evidence in the graphical descriptive statistics that this assumption might be suspect. For example, you have seen that the volatility of asset returns appears to change over time. In this section, I introduce *rolling descriptive statistics* to further investigate the appropriateness of the covariance stationary assumption of asset returns.

A commonly used technique to investigate whether certain asset return characteristics are constant over time is to compute sample descriptive statistics over rolling sub-sample windows of fixed length. To illustrate, suppose one is interested in determining if the expected return parameter of an asset

return μ is constant over time. Assume returns are observed over a sample of size T. An informal way to do this is to compute estimates of μ over rolling windows of length $n < T$:

$$\hat{\mu}_t(n) = \frac{1}{n}\sum_{k=0}^{n-1} r_{t-k} = \frac{1}{n}(r_t + r_{t-1} + \cdots + r_{t-n+1}), \tag{5.18}$$

for $t = n, \ldots, T$. Here $\hat{\mu}_n(n)$ is the sample mean of the returns $\{r_t\}_{t=1}^{n}$ over the first sub-sample window of size n. Then $\hat{\mu}_{n+1}(n)$ is the sample mean of the returns $\{r_t\}_{t=2}^{n+1}$ over the second sub-sample window of size n. Continuing in this fashion I roll the sample window forward one observation at a time and compute $\hat{\mu}_t(n)$ until I arrive at the last sub-sample window of size n. The end result is $T - n + 1$ rolling estimates $\{\hat{\mu}_n(n), \hat{\mu}_{n+1}(n), \ldots, \hat{\mu}_T(n)\}$.

If returns are covariance stationary then μ is constant over time and you would expect that $\hat{\mu}_n(n) \approx \hat{\mu}_{n+1}(n) \approx \cdots \approx \hat{\mu}_T(n)$. That is, the estimates of μ over each rolling window should be similar. I would not expect them to be exactly equal due to sampling uncertainty but they should not be drastically different. In contrast, if returns are not covariance stationary due to time variation in μ then you would expect to see meaningful time variation in the rolling estimates $\{\hat{\mu}_n(n), \hat{\mu}_{n+1}(n), \ldots, \hat{\mu}_T(n)\}$.

A natural way to see the time variation in $\{\hat{\mu}_t(n)\}_{t=n}^{T}$ is to make a time series plot of these rolling estimates as illustrated in the next example.

Example 5.26 (Rolling mean estimates for Microsoft S&P 500 Index and simulated GWN)**.** Consider computing rolling estimates of μ for Microsoft, Starbucks and the S&P 500 index returns. You can easily do this using the **zoo** function `rollapply()`, which applies a user-specified function to a time series over rolling windows of fixed size. The arguments to `rollapply()` are:

```
args(zoo:::rollapply.zoo)
#> function (data, width, FUN, ..., by = 1, by.column = TRUE, fill = if (na.pad) NA,
#>     na.pad = FALSE, partial = FALSE, align = c("center", "left",
#>         "right"), coredata = TRUE)
#> NULL
```

where `data` is either a `"zoo"` or `"xts"` object, `width` is the window width, `FUN` is the R function to be applied over the rolling windows, `...` are any optional arguments to be supplied to `FUN`, `by` specifies how many observations to increment the rolling windows, `by.column` specifies if `FUN` is to be applied to each column of data or to be applied to all columns of data, and `align` specifies the location of the time index in the window. The function returns an "xts/zoo" object containing the rolling estimates.

For example, to compute 24-month rolling estimates of μ for Microsoft, S&P 500 index, and simulated GWN data (that matches the mean and volatility of Microsoft returns), incremented by one

month with the time index aligned to the end of the window use:

```
roll.data = merge(msftMonthlyRetS,sp500MonthlyRetS,gwnMonthly)
colnames(roll.data) = c("MSFT", "SP500", "GWN")
roll.muhat = rollapply(roll.data, width=24, by=1, by.column=TRUE,
                       FUN=mean, align="right")
class(roll.muhat)
#> [1] "xts" "zoo"
```

The object `roll.muhat` is an "xts" object with three columns giving the rolling mean estimates for the GWN data, Microsoft, and the S&P 500 index. The first 23 months of data are NA as you need at least 24 months of data to compute the first rolling estimates:

```
head(roll.muhat, n=3)
#>          MSFT SP500 GWN
#> Feb 1998   NA    NA  NA
#> Mar 1998   NA    NA  NA
#> Apr 1998   NA    NA  NA
```

The first rolling estimates are for the 24-months just prior to January, 2000 and the last rolling estimates are for the 24-months just prior to May 2012.

```
head(na.omit(roll.muhat), n=3)
#>            MSFT  SP500       GWN
#> Jan 2000 0.0487 0.0161 -1.68e-03
#> Feb 2000 0.0394 0.0124 -9.85e-05
#> Mar 2000 0.0450 0.0143  1.39e-02
tail(na.omit(roll.muhat), n=3)
#>             MSFT   SP500     GWN
#> Mar 2012 0.00869 0.00894 -0.0307
#> Apr 2012 0.00661 0.00801 -0.0324
#> May 2012 0.00948 0.00882 -0.0298
```

The rolling means are shown in Figure 5.29 created using:

```
plot(roll.muhat, main="", multi.panel=FALSE, lwd=2,
     col=c("black", "red", "green"), lty=c("solid", "solid", "solid"),
     major.ticks="years", grid.ticks.on="years",
     legend.loc = "topright")
```

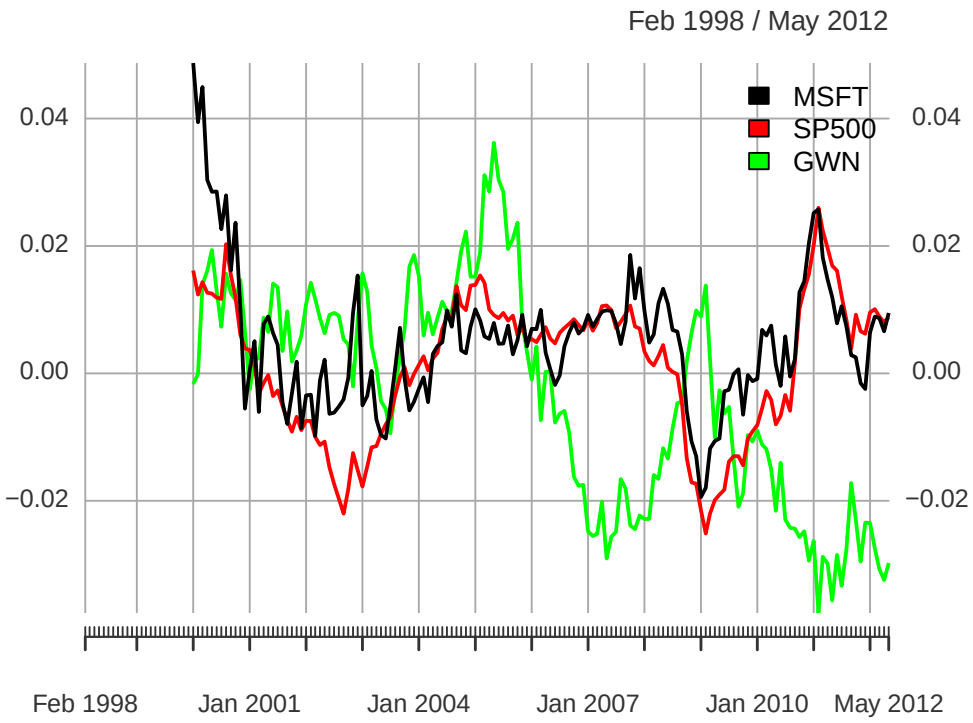


Figure 5.29: 24-month rolling estimates of μ

All of the rolling estimates show time variation over the sample, including the simulated GWN returns which have a constant mean by construction. The ranges of the rolling estimates are:

```
ans = apply(na.omit(roll.muhat),2,range)
rownames(ans) = c("Min", "Max")
ans
#>         MSFT   SP500     GWN
#> Min -0.0194 -0.0251 -0.0376
#> Max  0.0487  0.0260  0.0362
```

Interestingly, the rolling estimates for Microsoft and the S&P 500 index exhibit a similar pattern of time variation. The estimates start positive at the end of the dot-com boom, decrease from 2000 to mid 2002 (Microsoft and the S&P 500 index become negative), increase from 2003 to 2006 during the booming economy, fall sharply during the financial crisis period 2007 to 2009, and increase from 2009 to 2011, and decrease afterward. The Microsoft and S&P 500 estimates follow each other very closely due to their moderately strong positive correlation of 0.614. Somewhat surprisingly, the rolling means for the simulated GWN data also show substantial time variation over the sample. Since the true mean of the simulated GWN data is constant, the variation observed in the rolling means represents sampling uncertainty due to the small sample sizes of the rolling means. The magnitude of the variation in the rolling mean estimates for the GWN data should make us pause

in concluding there is substantial change in the mean estimates of Microsoft (and possibly the S&P 500). However, the fact that the rolling estimates for Microsoft and the S&P 500 index behave as expected during the booms and busts strongly suggests that the underlying mean values for these assets are not constant over the full sample. ■

To investigate time variation in return volatility, I compute rolling estimates of σ^2 and σ over fixed windows of size $n < T$:

$$\hat{\sigma}_t^2(n) = \frac{1}{n-1}\sum_{k=0}^{n-1}(r_{t-k} - \hat{\mu}_t(n))^2, \tag{5.19}$$

$$\hat{\sigma}_t(n) = \sqrt{\hat{\sigma}_t^2(n)}. \tag{5.20}$$

If return volatility is constant over time, then you would expect that $\hat{\sigma}_n(n) \approx \hat{\sigma}_{n+1}(n) \approx \cdots \approx \hat{\sigma}_T(n)$. That is, the estimates of σ over each rolling window should be similar. In contrast, if volatility is not constant over time then you would expect to see meaningful time variation in the rolling estimates $\{\hat{\sigma}_n(n), \hat{\sigma}_{n+1}(n), \ldots, \hat{\sigma}_T(n)\}$.

Example 5.27 (Rolling volatility estimates for Microsoft Starbucks and the S&P 500 Index)**.** To compute and plot 24-month rolling estimates of σ for Microsoft, S&P 500 index, and GWN returns incremented by one month with the time index aligned to the beginning of the window use:

```
roll.sigmahat = rollapply(roll.data,width=24, by=1, by.column=TRUE,
                          FUN=sd, align="right")
```

The rolling estimates are illustrated in Figure 5.30.

The rolling volatilities for Microsoft and the S&P 500 index show a similar pattern of time variation over the sample but the Microsoft volatilities are considerably more variable:

```
ans = apply(na.omit(roll.sigmahat),2,range)
rownames(ans) = c("Min", "Max")
ans
#>       MSFT  SP500    GWN
#> Min 0.0402 0.0186 0.0716
#> Max 0.1863 0.0706 0.1192
```

The rolling volatilities for Microsoft peak at about 18% per month at the end of the dot-com boom, fall sharply to about 4% per month at the beginning of 2005, then level off until the beginning of the financial crisis, and increase during the financial crisis to about 10% per month and fall slightly afterward. The S&P 500 volatilities have a similar pattern as the Microsoft volatilities but their movements are not as extreme. In contrast, the rolling volatilities of the GWN series are fairly stable bouncing between 7% and 12%. The time variation exhibited by the rolling volatilities of

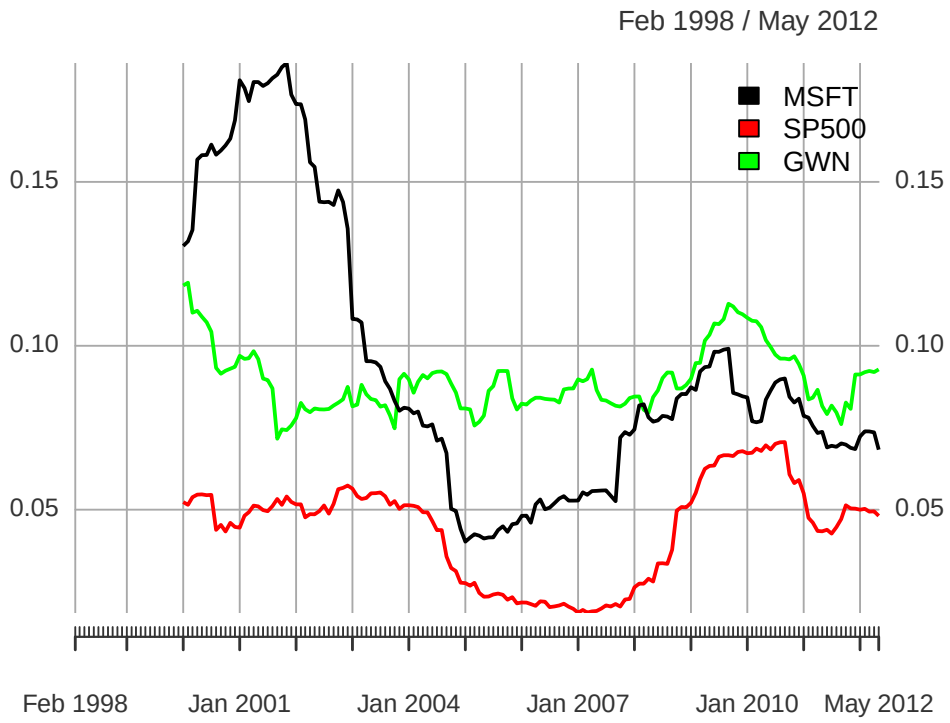


Figure 5.30: 24-month rolling estimates of σ.

Microsoft and the S&P 500 index places additional doubt on the covariance stationarity assumption for returns. ■

To investigate time variation in the correlation between two asset returns, ρ_{ij}, one can compute the rolling covariances and correlations over fixed windows of size $n < T$:

$$\hat{\sigma}_{ij,t}(n) = \frac{1}{n-1}\sum_{k=0}^{n-1}(r_{i,t-k} - \hat{\mu}_i(n))(r_{j,t-k} - \hat{\mu}_j(n)), \tag{5.21}$$

$$\hat{\rho}_{ij,t}(n) = \frac{\hat{\sigma}_{ij,t}(n)}{\hat{\sigma}_{i,t}(n)\hat{\sigma}_{j,t}(n)}. \tag{5.22}$$

Example 5.28 (Rolling correlation estimates for Microsoft Starbucks and the S&P 500 index)**.** A function to compute the pair-wise sample correlations between Microsoft, Starbucks and the S&P 500 index is

```
rhohat = function(x) {
   corhat = cor(x)
   corhat.vals = corhat[lower.tri(corhat)]
   names(corhat.vals) = c("MSFT.SP500", "MSFT.GWN", "SP500.GWN")
   corhat.vals
```

```
}
```

Using the function `rhohat()`, 24-month rolling estimates of ρ_{ij} for Microsoft, Starbucks and the S&P 500 index incremented by one month with the time index aligned to the beginning of the window can be computed using

```
roll.rhohat = rollapply(roll.data, width=24, FUN=rhohat,
   by=1, by.column=FALSE, align="right")
head(na.omit(roll.rhohat),n=3)
#>          MSFT.SP500 MSFT.GWN SP500.GWN
#> Jan 2000      0.638    0.392     0.278
#> Feb 2000      0.637    0.345     0.244
#> Mar 2000      0.663    0.415     0.377
```

Notice that `by.column=FALSE` in the call to `rollapply()`. This is necessary as the function `rhohat()` operates on all columns of the data object `roll.data`. The rolling estimates are illustrated in Figure 5.31.

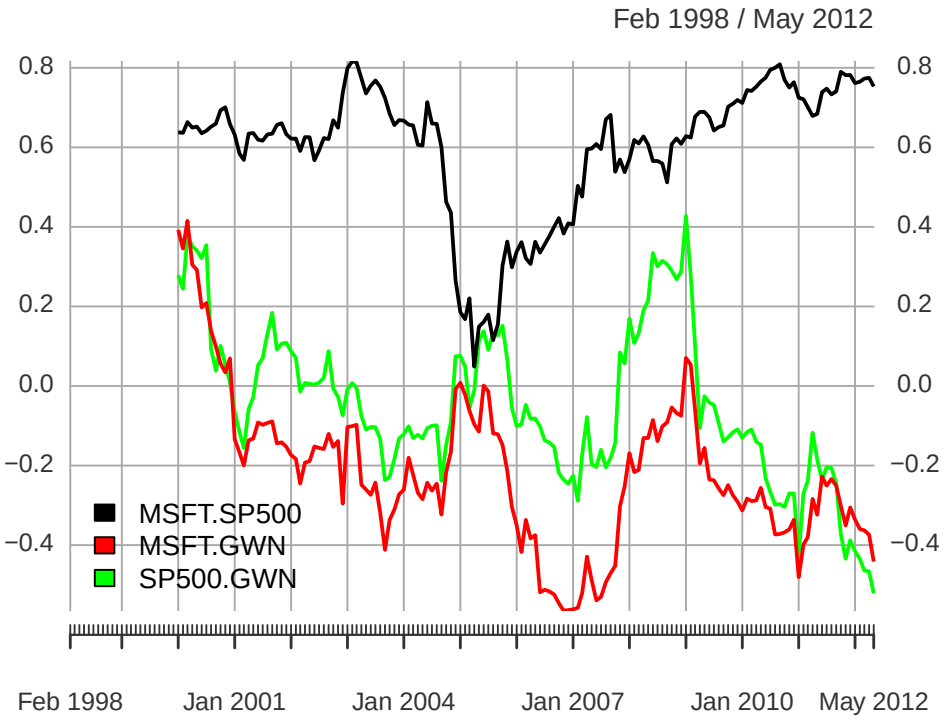


Figure 5.31: 24-month rolling estimates of ρ_{ij} for Microsoft, Starbucks and the S&P 500 index.

The ranges of the rolling correlations are:

```
ans = apply(na.omit(roll.rhohat),2,range)
rownames(ans) = c("Min", "Max")
ans
#>     MSFT.SP500 MSFT.GWN SP500.GWN
#> Min     0.0487   -0.565    -0.521
#> Max     0.8169    0.415     0.428
```

The rolling correlations between Microsoft and the S&P 500 index show an interesting pattern. They start moderately high, around 0.6, then plunge to about 0.05 in early 2005, and then gradually pick back up to about 0.8 at the end of the sample. The rolling correlations are highest during the financial crisis periods and lowest during the calm periods. Interestingly, the rolling correlations between the GWN series and Microsoft, and the GWN series and the S&P 500 index fluctuate somewhat randomly between -0.5 and 0.5 and show no particular pattern. There is compelling evidence that the correlation between Microsoft and the S&P 500 index is not constant over time. ■

It must be kept in mind that the rolling descriptive statistics are *informal diagnostics* for evaluating the assumption of covariance stationarity. As you have seen, there is considerably sampling variability in the rolling statistics due to the small window size which can make it difficult to untangle statistical noise from true changes in the underlying parameters. In Chapter 9 I will consider formal statistical tests for constant parameters that can give better evidence for or against the assumption of covariance stationarity.

5.4.1 Practical issues associated with rolling estimates

To be completed

- Discuss choice of window width. Small window widths tend to produce choppy estimates with a lot of estimation error. Large window widths produce smooth estimates with less estimation error
- Discuss choice of increment. For a given window width, the roll-by increment determines the number of overlapping returns in each window. Number of overlapping returns is $n - incr$ where n is the window width and $incr$ is the roll-by increment. Incrementing the window by 1 gives the largest number of over lapping returns. If $incr \geq n$ then the rolling windows are non-overlapping.

- A large positive or negative return in a window can greatly influence the estimate and cause a drop-off effect as the window rolls past the large observation.

5.5 Stylized facts for daily and monthly asset returns

A *stylized fact* is something that is generally true but not always. From the data analysis of the daily and monthly returns on Microsoft and the S&P 500 index I observe a number of stylized facts that tend to be true for other individual assets and portfolios of assets. For monthly data I observe the following stylized facts:

1. M1. Prices appear to be random walk non-stationary and returns appear to be mostly covariance stationary. There is evidence that return volatility changes over time.
2. M2. Returns appear to be approximately normally distributed. There is some negative skewness and excess kurtosis.
3. M3. Assets that have high average returns tend to have high standard deviations (volatilities) and vice-versa. This is the *no free lunch* principle.
4. M4. Returns on individual assets (stocks) have higher standard deviations than returns on diversified portfolios of assets (stocks).
5. M5. Returns on different assets tend to be positively correlated. It is unusual for returns on different assets to be negatively correlated.
6. M6. Returns are approximately uncorrelated over time. That is, there is little evidence of linear time dependence in asset returns.
7. M7. Returns do not exhibit dynamic feedback. That is, there are no strong lead-lag effects between pairs of returns.
8. M8 There is some informal evidence of non-constant volatilities and correlations, particularly during financial crisis periods.

Daily returns have some features in common with monthly returns and some not. The common features are M1 and M3-M7 above. The stylized facts that are specific to daily returns are:

1. D2. Returns are not normally distributed. Empirical distributions have much fatter tails than the normal distribution (excess kurtosis).
2. D8. Returns are not independent over time. Absolute and squared returns are positively auto correlated and the correlation dies out very slowly. Volatility appears to be auto correlated and, hence, predictable.

These stylized facts of daily and monthly asset returns are the main features of returns that models of assets returns should capture. A *good model* is one that can explain many stylized facts of the data. A bad model does not capture important stylized facts.

5.6 Further Reading: Descriptive Statistics

(Fama 1976) is the first book that provides a comprehensive descriptive statistical analysis of asset returns. Much of the material in this chapter, and this book, is motivated by Fama's book. Several

recent textbooks discuss exploratory data analysis of finance time series. This discussion in this chapter is similar to that presented in (Carmona 2014), (Jondeau, Poon, and Rockinger 2007), (Ruppert and Matteson 2015), (Tsay 2010), (Tsay 2014), and (Zivot 2016).

- Mention books by Carol Alexander
- Mention R packages for descriptive statistics: dataExplorer,

5.7 Problems: Descriptive Statistics

Exercise 5.1. Compute time plots with *ggplot2* function `autoplot()`

Exercise 5.2. In this question, you will compute descriptive statistics for the continuously compounded monthly return data on the Vanguard long term bond index fund (VBLTX), Fidelity Magellan stock mutual fund (FMAGX), and Starbucks stock (SBUX). You will use the R packages **corrplot**, **IntroCompFinR**, **PerformanceAnalytics**, **tseries** and **zoo**.

1. Downloading data and return calculations.
    a. Use the **tseries** function `get.hist.quote()` to download from Yahoo! the monthly adjusted closing prices on VBLTX, FMAGX and SBUX over the period 1998-01-01 through 2009-12-31. Download these prices as `"zoo"` objects.
    b. Use the **PerformanceAnalytics** function `Return.calculate()` to compute continuously compounded monthly returns and remove the first NA observation.
2. Univariate graphical analysis.
    a. Make time plots of the return data using the **zoo** function `plot.zoo()`. Comment on any relationships between the returns suggested by the plots. Pay particular attention to the behavior of returns toward the end of 2008 at the beginning of the financial crisis.
    b. Make a equity curve/cumulative return plot (future of $1 invested in each asset) and comment. Which assets gave the best and worst future values over the investment horizon?
    c. For each return series, use the **IntroCompFinR** function `fourPanelPlot()` to make a four-panel plot containing a histogram, boxplot, normal QQ-plot, and SACF plot. Do the return series look normally distributed? Briefly compare the return distributions. Do you see any evidence of time dependence in the returns?
3. Univariate numerical summary statistics and historical VaR.
    a. Compute numerical descriptive statistics for all assets using the R functions `summary()`, `mean()`, `var()`, `stdev()`, and the **PerformanceAnalytics** functions `skewness()` and `kurtosis()`. Compare and contrast the descriptive statistics for the three assets. Which asset appears to be the riskiest asset?

b. Using the mean monthly return for each asset, compute an estimate of the annual continuously compounded return (i.e., recall the relationship between the expected monthly cc return and the expected annual cc return). Convert this annual continuously compounded return into a simple annual return. Are there any surprises?
c. Using the estimate of the monthly return standard deviation for each asset, compute an estimate of the annual return standard deviation. Briefly comment on the magnitude of the annual standard deviations.
d. For each asset compute the empirical 1% and 5% quantiles of the cc returns. Using these quantiles compute the 1% and 5% historical (monthly) VaR values based on an initial $100,000 investment. Which asset has the highest and lowest VaR values? Are you surprised?

4. Bivariate graphical analysis and numerical summary statistics

a. Use the R `pairs()` function to create all pair-wise scatterplots of returns Comment on the direction and strength of the linear relationships in these plots.
b. Use the **corrplot** functions `corrplot()` and `corrplot.mixed()` to plot the correlation matrix of the returns on the three assets.
c. Use the R functions `var()`, `cov()`, and `cor()` to compute the sample covariance matrix and sample correlation matrix of the returns. Comment on the direction and strength of the linear relationships suggested by the values of the covariances and correlations.

Exercise 5.3. In this question, you will compute descriptive statistics for the continuously compounded weekly return data on the Vanguard long term bond index fund (VBLTX), Fidelity Magellan stock mutual fund (FMAGX), and Starbucks stock (SBUX). Repeat the analysis for the monthly returns but use instead weekly returns computed from end-of-week adjusted closing prices.

Exercise 5.4. In this question, you will compute descriptive statistics for the continuously compounded daily return data on the Vanguard long term bond index fund (VBLTX), Fidelity Magellan stock mutual fund (FMAGX), and Starbucks stock (SBUX). Repeat the analysis for the monthly returns but use instead daily returns computed from end-of-day adjusted closing prices.

Exercise 5.5. In this problem you will analyze the monthly simple returns on the Vanguard S&P 500 index (VFINX) and Amazon stock (AMZN) over the 5-year period January 2010, through January 2015. For this period there are T=60 monthly returns. The monthly prices for VFINX are in the **IntroCompFinR** object `VangardPrices` and the daily prices for AMZN are in the **IntroCompFinR** object `amznDailyPrices`.

1. Compute simple monthly returns on AMZN and VFINX and plot these returns together in a single plot. Do the monthly returns from the two assets look like realizations from a covariance stationary stochastic process? Why or why not?

2. Compare and contrast the return characteristics of the two assets. In addition, comment on any common features, if any, of the two return series.
3. Plot the cumulative simple returns (equity curve) for each fund, which represents the growth of $1 invested in each fund over the sample period, and plot these returns in the same graph. Which fund performed better over the sample?
4. Using the **IntroCompFinR** function `fourPanelPlot()`, create descriptive statistics plots for each asset. In addition, compute Based on statistics, do the returns on VFINX and AMZN look normally distributed? Is there any evidence of linear time dependence? Briefly justify your answer.
5. Which asset appears to be riskier? Briefly justify your answer.

Exercise 5.6 (Descriptive statistics for weekly data)**.**

- Create weekly time series for MSFT, SP500
- Repeat graphical and numerical descriptive statistics
- Deduce stylized facts for weekly data

Exercise 5.7 (Rolling descriptive statistics for daily data)**.** TBD

Exercise 5.8. Loading data and computing returns

Load the daily price data from **IntroCompFinR**, and create monthly returns over the period Jan 1998 through Dec 2014:

```
data(amznDailyPrices, baDailyPrices, costDailyPrices, jwnDailyPrices, sbuxDailyPrices)
fiveStocks = merge(amznDailyPrices, baDailyPrices, costDailyPrices, jwnDailyPrices, sbuxDail
fiveStocks = to.monthly(fiveStocks, OHLC=FALSE)
```

Let's look at the data

```
head(fiveStocks, n=3)
#>          AMZN   BA COST  JWN SBUX
#> Jan 1998 4.92 34.0 17.4  9.4 4.24
#> Feb 1998 6.42 38.9 19.6 10.6 4.59
#> Mar 1998 7.13 37.4 21.4 11.8 5.25
```

```
tail(fiveStocks, n=3)
#>          AMZN  BA COST JWN SBUX
#> Oct 2014  305 123  128  72 75.0
#> Nov 2014  339 134  137  76 80.9
#> Dec 2014  310 129  137  79 81.8
```

Next, let's compute monthly simple returns using the **PerformanceAnalytics** function `Return.Calculate()`

```
fiveStocksRet = na.omit(Return.calculate(fiveStocks, method = "simple"))
head(fiveStocksRet, n=3)
#>             AMZN      BA   COST    JWN   SBUX
#> Feb 1998 0.3049  0.1425 0.1267 0.1298 0.0825
#> Mar 1998 0.1106 -0.0393 0.0920 0.1121 0.1438
#> Apr 1998 0.0729 -0.0396 0.0468 0.0262 0.0629
```

I removed the missing January return using the function `na.omit()`.

Part I: Univariate Graphical Analysis

1. Make time plots of the return data using the **zoo** function `plot.zoo()`. Comment on any relationships between the returns suggested by the plots.
2. Make a cumulative return plot (equity curve) showing the future value of $1 invested in each asset. Which assets gave the best and worst future values over the investment horizon? Hint: you can use the `cumprod()` function or the **PerformanceAnalytics** function `chart.CumReturns()`.
3. For each return series, make a four panel plot containing a histogram, boxplot, normal QQ-plot and SACF. Do the return series look normally distributed? Is there any evidence of time dependence? Briefly compare the return distributions. Hint: the `fourPanelPlot()` function only works on one asset at a time. So call the function for each asset. For example, `fourPanelPlot(fiveStocksRet[, "AMZN])`

Part II: Univariate Numerical Summaries

1. Compute numerical descriptive statistics for all assets using the R functions `mean()`, `var()`, `stdev()`, and the **PerformanceAnalytics** functions `skewness()` and `kurtosis()`. Compare and contrast the descriptive statistics for the five assets. Which asset appears to be the riskiest asset?
2. Using the monthly mean return for each asset, compute an estimate of the mean annual return. Hint: use the square root of time rule for cc returns. (The square root of time rule is exact for continuously compounded returns but can be applied as an approximation to simple returns.)
3. Using the monthly return standard deviation for each asset, compute an estimate of the annual return standard deviation. Hint: use the square root of time rule for cc returns. (The square root of time rule is exact for continuously compounded returns but can be applied as an approximation to simple returns.)

Part III: Historical VaR

1. For each asset compute the empirical 1% and 5% quantiles of the returns. Using these quantiles compute the 1% and 5% historical (monthly) VaR values based on an initial $100,000 investment. Which asset has the highest and lowest VaR values? Are you surprised?

Part IV. Bivariate Graphical Analysis

1. Use the `pairs()` function to create all pair-wise scatter plots of returns. Comment on the direction and strength of the linear relationships in these plots.
2. Use the **corrplot** function `corrplot.mixed()` to plot the correlation matrix of the returns on the five assets. Comment on what you see.

Part V. Bivariate Numerical Summary Statistics

Use the R functions `cov()`, and `cor()` to compute the sample covariance matrix and sample correlation matrix of the returns. Comment on the direction and strength of the linear relationships suggested by the values of the covariances and correlations.

Part VI. Stylized Facts

1. Based on your analysis of the five Northwest stocks, what stylized facts do you see?

6 The Gaussian White Noise Return Model

Updated: September 4, 2026

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The first model of asset returns I consider is the very simple *Gaussian White Noise* (GWN) model . This model is motivated by the stylized facts for monthly asset returns discussed in the previous chapter. The GWN model assumes that an asset's return (simple or continuously compounded) over time is independent and identically normally distributed. The model allows for the returns on different assets to be contemporaneously correlated but does not allow for any lead-lag cross correlations. The GWN model is widely used in finance. For example, it is used in risk analysis (e.g., for computing Value-at-Risk) for assets and portfolios, in mean-variance portfolio analysis, in the *Capital Asset Pricing Model* (CAPM), and in the *Black-Scholes* option pricing model. Although this model is very simple, it provides important intuition about the statistical behavior of asset returns and prices and serves as a benchmark against which more sophisticated models can be compared and evaluated. It allows me to discuss and develop several important econometric topics such as Monte Carlo simulation, estimation, bootstrapping, hypothesis testing, forecasting, and model evaluation that are discussed in later chapters.

The outline of this chapter is as follows. In Section 6.1 I review the assumptions of the GWN model and presents several equivalent specifications of the model. I also discuss the application of the model to continuously compounded returns, simple returns, and portfolios. In Section 6.2 I illustrate Monte Carlo simulation of the GWN model as a first-step reality check of the model.

The R packages I use in this chapter are **introcompfinr**, **mvtnorm**, and **PerformanceAnalytics**. Make sure these packages are installed and loaded before running the R examples in the chapter.

6.1 GWN Model Assumptions

Let R_{it} denote the simple or continuously compounded (cc) return on asset i over the investment horizon between times $t-1$ and t (e.g., monthly returns). I make the following assumptions regarding the probability distribution of R_{it} for $i = 1, \dots, N$ assets for all times t.

Assumption GWN

1. *Covariance stationarity and ergodicity*: $\{R_{i1}, \ldots, R_{iT}\} = \{R_{it}\}_{t=1}^{T}$ is a covariance stationary and ergodic stochastic process with $E[R_{it}] = \mu_i$, $\text{var}(R_{it}) = \sigma_i^2$, $\text{cov}(R_{it}, R_{jt}) = \sigma_{ij}$ and $\text{cor}(R_{it}, R_{jt}) = \rho_{ij}$.
2. *Normality*: $R_{it} \sim N(\mu_i, \sigma_i^2)$ for all i and t, and all joint distributions are normal.
3. *No serial correlation*: $\text{cov}(R_{it}, R_{js}) = \text{cor}(R_{it}, R_{is}) = 0$ for $t \neq s$ and $i, j = 1, \ldots, N$.

Assumption GWN states that in every time period asset returns are jointly (multivariate) normally distributed, that the means and the variances of all asset returns, and all of the pairwise contemporaneous covariances and correlations between assets are constant over time. In addition, all of the asset returns are *serially uncorrelated*:

$$\text{cor}(R_{it}, R_{is}) = \text{cov}(R_{it}, R_{is}) = 0 \text{ for all } i \text{ and } t \neq s,$$

and the returns on all possible pairs of assets i and j are *cross lead-lag uncorrelated*:

$$\text{cor}(R_{it}, R_{js}) = \text{cov}(R_{it}, R_{js}) = 0 \text{ for all } i \neq j \text{ and } t \neq s.$$

In addition, under the normal distribution assumption lack of serial and cross lead-lag correlation implies time independence of returns. Clearly, these are very strong assumptions. However, they allow us to develop a straightforward probabilistic model for asset returns as well as statistical tools for estimating the parameters of the model, testing hypotheses about the parameter values and assumptions.

6.1.1 Regression model representation

A convenient mathematical representation or *model* of asset returns can be given based on Assumption GWN. This is the *GWN regression model*.

Definition 6.1 (GWN Regression Model)**.** For assets $i = 1, \ldots, N$ and time periods $t = 1, \ldots, T$, the GWN regression model is:

$$\begin{aligned} R_{it} &= \mu_i + \varepsilon_{it}, \\ \{\varepsilon_{it}\}_{t=1}^{T} &\sim \text{GWN}(0, \sigma_i^2), \\ \text{cov}(\varepsilon_{it}, \varepsilon_{js}) &= \begin{cases} \sigma_{ij} & t = s \\ 0 & t \neq s \end{cases}. \end{aligned} \tag{6.1}$$

The notation $\varepsilon_{it} \sim \text{GWN}(0, \sigma_i^2)$ stipulates that the stochastic process $\{\varepsilon_{it}\}_{t=1}^{T}$ is a GWN process with $E[\varepsilon_{it}] = 0$ and $\text{var}(\varepsilon_{it}) = \sigma_i^2$. In addition, the random error term ε_{it} is independent of ε_{js} for all assets $i \neq j$ and all time periods $t \neq s$.

Using the basic properties of expectation, variance and covariance, you can derive the following properties of returns in the GWN model:

$$\begin{aligned}
E[R_{it}] &= E[\mu_i + \varepsilon_{it}] = \mu_i + E[\varepsilon_{it}] = \mu_i, \\
\text{var}(R_{it}) &= \text{var}(\mu_i + \varepsilon_{it}) = \text{var}(\varepsilon_{it}) = \sigma_i^2, \\
\text{cov}(R_{it}, R_{jt}) &= \text{cov}(\mu_i + \varepsilon_{it}, \mu_j + \varepsilon_{jt}) = \text{cov}(\varepsilon_{it}, \varepsilon_{jt}) = \sigma_{ij}, \\
\text{cov}(R_{it}, R_{js}) &= \text{cov}(\mu_i + \varepsilon_{it}, \mu_j + \varepsilon_{js}) = \text{cov}(\varepsilon_{it}, \varepsilon_{js}) = 0, \ t \neq s.
\end{aligned}$$

Given that covariances and variances of returns are constant over time implies that the correlations between returns over time are also constant:

$$\begin{aligned}
\text{cor}(R_{it}, R_{jt}) &= \frac{\text{cov}(R_{it}, R_{jt})}{\sqrt{\text{var}(R_{it})\text{var}(R_{jt})}} = \frac{\sigma_{ij}}{\sigma_i \sigma_j} = \rho_{ij}, \\
\text{cor}(R_{it}, R_{js}) &= \frac{\text{cov}(R_{it}, R_{js})}{\sqrt{\text{var}(R_{it})\text{var}(R_{js})}} = \frac{0}{\sigma_i \sigma_j} = 0, \ i \neq j, t \neq s.
\end{aligned}$$

Finally, since $\{\varepsilon_{it}\}_{t=1}^T \sim \text{GWN}(0, \sigma_i^2)$ it follows that $\{R_{it}\}_{t=1}^T \sim iid\ N(\mu_i, \sigma_i^2)$. Hence, the GWN regression model Equation 6.1 for R_{it} is equivalent to the model implied by Assumption GWN.

6.1.1.1 Interpretation of the GWN regression model

The GWN model has a very simple form and is identical to the *measurement error model* in the statistics literature.[1] In words, the model states that each asset return is equal to a constant μ_i (the expected return) plus an *i.i.d.* normally distributed random variable ε_{it} with mean zero and constant variance. The random variable ε_{it} can be interpreted as representing the *unexpected news* concerning the value of the asset that arrives between time $t-1$ and time t. To see this, Equation 6.1 implies that:

$$\varepsilon_{it} = R_{it} - \mu_i = R_{it} - E[R_{it}],$$

so that ε_{it} is defined as the deviation of the random return from its expected value. If the news between times $t-1$ and t is good, then the realized value of ε_{it} is positive and the observed return is above its expected value μ_i. If the news is bad, then ε_{it} is negative and the observed return is less than expected. The assumption $E[\varepsilon_{it}] = 0$ means that news, on average, is neutral; neither good nor bad. The assumption that $\text{sd}(\varepsilon_{it}) = \sigma_i$ can be interpreted as saying that the volatility, or typical magnitude, of news arrival is constant over time. The random news variable affecting asset i, ε_{it}, is allowed to be contemporaneously correlated with the random news variable affecting asset j, ε_{jt}, to capture the idea that news about one asset may spill over and affect another asset. For

[1] In the measurement error model, r_{it} represents the tth measurement of some physical quantity μ_i and ε_{it} represents the random measurement error associated with the measurement device. The value σ_i represents the typical size of a measurement error.

example, if asset i is Microsoft stock and asset j is Apple Computer stock, then one interpretation of news in this context is general news about the computer industry and technology. Good news should lead to positive values of both ε_{it} and ε_{jt}. Hence these variables will be positively correlated due to a positive reaction to a common news component. Finally, the news on asset j at time s is unrelated to the news on asset i at time t for all times $t \neq s$. For example, this means that the news for Apple in January is not related to the news for Microsoft in February.

6.1.2 Location-Scale model representation

Sometimes it is convenient to re-express the regression form of the GWN model Equation 6.1 in *location-scale* form:

$$R_{it} = \mu_i + \varepsilon_{it} = \mu_i + \sigma_i \cdot Z_{it}, \ \{Z_{it}\}_{t=1}^{T} \sim \text{GWN}(0,1), \tag{6.2}$$

where I use the decomposition $\varepsilon_{it} = \sigma_i \cdot Z_{it}$. In this form, the random news shock is the *i.i.d.* standard normal random variable Z_{it} scaled by the "news" volatility σ_i.

This form is particularly convenient for Value-at-Risk calculations because the $\alpha \times 100\%$ quantile of the return distribution has the simple form:

$$q_\alpha^{R_i} = \mu_i + \sigma_i \times q_\alpha^Z,$$

where q_α^Z is the $\alpha \times 100\%$ quantile of the standard normal random distribution. Let W_0 be the initial amount of wealth to be invested from time $t-1$ to t. If R_{it} is the simple return then,

$$\text{VaR}_\alpha = W_0 \times q_\alpha^{R_i} = W_0 \times \left(\mu_i + \sigma_i \times q_\alpha^Z\right),$$

whereas if R_{it} is the continuously compounded return then,

$$\text{VaR}_\alpha = W_0 \times \left(e^{q_\alpha^{R_i}} - 1\right) = W_0 \times \left(e^{\mu_i + \sigma_i \times q_\alpha^Z} - 1\right).$$

6.1.3 The GWN model in matrix notation

Define the $N \times 1$ vectors $\mathbf{R}_t = (R_{1t}, \ldots, R_{Nt})'$, $\mu = (\mu_1, \ldots, \mu_N)'$, $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{Nt})'$ and the $N \times N$ symmetric covariance matrix:

$$\text{var}(\varepsilon_t) = \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_N^2 \end{pmatrix}.$$

Definition 6.2 (GWN Regression Model in Matrix Form)**.** The regression form of the GWN model in matrix notation is:

$$\mathbf{R}_t = \mu + \varepsilon_t, \varepsilon_t \sim \text{iid } N(\mathbf{0}, \Sigma), \tag{6.3}$$

which implies that $\mathbf{R}_t \sim iid\ N(\mu, \Sigma)$.

The location-scale form of the GWN model in matrix notation makes use of the matrix square root factorization $\Sigma = \Sigma^{1/2}\Sigma^{1/2\prime}$ where $\Sigma^{1/2}$ is the lower-triangular matrix square root (usually the Cholesky factorization). Then Equation 6.3 can be rewritten as:

$$\mathbf{R}_t = \mu + \Sigma^{1/2}\mathbf{Z}_t, \mathbf{Z}_t \sim \text{GWN}(\mathbf{0}, \mathbf{I}_N), \tag{6.4}$$

where $\mathbf{I}_N$ denotes the N-dimensional identity matrix.

6.1.4 The GWN model for continuously compounded returns

The GWN model is often used to describe continuously compounded (cc) returns defined as $R_{it} = \ln(P_{it}/P_{it-1})$ where P_{it} is the price of asset i at time t. An advantage of the GWN model for cc returns is that the model aggregates to any time horizon because multi-period cc returns are additive. This is particularly convenient for investment risk analysis. The GWN model for cc returns also gives rise to the *random walk model* for the logarithm of asset prices. The normal distribution assumption of the GWN model for cc returns implies that single-period simple returns are log-normally distributed.

A disadvantage of the GWN model for cc returns is that the model has some limitations for the analysis of portfolios because the cc return on a portfolio of assets is not a weighted average of the cc returns on the individual assets. As a result, for portfolio analysis the GWN model is typically applied to simple returns. However, if the investment horizon is short (e.g. daily, weekly or monthly) then simple returns are typically close to zero on average and there is little difference between simple and cc returns. Hence, the time aggregation results of the GWN model for cc returns can be applied to simple returns and the portfolio results of the GWN model for simple returns can be applied to cc returns.

6.1.4.1 Time Aggregation and the GWN Model

The GWN model for cc returns has the following nice aggregation property with respect to the interpretation of ε_{it} as news. For illustration purposes, suppose that t represents months so that R_{it} is the cc monthly return on asset i. Now, instead of the monthly return, suppose you are interested in the annual cc return $R_{it} = R_{it}(12)$. Since multi-period cc returns are additive, $R_{it}(12)$ is the

sum of 12 monthly cc returns:

$$R_{it} = R_{it}(12) = \sum_{k=0}^{11} R_{it-k} = R_{it} + R_{it-1} + \cdots + R_{it-11}.$$

Using the GWN regression model Equation 6.1 for the monthly return R_{it}, you may express the annual return $R_{it}(12)$ as:

$$R_{it}(12) = \sum_{t=0}^{11}(\mu_i + \varepsilon_{it}) = 12 \cdot \mu_i + \sum_{t=0}^{11} \varepsilon_{it} = \mu_i(12) + \varepsilon_{it}(12),$$

where $\mu_i(12) = 12 \cdot \mu_i$ is the annual expected return on asset i, and $\varepsilon_{it}(12) = \sum_{k=0}^{11} \varepsilon_{it-k}$ is the annual random news component. The annual expected return, $\mu_i(12)$, is simply 12 times the monthly expected return, μ_i. The annual random news component, $\varepsilon_{it}(12)$, is the accumulation of news over the year. As a result, the variance of the annual news component, $\sigma_i^2(12)$, is 12 times the variance of the monthly news component:

$$\begin{aligned}
\text{var}(\varepsilon_{it}(12)) &= \text{var}\left(\sum_{k=0}^{11} \varepsilon_{it-k}\right) \\
&= \sum_{k=0}^{11} \text{var}(\varepsilon_{it-k}) \text{ since } \varepsilon_{it} \text{ is uncorrelated over time} \\
&= \sum_{k=0}^{11} \sigma_i^2 \text{ since var}(\varepsilon_{it}) \text{ is constant over time} \\
&= 12 \cdot \sigma_i^2 = \sigma_i^2(12).
\end{aligned}$$

It follows that the standard deviation of the annual news is equal to $\sqrt{12}$ times the standard deviation of monthly news:

$$\text{sd}(\varepsilon_{it}(12)) = \sqrt{12} \times \text{sd}(\varepsilon_{it}) = \sqrt{12} \times \sigma_i.$$

Similarly, due to the additivity of covariances, the covariance between $\varepsilon_{it}(12)$ and $\varepsilon_{jt}(12)$ is 12 times the monthly covariance:

$$\begin{aligned}
\text{cov}(\varepsilon_{it}(12), \varepsilon_{jt}(12)) &= \text{cov}\left(\sum_{k=0}^{11} \varepsilon_{it-k}, \sum_{k=0}^{11} \varepsilon_{jt-k}\right) \\
&= \sum_{k=0}^{11} \text{cov}(\varepsilon_{it-k}, \varepsilon_{jt-k}) \text{ since } \varepsilon_{it} \text{ and } \varepsilon_{jt} \text{ are uncorrelated over time} \\
&= \sum_{k=0}^{11} \sigma_{ij} \text{ since cov}(\varepsilon_{it}, \varepsilon_{jt}) \text{ is constant over time} \\
&= 12 \cdot \sigma_{ij} = \sigma_{ij}(12).
\end{aligned}$$

The above results imply that the correlation between the annual errors $\varepsilon_{it}(12)$ and $\varepsilon_{jt}(12)$ is the same as the correlation between the monthly errors ε_{it} and ε_{jt}:

$$\begin{aligned}\text{cor}(\varepsilon_{it}(12), \varepsilon_{jt}(12)) &= \frac{\text{cov}(\varepsilon_{it}(12), \varepsilon_{jt}(12))}{\sqrt{\text{var}(\varepsilon_{it}(12)) \cdot \text{var}(\varepsilon_{jt}(12))}} \\ &= \frac{12 \cdot \sigma_{ij}}{\sqrt{12\sigma_i^2 \cdot 12\sigma_j^2}} \\ &= \frac{\sigma_{ij}}{\sigma_i \sigma_j} = \rho_{ij} = \text{cor}(\varepsilon_{it}, \varepsilon_{jt}).\end{aligned}$$

The above results generalize to aggregating returns to arbitrary time horizons as summarized in the following Proposition.

Proposition 6.1 (Square Root of Time Rule)**.** *Let R_{it} denote the cc return on asset i between times $t-1$ and t, where t represents the general investment horizon, and let $R_{it}(k) = \sum_{j=0}^{k-1} R_{it-j}$ denote the k-period cc return. Then the GWN model for $R_{it}(k)$ has the form:*

$$R_{it}(k) = \mu_i(k) + \varepsilon_{it}(k), \varepsilon_{it}(k) \sim \text{GWN}(0, \sigma_i^2(k)), \tag{6.5}$$

where $\mu_i(k) = k \times \mu_i$ is the k-period expected return, $\varepsilon_{it}(k) = \sum_{j=0}^{k-1} \varepsilon_{it-j}$ is the k-period error term, and $\sigma_i^2(k) = k \times \sigma_i^2$ is the k-period variance. The k-period volatility follows the square-root-of-time rule*: $\sigma_i(k) = \sqrt{k} \times \sigma_i$. The k-period covariance between asset i and asset j is $\sigma_{ij}(k) = k \times \sigma_{ij}$, and the k-period correlation is $\rho_{ij}(k) = \rho_{ij}$. This aggregation result is exact for cc returns but it is often used as an approximation for simple returns.*

6.1.4.2 The random walk model of asset prices

The GWN model for cc returns Equation 6.1 gives rise to the so-called *random walk* (RW) model for the *logarithm* of asset prices. To see this, recall that the continuously compounded return, R_{it}, is defined from asset prices via $R_{it} = \ln\left(\frac{P_{it}}{P_{it-1}}\right) = \ln(P_{it}) - \ln(P_{it-1})$. Letting $p_{it} = \ln(P_{it})$ and using the representation of R_{it} in the GWN model Equation 6.1, you can express the log-price as:

$$p_{it} = p_{it-1} + \mu_i + \varepsilon_{it}. \tag{6.6}$$

The representation in Equation 6.6 is known as the RW model for log-prices.[2]. It is a representation of the GWN model in terms of log-prices.

[2]The model Equation 6.6 is technically a random walk with drift μ_i A pure random walk has zero drift ($\mu_i = 0$)

In the RW model Equation 6.6, μ_i represents the expected change in the log-price (cc return) between months $t-1$ and t, and ε_{it} represents the unexpected change in the log-price. That is,

$$\begin{aligned} E[\Delta p_{it}] &= E[R_{it}] = \mu_i, \\ \varepsilon_{it} &= \Delta p_{it} - E[\Delta p_{it}]. \end{aligned}$$

where $\Delta p_{it} = p_{it} - p_{it-1}$. Further, in the RW model, the unexpected changes in the log-price, $\{\varepsilon_{it}\}$, are uncorrelated over time ($\text{cov}(\varepsilon_{it}, \varepsilon_{is}) = 0$ for $t \neq s$) so that future changes in the log-price cannot be predicted from past changes in the log-price.[3]

The RW model gives the following interpretation for the evolution of log prices. Let p_{i0} denote the initial log price of asset i. The RW model says that the log-price at time $t = 1$ is:

$$p_{i1} = p_{i0} + \mu_i + \varepsilon_{i1},$$

where ε_{i1} is the value of random news that arrives between times 0 and 1. The expected log-price at time $t = 1$ is:

$$E[p_{i1}] = p_{i0} + \mu_i + E[\varepsilon_{i1}] = p_{i0} + \mu_i,$$

which is the initial price plus the expected return between times 0 and 1. Similarly, by recursive substitution the log-price at time $t = 2$ is:

$$\begin{aligned} p_{i2} &= p_{i1} + \mu_i + \varepsilon_{i2} \\ &= p_{i0} + \mu_i + \mu_i + \varepsilon_{i1} + \varepsilon_{i2} \\ &= p_{i0} + 2 \cdot \mu_i + \sum_{t=1}^{2} \varepsilon_{it} \\ &= p_{i0} + 2 \cdot \mu_i + \varepsilon_{it}(2), \end{aligned}$$

which is equal to the initial log-price, p_{i0}, plus the two period expected return, $2 \cdot \mu_i$, plus the accumulated random news over the two periods, $\sum_{t=1}^{2} \varepsilon_{it} = \varepsilon_{it}(2)$. By repeated recursive substitution, the log price at time $t = T$ is,

$$p_{iT} = p_{i0} + T \cdot \mu_i + \sum_{t=1}^{T} \varepsilon_{it} = p_{i0} + T \cdot \mu_i + \varepsilon_{it}(T).$$

At time $t = 0$, the expected log-price at time $t = T$ is,

$$E[p_{iT}] = p_{i0} + T \cdot \mu_i,$$

which is the initial price plus the expected growth in prices over T periods. The actual price, p_{iT},

[3] The notion that future changes in asset prices cannot be predicted from past changes in asset prices is often referred to as the weak form of the efficient markets hypothesis.

deviates from the expected price by the accumulated random news:

$$p_{iT} - E[p_{iT}] = \sum_{t=1}^{T} \varepsilon_{it} = \varepsilon_{it}(T).$$

At time $t = 0$, the variance of the log-price at time T is,

$$\text{var}(p_{iT}) = \text{var}\left(\sum_{t=1}^{T} \varepsilon_{it}\right) = T \cdot \sigma_i^2$$

Hence, the RW model implies that the stochastic process of log-prices $\{p_{it}\}$ is non-stationary because the variance of p_{it} increases with t. Finally, because $\varepsilon_{it} \sim (0, \sigma_i^2)$ it follows that (conditional on p_{i0}) $p_{iT} \sim N(p_{i0} + T\mu_i, T\sigma_i^2)$.

The term *random walk* was originally used to describe the unpredictable movements of a drunken sailor staggering down the street. The sailor starts at an initial position, p_0, outside the bar. The sailor generally moves in the direction described by μ but randomly deviates from this direction after each step t by an amount equal to ε_t. After T steps the sailor ends up at position $p_T = p_0 + \mu \cdot T + \sum_{t=1}^{T} \varepsilon_t$. The sailor is expected to be at location μT, but where he actually ends up depends on the accumulation of the random changes in direction $\sum_{t=1}^{T} \varepsilon_t$. Because $\text{var}(p_T) = \sigma^2 T$, the uncertainty about where the sailor will be increases with each step.

The RW model for log-prices implies the following model for prices:

$$P_{it} = e^{p_{it}} = P_{i0} e^{\mu_i \cdot t + \sum_{s=1}^{t} \varepsilon_{is}} = P_{i0} e^{\mu_i t} e^{\sum_{s=1}^{t} \varepsilon_{is}},$$

where $p_{it} = p_{i0} + \mu_i t + \sum_{s=1}^{t} \varepsilon_s$. Since $P_{it} = e^{p_{it}}$, $P_{it} > 0$. The term $e^{\mu_i t}$ represents the expected exponential growth rate in prices between times 0 and time t, and the term $e^{\sum_{s=1}^{t} \varepsilon_{is}}$ represents the unexpected exponential growth in prices. Here, conditional on P_{i0}, P_{it} is log-normally distributed because $p_{it} = \ln P_{it}$ is normally distributed.

6.1.5 GWN model for simple returns

For simple returns, defined as $R_{it} = (P_{it} - P_{it-1})/P_{it-1}$, the GWN model is often used for the analysis of portfolios as discussed in chapters Chapter 10 - Chapter 14. The reason is that the simple return on a portfolio of N assets is weighted average of the simple returns on the individual assets. Hence, the GWN model for simple returns extends naturally to portfolios of assets.

6.1.5.1 GWN Model and Portfolios

Consider the GWN model in matrix form Equation 6.3 for the $N \times 1$ vector of simple returns $\mathbf{R}_t = (R_{1t}, \ldots, R_{Nt})'$. For a vector of portfolio weights $\mathbf{x} = (x_1, \ldots, x_N)$ such that $\mathbf{x}'\mathbf{1} = \sum_{i=1}^{N} x_i = 1$, the

simple return on the portfolio is:

$$R_{pt} = \mathbf{x}'\mathbf{R}_t = \sum_{i=1}^{N} x_i R_{it}.$$

Substituting in Equation 6.1 gives the GWN model for the portfolio returns:

$$R_{pt} = \mathbf{x}'(\mu + \varepsilon_t) = \mathbf{x}'\mu + \mathbf{x}'\varepsilon_t = \mu_p + \varepsilon_{pt} \tag{6.7}$$

where $\mu_p = \mathbf{x}'\mu = \sum_{i=1}^{N} x_i\mu_i$ is the portfolio expected return, and $\varepsilon_{pt} = \mathbf{x}'\varepsilon_t = \sum_{i=1}^{N} x_i\varepsilon_{it}$ is the portfolio error. The variance of R_{pt} is given by:

$$\text{var}(R_{pt}) = \text{var}(\mathbf{x}'\mathbf{R}_t) = \mathbf{x}'\Sigma\mathbf{x} = \sigma_p^2.$$

Therefore, the distribution of portfolio returns is normal,

$$R_{pt} \sim N(\mu_p, \sigma_p^2).$$

This result is exact for simple returns but is often used as an approximation for cc returns.

6.1.5.2 GWN Model for Multi-Period Simple Returns

The GWN model for single period simple returns does not extend exactly to multi-period simple returns because multi-period simple returns are not additive. Recall, the k-period simple return has a multiplicative relationship to single period returns:

$$\begin{aligned} R_t(k) &= (1 + R_t)(1 + R_{t-1}) \times \cdots \times (1 + R_{t-k+1}) - 1 \\ &= R_t + R_{t-1} + \cdots + R_{t-k+1} \\ &+ R_t R_{t-1} + R_t R_{t-2} + \cdots + R_{t-k+2} R_{t-k+1}. \end{aligned}$$

Even though single period returns are normally distributed in the GWN model, multi-period returns are not normally distributed because the product of two normally distributed random variables (e.g., $R_t R_{t-1}$) is not normally distributed. Hence, the GWN model does not exactly generalize to multi-period simple returns. However, if single period returns are small then all of the cross products of returns are approximately zero ($R_t R_{t-1} \approx \cdots \approx R_{t-k+2} R_{t-k+1} \approx 0$) and,

$$\begin{aligned} R_t(k) &\approx R_t + R_{t-1} + \cdots + R_{t-k+1} \\ &\approx \mu(k) + \varepsilon_t(k). \end{aligned}$$

where $\mu(k) = k\mu$ and $\varepsilon_t(k) = \sum_{j=0}^{k-1} \varepsilon_{t-j}$. Hence, the GWN model is approximately true for multi-period simple returns when single period simple returns are not too big.

Some exact returns can be derived for the mean and variance of multi-period simple returns. For simplicity, let $k = 2$ so that,

$$R_t(2) = (1 + R_t)(1 + R_{t-1}) - 1 = R_t + R_{t-1} + R_t R_{t-1}.$$

Substituting in Equation 6.1 then gives,

$$\begin{aligned} R_t(2) &= (\mu + \varepsilon_t) + (\mu + \varepsilon_{t-1}) + (\mu + \varepsilon_t)(\mu + \varepsilon_{t-1}) \\ &= 2\mu + \varepsilon_t + \varepsilon_{t-1} + \mu^2 + \mu\varepsilon_t + \mu\varepsilon_{t-1} + \varepsilon_t\varepsilon_{t-1} \\ &= 2\mu + \mu^2 + \varepsilon_t(1 + \mu) + \varepsilon_{t-1}(1 + \mu) + \varepsilon_t\varepsilon_{t-1}. \end{aligned}$$

The result for the expected return is easy,

$$\begin{aligned} E[R_t(2)] &= 2\mu + \mu^2 + (1 + \mu)E[\varepsilon_t] + (1 + \mu)E[\varepsilon_{t-1}] + E[\varepsilon_t\varepsilon_{t-1}] \\ &= 2\mu + \mu^2 = (1 + \mu)^2 - 1, \end{aligned}$$

The result uses the independence of ε_t and ε_{t-1} to get $E[\varepsilon_t\varepsilon_{t-1}] = E[\varepsilon_t]E[\varepsilon_{t-1}] = 0$. The result for the variance, however, is more work:

$$\begin{aligned} \text{var}(R_t(2)) &= \text{var}(\varepsilon_t(1 + \mu) + \varepsilon_{t-1}(1 + \mu) + \varepsilon_t\varepsilon_{t-1}) \\ &= (1 + \mu)^2\text{var}(\varepsilon_t) + (1 + \mu)^2\text{var}(\varepsilon_{t-1}) + \text{var}(\varepsilon_t\varepsilon_{t-1}) \\ &+ 2(1 + \mu)^2\text{cov}(\varepsilon_t, \varepsilon_{t-1}) + 2(1 + \mu)\text{cov}(\varepsilon_t, \varepsilon_t\varepsilon_{t-1}) \\ &+ 2(1 + \mu)\text{cov}(\varepsilon_{t-1}, \varepsilon_t\varepsilon_{t-1}) \end{aligned}$$

Now, $\text{var}(\varepsilon_t) = \text{var}(\varepsilon_{t-1}) = \sigma^2$ and $\text{cov}(\varepsilon_t, \varepsilon_{t-1}) = 0$. Next, note that,

$$\text{var}(\varepsilon_t\varepsilon_{t-1}) = E[\varepsilon_t^2\varepsilon_{t-1}^2] - (E[\varepsilon_t\varepsilon_{t-1}])^2 = E[\varepsilon_t^2]E[\varepsilon_{t-1}^2] - (E[\varepsilon_t]E[\varepsilon_{t-1}])^2 = 2\sigma^2.$$

Finally,

$$\begin{aligned} \text{cov}(\varepsilon_t, \varepsilon_t\varepsilon_{t-1}) &= E[\varepsilon_t(\varepsilon_t\varepsilon_{t-1})] - E[\varepsilon_t]E[\varepsilon_t\varepsilon_{t-1}] \\ &= E[\varepsilon_t^2]E[\varepsilon_{t-1}] - E[\varepsilon_t]E[\varepsilon_t]E[\varepsilon_{t-1}] \\ &= 0. \end{aligned}$$

Then,

$$\begin{aligned} \text{var}(R_t(2)) &= (1 + \mu)^2\sigma^2 + (1 + \mu)^2\sigma^2 + 2\sigma^2 \\ &= 2\sigma^2[(1 + \mu)^2 + 1]. \end{aligned}$$

If μ is close to zero then $E[R_t(2)] \approx 2\mu$ and $\text{var}(R_t(2)) \approx 2\sigma^2$ and so the square-root-of-time rule holds approximately.

6.1.5.3 Implied model for prices

Since $R_{it} = (P_{it} - P_{it-1})/P_{it-1}$, you can write

$$P_{it} = P_{it-1}(1 + R_{it})$$

Starting at $t = 1$ and assuming $P_{i0} > 0$ is fixed, by recursive substitution you get

$$P_{it} = P_{i0}(1 + R_{i1})(1 + R_{i2}) \dots (1 + R_{it-1})$$

Prices will be positive provided all returns are greater than -1. This model for prices is not a RW model but behaves similarly to the RW model if all simple returns are close to zero.

6.2 Monte Carlo Simulation of the GWN Model

A simple technique that can be used to understand the probabilistic behavior of a model involves using computer simulation methods to create pseudo data from the model. The process of creating such pseudo data is called *Monte Carlo simulation*.[4] Monte Carlo simulation of a model can be used as a *first step reality check* of the model. If simulated data from the model do not look like the data that the model is supposed to describe, then serious doubt is cast on the model. However, if simulated data look reasonably close to the actual data then the first step reality check is passed. Ideally, one should consider many simulated samples from the model because it is possible for a given simulated sample to look strange simply because of an unusual set of random numbers.

Monte Carlo simulations can be used to reveal properties of a model that are hard to derive analytically.

Monte Carlo simulation can also be used to create "what if?" type *scenarios* for a model. Different scenarios typically correspond with different model parameter values.

Finally, Monte Carlo simulation can be used to study properties of statistics computed from the pseudo data from the model. For example, Monte Carlo simulation can be used to illustrate the concepts of estimator bias and confidence interval coverage probabilities (see Chapter Chapter 9).

To illustrate the use of Monte Carlo simulation, consider creating pseudo return data from the GWN model Equation 6.1 for a single asset. The steps to create a Monte Carlo simulation from the GWN model are:

Monte Carlo Steps: Single Asset

1. Fix values for the GWN model parameters μ and σ.
2. Determine the number of simulated values, T, to create.

[4] Monte Carlo refers to the famous city in Monaco where gambling is legal.

3. Use a computer random number generator to simulate T *iid* values of ε_t from a $N(0, \sigma^2)$ distribution. Denote these simulated values as $\tilde{\varepsilon}_1, \ldots, \tilde{\varepsilon}_T$.
4. Create the simulated return data $\tilde{R}_t = \mu + \tilde{\varepsilon}_t$ for $t = 1, \ldots, T$.

Example 6.1 (Microsoft data to calibrate univariate Monte Carlo simulation of GWN model)**.** To motivate plausible values for μ and σ in the simulation, Figure 6.1 shows the monthly continuously compounded (cc) returns on Microsoft stock over the period January 1998 through May 2012. The data is the same as that used in Chapter Chapter 5 and is constructed from the **IntroCompFinR** `"xts"` object `msftDailyPrices` as follows:

```
data(msftDailyPrices)
msftPrices = to.monthly(msftDailyPrices, OHLC=FALSE)
smpl = "1998-01::2012-05"
msftPrices = msftPrices[smpl]
msftRetC = na.omit(Return.calculate(msftPrices, method="log"))
mu.msftRetC = mean(msftRetC)
sigma.msftRetC = sd(msftRetC)
c(mu.msftRetC, sigma.msftRetC)
#> [1] 0.00413 0.10021
```

The parameter $\mu = E[R_t]$ in the GWN model is the expected monthly return, and σ represents the typical size of a deviation about μ. In Figure 6.1, the returns seem to fluctuate up and down about a central value near 0 and the typical size of a return deviation about 0 is roughly 0.10, or 10% (see dashed lines in Figure 6.1). The sample mean turns out to be $\hat{\mu} = 0.004$ (0.4%) and the sample standard deviation is $\hat{\sigma} = 0.100$ (10%). Figure 6.2 shows three distribution summaries (histogram, boxplot and normal qq-plot) and the SACF computed from the sample returns. The returns look to have slightly fatter tails than the normal distribution and show little evidence of linear time dependence (autocorrelation).

■

Example 6.2 (Simulating observations from the GWN model)**.** To mimic the monthly return data on Microsoft in the Monte Carlo simulation, the values $\mu = 0.004$ and $\sigma = 0.10$ are used as the model's true parameters and $T = 172$ is the number of simulated values (sample size of actual data). Let $\{\tilde{\varepsilon}_1, \ldots, \tilde{\varepsilon}_{172}\}$ denote the 172 simulated values of the random news variable $\varepsilon_t \sim \text{GWN}(0, (0.10)^2)$. The simulated returns are then computed using:[5]

$$\tilde{R}_t = 0.004 + \tilde{\varepsilon}_t, \; t = 1, \ldots, 172 \tag{6.8}$$

[5] Alternatively, the returns can be simulated directly by simulating observations from a normal distribution with mean 0.0 and standard deviation 0.10.

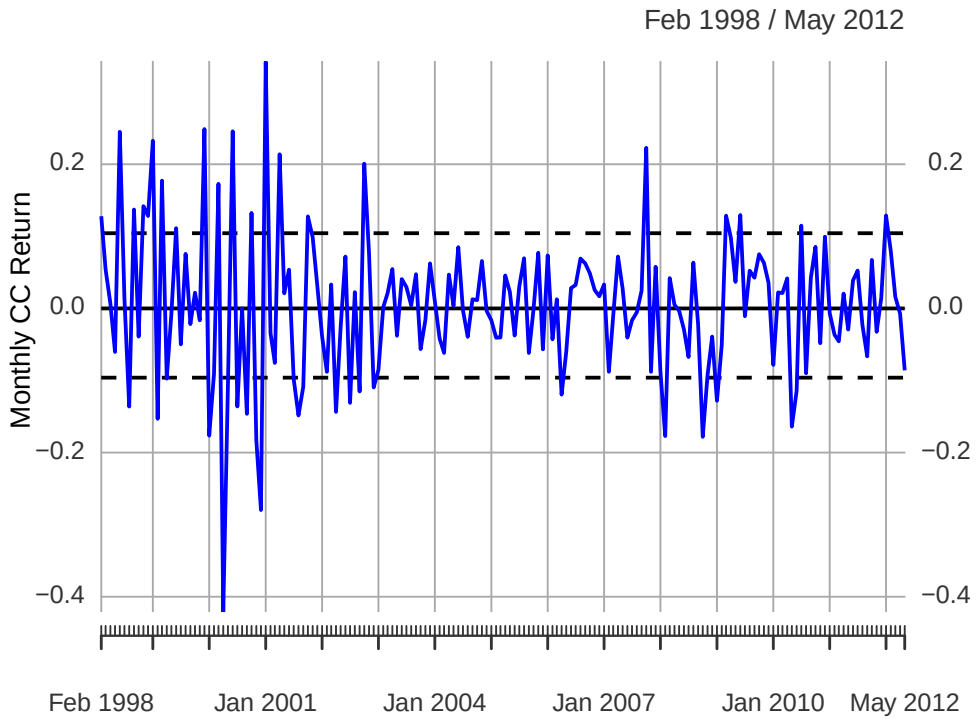


Figure 6.1: Monthly cc returns on Microsoft. Dashed lines indicate $\hat{\mu} \pm \hat{\sigma}$.

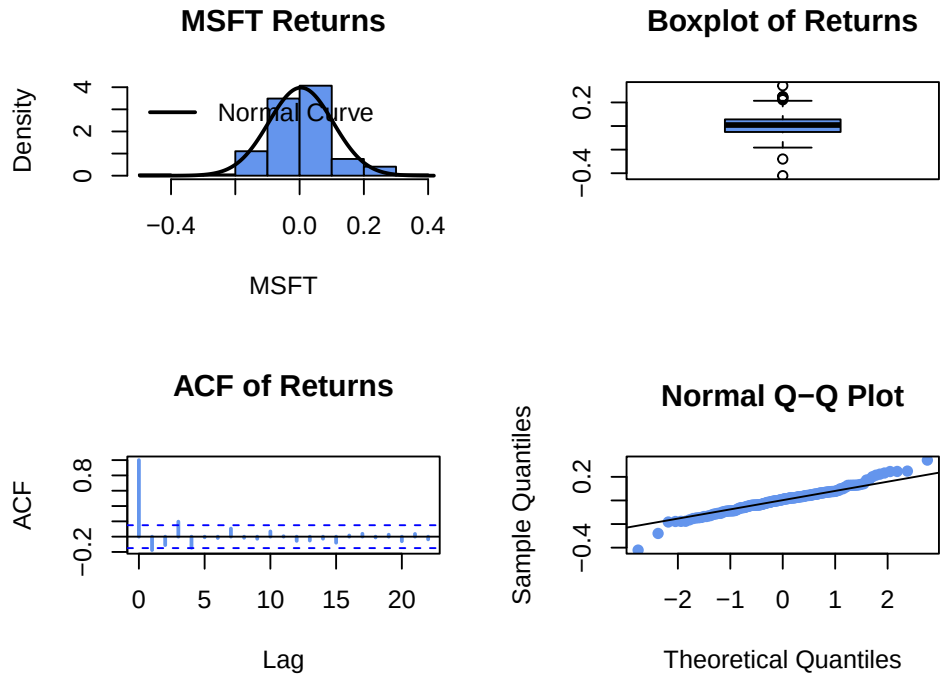


Figure 6.2: Graphical descriptive statistics for the monthly cc returns on Microsoft.

To create the simulated returns from Equation 6.8 use:

```
mu = 0.004
sd.e = 0.10
n.obs = length(msftRetC)
set.seed(111)
sim.e = rnorm(n.obs, mean=0, sd=sd.e)
sim.ret = mu + sim.e
sim.ret = xts(sim.ret, index(msftRetC))
```

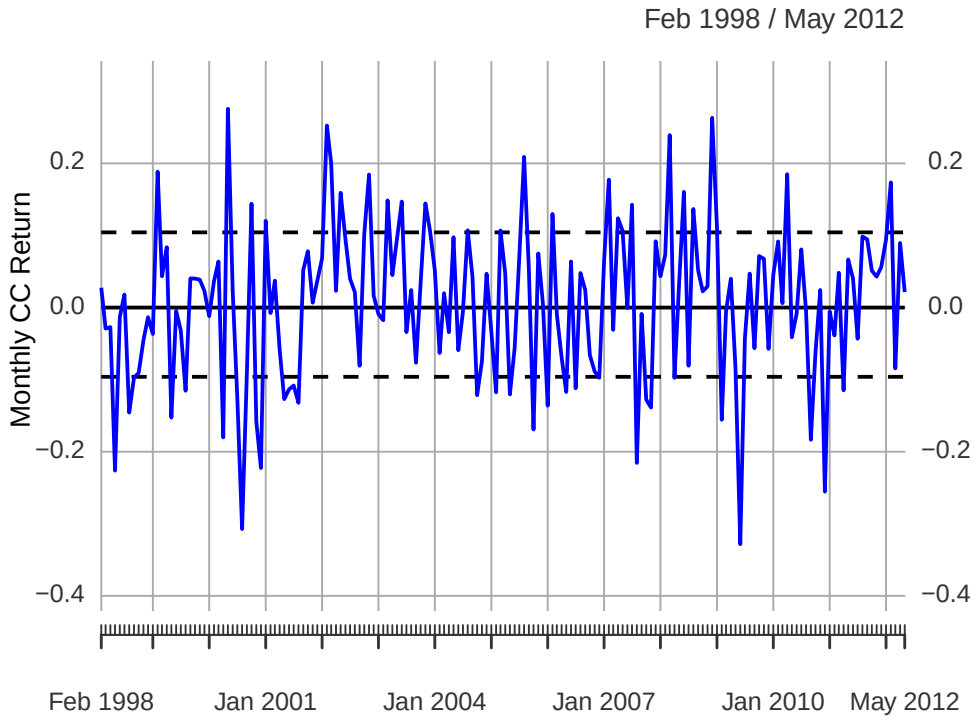


Figure 6.3: Monte Carlo simulated cc returns from the GWN model for Microsoft.

The simulated returns $\{\tilde{R}_t\}_{t=1}^{172}$ (with the same time index as the Microsoft returns) are shown in Figure 6.3. The simulated return data fluctuate randomly about $\mu = 0.004$, and the typical size of the fluctuation is approximately equal to $\sigma = 0.10$. The simulated return data look somewhat like the actual monthly return data for Microsoft. The main difference is that the return volatility for Microsoft appears to have decreased in the latter part of the sample whereas the simulated data has constant volatility over the entire sample. Figure 6.4 shows the distribution summaries (histogram, boxplot and normal qq-plot) and the SACF for the simulated returns. The simulated returns are normally distributed and show thinner tails than the actual returns. The simulated returns also show no evidence of linear time dependence (autocorrelation). ■

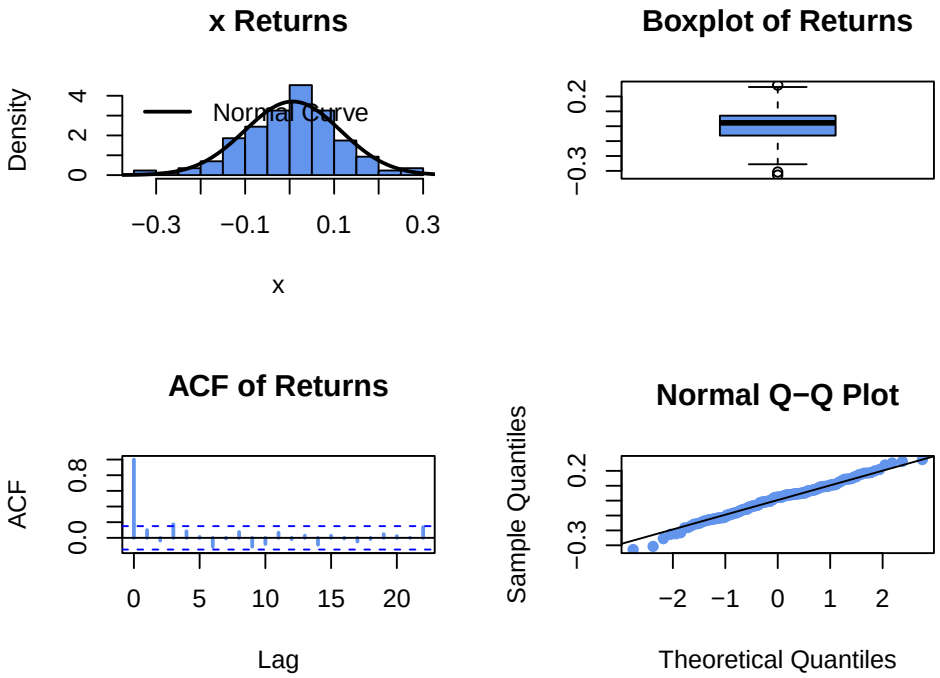


Figure 6.4: Graphical descriptive statistics for the Monte Carlo simulated returns on Microsoft.

Example 6.3 (Simulating log-prices and prices from the RW model)**.** The RW model for log-price based on the GWN model Equation 6.8 calibrated to Microsoft log prices is:

$$p_t = 2.592 + 0.004 \cdot t + \sum_{j=1}^{t} \varepsilon_j, \varepsilon_j \sim \text{GWN}(0, (0.10)^2).$$

where $p_0 = 2.586 = \ln(13.27)$ is the log of Microsoft Price at the end of January 1998. Given p_t, the price level is determined using $P_t = e^{p_t}$. A Monte Carlo simulation of this RW model with drift can be created in R using:

```
p0 = 2.58
sim.p = p0 + mu*seq(n.obs) + cumsum(sim.e)
sim.p = c(p0, sim.p)
sim.P = exp(sim.p)
sim.p = xts(sim.p, index(msftPrices))
sim.P = xts(sim.P, index(msftPrices))
```

Figure 6.5 shows the simulated log prices and prices. The top panel shows the simulated log price, $\tilde{p}_t$, and the bottom panel shows the simulated price levels $\tilde{P}_t = e^{\tilde{p}_t}$ For comparison, Figure 6.6 shows the actual log prices and price levels for Microsoft stock. Notice the similar behavior of the simulated random walk data and the actual data. ■

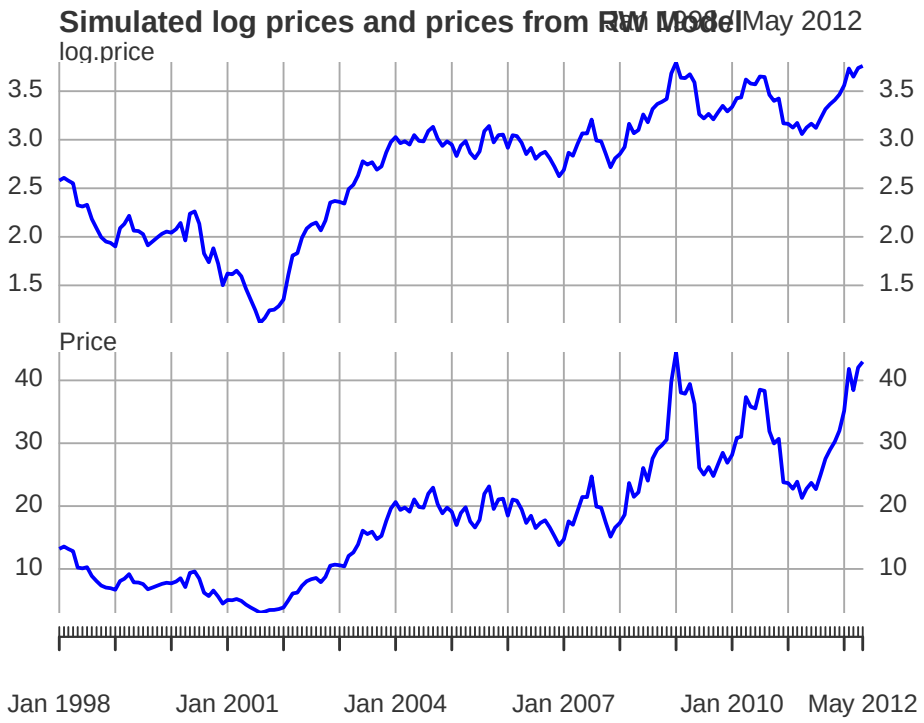


Figure 6.5: Top panel: Simulated log monthly prices of Microsoft from RW model. Bottom panel: Simulated monthly prices of Micorosft stock from RW model.

The previous examples show one Monte Carlo simulation from the GWN model for cc returns. The simulated returns were compared to the actual returns to do the first stage reality check. Here, the simulated returns looked similar to actual returns so the GWN model passed the first stage reality check. However, keep in mind that the simulated returns represent only *one random draw* of size T from the GWN model for cc returns. It is possible that you can, by chance, get a strange set of simulated returns which would lead you to reject the first stage reality check even if the GWN model is a good model for actual returns. To guard against this possibility, you should always create many Monte Carlo simulations from the proposed model and use the typical behavior from the simulations to do the first step reality check.

Example 6.4 (Creating many simulations from the GWN model)**.** To create ten simulated samples of size $T = 132$ from the GWN model for Microsoft cc returns use:

```
sim.e = matrix(0, n.obs, 10)
set.seed(111)
for (i in 1:10) {
sim.e[,i] = rnorm(n.obs, mean=0, sd=sd.e)
}
sim.ret = mu + sim.e
```

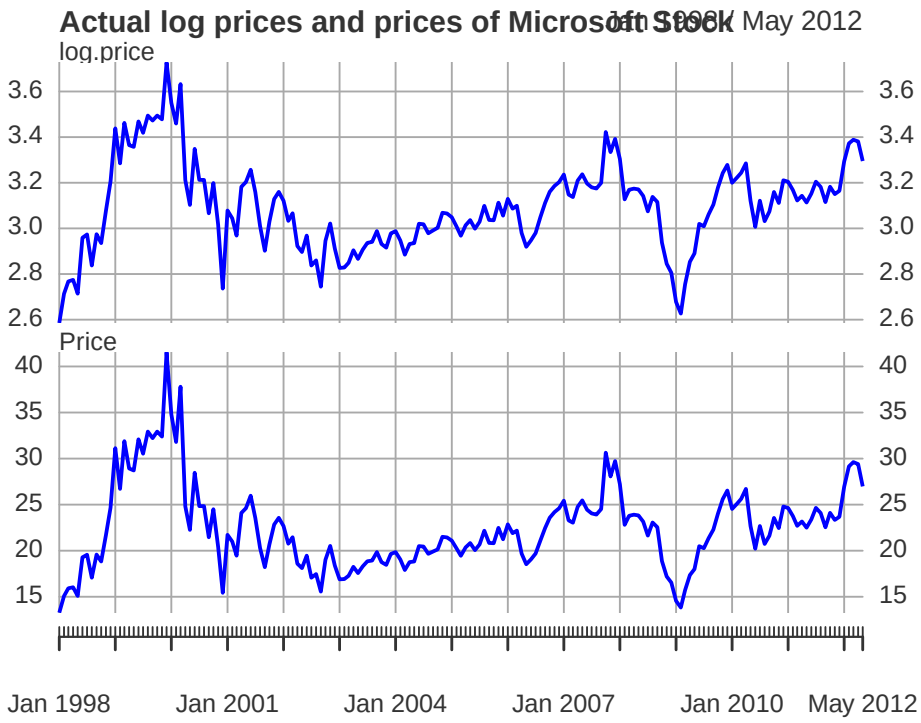


Figure 6.6: Top panel: Actual log monthly prices of Microsoft. Bottom panel: Actual monthly prices of Microsoft stock.

```
sim.ret = xts(sim.ret, index(msftRetC))
colnames(sim.ret) = paste("sim", 1:10, sep=".")
```

The ten simulations are shown in Figure 6.7.

Each panel shows an *alternative reality* for returns over the period February 1998 through May 2012. Each one of these alternative realities is equally likely to happen, and the typical behavior of the simulated returns is similar to actual returns.

Figure 6.8 shows the same ten simulations from the GWN model in terms of prices. Notice the large variability in prices at the end of the sample. ■

6.2.1 Simulating returns on more than one asset

Creating a Monte Carlo simulation of more than one return from the GWN model requires simulating observations from a multivariate normal distribution. This follows from the matrix representation of the GWN model given in Equation 6.3. The steps required to create a multivariate Monte Carlo simulation are:

Monte Carlo Steps: Multiple Assets

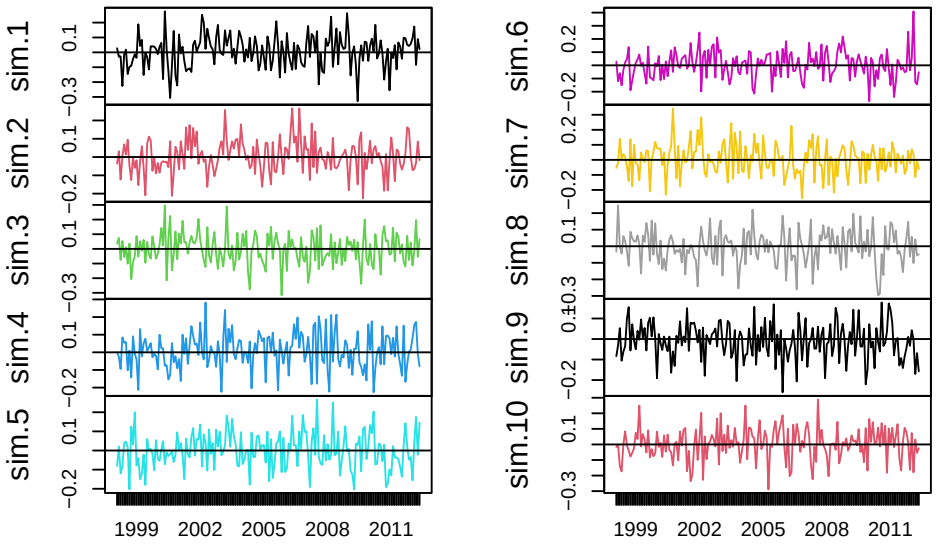


Figure 6.7: Ten Simulations from GWN Model for Microsoft CC Returns.

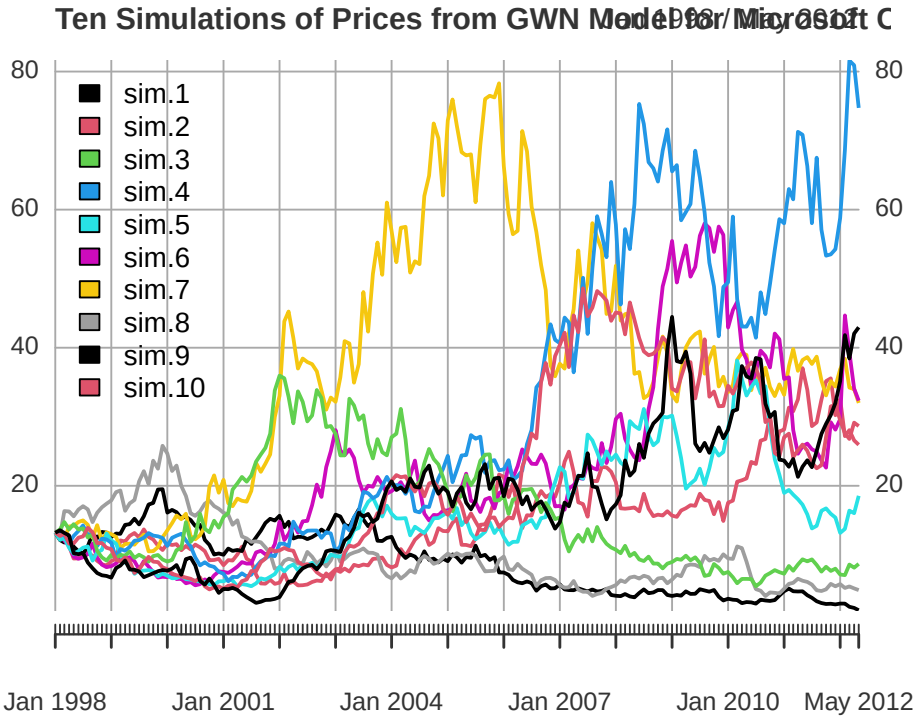


Figure 6.8: Ten Price Simulations from GWN Model for Microsoft CC Returns.

1. Fix values for the $N \times 1$ mean vector μ and the $N \times N$ covariance matrix Σ.
2. Determine the number of simulated values, T, to create.

3. Use a computer random number generator to simulate T *iid* values of the $N \times 1$ random vector ε_t from the multivariate normal distribution $N(\mathbf{0}, \Sigma)$. Denote these simulated vectors as $\tilde{\varepsilon}_1, \ldots, \tilde{\varepsilon}_T$.
4. Create the $N \times 1$ simulated return vector $\tilde{\mathbf{R}}_t = \mu + \tilde{\varepsilon}_t$ for $t = 1, \ldots, T$.
5. Repeat steps 1-4 to create multiple simulations (alternative realities).

Example 6.5 (Microsoft Starbucks and S&P 500 data to calibrate multivariate Monte Carlo simulation of GWN model)**.** To motivate the parameters for a multivariate simulation of the GWN model, consider the monthly cc returns for Microsoft, Starbucks and the S&P 500 index over the period January 1998 through May 2012 illustrated in Figures Figure 6.9 and Figure 6.10. The data is assembled using the R commands:

```
data(sp500DailyPrices, sbuxDailyPrices)
sp500Prices = to.monthly(sp500DailyPrices, OHLC=FALSE)
sbuxPrices = to.monthly(sbuxDailyPrices, OHLC=FALSE)
sp500Prices = sp500Prices[smpl]
sbuxPrices = sbuxPrices[smpl]
sp500RetC = na.omit(Return.calculate(sp500Prices, method="log"))
sbuxRetC = na.omit(Return.calculate(sbuxPrices, method="log"))
gwnRetC = merge(msftRetC, sbuxRetC, sp500RetC)
colnames(gwnRetC) = c("MSFT", "SBUX", "SP500")
```

The multivariate sample descriptive statistics (mean vector, standard deviation vector, covariance matrix and correlation matrix) are:

```
mean.vals = apply(gwnRetC, 2, mean)
sd.vals = apply(gwnRetC, 2, sd)
rbind(mean.vals, sd.vals)
#>              MSFT   SBUX   SP500
#> mean.vals 0.00413 0.0147 0.00169
#> sd.vals   0.10021 0.1116 0.04847
cov(gwnRetC)
#>          MSFT    SBUX   SP500
#> MSFT  0.01004 0.00381 0.00300
#> SBUX  0.00381 0.01246 0.00248
#> SP500 0.00300 0.00248 0.00235
cor(gwnRetC)
```

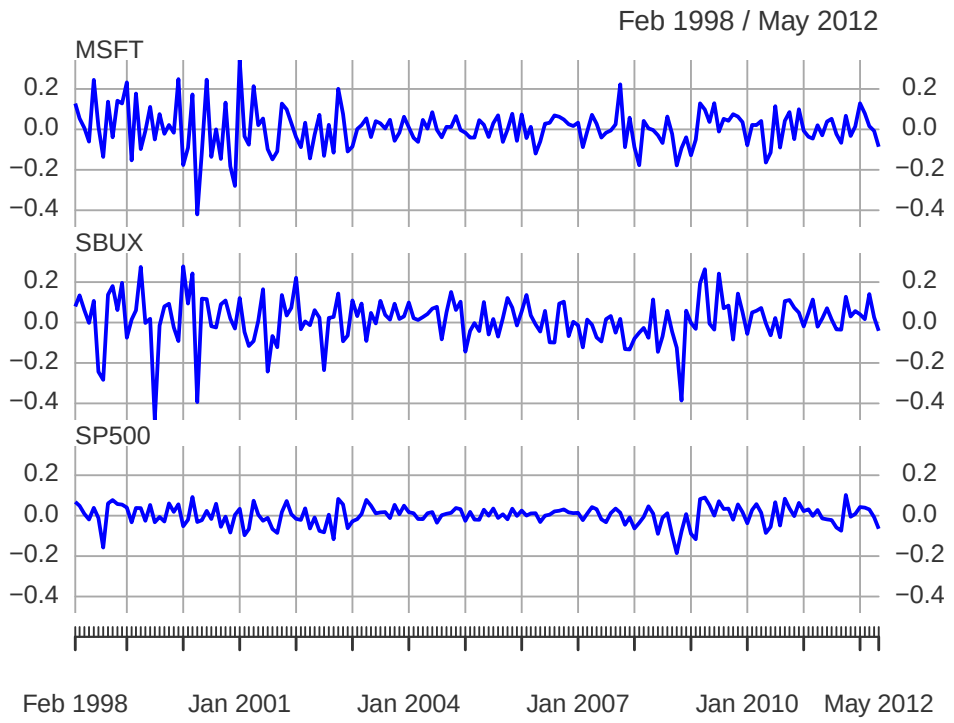


Figure 6.9: Monthly cc returns on Microsoft, Starbucks, and the S&P 500 Index.

```
#>        MSFT  SBUX SP500
#> MSFT  1.000 0.341 0.617
#> SBUX  0.341 1.000 0.457
#> SP500 0.617 0.457 1.000
```

All returns fluctuate around mean values close to zero. The volatilities of Microsoft and Starbucks are similar with typical magnitudes around 0.10, or 10%. The volatility of the S&P 500 index is considerably smaller at about 0.05, or 5%. The pairwise scatterplots show that all returns are positively related. The pairs (MSFT, SP500) and (SBUX, SP500) are the most correlated with sample correlation values around 0.5. The pair (MSFT, SBUX) has a moderate positive correlation around 0.3. ■

Example 6.6 (Monte Carlo simulation of GWN model for three assets)**.** Simulating values from the multivariate GWN model Equation 6.3 requires simulating multivariate normal random variables. In R, this can be done using the function `rmvnorm()` from the package **mvtnorm**. The function `rmvnorm()` requires a vector of mean values and a covariance matrix. Define:

$$\mathbf{R}_t = \begin{pmatrix} R_{\text{msft},t} \\ R_{\text{sbux},t} \\ R_{\text{sp500},t} \end{pmatrix}, \ \mu = \begin{pmatrix} \mu_{\text{msft},t} \\ \mu_{\text{sbux},t} \\ \mu_{\text{sp500},t} \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma^2_{\text{msft}} & \sigma_{\text{msft,sbux}} & \sigma_{\text{msft,sp500}} \\ \sigma_{\text{msft,sbux}} & \sigma^2_{\text{sbux}} & \sigma_{\text{sbux,sp500}} \\ \sigma_{\text{msft,sp500}} & \sigma_{\text{sbux,sp500}} & \sigma^2_{\text{sp500}} \end{pmatrix}$$

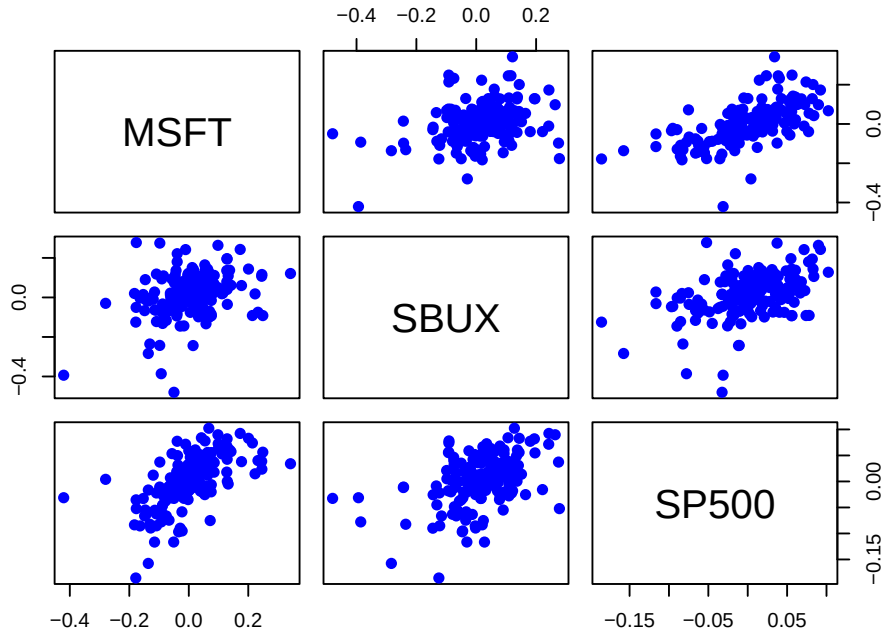


Figure 6.10: Pairwise scatterplots of the monthly returns on Microsoft, Starbucks, and the S&P 500 Index.

The parameters μ and Σ of the multivariate GWN model are set equal to the sample mean vector μ and sample covariance matrix Σ:

```
muVec = apply(gwnRetC, 2, mean)
covMat = cov(gwnRetC)
```

To create a Monte Carlo simulation from the GWN model calibrated to the month continuously returns on Microsoft, Starbucks and the S&P 500 index use:[6]

```
set.seed(123)
e.sim = rmvnorm(n.obs, sigma=covMat)
returns.sim = muVec + t(e.sim)
returns.sim = t(returns.sim)
colnames(returns.sim) = colnames(gwnRetC)
returns.sim = xts(returns.sim, index(gwnRetC))
```

The simulated returns are shown in Figure 6.11 and Figure 6.12. They look similar to the actual

[6] `rmvnorm` returns a matrix of dimension `n.obs` by 3. The simulated news shocks for each asset are in the columns of `e.sim`. Transposing `e.sim` allows the mean vector to be added row-wise. Transposing again creates a column vector of simulated returns.

returns shown in Figure 6.9 and Figure 6.10. The actual returns show periods of high and low volatility that the simulated returns do not. The sample statistics from the simulated returns, however, are close to the sample statistics of the actual data:

```
mean.vals = apply(returns.sim, 2, mean)
sd.vals = apply(returns.sim, 2, sd)
rbind(mean.vals, sd.vals)
#>              MSFT   SBUX   SP500
#> mean.vals 0.00671 0.0138 0.00508
#> sd.vals   0.09509 0.1051 0.04601
cov(returns.sim)
#>          MSFT    SBUX   SP500
#> MSFT  0.00904 0.00353 0.00246
#> SBUX  0.00353 0.01105 0.00194
#> SP500 0.00246 0.00194 0.00212
cor(returns.sim)
#>        MSFT  SBUX SP500
#> MSFT  1.000 0.354 0.563
#> SBUX  0.354 1.000 0.402
#> SP500 0.563 0.402 1.000
```

■

Example 6.7 (Multiple Monte Carlo simulations of GWN model for three assets)**.** TBD ■

6.3 Scenario Analysis

TBD

6.4 Further Reading: The GWN Return Model

The location-scale form of the GWN model Equation 6.2 is commonly used as a starting point for the distributional analysis of returns. In this chapter, and throughout most of the book, I use the standard normal distribution for the standardized news shock. However, many other standardized distributions that are additive (e.g., skew-normal, Student's t, skewed-t, etc.) can also be used as illustrated in (Jondeau, Poon, and Rockinger 2007), (Pfaff 2013), (Carmona 2014), and (Ruppert and Matteson 2015).

xxx gives as comprehensive non-technical description of the RW model for stock prices.

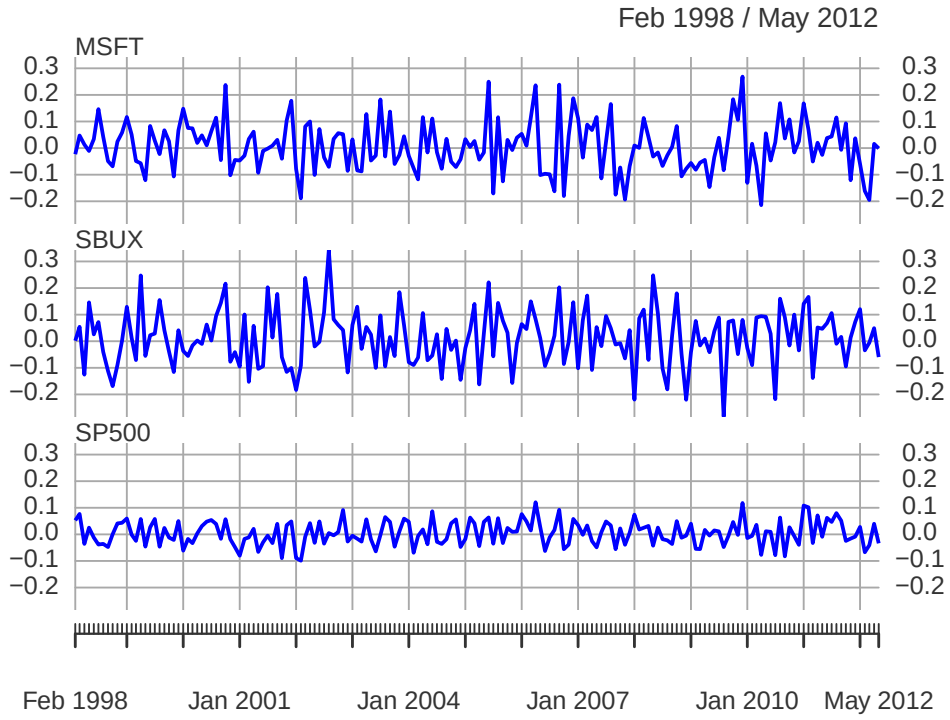


Figure 6.11: Simulated GWN model returns on Microsoft, Starbucks and the S&P 500 Index.

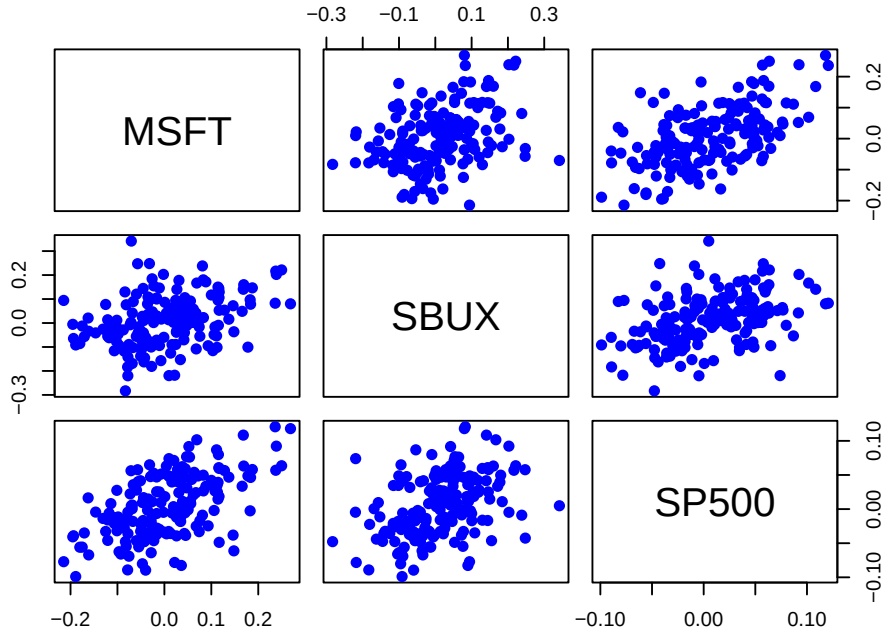


Figure 6.12: Pairwise scatterplot of simulated returns

6.5 Exercises

Exercise 6.1. Consider the GWN model for cc returns:

$$r_t = \mu + \epsilon_t,\ \epsilon_t \sim \text{GWN}(0, \sigma^2).$$

It implies that the log price follows a random walk with drift:

$$\ln P_t = \ln P_{t-1} + r_t = \ln P_0 + \mu t + \sum_{s=1}^{t} \epsilon_s.$$

Show that $E[\ln P_t]$ and $\text{var}[\ln P_t]$ depend on t so that $\ln P_t$ is non-stationary.

Exercise 6.2. Consider using Monte Carlo simulation to evaluate the GWN model for the monthly simple returns on Amazon stock and the S&P 500 index over the 5 year period January 2009, through January 2014. 1. Daily adjusted closing prices on Amazon stock and the S&P 500 index are in the **introcompfinr** `"xts"` objects `amznDailyPrices` and `sp500DailyPrices`, respectively. Using these prices, create monthly returns over the period January 2009, through January 2014. 2. Create a time plot showing the monthly returns on the two assets. Do the monthly returns from the two assets look like realizations from a covariance stationary stochastic process? Why or why not? 3. Compare and contrast the return characteristics of the two assets. In addition, comment on any common features, if any, of the two return series. 4. Using the **introcompfinr** function `four_panel_plot()`, create graphical descriptive statistics for the monthly returns. Also, create the scatterplot between the two return series. Do the returns look normally distributed? Is there any evidence for linear time dependence? Do the return appear to contemporaneously correlated? 5. Compute sample descriptive statistics (mean, standard deviation, skewness, kurtosis, and correlation between Amazon and the S&P 500 index). 6. In the GWN model for simple returns prices do not follow a random walk but instead evolve over time according to

$$P_t = P_{t-1}(1 + R_t)$$

7 Estimation of The GWN Model

Updated: October 22, 2026

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The GWN model of asset returns presented in the previous chapter gives you a simple framework for interpreting the time series behavior of asset returns and prices. At the beginning of time $t-1$, $\mathbf{R}_t$ is an $N \times 1$ random vector representing the returns (simple or continuously compounded) on assets $i = 1, \ldots, N$ to be realized at time t. The GWN model states that $\mathbf{R}_t \sim iid\ N(\mu, \Sigma)$. Your best guess for the return at t on asset i is $E[R_{it}] = \mu_i$, your measure of uncertainty about our best guess is captured by $\text{sd}(R_{it}) = \sigma_i$, and your measures of the direction and strength of linear association between R_{it} and R_{jt} are $\sigma_{ij} = \text{cov}(R_{it}, R_{jt})$ and $\rho_{ij} = \text{cor}(R_{it}, R_{jt})$, respectively. The GWN model assumes that the economic environment is constant over time (i.e., covariance stationary) so that the multivariate normal distribution characterizing returns is the same for all time periods t.

Your life would be very easy if you knew the exact values of μ_i, σ_i^2, σ_{ij} and ρ_{ij}: the parameters of the GWN model. Then you could use the GWN model for risk and portfolio analysis. In actuality, however, you do not know these values with certainty. Therefore, a key task in *financial econometrics* is estimating these values from a history of observed return data. Given estimates of the GWN model parameters you can then apply the model to risk and portfolio analysis.

Suppose you observe returns on N different assets over the sample $t = 1, \ldots, T$. Denote this sample $\{\mathbf{r}_1, \ldots, \mathbf{r}_T\} = \{\mathbf{r}_t\}_{t=1}^T$, where $\mathbf{r}_t = (r_{1t}, \ldots, r_{Nt})'$ is the $N \times 1$ vector of returns on N assets observed at time t. It is assumed that the observed returns are realizations of the random variables $\{\mathbf{R}_t\}_{t=1}^T$, where $\mathbf{R}_t = (R_{1t}, \ldots, R_{Nt})'$ is a vector of N asset returns described by the GWN model:

$$\mathbf{R}_t = \mu + \varepsilon_t,\ \varepsilon_t \sim \text{GWN}(\mathbf{0}, \Sigma). \tag{7.1}$$

Under these assumptions, you can use the observed returns $\{\mathbf{r}_t\}_{t=1}^T$ to estimate the unknown parameters in μ and Σ of the GWN model.

However, before I describe the estimation of the GWN model in detail, it is necessary to review some fundamental concepts in the statistical theory of estimation. This is important because estimates of parameters have *estimation error* and statistical theory allows you to evaluate characteristics of estimation error and to develop quantitative measures of the typical magnitude of estimation

error. As you shall see, some parameters of the GWN model are precisely estimated (have small estimation error) and some parameters are not (have large estimation error).

The chapter is outlined as follows. In Section 7.1 I review some of the statistical theory of estimation and discusses some important finite sample properties of estimators such as bias and precision. For many estimators it is difficult to derive exact finite sample properties which motivates the need to look at asymptotic properties of estimators. These are properties that hold exactly as the sample size goes to infinity but are used as approximations for a finite sample size. I present estimators of the GWN model parameters in Section 7.4 and I investigate the statistical properties of these estimators in Section 7.5. In Section 7.6 I illustrates how Monte Carlo simulation can be used to evaluate and understand the statistical properties of the GWN model estimators.

In this chapter I use the R packages **introcompfinr**, **mvtnorm**, and **PerformanceAnalytics**. Make sure these packages are installed and loaded prior to running the R examples in the chapter.

7.1 Estimators and Estimates

Let R_t be the return on a single asset (simple or continuously compounded) described by the GWN model and let θ denote some characteristic (parameter) of the GWN model you are interested in estimating. For simplicity, assume that $\theta \in \mathbb{R}$ is a single parameter. For example, if you are interested in the expected return on the asset, then $\theta = \mu$; if you are interested in the variance of the asset returns, then $\theta = \sigma^2$; if you are interested in the first lag autocorrelation of returns then $\theta = \rho_1$. The goal is to estimate θ based on a sample of size T of the observed data when you believe the data is generated from the GWN return model.

Definition 7.1 (Estimator)**.** Let $\{R_1, \ldots, R_T\}$ denote a collection of T random returns from the GWN model, and let θ denote some characteristic of the model. An *estimator* of θ, denoted $\hat{\theta}$, is a rule or algorithm for estimating θ as a function of the random variables $\{R_1, \ldots, R_T\}$. Here, $\hat{\theta} = \hat{\theta}(\{R_1, \ldots, R_T\})$ is a random variable.

Definition 7.2 (Estimate)**.** Let $\{r_1, \ldots, r_T\}$ denote an observed sample of size T from the GWN model, and let θ denote some characteristic of the model. An *estimate* of θ, denoted $\hat{\theta}$, is simply the value of the estimator for θ based on the observed sample $\{r_1, \ldots, r_T\}$. Here, $\hat{\theta} = \hat{\theta}(\{r_1, \ldots, r_T\})$ is a number.

Example 7.1 (The sample average as an estimator and an estimate)**.** Let R_t be the return on a single asset described by the GWN model, and suppose you are interested in estimating $\theta = \mu = E[R_t]$ from the sample of observed returns $\{r_t\}_{t=1}^T$. The sample average is an algorithm for computing an estimate of the expected return μ. Before the sample is observed, you can think of $\hat{\mu} = \frac{1}{T}\sum_{t=1}^T R_t$ as a simple linear function of the random variables $\{R_t\}_{t=1}^T$ and so is itself a

random variable. You can study the properties of $\hat{\mu}$ using the tools of probability theory presented in chapter 2.

After the sample is observed, the sample average can be evaluated using the observed data $\{r_t\}_{t=1}^T$ giving $\hat{\mu} = \frac{1}{T}\sum_{t=1}^T r_t$, which produces the numerical estimate of μ. For example, suppose $T = 5$ and the realized values of the returns are $r_1 = 0.1, r_2 = 0.05, r_3 = 0.025, r_4 = -0.1, r_5 = -0.05$. Then the estimate of μ using the sample average is:

$$\hat{\mu} = \frac{1}{5}(0.1 + 0.05 + 0.025 + -0.1 + -0.05) = 0.005.$$

■

The example above illustrates the distinction between an estimator and an estimate of a parameter θ. However, typically in the statistics literature you use the same symbol, $\hat{\theta}$, to denote both an estimator and an estimate. When $\hat{\theta}$ is treated as a function of the random returns it denotes an estimator and is a random variable. When $\hat{\theta}$ is evaluated using the observed data it denotes an estimate and is simply a number. The context in which you discuss $\hat{\theta}$ will determine how to interpret it.

7.2 Finite Sample Properties of Estimators

Consider $\hat{\theta}$ as a random variable. In general, the pdf of $\hat{\theta}$, $f(\hat{\theta})$, depends on the pdf's of the random variables $\{R_t\}_{t=1}^T$. The exact form of $f(\hat{\theta})$ may be very complicated. Sometimes you can use analytic calculations to determine the exact form of $f(\hat{\theta})$. This can be done, for example, when $\hat{\theta} = \hat{\mu}$. In general, the exact form of $f(\hat{\theta})$ is often too difficult to derive exactly. When $f(\hat{\theta})$ is too difficult to compute you can often approximate $f(\hat{\theta})$ using either *Monte Carlo simulation* techniques or the *Central Limit Theorem* (CLT).

In Monte Carlo simulation, you use the computer to simulate many different realizations of the random returns $\{R_t\}_{t=1}^T$ and on each simulated sample you evaluate the estimator $\hat{\theta}$. The Monte Carlo approximation of $f(\hat{\theta})$ is the empirical distribution of $\hat{\theta}$ over the different simulated samples. For a given sample size T, Monte Carlo simulation gives a very accurate approximation to $f(\hat{\theta})$ if the number of simulated samples is very large.

The CLT approximation of $f(\hat{\theta})$ is a normal distribution approximation that becomes more accurate as the sample size T gets very large. An advantage of the CLT approximation is that it is often easy to compute. The disadvantage is that the accuracy of the approximation depends on the estimator $\hat{\theta}$ and sample size T.

For analysis purposes, you often focus on certain characteristics of $f(\hat{\theta})$, like its expected value (center), variance and standard deviation (spread about expected value). The expected value of an estimator is related to the concept of *estimator bias*, and the variance/standard deviation of

an estimator is related to the concept of *estimator precision*. Different realizations of the random variables $\{R_t\}_{t=1}^T$ will produce different values of $\hat{\theta}$. Some values of $\hat{\theta}$ will be bigger than θ and some will be smaller. Intuitively, a *good estimator* of θ is one that is on average correct (unbiased) and never gets too far away from θ (small variance). That is, a good estimator will have small bias and high precision.

7.2.1 Bias

Bias concerns the location or center of $f(\hat{\theta})$ in relation to θ. If $f(\hat{\theta})$ is centered away from θ, then you say $\hat{\theta}$ is a *biased* estimator of θ. If $f(\hat{\theta})$ is centered at θ, then you say that $\hat{\theta}$ is an *unbiased* estimator of θ. Formally, you have the following definitions:

Definition 7.3 (Estimation Error)**.** The *estimation error* is the difference between the estimator and the parameter being estimated:

$$\text{error}(\hat{\theta}, \theta) = \hat{\theta} - \theta. \tag{7.2}$$

Definition 7.4 (bias)**.** The *bias* of an estimator $\hat{\theta}$ of θ is the expected estimation error:

$$\text{bias}(\hat{\theta}, \theta) = E[\text{error}(\hat{\theta}, \theta)] = E[\hat{\theta}] - \theta. \tag{7.3}$$

Definition 7.5 (Unbiased Estimator)**.** An estimator $\hat{\theta}$ of θ is *unbiased* if $\text{bias}(\hat{\theta}, \theta) = 0$; i.e., if $E[\hat{\theta}] = \theta$ or $E[\text{error}(\hat{\theta}, \theta)] = 0$.

Unbiasedness is a desirable property of an estimator. It means that the estimator produces the correct answer "on average", where "on average" means over many hypothetical realizations of the random variables $\{R_t\}_{t=1}^T$. It is important to keep in mind that an unbiased estimator for θ may not be very close to θ for a particular sample, and that a biased estimator may actually be quite close to θ. For example, consider two estimators of θ, $\hat{\theta}_1$ and $\hat{\theta}_2$. The pdfs of $\hat{\theta}_1$ and $\hat{\theta}_2$ are illustrated in Figure 7.1. $\hat{\theta}_1$ is an unbiased estimator of θ with a large variance, and $\hat{\theta}_2$ is a biased estimator of θ with a small variance. Consider first, the pdf of $\hat{\theta}_1$. The center of the distribution is at the true value $\theta = 0$, $E[\hat{\theta}_1] = 0$, but the distribution is very widely spread out about $\theta = 0$. That is, $\text{var}(\hat{\theta}_1)$ is large. On average (over many hypothetical samples) the value of $\hat{\theta}_1$ will be close to θ, but in any given sample the value of $\hat{\theta}_1$ can be quite a bit above or below θ. Hence, unbiasedness by itself does not guarantee a good estimator of θ. Now consider the pdf for $\hat{\theta}_2$. The center of the pdf is slightly higher than $\theta = 0$, i.e., $\text{bias}(\hat{\theta}_2, \theta) > 0$, but the spread of the distribution is small. Although the value of $\hat{\theta}_2$ is not equal to 0 *on average* you might prefer the estimator $\hat{\theta}_2$ over $\hat{\theta}_1$ because it is generally closer to $\theta = 0$ *on average* than $\hat{\theta}_1$.

While unbiasedness is a desirable property of an estimator $\hat{\theta}$ of θ, it, by itself, is not enough to determine if $\hat{\theta}$ is a good estimator. Being correct on average means that $\hat{\theta}$ is seldom exactly correct for any given sample. In some samples $\hat{\theta}$ is less than θ, and some samples $\hat{\theta}$ is greater than θ. More importantly, you need to know how far $\hat{\theta}$ typically is from θ. That is, you need to know about the magnitude of the spread of the distribution of $\hat{\theta}$ about its average value. This will tell you the precision of $\hat{\theta}$.

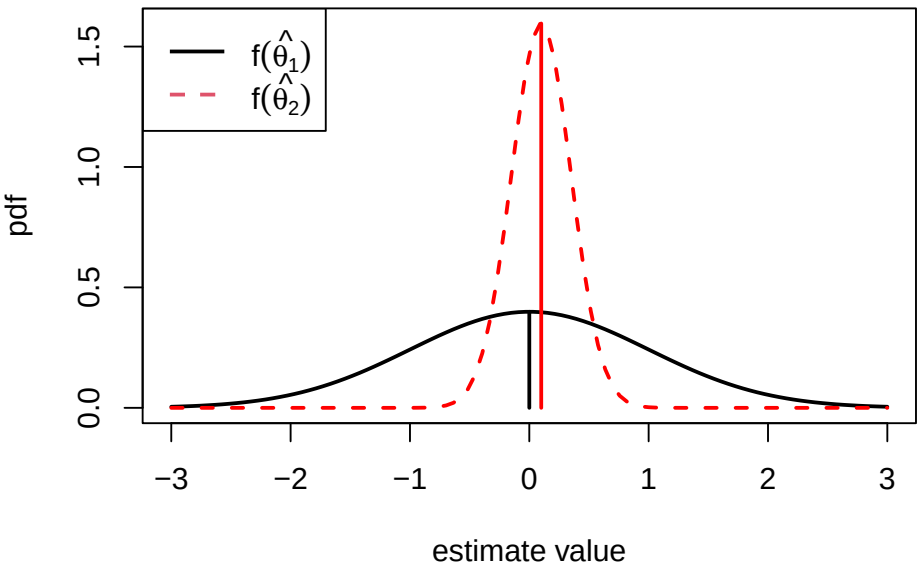


Figure 7.1: Distributions of competing estimators for $\theta = 0$. $\hat{\theta}_1$ is unbiased but has high variance, and $\hat{\theta}_2$ is biased but has low variance.

7.2.2 Precision

An estimate is, hopefully, our best guess of the true (but unknown) value of θ. Our guess most certainly will be wrong, but you hope it will not be too far off. A *precise estimate* is one in which the variability of the estimation error is small. The variability of the estimation error is captured by the *mean squared error*.

Definition 7.6 (Mean Squared Error)**.** The *mean squared error* of an estimator $\hat{\theta}$ of θ is given by:

$$\text{mse}(\hat{\theta}, \theta) = E[(\hat{\theta} - \theta)^2] = E[\text{error}(\hat{\theta}, \theta)^2] \tag{7.4}$$

The mean squared error measures the expected squared deviation of $\hat{\theta}$ from θ. If this expected

deviation is small, then you know that $\hat{\theta}$ will almost always be close to θ. Alternatively, if the mean squared is large then it is possible to see samples for which $\hat{\theta}$ is quite far from θ.

A useful decomposition of $\text{mse}(\hat{\theta}, \theta)$ is:

$$\text{mse}(\hat{\theta}, \theta) = E[(\hat{\theta} - E[\hat{\theta}])^2] + \left(E[\hat{\theta}] - \theta\right)^2 = \text{var}(\hat{\theta}) + \text{bias}(\hat{\theta}, \theta)^2$$

The derivation of this result is straightforward. Use the *add and subtract trick* and write

$$\hat{\theta} - \theta = \hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta.$$

Then square both sides giving

$$(\hat{\theta} - \theta)^2 = \left(\hat{\theta} - E[\hat{\theta}]\right)^2 + 2\left(\hat{\theta} - E[\hat{\theta}]\right)\left(E[\hat{\theta}] - \theta\right) + \left(E[\hat{\theta}] - \theta\right)^2.$$

Taking expectations of both sides yields,

$$\begin{aligned}
\text{mse}(\hat{\theta}, \theta) &= E\left[\left(\hat{\theta} - E[\hat{\theta}]\right)\right]^2 + 2\left(E[\hat{\theta}] - E[\hat{\theta}]\right)\left(E[\hat{\theta}] - \theta\right) + E\left[\left(E[\hat{\theta}] - \theta\right)^2\right] \\
&= E\left[\left(\hat{\theta} - E[\hat{\theta}]\right)\right]^2 + E\left[\left(E[\hat{\theta}] - \theta\right)^2\right] \\
&= \text{var}(\hat{\theta}) + \text{bias}(\hat{\theta}, \theta)^2.
\end{aligned}$$

The result states that for any estimator $\hat{\theta}$ of θ, $\text{mse}(\hat{\theta}, \theta)$ can be split into a variance component, $\text{var}(\hat{\theta})$, and a squared bias component, $\text{bias}(\hat{\theta}, \theta)^2$. Clearly, $\text{mse}(\hat{\theta}, \theta)$ will be small only if both components are small. If an estimator is unbiased then $\text{mse}(\hat{\theta}, \theta) = \text{var}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$ is just the squared deviation of $\hat{\theta}$ about θ. Hence, an unbiased estimator $\hat{\theta}$ of θ is *good,* if it has a small variance.

The $\text{mse}(\hat{\theta}, \theta)$ and $\text{var}(\hat{\theta})$ are based on squared deviations and so are not in the same units of measurement as θ. Measures of precision that are in the same units as θ are the *root mean squared error* and the *standard error*.

Definition 7.7 (Root Mean Squared Error and Standard Error)**.** The *root mean squared error* and the *standard error* of an estimator $\hat{\theta}$ are:

$$\text{rmse}(\hat{\theta}, \theta) = \sqrt{\text{mse}(\hat{\theta}, \theta)}, \tag{7.5}$$

$$\text{se}(\hat{\theta}) = \sqrt{\text{var}(\hat{\theta})}. \tag{7.6}$$

If $\text{bias}(\hat{\theta}, \theta) \approx 0$ then the precision of $\hat{\theta}$ is typically measured by $\text{se}(\hat{\theta})$.

With the concepts of bias and precision in hand, I can state what defines a good estimator.

Definition 7.8 (Good Estimator)**.** A good estimator $\hat{\theta}$ of θ has a small bias Equation 7.3 and a small standard error Equation 7.6.

7.2.3 Estimated standard errors

Standard error formulas $\text{se}(\hat{\theta})$ that I will derive in this chapter often depend on the unknown values of the GWN model parameters. For example, later on I will show that $\text{se}(\hat{\mu}) = \sigma^2/T$ which depends on the GWN parameter σ^2 which is unknown. Hence, the formula for $\text{se}(\hat{\mu})$ is a *practically useless* quantity as I cannot compute its value because I don't the know the value of σ^2. Fortunately, I can create a *practically useful* formula by replacing the unknown quantity σ^2 with a good estimate $\hat{\sigma}^2$ that I compute from the data (this is the so-called *plug-in principle* from statistics). This gives rise the *estimated standard error* $\widehat{\text{se}}(\hat{\mu}) = \hat{\sigma}^2/T$.

7.3 Asymptotic Properties of Estimators

Estimator bias and precision are finite sample properties. That is, they are properties that hold for a fixed sample size T. Very often you are also interested in properties of estimators when the sample size T gets very large. For example, analytic calculations may show that the bias and mse of an estimator $\hat{\theta}$ depend on T in a decreasing way. That is, as T gets very large the bias and mse approach zero. So for a very large sample, $\hat{\theta}$ is effectively unbiased with high precision. In this case you say that $\hat{\theta}$ is a *consistent* estimator of θ. In addition, for large samples the *Central Limit Theorem* (CLT) says that $f(\hat{\theta})$ can often be well approximated by a normal distribution. In this case, you say that $\hat{\theta}$ is *asymptotically normally distributed*. The word "asymptotic" means "in an infinitely large sample" or "as the sample size T goes to infinity". Of course, in the real world you don't have an infinitely large sample and so the asymptotic results are only approximations. How good these approximations are for a given sample size T depends on the context. Monte Carlo simulations (see Section 7.6) can often be used to evaluate asymptotic approximations in a given context.

7.3.1 Consistency

Let $\hat{\theta}$ be an estimator of θ based on the random returns $\{R_t\}_{t=1}^T$.

Definition 7.9 (Consistent Estimator)**.** $\hat{\theta}$ is consistent for θ (converges in probability to θ) if for any $\varepsilon > 0$:

$$\lim_{T\to\infty} \Pr(|\hat{\theta} - \theta| > \varepsilon) = 0.$$

Intuitively, consistency says that as you get enough data then $\hat{\theta}$ will eventually equal θ. In other words, if you have enough data then you know the truth.

Theorems in probability theory known as *Laws of Large Numbers* are used to determine if an estimator is consistent or not. In general, you have the following result.

Proposition 7.1 (Estimator Consistency)**.** *An estimator $\hat{\theta}$ is consistent for θ if:*

- $\text{bias}(\hat{\theta}, \theta) = 0$ *as* $T \to \infty$.
- $\text{se}(\hat{\theta}) = 0$ *as* $T \to \infty$.

Equivalently, $\hat{\theta}$ is consistent for θ if $\text{mse}(\hat{\theta}, \theta) \to 0$ *as* $T \to \infty$.

Intuitively, if $f(\hat{\theta})$ collapses to θ as $T \to \infty$ then $\hat{\theta}$ is consistent for θ.

7.3.2 Asymptotic Normality

Let $\hat{\theta}$ be an estimator of θ based on the random returns $\{R_t\}_{t=1}^T$.

Definition 7.10 (Asymptotically Normal Estimator)**.** An estimator $\hat{\theta}$ is *asymptotically normally distributed* if:

$$\hat{\theta} \sim N(\theta, \text{se}(\hat{\theta})^2) \tag{7.7}$$

for large enough T.

Asymptotic normality means that $f(\hat{\theta})$ is well approximated by a normal distribution with mean θ and variance $\text{se}(\hat{\theta})^2$. The justification for asymptotic normality comes from the famous *Central Limit Theorem*.

In the definition of an asymptotically normal estimator, the variance of the normal distribution, $\text{se}(\hat{\theta})^2$, often depends on unknown GWN model parameters and so is practically useless. Fortunately, you can create a practically useful result if you replace the unknown parameters in $\text{se}(\hat{\theta})^2$ with consistent estimates.

Proposition 7.2 (Practically Useful Asymptotic Normality)**.** *Supoose an estimator $\hat{\theta}$ is asymptotically normally distributed with variance* $\text{se}(\hat{\theta})^2$ *that depends on unknown GWN model parameters. If you replace these unknown parameters with consistent estimates, creating the estimated variance* $\widehat{\text{se}}(\hat{\theta})^2$*, then*

$$\hat{\theta} \sim N(\theta, \widehat{\text{se}}(\hat{\theta})^2) \tag{7.8}$$

for large enough T.

7.3.3 Central Limit Theorem

There are actually many versions of the CLT with different assumptions. [1] In its simplest form, the CLT says that the sample average of a collection of iid random variables $X_1, \ldots, X_T$ with $E[X_i] = \mu$ and $var(X_i) = \sigma^2$ is asymptotically normal with mean μ and variance σ^2/T. In particular, the CDF of the standardized sample mean:

$$\frac{\bar{X} - \mu}{\text{se}(\bar{X})} = \frac{\bar{X} - \mu}{\sigma/\sqrt{T}} = \sqrt{T}\left(\frac{\bar{X} - \mu}{\sigma}\right),$$

converges to the CDF of a standard normal random variable Z as $T \to \infty$. This result can be expressed as:

$$\sqrt{T}\left(\frac{\bar{X} - \mu}{\sigma}\right) \sim Z \sim N(0, 1),$$

for large enough T. Equivalently,

$$\bar{X} \sim \mu + \frac{\sigma}{\sqrt{T}} \times Z \sim N\left(\mu, \frac{\sigma^2}{T}\right) = N\left(\mu, \text{se}(\bar{X})^2\right),$$

for large enough T. This form shows that $\bar{X}$ is asymptotically normal with mean μ and variance $\text{se}(\bar{X})^2 = \sigma^2/T$.

Remarks

1. The CLT result is truly remarkable because the iid random variables X_i can come from *any distribution* that has mean μ and variance σ^2. For example, X_i could be binomial, Student's t with df > 2, chi-square, etc.

2. While the basic CLT shows that the sample average $\bar{X}$ is asymptotically normally distributed, it can be extended to general estimators of parameters of the GWN model because, as you shall see, these estimators are essentially averages of iid random variables.

3. The iid assumption of the basic CLT can be relaxed to allow $\{X_i\}$ to be covariance stationary and ergodic. This allows the asymptotic normality result to be extended to estimators computed from time series data that could be serially correlated.

4. The CLT result depends on knowing σ^2, which is practically useless because you almost never know σ^2. However, the CLT result holds if you replace σ^2 with a consistent estimate $\hat{\sigma}^2$.

[1] White (1984) gives a comprehensive discussion of CLTs useful in econometrics.

7.3.4 Asymptotic Confidence Intervals

For an asymptotically normal estimator $\hat{\theta}$ of θ, the precision of $\hat{\theta}$ is measured by $\widehat{\text{se}}(\hat{\theta})$ but is best communicated by computing a (asymptotic) *confidence interval* for the unknown value of θ. A confidence interval is an *interval estimate* of θ such that you can put an explicit probability statement about the likelihood that the interval covers θ.

The construction of an asymptotic confidence interval for θ uses the asymptotic normality result:

$$\frac{\hat{\theta} - \theta}{\widehat{\text{se}}(\hat{\theta})} = Z \sim N(0,1). \tag{7.9}$$

Then, for $\alpha \in (0,1)$, you compute a $(1-\alpha)\cdot 100\%$ confidence interval for θ using Equation 7.9 and the $1-\alpha/2$ standard normal quantile (critical value) $q^Z_{(1-\alpha/2)}$ to give:

$$\Pr\left(-q^Z_{(1-\alpha/2)} \leq \frac{\hat{\theta} - \theta}{\widehat{\text{se}}(\hat{\theta})} \leq q^Z_{(1-\alpha/2)}\right) = 1-\alpha,$$

which can be rearranged as,

$$\Pr\left(\hat{\theta} - q^Z_{(1-\alpha/2)} \cdot \widehat{\text{se}}(\hat{\theta}) \leq \mu_i \leq \hat{\theta} + q^Z_{(1-\alpha/2)} \cdot \widehat{\text{se}}(\hat{\theta})\right) = 1-\alpha.$$

Hence, the random interval,

$$[\hat{\theta} - q^Z_{(1-\alpha/2)} \cdot \widehat{\text{se}}(\hat{\theta}),\ \hat{\theta} + q^Z_{(1-\alpha/2)} \cdot \widehat{\text{se}}(\hat{\theta})] = \hat{\theta} \pm q^Z_{(1-\alpha/2)} \cdot \widehat{\text{se}}(\hat{\theta}) \tag{7.10}$$

covers the true unknown value of θ with probability $1-\alpha$.

In practice, typical values for α are 0.05 and 0.01 for which $q^Z_{(0.975)} = 1.96$ and $q^Z_{(0.995)} = 2.58$. Then, approximate 95% and 99% asymptotic confidence intervals for θ have the form $\hat{\theta} \pm 2 \cdot \widehat{\text{se}}(\hat{\theta})$ and $\hat{\theta} \pm 2.5 \cdot \widehat{\text{se}}(\hat{\theta})$, respectively.

7.4 Estimators for the Parameters of the GWN Model

Let $\{\mathbf{r}_t\}_{t=1}^T$ denote a sample of size T of observed returns on N assets from the GWN model Equation 7.1. To estimate the unknown GWN model parameters μ_i, σ_i^2, σ_{ij} and ρ_{ij} from $\{\mathbf{r}_t\}_{t=1}^T$ you can use the *plug-in principle* from statistics:

Definition 7.11 (Plug-in-Principle)**.** The *plug-in-principle* says to estimate model parameters using corresponding sample statistics.

Proposition 7.3 (Plug-in Principle Estimates for GWN Model Parameters)**.** *For the GWN model parameters, the plug-in principle estimates are the following sample descriptive statistics discussed in Chapter 5:*

$$\hat{\mu}_i = \frac{1}{T}\sum_{t=1}^{T} r_{it}, \tag{7.11}$$

$$\hat{\sigma}_i^2 = \frac{1}{T-1}\sum_{t=1}^{T}(r_{it} - \hat{\mu}_i)^2, \tag{7.12}$$

$$\hat{\sigma}_i = \sqrt{\hat{\sigma}_i^2}, \tag{7.13}$$

$$\hat{\sigma}_{ij} = \frac{1}{T-1}\sum_{t=1}^{T}(r_{it} - \hat{\mu}_i)(r_{jt} - \hat{\mu}_j), \tag{7.14}$$

$$\hat{\rho}_{ij} = \frac{\hat{\sigma}_{ij}}{\hat{\sigma}_i\hat{\sigma}_j}. \tag{7.15}$$

The plug-in principle is appropriate because the GWN model parameters μ_i, σ_i^2, σ_{ij} and ρ_{ij} are characteristics of the underlying distribution of returns that are naturally estimated using sample statistics.

The plug-in principle sample statistics Equation 7.11 - Equation 7.15 are given for a single asset and the statistics Equation 7.14 - Equation 7.15 are given for one pair of assets. However, these statistics can be computed for a collection of N assets using the matrix sample statistics:

$$\underset{(N\times 1)}{\hat{\mu}} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{r}_t = \begin{pmatrix} \hat{\mu}_1 \\ \vdots \\ \hat{\mu}_N \end{pmatrix}, \tag{7.16}$$

$$\underset{(N\times N)}{\hat{\Sigma}} = \frac{1}{T-1}\sum_{t=1}^{T}(\mathbf{r}_t - \hat{\mu})(\mathbf{r}_t - \hat{\mu})' = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \cdots & \hat{\sigma}_{1N} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 & \cdots & \hat{\sigma}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\sigma}_{1N} & \hat{\sigma}_{2N} & \cdots & \hat{\sigma}_N^2 \end{pmatrix}. \tag{7.17}$$

Here $\hat{\mu}$ is called the *sample mean vector* and $\hat{\Sigma}$ is called the *sample covariance matrix*. The sample variances are the diagonal elements of $\hat{\Sigma}$, and the sample covariances are the off diagonal elements of $\hat{\Sigma}$.

To get the sample correlations, define the $N \times N$ diagonal matrix:

$$\hat{\mathbf{D}} = \begin{pmatrix} \hat{\sigma}_1 & 0 & \cdots & 0 \\ 0 & \hat{\sigma}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{\sigma}_N \end{pmatrix}.$$

Then the sample correlation matrix $\hat{\mathbf{C}}$ is computed as:

$$\hat{\mathbf{C}} = \hat{\mathbf{D}}^{-1} \hat{\mathbf{D}}^{-1} = \begin{pmatrix} 1 & \hat{\rho}_{12} & \cdots & \hat{\rho}_{1N} \\ \hat{\rho}_{12} & 1 & \cdots & \hat{\rho}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\rho}_{1N} & \hat{\rho}_{2N} & \cdots & 1 \end{pmatrix}. \tag{7.18}$$

Here, the sample correlations are the off diagonal elements of $\hat{\mathbf{C}}$.

Example 7.2 (Estimating the GWN model parameters for Microsoft Starbucks and the S&P500 index)**.** To illustrate typical estimates of the GWN model parameters, I use data on monthly simple and continuously compounded (cc) returns for Microsoft, Starbucks and the S & P 500 index over the period January 1998 through May 2012 from the **introcompfinr** package. This is same data used in Chapter 5 and is constructed as follows:

```
gwnDailyPrices = merge(msftDailyPrices,
                       sbuxDailyPrices,
                       sp500DailyPrices)
gwnPrices = to.monthly(gwnDailyPrices, OHLC=FALSE)
gwnPrices = gwnPrices["1998-01::2012-05"]
gwnRetS = na.omit(Return.calculate(gwnPrices,
                                   method="simple"))
gwnRetC = log(1 + gwnRetS)
```

These data are illustrated in Figure 7.2 and Figure 7.3.

The estimates of μ_i (i = msft, sbux, sp500) using Equation 7.11 or Equation 7.16 can be computed using the R functions `apply()` and `mean()`:

```
muhat = apply(gwnRetC,2,mean)
muhat
#>    MSFT    SBUX   SP500
#> 0.00413 0.01466 0.00169
```

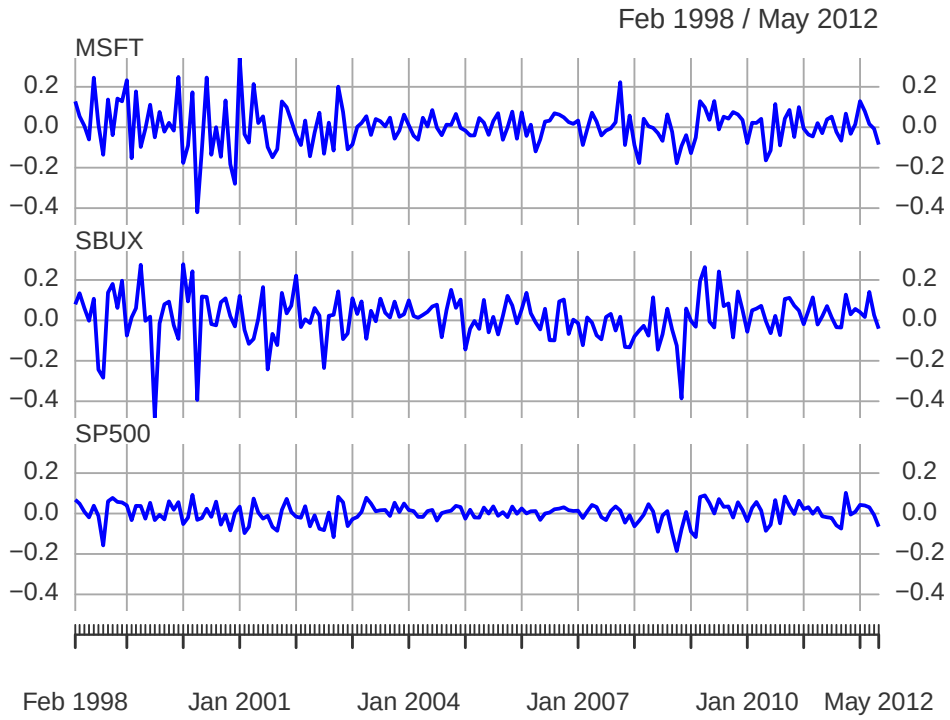


Figure 7.2: Monthly cc returns on Microsoft stock, Starbucks stock, and the S&P 500 index, over the period January 1998 through May 2012.

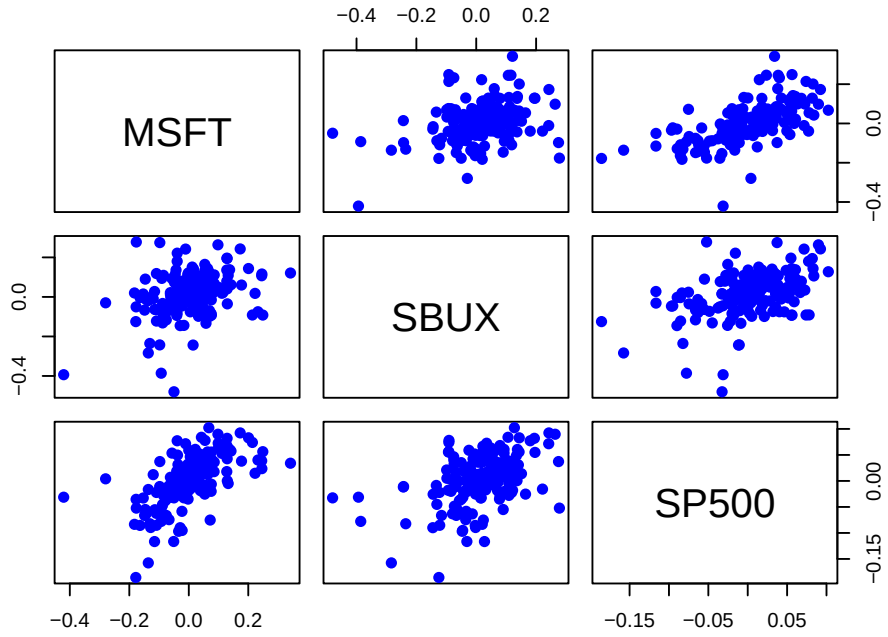


Figure 7.3: Monthly cc returns on Microsoft stock, Starbucks stock, and the S&P 500 index, over the period January 1998 through May 2012.

Starbucks has the highest average monthly return at 1.5% and the S&P 500 index has the lowest at 0.2%.

The estimates of the parameters σ_i^2, σ_i, using Equation 7.12 and Equation 7.13 can be computed using `apply()`, `var()` and `sd()`:

```
sigma2hat = apply(gwnRetC,2,var)
sigmahat = apply(gwnRetC,2,sd)
rbind(sigma2hat, sigmahat)
#>            MSFT   SBUX   SP500
#> sigma2hat 0.01 0.0125 0.00235
#> sigmahat  0.10 0.1116 0.04847
```

Starbucks has the most variable monthly returns at 11%, and the S&P 500 index has the smallest at 5%.

The scatterplots of the returns are illustrated in Figure 7.3. All returns appear to be positively related. The covariance and correlation matrix estimates using Equation 7.17 and Equation 7.18 can be computed using the functions `var()` (or `cov()`) and `cor()`:

```
covmat = var(gwnRetC)
covmat
#>           MSFT    SBUX   SP500
#> MSFT  0.01004 0.00381 0.00300
#> SBUX  0.00381 0.01246 0.00248
#> SP500 0.00300 0.00248 0.00235
cormat = cor(gwnRetC)
cormat
#>         MSFT  SBUX SP500
#> MSFT  1.000 0.341 0.617
#> SBUX  0.341 1.000 0.457
#> SP500 0.617 0.457 1.000
```

To extract the unique pairwise values of σ_{ij} and ρ_{ij} from the matrix objects `covmat` and `cormat` use:

```
covhat = covmat[lower.tri(covmat)]
rhohat = cormat[lower.tri(cormat)]
names(covhat) <- names(rhohat) <-
c("msft,sbux","msft,sp500","sbux,sp500")
covhat
```

```
#>  msft,sbux msft,sp500 sbux,sp500
#>    0.00381    0.00300    0.00248
rhohat
#>  msft,sbux msft,sp500 sbux,sp500
#>      0.341      0.617      0.457
```

The pairs (MSFT, SP500) and (SBUX, SP500) are the most correlated. These estimates confirm the visual results from the scatterplot matrix in Figure 7.3.

■

Example 7.3 (Estimating the GWN model parameters using the introcompfin function estimate_gwn_model)**.** To simplify the estimation of the GWN model parameters, you can use the `estimate_gwn_model()` function from the **introcompfinr** package:

```
gwn_fit = estimate_gwn_model(gwnRetS)
class(gwn_fit)
#> [1] "gwn_model"
```

The function `estimate_gwn_model()` takes an `xts` object of asset returns and creates a `gwn_model`. This object has the following components:

```
names(gwn_fit)
#>  [1] "call"         "date_range"   "muhat"        "se_muhat"     "Sigmahat"
#>  [6] "Rhat"         "sigma2hat"    "se_sigma2hat" "sigmahat"     "se_sigmahat"
#> [11] "covhat"       "se_cov"       "rhohat"       "se_rhohat"    "n.obs"
#> [16] "ehat"
```

You extract the components using `$`. For example,

```
gwn_fit$muhat
#>    MSFT    SBUX   SP500
#> 0.00915 0.02086 0.00285
```

You can extract the coefficient components using the `coef()` extractor function. For example, to extract the estimated correlations ρ_{ij} use:

```
coef(gwn_fit, which_coef="correlation")
#>  MSFT.SBUX MSFT.SP500 SBUX.SP500
#>      0.314      0.614      0.463
```

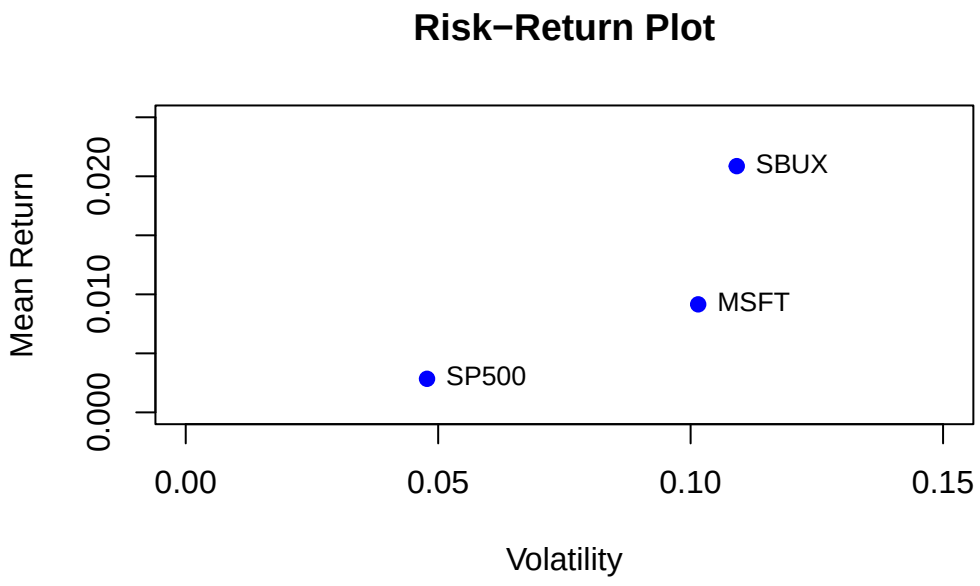
In R, most fitting functions have `print`, `summary` and `plot` methods. The `print` method (which is invoked by just typing the object name and hitting return) prints the estimated coefficients and some other basic information about the estimation:

```
gwn_fit
#> Call:
#> estimate_gwn_model(x = gwnRetS)
#>
#> Date range:  Feb 1998  to  May 2012
#> Number of observations:  172
#>
#>         Mean Volatility
#> MSFT  0.0092     0.1015
#> SBUX  0.0209     0.1091
#> SP500 0.0028     0.0478
```

Here, the `print` method only shows the estimate of μ and σ.

The `summary` method shows the estimates and estimated standard errors. I will come back to this later on in the chapter. The `plot` method provides some useful visualizations. For example, to create a risk-return scatterplot use:

```
plot(gwn_fit, which.plot="risk-return", ylim=c(0, 0.025), xlim=c(0, 0.15))
```



See the help files for `estimate_gwn_model` and the associated methods for more details. ■

7.5 Statistical Properties of the GWN Model Estimates

To determine the statistical properties of plug-in principle estimators $\hat{\mu}_i, \hat{\sigma}_i^2, \hat{\sigma}_i, \hat{\sigma}_{ij}$ and $\hat{\rho}_{ij}$ in the GWN model, you treat them as functions of the random variables $\{\mathbf{R}_t\}_{t=1}^T$ where $\mathbf{R}_t$ is assumed to be generated by the GWN model Equation 7.1.

7.5.1 Bias

Proposition 7.4 (Bias of GWN Model Estimators)**.** *Assume that returns are generated by the GWN model Equation 7.1. Then the estimators $\hat{\mu}_i$, $\hat{\sigma}_i^2$ and $\hat{\sigma}_{ij}$ are unbiased estimators:*[2]

$$\begin{aligned} E[\hat{\mu}_i] &= \mu_i \\ E[\hat{\sigma}_i^2] &= \sigma_i^2, \\ E[\hat{\sigma}_{ij}] &= \sigma_{ij}. \end{aligned}$$

The estimators $\hat{\sigma}_i$ and $\hat{\rho}_{ij}$ are biased estimators:

$$\begin{aligned} E[\hat{\sigma}_i] &\neq \sigma_i, \\ E[\hat{\rho}_{ij}] &\neq \rho_{ij}. \end{aligned}$$

It can be shown that the biases in $\hat{\sigma}_i$ and $\hat{\rho}_{ij}$, are very small and decreasing in T such that $\text{bias}(\hat{\sigma}_i, \sigma_i) = \text{bias}(\hat{\rho}_{ij}, \rho_{ij}) = 0$ as $T \to \infty$. The proofs of these results are beyond the scope of this book and may be found, for example, in Goldberger (1991). As you shall see, these results about bias can be easily verified using Monte Carlo methods.

It is instructive to illustrate how to derive the result $E[\hat{\mu}_i] = \mu_i$. Using $\hat{\mu}_i = \frac{1}{T}\sum_{t=1}^T R_{it}$ and results about the expectation of a linear combination of random variables, it follows that:

$$\begin{aligned} E[\hat{\mu}_i] &= E\left[\frac{1}{T}\sum_{t=1}^T R_{it}\right] \\ &= \frac{1}{T}\sum_{t=1}^T E[R_{it}] \text{ (by the linearity of } E[\cdot]) \\ &= \frac{1}{T}\sum_{t=1}^T \mu_i \text{ (since } E[R_{it}] = \mu_i,\ t = 1, \dots, T) \\ &= \frac{1}{T} T \cdot \mu_i = \mu_i. \end{aligned}$$

[2] The matrix sample statistics Equation 7.16 and Equation 7.17 are unbiased estimators of μ and Σ, respectively.

The derivation of the results $E[\hat{\sigma}_i^2] = \sigma_i^2$ and $E[\hat{\sigma}_{ij}] = \sigma_{ij}$ are similar but are considerably more involved and so are omitted.

7.5.2 Precision

Because the GWN model estimators are either unbiased or the bias is very small, the precision of these estimators is measured by their standard errors. Here, I give the mathematical formulas for their standard errors.

Proposition 7.5 (Standard error for $\hat{\mu}_i$)**.** *The standard error for* $\hat{\mu}_i$, $\text{se}(\hat{\mu}_i)$, *can be calculated exactly and is given by:*

$$\text{se}(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{T}}. \tag{7.19}$$

The derivation of this result is straightforward. Using the results for the variance of a linear combination of uncorrelated random variables, you have:

$$\begin{aligned}
\text{var}(\hat{\mu}_i) &= \text{var}\left(\frac{1}{T}\sum_{t=1}^{T} R_{it}\right) \\
&= \frac{1}{T^2}\sum_{t=1}^{T} \text{var}(R_{it}) \text{ (since } R_{it} \text{ is independent over time)} \\
&= \frac{1}{T^2}\sum_{t=1}^{T} \sigma_i^2 \text{ (since } \text{var}(R_{it}) = \sigma_i^2,\ t = 1, \dots, T) \\
&= \frac{1}{T^2} T\sigma_i^2 \\
&= \frac{\sigma_i^2}{T}.
\end{aligned}$$

Then $\text{se}(\hat{\mu}_i) = \text{sd}(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{T}}$.

i Remarks

1. The value of $\text{se}(\hat{\mu}_i)$ is in the same units as $\hat{\mu}_i$ and measures the precision of $\hat{\mu}_i$ as an estimate. If $\text{se}(\hat{\mu}_i)$ is small relative to $\hat{\mu}_i$ then $\hat{\mu}_i$ is a relatively precise estimate of μ_i because $f(\hat{\mu}_i)$ will be tightly concentrated around μ_i; if $\text{se}(\hat{\mu}_i)$ is large relative to μ_i then $\hat{\mu}_i$ is a relatively imprecise estimate of μ_i because $f(\hat{\mu}_i)$ will be spread out about μ_i.
2. The magnitude of $\text{se}(\hat{\mu}_i)$ depends positively on the volatility of returns, $\sigma_i = \text{sd}(R_{it})$. For a given sample size T, assets with higher return volatility have larger values of $\text{se}(\hat{\mu}_i)$ than assets with lower return volatility. In other words, estimates of expected return for high volatility assets are less precise than estimates of expected returns for low volatility assets.

3. For a given return volatility σ_i, $\text{se}(\hat{\mu}_i)$ is smaller for larger sample sizes T. In other words, $\hat{\mu}_i$ is more precisely estimated for larger samples.
4. $\text{se}(\hat{\mu}_i) \to 0$ as $T \to \infty$. This together with $E[\hat{\mu}_i] = \mu_i$ shows that $\hat{\mu}_i$ is consistent for μ_i.

The derivations of the standard errors for $\hat{\sigma}_i^2, \hat{\sigma}_i$, $\hat{\sigma}_{ij}$ and $\hat{\rho}_{ij}$ are complicated, and the exact results are extremely messy and hard to work out[3]. However, there are simple approximate formulas for the standard errors of $\hat{\sigma}_i^2, \hat{\sigma}_i$, $\hat{\sigma_{ij}}$, and $\hat{\rho}_{ij}$ based on the CLT under the GWN model that are valid if the sample size, T, is reasonably large.

Proposition 7.6 (Approximate standard error formulas for $\hat{\sigma}_i^2$, $\hat{\sigma}_i$, $\hat{\sigma}_{ij}$, and $\hat{\rho}_{ij}$)**.** *The approximate standard error formulas for $\hat{\sigma}_i^2, \hat{\sigma}_i$, $\hat{\sigma_{ij}}$, and $\hat{\rho}_{ij}$ are given by:*

$$\text{se}(\hat{\sigma}_i^2) \approx \frac{\sqrt{2}\sigma_i^2}{\sqrt{T}} = \frac{\sigma_i^2}{\sqrt{T/2}}, \tag{7.20}$$

$$\text{se}(\hat{\sigma}_i) \approx \frac{\sigma_i}{\sqrt{2T}}, \tag{7.21}$$

$$\text{se}(\hat{\sigma}_{ij}) \approx \frac{\sqrt{\sigma_i^2\sigma_j^2 + \sigma_{ij}^2}}{\sqrt{T}} = \frac{\sqrt{\sigma_i^2\sigma_j^2(1+\rho_{ij}^2)}}{\sqrt{T}} \tag{7.22}$$

$$\text{se}(\hat{\rho}_{ij}) \approx \frac{1-\rho_{ij}^2}{\sqrt{T}}, \tag{7.23}$$

where "$\approx$" denotes approximately equal. The approximations are such that the approximation error goes to zero as the sample size T gets very large.

i Remarks

1. As with the formula for the standard error of the sample mean, the formulas for $\text{se}(\hat{\sigma}_i^2)$ and $\text{se}(\hat{\sigma}_i)$ depend on σ_i^2. Larger values of σ_i^2 imply less precise estimates of $\hat{\sigma}_i^2$ and $\hat{\sigma}_i$.
2. The formula for $\text{se}(\hat{\sigma}_{ij})$ depends on σ_i^2, σ_j^2, and ρ_{ij}^2. Given σ_i^2 and σ_j^2, the standard error is smallest when $\rho_{ij} = 0$.
3. The formula for $\text{se}(\hat{\rho}_{ij})$, does not depend on σ_i^2 but rather depends on ρ_{ij}^2 and is smaller the closer ρ_{ij}^2 is to unity. Intuitively, this makes sense because as ρ_{ij}^2 approaches one the linear dependence between R_{it} and R_{jt} becomes almost perfect and this will be easily recognizable in the data (scatterplot will almost follow a straight line).
4. The formulas for the standard errors above are inversely related to the square root of the sample size, $\sqrt{T}$, which means that larger sample sizes imply smaller values of the standard errors.
5. Interestingly, $\text{se}(\hat{\sigma}_i)$ goes to zero the fastest and $\text{se}(\hat{\sigma}_i^2)$ goes to zero the slowest. Hence,

[3] See (Goldberger 1991) for details.

for a fixed sample size, these formulas suggest that σ_i is generally estimated more precisely than σ_i^2 and ρ_{ij}, and ρ_{ij} is estimated generally more precisely than σ_i^2.

The above formulas Equation 7.20 - Equation 7.23 are not practically useful, however, because they depend on the unknown quantities σ_i^2, σ_i and ρ_{ij}. Practically useful formulas make use of the plug-in principle and replace σ_i^2, σ_i and ρ_{ij} by the estimates $\hat{\sigma}_i^2, \hat{\sigma}_i$ and $\hat{\rho}_{ij}$ and give rise to the *estimated standard errors*:

$$\widehat{\text{se}}(\hat{\mu}_i) = \frac{\hat{\sigma}_i}{\sqrt{T}} \tag{7.24}$$

$$\widehat{\text{se}}(\hat{\sigma}_i^2) \approx \frac{\hat{\sigma}_i^2}{\sqrt{T/2}}, \tag{7.25}$$

$$\widehat{\text{se}}(\hat{\sigma}_i) \approx \frac{\hat{\sigma}_i}{\sqrt{2T}}, \tag{7.26}$$

$$\widehat{\text{se}}(\hat{\sigma}_{ij}) \approx \frac{\sqrt{\hat{\sigma}_i^2\hat{\sigma}_j^2(1+\hat{\rho}_{ij}^2)}}{\sqrt{T}} \tag{7.27}$$

$$\widehat{\text{se}}(\hat{\rho}_{ij}) \approx \frac{1-\hat{\rho}_{ij}^2}{\sqrt{T}}. \tag{7.28}$$

It is good practice to report estimates together with their estimated standard errors. In this way the precision of the estimates is transparent to the user. Typically, estimates are reported in a table with the estimates in one column and the estimated standard errors in an adjacent column.

Example 7.4 ($\widehat{\text{se}}(\hat{\mu}_i)$ values for Microsoft Starbucks and the S&P 500 index)**.** For Microsoft, Starbucks and S&P 500, the values of $\widehat{\text{se}}(\hat{\mu}_i)$ are easily computed in R using:

```
n.obs = nrow(gwnRetC)
seMuhat = sigmahat/sqrt(n.obs)
```

The values of $\hat{\mu}_i$ and $\widehat{\text{se}}(\hat{\mu}_i)$ shown together are:

```
cbind(muhat, seMuhat)
#>         muhat seMuhat
#> MSFT  0.00413 0.00764
#> SBUX  0.01466 0.00851
#> SP500 0.00169 0.00370
```

For Microsoft and Starbucks, the values of $\widehat{\text{se}}(\hat{\mu}_i)$ are similar because the values of $\hat{\sigma}_i$ are similar, and $\widehat{\text{se}}(\hat{\mu}_i)$ is smallest for the S&P 500 index. This occurs because $\hat{\sigma}_{\text{sp500}}$ is much smaller than

$\hat{\sigma}_{\text{msft}}$ and $\hat{\sigma}_{\text{sbux}}$. Hence, $\hat{\mu}_i$ is estimated more precisely for the S&P 500 index (a highly diversified portfolio) than it is for Microsoft and Starbucks stock (individual assets).

It is tempting to compare the magnitude of $\hat{\mu}_i$ to $\widehat{\text{se}}(\hat{\mu}_i)$ to evaluate if $\hat{\mu}_i$ is a precise estimate. A common way to do this is to compute the so-called *t-ratio*, $t = \frac{\hat{\mu}_i}{\widehat{\text{se}}(\hat{\mu}_i)}$, using:

```
muhat/seMuhat
#>  MSFT  SBUX SP500
#> 0.540 1.722 0.457
```

The t-ratio shows the number of $\widehat{\text{se}}(\hat{\mu}_i)$ values $\hat{\mu}_i$ is from zero. For example, $\hat{\mu}_{msft} = 0.004$ is 0.54 values of $\widehat{\text{se}}(\hat{\mu}_{msft}) = 0.008$ above zero. This is not very far from zero. In contrast, $\hat{\mu}_{sbux} = 0.015$ is 1.72 values of $\widehat{\text{se}}(\hat{\mu}_{sbux}) = 0.008$ above zero. This is moderately far from zero. Because, $\hat{\mu}_i$ represents an average rate of return on an asset it informative to know how far above zero it is likely to be. The farther $\hat{\mu}_i$ is from zero, in units of $\widehat{\text{se}}(\hat{\mu}_i)$, the more sure you are that the asset has a positive average rate of return. ■

Example 7.5 (Computing $\widehat{\text{se}}(\hat{\sigma}_i^2)$, $\widehat{\text{se}}(\hat{\sigma}_i)$, $\widehat{\text{se}}(\hat{\sigma}_{ij})$, and $\widehat{\text{se}}(\hat{\rho}_{ij})$ for Microsoft Starbucks and the S&P 500)**.** For Microsoft, Starbucks and S&P 500, the values of $\widehat{\text{se}}(\hat{\sigma}_i^2)$ and $\widehat{\text{se}}(\hat{\sigma}_i)$ (together with the estimates $\hat{\sigma}_i^2$ and $\hat{\sigma}_i$ are:

```
seSigma2hat = sigma2hat/sqrt(n.obs/2)
seSigmahat = sigmahat/sqrt(2*n.obs)
cbind(sigma2hat, seSigma2hat, sigmahat, seSigmahat)
#>       sigma2hat seSigma2hat sigmahat seSigmahat
#> MSFT    0.01004    0.001083   0.1002    0.00540
#> SBUX    0.01246    0.001344   0.1116    0.00602
#> SP500   0.00235    0.000253   0.0485    0.00261
```

Notice that σ^2 and σ for the S&P 500 index are estimated much more precisely than the values for Microsoft and Starbucks. Also notice that σ_i is estimated more precisely than μ_i for all assets: the values of $\widehat{\text{se}}(\hat{\sigma}_i)$ relative to $\hat{\sigma}_i$ are much smaller than the values of $\widehat{\text{se}}(\hat{\mu}_i)$ to $\hat{\mu}_i$.

The values of $\widehat{\text{se}}(\hat{\sigma}_{ij})$ and $\widehat{\text{se}}(\hat{\rho}_{ij})$ (together with the values of σ_{ij} and $\hat{\rho}_{ij}$) are:

```
seCovhat = sqrt(sigma2hat[c("MSFT","MSFT","SBUX")]*sigma2hat[c("SBUX","SP500","SP500")]*(1 +
seRhohat = (1-rhohat^2)/sqrt(n.obs)
cbind(covhat, seCovhat, rhohat, seRhohat)
#>              covhat seCovhat rhohat seRhohat
#> msft,sbux  0.00381 0.000901  0.341   0.0674
#> msft,sp500 0.00300 0.000435  0.617   0.0472
#> sbux,sp500 0.00248 0.000454  0.457   0.0603
```

The values of $\widehat{\text{se}}(\hat{\sigma}_{ij})$ and $\widehat{\text{se}}(\hat{\rho}_{ij})$ are moderate in size (relative to $\hat{\sigma}_{ij}$ and $\hat{\rho}_{ij}$). Notice that $\hat{\rho}_{\text{msft,sp500}}$ has the smallest estimated standard error because $\hat{\rho}^2_{\text{msft,sp500}}$ is closest to one. ■

7.5.3 Sampling Distributions and Confidence Intervals

7.5.3.1 Sampling Distribution for $\hat{\mu}_i$

In the GWN model, $R_{it} \sim iid\ N(\mu_i, \sigma_i^2)$ and since $\hat{\mu}_i = \frac{1}{T}\sum_{t=1}^{T} R_{it}$ is an average of these normal random variables, it is also normally distributed. Previously I showed that $E[\hat{\mu}_i] = \mu_i$ and $\text{var}(\hat{\mu}_i) = \frac{\sigma_i^2}{T}$. Therefore, the exact probability distribution of $\hat{\mu}_i$, $f(\hat{\mu}_i)$, for a fixed sample size T is the normal distribution $\hat{\mu}_i \sim\ N\left(\mu_i, \frac{\sigma_i^2}{T}\right)$. Therefore, you have an exact formula for $f(\hat{\mu}_i)$:

$$f(\hat{\mu}_i) = \left(\frac{2\pi\sigma_i^2}{T}\right)^{-1/2} \exp\left\{-\frac{1}{2\sigma_i^2/T}(\hat{\mu}_i - \mu_i)^2\right\}. \tag{7.29}$$

The probability curve $f(\hat{\mu}_i)$ is centered at the true value μ_i, and the spread about μ_i depends on the magnitude of σ_i^2, the variability of R_{it}, and the sample size, T. For a fixed sample size T, the uncertainty in $\hat{\mu}_i$ is larger (smaller) for larger (smaller) values of σ_i^2. Notice that the variance of $\hat{\mu}_i$ is inversely related to the sample size T. Given σ_i^2, $\text{var}(\hat{\mu}_i)$ is smaller for larger sample sizes than for smaller sample sizes. This makes sense since you expect to have a more precise estimator when you have more data. If the sample size is very large (as $T \to \infty$) then $\text{var}(\hat{\mu}_i)$ will be approximately zero and the normal distribution of $\hat{\mu}_i$ given by Equation 7.29 will be essentially a spike at μ_i. In other words, if the sample size is very large then you essentially know the true value of μ_i. Hence, you have established that $\hat{\mu}_i$ is a *consistent* estimator of μ_i as the sample size goes to infinity.

The exact sampling distribution for $\hat{\mu}$ assumes that σ^2 is known, which is not practically useful. Below, I provide a practically useful result which holds when you use $\hat{\sigma}^2$.

Example 7.6 (Sampling distribution of $\hat{\mu}$ with different sample sizes)**.** The distribution of $\hat{\mu}_i$, with $\mu_i = 0$ and $\sigma_i^2 = 1$ for various sample sizes is illustrated in Figure 7.4. Notice how fast the distribution collapses at $\mu_i = 0$ as T increases. ■

7.5.3.2 Confidence Intervals for μ_i

In practice, the precision of $\hat{\mu}_i$ is measured by $\widehat{\text{se}}(\hat{\mu}_i)$ but is best communicated by computing a *confidence interval* for the unknown value of μ_i. A confidence interval is an *interval estimate* of μ_i such that you can put an explicit probability statement about the likelihood that the interval covers μ_i.

Because you know the exact finite sample distribution of $\hat{\mu}_i$ in the GWN return model, you can construct an exact confidence interval for μ_i.

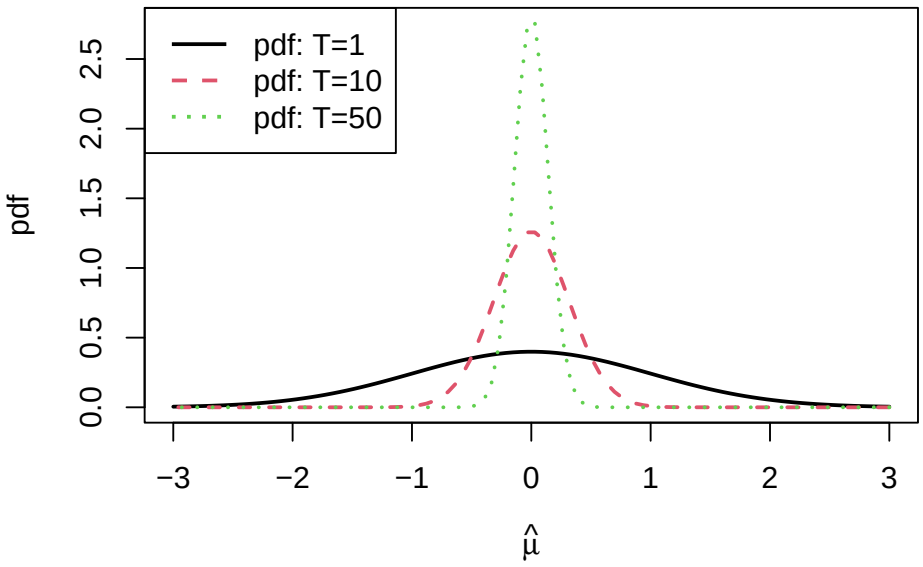


Figure 7.4: $N(0, 1/T)$ sampling distributions for $\hat{\mu}$ for $T = 1, 10$ and 50.

Proposition 7.7 (t-ratio for sample mean)**.** *Let* $\{R_{it}\}_{t=1}^{T}$ *be generated from the GWN model Equation 7.1. Define the t-ratio as:*

$$t_i = \frac{\hat{\mu}_i - \mu_i}{\widehat{\text{se}}(\hat{\mu}_i)} = \frac{\hat{\mu}_i - \mu_i}{\hat{\sigma}/\sqrt{T}}, \tag{7.30}$$

Then $t_i \sim t_{T-1}$ *where* t_{T-1} *denotes a Student's t random variable with* $T-1$ *degrees of freedom.*

The Student's t distribution with $v > 0$ degrees of freedom is a symmetric distribution centered at zero, like the standard normal. The tail-thickness (kurtosis) of the distribution is determined by the degrees of freedom parameter v. For values of v close to zero, the tails of the Student's t distribution are much fatter than the tails of the standard normal distribution. As v gets large, the Student's t distribution approaches the standard normal distribution.

For $\alpha \in (0, 1)$, you compute a $(1-\alpha) \cdot 100\%$ confidence interval for μ_i using Equation 7.30 and the $1 - \alpha/2$ quantile (critical value) $t_{T-1}(1 - \alpha/2)$ to give:

$$\Pr\left(-t_{T-1}(1-\alpha/2) \leq \frac{\hat{\mu}_i - \mu_i}{\widehat{\text{se}}(\hat{\mu}_i)} \leq t_{T-1}(1-\alpha/2)\right) = 1 - \alpha,$$

which can be rearranged as,

$$\Pr\left(\hat{\mu}_i - t_{T-1}(1-\alpha/2) \cdot \widehat{\text{se}}(\hat{\mu}_i) \leq \mu_i \leq \hat{\mu}_i + t_{T-1}(1-\alpha/2) \cdot \widehat{\text{se}}(\hat{\mu}_i)\right) = 1 - \alpha.$$

Hence, the interval,

$$[\hat{\mu}_i - t_{T-1}(1-\alpha/2)\cdot\widehat{\text{se}}(\hat{\mu}_i),\ \hat{\mu}_i + t_{T-1}(1-\alpha/2)\cdot\widehat{\text{se}}(\hat{\mu}_i)] = \hat{\mu}_i \pm t_{T-1}(1-\alpha/2)\cdot\widehat{\text{se}}(\hat{\mu}_i) \quad (7.31)$$

covers the true unknown value of μ_i with probability $1-\alpha$.

Example 7.7 (Computing 95% confidence intervals for μ_i)**.** Suppose you want to compute a 95% confidence interval for μ_i. In this case $\alpha = 0.05$ and $1-\alpha = 0.95$. Suppose further that $T-1 = 60$ (e.g., five years of monthly return data) so that $t_{T-1}(1-\alpha/2) = t_{60}(0.975) = 2$. This can be verified in R using the function `qt()`.

Then the 95% confidence for μ_i is given by:

$$\hat{\mu}_i \pm 2\cdot\widehat{\text{se}}(\hat{\mu}_i). \quad (7.32)$$

The above formula for a 95% confidence interval is often used as a rule of thumb for computing an approximate 95% confidence interval for moderate sample sizes. It is easy to remember and does not require the computation of the quantile $t_{T-1}(1-\alpha/2)$ from the Student's t distribution. It is also an approximate 95% confidence interval that is based the asymptotic normality of $\hat{\mu}_i$. Recall, for a normal distribution with mean μ and variance σ^2 approximately 95% of the probability lies between $\mu \pm 2\sigma$. ■

The coverage probability associated with the confidence interval for μ_i is based on the fact that the estimator $\hat{\mu}_i$ is a random variable. Since the confidence interval is constructed as $\hat{\mu}_i \pm t_{T-1}(1-\alpha/2)\cdot\widehat{\text{se}}(\hat{\mu}_i)$ it is also a random variable. An intuitive way to think about the coverage probability associated with the confidence interval is to think about the game of "horseshoes"[4]. The horseshoe is the confidence interval and the parameter μ_i is the post at which the horse shoe is tossed. Think of playing the game 100 times. If the thrower is 95% accurate (if the coverage probability is 0.95) then 95 of the 100 tosses should ring the post (95 of the constructed confidence intervals should contain the true value μ_i).

Example 7.8 (95% confidence intervals for μ_i for Microsoft Starbucks and the S&P 500 index)**.** Consider computing 95% confidence intervals for μ_i using Equation 7.31 based on the estimated results for the Microsoft, Starbucks and S&P 500 data. The degrees of freedom for the Student's t distribution is $T-1 = 171$. The 97.5% quantile, $t_{99}(0.975)$, can be computed using the R function `qt()`:

```
t.975 = qt(0.975, df=(n.obs-1))
t.975
```

[4]Horse shoes is a game commonly played at county fairs in the U.S. See (http://en.wikipedia.org/wiki/Horseshoes) for a complete description of the game.

```
#> [1] 1.97
```

Notice that this quantile is very close to 2. Then the exact 95% confidence intervals are given by:

```
lower = muhat - t.975*seMuhat
upper = muhat + t.975*seMuhat
width = upper - lower
ans= cbind(lower, upper, width)
colnames(ans) = c("2.5%", "97.5%", "Width")
ans
#>           2.5%   97.5%  Width
#> MSFT  -0.01096 0.01921 0.0302
#> SBUX  -0.00215 0.03146 0.0336
#> SP500 -0.00561 0.00898 0.0146
```

With probability 0.95, the above intervals will contain the true mean values assuming the GWN model is valid. The 95% confidence intervals for Microsoft and Starbucks are fairly wide (about 3%) and contain both negative and positive values. The confidence interval for the S&P 500 index is tighter but also contains negative and positive values. For Microsoft, the confidence interval is [−1.1% 1.9%]. This means that with probability 0.95, the true monthly expected return is somewhere between -1.1% and 1.9%. The economic implications of a -1.1% expected monthly return and a 1.9% expected return are vastly different. In contrast, the 95% confidence interval for the SP500 is about half the width of the intervals for Microsoft or Starbucks. The lower limit is near -0.5% and the upper limit is near 1%. This result clearly shows that the monthly mean return for the S&P 500 index is estimated much more precisely than the monthly mean returns for Microsoft or Starbucks. ■

7.5.3.3 Sampling Distributions for $\hat{\sigma}_i^2$, $\hat{\sigma}_i$, $\hat{\sigma}_{ij}$, and $\hat{\rho}_{ij}$

The exact distributions of $\hat{\sigma}_i^2$, $\hat{\sigma}_i$, $\hat{\sigma}_{ij}$, and $\hat{\rho}_{ij}$ based on a fixed sample size T are not normal distributions and are difficult to derive[5]. However, approximate normal distributions of the form Equation 7.7 based on the CLT are readily available:

$$\hat{\sigma}_i^2 \sim N\left(\sigma_i^2, \widehat{\text{se}}(\hat{\sigma}_i^2)^2\right) = N\left(\sigma_i^2, \frac{4\hat{\sigma}_i^4}{T}\right), \tag{7.33}$$

$$\hat{\sigma}_i \sim N\left(\sigma_i, \widehat{\text{se}}(\hat{\sigma}_i)^2\right) = N\left(\sigma_i, \frac{\hat{\sigma}_i^2}{2T}\right), \tag{7.34}$$

[5]For example, the exact sampling distribution of $(T-1)\hat{\sigma}_i^2/\sigma_i^2$ is chi-square with $T-1$ degrees of freedom.

$$\hat{\sigma}_{ij} \sim N\left(\sigma_{ij}, \widehat{\text{se}}(\hat{\sigma}_{ij})^2\right) = N\left(\sigma_{ij}, \frac{\hat{\sigma}_i^2\hat{\sigma}_j^2(1+\hat{\rho}_{ij}^2)}{T}\right) \tag{7.35}$$

$$\hat{\rho}_{ij} \sim N\left(\rho_{ij}, \widehat{\text{se}}(\hat{\rho}_{ij})^2\right) = N\left(\rho_{ij}, \frac{(1-\hat{\rho}_{ij}^2)^2}{T}\right). \tag{7.36}$$

These approximate normal distributions can be used to compute approximate confidence intervals for σ_i^2, σ_i, σ_{ij} and ρ_{ij}.

7.5.3.4 Approximate Confidence Intervals for σ_i^2, σ_i, σ_{ij}, and ρ_{ij}

Approximate 95% confidence intervals for σ_i^2, σ_i and ρ_{ij} are given by:

$$\hat{\sigma}_i^2 \pm 2 \cdot \widehat{\text{se}}(\hat{\sigma}_i^2) = \hat{\sigma}_i^2 \pm 2 \cdot \frac{\hat{\sigma}_i^2}{\sqrt{T/2}}, \tag{7.37}$$

$$\hat{\sigma}_i \pm 2 \cdot \widehat{\text{se}}(\hat{\sigma}_i) = \hat{\sigma}_i \pm 2 \cdot \frac{\hat{\sigma}_i}{\sqrt{2T}}, \tag{7.38}$$

$$\hat{\sigma}_{ij} \pm 2 \cdot \widehat{\text{se}}(\hat{\sigma}_{ij}) = \hat{\sigma}_{ij} \pm 2 \cdot \frac{\sqrt{\hat{\sigma}_i^2\hat{\sigma}_j^2(1+\hat{\rho}_{ij}^2)}}{\sqrt{T}}. \tag{7.39}$$

$$\hat{\rho}_{ij} \pm 2 \cdot \widehat{\text{se}}(\hat{\rho}_{ij}) = \hat{\rho}_{ij} \pm 2 \cdot \frac{(1-\hat{\rho}_{ij}^2)}{\sqrt{T}}. \tag{7.40}$$

Example 7.9 (Approximate 95% confidence intervals for σ_i^2, σ_i, σ_{ij} and ρ_{ij} for Microsoft Starbucks and the S&P 500)**.** Using Equation 7.37 - Equation 7.38, the approximate 95% confidence intervals for σ_i^2 and σ_i (i = Microsoft, Starbucks, S&P 500) are:

```
# 95% confidence interval for variance
lowerSigma2 = sigma2hat - 2*seSigma2hat
upperSigma2 = sigma2hat + 2*seSigma2hat
widthSigma2 = upperSigma2 - lowerSigma2
ans = cbind(lowerSigma2, upperSigma2, widthSigma2)
colnames(ans) = c("2.5%", "97.5%", "Width")
ans
#>          2.5%   97.5%   Width
#> MSFT  0.00788 0.01221 0.00433
#> SBUX  0.00978 0.01515 0.00538
#> SP500 0.00184 0.00286 0.00101
```

```
# 95% confidence interval for volatility
lowerSigma = sigmahat - 2*seSigmahat
upperSigma = sigmahat + 2*seSigmahat
widthSigma = upperSigma - lowerSigma
ans = cbind(lowerSigma, upperSigma, widthSigma)
colnames(ans) = c("2.5%", "97.5%", "Width")
ans
#>         2.5%  97.5%  Width
#> MSFT  0.0894 0.1110 0.0216
#> SBUX  0.0996 0.1237 0.0241
#> SP500 0.0432 0.0537 0.0105
```

The 95% confidence intervals for σ and σ^2 are larger for Microsoft and Starbucks than for the S&P 500 index. For all assets, the intervals for σ are fairly narrow (2% for Microsoft and Starbucks and 1% for S&P 500 index) indicating that σ is precisely estimated.

The approximate 95% confidence intervals for σ_{ij} are:

```
lowerCov = covhat - 2*seCovhat
upperCov = covhat + 2*seCovhat
widthCov = upperCov - lowerCov
ans = cbind(lowerCov, upperCov, widthCov)
colnames(ans) = c("2.5%", "97.5%", "Width")
ans
#>                2.5%   97.5%   Width
#> msft,sbux  0.00201 0.00562 0.00361
#> msft,sp500 0.00213 0.00387 0.00174
#> sbux,sp500 0.00157 0.00338 0.00181
```

The approximate 95% confidence intervals for ρ_{ij} are:

```
lowerRho = rhohat - 2*seRhohat
upperRho = rhohat + 2*seRhohat
widthRho = upperRho - lowerRho
ans = cbind(lowerRho, upperRho, widthRho)
colnames(ans) = c("2.5%", "97.5%", "Width")
ans
#>              2.5% 97.5% Width
#> msft,sbux   0.206 0.476 0.270
#> msft,sp500 0.523 0.712 0.189
```

```
#> sbux,sp500 0.337 0.578 0.241
```

The 95% confidence intervals for σ_{ij} ρ_{ij} are not too wide and all contain just positive values away from zero. The smallest interval is for $\rho_{\text{msft,sp500}}$ because $\hat{\rho}_{\text{msft,sp500}}$ is closest to 1. ■

Example 7.10 (Computing estmated standard errors and 95% confidence intervals from a gwn_model object)**.** If the GWN model is fit using the **introcompfinr** function `estimate_gwn_model()`, then estimated standard errors and 95% confidence intervals for all estimates is produced by the `summary()` method:

```
summary(gwn_fit, show_correlations=TRUE)
#> Call:
#> estimate_gwn_model(x = gwnRetS)
#>
#> Date range:  Feb 1998  to  May 2012
#> Number of observations:  172
#>
#>         Mean Std Error    2.5%  97.5% Volatility Std Error   2.5%  97.5%
#> MSFT  0.0092    0.0077 -0.0063 0.0246     0.1015    0.0055 0.0906 0.1124
#> SBUX  0.0209    0.0083  0.0042 0.0375     0.1091    0.0059 0.0974 0.1209
#> SP500 0.0028    0.0036 -0.0044 0.0101     0.0478    0.0026 0.0426 0.0529
#>
#> Estimated pairwise correlations:
#>            Correlation Std Error  2.5% 97.5%
#> MSFT.SBUX        0.314    0.0687 0.176 0.451
#> MSFT.SP500       0.614    0.0475 0.519 0.709
#> SBUX.SP500       0.463    0.0599 0.343 0.583
```

■

7.6 Using Monte Carlo Simulation to Understand the Statistical Properties of Estimators

Let R_t be the return on a single asset described by the GWN model, let θ denote some characteristic (parameter) of the GWN model you are interested in estimating, and let $\hat{\theta}$ denote an estimator for θ based on a sample of size T. The exact meaning of estimator bias, $\text{bias}(\hat{\theta}, \theta)$, the interpretation of $\text{se}(\hat{\theta})$ as a measure of precision, the sampling distribution $f(\hat{\theta})$, and the interpretation of the coverage probability of a confidence interval for θ, can all be a bit hard to grasp at first. If $\text{bias}(\hat{\theta}, \theta) = 0$ so that $E[\hat{\theta}] = \theta$ then over an infinite number of repeated samples of $\{R_{it}\}_{t=1}^{T}$ the

average of the $\hat{\theta}$ values computed over the infinite samples is equal to the true value θ. The value of se$(\hat{\theta})$ represents the standard deviation of these $\hat{\theta}$ values. The sampling distribution $f(\hat{\theta})$ is the smoothed histogram of these $\hat{\theta}$ values. And the 95% confidence intervals for θ will actually contain θ in 95% of the samples. You can think of these hypothetical samples as different Monte Carlo simulations of the GWN model. In this way you can approximate the computations involved in evaluating $E[\hat{\theta}]$, se$(\hat{\theta})$, $f(\hat{\theta})$, and the coverage probability of a confidence interval using a large, but finite, number of Monte Carlo simulations.

7.6.1 Evaluating the Statistical Properties of $\hat{\mu}$ Using Monte Carlo Simulation

Consider the GWN model:

$$\begin{aligned} R_t &= 0.05 + \varepsilon_t, t = 1, \dots, 100 \\ \varepsilon_t &\sim \text{GWN}(0, (0.10)^2). \end{aligned} \tag{7.41}$$

Here, the true parameter values are $\mu = 0.05$ and $\sigma = 0.10$. Using Monte Carlo simulation, you can simulate $N = 1000$ different samples of size $T = 100$ from Equation 7.41 giving the sample realizations $\{r_t^j\}_{t=1}^{100}$ for $j = 1, \dots, 1000$. The R code to perform the Monte Carlo simulation is:

```
mu = 0.05
sigma = 0.10
n.obs = 100
n.sim = 1000
set.seed(111)
sim.means = rep(0,n.sim)
mu.lower = rep(0,n.sim)
mu.upper = rep(0,n.sim)
qt.975 = qt(0.975, n.obs-1)
for (sim in 1:n.sim) {
    sim.ret = rnorm(n.obs,mean=mu,sd=sigma)
    sim.means[sim] = mean(sim.ret)
    se.muhat = sd(sim.ret)/sqrt(n.obs)
    mu.lower[sim] = sim.means[sim]-qt.975*se.muhat
    mu.upper[sim] = sim.means[sim]+qt.975*se.muhat
}
```

The first 10 of these simulated samples are illustrated in Figure 7.5. Notice that there is considerable variation in the appearance of the simulated samples, but that all of the simulated samples fluctuate about the true mean value of $\mu = 0.05$ and have a typical deviation from the mean of about 0.10.

For each of the 1000 simulated samples you can estimate $\hat{\mu}$ giving 1000 mean estimates $\{\hat{\mu}^1, \dots, \hat{\mu}^{1000}\}$. A histogram of these 1000 mean values is illustrated in Figure 7.6.

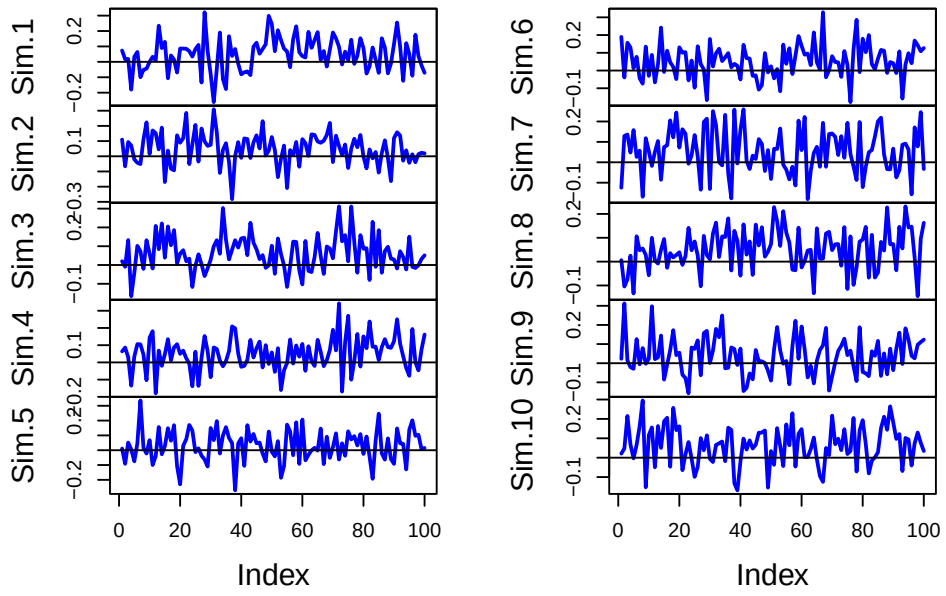


Figure 7.5: Ten simulated samples of size $T = 100$ from the GWN model $R_t = 0.05 + \varepsilon_t$, $\varepsilon_t \sim iid\ N(0, (0.10)^2)$.

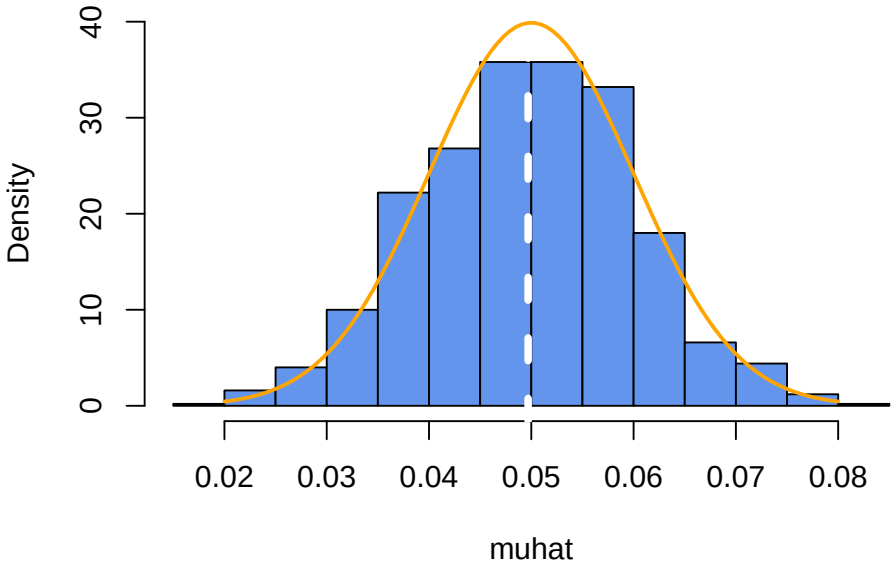


Figure 7.6: Distribution of $\hat{\mu}$ computed from 1000 Monte Carlo simulations from the GWN model. White dashed line is the average of the μ values, and orange curve is the true $f(\hat{\mu})$.

The histogram of the estimated means, $\hat{\mu}^j$, can be thought of as an estimate of the underlying pdf, $f(\hat{\mu})$, of the estimator $\hat{\mu}$ which you know from Equation 7.29 is a normal pdf centered at $E[\hat{\mu}] = \mu = 0.05$ with $\text{se}(\hat{\mu}_i) = 0.10/\sqrt{100} = 0.01$. This normal curve (solid orange line) is superimposed on the histogram in Figure 7.6. Notice that the center of the histogram (white dashed vertical line) is very close to the true mean value $\mu = 0.05$. That is, on average over the 1000 Monte Carlo samples the value of $\hat{\mu}$ is about 0.05. In some samples, the estimate is too big and in some samples the estimate is too small but on average the estimate is correct. In fact, the average value of $\{\hat{\mu}^1, \ldots, \hat{\mu}^{1000}\}$ from the 1000 simulated samples is:

$$\overline{\hat{\mu}} = \frac{1}{1000} \sum_{j=1}^{1000} \hat{\mu}^j = 0.0497,$$

which is very close to the true value 0.05. If the number of simulated samples is allowed to go to infinity then the sample average $\overline{\hat{\mu}}$ will be exactly equal to $\mu = 0.05$:

$$\lim_{N \to \infty} \frac{1}{N} \sum_{j=1}^{N} \hat{\mu}^j = E[\hat{\mu}] = \mu = 0.05.$$

The typical size of the spread about the center of the histogram represents $\text{se}(\hat{\mu}_i)$ and gives an indication of the precision of $\hat{\mu}_i$. The value of $\text{se}(\hat{\mu}_i)$ may be approximated by computing the sample standard deviation of the 1000 $\hat{\mu}^j$ values:

$$\hat{\sigma}_{\hat{\mu}} = \sqrt{\frac{1}{999} \sum_{j=1}^{1000} (\hat{\mu}^j - 0.04969)^2} = 0.0104.$$

Notice that this value is very close to the exact standard error: $\text{se}(\hat{\mu}_i) = 0.10/\sqrt{100} = 0.01$. If the number of simulated sample goes to infinity then:

$$\lim_{N \to \infty} \sqrt{\frac{1}{N-1} \sum_{j=1}^{N} (\hat{\mu}^j - \overline{\hat{\mu}})^2} = \text{se}(\hat{\mu}_i) = 0.01.$$

The coverage probability of the 95% confidence interval for μ can also be illustrated using Monte Carlo simulation. For each simulation j, the interval $\hat{\mu}^j \pm t_{100}(0.975) \times \widehat{\text{se}}(\hat{\mu}^j)$ is computed. The coverage probability is approximated by the fraction of intervals that contain (cover) the true $\mu = 0.05$. For the 1000 simulated samples, this fraction turns out to be 0.931. As the number of simulations goes to infinity, the Monte Carlo coverage probability will be equal to 0.95.

Here are the R results to support the above calculations. The mean and standard deviation of $\{\hat{\mu}^1, \ldots, \hat{\mu}^{1000}\}$ are:

```
ans = c(mean(sim.means), sd(sim.means))
names(ans) = c("Mu", "Sigma")
ans
#>     Mu  Sigma
#> 0.0497 0.0104
```

To evaluate the coverage probability of the 95% confidence intervals, you count the number of times each interval actually contains the true value of μ:

```
in.interval = mu >= mu.lower & mu <= mu.upper
sum(in.interval)/n.sim
#> [1] 0.931
```

7.6.2 Evaluating the Statistical Properties of $\hat{\sigma}_i^2$ and $\hat{\sigma}_i$ Using Monte Carlo simulation

You can evaluate the statistical properties of $\hat{\sigma}_i^2$ and $\hat{\sigma}_i$ by Monte Carlo simulation in the same way that you evaluated the statistical properties of $\hat{\mu}_i$. You use the simulation model Equation 7.41 and $N = 1000$ simulated samples of size $T = 100$ to compute the estimates $\{(\hat{\sigma}^2)^1, \dots, (\hat{\sigma}^2)^{1000}\}$ and $\{\hat{\sigma}^1, \dots, \hat{\sigma}^{1000}\}$. The R code for the simulation loop is:

```
mu = 0.05
sigma = 0.10
n.obs = 100
n.sim = 1000
set.seed(111)
sim.sigma2 = rep(0,n.sim)
sigma2.lower = rep(0,n.sim)
sigma2.upper = rep(0,n.sim)
sim.sigma = rep(0,n.sim)
sigma.lower = rep(0,n.sim)
sigma.upper = rep(0,n.sim)
for (sim in 1:n.sim) {
    sim.ret = rnorm(n.obs,mean=mu,sd=sigma)
    sim.sigma2[sim] = var(sim.ret)
    sim.sigma[sim] = sqrt(sim.sigma2[sim])
    sigma2.lower[sim] = sim.sigma2[sim] - 2*sim.sigma2[sim]/sqrt(n.obs/2)
    sigma2.upper[sim] = sim.sigma2[sim] + 2*sim.sigma2[sim]/sqrt(n.obs/2)
    sigma.lower[sim] = sim.sigma[sim] - 2*sim.sigma[sim]/sqrt(n.obs*2)
    sigma.upper[sim] = sim.sigma[sim] + 2*sim.sigma[sim]/sqrt(n.obs*2)
```

```
}
```

The histograms of these values, with the asymptotic normal distributions overlayed, are displayed in Figure 7.7. The histogram for the $\hat{\sigma}^2$ values is bell-shaped and slightly right skewed but is centered very close to $\sigma^2 = 0.010$. The histogram for the $\hat{\sigma}$ values is more symmetric and is centered near $\sigma = 0.10$.

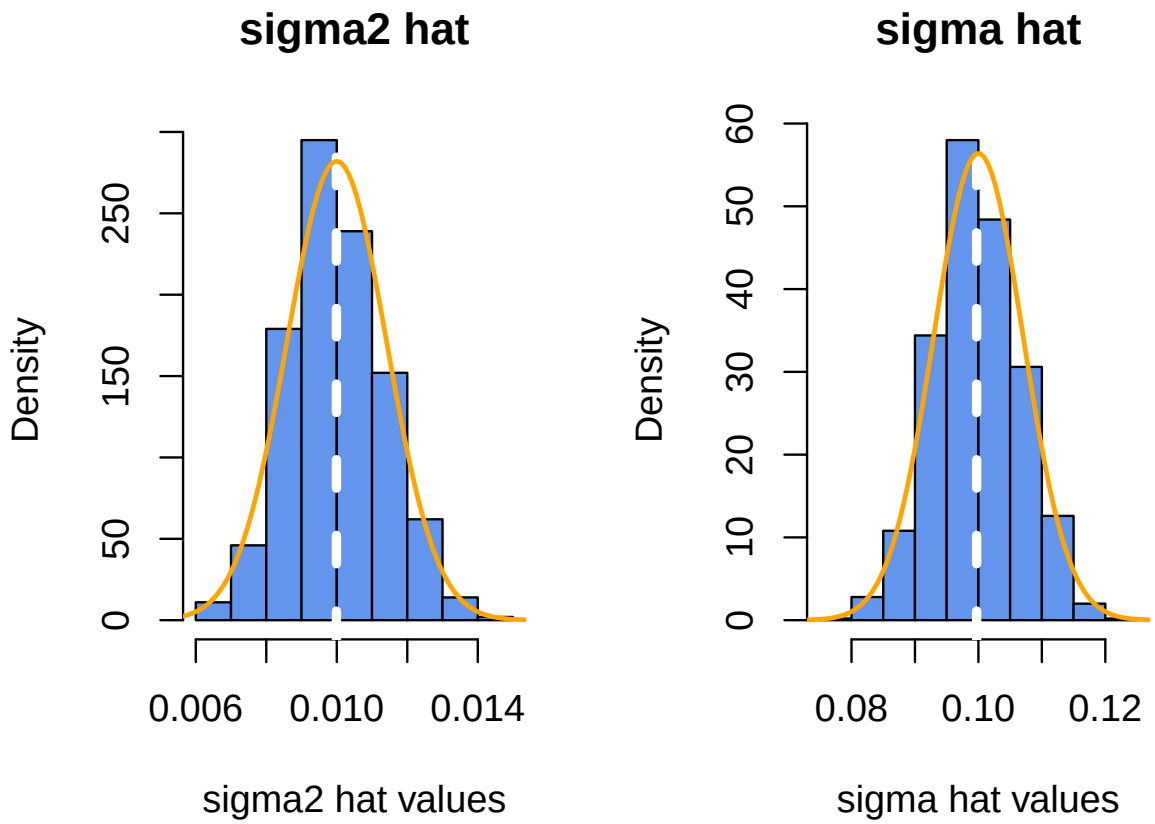


Figure 7.7: Histograms of $\hat{\sigma}^2$ and $\hat{\sigma}$ computed from $N = 1000$ Monte Carlo samples from GWN model.

The average values of $\hat{\sigma}^2$ and $\hat{\sigma}$ from the 1000 simulations are:

$$\overline{\hat{\sigma}^2} = \frac{1}{1000}\sum_{j=1}^{1000} (\hat{\sigma}^2)^j = 0.00999,$$

$$\overline{\hat{\sigma}} = \frac{1}{1000}\sum_{j=1}^{1000} \hat{\sigma}^j = 0.0997.$$

The Monte Carlo estimate of the bias for $\hat{\sigma}^2$ is $0.00999 - 0.01 = -0.0000$, and the estimate of bias for $\hat{\sigma}$ is $0.0997 - 0.010 = -0.0003$. This confirms that $\hat{\sigma}^2$ is unbiased and that the bias in $\hat{\sigma}$ is extremely

small. If the number of simulated samples, N, goes to infinity then $\overline{\hat{\sigma}^2} \to E[\hat{\sigma}^2] = \sigma^2 = 0.01$, and $\overline{\hat{\sigma}} \to E[\hat{\sigma}] = \sigma + \text{bias}(\hat{\sigma}, \sigma)$.

The sample standard deviation values of the Monte Carlo estimates of σ^2 and σ give approximations to $\text{se}(\hat{\sigma}^2)$ and $\text{se}(\hat{\sigma})$:

$$\hat{\sigma}_{\hat{\sigma}^2} = \sqrt{\frac{1}{999}\sum_{j=1}^{1000}((\hat{\sigma}^2)^j - 0.00999)^2} = 0.00135,$$
$$\hat{\sigma}_{\hat{\sigma}} = \sqrt{\frac{1}{999}\sum_{j=1}^{1000}(\hat{\sigma}^j - 0.0997)^2} = 0.00676.$$

The approximate values for $\text{se}(\hat{\sigma}^2)$ and $\text{se}(\hat{\sigma})$ based on the CLT are:

$$\text{se}(\hat{\sigma}^2) = \frac{(0.10)^2}{\sqrt{100/2}} = 0.00141,$$
$$\text{se}(\hat{\sigma}^2) = \frac{0.10}{\sqrt{2 \times 100}} = 0.00707.$$

Notice that the Monte Carlo estimates of $\text{se}(\hat{\sigma}^2)$ and $\text{se}(\hat{\sigma})$ are a bit different from the CLT based estimates. The reason is that the CLT based estimates are approximations that hold when the sample size T is large. Because $T = 100$ is not too large, the Monte Carlo estimates of $\text{se}(\hat{\sigma}^2)$ and $\text{se}(\hat{\sigma})$ are likely more accurate (and will be more accurate if the number of simulations is larger).

For each simulation j, the approximate 95% confidence intervals $(\hat{\sigma}^2)^j \pm 2 \times \widehat{\text{se}}((\hat{\sigma}^2)^j)$ and $\hat{\sigma}^j \pm 2 \times \widehat{\text{se}}(\hat{\sigma}^j)$ are computed. The coverage probabilities of these intervals is approximated by the fractions of intervals that contain (cover) the true values $\sigma^2 = 0.01$ and $\sigma = 0.10$ respectively. For the 1000 simulated samples, these fractions turn out to be 0.951 and 0.963, respectively. As the number of simulations and the sample size go to infinity, the Monte Carlo coverage probability will be equal to 0.95.

Here is the R code to support the above calculation. The mean values and biases of the estimates are:

```
ans = rbind(c(sigma^2,mean(sim.sigma2),mean(sim.sigma2) - sigma^2),
            c(sigma, mean(sim.sigma), mean(sim.sigma) - sigma))
colnames(ans) = c("True Value", "Mean", "Bias")
rownames(ans) = c("Variance", "Volatility")
round(ans, digits=5)
#>            True Value    Mean     Bias
#> Variance         0.01 0.00999 -0.00001
#> Volatility       0.10 0.09972 -0.00028
```

The Monte Carlo and analytic standard error estimates are:

```
ans = rbind(c(sd(sim.sigma2),sigma^2/sqrt(n.obs/2)),
            c(sd(sim.sigma), sigma/sqrt(2*n.obs)))
colnames(ans) = c("MC Std Error", "Analytic Std Error")
rownames(ans) = c("Variance", "Volatility")
ans
#>            MC Std Error Analytic Std Error
#> Variance        0.00135            0.00141
#> Volatility      0.00676            0.00707
```

The coverage probabilities of the Monte Carlo 95% Confidence intervals are:

```
in.interval.sigma2 = sigma^2 >= sigma2.lower & sigma^2 <= sigma2.upper
in.interval.sigma = sigma >= sigma.lower & sigma <= sigma.upper
ans = rbind(c(sum(in.interval.sigma2)/n.sim,0.95),
            c(sum(in.interval.sigma)/n.sim,0.95))
colnames(ans) = c("MC Coverage", "True Coverage")
rownames(ans) = c("Variance", "Volatility")
ans
#>            MC Coverage True Coverage
#> Variance         0.951          0.95
#> Volatility       0.963          0.95
```

7.6.3 Evaluating the Statistical Properties of $\hat{\sigma}_{ij}$ and $\hat{\rho}_{ij}$ by Monte Carlo simulation

To evaluate the statistical properties of $\hat{\sigma}_{ij}$ and $\hat{\rho}_{ij} = \text{cor}(R_{it}, R_{jt})$, you must simulate from the GWN model in matrix form Equation 7.1. For example, consider the bivariate GWN model:

$$\begin{pmatrix} R_{1t} \\ R_{2t} \end{pmatrix} = \begin{pmatrix} 0.05 \\ 0.03 \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix}, \; t = 1, \dots, 100, \tag{7.42}$$

$$\begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix} \sim iid \; N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} (0.10)^2 & (0.75)(0.10)(0.05) \\ (0.75)(0.10)(0.05) & (0.05)^2 \end{pmatrix} \right), \tag{7.43}$$

where $\mu_1 = 0.05$, $\mu_2 = 0.03$, $\sigma_1 = 0.10$, $\sigma_2 = 0.05$, and $\rho_{12} = 0.75$. You use the simulation model Equation 7.42 - Equation 7.43 with $N = 1000$ simulated samples of size $T = 100$ to compute the estimates $\{\hat{\rho}_{12}^1, \dots, \hat{\rho}_{12}^{1000}\}$. The R code to peform the Monte Carlo simulation is:

```
library(mvtnorm)
n.obs = 100
n.sim = 1000
```

```
set.seed(111)
sim.corrs = rep(0, n.sim)   # initialize vectors
sim.covs = rep(0, n.sim)
muvec = c(0.05, 0.03)
sigmavec = c(0.1, 0.05)
names(muvec) = names(sigmavec) = c("Asset.1", "Asset.2")
rho12 = 0.75
cov12 = rho12*sigmavec[1]*sigmavec[2]
Sigma = diag(sigmavec^2)
Sigma[1,2] = Sigma[2,1] = cov12
rho.lower = rep(0,n.sim)
rho.upper = rep(0,n.sim)
cov.lower = rep(0,n.sim)
cov.upper = rep(0,n.sim)
for (sim in 1:n.sim) {
    sim.ret = rmvnorm(n.obs,mean=muvec, sigma=Sigma)
    sim.corrs[sim] = cor(sim.ret)[1,2]
    sim.covs[sim] = cov(sim.ret)[1,2]
    sig1hat2 = cov(sim.ret)[1,1]
    sig2hat2 = cov(sim.ret)[2,2]
    se.rhohat = (1 - sim.corrs[sim]^2)/sqrt(n.obs)
    se.covhat = sqrt((sig1hat2*sig2hat2)*(1+sim.corrs[sim]^2))/sqrt(n.obs)
    rho.lower[sim] = sim.corrs[sim] - 2*se.rhohat
    rho.upper[sim] = sim.corrs[sim] + 2*se.rhohat
    cov.lower[sim] = sim.covs[sim] - 2*se.covhat
    cov.upper[sim] = sim.covs[sim] + 2*se.covhat
}
```

The histograms of these values, with the asymptotic normal distributions overlaid, is displayed in Figure 7.8.

The histogram for the $\hat{\sigma}_{12}$ values is bell-shaped with and centered at the true value $\sigma_{12} = 0.00375$. The sample mean and standard deviation values of $\hat{\sigma}_{12}$ across the 1000 simulations are, respectively,

$$\bar{\hat{\sigma}}_{12} = \frac{1}{1000}\sum_{j=1}^{1000} \hat{\sigma}_{12}^{j} = 0.00376,$$

$$\hat{\sigma}_{\hat{\sigma}_{12}} = \sqrt{\frac{1}{999}\sum_{j=1}^{1000} (\hat{\sigma}_{12}^{j} - 0.00376)^2} = 0.000628.$$

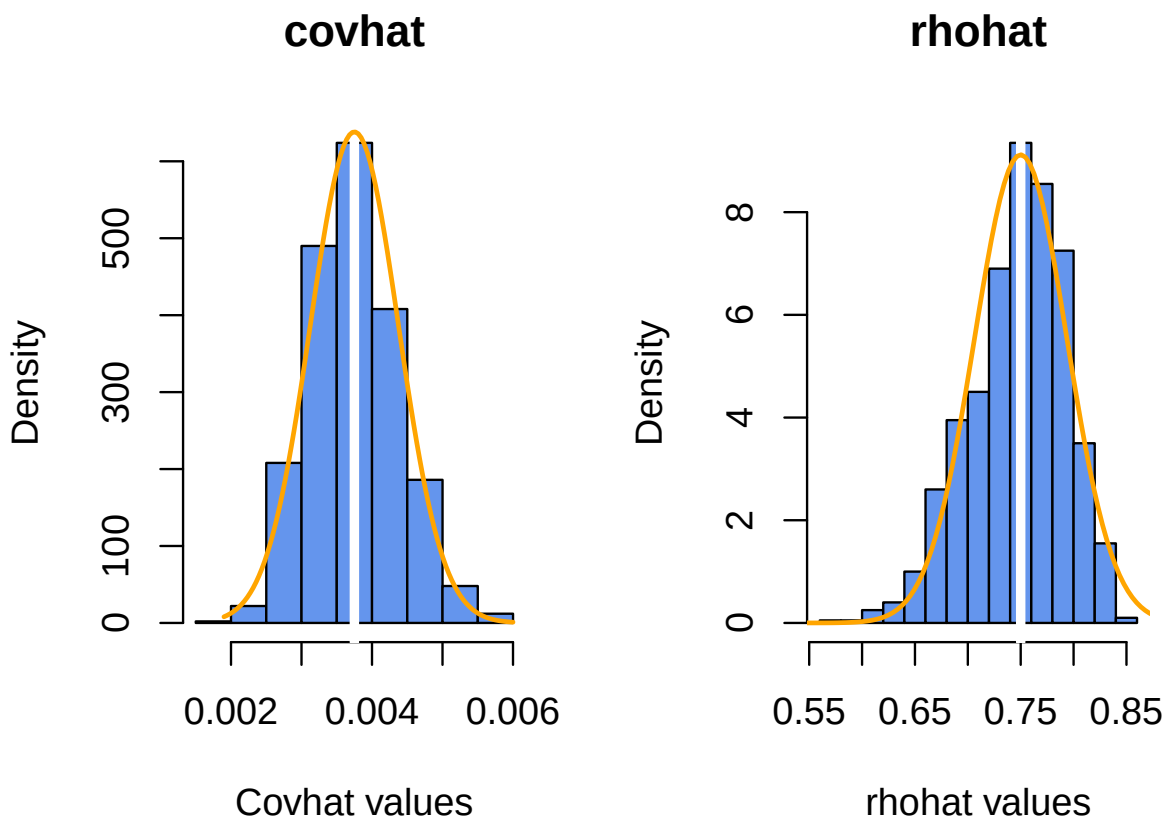


Figure 7.8: Histograms of $\hat{\rho}_{12}$ computed from $N = 1000$ Monte Carlo samples from the bivariate GWN model

Here, you see that $\widehat{\sigma_{12}}$ is essentially unbiased and the Monte Carlo standard error is almost identical to the analytic standard based the asymptotic distribution of $\hat{\sigma}_{12}$.

The histogram for the $\hat{\rho}_{12}$ values is bell-shaped with a mild left skewness and centered close to $\rho_{12} = 0.75$. The sample mean and standard deviation values of $\hat{\rho}_{12}$ across the 1000 simulations are, respectively,

$$\bar{\hat{\rho}}_{12} = \frac{1}{1000}\sum_{j=1}^{1000} \hat{\rho}_{12}^{j} = 0.747,$$
$$\hat{\sigma}_{\hat{\rho}_{12}} = \sqrt{\frac{1}{999}\sum_{j=1}^{1000}(\hat{\rho}_{12}^{j} - 0.747)^2} = 0.045.$$

There is very slight downward bias in $\hat{\rho}_{12}$. The Monte Carlo standard deviation, $\hat{\sigma}_{\hat{\rho}_{12}}$, is very close to the approximate standard error $\text{se}(\hat{\rho}_{12}) = (1 - 0.75^2)/\sqrt{100} = 0.044$.

For each simulation j, the approximate 95% confidence intervals

$$\hat{\sigma}_{12}^{j} \pm 2 \times \widehat{\text{se}}(\hat{\rho}_{12}^{j}),,$$
$$\hat{\rho}_{12}^{j} \pm 2 \times \widehat{\text{se}}(\hat{\rho}_{12}^{j}),$$

are computed. The coverage probabilities of these interval is approximated by the fractions of intervals that contain (cover) the true values $\sigma_{12} = 0.00375$ and $\rho_{12} = 0.75$, respectively. For the 1000 simulated samples, this fractions turns out to be 0.942 adn 0.952, respectively.

Here is the R code to support the above calculations. The Monte Carlo means, standard deviations, and analytic standard errors are:

```
secovhat.a = sqrt((sigmavec[1]^2*sigmavec[2]^2)*(1+rho12^2))/sqrt(n.obs)
serhohat.a = (1 - rho12^2)/sqrt(n.obs)
ans = rbind(c(cov12, mean(sim.covs),sd(sim.covs),secovhat.a),
            c(rho12, mean(sim.corrs), sd(sim.corrs), serhohat.a))
colnames(ans) = c("True Value", "MC Mean", "MC SE", "Analytic SE")
rownames(ans) = c("Covariance", "Correlation")
ans
#>             True Value MC Mean    MC SE Analytic SE
#> Covariance     0.00375 0.00376 0.000628    0.000625
#> Correlation    0.75000 0.74685 0.045036    0.043750
```

The fractions of confidence intervals that cover the true parameters are:

```
in.interval.rho = (rho12 >= rho.lower) & (rho12 <= rho.upper)
in.interval.cov = (cov12 >= cov.lower) & (cov12 <= cov.upper)
ans = rbind(c(sum(in.interval.cov)/n.sim,0.95),
            c(sum(in.interval.rho)/n.sim,0.95))
colnames(ans) = c("MC Coverage", "True Coverage")
rownames(ans) = c("Covariance", "Correlation")
ans
#>             MC Coverage True Coverage
#> Covariance        0.942          0.95
#> Correlation       0.952          0.95
```

7.7 Sampling distributions of $\hat{\mu}$ and $\hat{\Sigma}$

Consider the GWN model in matrix form Equation 7.1. The estimator of the $N \times 1$ matrix μ is the sample mean vector Equation 7.16 and the estimator of the $N \times N$ matrix Σ is the sample covariance matrix **?@eq-SigmaHat**. In this section I give the sampling distributions for the mean vector $\hat{\mu}$ and for the elements of the covariance matrix $\hat{\Sigma}$.

7.7.1 Sampling distribution of $\hat{\mu}$

The following proposition gives the sampling distribution of the $N \times 1$ vector $\hat{\mu}$.

Proposition 7.8 (Sampling distribution of $\hat{\mu}$)**.** *Under the GWN model and for large enough T, the sampling distribution for $\hat{\mu}$ is* multivariate normal*:*

$$\hat{\mu} \sim N\left(\mu,\ \frac{1}{T}\hat{\Sigma}\right).$$

This result shows that the elements of the vector $\hat{\mu}$ are correlated, with the correlations being the same as the asset return correlations.

Example 7.11 (Joint distribution of $\hat{\mu}_1$ and $\hat{\mu}_2$)**.** Consider the bivariate case ($N = 2$). Proposition 7.8 tells us the joint distribution between $\hat{\mu}_1$ and $\hat{\mu}_2$ is bivariate normal:

$$\begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix},\ \frac{1}{T}\begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 \end{pmatrix}\right).$$

Hence, $\widehat{\text{cov}}(\hat{\mu}_1, \hat{\mu}_2) = \frac{1}{T}\hat{\sigma}_{12}$ and $\widehat{\text{cor}}(\hat{\mu}_1, \hat{\mu}_2) = \hat{\rho}_{12}$.

The fact that the elements of $\hat{\mu}$ are correlated and the correlations are equal to the correlations between the corresponding assets is a useful result. For example, if assets 1 and 2 are strongly positively (negatively) correlated, then $\hat{\mu}_1$ and $\hat{\mu}_2$ are strongly positively (negatively) correlated. As a result, a positive (negative) estimate of $\hat{\mu}_1$ will tend to be associated with a positive (negative) estimate of $\hat{\mu}_2$. ■

7.7.2 Sampling distribution of the elements of $\hat{\Sigma}$

You can also determine the sampling distribution for the vector of unique elements of $\hat{\Sigma}$. This requires a bit of new matrix notation because the joint distribution of a $N \times N$ symmetric matrix of random variables is defined as the joint distribution of the $N(N+1)/2$ elements of the matrix stacked into a $N(N+1)/2 \times 1$ vector.

Define $\hat{\theta}$ as the $N(N+1)/2 \times 1$ vector of unique elements of Σ. These elements can be extracted using the vech$(\cdot)$ operator, which extracts the elements on and below the main diagonal of $\hat{\Sigma}$ column-wise and stacks them into a $N(N+1)/2$ vector. For example, if $N = 3$ then $\hat{\theta}$ is the 6×1 vector

$$\hat{\theta} = \text{vech}(\hat{\Sigma}) = \text{vech}\begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \hat{\sigma}_{13} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 & \hat{\sigma}_{23} \\ \hat{\sigma}_{13} & \hat{\sigma}_{23} & \hat{\sigma}_3^2 \end{pmatrix} = \begin{pmatrix} \hat{\sigma}_1^2 \\ \hat{\sigma}_{12} \\ \hat{\sigma}_{13} \\ \hat{\sigma}_2^2 \\ \hat{\sigma}_{23} \\ \hat{\sigma}_3^2 \end{pmatrix}.$$

Proposition 7.9 (Sampling distribution of the unique elements of $\hat{\Sigma}$)**.** *Under the GWN model and for large enough T, the sampling distribution for the $N(N+1)/2 \times 1$ vector $\hat{\theta} = \text{vech}(\hat{\Sigma})$ is multivariate normal:*

$$\hat{\theta} \sim N\left(\theta,\ \frac{1}{T}\hat{\Omega}\right),$$

where $\frac{1}{T}\hat{\Omega}$ is an $N(N+1)/2 \times N(N+1)/2$ matrix giving the estimated variances and covariances of the unique elements of $\hat{\Sigma}$. The element of $\frac{1}{T}\hat{\Omega}$ corresponding to $\widehat{\text{cov}}(\hat{\sigma}_{ij}, \hat{\sigma}_{lm})$ is given by $T^{-1}(\hat{\sigma}_{il}\hat{\sigma}_{jm} + \hat{\sigma}_{im}\hat{\sigma}_{jl})$ for all $i, j, l, m = 1, 2, \ldots, N$, including $i = j = l = m$. If $i = j$ and $l = m$ then

$$\begin{aligned} \widehat{\text{cov}}(\hat{\sigma}_{ij}, \hat{\sigma}_{lm}) &= \widehat{\text{cov}}(\hat{\sigma}_{ii}, \hat{\sigma}_{ll}) = \widehat{\text{cov}}(\hat{\sigma}_i^2, \hat{\sigma}_l^2) \\ &= \frac{\hat{\sigma}_{il}\hat{\sigma}_{il} + \hat{\sigma}_{il}\hat{\sigma}_{il}}{T} = \frac{\hat{\sigma}_{il}^2 + \hat{\sigma}_{il}^2}{T} = \frac{2\hat{\sigma}_{il}^2}{T} \end{aligned}$$

.

If $i = j = l = m$ then

$$\begin{aligned}\widehat{\text{cov}}(\hat{\sigma}_{ij}, \hat{\sigma}_{lm}) = \widehat{\text{cov}}(\hat{\sigma}_i^2, \hat{\sigma}_i^2) &= \widehat{\text{var}}(\hat{\sigma}_i^2) \\ &= \frac{\hat{\sigma}_i^2\hat{\sigma}_i^2 + \hat{\sigma}_i^2\hat{\sigma}_i^2}{T} = \frac{2\hat{\sigma}_i^4}{T}\end{aligned}$$

.

In Proposition 7.9, it is not easy to see what are the elements of $\hat{\Omega}$. A simple example can make things more clear. Consider the bivariate case $N = 2$. Then the result in Proposition 7.9 gives:

$$\begin{aligned}\hat{\theta} = \begin{pmatrix} \hat{\sigma}_1^2 \\ \hat{\sigma}_{12} \\ \hat{\sigma}_2^2 \end{pmatrix} &\sim N\left(\begin{pmatrix} \sigma_1^2 \\ \sigma_{12} \\ \sigma_2^2 \end{pmatrix}, \frac{1}{T}\begin{pmatrix} \hat{\omega}_{11} & \hat{\omega}_{12} & \hat{\omega}_{13} \\ \hat{\omega}_{21} & \hat{\omega}_{22} & \hat{\omega}_{23} \\ \hat{\omega}_{31} & \hat{\omega}_{32} & \hat{\omega}_{33} \end{pmatrix}\right) \\ &\sim N\left(\begin{pmatrix} \sigma_1^2 \\ \sigma_{12} \\ \sigma_2^2 \end{pmatrix}, \begin{pmatrix} \widehat{\text{var}}(\hat{\sigma}_1^2) & \widehat{\text{cov}}(\hat{\sigma}_1^2, \hat{\sigma}_{12}) & \widehat{\text{cov}}(\hat{\sigma}_1^2, \hat{\sigma}_2^2) \\ \widehat{\text{cov}}(\hat{\sigma}_{12}, \hat{\sigma}_1^2) & \widehat{\text{var}}(\hat{\sigma}_{12}) & \widehat{\text{cov}}(\hat{\sigma}_{12}, \hat{\sigma}_2^2) \\ \widehat{\text{cov}}(\hat{\sigma}_1^2, \hat{\sigma}_2^2) & \widehat{\text{cov}}(\hat{\sigma}_2^2, \hat{\sigma}_{12}) & \widehat{\text{var}}(\hat{\sigma}_2^2) \end{pmatrix}\right) \\ &\sim N\left(\begin{pmatrix} \sigma_1^2 \\ \sigma_{12} \\ \sigma_2^2 \end{pmatrix}, \frac{1}{T}\begin{pmatrix} 2\hat{\sigma}_1^4 & 2\hat{\sigma}_1^2\hat{\sigma}_{12} & 2\hat{\sigma}_{12}^2 \\ 2\hat{\sigma}_1^2\hat{\sigma}_{12} & \hat{\sigma}_1^2\hat{\sigma}_2^2 + \hat{\sigma}_{12}^2 & 2\hat{\sigma}_{12}\hat{\sigma}_2^2 \\ 2\hat{\sigma}_{12} & 2\hat{\sigma}_{12}\hat{\sigma}_2^2 & 2\hat{\sigma}_2^4 \end{pmatrix}\right).\end{aligned}$$

Notice that the results for the diagonal elements of $\frac{1}{T}\hat{\Omega}$ match those given earlier (see equations Equation 7.20 and Equation 7.22). What is new here are formulas for the off-diagonal elements.

7.7.3 Joint sampling distribution between $\hat{\mu}$ and $\text{vech}(\hat{\Sigma})$

When you study hypothesis testing in Chapter 9 and the statistical analysis of portfolios in Chapter 14 you will need to use results on the joint sampling distribution between $\hat{\mu}$ and $\text{vech}(\hat{\Sigma})$. The follow proposition states the result:

Proposition 7.10 (Joint sampling distribution between $\hat{\mu}$ and $\text{vech}(\hat{\Sigma})$)**.** *Under the GWN model and for large enough T, the joint sampling distribution for the $N + N(N+1)/2 \times 1$ vector $(\hat{\mu}', \hat{\theta}')'$, where $\hat{\theta} = \text{vech}(\hat{\Sigma})$ is multivariate normal*

$$\begin{pmatrix} \hat{\mu} \\ \hat{\theta} \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \theta \end{pmatrix}, \frac{1}{T}\begin{pmatrix} \hat{\Sigma} & 0 \\ 0 & \hat{\Omega} \end{pmatrix}\right).$$

Proposition 7.10 tells us that the vectors $\hat{\mu}$ and $\hat{\theta}$ are jointly multivariate normally distributed and that all elements of $\hat{\mu}$ are independent of all the elements of $\hat{\theta}$.

Example 7.12 (Joint distribution of $\hat{\mu}$, and $\hat{\sigma}$)**.** Consider the GWN model for a single asset and suppose you are interested in the joint distribution of the vector $(\hat{\mu}, \hat{\sigma}^2)'$. Then Proposition 7.10 gives us the result:

$$\begin{pmatrix} \hat{\mu} \\ \hat{\sigma}^2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}, \begin{pmatrix} \frac{\hat{\sigma}^2}{T} & 0 \\ 0 & \frac{2\hat{\sigma}^4}{T} \end{pmatrix} \right). \tag{7.44}$$

Here, you see that $\text{cov}(\hat{\mu}, \hat{\sigma}^2) = 0$, which implies that $\hat{\mu}$ and $\hat{\sigma}^2$ are independent since they are jointly normally distributed. Furthermore, since $\hat{\mu}$ and $\hat{\sigma}^2$ are independent any function of $\hat{\mu}$ is independent of any function of $\hat{\sigma}^2$. For example, $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ is independent of $\hat{\mu}$. Using Equation 7.34 you can then deduce the joint distribution of $(\hat{\mu}, \hat{\sigma})'$:

$$\begin{pmatrix} \hat{\mu} \\ \hat{\sigma} \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \sigma \end{pmatrix}, \begin{pmatrix} \frac{\hat{\sigma}^2}{T} & 0 \\ 0 & \frac{\hat{\sigma}^2}{2T} \end{pmatrix} \right). \tag{7.45}$$

■

7.8 Further Reading: Estimation of The GWN Model

- Good introductory statistical theory is given in (DeGroot and Schervish 2011). For an econometrics focus see (Goldberger 1991). More advanced books - (Casella and Berger 2002), Bickell and Dokson.
- R books on statistics, Monte Carlo and bootstrapping: (Vinod 2010) (bootstrapping)
- Surprisingly, the statistical properties of VaR are often not discussed much in standard textbook treatments. See, for example (Jorion 1997).

7.9 Problems: Estimation of The GWN Model

Exercise 7.1. Show that the matrix sample statistics Equation 7.16 and Equation 7.17 are unbiased estimators for μ and Σ, respectively.

8 Estimating Functions of GWN Model Parameters

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In the previous chapter, I considered estimation of the parameters of the GWN model for asset returns and the construction of estimated standard errors and confidence intervals for the estimated parameters. In this chapter, I consider estimation of functions of the parameters of the GWN model, and the construction of estimated standard errors and confidence intervals for these estimates. This is an important topic as you are often more interested in a function of the GWN model parameters (e.g., VaR_α) than the GWN model parameters themselves. In order to do this, I introduce some new statistical tools including the analytic delta method based on asymptotic distribution theory, and two computer intensive resampling methods called the jackknife and the bootstrap.

The R packages I use in this chapter are **boot**, **bootstrap**, **car**, **introcompfinr**, and **PerformanceAnalytics**. Have these packages installed and loaded in R before replicating the chapter examples.

8.1 Functions of GWN Model Parameters

Consider the GWN model for returns (simple or continuously compounded):

$$\begin{aligned} R_{it} &= \mu_i + \varepsilon_{it}, \\ \{\varepsilon_{it}\}_{t=1}^T &\sim \text{GWN}(0, \sigma^2), \\ \text{cov}(\varepsilon_{it}, \varepsilon_{js}) &= \begin{cases} \sigma_{ij} & t = s \\ 0 & t \neq s \end{cases}. \end{aligned}$$

Let θ denote a $k \times 1$ vector of GWN model parameters from which you want to compute a function of interest (I drop the asset subscript i to simplify notation). Let $f : R^k \rightarrow R$ be a continuous and

differentiable function of θ, and define $\eta = f(\theta)$ as the scalar parameter of interest. To make things concrete, let $\theta = (\mu, \sigma)'$. Consider the following functions of θ:

$$f_1(\theta) = \mu + \sigma \times q_\alpha^Z = q_\alpha^R \tag{8.1}$$

$$f_2(\theta) = -W_0 \times (\mu + \sigma \times q_\alpha^Z) = W_0 \times f_1(\theta) = \text{VaR}_\alpha^N \tag{8.2}$$

$$f_3(\theta) = -W_0 \times (\exp(\mu + \sigma \times q_\alpha^Z) - 1) = W_0 \times (\exp(f_1(\theta)) - 1) = \text{VaR}_\alpha^{LN} \tag{8.3}$$

The first function Equation 8.1 is the α-quantile, q_α^R, of the $N(\mu, \sigma^2)$ distribution for simple returns. The second function Equation 8.2 is VaR_α of an initial investment of W_0 based on the normal distribution for simple returns. The third function Equation 8.3 is VaR_α based on the log-normal distribution for simple returns (normal distribution for continuously compounded returns). The first two functions are linear functions of the elements of θ, and the third function is a nonlinear function of the elements of θ.

Given the plug-in estimate of θ, $\hat{\theta} = (\hat{\mu}, \hat{\sigma})'$, I want to estimate each of the above functions of θ, and construct estimated standard errors and confidence intervals.

Example 8.1 (Data for examples)**.** For the examples in this chapter, I use data on monthly continuously compounded and simple returns for Microsoft over the period January 1998, through May 2012, from the **introcompfinr** package. The data is the same as that used in Chapter 5 and is constructed as follows:

```
msftPrices = to.monthly(msftDailyPrices, OHLC=FALSE)
smpl = "1998-01::2012-05"
msftPrices = msftPrices[smpl]
msftRetS = na.omit(Return.calculate(msftPrices, method="discrete"))
msftRetC = log(1 + msftRetS)
```

The GWN model estimates for μ_{msft} and σ_{msft} for both the continuously compounded returns and the simple returns are computed using the **introcompfinr** function 'estimate_gwn_model():

```
gwn_fit_msft_c = estimate_gwn_model(msftRetC)
gwn_fit_msft_s = estimate_gwn_model(msftRetS)
```

The estimates of μ_{msft} and σ_{msft} together with their estimated standard errors and 95% confidence intervals are:

```
summary(gwn_fit_msft_c)
#> Call:
#> estimate_gwn_model(x = msftRetC)
```

```
#>
#> Date range:  Feb 1998  to  May 2012
#> Number of observations:  172
#>
#>        Mean Std Error    2.5%  97.5% Volatility Std Error   2.5% 97.5%
#> MSFT 0.0041    0.0076 -0.0112 0.0194        0.1    0.0054 0.0894 0.111
summary(gwn_fit_msft_s)
#> Call:
#> estimate_gwn_model(x = msftRetS)
#>
#> Date range:  Feb 1998  to  May 2012
#> Number of observations:  172
#>
#>        Mean Std Error    2.5%  97.5% Volatility Std Error   2.5% 97.5%
#> MSFT 0.0092    0.0077 -0.0063 0.0246      0.102    0.0055 0.0906 0.112
```

■

8.2 Estimation of Functions of GWN Model Parameters

In Chapter 7, I used the *plug-in principle* to motivate estimators of the GWN model parameters μ_i σ_i^2, σ_{ij}, and ρ_{ij}. I can also use the plug-in principle to estimate *functions* of the GWN model parameters.

Definition 8.1 (Plug-in principle for functions of GWN model parameters)**.** Let θ denote a $k \times 1$ vector of GWN model parameters, and let $\eta = f(\theta)$ where $f : \mathbb{R}^k \to \mathbb{R}$ is a scalar function of θ. Let $\hat{\theta}$ denote the plug-in estimator of θ. Then the plug-in estimator of η is $\hat{\eta} = f(\hat{\theta})$.

Example 8.2 (Plug-in estimates of example functions for example data)**.** Let $W_0 = \$100,000$, and $r_f = 0.03/12 = 0.0025$. Using the plug-in estimates of the GWN model parameters, the plug-in estimates of the functions Equation 8.1 - Equation 8.3 are:

```
W0 = 100000
r.f = 0.03/12
f1.hat = gwn_fit_msft_s$muhat + gwn_fit_msft_s$sigmahat*qnorm(0.05)
f2.hat = -W0*f1.hat
f3.hat = -W0*(exp(gwn_fit_msft_c$muhat + gwn_fit_msft_c$sigmahat*qnorm(0.05)) - 1)
fhat.vals = cbind(f1.hat, f2.hat, f3.hat)
colnames(fhat.vals) = c("f1", "f2", "f3")
```

```
rownames(fhat.vals) = "Estimate"
fhat.vals
#>              f1    f2    f3
#> Estimate -0.158 15780 14846
```

■

Plug-in estimators of functions are random variables and are subject to estimation error. Just as you studied the statistical properties of the plug-in estimators of the GWN model parameters you can study the statistical properties of the plug-in estimators of functions of GWN model parameters.

8.2.1 Bias

As discussed in Chapter 7, a desirable finite sample property of an estimator $\hat{\theta}$ of θ is *unbiasedness*: on average over many hypothetical samples $\hat{\theta}$ is equal to θ. When estimating $f(\theta)$, unbiasedness of $\hat{\theta}$ may or may not carry over to $f(\hat{\theta})$.

Suppose $f(\theta)$ is a linear function of the elements of θ so that

$$f(\theta) = a + b_1\theta_1 + b_2\theta_2 + \cdots + b_k\theta_k.$$

where $a, b_1, b_2, \ldots b_k$ are constants. For example, the functions Equation 8.1 and Equation 8.2 are linear functions of the elements of $\theta = (\mu, \sigma)'$. Let $\hat{\theta}$ denote an estimator of θ and let $f(\hat{\theta})$ denote the plug-in estimator of $f(\theta)$. If $E[\hat{\theta}] = \theta$ for all elements of θ ($\hat{\theta}_i$ is unbiased for θ_i, $i = 1, \ldots, k$), then

$$\begin{aligned} E[f(\hat{\theta})] &= a + b_1E[\hat{\theta}_1] + b_2E[\hat{\theta}_2] + \cdots + b_kE[\hat{\theta}_k] \\ &= a + b_1\theta_1 + b_2\theta_2 + \cdots + b_k\theta_k = f(\theta) \end{aligned}$$

and so $f(\hat{\theta})$ is unbiased for $f(\theta)$.

Now suppose $f(\theta)$ is a nonlinear function of the elements of θ. For example, the function Equation 8.3 is a nonlinear function of the elements of $\theta = (\mu, \sigma)'$. Then, in general, $f(\hat{\theta})$ is not unbiased for $f(\theta)$ even if $\hat{\theta}$ is unbiased for θ. The direction of the bias depends on the properties of $f(\theta)$. If $f(\cdot)$ is convex at the scalar parameter θ or concave at θ then you can say something about the direction of the bias in $f(\hat{\theta})$.

Definition 8.2 (Convex function)**.** Let θ be a scalar parameter. A scalar function $f(\theta)$ is *convex* (in two dimensions) if all points on a straight line connecting any two points on the graph of $f(\theta)$ is above or on that graph. More formally, $f(\theta)$ is convex if for all θ_1 and θ_2 and for all $0 \leq \alpha \leq 1$

$$f(\alpha\theta_1 + (1-\alpha)\theta_2) \leq \alpha f(\theta_1) + (1-\alpha)f(\theta_2)$$

For example, the function $f(\theta) = \theta^2$ is a convex function. Another common convex function is $\exp(\theta)$.

Definition 8.3 (Concave function)**.** Let θ be a scalar parameter. A scalar function $f(\theta)$ is *concave* if $-f(\theta)$ is convex.

For example, the function $f(\theta) = -\theta^2$ is concave. Other common concave functions are $log(\theta)$ and $\sqrt{\theta}$.

Proposition 8.1 (Jensen's inequality)**.** *Let θ be a scalar parameter and let $\widehat{\theta}$ be an unbiased estimate of θ so that $E[\widehat{\theta}] = \theta$. Let $f(\theta)$ be a convex function of θ. Then*

$$E[f(\widehat{\theta})] \geq f(E[\widehat{\theta}]) = f(\theta).$$

Jensen's inequality tells you that if f is convex (concave) and $\widehat{\theta}$ is unbiased then $f(\widehat{\theta})$ is positively (negatively) biased. For example, you know from Chapter 7 that $\hat{\sigma}^2$ is unbiased for σ^2. Now, $\hat{\sigma} = f(\hat{\sigma}^2) = \sqrt{\hat{\sigma}^2}$ is a concave function of $\hat{\sigma}$. From Jensen's inequality, you know that $\hat{\sigma}$ is negatively biased (i.e., too small).

■

8.2.2 Consistency

An important property of the plug-in estimators of the GWN model parameters is that they are *consistent*: as the sample size gets larger and larger the estimators get closer and closer to their true values and eventually equal their true values for an infinitely large sample.

Consistency also holds for plug-in estimators of functions of GWN model parameters. The justification comes from *Slutsky's Theorem*.

Proposition 8.2 (Slutsky's Theorem)**.** *Let $\widehat{\theta}$ be consistent for θ, and let $f : \mathbb{R}^k \to \mathbb{R}$ be continuous at θ. Then $f(\widehat{\theta})$ is consistent for $f(\theta)$.*

The key condition on $f(\theta)$ is continuity at θ.[1]

[1] Recall, the intuitive definition of continuity is that you can draw the function on paper without lifting up your pen.

Example 8.3 (Consistency of functions of GWN model parameters)**.** All of the functions Equation 8.1 - Equation 8.3 are continuous at $\theta = (\mu, \sigma)'$. By Slutsky's Theorem, all of the plug-in estimators of the functions are also consistent.

■

8.3 Asymptotic normality and the delta method

Let θ denote a scalar parameter of the GWN model for returns and let $\hat{\theta}$ denote the plug-in estimator of θ. In Chapter 7 you learned that $\hat{\theta}$ is asymptotically normally distributed

$$\hat{\theta} \sim N(\theta, \widehat{\text{se}}(\hat{\theta})^2),$$

for large enough T, where $\widehat{\text{se}}(\hat{\theta})$ denotes the estimated standard error of $\hat{\theta}$. This result is justified by the CLT. The following proposition establishes the asymptotic distribution of the plug-in estimator of a function of θ using a technique called the *delta method*.

Proposition 8.3 (The delta method)**.** *Let $f(\theta)$ denote a real-valued continuous and differentiable scalar function of θ. Using a technique called the* delta method*, you can show that the plug-in estimator $f(\hat{\theta})$ is also asymptotically normally distributed:*

$$f(\hat{\theta}) \sim N(f(\theta), \widehat{\text{se}}(f(\hat{\theta}))^2),$$

for large enough T, where $\widehat{\text{se}}(f(\hat{\theta}))$ is the estimated standard error for $f(\hat{\theta})$.

The delta method *shows you how to compute $\widehat{\text{se}}(f(\hat{\theta}))^2$:*

$$\widehat{\text{se}}(f(\hat{\theta}))^2 = f'(\hat{\theta})^2 \times \widehat{\text{se}}(\hat{\theta})^2,$$

where $f'(\hat{\theta})$ denotes the derivative of $f(\theta)$ evaluated at $\hat{\theta}$. Then,

$$\widehat{\text{se}}(f(\hat{\theta})) = \sqrt{f'(\hat{\theta})^2 \times \widehat{\text{se}}(\hat{\theta})^2}. \tag{8.4}$$

Example 8.4 (Delta method standard error for $\hat{\sigma}$)**.** The results from Chapter 7 showed, for large enough T,

$$\hat{\sigma}^2 \sim N(\sigma^2, \text{se}(\hat{\sigma}^2)^2), \; \hat{\sigma} \sim N(\sigma, \text{se}(\hat{\sigma})^2),$$

where $\text{se}(\sigma^2)^2 = 2\sigma^4/T$ and $\text{se}(\sigma)^2 = \sigma^2/2T$. The formula for $\text{se}(\hat{\sigma})^2$ is the result of the delta method for the function $f(\sigma^2) = \sqrt{\sigma^2} = \sigma$ applied to the asymptotic distribution of $\hat{\sigma}^2$. To see this, first note that by the chain rule

$$f'(\sigma^2) = \frac{1}{2}(\sigma^2)^{-1/2},$$

so that $f'(\sigma^2)^2 = \frac{1}{4}(\sigma^2)^{-1}$. Then,

$$\begin{aligned}\text{se}(f(\hat{\theta}))^2 = f'(\sigma^2)^2 \times \text{se}(\hat{\sigma}^2)^2 &= \frac{1}{4}(\sigma^2)^{-1} \times 2\sigma^4/T \\ &= \frac{1}{2}\sigma^2/T = \sigma^2/2T \\ &= \text{se}(\hat{\sigma})^2.\end{aligned}$$

The estimated squared standard error replaces σ^2 with its estimate $\hat{\sigma}^2$ and is given by

$$\widehat{\text{se}}(\hat{\sigma}^2)^2 = \hat{\sigma}^2/2T.$$

■

Now suppose θ is a $k \times 1$ vector of GWN model parameters, and $f : \mathbb{R}^k \to \mathbb{R}$ is a continuous and differentiable function of θ. Define the $k \times 1$ *gradient function* $\mathbf{g}(\theta)$ as:

$$\mathbf{g}(\theta) = \frac{\partial f(\theta)}{\partial \theta} = \begin{pmatrix} \frac{\partial f(\theta)}{\partial \theta_1} \\ \vdots \\ \frac{\partial f(\theta)}{\partial \theta_k} \end{pmatrix}.$$

Assume that $\hat{\theta}$ is asymptotically normally distributed:

$$\hat{\theta} \sim N(\theta, \widehat{\text{var}}(\hat{\theta})),$$

where $\widehat{\text{var}}(\hat{\theta})$ is the $k \times k$ estimated variance-covariance matrix of $\hat{\theta}$. Then the delta method shows you that $f(\hat{\theta})$ is asymptotically normally distributed:

$$f(\hat{\theta}) \sim N(f(\theta), \widehat{\text{se}}(f(\hat{\theta}))^2),$$

where

$$\widehat{\text{se}}(f(\hat{\theta}))^2 = \mathbf{g}(\hat{\theta})'\widehat{\text{var}}(\hat{\theta})\mathbf{g}(\hat{\theta}).$$

Example 8.5 (Delta method standard error for $\hat{q}_\alpha^Z$)**.** Let $\theta = (\mu, \sigma)'$ and consider $f_1(\theta) = q_\alpha^R = \mu + \sigma q_\alpha^Z$ (normal simple return quantile). The gradient function $\mathbf{g}_1(\theta)$ for $f_1(\theta)$ is:

$$\mathbf{g}_1(\theta) = \frac{\partial f_1(\theta)}{\partial \theta} = \begin{pmatrix} \frac{\partial f_1(\theta)}{\partial \mu} \\ \frac{\partial f_1(\theta)}{\partial \sigma} \end{pmatrix} = \begin{pmatrix} 1 \\ q_\alpha^Z \end{pmatrix}.$$

Regarding $\text{var}(\hat{\theta})$, recall from Chapter 7, in the GWN model:

$$\begin{pmatrix} \hat{\mu} \\ \hat{\sigma}^2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}, \begin{pmatrix} \text{se}(\hat{\mu})^2 & 0 \\ 0 & \text{se}(\hat{\sigma}^2)^2 \end{pmatrix}\right)$$

for large enough T. I show below, using the delta method, that

$$\begin{pmatrix} \hat{\mu} \\ \hat{\sigma} \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \sigma \end{pmatrix}, \begin{pmatrix} \text{se}(\hat{\mu})^2 & 0 \\ 0 & \text{se}(\hat{\sigma})^2 \end{pmatrix}\right)$$

Hence,

$$\text{var}(\hat{\theta}) = \begin{pmatrix} \text{se}(\hat{\mu})^2 & 0 \\ 0 & \text{se}(\hat{\sigma})^2 \end{pmatrix} = \begin{pmatrix} \frac{\sigma^2}{T} & 0 \\ 0 & \frac{\sigma^2}{2T} \end{pmatrix}$$

Then,

$$\begin{aligned} \mathbf{g}_1(\theta)'\text{var}(\hat{\theta})\mathbf{g}_1(\theta) &= \begin{pmatrix} 1 & q_\alpha^Z \end{pmatrix} \begin{pmatrix} \text{se}(\hat{\mu})^2 & 0 \\ 0 & \text{se}(\hat{\sigma})^2 \end{pmatrix} \begin{pmatrix} 1 \\ q_\alpha^Z \end{pmatrix} \\ &= \text{se}(\hat{\mu})^2 + (q_\alpha^Z)^2\text{se}(\hat{\sigma})^2 \\ &= \frac{\sigma^2}{T} + \frac{\left(q_\alpha^Z\right)^2 \sigma^2}{2T} \\ &= \frac{\sigma^2}{T}\left[1 + \frac{1}{2}\left(q_\alpha^Z\right)^2\right] \\ &= \text{se}(\hat{q}_\alpha^Z)^2. \end{aligned}$$

The standard error for $\hat{q}_\alpha^Z$ is:

$$\text{se}(\hat{q}_\alpha^R) = \sqrt{\text{var}(\hat{q}_\alpha^R)} = \frac{\sigma}{\sqrt{T}}\left[1 + \frac{1}{2}\left(q_\alpha^Z\right)^2\right]^{1/2}. \tag{8.5}$$

The formula Equation 8.6 shows that $\text{se}(\hat{q}_\alpha^R)$ increases with σ and q_α^Z, and decreases with T. In addition, for fixed T, $\text{se}(\hat{q}_\alpha^R) \to \infty$ as $\alpha \to 0$ or as $\alpha \to 1$ because $q_\alpha^Z \to -\infty$ as $\alpha \to 0$ and $q_\alpha^Z \to \infty$ as $\alpha \to 1$. Hence, there is very large estimation error in $\hat{q}_\alpha^R$ when α is very close to zero or one (for fixed values of σ and T). This makes intuitive sense as you have very few observations in the extreme tails of the empirical distribution of the data for a fixed sample size and so you expect a larger estimation error.

Replacing σ with its estimate $\hat{\sigma}$ gives the *practically useful* estimated standard error for $\hat{q}_\alpha^R$:

$$\widehat{\text{se}}(\hat{q}_\alpha^R) = \frac{\hat{\sigma}}{\sqrt{T}}\left[1 + \frac{1}{2}\left(q_\alpha^Z\right)^2\right]^{1/2}. \tag{8.6}$$

Using the above results, the sampling distribution of $\hat{q}_\alpha^R$ can be approximated by the normal distribution:

$$\hat{q}_\alpha^R \sim N(q_\alpha^R,\ \widehat{\text{se}}(\hat{q}_\alpha^R)^2), \tag{8.7}$$

for large enough T. ■

Example 8.6 (Applying the delta method to $\hat{q}_\alpha^R$ for Microsoft)**.** The estimates of q_α^R, $\text{se}(\hat{q}_\alpha^R)$, and 95% confidence intervals, for $\alpha = 0.05$, 0.01 and 0.001, from the simple monthly returns for Microsoft are:

```
qhat = gwn_fit_msft_s$muhat + gwn_fit_msft_s$sigmahat*qnorm(c(0.05,0.01,0.001))
seQhat = (gwn_fit_msft_s$sigmahat/sqrt(gwn_fit_msft_s$n.obs))*sqrt(1 + 0.5*qnorm(c(0.05,0.01
lowerQhat = qhat - 2*seQhat
upperQhat = qhat + 2*seQhat
width = upperQhat - lowerQhat
ans = cbind(qhat,seQhat,lowerQhat,upperQhat,width)
colnames(ans) = c("Estimates", "Std Errors", "2.5%", "97.5%", "Width")
rownames(ans) = c("q.05", "q.01", "q.001")
ans
#>       Estimates Std Errors   2.5%  97.5%  Width
#> q.05     -0.158     0.0119 -0.182 -0.134 0.0475
#> q.01     -0.227     0.0149 -0.257 -0.197 0.0596
#> q.001    -0.304     0.0186 -0.342 -0.267 0.0744
```

For Microsoft the values of $\widehat{\text{se}}(\hat{q}_\alpha^R)$ are about 1.2%, 1.5%, and 1.9% for $\alpha = 0.05$, $\alpha = 0.01$ and $\alpha = 0.001$, respectively. The 95% confidence interval widths increase with decreases in α indicating more uncertainty about the true value of q_α^R for smaller values of α. For example, the 95% confidence for interval $q_{0.01}^R$ indicates that the true loss with 1% probability (one month out of 100 months) is between 19.7% and 25.7%.

■

Example 8.7 (Delta method standard errors for $\widehat{\text{VaR}}_\alpha^N$ and $\widehat{\text{VaR}}_\alpha^{LN}$)**.** Let $\theta = (\mu, \sigma)'$ and consider the functions $f_2(\theta) = \text{VaR}_\alpha^N = -W_0(\mu + \sigma q_\alpha^Z)$ (normal VaR), and $f_3(\theta) = \text{VaR}_\alpha^{LN} = -W_0(\exp((\mu + \sigma q_\alpha^Z)) - 1)$ (log-normal VaR). It is straightforward to show (see end-of-chapter exercises) that the gradient functions are:

$$\begin{aligned}
\mathbf{g}_2(\theta) &= \frac{\partial f_2(\theta)}{\partial \theta} = \begin{pmatrix} -W_0 \\ -W_0 q_\alpha^Z \end{pmatrix}, \\
\mathbf{g}_3(\theta) &= \frac{\partial f_3(\theta)}{\partial \theta} = \begin{pmatrix} -W_0 \exp(\mu + \sigma q_\alpha^Z) \\ -W_0 \exp(\mu + \sigma q_\alpha^Z) q_\alpha^Z \end{pmatrix}.
\end{aligned}$$

The delta method variances are:

$$\begin{aligned}
\mathbf{g}_2(\theta)' \text{var}(\hat{\theta}) \mathbf{g}_2(\theta) &= W_0^2 \text{se}(\hat{q}_\alpha^R)^2 = \text{se}\left(\widehat{\text{VaR}}_\alpha^N\right)^2 \\
\mathbf{g}_3(\theta)' \text{var}(\hat{\theta}) \mathbf{g}_3(\theta) &= W_0^2 \exp(q_\alpha^R)^2 \text{se}(\hat{q}_\alpha^R)^2 = \text{se}\left(\widehat{\text{VaR}}_\alpha^{LN}\right)^2.
\end{aligned}$$

The corresponding standard errors are:

$$\begin{aligned}
\text{se}\left(\widehat{\text{VaR}}_\alpha^N\right) &= W_0 \text{se}(\hat{q}_\alpha^R) \\
\text{se}\left(\widehat{\text{VaR}}_\alpha^{LN}\right) &= W_0 \exp(q_\alpha^R) \text{se}(\hat{q}_\alpha^R).
\end{aligned}$$

where

$$\text{se}(\hat{q}_\alpha^Z) = \sqrt{\frac{\sigma^2}{T}\left[1 + \frac{1}{2}(q_\alpha^Z)^2\right]}.$$

Remarks:

1. $\text{se}(\widehat{\text{VaR}}_\alpha^N)$ and $\text{se}(\widehat{\text{VaR}}_\alpha^{LN})$ increase with W_0, σ, and q_α^Z, and decrease with T.
2. For fixed values of W_0, σ, and T, $\text{se}(\widehat{\text{VaR}}_\alpha^N)$ and $\text{se}(\widehat{\text{VaR}}_\alpha^{LN})$ become very large as α approaches zero. That is, for very small loss probabilities you will have very imprecise estimates of VaR_α^N and VaR_α^{LN}.
3. The practically useful estimated standard errors replace the unknown value of q_α^R with its estimated value $\hat{q}_\alpha^R$ and are given by:

$$\hat{\text{se}}\left(\widehat{\text{VaR}}_\alpha^N\right) = W_0 \hat{\text{se}}(\hat{q}_\alpha^R) \tag{8.8}$$

$$\hat{\text{se}}\left(\widehat{\text{VaR}}_\alpha^{LN}\right) = W_0 \exp(\hat{q}_\alpha^R)\hat{\text{se}}(\hat{q}_\alpha^R) \tag{8.9}$$

■

Example 8.8 (Applying the delta method to $\widehat{VaR}_\alpha^N$ and $\widehat{VaR}_\alpha^{LN}$ for Microsoft)**.** Consider a \$100,000 investment for one month in Microsoft. The estimates of VaR_α^N, its standard error, and 95% confidence intervals, for $\alpha = 0.05$, $\alpha = 0.01$, and $\alpha = 0.001$ are:

```
VaR.N = -qhat*W0
seVaR.N = W0*seQhat
lowerVaR.N = VaR.N - 2*seVaR.N
upperVaR.N = VaR.N + 2*seVaR.N
width = upperVaR.N - lowerVaR.N
ans = cbind(VaR.N, seVaR.N, lowerVaR.N, upperVaR.N, width)
colnames(ans) = c("Estimate", "Std Error", "2.5%", "97.5%", "Width")
rownames(ans) = c("VaR.N.05", "VaR.N.01", "VaR.N.001")
ans
#>           Estimate Std Error  2.5% 97.5% Width
#> VaR.N.05     15780      1187 13405 18154  4748
#> VaR.N.01     22696      1490 19717 25676  5959
#> VaR.N.001    30450      1860 26730 34169  7439
```

The estimated values of $\text{VaR}_{.05}^N$, $\text{VaR}_{.01}^N$, and $\text{VaR}_{.001}^N$ are \$15,780, \$22,696, and \$30,450 respectively, with estimated standard errors of \$1,187, \$1,490, and \$1,860. The estimated standard errors are fairly small compared with the estimated VaR values, so you can say that VaR is estimated fairly precisely here. Notice that the estimated standard error for $\text{VaR}_{.001}$ is about 56% larger than the estimated standard error for $\text{VaR}_{.05}^N$ indicating you have much less precision for estimating VaR for very small values of α.

The estimates of VaR_α^{LN} and its standard error for $\alpha = 0.05$ and $\alpha = 0.01$ are:

```
qhat.c = gwn_fit_msft_c$muhat + gwn_fit_msft_c$sigmahat*qnorm(c(0.05,0.01,0.001))
seQhat.c = (gwn_fit_msft_c$sigmahat/sqrt(gwn_fit_msft_c$n.obs))*sqrt(1 + 0.5*qnorm(c(0.05,0.
VaR.LN = -W0*(exp(qhat.c)-1)
seVaR.LN = W0*exp(qhat.c)*seQhat.c
lowerVaR.LN = VaR.LN - 2*seVaR.LN
upperVaR.LN = VaR.LN + 2*seVaR.LN
width = upperVaR.LN - lowerVaR.LN
ans = cbind(VaR.LN, seVaR.LN, lowerVaR.LN, upperVaR.LN, width)
colnames(ans) = c("Estimate", "Std Error", "2.5%", "97.5%", "Width")
rownames(ans) = c("VaR.LN.05", "VaR.LN.01", "VaR.LN.001")
ans
#>             Estimate Std Error  2.5% 97.5% Width
#> VaR.LN.05      14846       998 12850 16842  3992
#> VaR.LN.01      20467      1170 18128 22807  4680
#> VaR.LN.001     26329      1353 23623 29034  5411
```

The estimates of VaR_α^{LN} (and its standard error) are slightly smaller than the estimates of VaR_α^{N} (and its standard error) due to the positive skewness in the log-normal distribution. ■

Example 8.9 (Estimating VaR using the introcompfinr function estimate_var)**.** You can estimate normal and log-normal VaR and their standard errors using the **introcompfinr** function `estimate_var()`. The arguments expected by `estimate_var()` are:

```
args(estimate_var)
#> function (x, alpha = 0.05, w = 1e+05, var_method = c("normal",
#>     "empirical"), var_type = c("return", "dollar"), return_type = c("simple",
#>     "cc"))
#> NULL
```

where `x` is an `xts` object of returns, `alpha` is the loss probability, and `w` is the initial investment. The argument `var_mathod` is the method used to calculate the VaR, either "normal" or "empirical". If the argument `return_type="simple` and `var_method="normal"` then normal VaR is computed. If `return_type="cc"` and `var_method="normal"` then log-normal VaR is computed. If `var_type="return"` then the VaR is computed as $-\hat{q}_\alpha$, and if `var_type="dollar"` then the VaR is computed as $-\hat{q}_\alpha \times w$.

To estimate 5% normal VaR for Microsoft based on an initial investment of $\$100,000$ use:

```
VaR_normal_05 = estimate_var(msftRetS, alpha = 0.05, w = 100000, return_type = "simple",
                             var_method = "normal", var_type = "dollar")
summary(VaR_normal_05)
```

```
#>
#> VaR method:  normal
#> VaR type:  dollar
#> Return type:  simple
#> Probability level:  0.05
#> Initial investment:  1e+05
#>      Estimates Std Errors  2.5% 97.5% Width
#> MSFT     15780       1187 13405 18154  4748
```

To estimate 5% log-normal VaR use:

```
VaR_lognormal_05 = estimate_var(msftRetC, alpha = 0.05, w = 100000, return_type = "cc",
                          var_method = "normal", var_type = "dollar")
summary(VaR_lognormal_05)
#>
#> VaR method:  normal
#> VaR type:  dollar
#> Return type:  cc
#> Probability level:  0.05
#> Initial investment:  1e+05
#>      Estimates Std Errors  2.5% 97.5% Width
#> MSFT     14846       1172 12502 17190  4688
```

■

8.3.1 The numerical delta method

The delta method is an advanced statistical technique and requires a bit of calculus to implement, especially for nonlinear vector-valued functions. It can be easy to make mistakes when working out the math! Fortunately, in R there is an easy way to implement the delta method that does not require you to work out the math. The **car** function `deltaMethod` implements the delta method for scalar valued functions *numerically*, utilizing the **stats** function `D()` which implements symbolic derivatives. The arguments expected by `deltaMethod` are:

```
args(car:::deltaMethod.default)
#> function (object, g., vcov., func = g., constants, level = 0.95,
#>     rhs = NULL, singular.ok = FALSE, ..., envir = parent.frame())
#> NULL
```

Here, `object` is the vector of named estimated parameters, $\hat{\theta}$, `g.` is a quoted string that is the function of the parameter estimates to be evaluated, $f(\theta)$, and `vcov` is the named estimated covariance matrix of the coefficient estimates, $\widehat{\text{var}}(\hat{\theta})$. The function returns an object of class "deltaMethod" for which there is a `print` method. The "deltaMethod" object is essentially a `data.frame` with columns giving the parameter estimate(s), the delta method estimated standard error(s), and lower and upper confidence limits. The following example illustrates how to use `deltaMethod` for the example functions.

Example 8.10 (Numerical delta method standard errors for example data)**.** I first create the named inputs $\hat{\theta}$ and $\widehat{\text{var}}(\hat{\theta})$ from the GWN model for simple returns:

```
thetahatS = c(gwn_fit_msft_s$muhat, gwn_fit_msft_s$sigmahat)
names(thetahatS) = c("mu", "sigma")
var.thetahatS = matrix(c(gwn_fit_msft_s$se_muhat^2,0,0,gwn_fit_msft_s$se_sigmahat^2), 2, 2,
rownames(var.thetahatS) = colnames(var.thetahatS) = names(thetahatS)
thetahatS
#>      mu   sigma
#> 0.00915 0.10150
var.thetahatS
#>              mu    sigma
#> mu    5.99e-05 0.00e+00
#> sigma 0.00e+00 2.99e-05
```

I do the same for the inputs from the GWN model for continuously compounded returns:

```
thetahatC = c(gwn_fit_msft_c$muhat, gwn_fit_msft_c$sigmahat)
names(thetahatC) = c("mu", "sigma")
var.thetahatC = matrix(c(gwn_fit_msft_c$se_muhat^2,0,0,gwn_fit_msft_c$se_sigmahat^2), 2, 2,
rownames(var.thetahatC) = colnames(var.thetahatC) = names(thetahatC)
```

It is important to name the elements in `thetahat` and `var.thetahat` as these names will be used in defining $f(\theta)$. To implement the numerical delta method for the normal quantile, you need to supply a quoted string specifying the function $f_1(\theta) = \mu + \sigma q_\alpha^Z$. Here I use the string `"mu+sigma*q.05"`, where `q.05 = qnorm(0.05)`. For, $\alpha = 0.05$ the R code is:

```
q.05 = qnorm(0.05)
dm1 = deltaMethod(thetahatS, "mu+sigma*q.05", var.thetahatS)
class(dm1)
#> [1] "deltaMethod" "data.frame"
```

The returned object `dm1` is of class "deltaMethod", inhereting from "data.frame", and has named columns:

```
names(dm1)
#> [1] "Estimate" "SE"       "2.5 %"    "97.5 %"
```

The `print` method shows the results of applying the delta method:

```
dm1
#>                    Estimate      SE   2.5 % 97.5 %
#> mu + sigma * q.05  -0.1578  0.0119 -0.1811  -0.13
```

The results match the analytic calculations performed above.[2]

The delta method for the normal VaR function $f_2(\theta)$ is:

```
dm2 = deltaMethod(thetahatS, "-W0*(mu+sigma*q.05)", var.thetahatS)
dm2
#>                             Estimate    SE 2.5 % 97.5 %
#> -W0 * (mu + sigma * q.05)    15780  1187 13453  18106
```

The delta method for the log-normal VaR function $f_3(\theta)$ is:

```
dm3 = deltaMethod(thetahatC, "-W0*(exp(mu+sigma*q.05)-1)", var.thetahatC)
dm3
#>                                      Estimate    SE 2.5 % 97.5 %
#> -W0 * (exp(mu + sigma * q.05) - 1)    14846   998 12890  16802
```

■

8.3.2 The delta method for vector valued functions (advanced)

Let θ denote a $k \times 1$ vector of GWN model parameters. For example, $\theta = (\mu, \sigma^2)'$. Assume that $\hat{\theta}$ is asymptotically normally distributed:

$$\hat{\theta} \sim N(\theta, \widehat{\text{var}}(\theta)),$$

for large enough T where $\widehat{\text{var}}(\theta)$ is the $k \times k$ variance-covariance matrix of $\hat{\theta}$. Sometimes you need to consider the asymptotic distribution of a vector-valued function of θ. Let $\mathbf{f} : \mathbb{R}^k \to \mathbb{R}^j$ where

[2] The 95% confidence interval computed in `deltaMethod` uses 1.96 instead of the rule-of-thumb value of 2 used in the analytic calculations.

$j \leq k$ be a continuous and differentiable function, and define the $j \times 1$ vector $\eta = \mathbf{f}(\theta)$. It is useful to express η as:

$$\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_j \end{pmatrix} = \begin{pmatrix} f_1(\theta) \\ f_2(\theta) \\ \vdots \\ f_j(\theta) \end{pmatrix}.$$

Example 8.11 (Example vector functions of GWN model parameters)**.** Let $\theta = (\sigma_1^2, \sigma_2^2)'$, and define the 2×1 vector of volatilities: $\eta = \mathbf{f}(\theta) = (\sigma_1, \sigma_2)'$. Then, $\eta_1 = f_1(\theta) = \sqrt{\sigma_2^2}$ and $\eta_2 = f_2(\theta) = \sqrt{\sigma_2^2}$.

■

Define the $j \times k$ Jacobian matrix:

$$\mathbf{G}(\theta)' = \frac{\partial \mathbf{f}(\theta)}{\partial \theta'} = \begin{pmatrix} \frac{\partial f_1(\theta)}{\partial \theta_1} & \frac{\partial f_1(\theta)}{\partial \theta_2} & ... & \frac{\partial f_1(\theta)}{\partial \theta_k} \\ \frac{\partial f_2(\theta)}{\partial \theta_1} & \frac{\partial f_2(\theta)}{\partial \theta_2} & ... & \frac{\partial f_2(\theta)}{\partial \theta_k} \\ \vdots & \vdots & ... & \vdots \\ \frac{\partial f_j(\theta)}{\partial \theta_1} & \frac{\partial f_j(\theta)}{\partial \theta_2} & ... & \frac{\partial f_j(\theta)}{\partial \theta_k} \end{pmatrix}. \tag{8.10}$$

Then, by the delta method, the asymptotic normal distribution of $\hat{\eta} = \mathbf{f}(\hat{\theta})$ is:

$$\hat{\eta} \sim N(\eta, \widehat{\text{var}}(\hat{\eta})),$$

where

$$\widehat{\text{var}}(\hat{\eta}) = \mathbf{G}(\hat{\theta})' \widehat{\text{var}}(\hat{\theta}) \mathbf{G}(\hat{\theta}). \tag{8.11}$$

Example 8.12 (Asymptotic distributions of example functions)**.** Consider the first example function where $\theta = (\sigma_1^2, \sigma_2^2)'$ and $\eta = \mathbf{f}(\theta) = (\sigma_1, \sigma_2)'$. Now, using the chain-rule:

$$\frac{\partial f_1(\theta)}{\partial \sigma_1^2} = \frac{1}{2}\sigma_1^{-1}, \ \frac{\partial f_1(\theta)}{\partial \sigma_2^2} = 0,$$
$$\frac{\partial f_2(\theta)}{\partial \sigma_1^2} = 0, \ \frac{\partial f_2(\theta)}{\partial \sigma_2^2} = \frac{1}{2}\sigma_1^{-1}.$$

The Jacobian matrix is

$$\mathbf{G}(\theta)' = \begin{pmatrix} \frac{1}{2}\sigma_1^{-1} & 0 \\ 0 & \frac{1}{2}\sigma_2^{-1} \end{pmatrix}.$$

From Chapter 7, Proposition 7.9,

$$\widehat{\text{var}}(\hat{\theta}) = \frac{1}{T}\begin{pmatrix} 2\hat{\sigma}_1^4 & 2\hat{\sigma}_{12}^2 \\ 2\hat{\sigma}_{12}^2 & 2\hat{\sigma}_2^4 \end{pmatrix}.$$

Then

$$\begin{aligned}
\widehat{\text{var}}(\hat{\eta}) &= \mathbf{G}(\hat{\theta})'\widehat{\text{var}}(\hat{\theta})\mathbf{G}(\hat{\theta}) \\
&= \begin{pmatrix} \frac{1}{2}\hat{\sigma}_1^{-1} & 0 \\ 0 & \frac{1}{2}\hat{\sigma}_2^{-1} \end{pmatrix} \frac{1}{T}\begin{pmatrix} 2\hat{\sigma}_1^4 & 2\hat{\sigma}_{12}^2 \\ 2\hat{\sigma}_{12}^2 & 2\hat{\sigma}_2^4 \end{pmatrix} \begin{pmatrix} \frac{1}{2}\hat{\sigma}_1^{-1} & 0 \\ 0 & \frac{1}{2}\hat{\sigma}_2^{-1} \end{pmatrix} \\
&= \frac{1}{T}\begin{pmatrix} \frac{1}{2}\hat{\sigma}_1^2 & \frac{1}{2}\hat{\sigma}_{12}^2\hat{\sigma}_1^{-1}\hat{\sigma}_2^{-1} \\ \frac{1}{2}\hat{\sigma}_{12}^2\hat{\sigma}_1^{-1}\hat{\sigma}_2^{-1} & \frac{1}{2}\hat{\sigma}_2^2 \end{pmatrix} = \frac{1}{T}\begin{pmatrix} \frac{1}{2}\hat{\sigma}_1^2 & \frac{1}{2}\hat{\sigma}_{12}\hat{\rho}_{12} \\ \frac{1}{2}\hat{\sigma}_{12}\hat{\rho}_{12} & \frac{1}{2}\hat{\sigma}_2^2 \end{pmatrix}.
\end{aligned}$$

■

8.3.3 The numerical delta method for vector valued functions (advanced)

- TBD
- Use the delta method implemented in **metafor** and **msm**

8.3.4 The delta method explained

The delta method deduces the asymptotic distribution of $f(\hat{\theta})$ using a first order Taylor series approximation of $f(\hat{\theta})$ evaluated at the true value θ. For simplicity, assume that θ is a scalar parameter. Then, the first order Taylor series expansion gives

$$\begin{aligned}
f(\hat{\theta}) &= f(\theta) + f'(\theta)(\hat{\theta} - \theta) + \text{remainder} \\
&=\approx f(\theta) + f'(\theta)(\hat{\theta} - \theta) = f(\theta) + Z,
\end{aligned}$$

where $Z = f'(\theta)(\hat{\theta} - \theta)$. The first order Taylor series expansion shows that $f(\hat{\theta})$ is an approximate linear function of the random variable Z. The values θ, $f(\theta)$, and $f'(\theta)$ are constants. Assuming that, for large enough T, T,

$$\hat{\theta} \sim N(\theta, \widehat{\text{se}}(\hat{\theta})^2),$$

it follows, for large enough T, that

$$Z = f'(\theta)(\hat{\theta} - \theta) \sim N(0, f'(\theta)^2 \times \widehat{\text{se}}(\hat{\theta})^2).$$

Hence, for large enough T, the $f(\hat{\theta})$ is a linear function of the normally distributed random variable Z and so is also normally distributed with distribution:

$$f(\hat{\theta}) \sim N(f(\theta), f'(\theta)^2 \widehat{\text{se}}(\hat{\theta})^2).)$$

Since $\hat{\theta}$ is consistent for θ you can replace $f'(\theta)^2$ with $f'(\hat{\theta})^2$ giving the practically useful result

$$f(\hat{\theta}) \sim N(f(\theta), f'(\hat{\theta})^2 \widehat{\text{se}}(\hat{\theta})^2).$$

8.4 Resampling

As described in the previous sections, the delta method can be used to produce analytic expressions for standard errors of functions of estimated parameters that are asymptotically normally distributed. The formulas are specific to each function and are approximations that rely on large samples and the validity of the Central Limit Theorem to be accurate.

Alternatives to the analytic asymptotic distributions of estimators and the delta method are computer intensive *resampling methods* such as the *jackknife* and the *bootstrap*. These resampling methods can be used to produce numerical estimated standard errors for estimators and functions of estimators without the use of mathematical formulas. Most importantly, these jackknife and bootstrap standard errors are: (1) easy to compute, and (2) often numerically very close to the analytic standard errors based on CLT approximations. In some cases the bootstrap standard errors can even be more reliable than asymptotic approximations.

There are many advantages of the jackknife and bootstrap over analytic formulas. The most important are the following:

- *Fewer assumptions.* The jackknife and boostrap procedures typically requires fewer assumptions than are needed to derive analytic formulas. In particular, they do not require the data to be normally distributed or the sample size to be large enough so that the Central Limit Theorem holds.
- *Greater accuracy.* In small samples the bootstrap is often more reliable than analytic approximations based on the Central Limit Theorem. In large samples when the Central Limit Theorem holds the bootstrap can be even be more accurate.
- *Generality.* In many situations, the jackknife and bootstrap procedures are applied in the same way regardless of the estimator under consideration. That is, you don't need a different jackknife or bootstrap procedure for each estimator.

8.5 The Jackknife

Let θ denote a scalar parameter of interest. It could be a scalar parameter of the GWN model (e.g. μ or σ) or it could be a scalar function of the parameters of the GWN model (e.g., VaR). Let $\hat{\theta}$ denote the plug-in estimator of θ. The goal is to compute a numerical estimated standard error for $\hat{\theta}$, $\widehat{\text{se}}(\hat{\theta})$, using the jackknife resampling method.

The jackknife resampling method utilizes T resamples of the original data $\{R_t\}_{t=1}^T$. The resamples are created by leaving out a single observation. For example, the first jackknife resample leaves out the first observation and is given by $\{R_t\}_{t=2}^T$. This sample has $T-1$ observations. In general, the tth jackknife resample leaves out the tth return R_t for $t \leq T$. Now, on each of the $t = 1, \ldots, T$ jackknife resamples you compute an estimate of θ. Denote this jackknife estimate $\hat{\theta}_{-t}$ to indicate that θ is estimated using the resample that omits observation t. The estimator $\hat{\theta}_{-t}$ is called the *leave-one-out* estimator of θ. Computing $\hat{\theta}_{-t}$ for $t = 1, \ldots, T$ gives T leave-one-out estimators $\left\{\hat{\theta}_{-t}\right\}_{t=1}^T$.

Definition 8.4 (Jackknife standard error)**.** The *jackknife estimator* of $\widehat{\text{se}}(\hat{\theta})$ is the scaled standard deviation of $\left\{\hat{\theta}_{-t}\right\}_{t=1}^T$:

$$\widehat{\text{se}}_{jack}(\hat{\theta}) = \sqrt{\frac{T-1}{T}\sum_{t=1}^{T}(\hat{\theta}_{-t} - \bar{\theta})^2}, \tag{8.12}$$

where $\bar{\theta} = T^{-1}\sum_{t=1}^T \hat{\theta}_{-t}$ is the sample mean of $\left\{\hat{\theta}_{-t}\right\}_{t=1}^T$.

The estimator Equation 8.12 is easy to compute and does not require any analytic calculations from the asymptotic distribution of $\hat{\theta}$.

The intuition for the jackknife standard error formula can be illustrated by considering the jackknife standard error estimate for the sample mean $\hat{\theta} = \hat{\mu}$. You know that the analytic estimated squared standard error for $\hat{\mu}$ is

$$\widehat{\text{se}}(\hat{\mu})^2 = \frac{\hat{\sigma}^2}{T} = \frac{1}{T}\frac{1}{T-1}\sum_{t=1}^{T}(R_t - \hat{\mu})^2.$$

Now consider the jackknife squared standard error for $\hat{\mu}$:

$$\widehat{\text{se}}_{jack}(\hat{\mu})^2 = \frac{T-1}{T}\sum_{t=1}^{T}(\hat{\mu}_{-t} - \bar{\mu})^2,$$

where $\bar{\mu} = T^{-1}\sum_{t=1}^T \hat{\mu}_{-t}$. Here, I will show that $\widehat{\text{se}}(\hat{\mu})^2 = \widehat{\text{se}}_{jack}(\hat{\mu})^2$.

First, note that

$$\begin{aligned}\hat{\mu}_{-t} &= \frac{1}{T-1}\sum_{j\neq t} R_j = \frac{1}{T-1}\left(\sum_{j=1}^{T} R_j - R_t\right) \\ &= \frac{1}{T-1}\left(T\hat{\mu} - R_t\right) \\ &= \frac{T}{T-1}\hat{\mu} - \frac{1}{T-1}R_t.\end{aligned}$$

Next, plugging in the above to the definition of $\bar{\mu}$ I get

$$\begin{aligned}\bar{\mu} &= \frac{1}{T}\sum_{j=1}^{T}\hat{\mu}_{-t} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{T}{T-1}\hat{\mu} - \frac{1}{T-1}R_t\right) \\ &= \frac{1}{T-1}\sum_{t=1}^{T}\hat{\mu} - \frac{1}{T}\frac{1}{T-1}\sum_{t=1}^{T}R_t \\ &= \frac{T}{T-1}\hat{\mu} - \frac{1}{T-1}\hat{\mu} \\ &= \hat{\mu}\left(\frac{T}{T-1} - \frac{1}{T-1}\right) \\ &= \hat{\mu}.\end{aligned}$$

Hence, the sample mean of $\hat{\mu}_{-t}$ is $\hat{\mu}$. Then I can write

$$\begin{aligned}\hat{\mu}_{-t} - \bar{\mu} &= \hat{\mu}_{-t} - \hat{\mu} \\ &= \frac{T}{T-1}\hat{\mu} - \frac{1}{T-1}R_t - \hat{\mu} \\ &= \frac{T}{T-1}\hat{\mu} - \frac{1}{T-1}R_t - \frac{T-1}{T-1}\hat{\mu} \\ &= \frac{1}{T-1}\hat{\mu} - \frac{1}{T-1}R_t \\ &= \frac{1}{T-1}(\hat{\mu} - R_t).\end{aligned}$$

Using the above I can write

$$
\begin{aligned}
\widehat{\text{se}}_{jack}(\hat{\mu})^2 &= \frac{T-1}{T}\sum_{t=1}^{T}(\hat{\mu}_{-t} - \bar{\mu})^2 \\
&= \frac{T-1}{T}\left(\frac{1}{T-1}\right)^2 \sum_{t=1}^{T}(\hat{\mu} - R_t)^2 \\
&= \frac{1}{T}\frac{1}{T-1}\sum_{t=1}^{T}(\hat{\mu} - R_t)^2 \\
&= \frac{1}{T}\hat{\sigma}^2 \\
&= \widehat{\text{se}}(\hat{\mu})^2.
\end{aligned}
$$

Example 8.13 (Jackknife standard error for $\hat{\mu}$ for Microsoft)**.** You can compute the leave-one-out estimates $\{\hat{\mu}_{-t}\}_{t=1}^{T}$ using a simple `for` loop as follows:

```
n.obs = length(msftRetS)
muhat.loo = rep(0, n.obs)
for(i in 1:n.obs) {
  muhat.loo[i] = mean(msftRetS[-i])
}
```

In the above loop, `msftRetS[-i]` extracts all observations except observation i. The mean of the leave-one-out estimates is

```
mean(muhat.loo)
#> [1] 0.00915
```

which is the sample mean of `msftRetS`. It is informative to plot a histogram of the leave-one-out estimates:

The white dotted line is at the point $\hat{\mu} = 0.00915$. The jackknife estimated standard error is the scaled standard deviation of the T $\hat{\mu}_{-t}$ values:[3]

```
se.jack.muhat = sqrt(((n.obs - 1)^2)/n.obs)*sd(muhat.loo)
se.jack.muhat
#> [1] 0.00774
```

This value is exactly the same as the analytic standard error:

[3] the scaling of the standard deviation of $\hat{\mu}_{-t}$ is approximately $\sqrt{T}$.

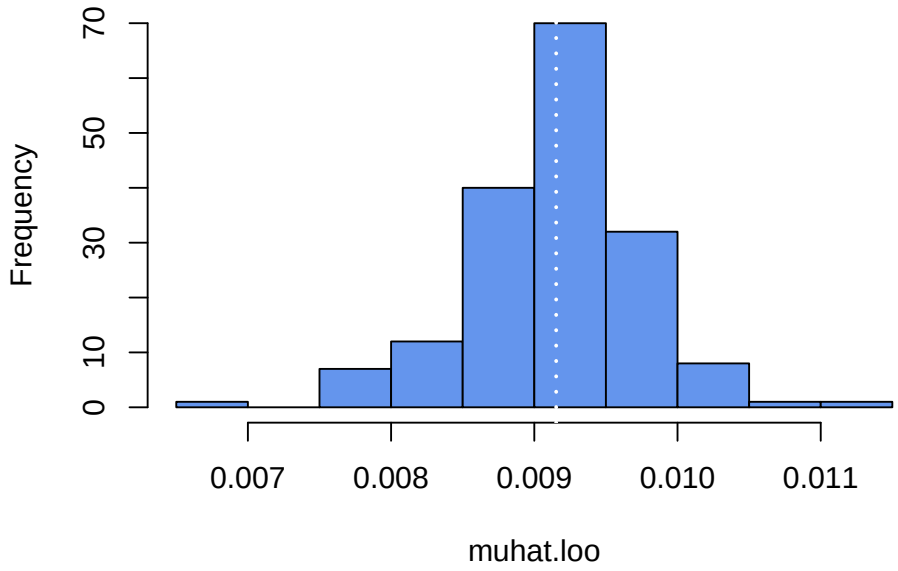


Figure 8.1: Histogram of leave-one-out estimator of μ for Microsoft.

```
gwn_fit_msft_s$se_muhat
#> [1] 0.00774
```

You can also compute jackknife estimated standard errors using the **bootstrap** function `jackknife`:

```
library(bootstrap)
muhat.jack = jackknife(coredata(msftRetS), mean)
names(muhat.jack)
#> [1] "jack.se"     "jack.bias"   "jack.values" "call"
```

By default, `jackknife()` takes as input a vector of data and the name of an R function and returns a list with components related to the jackknife procedure. The component `jack.values` contains the leave-one-out estimators, and the component `jack.se` gives you the jackknife estimated standard error:

```
muhat.jack$jack.se
#> [1] 0.00774
```

■

The fact that the jackknife standard error estimate for $\hat{\mu}$ equals the analytic standard error is

motivates using the jackknife to compute standard errors for general estimators. For estimated parameters other than $\hat{\mu}$, and general estimated functions of GWN model parameters, the jackknife standard errors are often numerically very close to the delta method standard errors. The following example illustrates this point for the example functions Equation 8.1 - Equation 8.3.

Example 8.14 (Jackknife standard errors for plug-in estimators of example functions)**.** Here, I use the **bootstrap** function `jackknife()` to compute jackknife estimated standard errors for the plug-in estimates of the example functions Equation 8.1 - Equation 8.3 and I compare the jackknife standard errors with the delta method standard errors. To use `jackknife()` with user-defined functions involving multiple parameters you must create the function such that the first input is an integer vector of index observations and the second input is the data. For example, a function for computing Equation 8.1 to be passed to `jackknife()` is:

```
theta.f1 = function(x, xdata) {
  mu.hat = mean(xdata[x, ])
  sigma.hat = sd(xdata[x, ])
  fun.hat = mu.hat + sigma.hat*qnorm(0.05)
  fun.hat
}
```

To compute the jackknife estimated standard errors for the estimate of Equation 8.1 use:

```
f1.jack = jackknife(1:n.obs, theta=theta.f1, coredata(msftRetS))
ans = cbind(f1.jack$jack.se, dm1$SE)
colnames(ans) = c("Jackknife", "Delta Method")
rownames(ans) = "f1"
ans
#>    Jackknife Delta Method
#> f1    0.0133       0.0119
```

Here, you see that the estimated jackknife standard error for $f_1(\hat{\theta})$ is very close to the delta method standard error computed earlier.

To compute the jackknife estimated standard error for $f_2(\hat{\theta})$ use:

```
theta.f2 = function(x, xdata, W0) {
  mu.hat = mean(xdata[x, ])
  sigma.hat = sd(xdata[x, ])
  fun.hat = -W0*(mu.hat + sigma.hat*qnorm(0.05))
  fun.hat
}
```

```
f2.jack = jackknife(1:n.obs, theta=theta.f2, coredata(msftRetS), W0)
ans = cbind(f2.jack$jack.se, dm2$SE)
colnames(ans) = c("Jackknife", "Delta Method")
rownames(ans) = "f2"
ans
#>    Jackknife Delta Method
#> f2      1329         1187
```

Again, the jackknife standard error is close to the delta method standard error.

To compute the jackknife estimated standard error for $f_3(\hat{\theta})$ use:

```
theta.f3 = function(x, xdata, W0) {
  mu.hat = mean(xdata[x, ])
  sigma.hat = sd(xdata[x, ])
  fun.hat = -W0*(exp(mu.hat + sigma.hat*qnorm(0.05)) - 1)
  fun.hat
}
f3.jack = jackknife(1:n.obs, theta=theta.f3, coredata(msftRetC), W0)
ans = cbind(f3.jack$jack.se, dm3$SE)
colnames(ans) = c("Jackknife", "Delta Method")
rownames(ans) = "f3"
ans
#>    Jackknife Delta Method
#> f3      1315          998
```

For log-normal VaR, the jackknife standard error is a bit larger than the delta method standard error.

Here, the jackknife and delta method standard errors are very close. ■

8.5.1 The jackknife for vector-valued estimates

The jackknife can also be used to compute an estimated variance matrix of a vector-valued estimate of parameters. Let θ denote a $k \times 1$ vector of GWN model parameters (e.g., $\theta = (\mu, \sigma)'$), and let $\hat{\theta}$ denote the plug-in estimate of θ. Let $\hat{\theta}_{-t}$ denote the leave-one-out estimate of θ, and $\bar{\theta} = T^{-1} \sum_{t=1}^{T} \hat{\theta}_{-t}$.

Definition 8.5 (Jackknife variance matrix)**.** The jackknife estimate of the $k \times k$ variance matrix $\widehat{\text{var}}(\hat{\theta})$ is the scaled sample covariance of $\{\hat{\theta}_{-t}\}_{t=1}^{T}$:

$$\widehat{\text{var}}_{jack}(\hat{\theta}) = \frac{T-1}{T}\sum_{t=1}^{T}(\hat{\theta}_{-t} - \bar{\theta})(\hat{\theta}_{-t} - \bar{\theta})'. \tag{8.13}$$

The jackknife estimated standard errors for the elements of $\hat{\theta}$ are given by the square root of the diagonal elements of $\widehat{\text{var}}_{jack}(\hat{\theta})$.

The same principle works for computing the estimated variance matrix of a vector of functions of GWN model parameters. Let $\mathbf{f} : R^k \to R^l$ with $l \leq k$ and define $\eta = \mathbf{f}(\theta)$ denote a $l \times 1$ vector of functions of GWN model parameters. Let $\hat{\eta} = f(\hat{\theta})$ denote the plug-in estimator of η. Let $\hat{\eta}_{-t}$ denote the leave-one-out estimate of η, and $\bar{\eta} = T^{-1}\sum_{t=1}^{T}\hat{\eta}_{-t}$. :::{#def-jackknife-estimator-of-variance-matrix-for-vector-valued-functions .definition name="Jackknife variance matrix for vector-valued functions"} The jackknife estimate of the $l \times l$ variance matrix $\widehat{\text{var}}(\hat{\eta})$ is the scaled sample covariance of $\{\hat{\eta}_{-t}\}_{t=1}^{T}$:

$$\widehat{\text{var}}_{jack}(\hat{\eta}) = \frac{T-1}{T}\sum_{t=1}^{T}(\hat{\eta}_{-t} - \bar{\eta})(\hat{\eta}_{-t} - \bar{\eta})'. \tag{8.14}$$

:::

Example 8.15 (Jackknife variance estimate for $\hat{\theta} = (\hat{\mu}, \hat{\sigma})'$ for Microsoft simple returns)**.** The **bootstrap** function `jackknife()` function only works for scalar-valued parameter estimates, so you have to compute the jackknife variance estimate by hand using a `for` loop. The R code is:

```
n.obs = length(msftRetS)
muhat.loo = rep(0, n.obs)
sigmahat.loo = rep(0, n.obs)
for(i in 1:n.obs) {
  muhat.loo[i] = mean(msftRetS[-i])
  sigmahat.loo[i] = sd(msftRetS[-i])
}
thetahat.loo = cbind(muhat.loo, sigmahat.loo)
colnames(thetahat.loo) = c("mu", "sigma")
varhat.jack = (((n.obs-1)^2)/n.obs)*var(thetahat.loo)
varhat.jack
#>              mu    sigma
#> mu    5.99e-05 1.49e-05
#> sigma 1.49e-05 6.12e-05
```

The jackknife variance estimate is somewhat close to the analytic estimate:

```
var.thetahatS
#>              mu    sigma
#> mu    5.99e-05 0.00e+00
#> sigma 0.00e+00 2.99e-05
```

Since the first element of $\hat{\theta}$ is $\hat{\mu}$, the (1,1) elements of the two variance matrix estimates match exactly. However, the `varhat.jack` has non-zero off diagonal elements and the (2,2) element corresponding to $\hat{\sigma}$ is considerably bigger than (2,2) element of the analytic estimate. Converting `var.thetahatS` to an estimated correlation matrix gives:

```
cov2cor(varhat.jack)
#>           mu sigma
#> mu    1.000 0.246
#> sigma 0.246 1.000
```

Here, you see that the jackknife shows a small positive correlation between $\hat{\mu}$ and $\hat{\sigma}$ whereas the analytic matrix shows a zero correlation. ■

8.5.2 Pros and Cons of the Jackknife

The jackknife gets its name because it is a useful statistical device.[4] The main advantages (pros) of the jacknife are:

- It is a good "quick and dirty" method for computing numerical estimated standard errors of estimates that does not rely on asymptotic approximations.
- The jackknife estimated standard errors are often close to the delta method standard errors.

This jackknife, however, has some limitations (cons). The main limitations are:

- It can be computationally burdensome for very large samples (e.g., if used with ultra high frequency data).
- It does not work well for non-smooth functions such as empirical quantiles.
- It is not well suited for computing confidence intervals.
- It is not as accurate as the bootstrap.

[4] A jackknife is also the name of a very useful folding knife.

8.6 The Nonparametric Bootstrap

In this section, I describe the easiest and most common form of the bootstrap: the *nonparametric bootstrap*. As you shall see, the nonparametric bootstrap procedure is very similar to a Monte Carlo simulation experiment. The main difference is how the random data is generated. In a Monte Carlo experiment, the random data are computer simulated observations from an assumed model (e.g., the GWN model). In the nonparametric bootstrap, the random data is created by resampling with replacement from the original data.

The procedure for the nonparametric bootstrap of an estimate $\hat{\theta}$ of a scalar parameter θ is as follows:

The Nonparametric Bootstrap Procedure

1. *Resample*. Create B bootstrap samples by sampling with replacement from the original data $\{r_1, \dots, r_T\}$.[5] Each bootstrap sample has T observations (same as the original sample)

$$\begin{aligned} \{r^*_{11}, r^*_{12}, \dots, r^*_{1T}\} &= \text{1st bootstrap sample} \\ &\vdots \\ \{r^*_{B1}, r^*_{B2}, \dots, r^*_{BT}\} &= \text{Bth boostrap sample} \end{aligned}$$

2. *Estimate* θ. From each bootstrap sample estimate θ and denote the resulting estimate $\hat{\theta}^*$. There will be B values of $\hat{\theta}^*$: $\left\{\hat{\theta}^*_1, \dots, \hat{\theta}^*_B\right\}$.
3. *Compute statistics*. Using $\left\{\hat{\theta}^*_1, \dots, \hat{\theta}^*_B\right\}$ compute estimate of bias, standard error, and approximate 95% confidence interval.

This procedure is very similar to the procedure used to perform a Monte Carlo experiment described in Chapter 7. The main difference is how the hypothetical samples are created. With the nonparametric bootstrap, the hypothetical samples are created by resampling with replacement from the original data. In this regard the bootstrap treats the sample as if it were the population. This ensures that the bootstrap samples inherit the same distribution as the original data – *whatever that distribution may be*. If the original data is normally distributed, then the nonparametric bootstrap samples will also be normally distributed. If the data is Student's t distributed, then the nonparametric bootstrap samples will be Student's t distributed. With Monte Carlo simulation, the hypothetical samples are simulated under an assumed model and distribution. This requires one to specify values for the model parameters and the distribution from which to simulate.

[5] You can also bootstrap data from multiple assets by defining r_t as a vector of asset returns.

8.6.1 Bootstrap bias estimate

The nonparametric bootstrap can be used to estimate the bias of an estimator $\hat{\theta}$ using the bootstrap estimates $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$. The key idea is to treat the empirical distribution (i.e., histogram) of the bootstrap estimates $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$ as an approximate to the unknown distribution of $\hat{\theta}$.

Recall, the bias of an estimator is defined as

$$E[\hat{\theta}] - \theta.$$

Definition 8.6 (Bootstrap estimate of bias)**.** The bootstrap estimate of bias is given by

$$\widehat{\text{bias}}_{boot}(\hat{\theta}, \theta) = \bar{\theta}^* - \hat{\theta} = \frac{1}{B}\sum_{j=1}^{B} \hat{\theta}_j^* - \hat{\theta}. \tag{8.15}$$

(bootstrap mean - estimate)

The bootstrap bias estimate Equation 8.15 is the difference between the mean of the bootstrap estimates of θ and the sample estimate of θ. This is similar to the Monte Carlo estimate of bias discussed in Chapter 7. However, the Monte Carlo estimate of bias is the difference between the mean of the Monte Carlo estimates of θ and the true value of θ. The bootstrap estimate of bias does not require knowing the true value of θ. Effectively, the bootstrap treats the sample estimate $\hat{\theta}$ as the population value θ and the bootstrap mean $\bar{\theta}^* = \frac{1}{B}\sum_{j=1}^{B} \hat{\theta}_j^*$ as an approximation to $E[\hat{\theta}]$. Here, $\widehat{\text{bias}}_{boot}(\hat{\theta}, \theta) = 0$ if the center of the histogram of $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$ is at $\hat{\theta}$.

Given the bootstrap estimate of bias Equation 8.15, you can compute a *bias-adjusted* estimate:

$$\hat{\theta}_{adj} = \hat{\theta} - (\bar{\theta}^* - \hat{\theta}) = 2\hat{\theta} - \bar{\theta}^* \tag{8.16}$$

if the bootstrap estimate of bias is large, it may be tempting to use the bias-adjusted estimate Equation 8.16 in place of the original estimate. This is generally not done in practice because the bias adjustment introduces extra variability into the estimator. So what use is the bootstrap bias estimate? It provides information to you that your estimate contains bias (or not) and this information can influence your decision making based on the estimate.

8.6.2 Bootstrap standard error estimate

Let $\hat{\theta}$ be an estimate of the scalar parameter θ.

Definition 8.7 (Bootstrap estimate of standard error)**.** The bootstrap estimate of $\widehat{\text{se}}(\hat{\theta})$ is given by the sample standard deviation of the bootstrap estimates $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$:

$$\widehat{\text{se}}_{boot}(\hat{\theta}) = \sqrt{\frac{1}{B-1}\sum_{j=1}^{B}\left(\hat{\theta}_j^* - \bar{\theta}^*\right)^2}. \tag{8.17}$$

where $\bar{\theta}^* = \frac{1}{B}\sum_{j=1}^{B}\hat{\theta}_j^*$.

Here, $\widehat{\text{se}}_{boot}(\hat{\theta})$ is the size of a typical deviation of a bootstrap estimate, $\hat{\theta}^*$, from the mean of the bootstrap estimates (typical deviation from the middle of the histogram of the bootstrap estimates). This is very closely related to the Monte Carlo estimate of $\widehat{\text{se}}(\hat{\theta})$, which is the sample standard deviation of the estimates of θ from the Monte Carlo samples.

Definition 8.8 (Bootstrap estimate of variance)**.** For a $k \times 1$ vector estimate $\hat{\theta}$, the bootstrap estimate of $\widehat{\text{var}}(\hat{\theta})$ is

$$\widehat{\text{var}}(\hat{\theta}) = \frac{1}{B-1}\sum_{j=1}^{B}(\hat{\theta}_j^* - \bar{\theta}^*)(\hat{\theta}_j^* - \bar{\theta}^*)'. \tag{8.18}$$

The bootstrap estimate of variance Equation 8.18 is the sample covariance matrix of the bootstrap estimates $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$. The bootstrap estimated standard errors for the elements of $\hat{\theta}$ are given by the square root of the diagonal elements of $\widehat{\text{var}}(\hat{\theta})$.

8.6.3 Bootstrap confidence intervals

In addition to providing standard error estimates, the bootstrap is commonly used to compute confidence intervals for scalar parameters. There are several ways of computing confidence intervals with the bootstrap and which confidence interval to use in practice depends on the characteristics of the distribution of the bootstrap estimates $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$.

If the bootstrap distribution appears to be normal then you can compute a 95% confidence interval for θ using two simple methods. The first method mimics the rule-of-thumb that is justified by the CLT to compute a 95% confidence interval but uses the bootstrap standard error estimate Equation 8.17:

$$\hat{\theta} \pm 2 \times \widehat{\text{se}}_{boot}(\hat{\theta}) = [\hat{\theta} - 2 \times \widehat{\text{se}}_{boot}(\hat{\theta}),\ \hat{\theta} + 2 \times \widehat{\text{se}}_{boot}(\hat{\theta})]. \tag{8.19}$$

The 95% confidence interval Equation 8.19 is called the *normal approximation* 95% confidence interval.

The second method directly uses the distribution of the bootstrap estimates $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$ to form a 95% confidence interval:

$$[\hat{q}_{.025}^*, \hat{q}_{.975}^*], \tag{8.20}$$

where $\hat{q}_{.025}^*$ and $\hat{q}_{.975}^*$ are the 2.5% and 97.5% empirical quantiles of $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$, respectively. By construction, 95% of the bootstrap estimates lie in the quantile-based interval Equation 8.20. This 95% confidence interval is called the *bootstrap percentile* 95% confidence interval.

The main advantage of Equation 8.20 over Equation 8.19 is *transformation invariance*. This means that if Equation 8.20 is a 95% confidence interval for θ and $f(\theta)$ is a monotonic function of θ then $[f(\hat{q}_{.025}^*), f(\hat{q}_{.975}^*)]$ is a 95% confidence interval for $f(\theta)$. The interval Equation 8.20 can also be asymmetric whereas Equation 8.19 is symmetric.

The bootstrap normal approximation and percentile 95% confidence intervals Equation 8.19 and Equation 8.20 will perform well (i.e., will have approximately correct coverage probability) if the bootstrap distribution looks normally distributed. However, if the bootstrap distribution is asymmetric then these 95% confidence intervals may have coverage probability different from 95%, especially if the asymmetry is substantial[6].

The basic shape of the bootstrap distribution can be visualized by the histogram of $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$, and the normality of the distribution can be visually evaluated using the normal QQ-plot of $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$. If the histogram is asymmetric and/or not bell-shaped or if the normal QQ-plot is not linear then normality is suspect and the confidence intervals Equation 8.20 and Equation 8.19 may be inaccurate.

If the bootstrap distribution is asymmetric then it is recommended to use Efron's *bias and skewness adjusted bootstrap percentile confidence* (BCA) interval. The details of how to compute the BCA interval are a bit complicated, and so are omitted, and the interested reader is referred to Chapter 10 of Hansen (2022) for a good explanation. Suffice it to say that the 95% BCA method adjusts the 2.5% and 97.5% quantiles of the bootstrap distribution for bias and skewness. The BCA confidence interval can be computed using the **boot** function `boot.ci()` with optional argument `type="bca"`.[7]

8.6.4 Performing the nonparametric bootstrap in R

The nonparametric bootstrap procedure is easy to perform in R. You can implement the procedure by "brute force" in very much the same way as you perform a Monte Carlo experiment. In this approach you program all parts of the bootstrapping procedure. Alternatively, you can use

[6] It is often incorrectly assumed that the bootstrap percentile 95% confidence interval is valid when the bootstrap distribution is asymmetric.

[7] The bootstrap percentile-t confidence interval is another bias and skewness adjusted confidence interval.

the R package **boot** which contains functions for automating certain parts of the bootstrapping procedure.

8.6.4.1 Brute force implementation

In the brute force implementation you program all parts of the bootstrap procedure. This typically involves four steps:

8.7 Brute force nonparametric bootstrap procedure

1. Sample with replacement from the original data using the R function `sample()` to create $\{r_1^*, r_2^*, \ldots, r_T^*\}$. Do this B times.
2. Compute B values of the statistic of interest $\hat{\theta}$ from each bootstrap sample giving $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$. Use $B = 1,000$ for routine applications of the bootstrap. To minimize simulation noise it may be required to have $B = 10,000$ or higher.
3. Compute the bootstrap bias and SE values from $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$ using the R functions `mean()` and `sd()`, respectively.
4. Compute the histogram and normal QQ-plot of $\{\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*\}$ using the R functions `hist()` and `qqnorm()`, respectively, to see if the bootstrap distribution looks normal.

Typically steps 1 and 2 are performed inside a `for` loop or within the R function `apply()`.

In step 1, sampling with replacement for the original data is performed with the R function `sample()`. To illustrate, consider sampling with replacement from the 5×1 return vector `r` created using:

```
r = c(0.1,0.05,-0.02,0.03,-0.04)
r
#> [1]  0.10  0.05 -0.02  0.03 -0.04
```

Recall, in R you can extract elements from a vector by subsetting based on their location, or index, in the vector. Since the vector `r` has five elements, its index is the integer vector `1:5`. Then to extract the first, second and fifth elements of `r` use the index vector `c(1, 2, 5)`:

```
r[c(1, 2, 5)]
#> [1]  0.10  0.05 -0.04
```

Using this idea, you can extract a random sample (of any given size) with replacement from `r` by creating a random sample with replacement of the integers $\{1, 2, \ldots, 5\}$ and using this set of integers to extract the sample from `r`. The R fucntion `sample()` can be used to do this process. When you pass a positive integer value `n` to `sample()`, with the optional argument `replace=TRUE`, it returns

a random sample from the set of integers from 1 to n. For example, to create a random sample with replacement of size 5 from the integers$\{1, 2, \ldots, 5\}$ use:[8]

```
set.seed(123)
idx = sample(5, replace=TRUE)
idx
#> [1] 3 3 2 2 3
```

You can then get a random sample with replacement from the vector `r` by subsetting using the index vector `idx`:

```
r[idx]
#> [1] -0.02 -0.02  0.05  0.05 -0.02
```

This two step process is automated when you pass a vector of observations to `sample()`:

```
set.seed(123)
sample(r, replace=TRUE)
#> [1] -0.02 -0.02  0.05  0.05 -0.02
```

Example 8.16 (Brute force nonparametric bootstrap of GWN model in R)**.** Consider using the nonparametric bootstrap to compute estimates of the bias and standard error for $\hat{\mu}$ in the GWN model for Microsoft. The R code for the brute force "for loop" to implement the nonparametric bootstrap is:

```
B = 1000
muhat.boot = rep(0, B)
n.obs = nrow(msftRetS)
set.seed(123)
for (i in 1:B) {
   boot.data = sample(msftRetS, n.obs, replace=TRUE)
   muhat.boot[i] = mean(boot.data)
}
```

The bootstrap bias estimate is:

```
mean(muhat.boot) - gwn_fit_msft_s$muhat
#> [1] 0.000483
```

which is very close to zero and confirms that $\hat{\mu}$ is unbiased. The bootstrap standard error estimate is

[8] I use `set.seed(123)` to initialize the random number generator in R and allow for reproducible results.

```
sd(muhat.boot)
#> [1] 0.00796
```

and is equal (to three decimals) to the analytic standard error estimate computed earlier. This confirms that the nonparametric bootstrap accurately computes $\widehat{\text{se}}(\hat{\mu})$. Figure 8.2 shows the histogram and normal QQ-plot of the bootstrap estimates created with:

```
par(mfrow=c(1,2))
hist(muhat.boot, col="cornflowerblue")
abline(v=gwn_fit_msft_s$muhat, col="white", lwd=2)
qqnorm(muhat.boot, col="cornflowerblue")
qqline(muhat.boot)
```

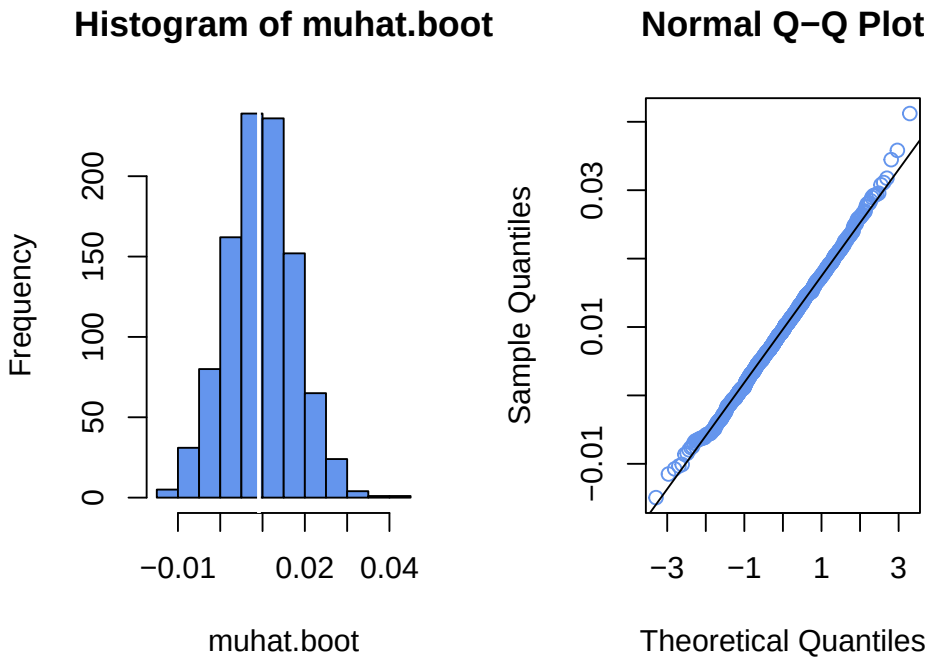


Figure 8.2: Histogram and normal QQ-plot of bootstrap estimates of μ for Microsoft.

The distribution of the bootstrap estimates looks normal. As a result, you can compute use the normal approximation 95% confidence interval Equation 8.19 for μ:

```
se.boot = sd(muhat.boot)
lower = gwn_fit_msft_s$muhat + se.boot
upper = gwn_fit_msft_s$muhat - se.boot
```

```
ans = cbind(gwn_fit_msft_s$muhat, se.boot, lower, upper)
colnames(ans)=c("Estimate", "Std Error", "2.5%", "97.5%")
rownames(ans) = "mu"
ans
#>    Estimate Std Error   2.5%  97.5%
#> mu  0.00915   0.00796 0.0171 0.0012
```

■

It is important to keep in mind that the bootstrap resampling algorithm draws *random samples* from the observed data. In the example above, the random number generator in R was initially set with `set.seed(123)`. If the seed is set to a different integer, say 345, then the $B = 1,000$ bootstrap samples will be different and you will get different numerical results for the bootstrap bias and estimated standard error. How different these results will be depends on number of bootstrap samples B and the statistic $\hat{\theta}$ being bootstrapped. Generally, the smaller is B the more variable the bootstraps results will be. It is good practice to repeat the bootstrap analysis with several different random number seeds to see how variable the results are. If the results vary considerably for different random number seeds then it is advised to increase the number of bootstrap simulations B. It is common to use $B = 1,000$ for routine calculations, and $B = 10,000$ for final results.

8.7.0.1 R package boot

The R package **boot** implements a variety of bootstrapping techniques including the basic non-parametric bootstrap described above. The **boot** package was written to accompany the textbook *Bootstrap Methods and Their Application* (Davison and Hinkley 1997). The two main functions in **boot** are `boot()` and `boot.ci()`, respectively. The `boot()` function implements the bootstrap for a statistic computed from a user-supplied function. The `boot.ci()` function computes bootstrap confidence intervals given the output from the `boot()` function.

The arguments to `boot()` are:

```
library(boot)
args(boot)
#> function (data, statistic, R, sim = "ordinary", stype = c("i",
#>     "f", "w"), strata = rep(1, n), L = NULL, m = 0, weights = NULL,
#>     ran.gen = function(d, p) d, mle = NULL, simple = FALSE, ...,
#>     parallel = c("no", "multicore", "snow"), ncpus = getOption("boot.ncpus",
#>         1L), cl = NULL)
#> NULL
```

where `data` is a data object (typically a vector or matrix but can be a time series object too), `statistic` is a use-specified function to compute the statistic of interest, and `R` is the number of bootstrap replications. The remaining arguments are not important for the basic non-parametric bootstrap. The function assigned to the argument `statistic` has to be written in a specific form, which is illustrated in the next example. The `boot()` function returns an object of class `boot` for which there are `print` and `plot` methods.

The arguments to `boot.ci()` are

```
args(boot.ci)
#> function (boot.out, conf = 0.95, type = "all", index = 1L:min(2L,
#>     length(boot.out$t0)), var.t0 = NULL, var.t = NULL, t0 = NULL,
#>     t = NULL, L = NULL, h = function(t) t, hdot = function(t) rep(1,
#>         length(t)), hinv = function(t) t, ...)
#> NULL
```

where `boot.out` is an object of class `boot`, `conf` specifies the confidence level, and `type` is a subset from `c("norm", "basic", "stud", "perc", "bca")` indicating the type of confidence interval to compute. The choices "norm", "perc", and "bca" compute the normal confidence interval Equation 8.19, the percentile confidence interval Equation 8.20, and the BCA confidence interval, respectively. The remaining arguments are not important for the computation of these bootstrap confidence intervals.

Example 8.17 (Nonparametric bootstrap of GWN model using the R package **boot**)**.** To use the `boot()` function to implement the bootstrap for $\hat{\mu}$, a function must be specified to compute $\hat{\mu}$ for each bootstrap sample. The function must have two arguments: `x` and `idx`. Here, `x` represents the original data and `idx` represents the random integer index (created internally by `boot()`) to subset `x` for each bootstrap sample. For example, a function to be passed to `boot()` for $\hat{\mu}$ is

```
mean.boot = function(x, idx) {
# arguments:
# x      data to be resampled
# idx    vector of scrambled indices created by boot() function
# value:
# ans   mean value computed using resampled data
    ans = mean(x[idx])
    ans
}
```

To implement the nonparametric bootstrap for $\hat{\mu}$ with $B = 1,000$ samples use:

```
set.seed(123)
muhat.boot = boot(msftRetS, statistic = mean.boot, R=1000)
class(muhat.boot)
#> [1] "boot"
names(muhat.boot)
#>  [1] "t0"        "t"         "R"         "data"      "seed"      "statistic"
#>  [7] "sim"       "call"      "stype"     "strata"    "weights"
```

The returned object `muhat.boot` is of class `boot`. The component `t0` is the sample estimate $\hat{\mu}$, and the component t is a $1,000 \times 1$ matrix containing the bootstrap estimates $\{\hat{\mu}^*_1, \ldots, \hat{\mu}^*_{1,000}\}$. The `print` method shows the sample estimate, the bootstrap bias and the bootstrap standard error:

```
muhat.boot
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot(data = msftRetS, statistic = mean.boot, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>     original   bias    std. error
#> t1*  0.00915 0.000483     0.00773
```

These statistics can be computed directly from the components of `muhat.boot`

```
c(muhat.boot$t0,mean(muhat.boot$t) - muhat.boot$t0, sd(muhat.boot$t))
#> [1] 0.009153 0.000483 0.007731
```

A histogram and nornal QQ-plot of the bootstrap values, shown in Figure 8.3, can be created using the `plot` method:

```
plot(muhat.boot)
```

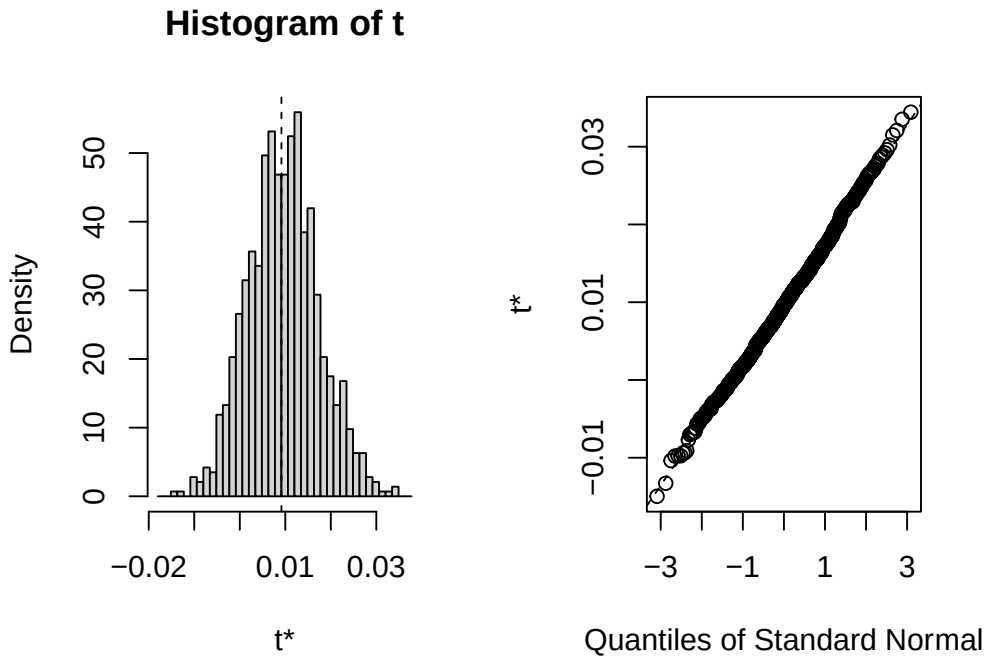


Figure 8.3: plot method for objects of class `boot`

Normal, percentile, and bias-adjusted (bca) percentile 95% confidence intervals can be computed using:

```
boot.ci(muhat.boot, conf=0.95, type=c("norm", "perc", "bca"))
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot.ci(boot.out = muhat.boot, conf = 0.95, type = c("norm",
#>     "perc", "bca"))
#>
#> Intervals :
#> Level      Normal             Percentile            BCa
#> 95%   (-0.0065,  0.0238 )   (-0.0048,  0.0255 )   (-0.0050,  0.0249 )
#> Calculations and Intervals on Original Scale
```

Because the bootstrap distribution looks normal, the normal, percentile, and bca confidence intervals are very similar.

The GWN model estimate $\hat{\sigma}$ can be "bootstrapped" in a similar fashion. First, you write a function to compute $\hat{\sigma}$ for each bootstrap sample:

```
sd.boot = function(x, idx) {
    ans = sd(x[idx])
    ans
}
```

Then you call `boot()` with `statistic=sd.boot`:

```
set.seed(123)
sigmahat.boot = boot(msftRetS, statistic = sd.boot, R=1000)
sigmahat.boot
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot(data = msftRetS, statistic = sd.boot, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>     original    bias    std. error
#> t1*    0.101 -0.000806     0.00764
```

The bootstrap distribution, shown in Figure 8.4, looks a bit non-normal. The three 95% confidences intervals for σ are computed using `boot.ci()` as follows:

```
boot.ci(sigmahat.boot, conf=0.95, type=c("norm", "perc", "bca"))
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot.ci(boot.out = sigmahat.boot, conf = 0.95, type = c("norm",
#>     "perc", "bca"))
#>
#> Intervals :
#> Level      Normal             Percentile            BCa
#> 95%   ( 0.0873,  0.1173 )   ( 0.0859,  0.1160 )   ( 0.0891,  0.1201 )
#> Calculations and Intervals on Original Scale
#> Some BCa intervals may be unstable
```

You see that all of these intervals are quite similar. ■

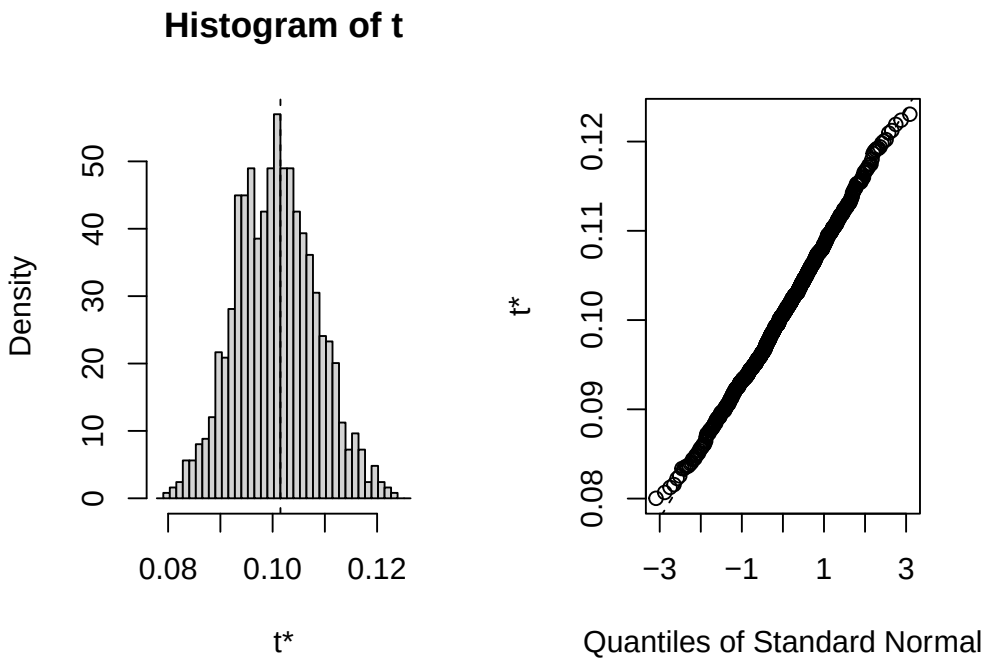


Figure 8.4: Bootstrap distribution for $\hat{\sigma}$

Example 8.18 (Nonparametric bootstrap for example functions)**.** The real power of the bootstrap procedure comes when you apply it to plug-in estimates of functions like Equation 8.1 - Equation 8.3 for which there are no easy analytic formulas for estimating bias and standard errors. In this situation, the bootstrap procedure easily computes numerical estimates of the bias, standard error, and 95% confidence interval. All you need is an R function for computing the plug-in function estimate in a form suitable for `boot()`. My R functions for the theree example functions are:

```
f1.boot = function(x, idx, alpha=0.05) {
  q = mean(x[idx]) + sd(x[idx])*qnorm(alpha)
  q
}

f2.boot = function(x, idx, alpha=0.05, w0=100000) {
  q = mean(x[idx]) + sd(x[idx])*qnorm(alpha)
  VaR = -w0*q
  VaR
}

f3.boot = function(x, idx, alpha=0.05, w0=100000) {
```

```
    q = mean(x[idx]) + sd(x[idx])*qnorm(alpha)
    VaR = -(exp(q) - 1)*w0
    VaR
}
```

The functions f_1, f_2, and f_3 have the additional argument `alpha` which specifies the tail probability for the quantile, and the functions f_2, and f_3 have the additional argument `w0` for the initial wealth invested.

To compute the nonparametric bootstrap for f_1 with $\alpha = 0.05$ use:

```
set.seed(123)
f1.bs = boot(msftRetS, statistic = f1.boot, R=1000)
f1.bs
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot(data = msftRetS, statistic = f1.boot, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>     original  bias    std. error
#> t1*   -0.158 0.00181      0.0135
```

Here, the bootstrap estimate of bias is small and the bootstrap estimated standard error is similar to the delta method and jackknife standard errors.

The bootstrap distribution looks normal, and the three methods for computing 95% confidence intervals are essentially the same:

```
boot.ci(f1.bs, type=c("norm", "perc", "bca"))
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot.ci(boot.out = f1.bs, type = c("norm", "perc", "bca"))
#>
#> Intervals :
#> Level      Normal             Percentile            BCa
```

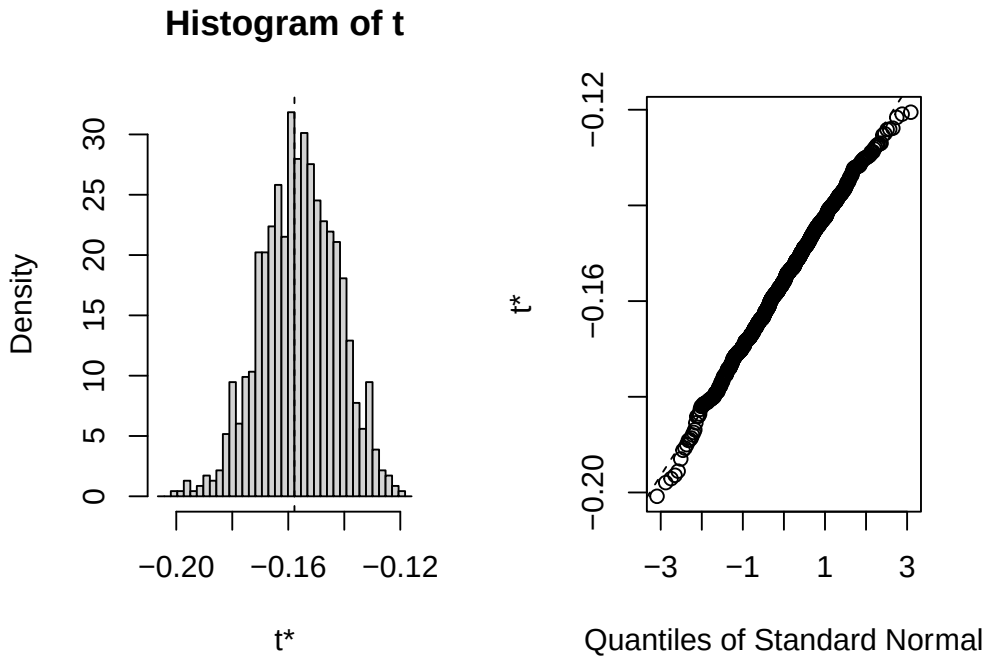


Figure 8.5: Bootstrap distribution for $\hat{f}_1$

```
#> 95%   (-0.186, -0.133 )   (-0.181, -0.130 )   (-0.191, -0.135 )
#> Calculations and Intervals on Original Scale
#> Some BCa intervals may be unstable
```

To compute the nonparametric bootstrap for f_2 with $\alpha = 0.05$ and $W_0 = \$100,000$ use:

```
set.seed(123)
f2.bs = boot(msftRetS, statistic = f2.boot, R=1000)
f2.bs
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot(data = msftRetS, statistic = f2.boot, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>     original  bias    std. error
```

```
#> t1*    15780    -181        1349
```

The bootstrap estimate of bias is small and the bootstrap estimated standard error is similar to the delta method and jackknife standard errors.

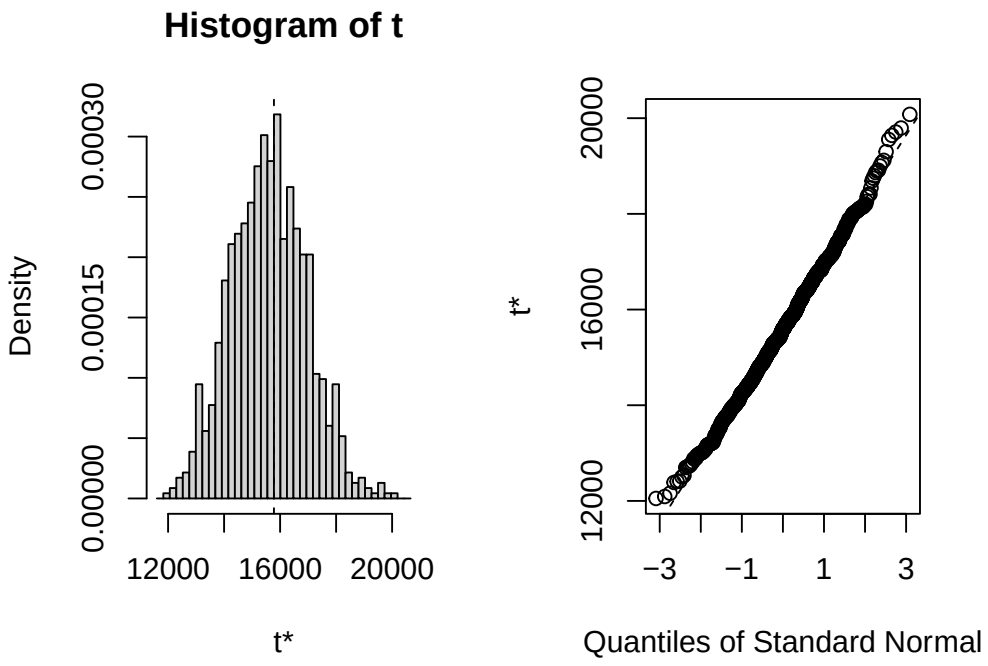


Figure 8.6: Bootstrap distribution for $\hat{f}_2$

The bootstrap distribution looks normal, and the three bootstrap 95% confidence intervals are all very similar:

```
boot.ci(f2.bs, type=c("norm", "perc", "bca"))
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot.ci(boot.out = f2.bs, type = c("norm", "perc", "bca"))
#>
#> Intervals :
#> Level      Normal             Percentile            BCa
#> 95%   (13317, 18603 )   (13017, 18150 )   (13529, 19137 )
#> Calculations and Intervals on Original Scale
#> Some BCa intervals may be unstable
```

To compute the nonparametric bootstrap for f_3 with $\alpha = 0.05$ and $W_0 = \$100,000$ use:

```
set.seed(123)
f3.bs = boot(msftRetC, statistic = f3.boot, R=1000)
f3.bs
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot(data = msftRetC, statistic = f3.boot, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>     original  bias    std. error
#> t1*    14846    -189        1319
```

The bias is small and the bootstrap estimated standard error is larger than the delta method standard error and is closer to the jackknife standard error.

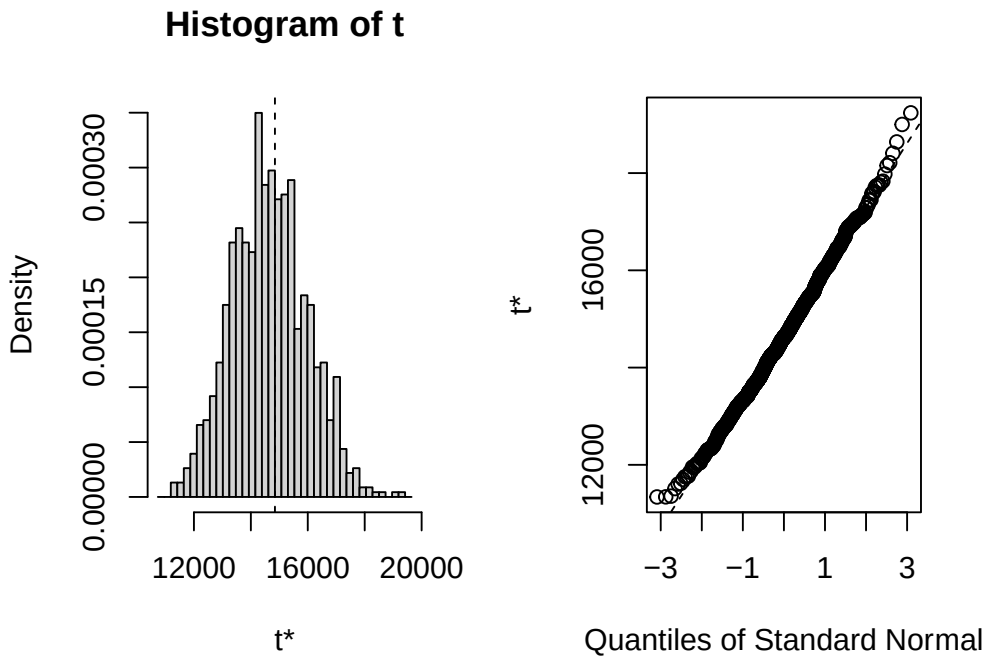


Figure 8.7: Bootstrap distribution for $\hat{f}_3$

The bootstrap distribution looks a little asymmetric and the BCA confidence interval is a bit

different from the normal and percentile intervals:

```
boot.ci(f3.bs, type=c("norm", "perc", "bca"))
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot.ci(boot.out = f3.bs, type = c("norm", "perc", "bca"))
#>
#> Intervals :
#> Level      Normal             Percentile            BCa
#> 95%   (12450, 17619 )   (12164, 17178 )   (12809, 18412 )
#> Calculations and Intervals on Original Scale
#> Some BCa intervals may be unstable
```

■

8.7.1 Pros and cons of the nonparametric bootstrap

The nonparametric bootstrap is extremely useful and powerful statistical technique. The main advantages (pros) are:

1. General procedure to estimate bias and standard errors, and to compute confidence intervals, that does not rely on asymptotic distributions.
2. Can be used for scalar and vector estimates.
3. Under general conditions, bootstrap distributions of estimators converge to the asymptotic distribution of the estimators.
4. In certain situations, bootstrap results can be more accurate than analytic results computed from asymptotic distributions.

There are some things to watch out for when using the bootstrap (cons):

1. It be computational intensive but this is really not much of an issue with today's fast computers.
2. With small B, bootstrap results can vary substantially across simulations with different random number seeds.
3. There are situations where the bootstrap does not work. A leading case is when the bootstrap is applied to a function that can be become unbounded (e.g. a ratio of means when the

denominator mean is close to zero). In these cases, a trimmed version of the bootstrap is recommended. See (**Hansen-2011?**) for a discussion of this issue.

4. You have to be careful applying the bootstrap to time series data. If there is serial correlation in the data then the nonparametric bootstrap may give poor results. There are versions of the bootstrap (e.g., the block bootstrap) that work with serially correlated data. For example, the *tseries* function `tsbootstrap()` is designed to to work with serially correlated time series data.

8.8 Using Monte Carlo Simulation to Evaluate Delta Method, Jackknife and Bootstrap Standard Errors

You have studied three ways of computing standard errors for functions of GWN model parameter: the delta method, the jackknife, and the bootstrap. All three ways give similar results for the example functions. In this section, I show how to use Monte Carlo simulation to evaluate the three methods when the GWN model is true. Because Monte Carlo simulation approximates the true finite sample distribution of any estimator, it can give the best numerical standard error with which you can compare to the delta method, jackknife and bootstrap standard errors.

You can evaluate the statistical properties of $f_j(\hat{\theta})$, for $j = 1, \dots, 3$, by Monte Carlo simulation in the same way that you evaluated the statistical properties of $\hat{\mu}$ in Chapter 7. You use the simulation model Equation 7.41 and $N = 1000$ simulated samples of size $T = 100$ to compute the estimates $\{\left(f_j(\hat{\theta})\right)^1, \dots, \left(f_j(\hat{\theta})\right)^{1000}\}$. The true values for $f_j(\theta)$ are:

$$
\begin{aligned}
f_1(\theta) &= 0.05 + (0.10) \times (-1.645) = -0.114, \\
f_2(\theta) &= -\$100,000 \times f_1(\theta) = \$11,449, \\
f_3(\theta) &= \$100,000 \times (\exp(f_1(\theta)) - 1) = \$10,818.
\end{aligned}
$$

The R code for the Monte Carlo simulation loop is:

```
mu = 0.05
sigma = 0.10
W0 = 100000
f1 = mu + sigma*qnorm(0.05)
f2 = -W0*f1
f3 = -W0*(exp(f1) - 1)
n.obs = 172
n.sim = 1000
set.seed(111)
```

```
sim.f = matrix(0,n.sim,3)
dm.f = matrix(0, n.sim, 3)
jack.f = matrix(0, n.sim, 3)
boot.f = matrix(0, n.sim, 3)
boot.bias = matrix(0, n.sim, 3)
colnames(sim.f) = colnames(dm.f) = colnames(jack.f) = c("f1", "f2","f3")
colnames(boot.f) = colnames(boot.bias) = c("f1", "f2","f3")

for (sim in 1:n.sim) {
sim.ret = rnorm(n.obs,mean=mu,sd=sigma)
muhat = mean(sim.ret)
sigmahat = sd(sim.ret)
sim.f[sim, "f1"] = muhat + sigmahat*qnorm(0.05)
sim.f[sim, "f2"] = -W0*sim.f[sim, "f1"]
sim.f[sim, "f3"] = -W0*(exp(sim.f[sim, "f1"]) - 1)
# delta method se
dm.f[sim, "f1"] = (sigmahat/sqrt(n.obs))*sqrt(1 + 0.5*qnorm(0.05)^2)
dm.f[sim, "f2"] = W0*dm.f[sim, "f1"]
dm.f[sim, "f3"] = W0*exp(sim.f[sim, "f1"])*dm.f[sim, "f1"]
# jackknife se
f1.jack = jackknife(1:n.obs, theta=theta.f1, as.matrix(sim.ret))
jack.f[sim, "f1"] = f1.jack$jack.se
f2.jack = jackknife(1:n.obs, theta=theta.f2, as.matrix(sim.ret), W0)
jack.f[sim, "f2"] = f2.jack$jack.se
f3.jack = jackknife(1:n.obs, theta=theta.f3, as.matrix(sim.ret), W0)
jack.f[sim, "f3"] = f3.jack$jack.se
# bootstrap se
f1.bs = boot(sim.ret, statistic = f1.boot, R=999)
boot.f[sim, "f1"] = sd(f1.bs$t)
f2.bs = boot(sim.ret, statistic = f2.boot, R=999)
boot.f[sim, "f2"] = sd(f2.bs$t)
f3.bs = boot(sim.ret, statistic = f3.boot, R=999)
boot.f[sim, "f3"] = sd(f3.bs$t)
# bootstrap bias
boot.bias[sim, "f1"] = summary(f1.bs)$bootBias
boot.bias[sim, "f2"] = summary(f2.bs)$bootBias
boot.bias[sim, "f3"] = summary(f3.bs)$bootBias
}
```

The histograms of $f_j(\hat{\theta})$ values are displayed in Figure 8.8. The dotted white lines indicate the true

values, $f_j(\theta)$.

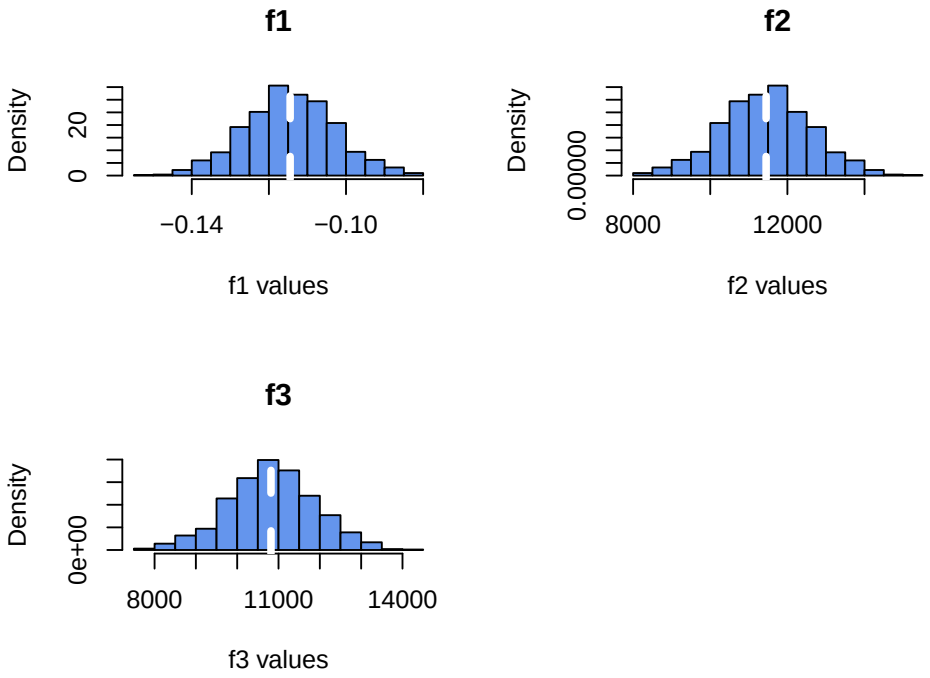


Figure 8.8: Histograms of $\hat{f}_j$ (for $j = 1, \ldots, 4$) computed from $N = 1000$ Monte Carlo samples from the GWN model.

The histograms for the $f_j(\hat{\theta})$ values are bell-shaped for $j = 1, 2, 3$ and the histogram for $j = 4$ is slightly right skewed. All of the histograms are centered very close to the respective true values. Recall, the standard deviation of the Monte Carlo estimates (spread of the histograms about true value) approximates the true standard error of the plug-in estimates.

The Monte Carlo estimates of bias are:

```
mc.bias = colMeans(sim.f) - c(f1, f2, f3)
round(mc.bias, digits=3)
#>    f1    f2    f3
#>  0.00 10.79  3.52
```

which confirm that f_j are essentially unbiased estimators. It is often useful to also look at the bias relative to the true value to eliminate units:

```
mc.bias/c(f1, f2, f3)
#>       f1       f2       f3
#> 0.000943 0.000943 0.000326
```

In relative terms, the largest bias is 0.09%, which occurs for the estimates of f_1 and f_2. This is extremely small.

The average bootstrap estimate of bias is:

```
boot.bias.vals = colMeans(boot.bias)
round(boot.bias.vals, digits=3)
#>      f1      f2      f3
#>   0.001 -69.435 -68.570
```

The average relative bias is

```
boot.bias.vals/c(f1, f2, f3)
#>       f1       f2       f3
#> -0.00619 -0.00606 -0.00634
```

The bootstrap bias values are very close to the Monte Carlo estimates of bias which indicates that the bootstrap estimate of bias is very accurate in this context.

The sample standard deviations of the Monte Carlo estimates approximate the true standard errors of the estimated functions:

```
mc.se = apply(sim.f, 2, sd)
round(mc.se, digits=3)
#>       f1       f2       f3
#>    0.012 1170.205 1043.851
```

The sample means of the delta method, jackknife, and bootstrap standard errors give the average performance of these estimated standard errors. It is informative to compare these average standard errors to the Monte Carlo standard errors:

```
dm.se = colMeans(dm.f)
jack.se = colMeans(jack.f)
boot.se = colMeans(boot.f)
ans = rbind(dm.se/mc.se, jack.se/mc.se, boot.se/mc.se)
rownames(ans) = c("Delta Method", "Jackknife", "Bootstrap")
ans
#>                 f1    f2    f3
#> Delta Method 0.998 0.998 0.998
#> Jackknife    0.997 0.997 0.997
#> Bootstrap    0.987 0.988 0.987
```

In the table above, a value of 1 means the estimated standard error is the same as the Monte Carlo standard error (on average), and a value greater than 1 means that the estimated standard error is larger than the Monte Carlo standard error (on average). Here, you see that all of the estimated standard errors are very close to the Monte Carlo standard errors, and overall the bootstrap estimated standard errors are the closest.

It is also useful to look at the standard deviations of the estimated standard errors over the Monte Carlo simulations. This tells you how variable the estimated standard errors are across different simulations:

```
dm.sd = apply(dm.f,2,sd)
jack.sd = apply(jack.f,2,sd)
boot.sd = apply(boot.f,2,sd)
ans = rbind(dm.sd, jack.sd, boot.sd)
rownames(ans) = c("Delta Method", "Jackknife", "Bootstrap")
ans
#>                     f1    f2   f3
#> Delta Method 0.000637  63.7 48.1
#> Jackknife    0.001148 114.8 98.1
#> Bootstrap    0.001145 115.1 96.8
```

Here, you see that, across the simulations, the delta method is the least variable and the jackknife and the bootstrap have similar variability that is about twice as large as the delta method variability. The variability of the bootstrap can be reduced by increasing the number of bootstrap simulations (here it is 1000).

Overall, the Monte Carlo experiment confirms that the delta method, jackknife and bootstrap all produce accurate standard error estimates for the example plug-in functions.

8.9 Further Reading: Delta Method and Resampling

Chernick and LaBudde (2011) give a comprehensive overview of the bootstrap with applications in R.

8.10 Exercises

Exercise 8.1 (Delta method)**.**

- TBD

Exercise 8.2 (Gradients for example functions)**.**

- TBD

Exercise 8.3 (Standard error of Sharpe ratio)**.**

- Look at determinants of se of Sharpe ratio. See discussion from Lo's paper

Exercise 8.4 (Correlation between estimated Sharpe ratios)**.**

- Look at the formula for the covariance between two Sharpe ratios

Exercise 8.5 (Bootstrap percentile confidence interval)**.** Illustrate the transformation preserving property of percentile interval using the example of computing the confidence interval for σ based on the percentile interval for σ^2. Show that asymptotic confidence interval is not transformation preserving.

Exercise 8.6 (Bootstrap variability)**.** Repeat the bootstrap simulations from Example 8.16 using 5 different random number seeds: 124, 125, 126, 127 and 128. Make a table to compare the bootstrap estimated standard errors for the example functions across the 5 bootstrap simulations. Briefly comment on your results.

Exercise 8.7 (Failure of the bootstrap)**.**

- Run the bootstrap for calculating a standard error for μ_i/μ_j.

Exercise 8.8 (Bootstrap accuracy)**.** Repeat the Monte Carlo simulation in Section 8.8 using 1,000 simulated samples and 10,000 bootstrap simulations for the bootstrap calculations. That is, set R=10000 in the call to `boot()` inside the Monte Carlo simulation loop.

1. How does the increase in simulations impact the bootstrap bias estimates?
2. How does the increase in simulations impact the bootstrap standard error estimates?

9 Hypothesis Testing in the GWN Model

Updated: September 7, 2026

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The core of statistical analysis involves estimation and inference. In the previous chapters I considered estimation of the GWN model parameters, and the computation of standard error estimates and confidence intervals. In this chapter I discuss issues of statistical inference within the GWN model. Statistical inference is about asking questions about a model and using statistical procedures to obtain answers to these questions with a given level of confidence. For example, you might find that our estimate of the monthly mean of an asset return is positive but due to large estimation error you might ask the question: is the true monthly mean return positive? As another example, you might find that a normal qq-plot of returns shows some evidence of non-normality and you ask the question: are returns normally distributed? As a third example, you might find that rolling estimates of the mean and volatility show some evidence of non-constant behavior and you might ask the question: are returns covariance stationary? Statistical hypothesis testing gives us a rigorous framework for answering such questions with a given level of confidence.

The R packages used in this chapter are **introcompfinr**, **PerformanceAnalytics**, **tseries**, and **zoo**. Make sure this packages are installed and loaded before replicating the examples.

1. Hypothesis Testing in the GWN Model
    a. Specification tests
        i. Normal distribution
            A. Tests for skewness and kurtosis
            B. JB test
            C. What to do if returns are not normally distributed?
        ii. No autocorrelation A. individual and joint tests
        iii. Constant parameters (covariance stationarity) A. Formal tests: testing same mean, volatility in sub-samples B. Informal diagnostics - standard error bands on rolling estimators
    b. Using Monte Carlo Simulation to Understand Hypothesis tests
        i. Size
        ii. Power

c. Using the bootstrap for hypothesis tests
    a. exploit duality between hypothesis tests and confidence intervals

2. Appendix: Distributions for test statistics
    a. Student's t distribution
    b. Chi-square distribution

9.1 Review of General Principles

Classical hypothesis testing is discussed in almost all introductory textbooks on statistics, and the reader is assumed be familiar with the basic concepts.[1] In this section I provide a brief and concise review of the general principles of classical hypothesis testing. In the following sections I apply these principles to answer questions about the GWN model parameters and assumptions.

9.1.1 Steps for hypothesis testing

The main steps for conducting a classical hypothesis test are as follows:

1. Specify the hypotheses to be tested:

$$H_0 : \text{null hypothesis vs. } H_1 : \text{alternative hypothesis.}$$

   The *null hypothesis* is the maintained hypothesis (what is assumed to be true) and is the hypothesis to be tested against the data. The *alternative hypothesis* specifies what is assumed to be true if the null hypothesis is false. This can be specific or vague depending on the context.

2. Specify the significance level of the test:

$$\alpha = \text{significance level} = \Pr(\text{Reject } H_0 | H_0 \text{ is true}).$$

   The *significance level* of a test specifies the probability of making a certain type of decision error: rejecting the null hypothesis when the null hypothesis is, in fact, true. In practice, the significance level α is chosen to be a small number like 0.01 or 0.05.

3. Construct a *test statistic*, S, from the observed data whose probability distribution is known when H_0 is true.

4. Use the test statistic S to evaluate the data evidence regarding the validity of H_0. Typically, if S is a big number then there is strong data evidence against H_0 and H_0 should be rejected; otherwise, there is not strong data evidence against H_0 and H_0 should not be rejected.

[1] In this book, I do not consider evaluating hypotheses from a Bayesian perspective.

- Decide to reject H_0 at the specified significance level α if the value of the test statistic S falls in the *rejection region* of the test. Usually, the rejection region for S is determined by a critical value cv_α such that:

$$\begin{aligned} S &> cv_\alpha \Rightarrow \text{reject } H_0, \\ S &\leq cv_\alpha \Rightarrow \text{do not reject } H_0. \end{aligned}$$

  Smaller values of α typically make cv_α larger and require more data evidence (i.e., larger S) to reject H_0.

- Alternatively, decide to reject H_0 at the specified significance level α if the p-value of the test statistic S is less than α. The *p-value* of the statistic S is defined as the significance level at which the test is just rejected.

9.1.2 Hypothesis tests and decisions

Hypothesis testing involves making a decision: reject H_0 or do not reject H_0. Notice that if the data evidence strongly favors H_0 then you say "I do not reject H_0" instead of "accept H_0". You can rarely know the truth with absolute certainty with a finite amount of data so it is more appropriate to say "do not reject H_0" than to say "accept H_0".[2]

Table 9.1: Decision table for hypothesis testing

		Reality	
		H_0 **is true**	H_0 **is false**
Decision	**Reject** H_0	Type I error	No error
	Do not reject H_0	No error	Type II error

Table 9.1 shows the 2×2 decision table associated with hypothesis testing. In reality there are two states of the world: either H_0 is true or it is not.[3] There are two decisions to make: reject H_0 or don't reject H_0. If the decision corresponds with reality then the correct decision is made and there is no decision error. This occurs in the off-diagonal elements of the table. If the decision and reality disagree then there is a decision error. These errors are in the diagonal elements of the table. Type I error results when the decision to reject H_0 occurs when, in fact, H_0 is true. Type II error happens when the decision not to reject H_0 occurs when H_0 is false. To put these types

[2] As an extreme example, consider testing H_0 : "all swans are white". To logically accept this hypothesis you would have to show that every swan that ever existed is white. In contrast, if you have a data set that only contains white swans then saying you cannot reject the null hypothesis that all swans are white is consistent with your data set.

[3] It is important to emphasize that there is a "true state of the world" in this context. That is, the null hypothesis H_0 is either true or it is not. As a result, $\Pr\{H_0 \text{ is true}\} = 1$ or 0 depending on whether H_0 is true or not.

of errors in context, consider a jury in a capital murder trial in the US.[4] In the US, a defendant is considered innocent until proven guilty. Here, the null hypothesis to be tested is:

$$H_0 : \text{defendant is innocent.}$$

The alternative is:

$$H_1 : \text{defendant is guilty.}$$

There is a true state of the world here: the defendant is either innocent or guilty. The jury must decide what the state of the world is based on evidence presented to them. The decision table for the jury is shown in Table 9.2. Type I error occurs when the jury convicts an innocent defendant and puts that person on "death row". Clearly, this is a terrible mistake. To avoid this type of mistake the jury should have a very small (close to zero) significance level for evaluating evidence so that Type I error very rarely occurs. This is why typical jury instructions are to convict only if evidence of guilt is presented beyond reasonable doubt. Type II error happens when the jury does not convict (acquits) a guilty defendant and sets that person free. This is also a terrible mistake, but perhaps it is not as terrible as convicting the innocent person. Notice that there is a conflict between Type I error and Type II error. In order to completely eliminate Type I error, you can never reject H_0. That is, to avoid ever sending innocent people to "death row" you must never convict anyone. But if you never convict anyone then you never convict guilty people either and this increases the occurrences of Type II errors.

9.1.3 Significance level and power

The *significance level* of a test is the probability of Type I error:

$$\alpha = \Pr(\text{Type I error}) = \Pr(\text{Reject } H_0 | H_0 \text{ is true}).$$

The *power* of a test is one minus the probability of Type II error:

$$\pi = 1 - \Pr(\text{Type II error}) = \Pr(\text{Reject } H_0 | H_0 \text{ is false}).$$

In classical hypothesis tests, the goal is to construct a test that has a small significance level (α close to zero) that you can control and has high power (π close to one). That is, a good test is one that has a small and controllable probability of rejecting the null when it is true and, at the same time, has a very high probability of rejecting the null when it is false. In general, an *optimal test* is one that has the highest possible power for a given significance level. In the jury example, an optimal test would be the one in which the jury has the highest probability of convicting a guilty defendant while at the same time has a very low probability of convicting an innocent defendant. A difficult problem indeed!

[4] A capital murder trial is one in which the defendant is eligible for the death penalty.

Table 9.2: Decision table for jury on capital murder trial

		Reality	
		Defendant is innocent	**Defendant is guilty**
Decision	**Convict**	Type I error	No error
	Acquit	No error	Type II error

9.2 Hypothesis Testing in the GWN Model

In this section, I discuss hypothesis testing in the context of the GWN model for asset returns:

$$\begin{aligned} R_{it} &= \mu_i + \varepsilon_{it},\ i = 1 \dots, N;\ t = 1, \dots, T \\ \{\varepsilon_{it}\}_{t=1}^{T} &\sim \text{GWN}(0, \sigma_i^2), \\ \text{cov}(\varepsilon_{it}, \varepsilon_{js}) &= \begin{cases} \sigma_{ij} & t = s \\ 0 & t \neq s \end{cases}. \end{aligned} \tag{9.1}$$

In Equation 9.1, R_{it} is the monthly return (continuously compounded or simple) on asset i in month t. Define $\mathbf{R}_t = (R_{1t}, \dots, R_{Nt})'$. The GWN model in matrix form is:

$$\begin{aligned} \mathbf{R}_t &= \mu + \varepsilon_t, \\ \varepsilon_t &\sim \text{iid } N(\mathbf{0}, \Sigma). \end{aligned} \tag{9.2}$$

I consider two types of hypothesis tests: (1) tests for model coefficients or functions of model coefficients; and (2) tests for model assumptions.[5] These tests can be specified for the GWN model for a single asset using Equation 9.1, or they can be specified for the GWN model for all assets using Equation 9.2.

9.2.1 Coefficient tests

Here, I assume that the GWM is the true model for our observed return data and I are interested in asking questions about the parameters of the model. The most common tests for model coefficients and functions of model coefficients for a single asset are tests for a specific value. Some typical hypotheses to be tested are:

[5] Tests for model assumptions are sometimes called model specification tests.

$$
\begin{aligned}
&H_0 : \mu_i = \mu_i^0 \text{ vs. } H_1 : \mu_i \neq \mu_i^0, \\
&H_0 : \sigma_i = \sigma_i^0 \text{ vs. } H_1 : \sigma_i \neq \sigma_i^0, \\
&H_0 : \rho_{ij} = \rho_{ij}^0 \text{ vs. } H_1 : \rho_{ij} \neq \rho_{ij}^0, \\
&H_0 : \text{SR}_i = \text{SR}_i^0 \text{ vs. } H_1 : \text{SR}_i \neq \text{SR}_i^0,
\end{aligned}
$$

where $\text{SR}_i = \frac{\mu_i - r_f}{\sigma_i}$ is the Sharpe ratio of asset i and r_f is the risk-free rate of interest. A hypothesis about the Sharpe ratio is a hypothesis about the ratio of two GWN model parameters. In the above hypotheses, the alternative hypotheses are *two-sided*. That is, under H_1 the model parameters are either larger or smaller than what is specified under H_0.

Also common are tests for specific sign. For example:

$$
\begin{aligned}
&H_0 : \mu_i = 0 \text{ vs. } H_1 : \mu_i > 0 \text{ or } H_1 : \mu_i < 0, \\
&H_0 : \rho_{ij} = 0 \text{ vs. } H_1 : \rho_{ij} > 0 \text{ or } H_1 : \rho_{ij} < 0, \\
&H_0 : \text{SR}_i = 0 \text{ vs. } H_1 : \text{SR}_i > 0 \text{ or } H_1 : \text{SR}_i < 0,
\end{aligned}
$$

Here, the alternative hypotheses are *one-sided*. That is, under H_1 the model parameter is either greater than zero or less than zero.

Sometimes you are interested in testing hypotheses about the relationship between model coefficients for two or more assets. For example:

$$
\begin{aligned}
&H_0 : \mu_1 = \mu_2 \text{ vs. } H_1 : \mu_1 \neq \mu_2, \\
&H_0 : \text{SR}_1 = \text{SR}_2 \text{ vs. } H_1 : \text{SR}_1 \neq \text{SR}_2.
\end{aligned}
$$

It is often convenient to re-express these hypotheses in terms of differences of coefficients under the null and alternative hypotheses:

$$
\begin{aligned}
&H_0 : \mu_1 - \mu_2 = 0 \text{ vs. } H_1 : \mu_1 - \mu_2 \neq 0, \\
&H_0 : \text{SR}_1 - \text{SR}_2 = 0 \text{ vs. } H_1 : \text{SR}_1 - \text{SR}_2 \neq 0.
\end{aligned}
$$

9.2.2 Model specification tests

In the GWN model Equation 9.1, asset returns are assumed to be normally distributed. The hypotheses to be tested for this assumption are:

$$H_0 : R_{it} \sim N(\mu_i, \sigma_i^2) \text{ vs. } H_1 : R_{it} \sim \text{ not normal.}$$

Notice that the distribution of R_t under H_1 is not specified (e.g., Student's t with 5 degrees of freedom).

In addition, it is assumed that asset returns are uncorrelated over time. One can test the hypothesis that each autocorrelation (up to some maximum lag J) is zero:

$$\begin{aligned} H_0 &: \rho_j = \text{cor}(r_{it}, r_{i,t-j}) = 0, \; j = 1, \dots, J \text{ vs.} \\ H_1 &: \rho_j = \text{cor}(r_{it}, r_{i,t-j}) \neq 0, \end{aligned}$$

or one can test the joint hypothesis that all autocorrelation coefficients (up to some maximum lag J) are zero:

$$\begin{aligned} H_0 : \rho_1 &= \rho_2 = \cdots = \rho_J = 0 \, vs. \\ H_1 : \rho_j &\neq 0 \text{ for some } j \leq J \end{aligned}$$

Lastly, it is assumed that asset returns are covariance stationary. This assumption implies that all model parameters are constant over time. The hypotheses of interest are then:

$$\begin{aligned} &H_0 : \mu_i, \sigma_i \text{ and } \rho_{ij} \text{ are constant over entire sample, vs.} \\ &H_1 : \mu_i \; \sigma_i \text{ or } \rho_{ij} \text{ changes in some sub-sample.} \end{aligned}$$

The following sections describe test statistics for the above listed hypotheses.

9.2.3 Data for hypothesis testing examples

The examples in this chapter use the monthly continuously compounded returns on Microsoft, Starbucks and the S&P 500 index over the period January 1998 through May 2012 created using:

```
gwnDailyPrices = merge(msftDailyPrices, sbuxDailyPrices, sp500DailyPrices)
gwnMonthlyPrices = to.monthly(gwnDailyPrices, OHLC=FALSE)
smpl = "1998-01::2012-05"
gwnMonthlyPrices = gwnMonthlyPrices[smpl]
gwnMonthlyRetC = na.omit(Return.calculate(gwnMonthlyPrices, method="log"))
```

The monthly continuously compounded returns, with a common y-axis, are illustrated in Figure 9.1 created using:

```
plot(gwnMonthlyRetC, multi.panel=TRUE, main="",
     col="blue", lwd=2, yaxis.same=TRUE,
     major.ticks="years", grid.ticks.on="years")
```

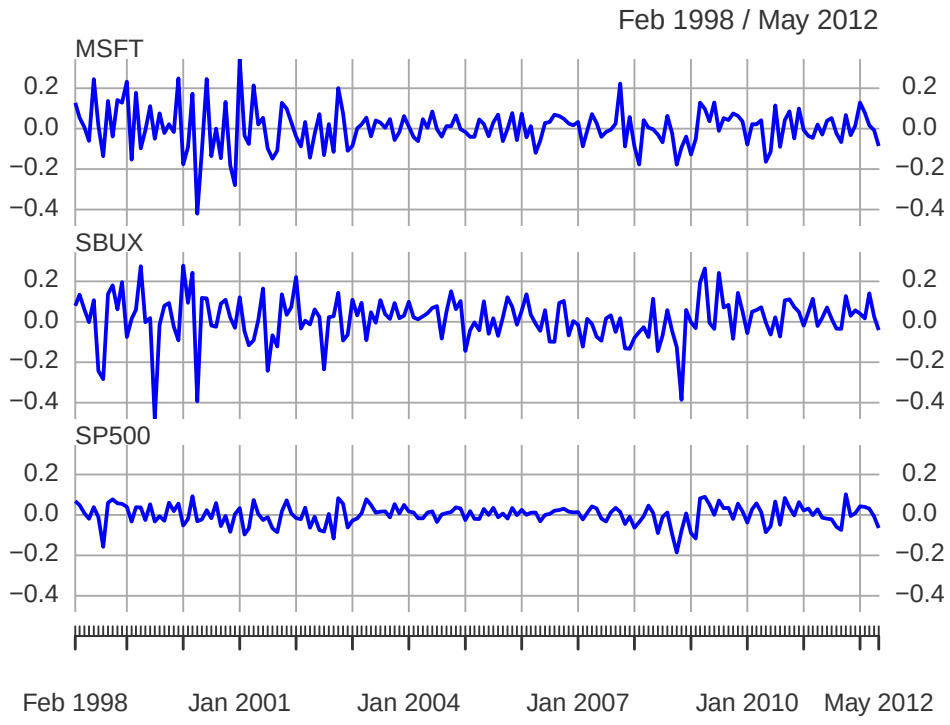


Figure 9.1: Monthly cc returns on Microsoft, Starbucks and the S&P 500 Index.

9.3 Tests for Individual Parameters: t-tests and z-scores

In this section, I present test statistics for testing hypotheses that certain model coefficients equal specific values. In some cases, I can derive *exact tests*. For an exact test, the pdf of the test statistic assuming the null hypothesis is true is known exactly for a finite sample size T. More generally, however, I rely on *asymptotic tests*. For an asymptotic test, the pdf of the test statistic assuming the null hypothesis is true is not known exactly for a finite sample size T, but can be approximated by a known pdf. The approximation is justified by the Central Limit Theorem (CLT) and the approximation becomes exact as the sample size becomes infinitely large.

9.3.1 Exact tests under normality of data

In the GWN model Equation 9.1, consider testing the hypothesis that the mean return, μ_i, is equal to a specific value μ_i^0:

$$H_0 : \mu_i = \mu_i^0. \tag{9.3}$$

For example, an investment analyst may have provided an expected return forecast of μ_i^0 and you would like to see if past data was consistent with such a forecast.

The alternative hypothesis can be either two-sided or one-sided. The two-sided alternative is:

$$H_1 : \mu_i \neq \mu_i^0. \tag{9.4}$$

Two-sided alternatives are used when you don't care about the sign of $\mu_i - \mu_i^0$ under the alternative. With one-sided alternatives, you care about the sign of $\mu_i - \mu_i^0$ under the alternative:

$$H_1 : \mu_i > \mu_i^0 \text{ or } H_1 : \mu_i < \mu_i^0. \tag{9.5}$$

How do you come up with a test statistic S for testing Equation 9.3 against Equation 9.4 or Equation 9.5? You generally use two criteria: (1) you know the pdf of S assuming Equation 9.3 is true; and (2) the value of S should be big if the alternative hypotheses Equation 9.4 or Equation 9.5 are true.

9.3.1.1 Two-sided test

To simplify matters, let's assume that the value of σ_i in Equation 9.1 is known and does not need to be estimated. This assumption is unrealistic and will be relaxed later. Consider testing Equation 9.3 against the two-sided alternative Equation 9.4 using a 5% significance level. Two-sided

alternatives are typically more commonly used than one-sided alternatives. Let $\{R_t\}_{t=1}^T$ denote a random sample from the GWN model. You estimate μ_i using the sample mean:

$$\hat{\mu}_i = \frac{1}{T}\sum_{t=1}^{T} R_{it}. \tag{9.6}$$

Under the null hypothesis Equation 9.3, it is assumed that $\mu_i = \mu_i^0$. Hence, if the null hypothesis is true then $\hat{\mu}_i$ computed from data should be close to μ_i^0. If $\hat{\mu}_i$ is far from μ_i^0 (either above or below μ_i^0) then this evidence casts doubt on the validity of the null hypothesis. Now, there is estimation error in Equation 9.6 which is measured by:

$$\text{se}(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{T}}. \tag{9.7}$$

Because you assume σ_i is known, you also know $\text{se}(\hat{\mu}_i)$. Because of estimation error, you don't expect $\hat{\mu}_i$ to equal μ_i^0 even if the null hypothesis Equation 9.3 is true. How far from μ_i^0 can $\hat{\mu}_i$ be if the null hypothesis is true? To answer this question, recall from Chapter Chapter 7 that the exact (finite sample) pdf of $\hat{\mu}_i$ is the normal distribution:

$$\hat{\mu}_i \sim N(\mu_i, \text{se}(\hat{\mu}_i)^2) = N\left(\mu_i, \frac{\sigma_i^2}{T}\right). \tag{9.8}$$

Under the null hypothesis Equation 9.3, the normal distribution for $\hat{\mu}_i$ is centered at μ_i^0:

$$\hat{\mu}_i \sim N(\mu_i^0, \text{se}(\hat{\mu}_i)^2) = N\left(\mu_i^0, \frac{\sigma_i^2}{T}\right). \tag{9.9}$$

From properties of the normal distribution you have:

$$\Pr\left(\mu_i^0 - 1.96 \times \text{se}(\hat{\mu}_i) \leq \hat{\mu}_i \leq \mu_i^0 + 1.96 \times \text{se}(\hat{\mu}_i)\right) = 0.95. \tag{9.10}$$

Hence, if the null hypothesis Equation 9.3 is true you only expect to see $\hat{\mu}_i$ more than 1.96 values of $\text{se}(\hat{\mu}_i)$ away from μ_i^0 with probability 0.05.[6] Therefore, it makes intuitive sense to base a test statistic for testing Equation 9.3 on a measure of distance between $\hat{\mu}_i$ and μ_i^0 relative to $\text{se}(\hat{\mu}_i)$. Such a statistic is the *z-score*:

$$z_{\mu=\mu^0} = \frac{\hat{\mu}_i - \mu_i^0}{\text{se}(\hat{\mu}_i)} = \frac{\hat{\mu}_i - \mu_i^0}{\sigma/\sqrt{T}}. \tag{9.11}$$

[6] This means that in an infinite number of hypothetical samples from the GWN model in which H_0 is true only 5 percent of the samples produce an estimate $\hat{\mu}_i$ that is more than 1.96 values of $\text{se}(\hat{\mu}_i)$ away from μ_i^0.

Assuming the null hypothesis Equation 9.3 is true, from Equation 9.9 it follows that:

$$z_{\mu=\mu^0} \sim N(0,1).$$

The intuition for using the z-score Equation 9.11 to test Equation 9.3 is straightforward. If $z_{\mu=\mu^0} \approx 0$ then $\hat{\mu}_i \approx \mu_i^0$ and Equation 9.3 should not be rejected. In contrast, if $z_{\mu=\mu^0} > 1.96$ or $z_{\mu=\mu^0} < -1.96$ then $\hat{\mu}_i$ is more than 1.96 values of $\text{se}(\hat{\mu}_i)$ away from μ_i^0. From Equation 9.10, this is very unlikely (less than 5% probability) if Equation 9.3 is true. In this case, there is strong data evidence against Equation 9.3 and you should reject it. Notice that the condition $z_{\mu=\mu^0} > 1.96$ or $z_{\mu=\mu^0} < -1.96$ can be simplified as the condition $\left|z_{\mu=\mu^0}\right| > 1.96$. When $\left|z_{\mu=\mu^0}\right|$ is big, larger than 1.96, you have data evidence against Equation 9.3. Using the value 1.96 to determine the rejection region for the test ensures that the significance level (probability of Type I error) of the test is exactly 5%:

$$\begin{aligned}\Pr(\text{Reject } H_0 | H_0 \text{ is true}) &= \Pr(S > 1.96 | \mu_i = \mu_i^0) \\ &= \Pr\left(\left|z_{\mu=\mu^0}\right| > 1.96 | \mu_i = \mu_i^0\right) = .05\end{aligned}$$

Hence, our formal test statistic for testing Equation 9.3 against Equation 9.4 is

$$S = \left|z_{\mu=\mu^0}\right|. \tag{9.12}$$

The 5% critical value is $cv_{.05} = 1.96$, and you reject Equation 9.3 at the 5% significance level if $S > 1.96$.

In summary, the steps for using $S = \left|z_{\mu=\mu^0}\right|$ to test the Equation 9.3 against Equation 9.4 are:

1. Set the significance level $\alpha = \Pr(\text{Reject } H_0 | H_0 \text{ is true})$ and determine the critical value cv_α such that $\Pr(S > cv_\alpha) = \alpha$. Now,

$$\begin{aligned}\Pr(S > cv_\alpha) &= \Pr\left(\left|z_{\mu=\mu^0}\right| > cv_\alpha\right) \\ &= \Pr\left(z_{\mu=\mu^0} > cv_\alpha\right) + \Pr\left(z_{\mu=\mu^0} < -cv_\alpha\right) = \alpha\end{aligned}$$

which implies that $cv_\alpha = -q^Z_{\alpha/2} = q^Z_{1-\alpha/2}$, where $q^Z_{\alpha/2}$ denotes the $\frac{\alpha}{2}$−quantile of $Z \sim N(0,1)$. For example, if $\alpha = .05$ then $cv_{.05} = -q^Z_{.025} = 1.96$.

2. Reject Equation 9.3 at the $\alpha \times 100\%$ significance level if $S > cv_\alpha$. For example, if $\alpha = .05$ then reject Equation 9.3 at the 5% level if $S > 1.96$.

3. Equivalently, reject Equation 9.3 at the $\alpha \times 100\%$ significance level if the p-value for S is less than α. Here, the *p-value* is defined as the significance level at which the test is just rejected. Let $Z \sim N(0,1)$. The p-value is computed as:

$$\text{p-value} = \Pr(|Z| > S) = \Pr(Z > S) + \Pr(Z < -S) = 2 \times \Pr(Z > S) = 2 \times (1 - \Pr(Z < S)). \tag{9.13}$$

Example 9.1 (Testing hypothesis with the z-score using simulated data: two-sided tests)**.** Assume returns follow the GWN model

$$\begin{aligned} R_t &= 0.05 + \epsilon_t, \, t = 1, \ldots 60, \\ \epsilon_t &\sim GWN(0, 0.10). \end{aligned}$$

and you are interested in testing the following hypotheses

$$\begin{aligned} H_0 : \mu = 0.05 \; vs. \; H_1 : \mu \neq .05 \\ H_0 : \mu = 0.06 \; vs. \; H_1 : \mu \neq .06 \\ H_0 : \mu = 0.10 \; vs. \; H_1 : \mu \neq .10 \end{aligned}$$

using the test statistic Equation 9.12 with a 5% significance level. The 5% critical value is $cv_{.05} = 1.96$. One hypothetical sample from the hypothesized model is simulated using:

```
set.seed(123)
n.obs = 60
mu = 0.05
sigma = 0.10
ret.sim = mu + rnorm(n.obs, sd = sigma)
```

The estimate of μ and the value of se($\hat{\mu}$) are:

```
muhat = mean(ret.sim)
se.muhat = sigma/sqrt(n.obs)
c(muhat, se.muhat)
#> [1] 0.0566 0.0129
```

The test statistic Equation 9.12 for testing $H_0 : \mu = 0.05$ computed from this sample is:

```
z.score.05 = (muhat - 0.05)/se.muhat
S = abs(z.score.05)
S
#> [1] 0.508
```

The z-score tells us that the estimated mean, $\hat{\mu} = 0.0566$, is 0.58 values of se($\hat{\mu}$) away from the hypothesized value $\mu^0 = 0.05$. This evidence does not contradict $H_0 : \mu = 0.05$. Since $S = 0.508 < 1.96$, you do not reject $H_0 : \mu = 0.05$ at the 5% significance level. The p-value of the test using Equation 9.13 is:

```
p.value = 2*(1 - pnorm(S))
p.value
#> [1] 0.611
```

Here, the p-value of 0.611 is less than the significance level $\alpha = 0.05$ so you do not reject the null at the 5% significance level. The p-value tells us that you would reject $H0 : \mu = 0.05$ at the 61.1% significance level.

The z-score information for testing $H_0 : \mu = 0.06$ is:

```
z.score.06 = (muhat - 0.06)/se.muhat
S = abs(z.score.06)
S
#> [1] 0.266
```

Here, the z-score indicates that $\hat{\mu} = 0.0566$ is just 0.266 values of se($\hat{\mu}$) away from the hypothesized value $\mu^0 = 0.06$. This evidence also does not contradict $H_0 : \mu = 0.06$. Since $S = 0.226 < 1.96$, you do not reject $H_0 : \mu = 0.06$ at the 5% significance level. The p-value of the test is:

```
p.value = 2*(1 - pnorm(S))
p.value
#> [1] 0.79
```

The large p-value shows that there is insufficient data to reject $H_0 : \mu = 0.06$.

Last, the z-score information for testing $H_0 : \mu = 0.10$ is:

```
z.score.10 = (muhat - 0.10)/se.muhat
S = abs(z.score.10)
S
#> [1] 3.36
```

Now, the z-score indicates that $\hat{\mu} = 0.0566$ is 3.36 values of se($\hat{\mu}$) away from the hypothesized value $\mu^0 = 0.10$. This evidence is not in support of $H_0 : \mu = 0.10$. Since $S = 3.36 > 1.96$, you reject $H_0 : \mu = 0.06$ at the 5% significance level. The p-value of the test is:

```
p.value = 2*(1 - pnorm(S))
p.value
#> [1] 0.000766
```

Here, the p-value is much less 5% supporting the rejection of $H_0 : \mu = 0.06$.
■

9.3.1.2 One-sided test

Here, I consider testing the null Equation 9.3 against the one-sided alternative Equation 9.5. For expositional purposes, consider $H_1 : \mu_i > \mu_i^0$. This alternative can be equivalently represented as $H_1 : \mu_i - \mu_i^0 > 0$. That is, under the alternative hypothesis the sign of the difference $\mu_i - \mu_i^0$ is positive. This is why a test against a one-sided alternative is sometimes called a *test for sign*. As in the previous sub-section, assume that σ_i is known and let the significance level be 5%.

The natural test statistic is the z-score Equation 9.11. The intuition is straightforward. If $z_{\mu=\mu^0} \approx 0$ then $\hat{\mu}_i \approx \mu_i^0$ and the null should not be rejected. However, if the one-sided alternative is true then you would expect to see $\hat{\mu}_i > \mu_i^0$ and $z_{\mu=\mu^0} > 0$. How big $z_{\mu=\mu^0}$ needs to be for us to reject the null depends on the significance level. Since under the null $z_{\mu=\mu^0} \sim N(0,1) = Z$, with a 5% significance level

$$\Pr(\text{Reject } H_0 | H_0 \text{ is true}) = \Pr(z_{\mu=\mu^0} > 1.645) = \Pr(Z > 1.645) = 0.05.$$

Hence, our 5% one-sided critical value is $cv_{.05} = q^Z_{.95} = 1.645$, and you reject the null at the 5% significance level if $z_{\mu=\mu^0} > 1.645$.

In general, the steps for using $z_{\mu=\mu^0}$ to test Equation 9.3 against the one-sided alternative $H_1 : \mu_i > \mu_i^0$ are:

1. Set the significance level $\alpha = \Pr(\text{Reject } H_0 | H_0 \text{ is true})$ and determine the critical value cv_α such that $\Pr(z_{\mu=\mu^0} > cv_\alpha) = \alpha$. Since under the null $z_{\mu=\mu^0} \sim N(0,1) = Z$, $cv_\alpha = q^Z_{1-\alpha}$.
2. Reject the null hypothesis Equation 9.3 at the $\alpha \times 100\%$ significance level if $z_{\mu=\mu^0} > cv_\alpha$.
3. Equivalently, reject Equation 9.3 at the $\alpha \times 100\%$ significance level if the one-sided p-value is less than α, where

$$\text{p-value} = \Pr(Z > z_{\mu=\mu^0}) = 1 - \Pr(Z \leq z_{\mu=\mu^0})$$

Example 9.2 (Testing hypothesis with the z-score using simulated data: one-sided tests)**.** Here, I use the simulated GWN return data from the previous example to test Equation 9.3 against the one-sided alternative $H_1 : \mu_i > \mu_i^0$ using a 5% significance level.

The z-scores and one-sided p-values for testing $H_0 : \mu = 0.05$, $H_0 : \mu = 0.06$, and $H_0 : \mu = 0.10$, computed from the simulated sample are:

```
z.score.05 = (muhat - 0.05)/se.muhat
p.value.05 = 1 - pnorm(z.score.05)
z.score.06 = (muhat - 0.06)/se.muhat
p.value.06 = 1 - pnorm(z.score.06)
z.score.10 = (muhat - 0.10)/se.muhat
p.value.10 = 1 - pnorm(z.score.10)
ans = rbind(c(z.score.05, p.value.05),
            c(z.score.06, p.value.06),
            c(z.score.10, p.value.10))
colnames(ans) = c("Z-score", "P-value")
rownames(ans) = c("H0:mu=0.05", "H0:mu=0.06","H0:mu=0.10")
ans
#>             Z-score P-value
#> H0:mu=0.05    0.508   0.306
#> H0:mu=0.06   -0.266   0.605
#> H0:mu=0.10   -3.365   1.000
```

Here, you do not reject any of the null hypotheses in favor of the one-sided alternative at the 5% significance level. ■

9.3.2 Exact tests with σ unknown

In practice, σ^2 is unknown and is estimated with $\hat{\sigma}^2$, and so the exact tests described above are not feasible. Fortunately, an exact test is still available. Instead of using the z-score Equation 9.11, we use the t-ratio (or t-score)

$$t_{\mu=\mu^0} = \frac{\hat{\mu}_i - \mu_i^0}{\widehat{\text{se}}(\hat{\mu}_i)} = \frac{\hat{\mu}_i - \mu_i^0}{\hat{\sigma}/\sqrt{T}}. \tag{9.14}$$

Assuming the null hypothesis Equation 9.3 is true, Proposition 7.7 tells us that Equation 9.14 is distributed Student's t with $T-1$ degrees of freedom and is denoted by the random variable t_{T-1}. The steps for using the z-score and the t-ratio are the same for evaluating Equation 9.3, but now we use critical values and p-values from t_{T-1}. Our test statistic for the two-sided alternative

$H_0 : \mu_i \neq \mu_i^0$ is:

$$S = \left| t_{\mu=\mu^0} \right|, \tag{9.15}$$

and our test statistic for the one-sided alternative $H_0 : \mu_i > \mu_i^0$ is

$$S = t_{\mu=\mu^0}, \tag{9.16}$$

Our critical values are determined from the quantiles of the Student's t with $T-1$ degrees of freedom, $t_{T-1}(1-\alpha/2)$. For example, if $\alpha = 0.05$ and $T-1 = 60$ then the two-sided critical value is $cv_{.05} = t_{60}(0.975) = 2$ (which can be verified using the R function `qt()`). The two-sided p-value is computed as:

$$\begin{aligned} \text{p-value} &= \Pr(|t_{T-1}| > S) = \Pr(t_{T-1} > S) + \Pr(t_{T-1} < -S) \\ &= 2 \times \Pr(t_{T-1} > S) = 2 \times (1 - \Pr(t_{T-1} < S)). \end{aligned}$$

The one-sided critical value is $cv_{.05} = t_{60}(0.95) = 1.67$, and the one-sided p-value is computed as:

$$\text{p-value} = \Pr(t_{T-1} > S) = 1 - \Pr(t_{T-1} \leq S).$$

As the sample size gets larger $\hat{\sigma}_i$ gets closer to σ_i and the Student's t distribution gets closer to the normal distribution. Decisions using the t-ratio and the z-score are almost the same for $T \geq 60$.

Example 9.3 (Testing hypothesis with the t-ratio using simulated data)**.** I repeat the hypothesis testing from the previous example this time use the t-ratio Equation 9.14 instead of the z-score. Using $T-1 = 59$ the 5% critical value for the two-sided test is

```
qt(0.975, df=59)
#> [1] 2
```

To compute the t-ratios we first estimate $\text{se}(\hat{\mu})$:

```
sigmahat = sd(ret.sim)
sehat.muhat = sigmahat/sqrt(n.obs)
c(sehat.muhat, se.muhat)
#> [1] 0.0118 0.0129
```

Here, $\widehat{\text{se}}(\hat{\mu}) = 0.0118 < 0.0129 = \text{se}(\hat{\mu})$ and so the t-ratios will be slightly larger than the z-scores. The t-ratios and test statistics for the three hypotheses are:

```
t.ratio.05 = (muhat - 0.05)/sehat.muhat
t.ratio.06 = (muhat - 0.06)/sehat.muhat
t.ratio.10 = (muhat - 0.10)/sehat.muhat
S.05 = abs(t.ratio.05)
S.06 = abs(t.ratio.06)
S.10 = abs(t.ratio.10)
ans = c(S.05, S.06, S.10)
names(ans) = c("S.05", "S.06", "S.07")
ans
#>  S.05  S.06  S.07
#> 0.558 0.293 3.696
```

As expected the test statistics computed from the t-ratios are slightly larger than the tests computed from the z-scores. The first two statistics are less than 2, and the third statistic is bigger than 2 and so we reach the same decisions as before. The p-values of the three tests are:

```
S.vals = c(S.05, S.06, S.10)
p.values = 2*(1 - pt(S.vals, df=59))
p.values
#> [1] 0.578747 0.770895 0.000482
```

Here, the p-values computed from t_{59} are very similar to those computed from $Z \sim N(0,1)$. ■

You can derive an exact test for testing hypothesis about the value of μ_i based on the z-score or the t-ratio, but you cannot derive exact tests for the values of σ_i or for the values of ρ_{ij} based on z-scores. Exact tests for these parameters are much more complicated. While t-ratios for the values of σ_i or for the values of ρ_{ij} do not have exact t-distributions in finite samples, as the sample size gets large the distributions of the t-ratios get closer and closer to the normal distribution due to the CLT. This motivates the use of so-called asymptotic z-scores discussed in the next sub-section.

9.3.3 Z-scores under asymptotic normality of estimators

Let $\hat{\theta}$ denote an estimator for θ. Here, I allow θ to be a GWN model parameter or a function of GWN model parameters. For example, in the GWN model θ could be μ_i, σ_i,, ρ_{ij}, q_α^R, VaR_α or SR_i. As you have seen, the CLT (and the delta method if needed) justifies the asymptotic normal distribution:

$$\hat{\theta} \sim N(\theta, \widehat{\text{se}}(\hat{\theta})^2), \tag{9.17}$$

for large enough sample size T, where $\widehat{\text{se}}(\hat{\theta})$ is the estimated standard error for $\hat{\theta}$. Consider testing:

$$H_0 : \theta = \theta_0 \text{ vs. } H_1 : \theta \neq \theta_0. \tag{9.18}$$

Under H_0, the asymptotic normality result Equation 9.17 implies that the z-score for testing Equation 9.18 has a standard normal distribution for large enough sample size T:

$$z_{\theta=\theta_0} = \frac{\hat{\theta} - \theta_0}{\widehat{\text{se}}(\hat{\theta})} \sim N(0,1) = Z. \tag{9.19}$$

The intuition for using the z-score Equation 9.19 is straightforward. If $z_{\theta=\theta_0} \approx 0$ then $\hat{\theta} \approx \theta_0$, and $H_0 : \theta = \theta_0$ should not be rejected. On the other hand, if $|z_{\theta=\theta_0}| > 2$, say, then $\hat{\theta}$ is more than 2 values of $\widehat{\text{se}}(\hat{\theta})$ away from θ_0. This is very unlikely if $\theta = \theta_0$ because $\hat{\theta} \sim N(\theta_0, \hat{\text{se}}(\hat{\theta})^2)$, so $H_0 : \theta \neq \theta_0$ should be rejected. Therefore, the test statistic for testing Equation 9.19 is $S = |z_{\theta=\theta_0}|$.

The steps for using the z-score Equation 9.19 with its critical value to test the hypotheses Equation 9.18 are:

1. Set the significance level α of the test and determine the two-sided critical value $cv_{\alpha/2}$. Using Equation 9.17, the critical value, $cv_{\alpha/2}$, is determined using:

$$\begin{aligned} &\Pr(|Z| \geq cv_{\alpha/2}) = \alpha \\ &\quad \Rightarrow cv_{\alpha/2} = -q^Z_{\alpha/2} = q^Z_{1-\alpha/2}, \end{aligned}$$

   where $q^Z_{\alpha/2}$ denotes the $\frac{\alpha}{2}-$quantile of $N(0,1)$. A commonly used significance level is $\alpha = 0.05$ and the corresponding critical value is $cv_{.025} = -q^Z_{.025} = q^Z_{.975} = 1.96 \approx 2$.

2. Reject Equation 9.18 at the $100 \times \alpha\%$ significance level if:

$$S = |z_{\theta=\theta_0}| = \left| \frac{\hat{\theta} - \theta^0}{\widehat{\text{se}}(\hat{\theta})} \right| > cv_{\alpha/2}.$$

   If the significance level is $\alpha = 0.05$, then reject Equation 9.18 at the 5% level using the rule-of-thumb:

$$S = |z_{\theta=\theta_0}| > 2.$$

The steps for using the z-score Equation 9.19 with its p-value to test the hypotheses Equation 9.18 are:

1. Determine the two-sided p-value. The p-value of the two-sided test is the significance level at which the test is just rejected. From Equation 9.17, the two-sided p-value is defined by

$$\text{p-value} = \Pr\left(|Z| > |z_{\theta=\theta_0}|\right) = 2 \times (1 - \Pr\left(Z \leq |z_{\theta=\theta_0}|\right). \tag{9.20}$$

2. Reject Equation 9.18 at the $100 \times \alpha\%$ significance level if the p-value Equation 9.20 is less than α.

Example 9.4 (Using the z-score to test hypothesis about ( ) in the GWN model)**.** Consider using the z-score Equation 9.19 to test $H_0 : \mu_i = 0 \, vs. \, H_1 : \mu_i \neq 0$ (i =Microsoft, Starbucks, S&P 500) using a 5% significance level. First, calculate the GWN model estimates for μ_i:

```
n.obs = nrow(gwnMonthlyRetC)
muhat.vals = apply(gwnMonthlyRetC, 2, mean)
muhat.vals
#>    MSFT    SBUX   SP500
#> 0.00413 0.01466 0.00169
```

Next, calculate the estimated standard errors:

```
sigmahat.vals = apply(gwnMonthlyRetC, 2, sd)
se.muhat = sigmahat.vals/sqrt(n.obs)
se.muhat
#>    MSFT    SBUX   SP500
#> 0.00764 0.00851 0.00370
```

Then calculate the z-scores and test statistics:

```
z.scores = muhat.vals/se.muhat
S.vals = abs(z.scores)
S.vals
#>  MSFT  SBUX SP500
#> 0.540 1.722 0.457
```

Since the absolute value of all of the z-scores are less than two, you do not reject $H_0 : \mu_i = 0$ at the 5% level for all assets.

The p-values for all of the test statistics are computed using:

```
2*(1-pnorm(S.vals))
#>   MSFT   SBUX  SP500
#> 0.5893 0.0851 0.6480
```

Since all p-values are greater than $\alpha = 0.05$, you reject $H_0 : \mu_i = 0$ at the 5% level for all assets. The p-value for Starbucks is the smallest at 0.0851. Here, you can reject $H_0 : \mu_{SBUX} = 0$ at the 8.51% level. ■

Example 9.5 (Using the z-score to test hypothesis about ρ in the GWN model)**.** Consider using the z-score Equation 9.19 to test the hypotheses:

$$H_0 : \rho_{ij} = 0.5 \, vs. \, H_1 : \rho_{ij} \neq 0.5. \tag{9.21}$$

using a 5% significance level. Here, you use the result from the GWN model that for large enough T:

$$\hat{\rho}_{ij} \sim N\left(\rho_{ij}, \widehat{\text{se}}(\hat{\rho}_{ij})^2\right), \widehat{\text{se}}(\hat{\rho}_{ij}) = \frac{1 - \hat{\rho}_{ij}^2}{\sqrt{T}}.$$

Then the z-score for testing Equation 9.21 has the form:

$$z_{\rho_{ij}=0.5} = \frac{\hat{\rho}_{ij} - 0.5}{\widehat{\text{se}}(\hat{\rho}_{ij})} = \frac{\hat{\rho}_{ij} - 0.5}{\left(1 - \hat{\rho}_{ij}^2\right)/\sqrt{T}}. \tag{9.22}$$

To compute the z-scores, first, calculate the GWN model estimates for ρ_{ij}:

```
corhat.mat = cor(gwnMonthlyRetC)
rhohat.vals = corhat.mat[lower.tri(corhat.mat)]
names(rhohat.vals) = c("MSFT.SBUX", "MSFT.SP500", "SBUX.SP500")
rhohat.vals
#>  MSFT.SBUX MSFT.SP500 SBUX.SP500
#>      0.341      0.617      0.457
```

Next, calculate estimated standard errors:

```
se.rhohat = (1 - rhohat.vals^2)/sqrt(n.obs)
se.rhohat
#>  MSFT.SBUX MSFT.SP500 SBUX.SP500
#>     0.0674     0.0472     0.0603
```

Then calculate the z-scores Equation 9.22 and test statistics:

```
z.scores = (rhohat.vals - 0.5)/se.rhohat
S.vals = abs(z.scores)
S.vals
#>  MSFT.SBUX MSFT.SP500 SBUX.SP500
#>      2.361      2.482      0.706
```

Here, the absolute value of the z-scores for $\rho_{MSFT,SBUX}$ and $\rho_{MSFT,SP500}$ are greater than 2 whereas the absolute value of the z-score for $\rho_{SBUX,SP500}$ is less than 2. Hence, for the pairs (MSFT, SBUX) and (MSFT, SP500) you cannot reject the null Equation 9.21 at the 5% level but

for the pair (SBUX, SP500) you can reject the null at the 5% level. The p-values for the test statistics are:

```
2*(1-pnorm(S.vals))
#>  MSFT.SBUX MSFT.SP500 SBUX.SP500
#>     0.0182     0.0130     0.4804
```

Here, the p-values for the pairs (MSFT,SBUX) and (MSFT,SP500) are less than 0.05 and the p-value for the pair (SBUX, SP500) is much greater than 0.05. ■

Example 9.6 (Using the z-score to test hypothesis about asset Sharpe ratios in the GWN model)**.** Consider using the z-score to test the hypotheses:

$$H_0 : \text{SR}_i = \frac{\mu_i - r_f}{\sigma_i} = 0 \text{ vs. } H_1 : \text{SR}_i > 0$$

using a 5% significance level. In Chapter Chapter 8 I used the delta method to show that, for large enough T:

$$\widehat{\text{SR}}_i \sim N\left(\text{SR}_i, \widehat{\text{se}}\left(\widehat{\text{SR}}_i\right)^2\right), \ \widehat{\text{se}}\left(\widehat{\text{SR}}_i\right) = \frac{1}{\sqrt{T}}\sqrt{1 + \frac{1}{2}\widehat{\text{SR}}_i^2}$$

Then, the z-score has the form

$$z_{\text{SR}=0} = \frac{\widehat{\text{SR}}_i - 0}{\widehat{\text{se}}\left(\widehat{\text{SR}}_i\right)} = \frac{\widehat{\text{SR}}_i}{\frac{1}{\sqrt{T}}\sqrt{1 + \frac{1}{2}\widehat{\text{SR}}_i^2}} = \frac{\sqrt{T} \times \widehat{\text{SR}}_i}{\sqrt{1 + \frac{1}{2}\widehat{\text{SR}}_i^2}}$$

■

9.3.4 Relationship between hypothesis tests and confidence intervals

Consider testing the hypotheses Equation 9.18 at the 5% significance level using the z-score Equation 9.19. The rule-of-thumb decision rule is to reject $H_0 : \theta = \theta_0$ if $|z_{\theta=\theta_0}| > 2$. This implies that:

$$\frac{\hat{\theta} - \theta_0}{\widehat{\text{se}}(\hat{\theta})} > 2 \text{ or } \frac{\hat{\theta} - \theta_0}{\widehat{\text{se}}(\hat{\theta})} < -2,$$

which further implies that:

$$\theta_0 < \hat{\theta} - 2 \times \widehat{\text{se}}(\hat{\theta}) \text{ or } \theta_0 > \hat{\theta} + 2 \times \widehat{\text{se}}(\hat{\theta}).$$

Recall the definition of an approximate 95% confidence interval for θ:

$$\hat{\theta} \pm 2 \times \widehat{\text{se}}(\hat{\theta}) = \left[\hat{\theta} - 2 \times \widehat{\text{se}}(\hat{\theta}),\ \hat{\theta} + 2 \times \widehat{\text{se}}(\hat{\theta})\right].$$

Notice that if $|z_{\theta=\theta_0}| > 2$ then θ_0 does not lie in the approximate 95% confidence interval for θ. Hence, you can reject $H_0 : \theta = \theta_0$ at the 5% significance level if θ_0 does not lie in the 95% confidence for θ. This result allows you to have a deeper understanding of the 95% confidence interval for θ: it contains all values of θ_0 for which you cannot reject $H_0 : \theta = \theta_0$ at the 5% significance level.

This duality between two-sided hypothesis tests and confidence intervals makes hypothesis testing for individual parameters particularly simple. As part of estimation, you calculate estimated standard errors and form approximate 95% confidence intervals. Then when you look at the approximate 95% confidence interval, it gives us all values of θ_0 for which you cannot reject $H_0 : \theta = \theta_0$ at the 5% significance level (approximately).

Example 9.7 (Hypothesis testing for μ_i, ρ_{ij}, and SR_i using 95% confidence intervals)**.** For the example data, the approximate 95% confidence intervals for μ_i for Microsoft, Starbucks and the S&P 500 index are:

```
lower = muhat.vals - 2*se.muhat
upper = muhat.vals + 2*se.muhat
cbind(lower, upper)
#>          lower   upper
#> MSFT  -0.01116 0.01941
#> SBUX  -0.00237 0.03168
#> SP500 -0.00570 0.00908
```

Consider testing $H_0 : \mu_i = 0$ at the 5% level. Here you see that $\mu_i^0 = 0$ lies in all of the 95% confidence intervals and so you do not reject $H_0 : \mu_i = 0$ at the 5% level for any asset. Each interval gives the values of μ_i^0 for which you cannot reject $H_0 : \mu_i = \mu_i^0$ at the 5% level. For example, for Microsoft you cannot reject the hypothesis (at the 5% level) that μ_{MSFT} is as small as -0.011 or as large as 0.019.

Next, the approximate 95% confidence intervals for ρ_{ij} for the pairs $\rho_{MSFT,SBUX}$, $\rho_{MSFT,SP500}$, and $\rho_{SBUX,SP500}$ are:

```
lower = rhohat.vals - 2*se.rhohat
upper = rhohat.vals + 2*se.rhohat
cbind(lower, upper)
#>            lower upper
#> MSFT.SBUX  0.206 0.476
#> MSFT.SP500 0.523 0.712
```

```
#> SBUX.SP500 0.337 0.578
```

Consider testing $H_0 : \rho_{ij} = 0.5$ for all pairs of assets. Here, you see that $\rho_{ij}^0 = 0.5$ is not in the approximate 95% confidence intervals for $\rho_{MSFT,SBUX}$ and $\rho_{MSFT,SP500}$ but is in the approximate 95% confidence interval for $\rho_{SBUX,SP500}$. Hence, you can reject $H_0 : \rho_{ij} = 0.5$ at the 5% significance level for $\rho_{MSFT,SBUX}$ and $\rho_{MSFT,SP500}$ but not for $\rho_{SBUX,SP500}$.

- Add SR info here

■

9.4 Coefficient tests between model coefficients for two assets

In this section I consider hypothesis tests for coefficients in the bivariate GWN model for asset returns.

9.4.1 Test for equal means between two assets

Often it is of interest to test if the expected returns on two assets are the same:

$$H_0 : \mu_1 = \mu_2 \text{ vs. } H_1 : \mu_1 \neq \mu_2.$$

Notice that this hypothesis is not of the form that the two coefficients equal a specific value (e.g. $\mu_1 = \mu_2 = 0.05$). However, you can transform the hypothesis into one in which a coefficient equals a specific value by considering the differences in the expected returns. In particular, the relation $\mu_1 = \mu_2$ is equivalent to the relation $\delta = \mu_1 - \mu_2 = 0$. Hence, you can reformulate the hypothesis of equal expected returns to the hypothesis that the difference in expected returns is equal to zero:

$$H_0 : \delta = \mu_1 - \mu_2 = 0 \text{ vs. } H_1 : \delta = \mu_1 - \mu_2 \neq 0. \tag{9.23}$$

Now, you have test in the form of the coefficient tests you considered earlier.

The parameter $\delta = \mu_1 - \mu_2$ can be estimated with the plug-in estimator:

$$\hat{\delta} = \hat{\mu}_1 - \hat{\mu}_2.$$

To form a z-score, you require an estimate of $\text{se}(\hat{\delta})$. Now,

$$\text{var}(\hat{\delta}) = \text{var}(\hat{\mu}_1 - \hat{\mu}_2) = \text{var}(\hat{\mu}_1) + \text{var}(\hat{\mu}_2) - 2\text{cov}(\hat{\mu}_1, \hat{\mu}_2). \tag{9.24}$$

To evaluate Equation 9.24, you use the result from Chapter 7 that the joint distribution of the vector $(\hat{\mu}_1, \hat{\mu}_2)'$ is multivariate normal:

$$\begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} = N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \frac{1}{T} \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 \end{pmatrix} \right),$$

for large enough T. Then

$$\widehat{\text{var}}(\hat{\delta}) = \frac{1}{T}(\hat{\sigma}_1^2 + \hat{\sigma}_2^2 - 2\hat{\sigma}_{12}),$$

and

$$\widehat{\text{se}}(\hat{\delta}) = \sqrt{\frac{1}{T}(\hat{\sigma}_1^2 + \hat{\sigma}_2^2 - 2\hat{\sigma}_{12})}.$$

you can test Equation 9.23 using the z-score:

$$z_{\delta=0} = \frac{\hat{\delta}}{\widehat{\text{se}}(\hat{\delta})} = \frac{\hat{\mu}_1 - \hat{\mu}_2}{\sqrt{\frac{1}{T}(\hat{\sigma}_1^2 + \hat{\sigma}_2^2 - 2\hat{\sigma}_{12})}}, \tag{9.25}$$

with the test statistic $S = |z_{\delta=0}|$.

The estimate $\widehat{\text{var}}(\hat{\delta})$ can also be computed using the delta method. Define $\theta = (\mu_1, \mu_2)'$ and $f(\theta) = \delta = \mu_1 - \mu_2$. The gradient of $f(\theta)$ is:

$$g(\theta) = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

and $\widehat{\text{var}}(\hat{\delta})$ can be computed as:

$$\begin{aligned} \widehat{\text{var}}(\hat{\delta}) = g(\hat{\theta})'\widehat{\text{var}}(\hat{\theta})g(\hat{\theta}) &= (1, -1)' \begin{pmatrix} \frac{\hat{\sigma}_1^2}{T} & \frac{\hat{\sigma}_{12}}{T} \\ \frac{\hat{\sigma}_{12}}{T} & \frac{\hat{\sigma}_2^2}{T} \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \frac{1}{T}(\hat{\sigma}_1^2 + \hat{\sigma}_2^2 - 2\hat{\sigma}_{12}). \end{aligned}$$

Example 9.8 (Testing that Microsoft and Starbucks have the same expected return.)**.** Let's use the z-score Equation 9.25 to test the hypothesis that the expected returns on Microsoft and Starbucks are the same using a 5% significance level. The R code to compute the z-score and test statistic is:

```
delta.hat = muhat.vals["MSFT"] - muhat.vals["SBUX"]
sig2hat.vals = sigmahat.vals^2
covhat = cov(gwnMonthlyRetC[, c("MSFT", "SBUX")])[1,2]
sehat.deltahat = sqrt((sig2hat.vals["MSFT"]+sig2hat.vals["SBUX"]-2*covhat)/n.obs)
z.score.delta = delta.hat/sehat.deltahat
S= abs(z.score.delta)
ans = c(delta.hat, sehat.deltahat, S)
names(ans) = c("Estimate", "Std Error", "Test Statistic")
ans
#>       Estimate      Std Error Test Statistic
#>        -0.0105         0.0093         1.1323
```

Here, $\hat{\delta} = -0.0105$, $\widehat{\text{se}}(\hat{\delta}) = 0.0093$ and $S = 1.1323$. Since $S < 2$ you do not reject $H_0 : \delta = \mu_{msft} - \mu_{sbux} = 0$ at the 5% level. ■

9.4.2 Test for equal Sharpe ratios between two assets

In the previous section, I considered testing if two assets have the same expected return. It is important to recognize that even if two assets have the same mean return they may have different standard deviations (volatilities) and, hence, different risks. Often, it is more interesting to test the hypothesis that two assets have the same reward-to-risk (i.e., Sharpe) ratios:

$$H_0 : \text{SR}_1 = \frac{\mu_1 - r_f}{\sigma_1} = \text{SR}_2 = \frac{\mu_2 - r_f}{\sigma_2} \text{ vs. } H_1 : \text{SR}_1 \neq \text{SR}_2.$$

where r_f denotes the risk-free rate over the investment horizon. As with testing the equality between two means, it is convenient to re-formulate the hypothesis of equal Sharpe ratio to the hypothesis that the difference in the Sharpe ratios is equal to zero:

$$H_0 : \delta = \text{SR}_1 - \text{SR}_2 = 0 \text{ vs. } H_1 : \delta = \text{SR}_1 - \text{SR}_2 \neq 0. \tag{9.26}$$

The plug-in estimator for δ is:

$$\hat{\delta} = \widehat{\text{SR}}_1 - \widehat{\text{SR}}_2 = \frac{\hat{\mu}_1 - r_f}{\hat{\sigma}_1} - \frac{\hat{\mu}_2 - r_f}{\hat{\sigma}_2}$$

To determine $\widehat{\text{se}}(\hat{\delta})$ from Equation 9.24, I utilize the result from Chapter Chapter 8 that the joint distribution of the vector $(\widehat{\text{SR}}_1, \widehat{\text{SR}}_2)$ is multivariate normal:

$$\begin{pmatrix}\widehat{\text{SR}}_1\\ \widehat{\text{SR}}_2\end{pmatrix} \sim N\left(\begin{pmatrix}\text{SR}_1\\ \text{SR}_1\end{pmatrix}, \begin{pmatrix}\frac{1}{T}(1+\frac{1}{2}\widehat{\text{SR}}_1^2) & \frac{\hat{\rho}_{12}}{T}(1+\frac{1}{2}\hat{\rho}_{12}\widehat{\text{SR}}_1\widehat{\text{SR}}_2)\\ \frac{\hat{\rho}_{12}}{T}(1+\frac{1}{2}\hat{\rho}_{12}\widehat{\text{SR}}_1\widehat{\text{SR}}_2) & \frac{1}{T}(1+\frac{1}{2}\widehat{\text{SR}}_2^2)\end{pmatrix}\right),$$

for large enough T. Then,

$$\begin{aligned}\widehat{\text{var}}(\hat{\delta}) &= \widehat{\text{var}}(\widehat{\text{SR}}_1) + \widehat{\text{var}}(\widehat{\text{SR}}_2) - 2\widehat{\text{cov}}(\widehat{\text{SR}}_1, \widehat{\text{SR}}_2)\\ &= \frac{1}{T}(1+\frac{1}{2}\widehat{\text{SR}}_1^2) + \frac{1}{T}(1+\frac{1}{2}\widehat{\text{SR}}_2^2)\\ &- 2\frac{\hat{\rho}_{12}}{T}(1+\frac{1}{2}\hat{\rho}_{12}\widehat{\text{SR}}_1\widehat{\text{SR}}_2)\\ &= \frac{1}{T}\left(2+\frac{1}{2}(\widehat{\text{SR}}_1^2+\widehat{\text{SR}}_2^2) - 2\hat{\rho}_{12}(1+\frac{1}{2}\hat{\rho}_{12}\widehat{\text{SR}}_1\widehat{\text{SR}}_2)\right),\end{aligned}$$

and,

$$\widehat{\text{se}}(\hat{\delta}) = \sqrt{\frac{1}{T}\left(2+\frac{1}{2}(\widehat{\text{SR}}_1^2+\widehat{\text{SR}}_2^2) - 2\hat{\rho}_{12}(1+\frac{1}{2}\hat{\rho}_{12}\widehat{\text{SR}}_1\widehat{\text{SR}}_2)\right)}.$$

You can test Equation 9.26 using the z-score:

$$z_{\delta=0} = \frac{\hat{\delta}}{\widehat{\text{se}}(\hat{\delta})} = \frac{\widehat{\text{SR}}_1 - \widehat{\text{SR}}_2}{\sqrt{\frac{1}{T}\left(2+\frac{1}{2}(\widehat{\text{SR}}_1^2+\widehat{\text{SR}}_2^2) - 2\hat{\rho}_{12}(1+\frac{1}{2}\hat{\rho}_{12}\widehat{\text{SR}}_1\widehat{\text{SR}}_2)\right)}}, \tag{9.27}$$

with the test statistic $S = |z_{\delta=0}|$.

Example 9.9 (Testing equal Sharpe ratios between Microsoft and Starbucks)**.** Let's use the z-score Equation 9.27 to test the hypothesis of equal Sharpe ratios between Microsoft and Starbucks using a 5% significance level. The R code to compute the z-score and test statistic is:

```
r.f = 0.03/12
SR.msft = (muhat.vals["MSFT"]-r.f)/sigmahat.vals["MSFT"]
SR.sbux = (muhat.vals["SBUX"]-r.f)/sigmahat.vals["SBUX"]
corhat = cor(gwnMonthlyRetC[, c("MSFT", "SBUX")])[1,2]
delta.hat = SR.msft - SR.sbux
var.deltahat = (2 + 0.5*(SR.msft^2 + SR.sbux^2)
                -2*corhat*(1+0.5*SR.msft*SR.sbux))
se.deltahat = sqrt(var.deltahat)
z.score.delta = delta.hat/se.deltahat
S = abs(z.score.delta)
```

```
ans = c(SR.msft, SR.sbux, delta.hat, se.deltahat, S)
names(ans) = c("SR.msft", "S.sbux", "deltahat", "se.deltahat", "S")
ans
#>     SR.msft      S.sbux    deltahat se.deltahat           S
#>      0.0162      0.1089     -0.0927      1.1505      0.0805
```

Although the Sharpe ratio for Starbucks is much larger than the Sharpe ratio for Microsoft, there is substantial estimation error in $\hat{\delta} = \text{SR}_{msft} - \text{SR}_{sbux}$ which produces a small test statistic $S = 0.0805$. Since $S < 2$ you do not reject the null hypothesis of equal Sharpe ratios at the 5% level of significance. ■

9.5 Wald tests for general linear hypotheses (advanced)

- testing equal means across many assets
- testing equal Sharpe ratios across many assets (mention result from Jobson and Korke etc here)

9.6 Model Specification Tests

- Review GWN model assumptions
- Discuss why it is interesting and useful to test these assumptions
- What to do if certain assumptions are violated?

9.6.1 Tests for normality

- Why do you care about normality?
  - Return distribution only depends on mean and volatility (and correlation in multivariate case).
  - Justifies volatility as an appropriate measure of risk.
  - Gives formula for Value-at-risk calculations.
  - Justifies mean-variance portfolio theory and Black-Scholes option pricing formula.
  - GWN model parameter estimates and standard error formula do not require the data to be normally distributed. These estimates are consistent and asymptotically normally distributed for many non-normal distributions. There may be estimators that are more accurate (i.e., have smaller standard errors)
- What to do if returns are found to not be normally distributed?

- Return distribution characteristics in addition to mean and volatility may be important to investors. Volatility may not be the best measure of asset return risk. For example, skewness and kurtosis. If many returns are non-normal then this casts doubt on the appropriateness of mean-variance portfolio theory.
- Standard practice is to compute VaR based on a normal distribution. If returns are not normally distributed this is an incorrect way to compute VaR. Instead of using the normal distribution quantile to compute VaR you can use the empirical quantile to compute VaR. If returns are substantially non-normal in the tails of the distribution then the empirical quantile will be more appropriate than the normal quantile for computing VaR
- How are returns non-normal? This is important.

In chapter Chapter 5, I used histograms, normal QQ-plots, and boxplots to assess whether the normal distribution is a good characterization of the empirical distribution of daily and monthly returns. For example, Figure 9.2 shows the four-panel plot for the monthly returns on Microsoft that I used to summarize the empirical distribution of returns. The histogram and smoothed histogram show that the empirical distribution of returns is bell-shaped like the normal distribution, whereas the boxplot and normal QQ-plot shows that returns have fatter tails than the normal distribution. Overall, the graphical descriptive statistics suggest that distribution of monthly Microsoft returns is not normally distributed. However, graphical diagnostics are not formal statistical tests. There are estimation errors in the statistics underlying the graphical diagnostics and it is possible that the observed non-normality of the data is due to sampling uncertainty. To be more definitive, I should use an appropriate test statistic to test the null hypothesis that the random return, R_t, is normally distributed. Formally, the null hypothesis to be tested is:

$$H_0 : R_t \text{ is normally distributed} \tag{9.28}$$

The alternative hypothesis is that R_t is not normally distributed:

$$H_1 : R_t \text{ is not normally distributed} \tag{9.29}$$

Here, I do not specify a particular distribution under the alternative (e.g., a Student's t distribution with a 5 degrees of freedom). The goal is to have a test statistic that will reject the null hypothesis Equation 9.28 for a wide range of non-normal distributions.

9.6.1.1 JB test based on skewness and kurtosis

In Chapter Chapter 2, I showed that the skewness and kurtosis of a normally distributed random variable are equal to zero and three, respectively. I can use this information to develop a simple

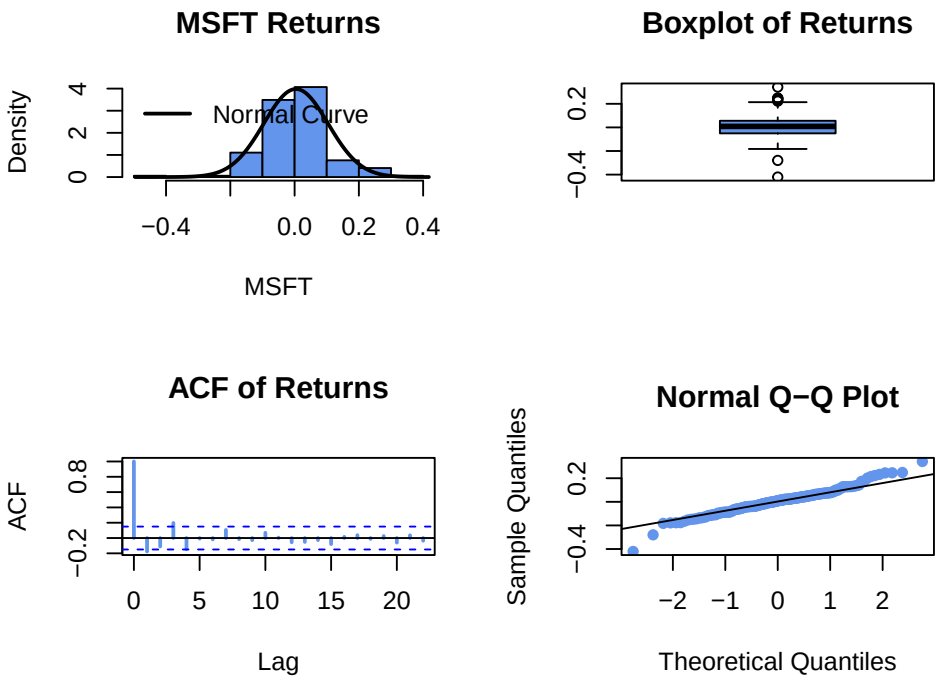


Figure 9.2: Four panel distribution plot for Microsoft monthly returns.

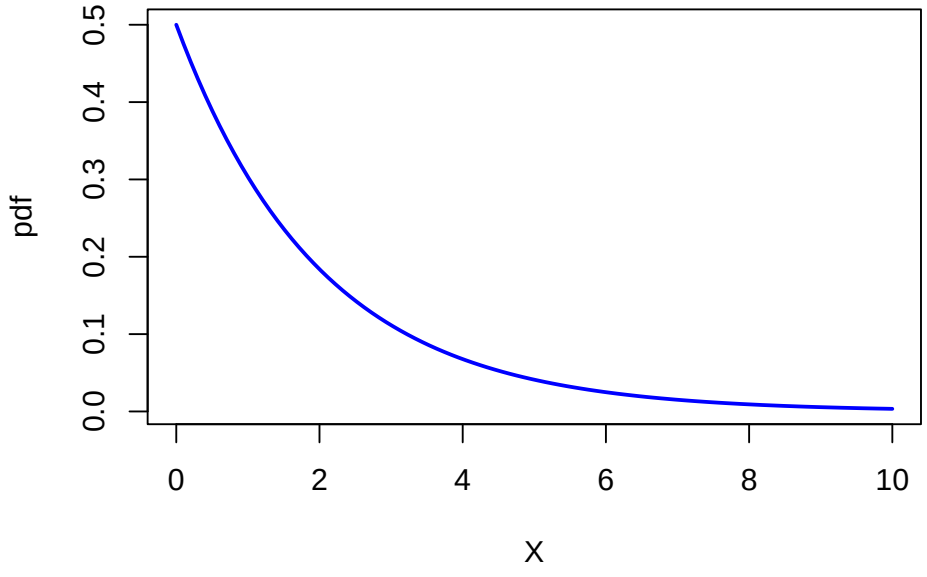


Figure 9.3: pdf of $\chi^2(2)$

test of Equation 9.28 against Equation 9.29 based on testing the hypotheses:

$$H_0 : \text{skew}(R_t) = 0 \text{ and } \text{kurt}(R_t) = 3 \tag{9.30}$$

versus

$$H_1 : \text{skew}(R_t) \neq 0 \text{ or } \text{kurt}(R_t) \neq 3 \text{ orboth.} \tag{9.31}$$

To develop a test statistic for testing Equation 9.30 against Equation 9.31, Jarque and Bera (1981) used the result (see Kendall and Stuart, 1977) that if $\{R_t\} \sim \text{iid}\, N(\mu, \sigma^2)$ then by the CLT it can be shown that:

$$\begin{pmatrix} \widehat{\text{skew}}_r \\ \widehat{\text{kurt}}_r - 3 \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{6}{T} & 0 \\ 0 & \frac{24}{T} \end{pmatrix} \right),$$

for large enough T, where $\widehat{\text{skew}}_r$ and $\widehat{\text{kurt}}_r$ denote the sample skewness and kurtosis of the returns $\{r_t\}_{t=1}^T$, respectively. This motivates Jarque and Bera's *JB statistic*:

$$\text{JB} = T \times \left[\frac{\left(\widehat{\text{skew}}_r\right)^2}{6} + \frac{\left(\widehat{\text{kurt}}_r - 3\right)^2}{24} \right], \tag{9.32}$$

to test Equation 9.28 against Equation 9.29. If returns are normally distributed then $\widehat{\text{skew}}_r \approx 0$ and $\widehat{\text{kurt}}_r - 3 \approx 0$ so that JB ≈ 0. In contrast, if returns are not normally distributed then $\widehat{\text{skew}}_r \neq 0$ or $\widehat{\text{kurt}}_r - 3 \neq 0$ or both so that JB > 0. Hence, if JB is big then there is data evidence against the null hypothesis that returns are normally distributed.

To determine the critical value for testing Equation 9.28 at a given significance level I need the distribution of the JB statistic Equation 9.32 under the null hypothesis Equation 9.28. To this end, Jarque and Bera show, assuming Equation 9.28 and large enough T, that:

$$\text{JB} \sim \chi^2(2),$$

where $\chi^2(2)$ denotes a chi-square distribution with two degrees of freedom (see Appendix). Hence, the critical values for the JB statistic are the upper quantiles of $\chi^2(2)$. Figure 9.3 shows the pdf of $\chi^2(2)$. The 90%, 95% and 99% quantiles can be computed using the R function `qchisq()`:

```
qchisq(c(0.90, 0.95, 0.99), 2)
#> [1] 4.61 5.99 9.21
```

Since the 95% quantile of $\chi^2(2)$ is $5.99 \approx 6$, a simple rule of thumb is to reject the null hypothesis Equation 9.28 at the 5% significance level if the JB statistic Equation 9.32 is greater than 6.

Let $X \sim \chi^2(2)$. The p-value of the JB test is computed using:

$$p - value = \Pr(X > JB).$$

If the p-value() is less than α then the null () is rejected at the

Example 9.10 (Using the JB statistic to test the normality assumption for returns on Microsoft Starbucks and the S&P 500 index)**.** The sample skewness and excess kurtosis values for the monthly returns on Microsoft, Starbucks, and the S&P 500 index are

```
skewhat = skewness(gwnMonthlyRetC)
ekurthat = kurtosis(gwnMonthlyRetC)
rbind(skewhat, ekurthat)
#>                    MSFT   SBUX SP500
#> Skewness        -0.0901 -0.987 -0.74
#> Excess Kurtosis  2.0802  3.337  1.07
```

The sample skewness values for the assets are not too large but the sample excess kurtosis values are fairly large. The JB statistics for the three assets are easy to compute

```
n.obs = nrow(gwnMonthlyRetC)
JB = n.obs*((as.numeric(skewhat)^2)/6 + (as.numeric(ekurthat)^2)/24)
names(JB) = colnames(gwnMonthlyRetC)
JB
#>  MSFT  SBUX SP500
#>  31.2 107.7  23.9
```

Since all JB values are greater than 6, you reject the null hypothesis that the monthly returns are normally distributed.

You can also use the **tseries** function `jarque.bera.test()` to compute the JB statistic:

```
jarque.bera.test(gwnMonthlyRetC[, 1])
#>
#>  Jarque Bera Test
#>
#> data:  gwnMonthlyRetC[, 1]
#> X-squared = 31, df = 2, p-value = 2e-07
jarque.bera.test(gwnMonthlyRetC[, 2])
#>
#>  Jarque Bera Test
#>
#> data:  gwnMonthlyRetC[, 2]
#> X-squared = 108, df = 2, p-value <2e-16
jarque.bera.test(gwnMonthlyRetC[, 3])
```

```
#>
#>  Jarque Bera Test
#>
#> data:  gwnMonthlyRetC[, 3]
#> X-squared = 24, df = 2, p-value = 7e-06
```

The test statistics are slightly different from the direct calculation above due to differences in how the sample skewness and sample kurtosis is computed within the function `jarque.bera.test()`. ■

9.6.1.2 Shipiro Wilks test

To be complete...

9.6.2 Tests for no autocorrelation

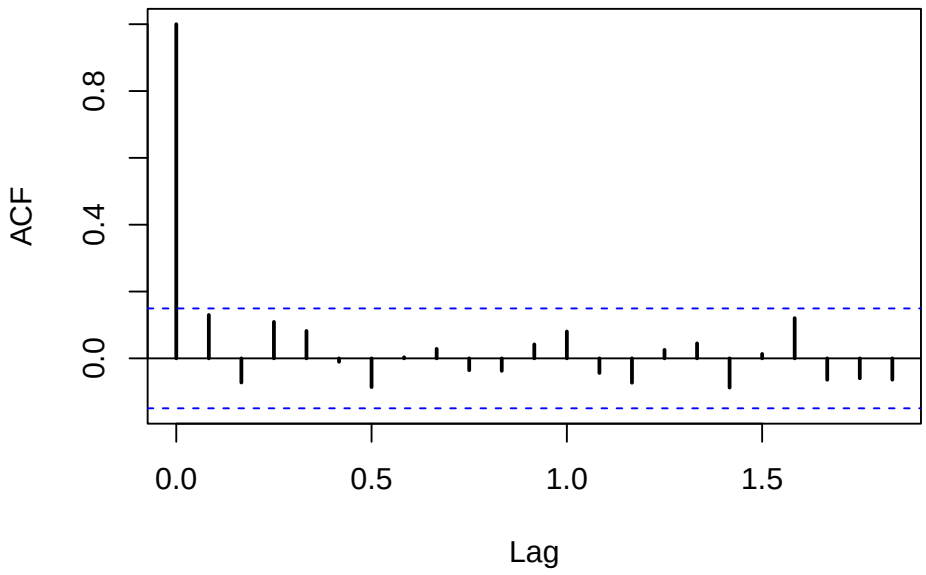


Figure 9.4: SACF for monthly returns on the S&P 500 index.

A key assumption of the GWN model is that returns are uncorrelated over time (not autocorrelated). In efficient financial markets where prices fully reflect true asset value, returns of actively traded assets should not be autocorrelated. In addition, if returns are autocorrelated then they

are predictable. That is, future returns can be predicted from the past history of returns. If this is the case then trading strategies can be easily developed to exploit this predictability. But if many market participants act on these trading strategies then prices will adjust to eliminate any predictability in returns. Hence, for actively traded assets in well functioning markets I expect to see that the returns on these assets are uncorrelated over time.

For less frequently traded assets returns may appear to be autocorrelated due to stale pricing.

As you saw in Chapter Chapter 6, this allows us to aggregate the GWN model over time and, in particular, derive the square-root-of time rule for aggregating return volatility. If returns are substantially correlated over time (i.e., autocorrelated) then the square-root-of-time rule will not be an accurate way to aggregate volatility.

To measure autocorrelation in asset returns $\{R_t\}$ I use the jth lag autocorrelation coefficient

$$\rho_j = \text{cor}(R_t, R_{t-j}) = \frac{\text{cov}(R_t, R_{t-j})}{\text{var}(R_t)},$$

which is estimated using the sample autocorrelation

$$\hat{\rho}_j = \frac{\frac{1}{T-1}\sum_{t=j+1}^{T}(r_t - \hat{\mu})(r_{t-j} - \hat{\mu})}{\frac{1}{T-1}\sum_{t=1}^{T}(r_t - \hat{\mu})^2}.$$

In chapter Chapter 5, I used the sample autocorrelation function (SACF) to graphically evaluate the data evidence on autocorrelation for daily and monthly returns. For example, Figure 9.4 shows the SACF for the monthly returns on the S&P 500 index. Here you see that none of the $\hat{\rho}_j$values are very large. You also see a pair of blue dotted lines on the graph. As you shall see, these dotted lines are related to critical values for testing the null hypothesis that $\rho_j = 0$. Hence, if

9.6.2.1 Individual tests

Individual tests for autocorrelation are similar to coefficient tests discussed earlier and are based on testing the hypotheses

$$H_0 : \rho_j = 0 \; vs. \; H_1 : \rho_j \neq 0,$$

for a given lag j. The tests rely on the result, justified by the CLT if T is large enough, that if returns are covariance stationary and uncorrelated then

$$\hat{\rho}_j \sim N\left(0, \frac{1}{T}\right) \text{ for all } j \geq 1 \text{ and } \text{se}(\hat{\rho}_j) = \frac{1}{\sqrt{T}}.$$

This motivates the test statistics

$$t_{\rho_{j=0}} = \frac{\hat{\rho}_j}{\text{se}(\hat{\rho}_j)} = \frac{\hat{\rho}_j}{1/\sqrt{T}} = \sqrt{T}\hat{\rho}_j, \; j \geq 1$$

and the 95% confidence intervals

$$\hat{\rho}_j \pm 2 \cdot \frac{1}{\sqrt{T}}.$$

If you use a 5% significance level, then you reject the null hypothesis $H_0 : \rho_j = 0$ if

$$|t_{\rho_{j=0}}| = \left|\sqrt{T}\hat{\rho}_j\right| > 2.$$

That is, reject if

$$\hat{\rho}_j > \frac{2}{\sqrt{T}} \text{ or } \hat{\rho}_j < \frac{-2}{\sqrt{T}}.$$

Here, note that the dotted lines on the SACF are at the points $\pm 2 \cdot \frac{1}{\sqrt{T}}$. If the sample autocorrelation $\hat{\rho}_j$ crosses either above or below the dotted lines then you can reject the null hypothesis $H_0 : \rho_j = 0$ at the 5% level.

9.6.2.2 Joint tests

- Have to be careful how to interpret the individual tests. Cannot simply use 20 individual tests at the 5% level to test a joint hypothesis at the 5% level. With independent individual tests each test has a 5% probability of rejecting the null when it is true. So if you do 20 individual tests it would not be surprising if one tests rejects and this could be purely by chance (a type I error) if the null of no autocorrelation was true.

To be complete

9.6.3 Tests for constant parameters

A key assumption of the GWN model is that returns are generated from a covariance stationary time series process. As a result, all of the parameters of the GWN model are constant over time. However, as shown in Chapter Chapter 5 there is evidence to suggest that the assumption of covariance stationarity may not always be appropriate. For example, the volatility of returns for some assets does not appear to be constant over time.

A natural null hypothesis associated with the covariance stationarity assumption for the GWN model for an individual asset return has the form:

$$H_0 : \mu_i, \sigma_i, \rho_{ij} \text{ are constant over time.}$$

To reject this null implied by covariance stationarity, one only needs to find evidence that at least one parameter is not constant over time.

9.6.3.1 Formal tests (two parameter structural change)

To be completed

- Do two-sample structural change tests based on known segments. Analysis is very similar to testing equality of parameters across two assets.

9.6.4 Tests for constant parameters based on rolling estimators

In Chapter Chapter 5 I introduced rolling descriptive statistics to informally investigate the assumption of covariance stationarity. Given any parameter θ of the GWN model (or any function of the parameters), you can compute a set of rolling estimates of θ over windows of length $n < T$: $\{\hat{\theta}_t(n)\}_{t=n}^T$. A time plot of these rolling estimates may show evidence of non-stationary behavior in returns.

However, you must always keep in mind that estimates have estimation error and part of the observed time variation in the rolling estimates $\{\hat{\theta}_t(n)\}_{t=n}^T$ is due to random estimation error. To account for this estimation error, rolling estimates are often displayed with estimated standard error bands of the form:

$$\hat{\theta}_t(n) \pm c \times \widehat{\text{se}}\left(\hat{\theta}_t(n)\right), \tag{9.33}$$

where c is a constant representing some multiple of the estimated standard error. Typically, c is chosen so that Equation 9.33 represents an approximate confidence interval for θ. For example, setting $c = 2$ gives an approximate 95% confidence interval.

Example 9.11. As in Chapter Chapter 5, you can compute 24-month rolling estimates of μ and σ for Microsoft, Starbucks and the S&P 500 index using the **zoo** function `rollapply()`:

```
roll.muhat = rollapply(gwnMonthlyRetC, width = 24, by = 1,
                       by.column = TRUE, FUN = mean, align = "right")
roll.sigmahat = rollapply(gwnMonthlyRetC, width = 24, by = 1,
                          by.column = TRUE, FUN = sd, align = "right")
```

Estimated standard errors for $\hat{\mu}_t(n)$ and $\hat{\sigma}_t(n)$ are computed using:

$$\widehat{\text{se}}(\hat{\mu}_t(n)) = \frac{\hat{\sigma}_t(n)}{\sqrt{T}}, \ \widehat{\text{se}}(\hat{\sigma}_t(n)) = \frac{\hat{\sigma}_t(n)}{\sqrt{2T}}$$

Figure 9.5 shows the rolling estimates of μ (top panel) and σ (bottom panel) for Starbucks together with 95% confidence intervals ($\pm 2\times$ standard error bands) computed using:

```
se.muhat.SBUX = roll.sigmahat[, "SBUX"]/sqrt(24)
se.sigmahat.SBUX = roll.sigmahat[, "SBUX"]/sqrt(2 * 24)
lower.muhat.SBUX = roll.muhat[, "SBUX"] - 2 * se.muhat.SBUX
upper.muhat.SBUX = roll.muhat[, "SBUX"] + 2 * se.muhat.SBUX
lower.sigmahat.SBUX = roll.sigmahat[, "SBUX"] - 2 * se.sigmahat.SBUX
upper.sigmahat.SBUX = roll.sigmahat[, "SBUX"] + 2 * se.sigmahat.SBUX
# plot estimates with standard error bands
dataToPlot = merge(roll.muhat[, "SBUX"], lower.muhat.SBUX,
                   upper.muhat.SBUX, roll.sigmahat[,"SBUX"],
                   lower.sigmahat.SBUX, upper.sigmahat.SBUX)
my.panel <- function(...) {
  lines(...)
  abline(h = 0)
}
plot.zoo(dataToPlot, main = "", screens = c(1, 1, 1, 2, 2, 2),
         lwd = 2, col = c("black","red", "red", "black", "red", "red"),
         lty = c("solid", "dashed", "dashed", "solid", "dashed", "dashed"),
         panel = my.panel, ylab = list("muhat", "sigmahat"))
```

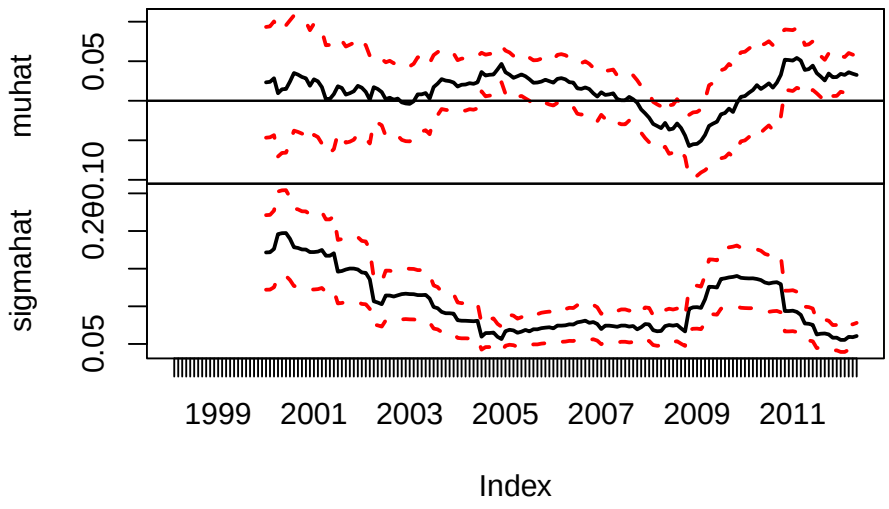


Figure 9.5: 24-month rolling estimates of μ and σ for Starbucks with estimated standard error bands

The standard error bands for $\hat{\mu}_t(n)$ are fairly wide, especially during the high volatility periods of the dot-com boom-bust and the financial crisis. During the dot-com period the 95% confidence intervals for μ are wide and contain both positive and negative values reflecting a high degree of uncertainty about the value of μ. However, during the economic boom around 2005, where $\hat{\mu}_t(n)$ is large and $\hat{\sigma}_t(n)$ is small, the 95% confidence interval for μ does not contain zero and one can infer that the mean is significantly positive. Also, at the height of the financial crisis, where $\hat{\mu}_t(n)$ is the largest negative value, the 95% confidence interval for μ contains all negative values and one can infer that the mean is significantly negative. At the end of the sample, the 95% confidence intervals for μ contain all positive values.[7]

The standard error bands for $\hat{\sigma}_t(n)$ are smaller than the bands for $\hat{\mu}_t(n)$ and one can see more clearly that $\hat{\sigma}_t(n)$ significantly larger during the dot-com period and the financial crisis period than during the economic boom period and the period after the financial crisis. ■

Plotting rolling estimates with estimated standard error bands is good practice as it reminds you that there is estimation error in the rolling estimates and part of any observed time variation is this estimation error.

9.7 Using Monte Carlo Simulation to Understand Hypothesis Testing

In chapter Chapter 7 I used Monte Carlo simulation to understand the statistical properties of estimators. Now, I will use Monte Carlo Simulation to understand hypothesis testing. To fix ideas, let R_t be the return on a single asset described by the GWN model, let θ denote some parameter of the GWN model you are interested in testing an hypothesis about, let $\hat{\theta}$ denote an estimator for θ based on a sample of size T, and let $\widehat{\text{se}}(\hat{\theta})$ denote the estimated standard error for $\hat{\theta}$. Suppose you are interested in testing $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ using the z-score $z_{\theta=\theta_0} = \frac{\hat{\theta}-\theta_0}{\widehat{\text{se}}(\hat{\theta})}$.

To be completed

9.8 Using the Bootstrap for Hypothesis Testing

To be completed

- duality between HT and CIs can be exploited by the bootstrap. Use bootstrap to compute CI and see if hypothesized values lies in the interval. Illustrate with testing equality of 2 Sharpe ratios

[7] You have to be a little careful about interpreting the implicit hypotheses tests underlying the rolling 95% confidence intervals. Each confidence interval represents a single hypothesis test with a 5% significane level. However, we have many confidence intervals and so we are really doing many hypothesis tests. The joint significance level for all of these tests will be larger than the significance level for a single test.

9.9 Further Reading: Hypothesis Testing in the GWN Model

- Mention books by Carol Alexander for discussion of rolling and EWMA estimates
- Formal tests for structural change require advanced probability and statistics. See (Zivot 2016) for discussion. Many tests are implemented in the **strucchange** package.

9.10 Exercises

To be completed…

10 Introduction to Portfolio Theory

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Chapters Chapter 7 through Chapter 9 focused on the development and estimation of, and the statistical inference for, the GWN model for asset returns. In this chapter, I use the GWN model for asset returns as the basis for the quantitative analysis of portfolios and asset allocation. Specifically, I introduce the mean-variance portfolio analysis framework originally developed by Harry Markowitz.[1] This framework is the widely used in academia and industry and is covered in most standard textbooks on investment analysis such as (Bodie, Kane, and Marcus 2013), (Elton et al. 2014), and (Sharpe 1999). In this framework, asset returns are assumed to be normally distributed and so investor preferences over portfolios of assets depend only on the expected return and variance of the portfolio returns. Investors are assumed to like portfolio with high mean returns but dislike portfolio with high return variances. Since portfolios that have high (low) expected returns tend to have high (low) variances investors face a mean-variance tradeoff between different portfolios. Markowitz's mean-variance portfolio theory quantifies this mean-variance tradeoff and provides a methodology that investors can use to determine optimal portfolio choice.

This chapter introduces Markowitz's mean-variance portfolio theory in a simplified setting where there are only two risky assets and a single risk-free asset. This allows us to present the main results of mean-variance portfolio theory graphically and with simple algebra. However, as you shall see in the next chapter, many of the analytic results I derive for portfolios of two risky assets, and for portfolios of two risky assets and a single risk-free asset, can be applied to situations in which there are many risky assets and a single risk-free asset. Therefore, it is highly beneficial to study and understand well the portfolio theory results in the simplified framework of this chapter.

This chapter is organized as follows. In Section 10.2 I review expected return, variance and value-at-risk calculations for portfolios of two risky assets. In Section 10.3 I introduce mean-variance portfolio analysis for portfolios of two risky assets. The portfolio risk-return frontier is introduced, efficient portfolios are defined, and the global minimum variance portfolio is defined. In Section 10.4 I present the analysis of portfolios with a single risky asset and a risk-free asset. I also define the Sharpe ratio/slope of a risky asset and illustrate it graphically. I discuss portfolios of two risky assets and a risk-free asset, define the The tangency portfolio, present the Mutual Fund Separation

[1] The mean-variance frame is described in detail in (Markowitz 1959), (Markowitz 1987). Harry Markowitz was awarded the Nobel Prize in Economics in 1990, along with Merton Miller and William Sharpe, for his work on portfolio theory.

Theorem in Section 10.5 I give a brief illustration of the theory to a real world two asset portfolio of stocks and bonds in Section 10.6. In the Appendix I review some basic results on constrained optimization of functions that I use in the chapter.

The R packages I use in this chapter are **introcompfinr**, **kableExtra**, **knitr**, and **PerformanceAnalytics**. Make sure these packages are installed and loaded in R before replicating the chapter examples.

10.1 Assumptions About Returns and Investor Risk Preferences

To develop a quantitative methodology for investment analysis, I make the following assumptions regarding the probability distribution of asset returns and the behavior of investors:

1. Returns are covariance stationary and ergodic, and jointly normally distributed over the investment horizon. This implies that means, variances and covariances of returns are constant over the investment horizon and completely characterize the joint distribution of returns.
2. Investors know the values of asset return means, variances and covariances.
3. Investors only care about portfolio expected return and portfolio variance. Investors like portfolios with high expected return but dislike portfolios with high return variance.

The first assumption is supported by the stylized facts of asset returns for investment horizons of at least a month as discussed in Chapter Chapter 5. The second assumption can be thought of as assuming that investors have access to high quality information that they can use to determine expected returns, variances, and covariances. For example, investors can download asset price data from `finance.yahoo.com` and compute estimates of expected returns, variances, and covariances using sample statistics. Two points are relevant for the third assumption. First, if the distribution of assets returns is jointly normal then the probability distribution of returns is completely characterized by means, variances, and covariances. Second, the third assumption describes the mean-variance trade-off investors face. All else equal, investors prefer assets that have the highest expected return. However, as you have seen, assets with high expected returns tend to also have high variances. High variance represents more uncertainty about future return and it is assumed that investors do not like uncertainty about future return. That is, I assume that investors are *risk averse*.

10.2 Portfolios of Two Risky Assets

Consider the following investment problem. You can invest in two assets A (Amazon stock) and B (Boeing stock) over the next year, and you want to know how much to invest in each asset to make the investor most happy given the assumptions about return distributions and risk preferences from

the previous sub-section. Let R_A and R_B denote the annual simple returns on assets A and B, respectively.[2] These returns are to be treated as random variables because the returns will not be realized until the end-of-the year. I assume that the returns R_A and R_B are jointly normally distributed, and that I have the following information about the means, variances, and covariances of the probability distribution of the two returns:

$$\mu_A = E[R_A],\ \sigma_A^2 = \text{var}(R_A),\ \mu_B = E[R_B],\ \sigma_B^2 = \text{var}(R_B), \tag{10.1}$$

$$\sigma_{AB} = \text{cov}(R_A, R_B),\ \rho_{AB} = \text{cor}(R_A, R_B) = \frac{\sigma_{AB}}{\sigma_A \sigma_B}. \tag{10.2}$$

I assume that these values are taken as given. In practice, they can be estimated from historical return data for the two stocks, or they can be subjective guesses by the investor.

The expected returns, μ_A and μ_B, are our best guesses for the annual returns on each of the assets. However, because the investment returns are random variables you must recognize that the realized returns may be different from our expectations. The variances, σ_A^2 and σ_B^2, provide measures of the uncertainty associated with these annual returns. You can also think of the variances as measuring the risk associated with the investments. Assets with high return variability (or volatility) are often thought to be risky, and assets with low return volatility are often thought to be safe. The covariance σ_{AB} gives us information about the direction of any linear dependence between returns. If $\sigma_{AB} > 0$ then the two returns tend to move in the same direction; if $\sigma_{AB} < 0$ the returns tend to move in opposite directions; if $\sigma_{AB} = 0$ then the returns tend to move independently. The strength of the dependence between the returns is measured by the correlation coefficient ρ_{AB}. If ρ_{AB} is close to one in absolute value then returns mimic each other extremely closely, whereas if ρ_{AB} is close to zero then the returns may show very little relationship.

Example 10.1 (Two risky asset portfolio information)**.** Table 10.1 gives annual return distribution parameters for the two hypothetical assets A and B that will be used in subsequent examples. Asset A is the high risk asset with an annual return of $\mu_A = 17.5\%$ and annual standard deviation of $\sigma_A = 25.8\%$. Asset B is a lower risk asset with annual return $\mu_B = 5.5\%$ and annual standard deviation of $\sigma_B = 11.5\%$. The assets are assumed to be slightly negatively correlated with correlation coefficient $\rho_{AB} = -0.164$.[3] Given the standard deviations and the correlation, the covariance can be determined from $\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B = (-0.164)(0.0258)(0.115) = -0.004875$. In R, the example data is:[4]

[2] If the stocks pay a dividend then returns are computed as total returns.

[3] The small negative correlation between the two stock returns is a bit unusual, as most stock returns are positively correlated. The negative correlation is assumed so that certain figures produced later look nice and are easy to read.

[4] I use the **kableExtra** function `kbl()` to print out nice formatted tables from a `data.frame` of information.

Table 10.1: Example data for two asset portfolio

\(\mu_{A}\)	\(\mu_{B}\)	\(\sigma^{2}_{A}\)	\(\sigma^{2}_{B}\)	\(\sigma_{A}\)	\(\sigma_{
0.175	0.055	0.067	0.013	0.258	

```
mu.A = 0.175
sig.A = 0.258
sig2.A = sig.A^2
mu.B = 0.055
sig.B = 0.115
sig2.B = sig.B^2
rho.AB = -0.164
sig.AB = rho.AB*sig.A*sig.B
```

■

10.2.1 The Portfolio Problem

The portfolio problem is set-up as follows. The investor has a given amount of initial wealth W_0 to be invested for one period (e.g., one month or one year). The investor can only invest in the two risky assets A and B and all wealth must be invested in the two assets. The investor has to decide how much wealth to put in asset A and how much to put in asset B. Let x_A and x_B denote the shares of wealth invested in assets A and B, respectively. The values of x_A and x_B can be positive or negative. Positive values denote *long* positions (purchases) in the assets. Negative values denote *short* positions (sales).[5] Since all wealth is put into the two investments it follows that $x_A + x_B = 1$. If asset A is shorted, then it is assumed that the proceeds of the short sale are used to purchase more of asset B and vice-versa.

The investment in the two assets forms a *portfolio,* and the shares x_A and x_B are referred to as *portfolio shares* or *portfolio weights*. The return on the portfolio over the next year is a random variable, and is given by:

$$R_p = x_A R_A + x_B R_B, \tag{10.3}$$

which is a linear combination or weighted average of the random variables R_A and R_B. Since R_A and R_B are assumed to be normally distributed, R_p is also normally distributed.

[5]To short an asset one borrows the asset, usually from a broker, and then sells it. The proceeds from the short sale are usually kept on account with a broker and there may be restrictions on the use of these funds for the purchase of other assets. The short position is closed out when the asset is repurchased and then returned to original owner. If the asset drops in value then a gain is made on the short sale and if the asset increases in value a loss is made. See Chapter Chapter 12 for more details on short selling assets

Using the properties of linear combinations of random variables from Chapter Chapter 2 it follows that

$$\mu_p = E[R_p] = x_A\mu_A + x_B\mu_B, \tag{10.4}$$

$$\sigma_p^2 = \text{var}(R_p) = x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_Ax_B\sigma_{AB}, \tag{10.5}$$

$$\sigma_p = \text{SD}(R_p) = \sqrt{x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_Ax_B\sigma_{AB}}. \tag{10.6}$$

Hence, the portfolio return has the following normal distribution:

$$R_p \sim N(\mu_p, \sigma_p^2).$$

Given the portfolio return distribution, end-of-period wealth is

$$W_1 = W_0 \times (1 + R_p) = W_0 + W_0 \times R_p,$$

which is a linear function of the random portfolio return R_p. It follows that

$$\begin{aligned} E[W_1] &= W_0 + W_0 \times E[R_p] = W_0(1 + \mu_p), \\ \text{var}(W_1) &= W_0^2 \times \text{var}(R_p) = W_0^2 \times \sigma_p^2, \end{aligned}$$

so that the distribution of W_1 is normal:

$$W_1 \sim N(W_0(1 + \mu_p), W_0^2 \times \sigma_p^2).$$

The investor cares most about the distribution of end-of-period wealth W_1, which depends on the values of W_0, μ_p, and σ_p^2. Since μ_p and σ_p^2 are functions of x_A and x_B, the portfolio problem is to find the values of x_A and x_B that makes the investor most happy with respect to the probability distribution of end-of-period wealth.[6]

The results Equation 10.4 and Equation 11.29 are so important to portfolio theory that it is worthwhile to review the derivations.[7] For the first result Equation 10.4, you have:

$$E[R_p] = E[x_AR_A + x_BR_B] = x_AE[R_A] + x_BE[R_B] = x_A\mu_A + x_B\mu_B,$$

[6] In formal terms, the portfolio problem is to find the values of x_A and x_B that maximizes the investor's expected utility over end-of-period wealth

[7] See chapter Chapter 2 section Section 2.1.

by the linearity of the expectation operator. For the second result Equation 11.29, you have:

$$\begin{aligned}
\text{var}(R_p) &= E[(R_p - \mu_p)^2] = E[(x_A(R_A - \mu_A) + x_B(R_B - \mu_B))^2] \\
&= E[x_A^2(R_A - \mu_A)^2 + x_B^2(R_B - \mu_B)^2 + 2x_A x_B(R_A - \mu_A)(R_B - \mu_B)] \\
&= x_A^2 E[(R_A - \mu_A)^2] + x_B^2 E[(R_B - \mu_B)^2] + 2x_A x_B E[(R_A - \mu_A)(R_B - \mu_B)] \\
&= x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B \sigma_{AB}.
\end{aligned}$$

The third line uses linearity of the expectations operator, and the fourth line uses the definition of variance and covariance.

Notice that the variance of the portfolio is a weighted average of the variances of the individual assets plus two times the product of the portfolio weights times the covariance between the assets. If the portfolio weights are both positive then a positive covariance will tend to increase the portfolio variance, because both returns tend to move in the same direction, and a negative covariance will tend to reduce the portfolio variance as returns tend to move in opposite directions. Thus finding assets with negatively correlated returns can be very beneficial when forming portfolios because risk, as measured by portfolio standard deviation, can be reduced. What is perhaps surprising is that forming portfolios with positively correlated assets can also reduce risk as long as the correlation is not too large. I illustrate these results with the following examples.

Example 10.2 (Two asset portfolios)**.** Consider creating some portfolios using the asset information in Table 10.1. The first portfolio is an equally weighted portfolio with $x_A = x_B = 0.5$. Using Equation 10.4-Equation 10.6, you have:

$$\begin{aligned}
\mu_{p_1} &= (0.5)\cdot(0.175) + (0.5)\cdot(0.055) = 0.115 \\
\sigma_{p_1}^2 &= (0.5)^2\cdot(0.067) + (0.5)^2\cdot(0.013) \\
&\quad + 2\cdot(0.5)(0.5)(-0.004866) \\
&= 0.01751 \\
\sigma_{p_1} &= \sqrt{0.01751} = 0.1323
\end{aligned}$$

This portfolio has expected return half-way between the expected returns on assets A and B, but the portfolio standard deviation is less than half-way between the asset standard deviations. This reflects risk reduction via *diversification*. Here, diversification means investing wealth in two assets instead of investing wealth in a single asset. In R, the portfolio parameters are computed using:

```
x.A = 0.5
x.B = 0.5
mu.p1 = x.A*mu.A + x.B*mu.B
sig2.p1 = x.A^2 * sig2.A + x.B^2 * sig2.B + 2*x.A*x.B*sig.AB
```

$\mu_{p_{1}}$	$\sigma_{p_{1}}$	$0.5\sigma_{A}+0.5\sigma_{B}$
0.115	0.132	0.186

$\mu_{p_{2}}$	$\sigma_{p_{2}}$	$1.5\sigma_{A}-0.5\sigma_{B}$
0.235	0.4	0.33

```
sig.p1 = sqrt(sig2.p1)
```

Next, consider a long-short portfolio with $x_A = 1.5$ and $x_B = -0.5$. In this portfolio, asset B is sold short and the proceeds of the short sale are used to *leverage* the investment in asset A.[8] The portfolio characteristics are:

$$\begin{aligned}
\mu_{p_2} &= (1.5) \cdot (0.175) + (-0.5) \cdot (0.055) = 0.235 \\
\sigma^2_{p_2} &= (1.5)^2 \cdot (0.067) + (-0.5)^2 \cdot (0.013) \\
&\quad + 2 \cdot (1.5)(-0.5)(-0.004866) \\
&= 0.1604 \\
\sigma_{p_2} &= \sqrt{0.1604} = 0.4005
\end{aligned}$$

This portfolio has both a higher expected return and standard deviation than asset A. The high standard deviation is due to the short sale, which is a type of leverage, and the negative correlation between assets A and B. In R, the portfolio parameters are computed using:

```
x.A = 1.5
x.B = -0.5
mu.p2 = x.A*mu.A + x.B*mu.B
sig2.p2 = x.A^2 * sig2.A + x.B^2 * sig2.B + 2*x.A*x.B*sig.AB
sig.p2 = sqrt(sig2.p2)
```

■

In the above example, the equally weighted portfolio has expected return half way between the expected returns on assets A and B, but has standard deviation (volatility) that is less than half way between the standard deviations of the two assets. For long-only portfolios, you can show that this is a general result as long as the correlation between the two assets is not perfectly positive.

Proposition 10.1. *Consider a portfolio of the two assets A and B with portfolios weights* $x_A \geq 0$ *and* $x_B \geq 0$ *such that* $x_A + x_B = 1$. *Then if* $\rho_{AB} \neq 1$

[8] Leverage refers to an investment that is financed through borrowing. Here, the short sale of asset B produces borrowed funds used to purchase more of asset A. This increases the risk (portfolio standard deviation) of the investment.

$$\sigma_p = (x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B \sigma_{AB})^{1/2} < x_A\sigma_A + x_B\sigma_B.$$

If $\rho_{AB} = 1$ *then*

$$\sigma_p = x_A\sigma_A + x_B\sigma_B.$$

The proof is straightforward. Use $\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B$ and write

$$\sigma_p^2 = x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B\sigma_{AB} = x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B\rho_{AB}\sigma_A\sigma_B.$$

Now add and subtract $2x_A x_B\sigma_A\sigma_B$ from the right-hand side to give

$$\begin{aligned}\sigma_p^2 &= (x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B\sigma_A\sigma_B) - 2x_A x_B\sigma_A\sigma_B + 2x_A x_B\rho_{AB}\sigma_A\sigma_B \\ &= (x_A\sigma_A + x_B\sigma_B)^2 - 2x_A x_B\sigma_A\sigma_B(1-\rho_{AB}).\end{aligned}$$

If $\rho_{AB} \neq 1$ then $2x_A x_B\sigma_A\sigma_B(1-\rho_{AB}) > 0$ and so

$$\sigma_p^2 < (x_A\sigma_A + x_B\sigma_B)^2 \Rightarrow \sigma_p < x_A\sigma_A + x_B\sigma_B.$$

If $\rho_{AB} = 1$ then $2x_A x_B\sigma_A\sigma_B(1-\rho_{AB}) = 0$ and

$$\sigma_p^2 = (x_A\sigma_A + x_B\sigma_B) \Rightarrow \sigma_p = x_A\sigma_A + x_B\sigma_B$$

This result shows that the volatility of a long-only two asset portfolio is a weighted average of individual asset volatility only when the two assets have perfectly positively correlated returns. Hence, in this case there is no risk reduction benefit from forming a portfolio. However, if assets are not perfectly correlated then there can be a risk reduction benefit from forming a portfolio. How big the risk reduction benefit is depends on the magnitude of the asset return correlation ρ_{AB}. In general, the closer ρ_{AB} is to -1 the larger is the risk reduction benefit.

Figure 10.1 shows the risk-return characteristics of assets A and B as well as the equally weighted and long-short portfolios (labeled P1 and P2, respectively). The dotted line connecting the points A and B represents linear risk-return trade-offs of portfolios of assets A and B, respectively, that occurs if $\rho_{AB} = 1$. The point P1, which represents the equally weighted long-only portfolio, has expected return half-way between the expected returns on assets A and B but has standard deviation less than half-way between the standard deviations of assets A and B. As a result, P1 lies to the left of the dotted line indicating a risk reduction benefit to the portfolio. In contrast, the point P2, which represents the long-short portfolio, lies to the right of the dotted line indicating a risk inflation cost of the portfolio due to leverage.

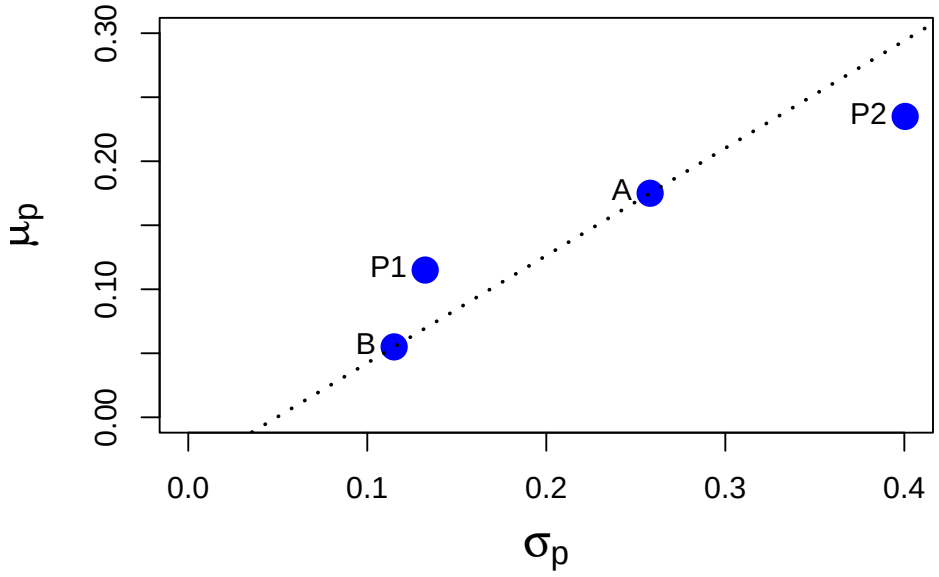


Figure 10.1: Risk-return characteristics of assets A and B, the equally weighted portfolio P1 and the long-short portfolio P2.

10.2.2 Portfolio Value-at-Risk

Consider an initial investment of $\$W_0$ in the portfolio of assets A and B with return given by Equation 10.3, expected return given by Equation 10.4) and variance given by Equation 11.29. Then $R_p \sim N(\mu_p, \sigma_p^2)$. For $\alpha \in (0,1)$, the $\alpha \times 100\%$ portfolio value-at-risk is given by:

$$\text{VaR}_{p,\alpha} = -q_\alpha^{R_p} W_0, \tag{10.7}$$

where $q_\alpha^{R_p}$ is the α-quantile of the distribution of R_p and is given by

$$q_\alpha^{R_p} = \mu_p + \sigma_p q_\alpha^z, \tag{10.8}$$

where q_α^z is the α-quantile of the standard normal distribution.[9]

What is the relationship between portfolio VaR and the individual asset VaRs? Is portfolio VaR a weighted average of the individual asset VaRs? In general, portfolio VaR is not a weighted average of the asset VaRs. To see this consider the portfolio weighted average of the individual asset return quantiles:

[9] If R_p is a continuously compounded return then the implied simple return quantile is $q_\alpha^{R_p} = \exp(\mu_p + \sigma_p q_\alpha^z) - 1$.

$$
\begin{aligned}
x_A q_\alpha^{R_A} + x_B q_\alpha^{R_B} &= x_A(\mu_A + \sigma_A q_\alpha^z) + x_B(\mu_B + \sigma_B q_\alpha^z) \\
&= x_A\mu_A + x_B\mu_B + (x_A\sigma_A + x_B\sigma_B)q_\alpha^z \\
&= \mu_p + (x_A\sigma_A + x_B\sigma_B)q_\alpha^z.
\end{aligned} \tag{10.9}
$$

The weighted asset quantile Equation 10.9 is not equal to the portfolio quantile Equation 10.8 unless $\rho_{AB} = 1$. Hence, weighted asset VaR is in general not equal to portfolio VaR because the quantile Equation 10.9 assumes a perfectly positive correlation between R_A and R_B.

Example 10.3 (Portfolio VaR)**.** Consider an initial investment of $W_0 = \$100,000$. Assuming that returns are simple, the 5% VaRs on assets A and B are:

$$
\begin{aligned}
\text{VaR}_{A,0.05} &= -q_{0.05}^{R_A} W_0 = -\left(\mu_A + \sigma_A q_{.05}^z\right) W_0 \\
&= -(0.175 + 0.258(-1.645)) \cdot 100,000 = 24,937, \\
\text{VaR}_{B,0.05} &= -q_{0.05}^{R_B} W_0 = -\left(\mu_B + \sigma_B q_{.05}^z\right) W_0 \\
&= -(0.055 + 0.115(-1.645)) \cdot 100,000 = 13,416.
\end{aligned}
$$

The 5% VaR on the equal weighted portfolio with $x_A = x_B = 0.5$ is:

$$
\begin{aligned}
\text{VaR}_{p,0.05} &= -q_{0.05}^{R_p} W_0 = -\left(\mu_p + \sigma_p q_{.05}^z\right) W_0 \\
&= -(0.115 + 0.1323(-1.645)) \cdot 100,000 = 10,268,
\end{aligned}
$$

and the weighted average of the individual asset VaRs is,

$$
x_A\text{VaR}_{A,0.05} + x_B\text{VaR}_{B,0.05} = 0.5(24,937) + 0.5(13,416) = 19,177.
$$

The 5% VaR on the long-short portfolio with $x_A = 1.5$ and $x_B = -0.5$ is:

$$
\text{VaR}_{p,0.05} = -q_{0.05}^{R_p} W_0 = -(0.235 + 0.4005(-1.645)) \cdot 100,000 = 42,371,
$$

and the weighted average of the individual asset VaRs is,

$$
x_A\text{VaR}_{A,0.05} + x_B\text{VaR}_{B,0.05} = 1.5(24,937) - 0.5(13,416) = 30,698.
$$

Notice that $\text{VaR}_{p,0.05} \neq x_A\text{VaR}_{A,0.05} + x_B\text{VaR}_{B,0.05}$ because $\rho_{AB} \neq 1$.

Using R, these computations are:

$VaR_{A,0.05}$	$VaR_{B,0.05}$	$VaR_{p_{1},0.05}$	$VaR_{p_{2},0.05}$
24937	13416	10268	42371

$0.5VaR_{A,0.05}+0.5VaR_{B,0.05}$	$1.5VaR_{A,0.05}-0.5VaR_{B,0.05}$
19177	30698

```
w0 = 100000
VaR.A = -(mu.A + sig.A*qnorm(0.05))*w0
VaR.B = -(mu.B + sig.B*qnorm(0.05))*w0
VaR.p1 = -(mu.p1 + sig.p1*qnorm(0.05))*w0
VaR.p2 = -(mu.p2 + sig.p2*qnorm(0.05))*w0
```

■

Example 10.4 (Create R function to compute normal VaR)**.** The previous example used repetitive R calculations to compute the 5% VaR for a variety of investments. An alternative approach is to create an R function to compute the VaR given μ, σ, α (VaR probability) and W_0, and then apply the function using the inputs of the different assets. A simple function to compute VaR based on normally distributed asset returns is:

```
normalVaR = function(mu, sigma, w0, alpha = 0.05, invert=TRUE) {
## compute normal VaR for collection of assets given mean and sd vector
## inputs:
## mu          n x 1 vector of expected returns
## sigma       n x 1 vector of standard deviations
## w0          scalar initial investment in $
## alpha       scalar tail probability
## invert      logical. If TRUE report VaR as positive number
## output:
## VaR         n x 1 vector of left tail return quantiles
  if ( length(mu) != length(sigma) )
    stop("mu and sigma must have same number of elements")
  if ( alpha < 0 || alpha > 1)
    stop("alpha must be between 0 and 1")
  VaR = w0*(mu + sigma*qnorm(alpha))
  if (invert) {
    VaR = -VaR
  }
  return(VaR)
}
```

You create an R function using `function()`. Every R function has a set of arguments or inputs. Here, the arguments are the $n \times 1$ vectors `mu` and `sigma`, the scalars `w0` and `alpha`, and the logical value `invert`. Default values for `alpha` and `invert` are specified in the definition of the function. The body of the function is defined within the braces `{}`, and the return value of the function is specified using `return()`. Here, you first check that the inputs `mu` and `sigma` are of the same length, and that the tail probability α is between 0 and 1. Then, you compute normal VaR using Equation 10.7. If `invert=TRUE` you multiply the result by -1. Finally, you return the vector of VaR values. After running the code that defines the function `normalVaR()`, it will be available in the R session for use like any other R function.

Using the `normalVaR()` function, the 5% VaR values of asset A, B and equally weighted portfolio are:

```
normalVaR(mu=c(mu.A, mu.B, mu.p1),
          sigma=c(sig.A, sig.B, sig.p1),
          w0=100000, alpha=0.05)
#> [1] 24937 13416 10268
```

■

10.2.3 The set of feasible portfolios

The collection of all *feasible* portfolios, or the *investment possibilities set*, in the case of two assets is simply all possible portfolios that can be formed by varying the portfolio weights x_A and x_B such that the weights sum to one ($x_A + x_B = 1$). I summarize the expected return-risk (mean-volatility) properties of the feasible portfolios in a plot with portfolio expected return, μ_p, on the vertical axis and portfolio standard deviation, σ_p, on the horizontal axis. The portfolio standard deviation is used instead of variance because standard deviation is measured in the same units as the expected value (recall, variance is the average squared deviation from the mean).

Example 10.5 (Investment possibilities set for example data)**.** The investment possibilities set for the data in Table 10.1 is illustrated in Figure 10.2. Here the portfolio weight on asset A, x_A, is varied from -0.4 to 1.4 in increments of 0.1 and, since $x_B = 1 - x_A$, the weight on asset B then varies from 1.4 to -0.4. This gives us 18 portfolios with weights $(x_A, x_B) = (-0.4, 1.4), (-0.3, 1.3), ..., (1.3, -0.3), (1.4, -0.4)$. For each of these portfolios I use the formulas Equation 10.4 and Equation 10.6 to compute μ_p and σ_p. In R, the calculations are:

```
x.A = seq(from=-0.4, to=1.4, by=0.1)
x.B = 1 - x.A
mu.p = x.A*mu.A + x.B*mu.B
```

	x_{A}	x_{B}	μ_{p}	σ_{p}
portfolio 1	-0.4	1.4	0.007	0.205
portfolio 2	-0.3	1.3	0.019	0.179
portfolio 3	-0.2	1.2	0.031	0.155
portfolio 4	-0.1	1.1	0.043	0.133
portfolio 5	0.0	1.0	0.055	0.115
portfolio 6	0.1	0.9	0.067	0.102
portfolio 7	0.2	0.8	0.079	0.098
portfolio 8	0.3	0.7	0.091	0.102
portfolio 9	0.4	0.6	0.103	0.114
portfolio 10	0.5	0.5	0.115	0.132
portfolio 11	0.6	0.4	0.127	0.154
portfolio 12	0.7	0.3	0.139	0.178
portfolio 13	0.8	0.2	0.151	0.204
portfolio 14	0.9	0.1	0.163	0.231
portfolio 15	1.0	0.0	0.175	0.258
portfolio 16	1.1	-0.1	0.187	0.286
portfolio 17	1.2	-0.2	0.199	0.314
portfolio 18	1.3	-0.3	0.211	0.343
portfolio 19	1.4	-0.4	0.223	0.372

```
sig2.p = x.A^2 * sig2.A + x.B^2 * sig2.B + 2*x.A*x.B*sig.AB
sig.p = sqrt(sig2.p)
port.names = paste("portfolio", 1:length(x.A), sep=" ")
data.tbl = as.data.frame(cbind(x.A, x.B, mu.p, sig.p))
rownames(data.tbl) = port.names
col.names = c("$x_{A}$","$x_{B}$", "$\\mu_{p}$", "$\\sigma_{p}$" )
tbl.data = kbl(data.tbl, col.names=col.names) %>%
   kable_styling(full_width=FALSE)
cat(tbl.data)
```

Portfolios 1-4 and 16-19 are the long-short portfolios and portfolios 5-15 are the long-only portfolios. In R, the risk-return properties of this set of feasible portfolios can be visualized using:

```
cex.val = 1.5
plot(sig.p, mu.p, type="b", pch=16, cex=cex.val,
     ylim=c(0, max(mu.p)), xlim=c(0, max(sig.p)),
     xlab=expression(sigma[p]), ylab=expression(mu[p]),
     cex.lab=1.5, col=c(rep("blue", 4), rep("black", 2),
                        "green", rep("black", 8), rep("blue", 4)))
text(x=sig.A, y=mu.A, labels="Asset A", pos=4)
text(x=sig.B, y=mu.B, labels="Asset B", pos=4)
```

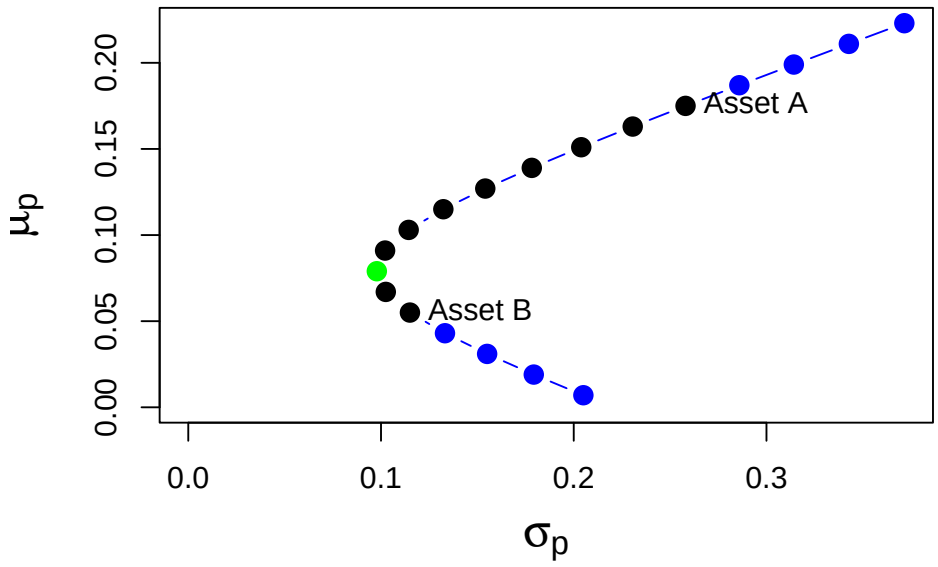


Figure 10.2: Portfolio frontier of example data.

■

Notice that the plot in (μ_p, σ_p)-space looks like a parabola turned on its side (in fact, it is one side of a hyperbola). The plot is often referred to as the *Markowitz bullet* as it resembles the shape of a bullet speeding toward the y-axis. The black dots plus the green dot represent the long-only portfolios, and the blue dots represent long-short portfolios. The black dot labeled "Asset A" is the portfolio $(x_A = 1, x_B = 0)$, and the black dot labeled "Asset B" is the portfolio $(x_A = 0, x_B = 1)$.

The bullet shape of the feasible set shows that portfolio risk can be reduced through diversification. Risk reduction through diversification can be understood as follows. Suppose an investor is initially invested 100% in asset B (i.e., portfolio 5). Diversification involves rebalancing the portfolio to include asset A. For example, portfolio 6 is $(x_A = 0.1, x_B = 0.9)$. This portfolio has both a higher expected return and a lower standard deviation than portfolio 5. This is the benefit from diversification. Portfolio expected return increases and risk decreases until portfolio 7 is reached, which is at the tip of the Markowitz bullet (green dot). This portfolio has the property that it has the smallest variance/volatility among all feasible portfolios. Accordingly, this portfolio is called the *global minimum variance portfolio*. Continuing from portfolio 8 through portfolio 19, you see that both expected return and risk increase.

10.2.4 Computing the global minimum variance portfolio

The global minimum variance portfolio plays a key role in mean-variance portfolio theory and it is important to know how to determine its weights. Here, I show that it is a simple exercise in calculus to find the global minimum variance portfolio weights. I solve the *constrained optimization* problem[10]:

$$\begin{aligned} \min_{x_A, x_B} \sigma_p^2 &= x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B \sigma_{AB} \\ s.t.\ x_A + x_B &= 1. \end{aligned}$$

This constrained optimization problem can be solved using two methods. The first method, called the *method of substitution*, uses the constraint to substitute out one of the variables to transform the constrained optimization problem in two variables into an unconstrained optimization problem in one variable. The second method, called the *method of Lagrange multipliers*, introduces an auxiliary variable called the *Lagrange multiplier* and transforms the constrained optimization problem in two variables into an unconstrained optimization problem in three variables.

The substitution method is straightforward. Substituting $x_B = 1 - x_A$ into the formula for σ_p^2 reduces the problem to:

$$\min_{x_A} \sigma_p^2 = x_A^2\sigma_A^2 + (1 - x_A)^2\sigma_B^2 + 2x_A(1 - x_A)\sigma_{AB}.$$

The first order conditions for a minimum, via the chain rule, are:

$$0 = \frac{d\sigma_p^2}{dx_A} = 2x_A^{\min}\sigma_A^2 - 2(1 - x_A^{\min})\sigma_B^2 + 2\sigma_{AB}(1 - 2x_A^{\min}),$$

and straightforward calculations yield,

$$x_A^{\min} = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}},\ x_B^{\min} = 1 - x_A^{\min}. \tag{10.10}$$

The method of Lagrange multipliers involves two steps. In the first step, the constraint $x_A + x_B = 1$ is put into homogenous form $x_A + x_B - 1 = 0$. In the second step, the Lagrangian function is formed by adding to σ_p^2 the homogenous constraint multiplied by an auxiliary variable λ (the Lagrange multiplier) giving:

$$L(x_A, x_B, \lambda) = x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B\sigma_{AB} + \lambda(x_A + x_B - 1).$$

[10] A review of optimization and constrained optimization is given in the appendix to this chapter.

This function is then minimized with respect to x_A, x_B, and λ. The first order conditions are:

$$\begin{aligned}
0 &= \frac{\partial L(x_A, x_B, \lambda)}{\partial x_A} = 2x_A\sigma_A^2 + 2x_B\sigma_{AB} + \lambda, \\
0 &= \frac{\partial L(x_A, x_B, \lambda)}{\partial x_B} = 2x_B\sigma_B^2 + 2x_A\sigma_{AB} + \lambda, \\
0 &= \frac{\partial L(x_A, x_B, \lambda)}{\partial \lambda} = x_A + x_B - 1.
\end{aligned}$$

The first two equations can be rearranged to give:

$$x_B = x_A\left(\frac{\sigma_A^2 - \sigma_{AB}}{\sigma_B^2 - \sigma_{AB}}\right).$$

Substituting this value for x_B into the third equation and rearranging gives the solution Equation 10.10.

Example 10.6 (Global minimum variance portfolio for example data)**.** Using the data in Table 10.1 and Equation 10.10 you have:

$$x_A^{\min} = \frac{0.01323 - (-0.004866)}{0.06656 + 0.01323 - 2(-0.004866)} = 0.2021, \; x_B^{\min} = 0.7979.$$

Hence, the global minimum variance portfolio is essentially portfolio 7 in Figure 10.2 and indeed lies at the tip of the Markowitz bullet.

The expected return, variance and standard deviation of this portfolio are:

$$\begin{aligned}
\mu_p &= (0.2021)\cdot(0.175) + (0.7979)\cdot(0.055) = 0.07925 \\
\sigma_p^2 &= (0.2021)^2\cdot(0.067) + (0.7979)^2\cdot(0.013) \\
&\quad + 2\cdot(0.2021)(0.7979)(-0.004875) \\
&= 0.00975 \\
\sigma_p &= \sqrt{0.00975} = 0.09782.
\end{aligned}$$

In R, the calculations to compute the global minimum variance portfolio weights and its expected return and volatility are:

```
xA.min = (sig2.B - sig.AB)/(sig2.A + sig2.B - 2*sig.AB)
xB.min = 1 - xA.min
mu.p.min = xA.min*mu.A + xB.min*mu.B
```

$x_{A}^{\min}$	$x_{B}^{\min}$	$\mu_{p}^{\min}$	$\sigma_{p}^{\min}$
0.202	0.798	0.079	0.098

```
sig2.p.min = xA.min^2 * sig2.A + xB.min^2 * sig2.B + 2*xA.min*xB.min*sig.AB
sig.p.min = sqrt(sig2.p.min)
data.tbl = as.data.frame(t(c(xA.min, xB.min, mu.p.min, sig.p.min)))
col.names = c("$x_{A}^{\\min}$","$x_{B}^{\\min}$", "$\\mu_{p}^{\\min}$",
              "$\\sigma_{p}^{\\min}$" )
tbl.data = kbl(data.tbl, col.names=col.names) %>%
   kable_styling(full_width=FALSE)
cat(tbl.data)
```

■

10.2.5 Correlation and the shape of the portfolio frontier

The shape of the investment possibilities set is very sensitive to the correlation between assets A and B given the other parameters. I illustrate this sensitivity by computing portfolio frontiers for the example data for $\rho_{AB} = -0.9, -0.5, 0, 0.5, 0.9$. Figure 10.3 shows these portfolio frontiers.

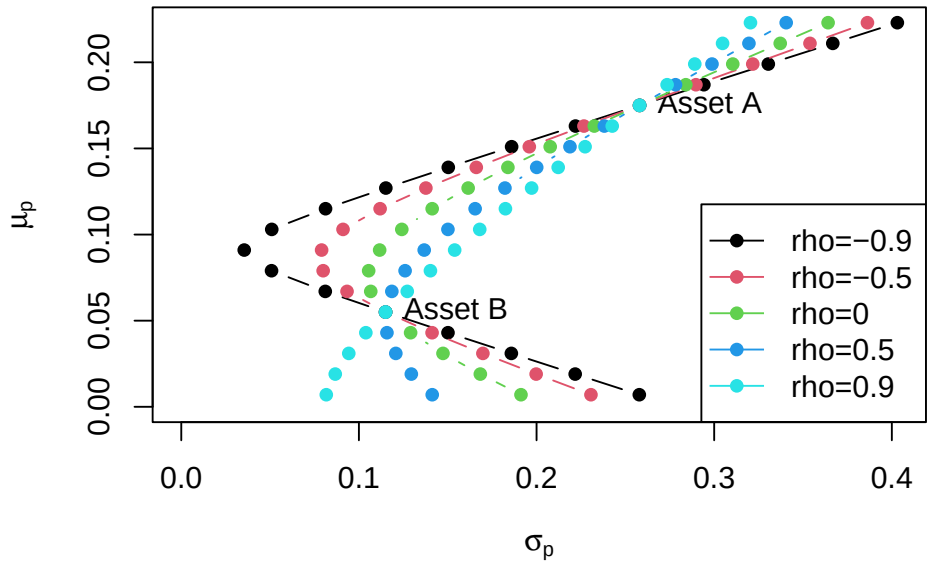


Figure 10.3: Portfolio frontier as a function of correlation.

The curvature of the portfolio frontier is determined by the value of ρ_{AB}. The closer ρ_{AB} is to -1 the more curved is the frontier toward the y-axis and the higher is the possible diversification benefit. What is perhaps surprising from Figure 10.3 is that there is noticeable curvature even for positive values of ρ_{AB}.

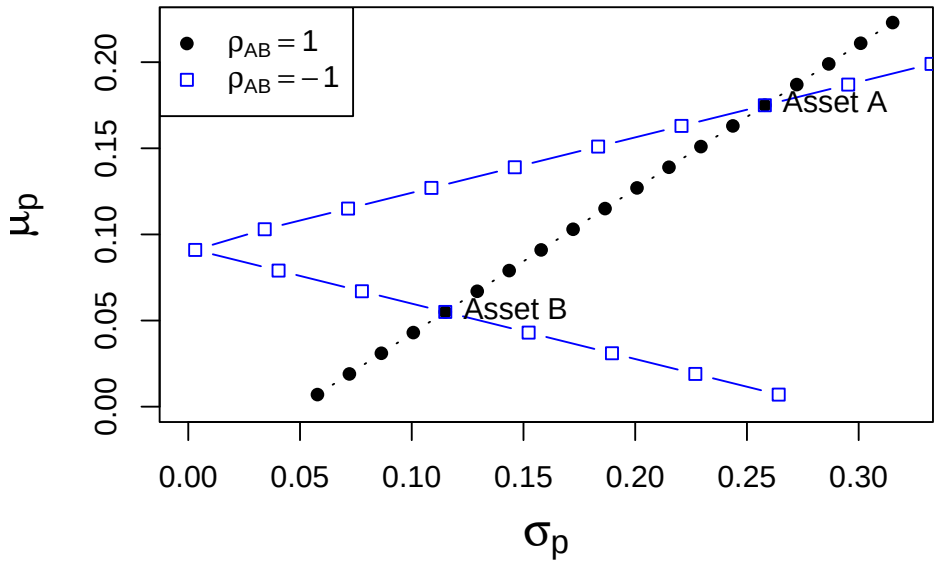


Figure 10.4: Portfolios with $\rho_{AB} = 1$ and $\rho_{AB} = -1$.

If ρ_{AB} is close to 1 then the portfolio frontier approaches a straight line connecting the portfolio with all wealth invested in asset B, $(x_A, x_B) = (0, 1)$, to the portfolio with all wealth invested in asset A, $(x_A, x_B) = (1, 0)$. If $\rho_{AB} = -1$ then the set actually touches the μ_p axis. What this means is that if assets A and B are perfectly negatively correlated then there exists a portfolio of A and B that has positive expected return and zero variance! These cases are illustrated in Figure 10.4.

To find the portfolio with $\sigma_p^2 = 0$ when $\rho_{AB} = -1$ I use Equation 10.10 and the fact that $\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B$ to give:

$$x_A^{\min} = \frac{\sigma_B}{\sigma_A + \sigma_B}, \; x_B^{\min} = 1 - x_A.$$

10.3 Efficient portfolios with two risky assets

Given the assumptions on returns and investor risk preferences, I now characterize the set of *efficient portfolios* and *inefficient portfolios*.

Definition 10.1 (Efficient Portfolios)**.** *Efficient portfolios* are the feasible portfolios that have the highest expected return for a given level of risk as measured by portfolio standard deviation. These are the portfolios that investors are most interested in holding. Graphically, these are the portfolios starting with the global minimum variance portfolio that are above and to the right of the global minimum variance portfolio.

Definition 10.2 (Inefficient Portfolios)**.** *Inefficient portfolios* are the portfolios such that there is another feasible portfolio that has the same risk but a higher expected return. Graphically, these are the portfolios that are below and to the right of the global minimum variance portfolio.

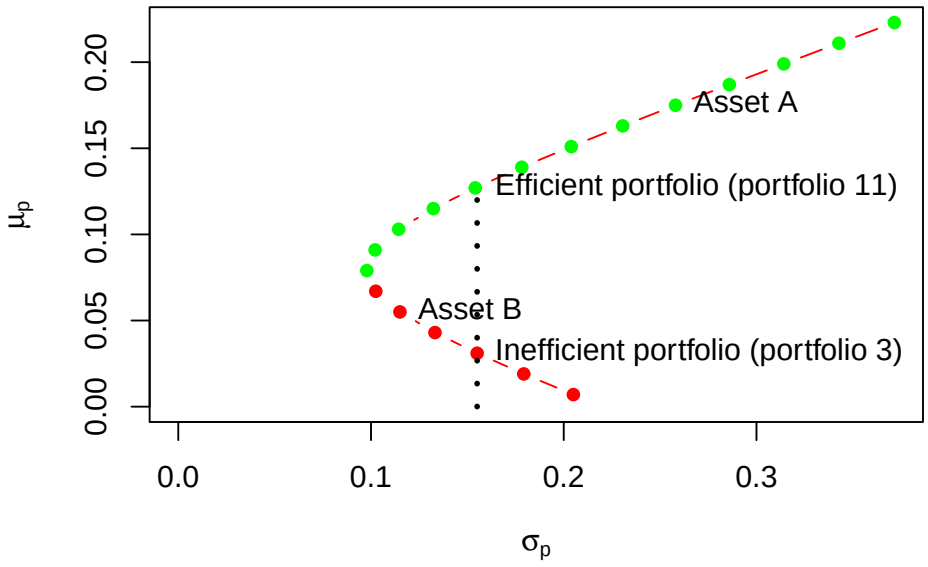


Figure 10.5: Efficient and inefficient portfolios for example data.

Example 10.7 (Efficient and inefficient portfolios for example data)**.** Figure 10.5 shows the efficient and inefficient portfolios from the example data in Table 10.1. The efficient portfolios are those that have the highest expected return for a given standard deviation value. These portfolios are the green dots starting with the global minimum variance portfolio at the tip of the Markowitz bullet. The inefficient portfolios are the red dots below the global minimum variance portfolio. For example, Figure 10.5 shows two feasible portfolios, portfolio 3 and portfolio 11, with the same standard deviation value but different expected return values. Portfolio 11 has the highest expected return and so is the efficient portfolio. Portfolio 3, with the lower expected return, is the inefficient portfolio.

■

10.3.1 Optimal portfolios

Given the efficient set of portfolios as described in Figure 10.2, which portfolio will an investor choose? Of the efficient portfolios, investors will choose the one that accords with their *risk preferences*. Very risk averse investors will want a portfolio that has low volatility (risk) and will choose a portfolio very close to the global minimum variance portfolio. In contrast, very risk tolerant investors will ignore volatility and seek portfolios with high expected returns. Hence, these investors will choose portfolios with large amounts of asset A which may involve short-selling asset B.

10.4 Efficient portfolios with a risk-free asset

In the preceding section I constructed the efficient set of portfolios involving two risky assets. Now I consider what happens when you introduce a *risk-free* asset. In the present context, a risk-free asset is equivalent to a default-free pure discount bond that matures at the end of the assumed investment horizon.[11] The *risk-free rate*, r_f, is then the nominal return on the bond. For example, if the investment horizon is one month then the risk-free asset is a 30-day U.S. Treasury bill (T-bill) and the risk-free rate is the nominal rate of return on the T-bill.[12] If our holdings of the risk-free asset is positive then you are "lending money" to the government at the risk-free rate, and if our holdings are negative then you are "borrowing" from the government at the risk-free rate.

The assumption that investors can borrow and lend at the same risk-free rate is unrealistic in the real world due to differing credit worthiness of investors. Investors can certainly buy T-Bills and receive the risk-free rate. However, most investors can only borrow funds at rates much higher than the T-Bill rate.

10.4.1 Portfolios with one risky asset and one risk-free asset

In this sub-section, I consider portfolios of a single risky asset with random return $R \sim N(\mu, \sigma^2)$ and a risk-free asset with non-random return r_f. Since the risk-free rate is fixed over the investment horizon it is not a random variable. As a result, it has some special properties summarized in the following Proposition.

[11] In the finance industry the risk-free asset is often referred to as "cash". In general, "cash" doesn't refer to holding dollars in your pocket but rather investing dollars in a very low risk asset that pays a stable rate of return.

[12] The default-free assumption of U.S. debt is often taken for granted. However, this assumption has been questioned due to the inability of the U.S. congress to address the long-term debt problems of the U.S. government.

Proposition 10.2 (Properties of Risk-Free Rate)**.** *Let r_f denote the fixed non-random risk-free rate, and let R denote the random return on a risky asset with $E[R] = \mu$ and $\text{var}(R) = \sigma^2$. Then*

$$\begin{aligned} E[r_f] &= r_f, \\ \text{var}(r_f) &= E[(r_f - E[r_f])^2] = 0, \\ \text{cov}(R, r_f) &= E[(r_f - E[r_f])(R - E[R])] = 0. \end{aligned}$$

Consider an investment in the risky asset and the risk-free asset (henceforth referred to as a T-bill). Let x denote the share of wealth in the risky asset, and x_f denote the share of wealth in T-bills such that $x + x_f = 1$. Using $x_f = 1 - x$, the portfolio return can be written as:

$$R_p = x_f r_f + xR = (1 - x)r_f + xR = r_f + x(R - r_f). \tag{10.11}$$

The quantity $R - r_f$ is called the *excess return* (over the return on T-bills) on the risky asset. The portfolio expected return is then:

$$\mu_p = r_f + x(E[R] - r_f) = r_f + x(\mu - r_f), \tag{10.12}$$

where the quantity $(\mu - r_f)$ is called the *expected excess return* or *risk premium* on the risky asset. The risk premium is typically positive indicating that investors expect a higher return on the risky asset than the safe asset (otherwise, why would investors hold the risky asset?). You may express the risk premium on the portfolio in terms of the risk premium on the risky asset:

$$\mu_p - r_f = x(\mu - r_f).$$

The more you invest in the risky asset the higher the risk premium on the portfolio of T-Bills and the risky asset.

Because the risk-free rate is constant, the portfolio variance only depends on the variability of the risky asset and is given by:

$$\sigma_p^2 = \text{var}(\text{R}_\text{p}) = \text{var}(\text{xR}) = x^2\sigma^2.$$

The portfolio standard deviation is therefore proportional to the standard deviation on the risky asset:[13]

$$\sigma_p = |x|\sigma. \tag{10.13}$$

Example 10.8 (Portfolios of T-Bills and asset B for example data)**.** To illustrate the risk-return properties of portfolios of a T-Bill and a single risky asset, consider forming portfolios of T-Bills and

[13] Here, you can allow x to be positive or negative but σ is always positive.

asset B. Let x_B and $1-x_B$ denote the shares of wealth in asset B and T-Bills, respectively. Assume the T-Bill rate is $r_f = 0.03$. Then the risk premium on asset B is $\mu_B - r_r = 0.055 - 0.03 = 0.025$, the expected return on portfolios of T-Bills and asset B is $\mu_p = 0.03 + x_B \times 0.025$, and the portfolio standard deviation is $\sigma_p = |x_B| \times 0.115$. Figure 10.6 plots μ_p vs. σ_p for portfolios with $x_B = -1.4, -1.3, \ldots 1.4$.

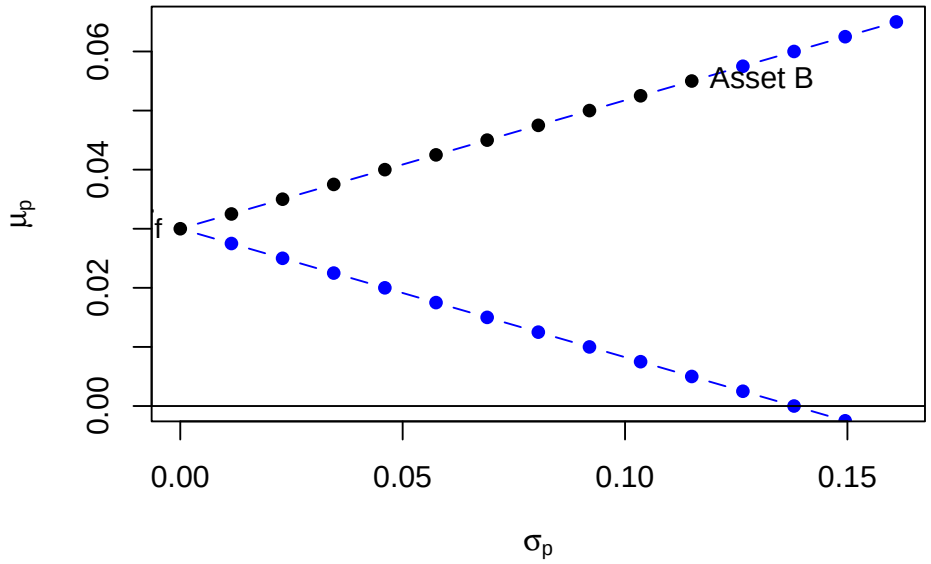


Figure 10.6: Portfolios of T-Bills and asset B for example data.

Notice that the set of feasible portfolios lie on two straight lines, one with a positive slope and one with a negative slope. The line with a positive slope represents portfolios with long positions in asset B, and the line with a negative slope represents portfolios with short positions in asset B. The blue dots represent portfolios with short positions and the black dots represent long-only portfolios. The blue dots above the point labeled "Asset B" represent long-short portfolios where the T-Bill is shorted to finance additional investment in asset B. ■

If you assume that $x > 0$ (do not short the risky asset), you can use Equation 10.13 to solve for x:

$$x = \frac{\sigma_p}{\sigma}. \tag{10.14}$$

Substituting Equation 10.14 into Equation 10.12 gives the following equation:

$$\mu_p = r_f + \frac{\mu - r_f}{\sigma} \cdot \sigma_p, \tag{10.15}$$

Equation Equation 10.15 describes a straight line in (μ_p, σ_p)-space with intercept r_f and slope $(\mu - r_f)/\sigma$. This line characterizes the linear risk-return tradeoff of portfolios of T-Bills and a risky asset when the risky asset is not shorted. This line has a positive slope provided the risk premium is positive.

This line described by Equation 10.15 is often called the *capital allocation line* (CAL) as it describes how investment capital is allocated between T-Bills and the risky asset. Points on the line close to the y-axis represent portfolios mostly allocated to T-Bills, and points far away from the y-axis represent portfolios mostly allocated to the risky asset. The slope of the CAL is called the *Sharpe ratio* (SR) (named after the economist William Sharpe), and it measures the risk premium on the asset per unit of risk (as measured by the standard deviation of the asset). From Equation 10.15 you get

$$\frac{d\mu_p}{d\sigma_p} = \frac{\mu - r_f}{\sigma} = SR.$$

Example 10.9 (Portfolios of T-Bills and risky assets for example data)**.** Consider creating portfolios of asset A and T-bills and asset B and T-Bills with $r_f = 0.03$:

```
r.f = 0.03
x.A = seq(from=0, to=1.4, by=0.1)
mu.p.A = r.f + x.A*(mu.A - r.f)
sig.p.A = x.A*sig.A
x.B = seq(from=0, to=1.4, by=0.1)
mu.p.B = r.f + x.B*(mu.B - r.f)
sig.p.B = x.B*sig.B
```

The risk-return properties of these portfolios are illustrated in Figure 10.7.

In Figure 10.7 the point labeled "r_f" represents a portfolio that is 100% invested in T-bills. The points labeled "Asset A" and "Asset B" represent portfolios that are 100% invested in assets A and B, respectively. A point half-way between "r_f" and "Asset A" is a portfolio that is 50% invested in T-bills and 50% invested in asset A. A point above and to the right of "Asset A" is a levered portfolio where more than 100% of asset A is financed by borrowing at the T-bill rate. Notice that as leverage increases (i.e, borrowing more at the T-Bill rate), portfolio volatility increases.

The portfolios which are combinations of asset A and T-bills have expected returns uniformly higher than the portfolios consisting of asset B and T-bills. This occurs because the Sharpe ratio for asset A is higher than the ratio for asset B:

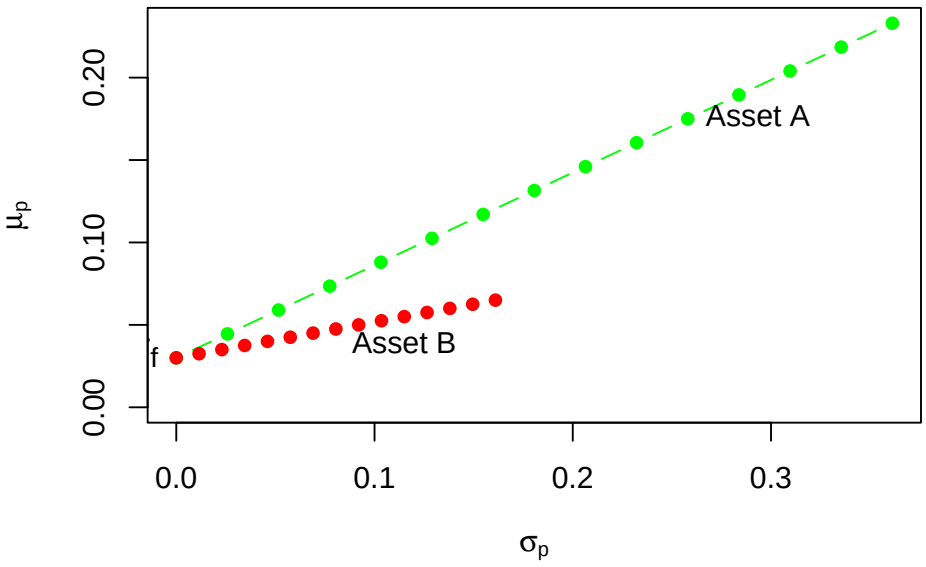


Figure 10.7: Portfolios of T-Bills and risky assets.

$$\text{SR}_A = \frac{\mu_A - r_f}{\sigma_A} = \frac{0.175 - 0.03}{0.258} = 0.562,$$
$$\text{SR}_B = \frac{\mu_B - r_f}{\sigma_B} = \frac{0.055 - 0.03}{0.115} = 0.217.$$

Hence, portfolios of asset A and T-bills are *efficient* relative to portfolios of asset B and T-bills. ■

The previous example shows that the Sharpe ratio can be used to rank the risk return properties of individual assets. Assets with a high Sharpe ratio have a better risk-return tradeoff than assets with a low Sharpe ratio. Accordingly, investment analysts routinely rank assets based on their Sharpe ratios.

10.4.2 Portfolios with one risky asset and one risk-free asset with different borrowing and lending rates

TBD

10.5 Efficient portfolios with two risky assets and a risk-free asset

Now I expand on the previous results by allowing our investor to form portfolios of assets A, B and T-bills. The efficient set of portfolios in this case will still be a straight line in (μ_p, σ_p)-space with intercept r_f. The slope of the efficient set, the maximum Sharpe ratio, is such that it is tangent to the portfolio frontier constructed just using the two risky assets A and B. Figure 10.8 illustrates why this is so.

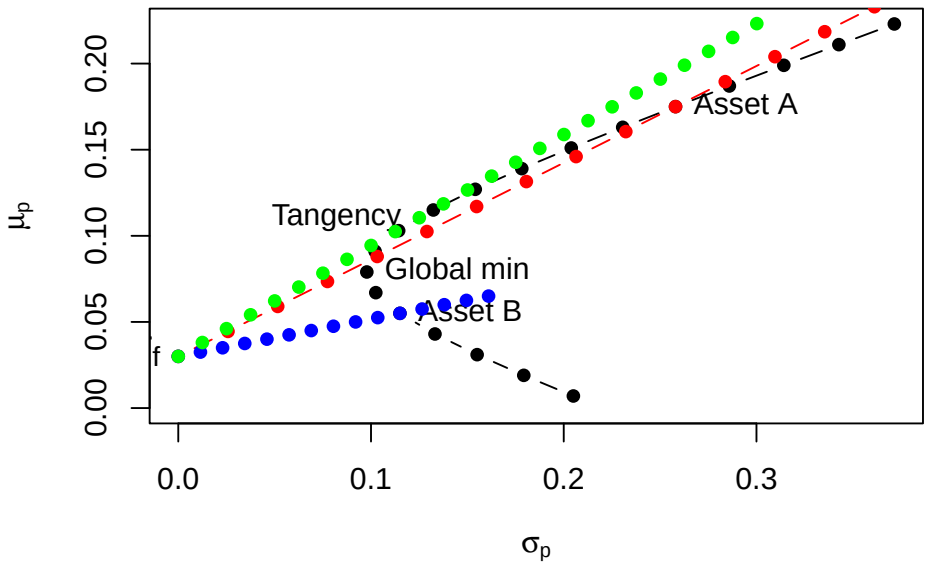


Figure 10.8: Efficient portfolios of two risky assets and T-bills. Black dots represent portfolios of assets A and B; blue dots represent portfolios of T-bills and asset B; red dots represent portfolios of T-bills and asset A; green dots represent portfolios of the tangency portfolio and T-bills.

If you invest in only in asset B and T-bills then the Sharpe ratio is $\text{SR}_B = \frac{\mu_B - r_f}{\sigma_B} = 0.217$, and the CAL intersects the set of risky asset portfolios at point B. This is clearly not the efficient set of portfolios. For example, you could do uniformly better if you instead invest only in asset A and T-bills. This gives us a Sharpe ratio of $\text{SR}_A = \frac{\mu_A - r_f}{\sigma_A} = 0.562$, and the new CAL intersects the set of risky asset portfolios at point A. However, you could do better still if you invest in T-bills and some combination of assets A and B. Geometrically, it is easy to see that the best you can do is obtained for the combination of assets A and B such that the CAL is just tangent to the set of risky asset portfolios. This point is labeled "Tangency" on the graph and represents the *tangency portfolio* of assets A and B. Portfolios of T-Bills and the tangency portfolio are the set of efficient portfolios consisting of T-Bills, asset A and asset B.

10.5.1 Solving for the Tangency Portfolio

You can determine the proportions of each asset in the tangency portfolio by finding the values of x_A and x_B that maximize the Sharpe ratio of a portfolio that is on the envelope of the Markowitz bullet. Formally, you solve the constrained maximization problem:

$$\begin{aligned}
\max_{x_A, x_B} \text{SR}_p = \frac{\mu_p - r_f}{\sigma_p} \; s.t. \\
\mu_p &= x_A \mu_A + x_B \mu_B, \\
\sigma_p^2 &= x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2 x_A x_B \sigma_{AB}, \\
1 &= x_A + x_B.
\end{aligned}$$

Using the method of substitution with $x_B = 1 - x_A$ and rearranging, the above problem can be reduced to solving the unconstrained maximization:

$$\max_{x_A} \frac{x_A(\mu_A - r_f) + (1 - x_A)(\mu_B - r_f)}{\left(x_A^2 \sigma_A^2 + (1 - x_A)^2 \sigma_B^2 + 2 x_A (1 - x_A) \sigma_{AB}\right)^{1/2}}.$$

This is a straightforward, albeit very tedious, calculus problem and the solution can be shown to be (see Exercises):

$$\begin{aligned}
x_A^{tan} &= \frac{(\mu_A - r_f)\sigma_B^2 - (\mu_B - r_f)\sigma_{AB}}{(\mu_A - r_f)\sigma_B^2 + (\mu_B - r_f)\sigma_A^2 - (\mu_A - r_f + \mu_B - r_f)\sigma_{AB}}, \\
x_B^{tan} &= 1 - x_T^A.
\end{aligned} \tag{10.16}$$

Example 10.10 (Tangency portfolio for example data)**.** For the example data in Table 10.1 using Equation 10.16 with $r_f = 0.03$, you get $x_A^{tan} = 0.4625$ and $x_B^{tan} = 0.5375$. The expected return, variance, standard deviation, and Sharpe ratio of the tangency portfolio are:

$$\begin{aligned}
\mu_{tan} &= x_A^{tan} \mu_A + x_B^{tan} \mu_B \\
&= (0.4625)(0.175) + (0.5375)(0.055) = 0.1105, \\
\sigma_{tan}^2 &= \left(x_A^{tan}\right)^2 \sigma_A^2 + \left(x_B^{tan}\right)^2 \sigma_B^2 + 2 x_A^{tan} x_B^{tan} \sigma_{AB} \\
&= (0.4625)^2(0.06656) + (0.5375)^2(0.01323) + 2(0.4625)(0.5375)(-0.004866) = 0.01564, \\
\sigma_{tan} &= \sqrt{0.01564} = 0.1251, \\
SR_{tan} &= \frac{\mu_{tan} - r_f}{\sigma_{tan}} = \frac{0.1105 - 0.03}{0.1251} = 0.6434
\end{aligned}$$

x_{A}^{tan}	x_{B}^{tan}	μ_{tan}	σ_{tan}	SR_{tan}
0.463	0.537	0.111	0.125	0.644

In R, the computations to compute the tangency portfolio are:

```
# compute portfolio weights
top = (mu.A - r.f)*sig2.B - (mu.B - r.f)*sig.AB
bot = (mu.A - r.f)*sig2.B + (mu.B - r.f)*sig2.A - (mu.A - r.f + mu.B - r.f)*sig.AB
x.A.tan = top/bot
x.B.tan = 1 - x.A.tan
mu.p.tan = x.A.tan*mu.A + x.B.tan*mu.B
sig2.p.tan = x.A.tan^2 * sig2.A + x.B.tan^2 * sig2.B + 2*x.A.tan*x.B.tan*sig.AB
sig.p.tan = sqrt(sig2.p.tan)
SR.tan = (mu.p.tan - r.f)/sig.p.tan
data.tbl = as.data.frame(t(c(x.A.tan,x.B.tan,c(mu.p.tan,sig.p.tan,SR.tan))))
col.names = c("$x_{A}^{tan}$","$x_{B}^{tan}$", "$\\mu_{tan}$",
              "$\\sigma_{tan}$","$\\mathrm{SR}_{tan}$" )
tbl.data = kbl(data.tbl, col.names=col.names) %>%
   kable_styling(full_width=FALSE)
cat(tbl.data)
```

■

10.5.2 Mutual Fund Separation

The efficient portfolios are combinations of the tangency portfolio (risky asset) and the T-bill. Accordingly, using Equation 10.12 and Equation 10.13 the expected return and standard deviation of any efficient portfolio are given by:

$$\mu_p^e = r_f + x_T(\mu_{tan} - r_f), \tag{10.17}$$

$$\sigma_p^e = x_T\sigma_{tan}, \tag{10.18}$$

where x_T represents the fraction of wealth invested in the tangency portfolio, $1 - x_T$ represents the fraction of wealth invested in T-Bills, and μ_{tan} and σ_{tan} are the expected return and standard deviation of the tangency portfolio, respectively. This important result is known as the *mutual fund separation theorem*. The tangency portfolio can be considered as a mutual fund (i.e. portfolio) of the two risky assets, where the shares of the two assets in the mutual fund are determined by the tangency portfolio weights (x_A^{tan} and x_B^{tan} determined from Equation 10.16, and the T-bill can be considered as a mutual fund of risk-free assets. The expected return-risk trade-off of these portfolios

is given by the line connecting the risk-free rate to the tangency point on the efficient frontier of risky asset only portfolios.

The optimal combination of the tangency portfolio and the T-bill an investor will choose depends on the investor's risk preferences. If the investor is very risk averse, then she will choose a portfolio with low volatility which will be a portfolio with very little weight in the tangency portfolio and a lot of weight in the T-bill. This will produce a portfolio with an expected return close to the risk-free rate and a variance that is close to zero. If the investor can tolerate a large amount of risk, then she would prefer a portfolio with highest expected return regardless of the volatility. This portfolio may involve borrowing at the risk-free rate (leveraging) and investing the proceeds in the tangency portfolio to achieve a high expected return.

Example 10.11 (Efficient portfolios chosen by risk averse and risk tolerant investors)**.** A highly risk averse investor may choose to put 10% of her wealth in the tangency portfolio and 90% in the T-bill. Then she will hold $(10\%) \times (46.25\%) = 4.625\%$ of her wealth in asset A, $(10\%) \times (53.75\%) = 5.375\%$ of her wealth in asset B, and 90% of her wealth in the T-bill. The expected return on this portfolio is:

$$\mu_p^e = r_f + 0.10(\mu_{tan} - r_f) = 0.03 + 0.10(0.1105 - 0.03) = 0.03805,$$

and the standard deviation is,

$$\sigma_p^e = 0.10\sigma_{tan} = 0.10(0.1251) = 0.01251.$$

In Figure 10.9, this efficient portfolio is labeled "Safe" . A very risk tolerant investor may actually borrow at the risk-free rate and use these funds to leverage her investment in the tangency portfolio. For example, suppose the risk tolerant investor borrows 100% of her wealth at the risk-free rate and uses the proceed to purchase 200% of her wealth in the tangency portfolio. Then she would hold $(200\%) \times (46.25\%) = 92.50\%$ of her wealth in asset A, $(200\%) \times (53.75\%) = 107.5\%$ in asset B, and she would owe 100% of her wealth to her lender. The expected return and standard deviation on this portfolio is:

$$\begin{aligned} \mu_p^e &= 0.03 + 2(0.1105 - 0.03) = 0.1910, \\ \sigma_p^e &= 2(0.1251) = 0.2501. \end{aligned}$$

In Figure 10.9, this efficient portfolio is labeled "Risky" . ■

10.5.3 Interpreting Efficient Portfolios

As you have seen, efficient portfolios are those portfolios that have the highest expected return for a given level of risk as measured by portfolio standard deviation. For portfolios with expected returns above the T-bill rate, efficient portfolios can also be characterized as those portfolios that

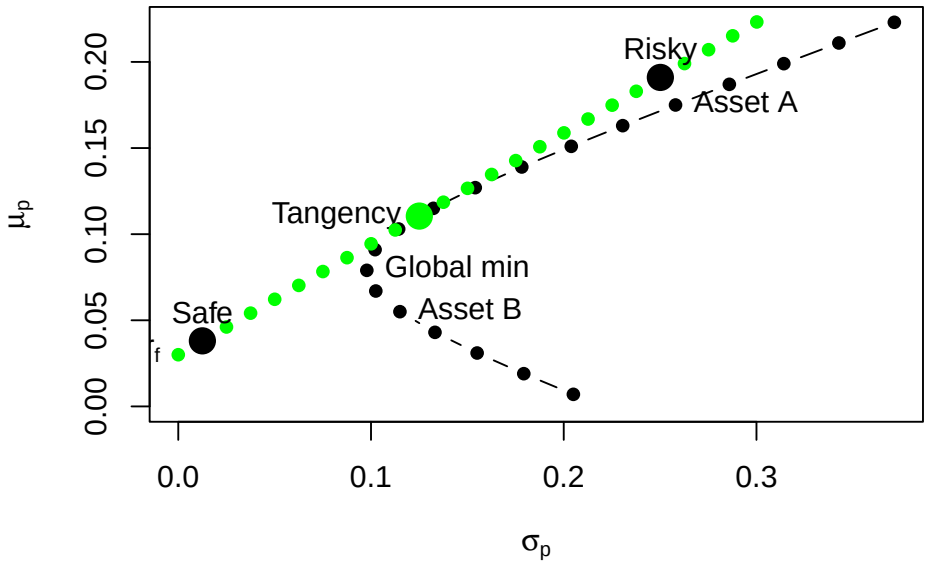


Figure 10.9: The efficient portfolio labeled safe has 10% invested in the tangency portfolio and 90% invested in T-Bills; the efficient portfolio labeled risky has 200% invested in the tangency portfolio and -100% invested in T-Bills.

have minimum risk (as measured by portfolio standard deviation) for a given target expected return.

Example 10.12 (Finding efficient portfolios with target expected return and target risk)**.** To illustrate, consider Figure 10.10 which shows the portfolio frontier for two risky assets and the efficient frontier for two risky assets plus T-Bills. Suppose an investor initially holds all of his wealth in asset B. The expected return on this portfolio is $\mu_B = 0.055$, and the standard deviation (risk) is $\sigma_B = 0.115$. An efficient portfolio (combinations of the tangency portfolio and T-bills) that has the same standard deviation (risk) as asset B is given by the portfolio on the efficient frontier that is directly above $\sigma_B = 0.115$. To find the shares in the tangency portfolio and T-bills in this portfolio recall from Equation 10.18 that the standard deviation of an efficient portfolio with x_T invested in the tangency portfolio and $1 - x_T$ invested in T-bills is $\sigma_p^e = x_T \sigma_{tan}$. Since you want to find the efficient portfolio with $\sigma_p^e = \sigma_B = 0.115$, you solve:

$$x_T = \frac{\sigma_B}{\sigma_{tan}} = \frac{0.115}{0.1251} = 0.9195, \; x_f = 1 - x_T = 0.08049.$$

That is, if you invest 91.95% of our wealth in the tangency portfolio and 8.049% in T-bills you will have a portfolio with the same standard deviation as asset B. Since this is an efficient portfolio, the expected return should be higher than the expected return on asset B. Indeed it is since:

$$\mu_p^e = r_f + x_T(\mu_{tan} - r_f) = 0.03 + 0.9195(0.1105 - 0.03) = 0.1040.$$

Notice that by diversifying our holding into assets A, B and T-bills you can obtain a portfolio with the same risk as asset B but with almost twice the expected return! By moving from asset B into an efficient portfolio you raise the Sharpe ratio of the investment from $SR_B = 0.217$ to $SR_{tan} = 0.6434$.

Next, consider finding an efficient portfolio that has the same expected return as asset B. Visually, this involves finding the combination of the tangency portfolio and T-bills that corresponds with the intersection of a horizontal line with intercept $\mu_B = 0.055$ and the line representing efficient combinations of T-bills and the tangency portfolio. To find the shares in the tangency portfolio and T-bills in this portfolio recall from Equation 10.17 that the expected return of an efficient portfolio with x_T invested in the tangency portfolio and $1 - x_T$ invested in T-bills has expected return equal to $\mu_p^e = r_f + x_T(\mu_{tan} - r_f)$. Since you want to find the efficient portfolio with $\mu_p^e = \mu_B = 0.055$ you solve:

$$x_T = \frac{\mu_p^e - r_f}{\mu_{tan} - r_f} = \frac{0.055 - 0.03}{0.1105 - 0.03} = 0.3105, \; x_f = 1 - x_T = 0.6895.$$

That is, if you invest 31.05% of wealth in the tangency portfolio and 68.95% of our wealth in T-bills you have a portfolio with the same expected return as asset B. Since this is an efficient portfolio, the standard deviation (risk) of this portfolio should be lower than the standard deviation on asset

B. Indeed it is since:

$$\sigma_p^e = x_T \sigma_{tan} = 0.3105(0.124) = 0.03884.$$

Notice how large the risk reduction is by forming an efficient portfolio. The standard deviation on the efficient portfolio is almost three times smaller than the standard deviation of asset B! Again, by moving from asset B into an efficient portfolio you raise the Sharpe ratio of the investment from $SR_B = 0.217$ to $SR_{tan} = 0.6434$ ■

The above example illustrates two ways to interpret the benefits from forming efficient portfolios. Starting from some benchmark portfolio, you can fix standard deviation (risk) at the value for the benchmark and then determine the gain in expected return from forming a diversified portfolio[14]. The gain in expected return has concrete meaning. Alternatively, you can fix expected return at the value for the benchmark and then determine the reduction in standard deviation (risk) from forming a diversified portfolio. The meaning to an investor of the reduction in standard deviation is not as clear as the meaning to an investor of the increase in expected return. It would be helpful if the risk reduction benefit can be translated into a number that is more interpretable than the standard deviation. The concept of Value-at-Risk (VaR) provides such a translation.

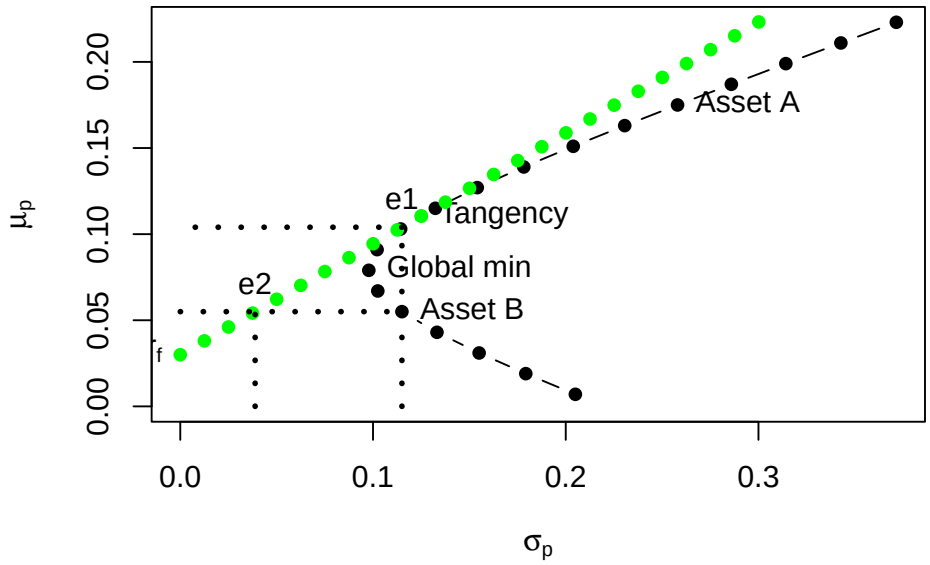


Figure 10.10: The point e1 represents an efficient portfolio with the same standard deviation as asset B; the point e2 represents an efficient portfolio with the same expected returns as asset B.

[14]The gain in expected return by investing in an efficient portfolio abstracts from the costs associated with selling the benchmark portfolio and buying the efficient portfolio.

10.5.4 Efficient Portfolios and Value-at-Risk

Recall, the VaR of an investment is the (lower bound of) loss in investment value over a given horizon with a stated probability. For example, consider an investor who invests $W_0 = \$100,000$ in asset B over the next year. Assuming that $R_B \sim N(0.055, (0.115)^2)$ represents the annual simple return on asset B, the 5% VaR is:

$$\text{VaR}_{B,0.05} = -q^{R_B}_{0.05} W_0 = (0.055 + 0.115(-1.645)) \cdot \$100,000 = \$13,416.$$

If an investor holds \$100,000 in asset B over the next year, then there is a 5% probability that he will lose \$13,416 or more.

Now suppose the investor chooses to hold an efficient portfolio with the same expected return as asset B. This portfolio consists of 31.05% in the tangency portfolio and 68.95% in T-bills and has a standard deviation equal to 0.03884. Then $R_p \sim N(0.055, 0.03884)$ and the 5% VaR on the portfolio is:

$$\text{VaR}_{p,0.05} = -q^{R_p}_{0.05} W_0 = (0.055 + 0.03884(-1.645)) \cdot \$100,000 = \$884.$$

Notice that the 5% VaR for the efficient portfolio is almost fifteen times smaller than the 5% VaR for the investment in asset B. Since VaR translates risk into a dollar figure, it is more interpretable than standard deviation as a measure of investment risk.

10.6 Application to Real World Portfolios

The portfolio theory for two risky assets and a risk-free asset developed in this chapter is not as limited as you might think if you realize that a single risky asset can also be a portfolio of many assets, such as a mutual fund of stocks or bonds, or an exchange traded fund (ETF) of stocks or bonds. In particular, a very common asset allocation for retirement portfolios is a simple two asset portfolio of in which one risky asset is a mutual fund of stocks and the other (less) risky asset is a mutual fund of U.S. Treasury bonds.

In the **introcompfinr** package there is a data object called `VanguardPrices` that contains monthly adjusted closing prices on six Vanguard mutual funds over the period January, 1995 through December, 2014.[15] The fund VFINX is a low cost fund designed to track the performance of the S&P 500 index. The S&P 500 index is a value-weighted index of the 500 largest stocks (by market capitalization) traded on the NYSE.[16] I will use this as my stock portfolio. The fund VBLTX is a low cost fund constructed to track the performance of a portfolio of long term (maturities greater

[15] Details about these funds are available at (finance.yahoo.com).
[16] ETF alternatives to VFINX include State Street's SPY, Vanguard's VOO, and iShares IVV. See `www.eft.com`.

than 10 years) U.S. Government Bonds.[17] I will use this as my bond portfolio. I will use the U.S. 3-month T-Bill rate as the risk-free asset. The following code creates the monthly simple return data for this example:

```
VanguardPrices = as.xts(VanguardPrices)
stockBondPrices = VanguardPrices[, c("vfinx", "vbltx")]
stockBondRetS = na.omit(Return.calculate(stockBondPrices, method="simple"))
head(stockBondRetS, n=3)
#>            vfinx  vbltx
#> Feb 1995 0.0389 0.0280
#> Mar 1995 0.0295 0.0102
#> Apr 1995 0.0293 0.0202
```

I will construct efficient portfolios using the GWN model estimates of expected returns, variances, and covariances over four sub-periods: January, 1995 to January 2000; January, 2000 to January, 2005; January, 2005 to January, 2010; January, 2010 to December, 2014. The approximate annualized T-Bill rates at the end of these periods are 0.05, 0.025, 0.001, and 0.001, respectively.[18]

```
# Define sub-periods
smpl1 = "1995-1::2000-1"
smpl2 = "2000-1::2005-1"
smpl3 = "2005-1::2010-1"
smpl4 = "2010-1::2014-12"
# Monthly risk-free rates for sub-periods
r.f = c(0.05, 0.025, 0.001, 0.001)/12
```

Historical performance of the stock and bond portfolios over the four sub-periods is illustrated in Figure 10.11, created using:[19]

```
par(mfrow=c(2,2))
chart.CumReturns(stockBondRetS[smpl1], main="", wealth.index = TRUE,
                 legend.loc = "topleft")
chart.CumReturns(stockBondRetS[smpl2], main="", wealth.index = TRUE)
chart.CumReturns(stockBondRetS[smpl3], main="", wealth.index = TRUE)
chart.CumReturns(stockBondRetS[smpl4], main="", wealth.index = TRUE)
```

[17] ETF alternatives to VBLTX include iShares TLT and GOVT, and State Street SPTL. See `www.eft.com`.
[18] Historical data on U.S. T-Bill rates is obtained from the Federal Reserve Economic Data (FRED) available at the St. Louis Fed. See `fred.stlouisfed.org`
[19] The **PerformanceAnalytics** function `chart.CumReturns()` is used to compute the equity curves for each asset.

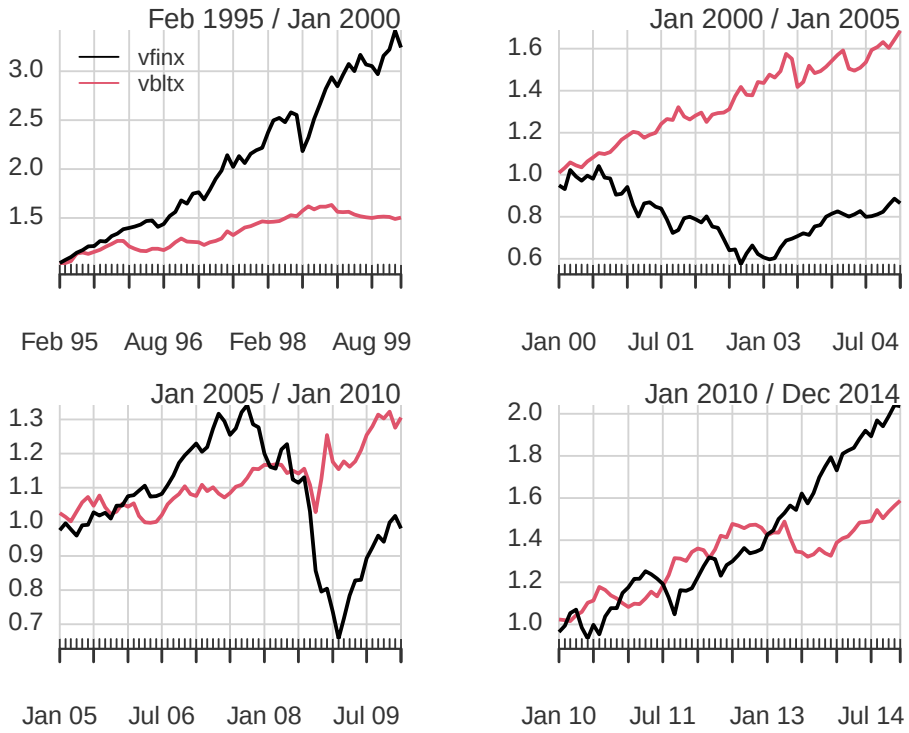


Figure 10.11: Equity curves for stock (vfinx) and bond (vbltx) portfolios over sub-periods.

The performance of stocks and bonds varies substantially over the four sub-periods. In the first and fourth sub-periods, which cover the dot-com boom and the recovery from the 2009 financial crisis, stocks substantially outperformed bonds. However, in the second and third sub-periods, which cover the dot-com bust and the 2009 financial crisis, bonds outperformed stocks.

The GWN model estimates of the monthly expected returns, volatilities, and Sharpe ratios on the stock and bond portfolios are computed and annualized using the square-root-of-time rule with the **PerformanceAnalytics** function `table.AnnualizedReturns()`.

```
# smpl1 = "1995-1::2000-1"
table.AnnualizedReturns(stockBondRetS[smpl1], Rf = r.f[1], geometric = FALSE)
#>                            vfinx  vbltx
#> Annualized Return          0.248 0.0846
#> Annualized Std Dev         0.143 0.0756
#> Annualized Sharpe (Rf=5%) 1.381 0.4581
# smpl2 = "2000-1::2005-1"
table.AnnualizedReturns(stockBondRetS[smpl2], Rf = r.f[2], geometric = FALSE)
#>                              vfinx  vbltx
#> Annualized Return          -0.0155 0.1069
#> Annualized Std Dev          0.1628 0.0871
```

```
#> Annualized Sharpe (Rf=2.5%) -0.2490 0.9399
# smpl3 = "2005-1::2010-1"
table.AnnualizedReturns(stockBondRetS[smpl3], Rf = r.f[3], geometric = FALSE)
#>                              vfinx  vbltx
#> Annualized Return           0.0092 0.0576
#> Annualized Std Dev          0.1600 0.1021
#> Annualized Sharpe (Rf=0.1%) 0.0514 0.5547
# smpl4 = "2010-1::2014-12"
table.AnnualizedReturns(stockBondRetS[smpl4], Rf = r.f[4], geometric = FALSE)
#>                             vfinx  vbltx
#> Annualized Return           0.151 0.0965
#> Annualized Std Dev          0.130 0.0864
#> Annualized Sharpe (Rf=0.1%) 1.157 1.1050
```

The annualized estimates confirm the relative performance of bonds and stocks in the four sub-periods. In general, stock returns are about twice as volatile as bond returns. Bonds performed well in all sub-periods, with annualized mean returns between 6% and 11%. In contrast, stocks only performed well in sub-periods 1 and 4 and had negative annualized mean returns in periods 2 and 3. In terms of Sharpe ratios, stocks had annualized Sharpe ratios a bit over 1 in sub-periods 1 and 4 and bonds had annualized Sharpe ratios of 0.9 and 0.55 in sub-periods 2 and 3.[20]

The sample correlations between the monthly stock and bond returns, by sub-period, are:

```
# smpl1 = "1995-1::2000-1"
cor(stockBondRetS[smpl1])[1,2]
#> [1] 0.2
# smpl2 = "2000-1::2005-1"
cor(stockBondRetS[smpl2])[1,2]
#> [1] -0.185
# smpl3 = "2005-1::2010-1"
cor(stockBondRetS[smpl3])[1,2]
#> [1] 0.24
# smpl4 = "2010-1::2014-12"
cor(stockBondRetS[smpl4])[1,2]
#> [1] -0.435
```

Interestingly, the estimated correlation between stocks and bonds also varies by sub-period. In sub-periods one and three, the estimated correlations are positive, and, in sub-periods two and four, the estimated correlations are negative. The observed low to negative correlation between

[20] Investments with annualized Sharpe ratios greater than one are typically considered good investments.

Table 10.2: Sub-period global minimum variance portfolios

	\(x_{stock}\)	\(x_{bond}\)	\(\mu_{min}\)	\(\sigma_{min}\)	\(SR_{min}\)
1995-1::2000-1	0.162	0.838	0.111	0.072	0.851
2000-1::2005-1	0.260	0.740	0.075	0.070	0.713
2005-1::2010-1	0.231	0.769	0.046	0.094	0.481
2010-1::2014-12	0.362	0.638	0.116	0.055	2.108

stock and bond returns suggests good risk reduction possibilities through diversification.

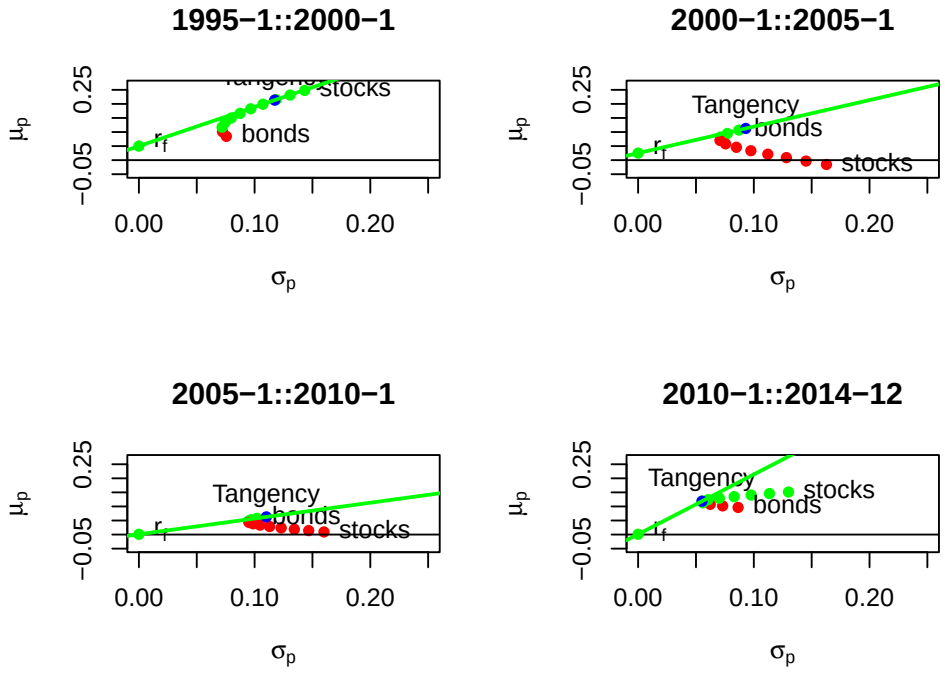


Figure 10.12: Mean-variance efficient portfolios of stocks and bonds by sub-period. Efficient portfolios are shown in green and inefficient portfolios are shown in red.

Using the sub-period GWN model estimates, I compute risky asset only efficient portfolios and efficient portfolios of T-Bills (risk-free asset) and the tangency portfolios. The resulting portfolios are illustrated in Figure 10.12. In sub-periods one and four, stocks are on the risky asset only efficient frontier and, in sub-periods two and three, bonds are on the risky asset only efficient frontier. The sub-period global minimum variance and tangency portfolios are listed in Tables Table 10.2 and Table 10.3. The global minimum variance portfolios are similar across sub-periods, with higher weights in bonds than stocks. The tangency portfolios, however, are quite different across sub-periods. In sub-periods one and four, stocks have weights 79% and 40%, respectively. In sub-periods two and three bonds essentially have 100% weights, respectively, and stocks are slighly shorted.

This example shows the clear gains that can be made by forming efficient portfolios relative to

Table 10.3: Sub-period tangency portfolios

	\(x_{stock}\)	\(x_{bond}\)	\(\mu_{tan}\)	\(\sigma_{tan}\)	\(SR_{tan}\)
1995-1::2000-1	0.789	0.211	0.213	0.117	1.393
2000-1::2005-1	-0.047	1.047	0.113	0.093	0.943
2005-1::2010-1	-0.107	1.107	0.063	0.110	0.561
2010-1::2014-12	0.404	0.596	0.119	0.055	2.128

holding stock or bonds by themselves. The results, however, are a bit misleading because the efficient portfolios formed in each sub-period use all of the data over the sub-period to estimate expected returns, volatilities, and covariances. This means that the computed efficient portfolios cannot be computed at the beginning of each sub-period. A more practical exercise would use the previous sub-period to compute GWN model estimates to be used in the subsequent sub-period. This method is explored in the end-of-chapter exercises.

10.7 Further Reading: Introduction to Portfolio Theory

Markowitz's mean-variance portfolio theory is described in detail in (Markowitz 1959), (Markowitz 1987). Most MBA level corporate finance and investments textbooks have chapters on mean-variance portfolio theory. Good intermediate level treatments of portfolio theory from a quantitative perspective are given in (Benninga 2000), (Bodie, Kane, and Marcus 2013) and (Elton et al. 2014).

10.8 Appendix: Review of Optimization and Constrained Optimization

Consider the function of a single variable:

$$y = f(x) = x^2.$$

Clearly the minimum of this function occurs at the point $x = 0$. Using calculus, you find the minimum by solving:

$$\min_x \ y = x^2.$$

The first order (necessary) condition for a minimum is:

$$0 = \frac{d}{dx} f(x) = \frac{d}{dx} x^2 = 2x$$

and solving for x gives $x = 0$. The second order condition for a minimum is,

$$0 < \frac{d^2}{dx} f(x),$$

and this condition is clearly satisfied for $f(x) = x^2$.

Next, consider the function of two variables:

$$y = f(x, z) = x^2 + z^2 \tag{10.19}$$

This function looks like a salad bowl whose bottom is at $x = 0$ and $z = 0$. To find the minimum of Equation 10.19, you solve:

$$\min_{x,z} y = x^2 + z^2,$$

and the first order necessary conditions are,

$$0 = \frac{\partial y}{\partial x} = 2x,$$
$$0 = \frac{\partial y}{\partial z} = 2z.$$

Solving these two linear equations gives $x = 0$ and $z = 0$.

Now suppose you want to minimize Equation 10.19 subject to the linear constraint:

$$x + z = 1. \tag{10.20}$$

The minimization problem is now a * constrained minimization*:

$$\min_{x,z} y = x^2 + z^2 \text{ subject to } (s.t.)$$
$$x + z = 1.$$

Given the constraint $x + z = 1$, the function Equation 10.19 is no longer minimized at the point $(x, z) = (0, 0)$ because this point does not satisfy $x + z = 1$. One simple way to solve this problem is to substitute the restriction Equation 10.20 into the function Equation 10.19 and reduce the problem to a minimization over one variable. To illustrate, use the restriction Equation 10.20 to solve for z as:

$$z = 1 - x. \tag{10.21}$$

Now substitute Equation 10.21 into Equation 10.19 giving:

$$y = f(x, z) = f(x, 1 - x) = x^2 + (1 - x)^2. \tag{10.22}$$

The function Equation 10.22 satisfies the restriction Equation 10.20 by construction. The constrained minimization problem now becomes:

$$\min_{x} y = x^2 + (1 - x)^2.$$

The first order conditions for a minimum are:

$$0 = \frac{d}{dx}(x^2 + (1-x)^2) = 2x - 2(1-x) = 4x - 2,$$

and solving for x gives $x = 1/2$. To solve for z, use Equation 10.21 to give $z = 1 - (1/2) = 1/2$. Hence, the solution to the constrained minimization problem is $(x, z) = (1/2, 1/2)$.

Another way to solve the constrained minimization is to use the method of *Lagrange multipliers*. This method augments the function to be minimized with a linear function of the constraint in homogeneous form. The constraint Equation 10.20 in homogenous form is:

$$x + z - 1 = 0.$$

The augmented function to be minimized is called the *Lagrangian* and is given by:

$$L(x, z, \lambda) = x^2 + z^2 - \lambda(x + z - 1).$$

The coefficient on the constraint in homogeneous form, λ, is called the *Lagrange multiplier*. It measures the cost, or shadow price, of imposing the constraint relative to the unconstrained problem. The constrained minimization problem to be solved is now:

$$\min_{x,z,\lambda} \; L(x, z, \lambda) = x^2 + z^2 + \lambda(x + z - 1).$$

The first order conditions for a minimum are:

$$\begin{aligned} 0 &= \frac{\partial L(x, z, \lambda)}{\partial x} = 2x + \lambda, \\ 0 &= \frac{\partial L(x, z, \lambda)}{\partial z} = 2z + \lambda, \\ 0 &= \frac{\partial L(x, z, \lambda)}{\partial \lambda} = x + z - 1. \end{aligned}$$

The first order conditions give three linear equations in three unknowns. Notice that the first order condition with respect to λ imposes the constraint. The first two conditions give:

$$2x = 2z = -\lambda,$$

or,

$$x = z.$$

Substituting $x = z$ into the third condition gives:

$$2z - 1 = 0$$

or,

$$z = 1/2.$$

The final solution is $(x, y, \lambda) = (1/2, 1/2, -1)$.

The Lagrange multiplier, λ, measures the marginal cost, in terms of the value of the objective function, of imposing the constraint. Here, $\lambda = -1$ which indicates that imposing the constraint $x + z = 1$ reduces the objective function. To understand the role of the Lagrange multiplier better, consider imposing the constraint $x + z = 0$. Notice that the unconstrained minimum achieved at $x = 0, z = 0$ satisfies this constraint. Hence, imposing $x + z = 0$ does not cost anything and so the Lagrange multiplier associated with this constraint should be zero. To confirm this, you solve the problem:

$$\min_{x,z,\lambda} \; L(x, z, \lambda) = x^2 + z^2 + \lambda(x + z - 0).$$

The first order conditions for a minimum are:

$$\begin{aligned} 0 &= \frac{\partial L(x, z, \lambda)}{\partial x} = 2x - \lambda, \\ 0 &= \frac{\partial L(x, z, \lambda)}{\partial z} = 2z - \lambda, \\ 0 &= \frac{\partial L(x, z, \lambda)}{\partial \lambda} = x + z. \end{aligned}$$

The first two conditions give:

$$2x = 2z = -\lambda$$

or,

$$x = z.$$

Substituting $x = z$ into the third condition gives:

$$2z = 0,$$

or,

$$z = 0.$$

The final solution is $(x, y, \lambda) = (0, 0, 0)$. Notice that the Lagrange multiplier, λ, is equal to zero in this case.

10.9 Exercises

Exercise 10.1. Annual estimates of the GWN model parameters for Boeing and Microsoft are given in the table below:

	Boeing	Microsoft
μ	0.149	0.331
σ^2	0.069	0.136
ρ	-0.008	-0.008

1. Create the following portfolios.
   - Combinations of Boeing and Microsoft (with $x_{\text{boeing}} = -1, -0.9, ..., 2$ and $x_{\text{msft}} = 1 - x_{\text{boeing}}$)
   - Combinations of Boeing and T-bills (with $x_{\text{boeing}} = 0, 0.1, ..., 2$)
   - Combinations of Microsoft and T-bills (with $x_{\text{msft}} = 0, 0.1, ..., 2$)
   - Use an annual risk-free rate of 3% per year for the T-bill. For each portfolio, compute $E(R_p)$, $\text{var}(R_p)$ and $\text{SD}(R_p)$ using the appropriate formulas. For each portfolio, plot $E(R_p)$ vs. $\text{SD}(R_p)$ and put these values on the same graph. Compute the Sharpe's slope for Boeing and Microsoft. Which asset has the highest slope value?
2. Compute the global minimum variance portfolio using the analytical formula Equation 10.10.
   - Make a bar chart showing the weights of Boeing and Microsoft in global minimum variance portfolio.
   - Compute $E(R_p)$, $\text{var}(R_p)$ and $\text{SD}(R_p)$ for the tangency portfolio.
   - Compute Sharpe's slope for the tangency portfolio.
   - Indicate the location of the tangency portfolio on the graph you created previously in question 1.
3. Using a risk-free rate of 3% per year for the T-bill, compute the tangency portfolio using the analytic formula Equation 10.16.
   - Make a bar chart showing the weights of Boeing and Microsoft in the tangency portfolio.
   - Compute $E(R_p)$, $\text{var}(R_p)$ and $\text{SD}(R_p)$ for the tangency portfolio.
   - Compute the Sharpe's slope for the tangency portfolio.
   - Indicate the location of the tangency portfolio on the graph you created previously in question 1.
4. Consider a portfolio that has 10% in the tangency portfolio and 90% in T-bills.
   - In this portfolio, what is the percent invested in Boeing and what is the percent invested in Microsoft? Give a bar chart showing the percents invested in T-bills, Boeing and Microsoft.
   - Compute $E(R_p)$, $\text{var}(R_p)$ and $\text{SD}(R_p)$ for this portfolio.
   - Compute Sharpe's slope for this portfolio.
   - Indicate the location of the tangency portfolio on the graph you created previously in question 1.

5. Find the efficicient portfolio (combination of T-bills and tangency portfolio) that has the same risk (SD) as Microsoft.

    - In this portfolio, what is the percent invested in Boeing and what is the percent invested in Microsoft? Give a bar chart showing the percent invested in T-bills, Boeing and Microsoft.
    - Compute $E(R_p)$, $\text{var}(R_p)$ and $\text{SD}(R_p)$ for this portfolio.
    - Compute Sharpe's slope for this portfolio.
    - Indicate the location of the tangency portfolio on the graph you created previoulsy in question 1.

Exercise 10.2 (Real world portfolios revisited)**.**

- Re-do the real-world portfolio analysis using forward looking estimates: For each 5-year period use the previous 5-year period to estimate GWN model parameters and then do asset allocation.

11 Portfolio Theory with Matrix Algebra

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In this chapter, I extend the mean-variance portfolio theory introduced in chapter Chapter 10 to handle more than two risky assets. This extension allows the theory to be applied to the analysis of real world portfolios that consist of many risky assets as well as a risk-free asset. When working with large portfolios, the algebra of representing portfolio expected returns and variances becomes extremely cumbersome. As shown in chapter Chapter 3, the use of matrix algebra can greatly simplify many of these computations. Matrix algebra formulations are also very useful when it comes to do actual computations on the computer. The matrix algebra formulas are easy to translate into matrix programming languages like R. Popular spreadsheet programs like Microsoft Excel, which are the workhorse programs of many financial houses, can also handle basic matrix calculations but with more limited functionality. All of this makes it worthwhile to become familiar with matrix techniques for portfolio calculations.

This chapter is laid out as follows. In Section 11.1 I describe portfolios with N risky assets using matrix algebra. I elaborate the concept of portfolio risk diversification using calculations for large portfolios. In Section 11.2 through @ I cover the matrix algebra calculations required for determining mean-variance efficient portfolios. I give Explicit formulas for the global minimum variance portfolio, a minimum variance portfolio that achieves a specified target expected return, and the tangency portfolio. I also present a simple algorithm to easily trace out the risky asset only efficient frontier. In Section 6 I discuss some practical problems associated with very large portfolios, and in section 7 I summarize the **introcompfinr** functions for portfolio analysis. In **?@sec-Real-World-Application-to-Vanguard-Mutual-Funds** I give a real-world application of the theory to asset allocation among Vanguard mutual funds.

The examples I show in this chapter use the **corrplot**, **introcompfinr** and **PerformanceAnalytics** packages. Make sure these packages are installed and loaded in R before replicating the chapter examples.

11.1 Portfolios with N Risky Assets

Consider a portfolio with N risky assets, where N can be a large number (e.g., $N = 1,000$). Let R_i $(i = 1, \dots, N)$ denote the simple return on asset i (over some fixed time horizon such as one year)

and assume that the constant expected return (GWN) model holds for all assets:

$$\begin{aligned} R_i &\sim GWN(\mu_i, \sigma_i^2), \\ \text{cov}(R_i, R_j) &= \sigma_{ij}. \end{aligned}$$

Let x_i denote the share of wealth invested in asset i $(i = 1, \ldots, N)$, and assume that all wealth is invested in the N assets so that $\sum_{i=1}^{N} x_i = 1$. Assume that short sales are allowed so that some values of x_i can be negative. The portfolio return, $R_{p,x}$, is the random variable $R_{p,x} = \sum_{i=1}^{N} x_i R_i$. The subscript "$x$" indicates that the portfolio is constructed using the "x-weights" $x_1, x_2, \ldots, x_N$. The expected return on the portfolio is:

$$\mu_{p,x} = \sum_{i=1}^{N} x_i \mu_i,$$

and the variance of the portfolio return is:

$$\sigma_{p,x}^2 = \sum_{i=1}^{N} x_i^2 \sigma_i^2 + 2 \sum_{i=1}^{N} \sum_{j \neq i} x_i x_j \sigma_{ij}.$$

Notice that variance of the portfolio return depends on N variance terms and $N(N-1)$ covariance terms. Hence, with N assets there are many more covariance terms than variance terms contributing to portfolio variance. For example, with $N = 100$ there are 100 variance terms and $100 \times 99 = 9900$ covariance terms. With N assets, the algebra representing the portfolio return and risk characteristics is cumbersome, especially for the variance. You can greatly simplify the portfolio algebra using matrix notation, and this was previewed in Chapter 3.

11.1.1 Portfolio return and risk characteristics using matrix notation

Define the following $N \times 1$ column vectors containing the asset returns and portfolio weights:

$$\mathbf{R} = \begin{pmatrix} R_1 \\ \vdots \\ R_N \end{pmatrix}, \ \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}.$$

In matrix notation we can lump multiple returns in a single vector which we denote by $\mathbf{R}$. Since each of the elements in $\mathbf{R}$ is a random variable we call $\mathbf{R}$ a *random vector*. The probability distribution of the random vector $\mathbf{R}$ is simply the joint distribution of the elements of $\mathbf{R}$. In the GWN model all returns are jointly normally distributed and this joint distribution is completely characterized by the means, variances and covariances of the returns. We can easily express these values using

matrix notation as follows. The $N \times 1$ vector of portfolio expected values is:

$$E[\mathbf{R}] = E\left[\begin{pmatrix} R_1 \\ \vdots \\ R_N \end{pmatrix}\right] = \begin{pmatrix} E[R_1] \\ \vdots \\ E[R_N] \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_N \end{pmatrix} = \mu,$$

and the $N \times N$ covariance matrix of returns is:

$$\begin{aligned} \text{var}(\mathbf{R}) &= \begin{pmatrix} \text{var}(R_1) & \text{cov}(R_1, R_2) & \cdots & \text{cov}(R_1, R_N) \\ \text{cov}(R_1, R_2) & \text{var}(R_2) & \cdots & \text{cov}(R_2, R_N) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(R_1, R_N) & \text{cov}(R_2, R_N) & \cdots & \text{var}(R_N) \end{pmatrix} \\ &= \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_N^2 \end{pmatrix} = \Sigma. \end{aligned}$$

Notice that the covariance matrix is symmetric (elements off the diagonal are equal so that $\Sigma = \Sigma'$, where Σ' denotes the transpose of Σ) since $\text{cov}(R_i, R_j) = \text{cov}(R_j, R_i)$ for $i \neq j$. It will be positive definite provided no pair of assets is perfectly correlated ($|\rho_{ij}| \neq 1$ for all $i \neq j$) and no asset has a constant return ($\sigma_i^2 > 0$ for all i).

The return on the portfolio using matrix notation is:

$$R_{p,x} = \mathbf{x}'\mathbf{R} = (x_1, \cdots, x_N) \cdot \begin{pmatrix} R_1 \\ \vdots \\ R_N \end{pmatrix} = x_1 R_1 + \cdots + x_N R_N.$$

Similarly, the expected return on the portfolio is:

$$\mu_{p,x} = E[\mathbf{x}'\mathbf{R}] = \mathbf{x}' E[\mathbf{R}] = \mathbf{x}'\mu = (x_1, \ldots, x_N) \cdot \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_N \end{pmatrix} = x_1\mu_1 + \cdots + x_N\mu_N.$$

The variance of the portfolio is:

$$\sigma_{p,x}^2 = \text{var}(\mathbf{x}'\mathbf{R}) = \mathbf{x}'\Sigma\mathbf{x} = (x_1, \dots, x_N) \cdot \begin{pmatrix} \sigma_1^2 & \cdots & \sigma_{1N} \\ \vdots & \ddots & \vdots \\ \sigma_{1N} & \cdots & \sigma_N^2 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}$$
$$= \sum_{i=1}^{N} x_i^2 \sigma_i^2 + 2 \sum_{i=1}^{N} \sum_{j \neq i} x_i x_j \sigma_{ij}.$$

The condition that the portfolio weights sum to one can be expressed as:

$$\mathbf{x}'\mathbf{1} = (x_1, \dots, x_N) \cdot \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = x_1 + \cdots + x_N = 1,$$

where $\mathbf{1}$ is a $N \times 1$ vector with each element equal to 1.

Consider another portfolio with weights $\mathbf{y} = (y_1, \dots, y_N)' \neq \mathbf{x}$. The return on this portfolio is $R_{p,y} = \mathbf{y}'\mathbf{R}$. Later on you will need to compute the covariance between the return on portfolio $\mathbf{x}$ and the return on portfolio $\mathbf{y}$, $\text{cov}(R_{p,x}, R_{p,y})$. Using matrix algebra, this covariance can be computed as:

$$\sigma_{xy} = \text{cov}(R_{p,x}, R_{p,y}) = \text{cov}(\mathbf{x}'\mathbf{R}, \mathbf{y}'\mathbf{R}) = \mathbf{x}'\Sigma\mathbf{y} = (x_1, \dots, x_N) \cdot \begin{pmatrix} \sigma_1^2 & \cdots & \sigma_{1N} \\ \vdots & \ddots & \vdots \\ \sigma_{1N} & \cdots & \sigma_N^2 \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}.$$

Example 11.1 (Three asset example data)**.** To illustrate portfolio calculations in R, table Table 11.1 gives example values on monthly means, variances and covariances for the simple returns on Microsoft, Nordstrom and Starbucks stock based on sample statistics computed over the five-year period January, 1995 through January, 2000. The risk-free asset is the monthly T-Bill with rate $r_f = 0.005$.

Table 11.1: Three asset example data.

Asset i	μ_i	σ_i	Pair (i,j)			σ_{ij}
MSFT	0.0427	0.1000	(MSFT,NORD)			0.0018
NORD	0.0015	0.1044	(MSFT,SBUX)			0.0011
SBUX	0.0285	0.1411	(NORD,SBUX)			0.0026
T-Bill	0.005	0	(T-Bill,MSFT)	(T-Bill,NORD)	(T-Bill,SBUX)	0

The example data in matrix notation is:

$$\mu = \begin{pmatrix} \mu_{MSFT} \\ \mu_{NORD} \\ \mu_{SBUX} \end{pmatrix} = \begin{pmatrix} 0.0427 \\ 0.0015 \\ 0.0285 \end{pmatrix},$$

$$\Sigma = \begin{pmatrix} \text{var}(R_{MSFT}) & \text{cov}(R_{MSFT,NORD}) & \text{cov}(R_{MSFT,SBUX}) \\ \text{cov}(R_{MSFT,NORD}) & \text{var}(R_{NORD}) & \text{cov}(R_{NORD,SBUX}) \\ \text{cov}(R_{MSFT,SBUX}) & \text{cov}(R_{NORD,SBUX}) & \text{var}(R_{SBUX}) \end{pmatrix}$$

$$= \begin{pmatrix} 0.0100 & 0.0018 & 0.0011 \\ 0.0018 & 0.0109 & 0.0026 \\ 0.0011 & 0.0026 & 0.0199 \end{pmatrix}.$$

In R, the example data is created using:

```
asset.names <- c("MSFT", "NORD", "SBUX")
mu.vec = c(0.0427, 0.0015, 0.0285)
names(mu.vec) = asset.names
sigma.mat = matrix(c(0.0100, 0.0018, 0.0011,
           0.0018, 0.0109, 0.0026,
           0.0011, 0.0026, 0.0199),
         nrow=3, ncol=3)
dimnames(sigma.mat) = list(asset.names, asset.names)
r.f=0.005
```

The values of μ_i and σ_i (i = MSFT, NORD, SBUX) are shown in Figure 11.1, created with:

```
sd.vec = sqrt(diag(sigma.mat))
cex.val = 1.5
plot(sd.vec, mu.vec,  ylim=c(0, 0.06), xlim=c(0, 0.20),
     ylab=expression(mu[p]), xlab=expression(sigma[p]),
     pch=16, col="blue", cex=cex.val, cex.lab=1.5)
text(sd.vec, mu.vec, labels=asset.names, pos=4, cex = cex.val)
text(0, r.f, labels=expression(r[f]), pos=4, cex = cex.val)
```

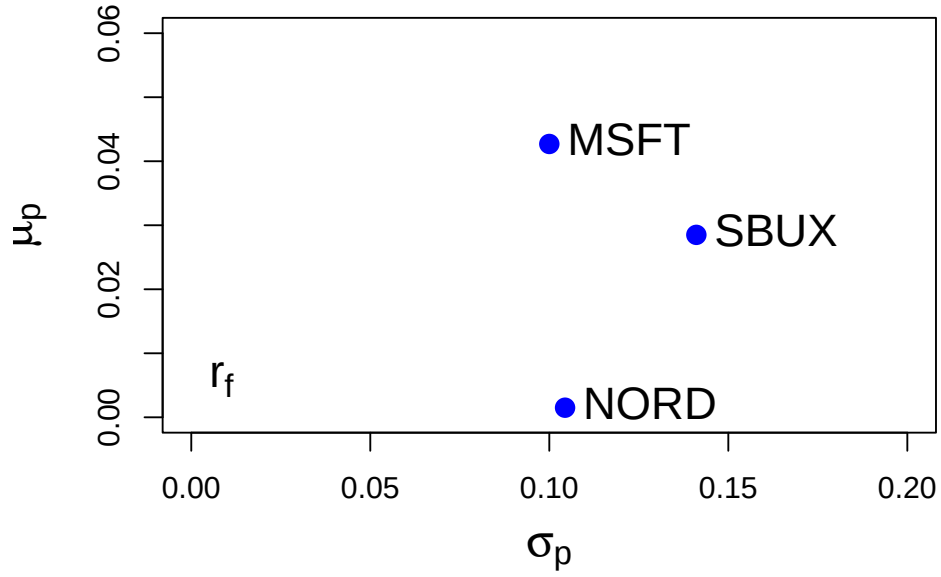


Figure 11.1: Risk-return characteristics of example data.

Clearly, Microsoft provides the best risk-return trade-off (i.e., has the highest Sharpe ratio) and Nordstorm provides with worst. ■

Example 11.2 (Portfolio computations in R)**.** Consider an equally weighted portfolio with $x_{MSFT} = x_{NORD} = x_{SBUX} = 1/3$. This portfolio has return $R_{p,x} = \mathbf{x}'\mathbf{R}$ where $\mathbf{x} = (1/3, 1/3, 1/3)'$. Using R, the portfolio mean and variance are:

```
x.vec = rep(1,3)/3
names(x.vec) = asset.names
mu.p.x = crossprod(x.vec,mu.vec)
sig2.p.x = t(x.vec)%*%sigma.mat%*%x.vec
sig.p.x = sqrt(sig2.p.x)
c(mu.p.x, sig.p.x)
#> [1] 0.0242 0.0759
```

Next, consider another portfolio with weight vector $\mathbf{y} = (y_{MSFT}, y_{NORD}, y_{SBUX})' = (0.8, 0.4, -0.2)'$ and return $R_{p,y} = \mathbf{y}'\mathbf{R}$. The mean and volatility of this portfolio are:

```
y.vec = c(0.8, 0.4, -0.2)
names(y.vec) = asset.names
```

```
mu.p.y = crossprod(y.vec,mu.vec)
sig2.p.y = t(y.vec)%*%sigma.mat%*%y.vec
sig.p.y = sqrt(sig2.p.y)
c(mu.p.y, sig.p.y)
#> [1] 0.0291 0.0966
```

The covariance and correlation between $R_{p,x}$ and $R_{p,y}$ are:

```
sig.xy = t(x.vec)%*%sigma.mat%*%y.vec
rho.xy = sig.xy/(sig.p.x*sig.p.y)
c(sig.xy,rho.xy)
#> [1] 0.00391 0.53329
```

The return and risk characteristics of the three assets together with portfolios $\mathbf{x}$ and $\mathbf{y}$ are illustrated in Figure 11.2. Notice that the equally weighted portfolio, portfolio $\mathbf{x}$, has the smallest volatility.

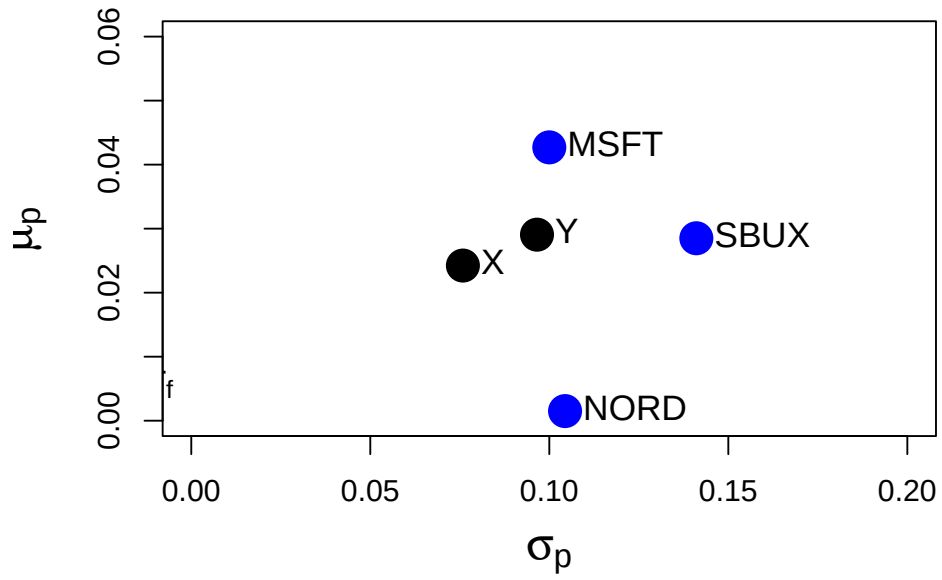


Figure 11.2: Risk-return characteristics of example data and portfolios with weight vectors $\mathbf{x} = (0.333, 0.333, 0.333)'$ and $\mathbf{y} = (0.8, 0.4, -0.2)'$.

■

Example 11.3 (Risk-return characteristics of random portfolios)**.** Figure 11.2 shows that the risk-return points associated with portfolios of three assets do not fall nicely on a bullet-shaped line,

as they do with two asset portfolios. In fact, the set of risk-return points generated by varying the three portfolio weights is a solid area. To see this, consider creating a set of 200 randomly generated three-asset portfolios whose weights sum to one:

```
set.seed(123)
x.msft = runif(200, min=-1.5, max=1.5)
x.nord = runif(200, min=-1.5, max=1.5)
x.sbux = 1 - x.msft - x.nord
head(cbind(x.msft, x.nord, x.sbux))
#>      x.msft  x.nord  x.sbux
#> [1,] -0.637 -0.7838  2.4211
#> [2,]  0.865  1.3871 -1.2520
#> [3,] -0.273  0.3041  0.9690
#> [4,]  1.149  0.0451 -0.1941
#> [5,]  1.321 -0.2923 -0.0291
#> [6,] -1.363  1.1407  1.2226
```

Figure 11.3 shows the means and volatilities, as grey dots, of these 200 random portfolios of the three assets. Notice that the grey dots outline a bullet-shaped area (Markowitz bullet) that has an outer boundary that looks like the line characterizing the two-asset portfolio frontier. If we let the number of random portfolios get very large then the grey dots will fill in a solid area. At the tip of the Markowitz bullet is the minimum variance portfolio.

Recall the definition of a mean-variance efficient portfolio given in Chapter Chapter 10. For a fixed volatility $\sigma_{p,0}$ the efficient portfolio is the one with the highest expected return. From Figure 11.3, we can see that efficient portfolios will lie on the outer boundary (above the minimum variance portfolio at the tip of the Markowitz bullet) of the set of feasible portfolios.

■

11.1.2 Large portfolios and diversification

One of the key benefits of forming large portfolios is risk reduction due to *portfolio diversification*. Generally, a diversified portfolio is one in which wealth is spread across many assets.

To illustrate the impact of diversification on the risk of large portfolios, consider an equally weighted portfolio of N risky assets where N is a large number (e.g., $N > 100$). Here, $x_i = 1/N$ and the $N \times 1$ portfolio weight vector is $\mathbf{x} = (1/N, 1/N, \dots, 1/N)' = (1/N)(1, 1, \dots, 1)' = (1/N)\mathbf{1}$ where $\mathbf{1}$ is an $N \times 1$ vector of ones. Let the $N \times 1$ return vector $\mathbf{R}$ be described by the GWN model so that $\mathbf{R} \sim N(\mu, \Sigma)$. The portfolio return is $R_{p,x} = \mathbf{x}'\mathbf{R} = (1/N)\mathbf{1}'\mathbf{R} = (1/N)\sum_{i=1}^{N} R_i$. The variance of

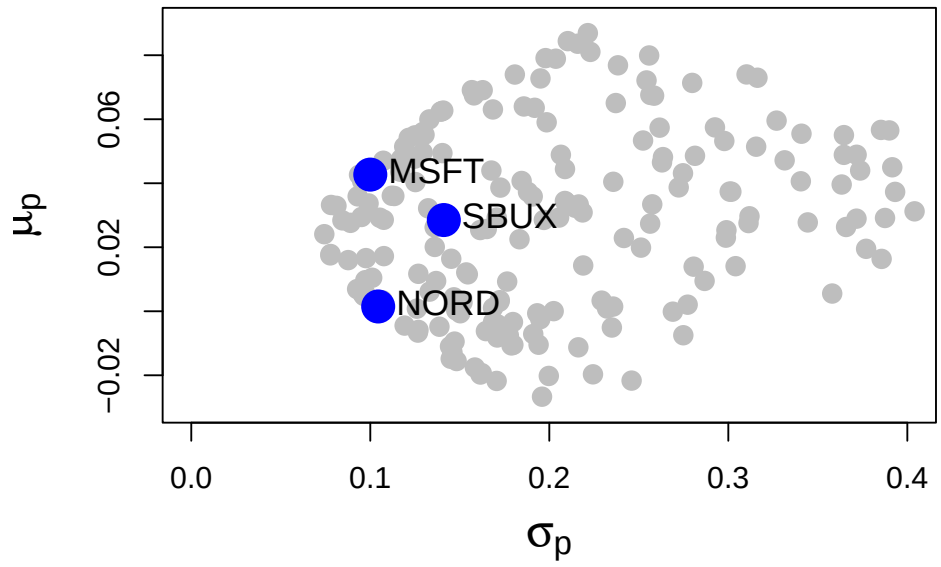


Figure 11.3: Risk-return characteristics of 200 random portfolios of three assets.

this portfolio is:

$$\text{var}(R_{p,x}) = \text{var}(\mathbf{x}'\mathbf{R}) = \text{var}\left(\frac{1}{N}\mathbf{1}'\mathbf{R}\right) = \frac{1}{N^2}\mathbf{1}'\Sigma\mathbf{1}.$$

Now,

$$\begin{aligned}\mathbf{1}'\Sigma\mathbf{1} &= \begin{pmatrix} 1 & 1 & \dots & 1 \end{pmatrix}' \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \dots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \dots & \sigma_N^2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \\ &= \text{ sum of all elements in } \Sigma.\end{aligned}$$

As a result, we can write:

$$\mathbf{1}'\Sigma\mathbf{1} = \sum_{i=1}^{N}\sigma_i^2 + \sum_{j\neq i}\sigma_{ij},$$

which is the sum of the N diagonal variance terms plus the $N(N-1)$ off-diagonal covariance terms. Define the average variance as the average of the diagonal terms of Σ:

$$\overline{\text{var}} = \frac{1}{N}\sum_{i=1}^{N}\sigma_i^2,$$

and the average covariance as the average of the off-diagonal terms of Σ:

$$\overline{\text{cov}} = \frac{1}{N(N-1)} \sum_{j \neq i} \sigma_{ij}.$$

Then $\mathbf{1}'\Sigma\mathbf{1} = N\overline{\text{var}} + N(N-1)\overline{\text{cov}}$ and $\text{var}(R_{p,x})$ reduces to:

$$\text{var}(R_{p,x}) = \frac{1}{N^2}\mathbf{1}'\Sigma\mathbf{1} = \frac{1}{N}\overline{\text{var}} + \frac{N(N-1)}{N^2}\overline{\text{cov}} = \frac{1}{N}\overline{\text{var}} + \left(1 - \frac{1}{N}\right)\overline{\text{cov}}. \tag{11.1}$$

If N is reasonably large (e.g. $N > 100$) then

$$\frac{1}{N}\overline{\text{var}} \approx 0 \,\text{and}\, \frac{1}{N}\overline{\text{cov}} \approx 0,$$

so that

$$\text{var}(R_{p,x}) \approx \overline{\text{cov}}.$$

Hence, in a large diversified portfolio the portfolio variance is approximately equal to the average of the pairwise covariances. So what matters for portfolio risk is not the individual asset variances but rather the average of the asset covariances. In particular, portfolios with highly positively correlated returns will have higher portfolio variance than portfolios with less correlated returns.

Equation 11.1 also shows that portfolio variance should be a decreasing function of the number of assets in the portfolio, and that this function should level off at $\overline{\text{cov}}$ for large N.

11.1.2.1 Large portfolio with perfectly positively correlated assets

In chapter Chapter 10 I showed that the volatility of a long-only portfolio of two risky assets with perfectly positively correlated returns ($\rho_{12} = 1$) is equal to a share weighted average of the asset volatilities: $\sigma_p = x_1\sigma_1 + x_2\sigma_2$. Here, we show using matrix algebra that this result generalizes to the case of N assets. Let $\mathbf{R}$ be an $N \times 1$ vector of asset returns and assume that all returns are perfectly positively correlated ($\rho_{ij} = 1$ for all i, j). The $N \times N$ return correlation matrix is then:

$$\mathbf{C} = \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & \dots & 1 \end{pmatrix} = \mathbf{1}\mathbf{1}',$$

and the $N \times N$ return covariance matrix is:

$$\Sigma = \mathbf{DCD} = \mathbf{D1}\mathbf{1}'\mathbf{D},$$

where $\mathbf{D} = \text{diag}(\sigma_1, \sigma_2, ..., \sigma_N)$. Let $\mathbf{x} = (x_1, x_2, ..., x_N)'$ be a portfolio weight vector and let $R_{p,x} = \mathbf{x}'\mathbf{R}$ denote the portfolio return. Then:

$$\sigma_p^2 = \text{var}(\mathbf{x}'\mathbf{R}) = \mathbf{x}'\Sigma\mathbf{x} = \mathbf{x}'\mathbf{D}\mathbf{1}\mathbf{1}'\mathbf{D}\mathbf{x} = (\mathbf{x}'\mathbf{D}\mathbf{1})^2.$$

Now,

$$\begin{aligned}
\mathbf{x}'\mathbf{D}\mathbf{1} &= \begin{pmatrix} x_1 & x_2 & ... & x_N \end{pmatrix} \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_N \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} x_1\sigma_1 & 0 & \cdots & 0 \\ 0 & x_2\sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & x_N\sigma_N \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \\
&= \sum_{i=1}^{N} x_i\sigma_i.
\end{aligned}$$

Therefore,

$$\sigma_p^2 = (\mathbf{x}'\mathbf{D}\mathbf{1})^2 = \left(\sum_{i=1}^{N} x_i\sigma_i\right)^2$$

and so:

$$\sigma_p = \sum_{i=1}^{N} x_i\sigma_i.$$

When all assets have perfectly positively correlated returns, portfolio volatility is a share weighted average of individual asset volatilities. You can think of this as the risk of a (long-only) portfolio when there are no diversification benefits. When assets are not all perfectly positively correlated, there are some diversification benefits and so σ_p will be less than $\sum_{i=1}^{N} x_i\sigma_i$.

11.2 Determining the Global Minimum Variance Portfolio Using Matrix Algebra

The global minimum variance portfolio $\mathbf{m} = (m_1, ..., m_N)'$ for the N asset case solves the constrained minimization problem:

$$\min_{\mathbf{m}} \ \sigma_{p,m}^2 = \mathbf{m}' \ \mathbf{m}, \ s.t. \ \mathbf{m}'\mathbf{1} = 1. \tag{11.2}$$

The Lagrangian for this problem is:

$$L(\mathbf{m}, \lambda) = \mathbf{m}'\Sigma\mathbf{m} + \lambda(\mathbf{m}'\mathbf{1} - 1),$$

and the first order conditions (FOCs) for a minimum are:

$$\mathbf{0} = \frac{\partial L}{\partial \mathbf{m}} = \frac{\partial \mathbf{m}'\Sigma\mathbf{m}}{\partial \mathbf{m}} + \frac{\partial \lambda(\mathbf{m}'\mathbf{1} - 1)}{\partial \mathbf{m}} = 2\Sigma\mathbf{m} + \lambda\mathbf{1}, \tag{11.3}$$

$$0 = \frac{\partial L}{\partial \lambda} = \frac{\partial \mathbf{m}'\Sigma\mathbf{m}}{\partial \lambda} + \frac{\partial \lambda(\mathbf{m}'\mathbf{1} - 1)}{\partial \lambda} = \mathbf{m}'\mathbf{1} - 1. \tag{11.4}$$

The FOCs Equation 11.3 - Equation 11.4 give $N + 1$ linear equations in $N + 1$ unknowns which can be solved to find the global minimum variance portfolio weight vector $\mathbf{m}$ as follows. The $N+1$ linear equations describing the first order conditions have the matrix representation:

$$\begin{pmatrix} 2\Sigma & \mathbf{1} \\ \mathbf{1}' & 0 \end{pmatrix} \begin{pmatrix} \mathbf{m} \\ \lambda \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}. \tag{11.5}$$

The system Equation 11.5 is of the form $\mathbf{A}_m\mathbf{z}_m = \mathbf{b}$, where:

$$\mathbf{A}_m = \begin{pmatrix} 2\Sigma & \mathbf{1} \\ \mathbf{1}' & 0 \end{pmatrix}, \ \mathbf{z}_m = \begin{pmatrix} \mathbf{m} \\ \lambda \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}.$$

Provided $\mathbf{A}_m$ is invertible, the solution for $\mathbf{z}_m$ is:[1]

$$\mathbf{z}_m = \mathbf{A}_m^{-1}\mathbf{b}. \tag{11.6}$$

The first N elements of $\mathbf{z}_m$ is the portfolio weight vector $\mathbf{m}$ for the global minimum variance portfolio with expected return $\mu_{p,m} = \mathbf{m}'\mu$ and variance $\sigma^2_{p,m} = \mathbf{m}'\Sigma\mathbf{m}$.

Example 11.4 (Global minimum variance portfolio for example data)**.** Using the data in Table Table 11.1, you can use R to compute the global minimum variance portfolio weights from Equation 11.6 as follows:

```
top.mat = cbind(2*sigma.mat, rep(1, 3))
bot.vec = c(rep(1, 3), 0)
Am.mat = rbind(top.mat, bot.vec)
b.vec = c(rep(0, 3), 1)
z.m.mat = solve(Am.mat)%*%b.vec
m.vec = z.m.mat[1:3,1]
m.vec
```

[1] For $\mathbf{A}_m$ to be invertible we need Σ to be invertible, which requires $|\rho_{ij}| \neq 1$ for $i \neq j$ and $\sigma_i^2 > 0$ for all i.

```
#>  MSFT  NORD  SBUX
#> 0.441 0.366 0.193
```

Hence, the global minimum variance portfolio has portfolio weights $m_{\text{msft}} = 0.441$, $m_{\text{nord}} = 0.366$ and $m_{\text{sbux}} = 0.193$, and is given by the vector $\mathbf{m} = (0.441, 0.366, 0.193)'$. The expected return on this portfolio, $\mu_{p,m} = \mathbf{m}'\mu$, is:

```
mu.gmin = as.numeric(crossprod(m.vec, mu.vec))
mu.gmin
#> [1] 0.0249
```

The portfolio variance, $\sigma^2_{p,m} = \mathbf{m}'\Sigma\mathbf{m}$, and standard deviation, $\sigma_{p,m}$, are:

```
sig2.gmin = as.numeric(t(m.vec)%*%sigma.mat%*%m.vec)
sig.gmin = sqrt(sig2.gmin)
c(sig2.gmin, sig.gmin)
#> [1] 0.00528 0.07268
```

In Figure 11.4, this portfolio is labeled "GMIN" .

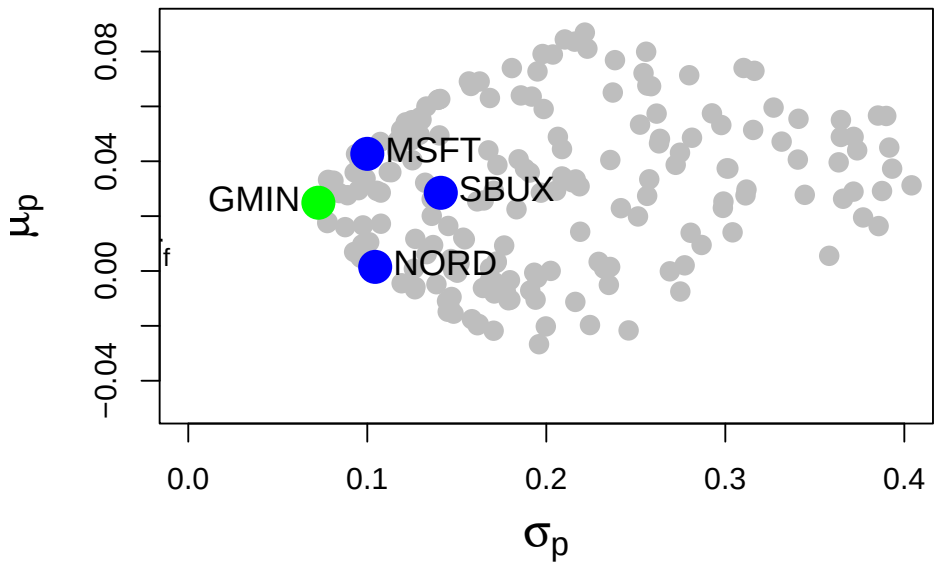


Figure 11.4: Global minimum variance portfolio from example data.

■

11.2.1 Alternative derivation of global minimum variance portfolio

We can use the first order conditions Equation 11.3 - Equation 11.4 to give an explicit solution for the global minimum variance portfolio $\mathbf{m}$ as follows. First, use Equation 11.3 to solve for $\mathbf{m}$ as a function of λ:

$$\mathbf{m} = -\frac{1}{2} \cdot \lambda \Sigma^{-1} \mathbf{1}. \tag{11.7}$$

Next, multiply both sides of Equation 11.7 by $\mathbf{1}'$ and use Equation 11.4 to solve for λ:

$$\begin{aligned} 1 = \mathbf{1}'\mathbf{m} &= -\frac{1}{2} \cdot \lambda \mathbf{1}'\Sigma^{-1}\mathbf{1} \\ \Rightarrow \lambda &= -2 \cdot \frac{1}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}. \end{aligned}$$

Finally, substitute the value for λ back into Equation 11.7 to solve for $\mathbf{m}$:

$$\mathbf{m} = -\frac{1}{2}(-2)\frac{1}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}\Sigma^{-1}\mathbf{1} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}. \tag{11.8}$$

Notice that Equation 11.8 shows that a solution for $\mathbf{m}$ exists as long as Σ is invertible.

Example 11.5 (Finding global minimum variance portfolio for example data)**.** Using the data in Table Table 11.1, we can use R to compute the global minimum variance portfolio weights from Equation 11.8 as follows:

```
one.vec = rep(1, 3)
sigma.inv.mat = solve(sigma.mat)
top.mat = sigma.inv.mat%*%one.vec
bot.val = as.numeric((t(one.vec)%*%sigma.inv.mat%*%one.vec))
m.mat = top.mat/bot.val
m.mat[,1]
#>  MSFT  NORD  SBUX
#> 0.441 0.366 0.193
```

■

11.3 Determining Mean-Variance Efficient Portfolios Using Matrix Algebra

The investment opportunity set is the set of portfolio expected return, μ_p, and portfolio standard deviation, σ_p, values for all possible portfolios whose weights sum to one. As in the two risky asset case, this set can be described in a graph with μ_p on the vertical axis and σ_p on the horizontal

axis. With two assets, the investment opportunity set in (μ_p, σ_p)-space lies on a curve (one side of a hyperbola). With three or more assets, the investment opportunity set in (μ_p, σ_p)-space is described by set of values whose general shape is complicated and depends crucially on the covariance terms σ_{ij}.[2] However, we do not have to fully characterize the entire investment opportunity set. If we assume that investors choose portfolios to maximize expected return subject to a target level of risk, or, equivalently, to minimize risk subject to a target expected return, then we can simplify the asset allocation problem by only concentrating on the set of efficient portfolios. These portfolios lie on the boundary of the investment opportunity set above the global minimum variance portfolio. This is the framework originally developed by Harry Markowitz, the father of portfolio theory and winner of the Nobel Prize in economics.

Following Markowitz, we assume that investors wish to find portfolios that have the best expected return-risk trade-off. We can characterize these efficient portfolios in two equivalent ways. In the first way, investors seek to find portfolios that maximize portfolio expected return for a given level of risk as measured by portfolio variance. Let $\sigma^2_{p,0}$ denote a target level of risk. Then the constrained maximization problem to find an efficient portfolio is:

$$\begin{aligned} &\max_{\mathbf{x}} \mu_p = \mathbf{x}'\mu \\ &\text{s.t. } \sigma^2_p = \mathbf{x}'\Sigma\mathbf{x} = \sigma^2_{p,0} \text{ and } \mathbf{x}'\mathbf{1} = 1. \end{aligned} \tag{11.9}$$

The investor's problem of maximizing portfolio expected return subject to a target level of risk has an equivalent dual representation in which the investor minimizes the risk of the portfolio (as measured by portfolio variance) subject to a target expected return level. Let $\mu_{p,0}$ denote a target expected return level. Then the dual problem is the constrained minimization problem:[3]

$$\begin{aligned} &\min_{\mathbf{x}} \ \sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x} \\ &\text{s.t. } \mu_p = \mathbf{x}'\mu = \mu_{p,0}, \text{ and } \mathbf{x}'\mathbf{1} = 1. \end{aligned} \tag{11.10}$$

To find efficient portfolios of risky assets in practice, the dual problem Equation 11.10 is most often solved. This is partially due to computational conveniences and partly due to investors being more willing to specify target expected returns rather than target risk levels. The efficient portfolio frontier is a graph of μ_p versus σ_p values for the set of efficient portfolios generated by solving Equation 11.10 for all possible target expected return levels $\mu_{p,0}$ above the expected return on the global minimum variance portfolio. Just as in the two asset case, the resulting efficient frontier will resemble one side of an hyperbola and is often called the "Markowitz bullet". This frontier is illustrated in Figure 11.3 as the boundary of the set generated by random portfolios above the global minimum variance portfolio.

To solve the constrained minimization problem Equation 11.10, first form the Lagrangian func-

[2] This set is partially visualized in Figure 11.3 using random portfolios.
[3] This duality is similar to the duality in economics of profit maximization and cost minimization.

tion:

$$L(x, \lambda_1, \lambda_2) = \mathbf{x}'\Sigma\mathbf{x} + \lambda_1(\mathbf{x}'\mu - \mu_{p,0}) + \lambda_2(\mathbf{x}'\mathbf{1} - 1).$$

Because there are two constraints ($\mathbf{x}'\mu = \mu_{p,0}$ and $\mathbf{x}'\mathbf{1} = 1$) there are two Lagrange multipliers λ_1 and λ_2. The FOCs for a minimum are the linear equations:

$$\frac{\partial L(\mathbf{x}, \lambda_1, \lambda_2)}{\partial \mathbf{x}} = 2\Sigma\mathbf{x} + \lambda_1\mu + \lambda_2\mathbf{1} = \mathbf{0}, \tag{11.11}$$

$$\frac{\partial L(\mathbf{x}, \lambda_1, \lambda_2)}{\partial \lambda_1} = \mathbf{x}'\mu - \mu_{p,0} = 0, \tag{11.12}$$

$$\frac{\partial L(\mathbf{x}, \lambda_1, \lambda_2)}{\partial \lambda_2} = \mathbf{x}'\mathbf{1} - 1 = 0. \tag{11.13}$$

These FOCs consist of $N + 2$ linear equations in $N + 2$ unknowns ($\mathbf{x}$, λ_1, λ_2). We can represent the system of linear equations using matrix algebra as:

$$\begin{pmatrix} 2\Sigma & \mu & \mathbf{1} \\ \mu' & 0 & 0 \\ \mathbf{1}' & 0 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mu_{p,0} \\ 1 \end{pmatrix},$$

which is of the form $\mathbf{A}\mathbf{z}_x = \mathbf{b}_0$, where

$$\mathbf{A} = \begin{pmatrix} 2\Sigma & \mu & \mathbf{1} \\ \mu' & 0 & 0 \\ \mathbf{1}' & 0 & 0 \end{pmatrix}, \ \mathbf{z}_x = \begin{pmatrix} \mathbf{x} \\ \lambda_1 \\ \lambda_2 \end{pmatrix} \text{ and } \mathbf{b}_0 = \begin{pmatrix} \mathbf{0} \\ \mu_{p,0} \\ 1 \end{pmatrix}.$$

The solution for $\mathbf{z}_x$ is then:

$$\mathbf{z}_x = \mathbf{A}^{-1}\mathbf{b}_0. \tag{11.14}$$

The first N elements of $\mathbf{z}_x$ are the portfolio weights $\mathbf{x}$ for the minimum variance portfolio with expected return $\mu_{p,x} = \mu_{p,0}$. If $\mu_{p,0}$ is greater than or equal to the expected return on the global minimum variance portfolio, then $\mathbf{x}$ is an efficient (frontier) portfolio. Otherwise, it is an inefficient (frontier) portfolio.

Example 11.6 (Efficient portfolio with the same expected return as Microsoft)**.** Using the data in Table Table 11.1, consider finding a minimum variance portfolio with the same expected return as Microsoft. This will be an efficient portfolio because $\mu_{\text{msft}} = 0.043 > \mu_{p,m} = 0.025$. Call this portfolio $\mathbf{x} = (x_{\text{msft}}, x_{\text{nord}}, x_{\text{sbux}})'$. That is, consider solving Equation 11.10 with target expected return $\mu_{p,0} = \mu_{\text{msft}} = 0.043$ using Equation 11.14. The R calculations to create the matrix $\mathbf{A}_x$ and the vectors $\mathbf{z}_x$ and $\mathbf{b}_{\text{msft}}$ are:

```
top.mat = cbind(2*sigma.mat, mu.vec, rep(1, 3))
mid.vec = c(mu.vec, 0, 0)
bot.vec = c(rep(1, 3), 0, 0)
Ax.mat = rbind(top.mat, mid.vec, bot.vec)
bmsft.vec = c(rep(0, 3), mu.vec["MSFT"], 1)
```

and the R code to solve for $\mathbf{x}$ using Equation 11.14 is:

```
z.mat = solve(Ax.mat)%*%bmsft.vec
x.vec = z.mat[1:3,]
x.vec
#>    MSFT    NORD    SBUX
#>  0.8275 -0.0907  0.2633
```

The efficient portfolio with the same expected return as Microsoft has portfolio weights $x_{\text{msft}} = 0.8275$, $x_{\text{nord}} = -0.0907$ and $x_{\text{sbux}} = 0.2633$, and is given by the vector $\mathbf{x} = (0.8275, -0.0907, 0.2633)'$. The expected return on this portfolio, $\mu_{p,x} = \mathbf{x}'\mu$, is equal to the target return μ_{msft}:

```
mu.px = as.numeric(crossprod(x.vec, mu.vec))
mu.px
#> [1] 0.0427
```

The portfolio variance, $\sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x}$, and standard deviation, $\sigma_{p,x}$, are:

```
sig2.px = as.numeric(t(x.vec)%*%sigma.mat%*%x.vec)
sig.px = sqrt(sig2.px)
c(sig2.px, sig.px)
#> [1] 0.0084 0.0917
```

and are smaller than the corresponding values for Microsoft (see Table Table 11.1). This efficient portfolio is labeled "E1" in Figure 11.5. ■

Example 11.7 (Efficient portfolio with the same expected return as Starbucks)**.** To find a minimum variance portfolio $\mathbf{y} = (y_{\text{msft}}, y_{\text{nord}}, y_{\text{sbux}})'$ with the same expected return as Starbucks we use Equation 11.14 with $\mathbf{b}_{\text{sbux}} = (\mathbf{0}', \mu_{\text{sbux}}, 1)'$:

```
bsbux.vec = c(rep(0, 3), mu.vec["SBUX"], 1)
z.mat = solve(Ax.mat)%*%bsbux.vec
y.vec = z.mat[1:3,]
y.vec
```

```
#>  MSFT  NORD  SBUX
#> 0.519 0.273 0.207
```

The portfolio $\mathbf{y} = (0.519, 0.273, 0.207)'$ is an efficient portfolio on the outer boundary because $\mu_{\text{sbux}} = 0.0285 > \mu_{p,m} = 0.0249$. The portfolio expected return and standard deviation are:

```
mu.py = as.numeric(crossprod(y.vec, mu.vec))
sig2.py = as.numeric(t(y.vec)%*%sigma.mat%*%y.vec)
sig.py = sqrt(sig2.py)
c(mu.py,sig.py)
#> [1] 0.0285 0.0736
```

This efficient portfolio is labeled "E2" in Figure 11.5. It has the same expected return as SBUX but a smaller standard deviation. ■

The covariance and correlation values between the portfolio returns $R_{p,x} = \mathbf{x}'\mathbf{R}$ and $R_{p,y} = \mathbf{y}'\mathbf{R}$ are given by:

```
sigma.xy = as.numeric(t(x.vec)%*%sigma.mat%*%y.vec)
rho.xy = sigma.xy/(sig.px*sig.py)
c(sigma.xy, rho.xy)
#> [1] 0.00591 0.87722
```

This covariance will be used later on when constructing the entire frontier of efficient portfolios.

11.3.1 Alternative derivation of an efficient portfolio

Equation 11.14 does not give an explicit solution for the minimum variance portfolio $\mathbf{x}$. As with the global minimum variance portfolio Equation 11.8, an explicit solution for $\mathbf{x}$ can also be found. Consider the first order conditions Equation 11.11 - Equation 11.13 from the optimization problem Equation 11.10. First, use Equation 11.11 to solve for the $N \times 1$ vector $\mathbf{x}$:

$$\mathbf{x} = -\frac{1}{2}\lambda_1 \Sigma^{-1}\mu - \frac{1}{2}\lambda_2 \Sigma^{-1}\mathbf{1}. \tag{11.15}$$

Define the $N \times 2$ matrix $\mathbf{M} = [\mu \vdots \mathbf{1}]$ and the 2×1 vector $\lambda = (\lambda_1, \lambda_2)'$. Then we can rewrite Equation 11.15 in matrix form as:

$$\mathbf{x} = -\frac{1}{2}\Sigma^{-1}\mathbf{M}\lambda. \tag{11.16}$$

Next, to find the values for λ_1 and λ_2, pre-multiply Equation 11.15 by μ' and use Equation 11.12 to give:

$$\mu_0 = \mu'\mathbf{x} = \frac{1}{2}\lambda_1 \mu'\Sigma^{-1}\mu - \frac{1}{2}\lambda_2 \mu'\Sigma^{-1}\mathbf{1}. \tag{11.17}$$

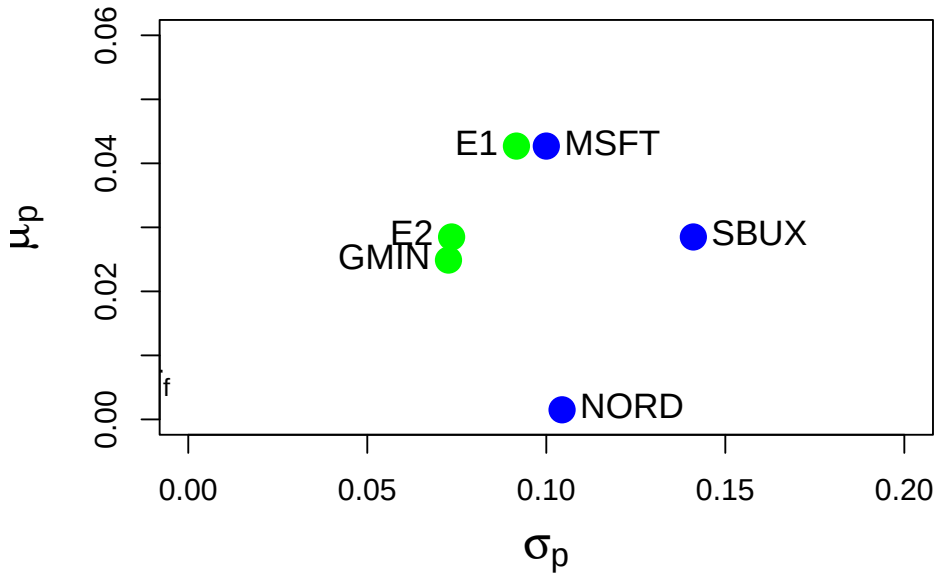


Figure 11.5: Minimum variance efficient portfolios from example data. Portfolio E1 has the same expected return as Microsoft, and portfolio E2 has the same expected returns as Starbucks.

Similarly, pre-multiply Equation 11.15 by $\mathbf{1}'$ and use Equation 11.13 to give:

$$1 = \mathbf{1}'\mathbf{x} = -\frac{1}{2}\lambda_1 \mathbf{1}'\Sigma^{-1}\mu - \frac{1}{2}\lambda_2 \mathbf{1}'\Sigma^{-1}\mathbf{1}. \tag{11.18}$$

Now, we have two linear equations Equation 11.17 and Equation 11.18 involving λ_1 and λ_2 which we can write in matrix notation as:

$$-\frac{1}{2}\begin{pmatrix} \mu'\Sigma^{-1}\mu & \mu'\Sigma^{-1}\mathbf{1} \\ \mu'\Sigma^{-1}\mathbf{1} & \mathbf{1}'\Sigma^{-1}\mathbf{1} \end{pmatrix}\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} \mu_0 \\ 1 \end{pmatrix}. \tag{11.19}$$

Define,

$$\begin{pmatrix} \mu'\Sigma^{-1}\mu & \mu'\Sigma^{-1}\mathbf{1} \\ \mu'\Sigma^{-1}\mathbf{1} & \mathbf{1}'\Sigma^{-1}\mathbf{1} \end{pmatrix} = \mathbf{M}'\Sigma^{-1}\mathbf{M} = \mathbf{B},$$

$$\tilde{\mu}_0 = \begin{pmatrix} \mu_0 \\ 1 \end{pmatrix},$$

so that we can rewrite Equation 11.19 as,

$$-\frac{1}{2}\mathbf{B}\lambda = \tilde{\mu}_0.$$

Provided $\mathbf{B}$ is invertible, the solution for $\lambda = (\lambda_1, \lambda_2)'$ is

$$\lambda = -2\mathbf{B}^{-1}\tilde{\mu}_0. \tag{11.20}$$

Substituting Equation 11.20 back into Equation 11.16 gives an explicit expression for the efficient portfolio weight vector $\mathbf{x}$:

$$\mathbf{x} = -\frac{1}{2}\Sigma^{-1}\mathbf{M}\lambda = -\frac{1}{2}\Sigma^{-1}\mathbf{M}\left(-2\mathbf{B}^{-1}\tilde{\mu}_0\right) = \Sigma^{-1}\mathbf{M}\mathbf{B}^{-1}\tilde{\mu}_0. \tag{11.21}$$

Example 11.8 (Alternative solution for efficient portfolio with the same expected return as Microsoft)**.** The R code to compute the efficient portfolio with the same expected return as Microsoft using Equation 11.21 is:

```
M.mat = cbind(mu.vec, one.vec)
B.mat = t(M.mat)%*%solve(sigma.mat)%*%M.mat
mu.tilde.msft = c(mu.vec["MSFT"], 1)
x.vec.2 = solve(sigma.mat)%*%M.mat%*%solve(B.mat)%*%mu.tilde.msft
t(x.vec.2)
#>       MSFT    NORD  SBUX
#> [1,] 0.827 -0.0907 0.263
```

■

11.4 Computing the Mean-Variance Efficient Frontier

The analytic expression for a minimum variance portfolio Equation 11.21 can be used to show that any minimum variance portfolio can be created as a linear combination of any two minimum variance portfolios with different target expected returns. If the expected return on the resulting portfolio is greater than the expected return on the global minimum variance portfolio, then the portfolio is an efficient frontier portfolio. Otherwise, the portfolio is an inefficient frontier portfolio. As a result, to compute the portfolio frontier in (μ_p, σ_p) space (Markowitz bullet) we only need to find two efficient portfolios. The remaining frontier portfolios can then be expressed as linear combinations of these two portfolios. The following proposition describes the process for the three risky asset case using matrix algebra.

Proposition 11.1 (Creating a frontier portfolio from two minimum variance portfolios)**.** *Let* $\mathbf{x}$ *and* $\mathbf{y}$ *be any two minimum variance portfolios with different target expected returns* $\mathbf{x}'\mu = \mu_{p,0} \neq \mathbf{y}'\mu = \mu_{p,1}$*. That is, portfolio* $\mathbf{x}$ *solves:*

$$\begin{aligned} \min_{\mathbf{x}} \ & \sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x} \\ s.t. \ & \mathbf{x}'\mu = \mu_{p,0} \text{ and } \mathbf{x}'\mathbf{1} = 1, \end{aligned}$$

and portfolio $\mathbf{y}$ *solves,*

$$\begin{aligned} \min_{\mathbf{y}} & \sigma^2_{p,y} = \mathbf{y}'\Sigma\mathbf{y} \\ s.t. \ & \mathbf{y}'\mu = \mu_{p,1} \text{ and } \mathbf{y}'\mathbf{1} = 1. \end{aligned}$$

Let α *be any constant and define the portfolio* $\mathbf{z}$ *as a linear combination of portfolios* $\mathbf{x}$ *and* $\mathbf{y}$*:*

$$\begin{aligned} \mathbf{z} &= \alpha \cdot \mathbf{x} + (1-\alpha) \cdot \mathbf{y} \\ &= \begin{pmatrix} \alpha x_1 + (1-\alpha) y_1 \\ \vdots \\ \alpha x_N + (1-\alpha) y_N \end{pmatrix}. \end{aligned} \tag{11.22}$$

Then the following results hold:

1. *The portfolio* $\mathbf{z}$ *is a minimum variance portfolio with expected return and variance given by:*

$$\mu_{p,z} = \mathbf{z}'\mu = \alpha \cdot \mu_{p,x} + (1-\alpha) \cdot \mu_{p,y}, \tag{11.23}$$

$$\sigma^2_{p,z} = \mathbf{z}'\Sigma\mathbf{z} = \alpha^2\sigma^2_{p,x} + (1-\alpha)^2\sigma^2_{p,y} + 2\alpha(1-\alpha)\sigma_{xy}, \tag{11.24}$$

where $\sigma_{p,x}^2 = \mathbf{x}'\Sigma\mathbf{x}, \sigma_{p,y}^2 = \mathbf{y}'\Sigma\mathbf{y}, \sigma_{xy} = \mathbf{x}'\Sigma\mathbf{y}$.

2. *If* $\mu_{p,z} \geq \mu_{p,m}$*, where* $\mu_{p,m}$ *is the expected return on the global minimum variance portfolio, then portfolio* $\mathbf{z}$ *is an efficient frontier portfolio. Otherwise,* $\mathbf{z}$ *is an inefficient frontier portfolio.*

The proof of (**creating-a-frontier-portfolio-from-two-minimum-variance-portfolios?**) a follows directly from applying Equation 11.21 to portfolios $\mathbf{x}$ and $\mathbf{y}$:

$$\begin{aligned}\mathbf{x} &= \Sigma^{-1}\mathbf{M}\mathbf{B}^{-1}\tilde{\mu}_x,\\ \mathbf{y} &= \Sigma^{-1}\mathbf{M}\mathbf{B}^{-1}\tilde{\mu}_y,\end{aligned}$$

where $\tilde{\mu}_x = (\mu_{p,x}, 1)'$ and $\tilde{\mu}_y = (\mu_{p,y}, 1)'$. Then, for portfolio $\mathbf{z}$:

$$\begin{aligned}\mathbf{z} &= \alpha \cdot \mathbf{x} + (1-\alpha)\cdot\mathbf{y}\\ &= \alpha \cdot \Sigma^{-1}\mathbf{M}\mathbf{B}^{-1}\tilde{\mu}_x + (1-\alpha)\cdot\Sigma^{-1}\mathbf{M}\mathbf{B}^{-1}\tilde{\mu}_y\\ &= \Sigma^{-1}\mathbf{M}\mathbf{B}^{-1}(\alpha\cdot\tilde{\mu}_x + (1-\alpha)\cdot\tilde{\mu}_y)\\ &= \Sigma^{-1}\mathbf{M}\mathbf{B}^{-1}\tilde{\mu}_z,\end{aligned}$$

where $\tilde{\mu}_z = \alpha\cdot\tilde{\mu}_x + (1-\alpha)\cdot\tilde{\mu}_y = (\mu_{p,z}, 1)'$. Result 2. follows from the definition of an efficient portfolio.

Example 11.9 (Create an arbitrary efficient frontier portfolio from two efficient minimum variance portfolios)**.** Consider the data in Table 1 and the previously computed minimum variance portfolios that have the same expected return as Microsoft and Starbucks, respectively, and let $\alpha = 0.5$. From Equation 11.22, the frontier portfolio $\mathbf{z}$ is constructed using:

$$\begin{aligned}\mathbf{z} &= \alpha\cdot\mathbf{x} + (1-\alpha)\cdot\mathbf{y}\\ &= 0.5\cdot\begin{pmatrix}0.8275\\ -0.0907\\ 0.2633\end{pmatrix} + 0.5\cdot\begin{pmatrix}0.519\\ 0.273\\ 0.207\end{pmatrix}\\ &= \begin{pmatrix}(0.5)(0.8275)\\ (0.5)(-0.0907)\\ (0.5)(0.2633)\end{pmatrix} + \begin{pmatrix}(0.5)(0.519)\\ (0.5)(0.273)\\ (0.5)(0.207)\end{pmatrix}\\ &= \begin{pmatrix}0.6734\\ 0.0912\\ 0.2354\end{pmatrix} = \begin{pmatrix}z_A\\ z_B\\ z_C\end{pmatrix}.\end{aligned}$$

In R, the new frontier portfolio is computed using:

```
a = 0.5
z.vec = a*x.vec + (1-a)*y.vec
z.vec
#>   MSFT   NORD   SBUX
#> 0.6734 0.0912 0.2354
```

Using $\mu_{p,z} = \mathbf{z}'\mu$ and $\sigma^2_{p,z} = \mathbf{z}'\Sigma\mathbf{z}$, the expected return, variance and standard deviation of this portfolio are:

```
mu.pz = as.numeric(crossprod(z.vec, mu.vec))
sig2.pz = as.numeric(t(z.vec)%*%sigma.mat%*%z.vec)
sig.pz = sqrt(sig2.pz)
c(mu.pz, sig.pz)
#> [1] 0.0356 0.0801
```

Equivalently, using $\mu_{p,z} = \alpha\mu_{p,x} + (1-\alpha)\mu_{p,y}$ and $\sigma^2_{p,z} = \alpha^2\sigma^2_{p,x} + (1-\alpha)^2\sigma^2_{p,y} + 2\alpha(1-\alpha)\sigma_{xy}$ the expected return, variance and standard deviation of this portfolio are:

```
mu.pz = a*mu.px + (1-a)*mu.py
sig.xy = as.numeric(t(x.vec)%*%sigma.mat%*%y.vec)
sig2.pz = a^2 * sig2.px + (1-a)^2 * sig2.py + 2*a*(1-a)*sig.xy
sig.pz = sqrt(sig2.pz)
c(mu.pz, sig.pz)
#> [1] 0.0356 0.0801
```

Because $\mu_{p,z} = 0.0356 > \mu_{p,m} = 0.0249$ the frontier portfolio $\mathbf{z}$ is an efficient frontier portfolio. The three efficient frontier portfolios $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ are illustrated in Figure 11.6 and are labeled "E1" , "E2" and "E3" , respectively. ■

Example 11.10 (Create an efficient frontier portfolio with a given expected return from two efficient minimum variance portfolios)**.** Given the two minimum variance portfolios with target expected returns equal to the expected returns on Microsoft and Starbucks, respectively, consider creating a frontier portfolio with target expected return equal to 0.05. To determine the value of α that corresponds with this portfolio we use the equation

$$\mu_{p,z} = \alpha_{.05}\mu_{p,x} + (1-\alpha_{.05})\mu_{p,y} = 0.05.$$

We can then solve for $\alpha_{.05}$:

$$\alpha_{.05} = \frac{0.05 - \mu_{p,y}}{\mu_{p,x} - \mu_{p,y}} = \frac{0.05 - 0.0285}{0.0427 - 0.0285} = 1.51.$$

Given $\alpha_{.05}$ we can solve for the portfolio weights using:

$$\begin{aligned}
\mathbf{z}_{.05} &= \alpha_{.05}\mathbf{x} + (1-\alpha_{.05})\mathbf{y} \\
&= 1.51 \times \begin{pmatrix} 0.8275 \\ -0.0907 \\ 0.2633 \end{pmatrix} - 0.514 \times \begin{pmatrix} 0.519 \\ 0.273 \\ 0.207 \end{pmatrix} \\
&= \begin{pmatrix} 0.986 \\ -0.278 \\ 0.292 \end{pmatrix}.
\end{aligned}$$

We can then compute the mean and variance of this portfolio using $\mu_{p,z_{.05}} = \mathbf{z}'_{.05}\mu$ and $\sigma^2_{p,z_{.05}} = \mathbf{z}'_{.05}\Sigma\mathbf{z}_{.05}$. Using R, the calculations are:

```
# compute alpha
a.05 = (0.05 - mu.py)/(mu.px - mu.py)
a.05
#> [1] 1.51
# compute weights
z.05 = a.05*x.vec + (1 - a.05)*y.vec
z.05
#>   MSFT   NORD   SBUX
#>  0.986 -0.278  0.292
# compute mean and volatility
mu.pz.05 = as.numeric(crossprod(z.05,mu.vec))
sig.pz.05 = as.numeric(sqrt(t(z.05)%*%sigma.mat%*%z.05))
c(mu.pz.05,sig.pz.05)
#> [1] 0.050 0.107
```

This portfolio is labeled "E4" in Figure 11.6. ■

Example 11.11 (Create an inefficient frontier portfolio with a given expected return from two minimum variance portfolios)**.** Given the two minimum variance portfolios with target expected returns equal to the expected returns on Microsoft and Starbucks, respectively, consider creating a frontier portfolio with target expected return equal to the expected return on Nordstrom. Then,

$$\mu_{p,z} = \alpha_{nord}\mu_{p,x} + (1-\alpha_{nord})\mu_{p,y} = \mu_{\text{nord}} = 0.0015,$$

and we can solve for α_{nord} using:

$$\alpha_{nord} = \frac{\mu_{\text{nord}} - \mu_{p,y}}{\mu_{p,x} - \mu_{p,y}} = \frac{0.0015 - 0.0285}{0.0427 - 0.0285} = -1.901.$$

The portfolio weights are:

$$\begin{aligned}
\mathbf{z}_{nord} &= \alpha_{nord}\mathbf{x} + (1-\alpha_{nord})\mathbf{y} \\
&= -1.901 \times \begin{pmatrix} 0.8275 \\ -0.0907 \\ 0.2633 \end{pmatrix} + 2.9 \times \begin{pmatrix} 0.519 \\ 0.273 \\ 0.207 \end{pmatrix} \\
&= \begin{pmatrix} -0.0064 \\ 0.9651 \\ 0.1013 \end{pmatrix}.
\end{aligned}$$

Using R, the calculations are:

```
# compute alpha
a.nord = (mu.vec["NORD"] - mu.py)/(mu.px - mu.py)
a.nord
#> NORD
#> -1.9
# compute weights
z.nord = a.nord*x.vec + (1 - a.nord)*y.vec
z.nord
#>    MSFT    NORD    SBUX
#> -0.0664  0.9651  0.1013
# compute mean and volatility
mu.pz.nord = as.numeric(crossprod(z.nord,mu.vec))
sig.pz.nord = as.numeric(sqrt(t(z.nord)%*%sigma.mat%*%z.nord))
c(mu.pz.nord,sig.pz.nord)
#> [1] 0.0015 0.1033
```

Because $\mu_{p,z} = 0.0015 < \mu_{p,m} = 0.02489$ the frontier portfolio $\mathbf{z}$ is an inefficient frontier portfolio. This portfolio is labeled "IE1" in Figure 11.6. ■

11.4.1 Algorithm for computing efficient frontier

The efficient frontier of portfolios, i.e., those frontier portfolios with expected return greater than the expected return on the global minimum variance portfolio, can be conveniently created using Equation 11.22 with two specific efficient portfolios. The first efficient portfolio is the global minimum variance portfolio Equation 11.2. The second efficient portfolio is the efficient portfolio whose target expected return is equal to the highest expected return among all of the assets under consideration. The choice of these two efficient portfolios makes it easy to produce a nice plot of the efficient frontier. The steps for constructing the efficient frontier are:

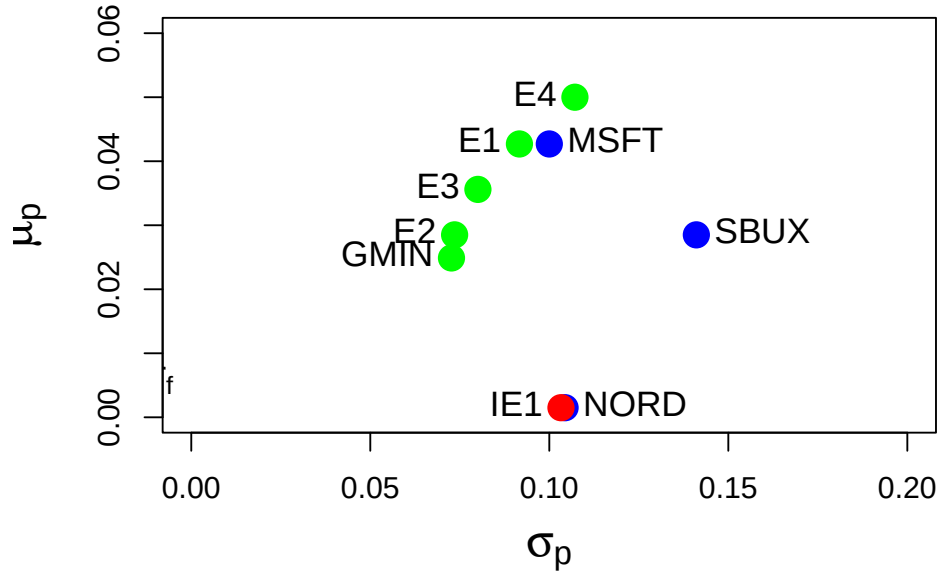


Figure 11.6: Minimum variance portfolios created as convex combinations of two minimum variance portfolios. Portfolios E3, E4 and IE1 are created from portfolios E1 and E2.

1. Compute the global minimum variance portfolio $\mathbf{m}$ by solving Equation 11.2, and compute $\mu_{p,m} = \mathbf{m}'\mu$ and $\sigma^2_{p,m} = \mathbf{m}'\Sigma\mathbf{m}$.
2. Compute the efficient portfolio $\mathbf{x}$ with target expected return equal to the maximum expected return of the assets under consideration. That is, solve Equation 11.10 with $\mu_0 = \max\{\mu_1, \dots, \mu_N\}$, and compute $\mu_{p,x} = \mathbf{x}'\mu$ and $\sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x}$.
3. Compute $\text{cov}(R_{p,m}, R_{p,x}) = \sigma_{mx} = \mathbf{m}'\Sigma\mathbf{x}$.
4. Create an initial grid of α values $\{0, 0.1, \dots 0.9, 1\}$, and compute the frontier portfolios $\mathbf{z}$ using

$$\mathbf{z} = \alpha \times \mathbf{x} + (1 - \alpha) \times \mathbf{m},$$

   and compute their expected returns and variances using Equation 11.22, Equation 11.23 and Equation 11.24, respectively.
5. Plot $\mu_{p,z}$ against $\sigma_{p,z}$ and adjust the grid of α values appropriately to create a nice plot. Negative values of α will give inefficient frontier portfolios. Values of α greater than one will give efficient frontier portfolios with expected returns greater than μ_0.

Example 11.12 (Compute and plot the efficient frontier of risky assets (Markowitz bullet))**.** To compute the efficient frontier from the three risky assets in Table Table 11.1 in R use:

```
a = seq(from=0, to=1, by=0.1)
n.a = length(a)
z.mat = matrix(0, n.a, 3)
colnames(z.mat) = names(mu.vec)
mu.z = rep(0, n.a)
sig2.z = rep(0, n.a)
sig.mx = t(m.vec)%*%sigma.mat%*%x.vec
for (i in 1:n.a) {
   z.mat[i, ] = a[i]*x.vec + (1-a[i])*m.vec
   mu.z[i] = a[i]*mu.px + (1-a[i])*mu.gmin
   sig2.z[i] = a[i]^2 * sig2.px + (1-a[i])^2 * sig2.gmin +
      2*a[i]*(1-a[i])*sig.mx
}
```

The variables `z.mat`, `mu.z` and `sig2.z` contain the weights, expected returns and variances, respectively, of the efficient frontier portfolios for a grid of α values between 0 and 1. The resulting efficient frontier is illustrated in Figure 11.7 created with:

```
plot(sqrt(sig2.z), mu.z, type="b", ylim=c(0, 0.06), xlim=c(0, 0.16),
     pch=16, col="green", cex = cex.val, ylab=expression(mu[p]),
     xlab=expression(sigma[p]),cex.lab=1.5)
points(sd.vec, mu.vec, pch=16, cex=2, lwd=2, col="blue")
points(sig.gmin, mu.gmin, pch=16, col="green", cex=2)
points(sig.px, mu.px, pch=16, col="green", cex=2)
text(sig.gmin, mu.gmin, labels="GLOBAL MIN", pos=2, cex = cex.val)
text(sd.vec, mu.vec, labels=asset.names, pos=4, cex = cex.val)
text(sig.px, mu.px, labels="E1", pos=2, cex = cex.val)
```

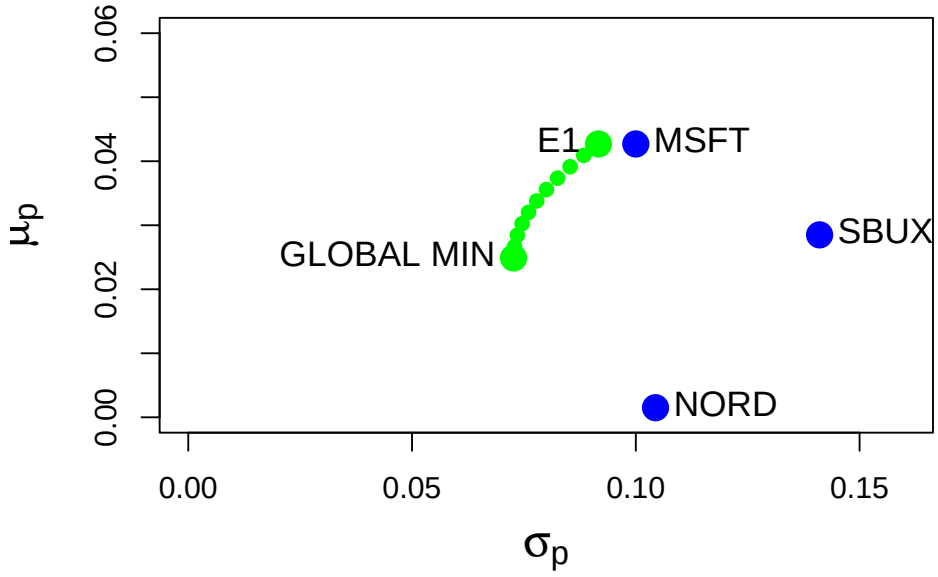


Figure 11.7: Efficient frontier computed from example data.

Each point on the efficient frontier is a portfolio of Microsoft, Nordstrom and Starbucks. It is instructive to visualize the weights in these portfolios as we move along the frontier from the global minimum variance portfolio to the efficient portfolio with expected return equal to the expected return on Microsoft. We can do this easily using the **PerformanceAnalytics** function `chart.StackedBar()`:

```
chart.StackedBar(z.mat, xaxis.labels=round(sqrt(sig2.z),digits=3),
                 legend.loc="upper",
                 colorset=rainbow8equal[1:3], ylab="Weights")
```

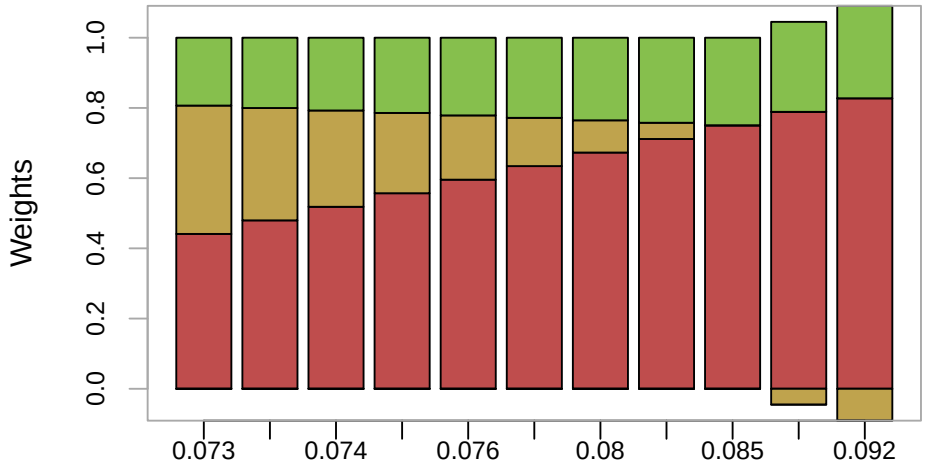


Figure 11.8: Portfolio weights in efficient frontier portfolios. x-axis is portfolio standard deviation

The resulting plot is shown in Figure 11.8. As we move along the frontier, the allocation to Microsoft increases and the allocation to Nordstrom decreases whereas the allocation to Starbucks stays about the same. ■

11.5 Computing Efficient Portfolios of N risky Assets and a Risk-Free Asset Using Matrix Algebra

In Chapter Chapter 10, we showed that efficient portfolios of two risky assets and a single risk-free (T-Bill) asset are portfolios consisting of the highest Sharpe ratio portfolio (tangency portfolio) and the T-Bill. With three or more risky assets and a T-Bill the same result holds.

11.5.1 Computing the tangency portfolio using matrix algebra

The tangency portfolio is the portfolio of risky assets that has the highest Sharpe ratio. The tangency portfolio, denoted $\mathbf{t} = (t_1, \ldots, t_N)'$, solves the constrained maximization problem:

$$\max_{\mathbf{t}} \frac{\mathbf{t}'\mu - r_f}{(\mathbf{t}'\Sigma\mathbf{t})^{\frac{1}{2}}} = \frac{\mu_{p,t} - r_f}{\sigma_{p,t}} \text{ s.t. } \mathbf{t}'\mathbf{1} = 1, \tag{11.25}$$

where $\mu_{p,t} = \mathbf{t}'\mu$ and $\sigma_{p,t} = (\mathbf{t}'\Sigma\mathbf{t})^{\frac{1}{2}}$. The Lagrangian for this problem is:

$$L(\mathbf{t}, \lambda) = \left(\mathbf{t}'\mu - r_f\right)(\mathbf{t}'\Sigma\mathbf{t})^{-\frac{1}{2}} + \lambda(\mathbf{t}'\mathbf{1} - 1).$$

Using the chain rule, the first order conditions are:

$$\begin{aligned}
\frac{\partial L(\mathbf{t}, \lambda)}{\partial \mathbf{t}} &= \mu(\mathbf{t}'\Sigma\mathbf{t})^{-\frac{1}{2}} - \left(\mathbf{t}'\mu - r_f\right)(\mathbf{t}'\Sigma\mathbf{t})^{-3/2}\Sigma\mathbf{t} + \lambda\mathbf{1} = \mathbf{0}, \\
\frac{\partial L(\mathbf{t}, \lambda)}{\partial \lambda} &= \mathbf{t}'\mathbf{1} - 1 = 0.
\end{aligned}$$

After much tedious algebra, it can be shown that the solution for $\mathbf{t}$ has a nice simple expression:

$$\mathbf{t} = \frac{\Sigma^{-1}(\mu - r_f \cdot \mathbf{1})}{\mathbf{1}'\Sigma^{-1}(\mu - r_f \cdot \mathbf{1})}. \tag{11.26}$$

The formula for the tangency portfolio Equation 11.26 looks similar to the formula for the global minimum variance portfolio Equation 11.8. Both formulas have Σ^{-1} in the numerator and $\mathbf{1}'\Sigma^{-1}$ in the denominator.

11.6 Remark

The location of the tangency portfolio, and the sign of the Sharpe ratio, depends on the relationship between the risk-free rate r_f and the expected return on the global minimum variance portfolio $\mu_{p,m}$. If $\mu_{p,m} > r_f$, which is the usual case, then the tangency portfolio will have a positive Sharpe ratio. If $\mu_{p,m} < r_f$, which could occur when stock prices are falling and the economy is in a recession, then the tangency portfolio will have a negative Sharpe slope. In this case, efficient portfolios involve shorting the tangency portfolio and investing the proceeds in T-Bills.[4]

Example 11.13 (Computing the tangency portfolio)**.** Suppose $r_f = 0.005$. To compute the tangency portfolio Equation 11.26 in R for the three risky assets in Table Table 11.1 use:

```
#>    MSFT    NORD    SBUX
#>   1.027 -0.326   0.299
```

The tangency portfolio has weights $t_{\text{msft}} = 1.027$, $t_{\text{nord}} = -0.326$ and $t_{\text{sbux}} = 0.299$, and is given by the vector $\mathbf{t} = (1.027, -0.326, 0.299)'$. Notice that Nordstrom, which has the lowest mean return, is sold short in the tangency portfolio. The expected return on the tangency portfolio, $\mu_{p,t} = \mathbf{t}'\mu$, is:

[4] For a mathematical proof of these results, see Ingersoll (1987).

```
mu.t = as.numeric(crossprod(t.vec, mu.vec))
mu.t
#> [1] 0.0519
```

The portfolio variance, $\sigma^2_{p,t} = \mathbf{t}'\Sigma\mathbf{t}$, and standard deviation, $\sigma_{p,t}$, are:

```
sig2.t = as.numeric(t(t.vec)%*%sigma.mat%*%t.vec)
sig.t = sqrt(sig2.t)
c(sig2.t, sig.t)
#> [1] 0.0125 0.1116
```

Because $r_f = 0.005 < \mu_{p,m} = 0.0249$ the tangency portfolio has a positive Sharpe's ratio/slope given by:

```
SR.t = (mu.t - r.f)/sig.t
SR.t
#> [1] 0.42
```

The tangency portfolio is illustrated in Figure 11.9. It is the portfolio on the efficient frontier of risky assets in which a straight line drawn from the risk-free rate to the tangency portfolio (green line) is just tangent to the efficient frontier (blue dots).

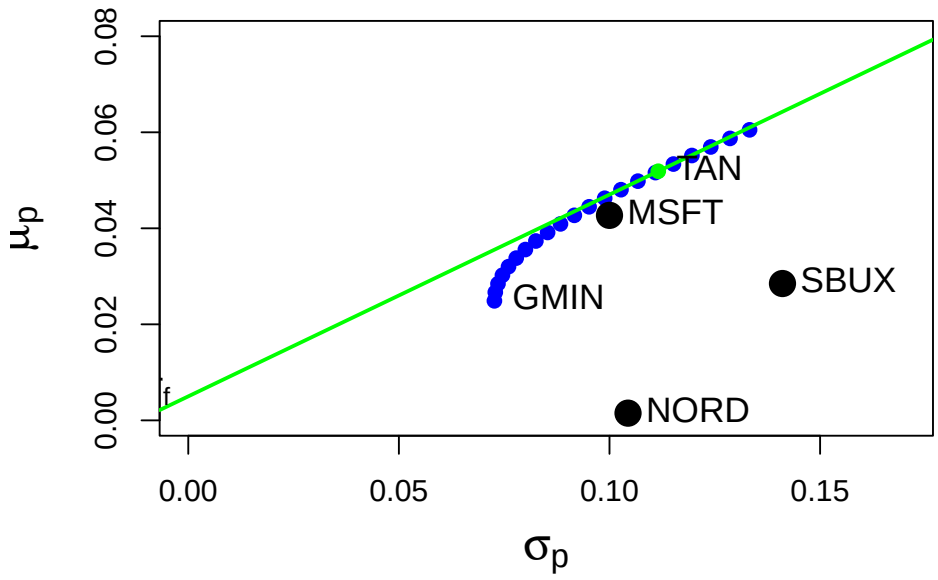


Figure 11.9: Tangency portfolio from example data.

■

11.6.1 Alternative derivation of the tangency portfolio

The derivation of tangency portfolio formula Equation 11.26 from the optimization problem Equation 11.25 is a very tedious problem. It can be derived in a different way as follows. Consider forming portfolios of N risky assets with return vector $\mathbf{R}$ and T-bills (risk-free asset) with constant return r_f. Let $\mathbf{x}$ denote the $N \times 1$ vector of risky asset weights and let x_f denote the safe asset weight and assume that $\mathbf{x}'\mathbf{1} + x_f = 1$ so that all wealth is allocated to these assets. The portfolio return is:

$$R_{p,x} = \mathbf{x}'\mathbf{R} + x_f r_f = \mathbf{x}'\mathbf{R} + (1 - \mathbf{x}'\mathbf{1})r_f = r_f + \mathbf{x}'(\mathbf{R} - r_f \cdot \mathbf{1}).$$

The portfolio excess return is:

$$R_{p,x} - r_f = \mathbf{x}'(\mathbf{R} - r_f \cdot \mathbf{1}). \tag{11.27}$$

The expected portfolio excess return (risk premium) and portfolio variance are:

$$\mu_{p,x} - r_f = \mathbf{x}'(\mu - r_f \cdot \mathbf{1}), \tag{11.28}$$

$$\sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x}. \tag{11.29}$$

For notational simplicity, define $\tilde{\mathbf{R}} = \mathbf{R} - r_f \cdot \mathbf{1}$, $\tilde{\mu} = \mu - r_f \cdot \mathbf{1}$, $\tilde{R}_{p,x} = R_{p,x} - r_f$, and $\tilde{\mu}_{p,x} = \mu_{p,x} - r_f$. Then Equation 11.27 and Equation 11.28 can be re-expressed as:

$$\tilde{R}_{p,x} = \mathbf{x}'\tilde{\mathbf{R}}, \tag{11.30}$$

$$\tilde{\mu}_{p,x} = \mathbf{x}'\tilde{\mu}. \tag{11.31}$$

To find the minimum variance portfolio of risky assets and a risk free asset that achieves the target excess return $\tilde{\mu}_{p,0} = \mu_{p,0} - r_f$ we solve the minimization problem:

$$\min_{\mathbf{x}} \ \sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x} \text{ s.t. } \tilde{\mu}_{p,x} = \tilde{\mu}_{p,0}.$$

Note that $\mathbf{x}'\mathbf{1} = 1$ is not a constraint because wealth need not all be allocated to the risky assets; some wealth may be held in the riskless asset. The Lagrangian is:

$$L(\mathbf{x}, \lambda) = \mathbf{x}'\Sigma\mathbf{x} + \lambda(\mathbf{x}'\tilde{\mu} - \tilde{\mu}_{p,0}).$$

The first order conditions for a minimum are:

$$\frac{\partial L(\mathbf{x}, \lambda)}{\partial \mathbf{x}} = 2\Sigma\mathbf{x} + \lambda\tilde{\mu} = \mathbf{0}, \tag{11.32}$$

$$\frac{\partial L(\mathbf{x}, \lambda)}{\partial \lambda} = \mathbf{x}'\tilde{\mu} - \tilde{\mu}_{p,0} = 0. \tag{11.33}$$

Using the Equation 11.32, we can solve for $\mathbf{x}$ in terms of λ:

$$\mathbf{x} = -\frac{1}{2}\lambda\Sigma^{-1}\tilde{\mu}. \tag{11.34}$$

Equation 11.33 implies that $\mathbf{x}'\tilde{\mu} = \tilde{\mu}'\mathbf{x} = \tilde{\mu}_{p,0}$. Then pre-multiplying Equation 11.34 by $\tilde{\mu}'$ gives:

$$\tilde{\mu}'\mathbf{x} = -\frac{1}{2}\lambda\tilde{\mu}'\Sigma^{-1}\tilde{\mu} = \tilde{\mu}_{p,0},$$

which we can use to solve for λ:

$$\lambda = -\frac{2\tilde{\mu}_{p,0}}{\tilde{\mu}'\Sigma^{-1}\tilde{\mu}}. \tag{11.35}$$

Plugging Equation 11.35 into Equation 11.34 then gives the solution for $\mathbf{x}$:

$$\mathbf{x} = -\frac{1}{2}\lambda\Sigma^{-1}\tilde{\mu} = -\frac{1}{2}\left(-\frac{2\tilde{\mu}_{p,0}}{\tilde{\mu}'\Sigma^{-1}\tilde{\mu}}\right)\Sigma^{-1}\tilde{\mu} = \tilde{\mu}_{p,0}\cdot\frac{\Sigma^{-1}\tilde{\mu}}{\tilde{\mu}'\Sigma^{-1}\tilde{\mu}}. \tag{11.36}$$

The solution for x_f is then $1 - \mathbf{x}'1$.

Now, the tangency portfolio $\mathbf{t}$ is 100% invested in risky assets so that $\mathbf{t}'\mathbf{1} = \mathbf{1}'\mathbf{t} = 1$. Using Equation 11.36, the tangency portfolio satisfies:

$$\mathbf{1}'\mathbf{t} = \tilde{\mu}_{p,t}\cdot\frac{\mathbf{1}'\Sigma^{-1}\tilde{\mu}}{\tilde{\mu}'\Sigma^{-1}\tilde{\mu}} = 1,$$

which implies that,

$$\tilde{\mu}_{p,t} = \frac{\tilde{\mu}'\Sigma^{-1}\tilde{\mu}}{\mathbf{1}'\Sigma^{-1}\tilde{\mu}}. \tag{11.37}$$

Plugging Equation 11.37 back into Equation 11.36 then gives an explicit solution for $\mathbf{t}$:

$$\begin{aligned}\mathbf{t} &= \left(\frac{\tilde{\mu}'\Sigma^{-1}\tilde{\mu}}{\mathbf{1}'\Sigma^{-1}\tilde{\mu}}\right)\frac{\Sigma^{-1}\tilde{\mu}}{\tilde{\mu}'\Sigma^{-1}\tilde{\mu}} = \frac{\Sigma^{-1}\tilde{\mu}}{\mathbf{1}'\Sigma^{-1}\tilde{\mu}} \\ &= \frac{\Sigma^{-1}(\mu - r_f\cdot\mathbf{1})}{\mathbf{1}'\Sigma^{-1}(\mu - r_f\cdot\mathbf{1})},\end{aligned}$$

which is the result Equation 11.26 we got from finding the portfolio of risky assets that has the maximum Sharpe ratio.

11.6.2 Mutual fund separation theorem again

When there is a risk-free asset (T-bill) available, the efficient frontier of T-bills and risky assets consists of portfolios of T-bills and the tangency portfolio. The expected return and standard

deviation values of any such efficient portfolio are given by:

$$\mu_p^e = r_f + x_t(\mu_{p,t} - r_f), \tag{11.38}$$

$$\sigma_p^e = x_t \sigma_{p,t}, \tag{11.39}$$

where x_t represents the fraction of wealth invested in the tangency portfolio ($1 - x_t$ represents the fraction of wealth invested in T-Bills), and $\mu_{p,t} = \mathbf{t}'\mu$ and $\sigma_{p,t} = (\mathbf{t}'\Sigma\mathbf{t})^{1/2}$ are the expected return and standard deviation on the tangency portfolio, respectively. Recall, this result is known as the *mutual fund separation theorem*. The tangency portfolio can be considered as a mutual fund of the risky assets, where the shares of the assets in the mutual fund are determined by the tangency portfolio weights, and the T-bill can be considered as a mutual fund of risk-free assets. The expected return-risk trade-off of these portfolios is given by the line connecting the risk-free rate to the tangency point on the efficient frontier of risky asset only portfolios. Which combination of the tangency portfolio and the T-bill an investor will choose depends on the investor's risk preferences. If the investor is very risk averse and prefers portfolios with very low volatility, then she will choose a combination with very little weight in the tangency portfolio and a lot of weight in the T-bill. This will produce a portfolio with an expected return close to the risk-free rate and a variance that is close to zero. If the investor can tolerate a large amount of volatility, then she will prefer a portfolio with a high expected return regardless of volatility. This portfolio may involve borrowing at the risk-free rate (leveraging) and investing the proceeds in the tangency portfolio to achieve a high expected return.

Example 11.14 (Efficient portfolios of three risky assets and T-bills chosen by risk averse and risk tolerant investors)**.** Consider the tangency portfolio computed from the example data in Table Table 11.1 with $r_f = 0.005$. This portfolio is:

```
ans = c(t.vec, mu.t, sig.t)
names(ans)[4:5] = c("mu.t", "sigma.t")
ans
#>    MSFT    NORD    SBUX    mu.t sigma.t
#>  1.0268 -0.3263  0.2994  0.0519  0.1116
```

The efficient portfolios of T-Bills and the tangency portfolio is illustrated in Figure 11.10.

We want to compute an efficient portfolio that would be preferred by a highly risk averse investor, and a portfolio that would be preferred by a highly risk tolerant investor. A highly risk averse investor might have a low volatility (risk) target for his efficient portfolio. For example, suppose the volatility target is $\sigma_p^e = 0.02$ or 2%. Using Equation 11.39 and solving for x_t, the weights in the tangency portfolio and the T-Bill are:

```
x.t.02 = 0.02/sig.t
c(x.t.02, 1-x.t.02)
#> [1] 0.179 0.821
```

In this efficient portfolio, the weights in the risky assets are proportional to the weights in the tangency portfolio:

```
x.t.02*t.vec
#>    MSFT    NORD    SBUX
#>  0.1840 -0.0585  0.0537
```

The expected return and volatility values of this portfolio are:

```
mu.t.02 = x.t.02*mu.t + (1-x.t.02)*r.f
sig.t.02 = x.t.02*sig.t
c(mu.t.02, sig.t.02)
#> [1] 0.0134 0.0200
```

These values are illustrated in Figure 11.10 as the portfolio labeled "E1" .

A highly risk tolerant investor might have a high expected return target for his efficient portfolio. For example, suppose the expected return target is $\mu_p^e = 0.07$ or 7%. Using Equation 11.38 and solving for the x_t, the weights in the tangency portfolio and the T-Bill are:

```
x.t.07 = (0.07 - r.f)/(mu.t - r.f)
c(x.t.07, 1-x.t.07)
#> [1]  1.386 -0.386
```

Notice that this portfolio involves borrowing at the T-Bill rate (leveraging) and investing the proceeds in the tangency portfolio. In this efficient portfolio, the weights in the risky assets are:

```
x.t.07*t.vec
#>   MSFT   NORD   SBUX
#>  1.423 -0.452  0.415
```

The expected return and volatility values of this portfolio are:

```
mu.t.07 = x.t.07*mu.t + (1-x.t.07)*r.f
sig.t.07 = x.t.07*sig.t
c(mu.t.07, sig.t.07)
#> [1] 0.070 0.155
```

In order to achieve the target expected return of 7%, the investor must tolerate a 15.47% volatility. These values are illustrated in Figure 11.10 as the portfolio labeled "E2" . ■

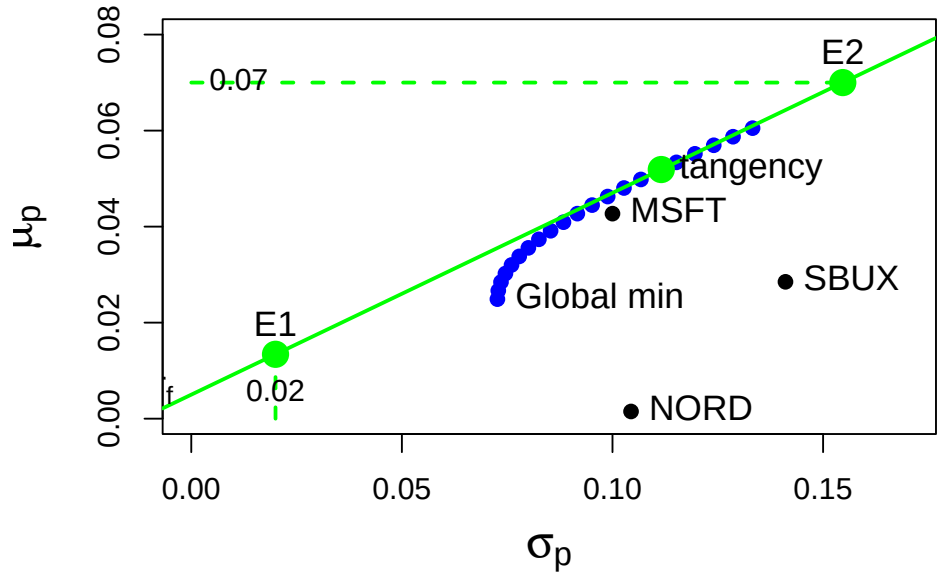


Figure 11.10: Tangency portfolio

11.7 Computational Problems with Very Large Portfolios

In principle, mean-variance portfolio analysis can be applied in situations in which there is a very large number of risky assets (e.g., $N = 5,000$). However, there are a number of practical problems that can arise. First, the computation of efficient portfolios requires inverting the $N \times N$ asset return covariance matrix Σ. When N is very large, inverting Σ can be computationally burdensome. Second, the practical application of the theory requires the estimation of Σ. Recall, there are N variance terms and $N(N-1)/2$ unique covariance terms in Σ. When $N = 5,000$, there are $12,502,500$ unique elements of Σ to estimate. And since each estimated element of Σ has estimation error, there is a tremendous amount of estimation error in the estimate of Σ. There is an additional problem with the estimation of Σ using the sample covariance matrix of asset returns when N is very large. If the number of assets, N, is greater than the number of sample observations,

T, then the $N \times N$ sample covariance matrix:

$$\begin{aligned} \hat{\Sigma} &= \frac{1}{T-1}\sum_{t=1}^{T}(\mathbf{R}_t - \hat{\mu})(\mathbf{R}_t - \hat{\mu})', \\ \hat{\mu} &= \frac{1}{T}\sum_{t=1}^{T}\mathbf{R}_t, \end{aligned}$$

is only positive semi-definite and less than full rank N. This means that $\hat{\Sigma}$ is not invertible and so mean-variance efficient portfolios cannot be uniquely computed. This problem can happen often. For example, suppose $N = 5,000$. For the sample covariance matrix to be full rank, you need at least $T = 5,000$ sample observations. For daily data, this means you would need $5,000/250 = 20$ years of daily data.[5] For weekly data, you would need $5000/52 = 96.2$ years of weekly data. For monthly data, you would need $5,000/12 = 417$ years of monthly data.

Example 11.15 (Nonsingular sample return covariance matrix)**.** To illustrate the rank failure of $\hat{\Sigma}$ that occurs when the number of assets N is greater than the number of data observations T, consider computing $\hat{\Sigma}$ for the six Vanguard mutual funds in the **introcompfinr** data object `VanguardPrices` using only five monthly observations:

```
colnames(VanguardPrices)
#> [1] "vfinx" "veurx" "veiex" "vbltx" "vbisx" "vpacx"
range(index(VanguardPrices))
#> [1] "Jan 1995" "Dec 2014"
VanguardRetS = na.omit(Return.calculate(VanguardPrices,
                                        method="simple"))
covhat = cov(VanguardRetS[1:5, ])
```

A quick way to determine if $\hat{\Sigma}$ is full rank (and invertible) is to compute the Cholesky decomposition $\hat{\Sigma} = \hat{\mathbf{C}}\hat{\mathbf{C}}'$, where $\hat{\mathbf{C}}$ is a lower triangular matrix with non-negative diagonal elements. If all of the diagonal elements of $\hat{\mathbf{C}}$ are positive then $\hat{\Sigma}$ is positive definite, full rank, and invertible. In R, we compute $\hat{\mathbf{C}}$ using the function `chol()`:

```
# chol(covhat) # uncomment this one to see the result
```

Here, `chol()` returns an error that indicates $\hat{\Sigma}$ is not positive definite and less than full rank. If we try to invert $\hat{\Sigma}$ using `solve()` we will also get an error indicating $\hat{\Sigma}$ is not invertible:[6]

```
# solve(covhat) # uncomment this one to see the result
```

■

[5] Recall, there are approximately 250 trading days per year.
[6] The **Matrix** function `rankMatrix()` can be used to compute the exact rank of a matrix.

Due to these practical problems of using the sample covariance matrix $\hat{\Sigma}$ to compute mean-variance efficient portfolios when N is large, there is a need for alternative methods for estimating Σ when N is large. One such method based on the *Single Index Model* for returns is presented in Chapter Chapter 15.

11.8 Portfolio Analysis Functions in R

The package **IntroCompFinR** contains a few R functions for computing Markowitz mean-variance efficient portfolios allowing for short sales using matrix algebra computations. These functions allow for the easy computation of the global minimum variance portfolio, an efficient portfolio with a given target expected return, the tangency portfolio, and the efficient frontier. These functions are summarized in Table Table 11.2.

Table 11.2: **introcompfinr** functions for computing mean-variance efficient portfolios

Function	Description
`get_portfolio`	create portfolio object
`globalmin_portfolio`	compute global minimum variance portfolio
`efficient_portfolio`	compute minimum variance portfolio subject to target return
`tangency_portfolio`	compute tangency portfolio
`efficient_frontier`	compute efficient frontier of risky assets

The following examples illustrate the use of the functions in Table Table 11.2 using the example data in Table Table 11.1:

```
asset.names <- c("MSFT", "NORD", "SBUX")
mu.vec = c(0.0427, 0.0015, 0.0285)
sigma.mat = matrix(c(0.0100, 0.0018, 0.0011,
           0.0018, 0.0109, 0.0026,
           0.0011, 0.0026, 0.0199),
         nrow=3, ncol=3)
r.f=0.005
mu.vec
#> [1] 0.0427 0.0015 0.0285
sigma.mat
#>        [,1]   [,2]   [,3]
#> [1,] 0.0100 0.0018 0.0011
#> [2,] 0.0018 0.0109 0.0026
#> [3,] 0.0011 0.0026 0.0199
```

```
r.f
#> [1] 0.005
```

To specify a `portfolio` object, you need an expected return vector and covariance matrix for the assets under consideration as well as a vector of portfolio weights. To create an equally weighted portfolio use:

```
ew = rep(1,3)/3
equalWeight.portfolio = get_portfolio(er_vec=mu.vec,cov_mat=sigma.mat,weight_vec=ew)
class(equalWeight.portfolio)
#> [1] "portfolio"
```

Portfolio objects have the following components:

```
names(equalWeight.portfolio)
#> [1] "call"       "er_port"    "sd_port"    "weight_vec" "er_vec"
#> [6] "cov_mat"
```

There are `print()`, `summary()` and `plot()` methods for portfolio objects. The `print()` method gives:

```
equalWeight.portfolio
#> Call:
#> get_portfolio(er_vec = mu.vec, cov_mat = sigma.mat, weight_vec = ew)
#>
#> Portfolio expected return:     0.0242
#> Portfolio standard deviation:  0.0759
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3
#>   0.333   0.333   0.333
```

The `plot()` method shows a bar chart of the portfolio weights:

```
plot(equalWeight.portfolio, col="cornflowerblue")
```

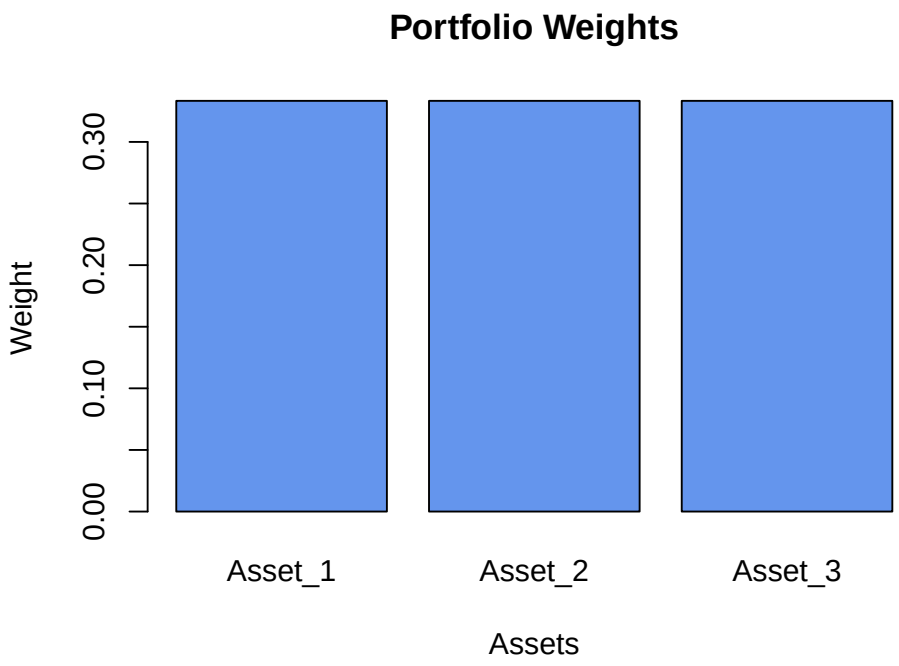


Figure 11.11: Plot method for objects of class portfolio

The resulting plot is shown in Figure 11.11. The global minimum variance portfolio (allowing for short sales) $\mathbf{m}$ solves the optimization problem Equation 11.2 and is computed using Equation 11.6. To compute this portfolio use the function `globalMin.portfolio()`:

```
gmin.port = globalmin_portfolio(mu.vec, sigma.mat)
class(gmin.port)
#> [1] "portfolio"
gmin.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.0249
#> Portfolio standard deviation:  0.0727
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3
#>   0.441   0.366   0.193
```

A mean-variance efficient portfolio $\mathbf{x}$ that achieves the target expected return μ_0 solves the optimization problem Equation 11.10 and is computed using Equation 11.14. To compute this portfolio for the target expected return $\mu_0 = E[R_{\text{msft}}] = 0.0427$ use the `efficient.portfolio()` function:

```
target.return = mu.vec[1]
e.port.msft = efficient_portfolio(mu.vec, sigma.mat, target.return)
class(e.port.msft)
#> [1] "portfolio"
e.port.msft
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w)
#>
#> Portfolio expected return:     0.0427
#> Portfolio standard deviation:  0.0917
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3
#>  0.8275 -0.0907  0.2633
```

The tangency portfolio $\mathbf{t}$ is the portfolio of risky assets with the highest Sharpe's slope and has solutions given by Equation 11.26. To compute this portfolio with $r_f = 0.005$ use the `tangency.portfolio()` function:

```
tan.port = tangency_portfolio(mu.vec, sigma.mat, r.f)
class(tan.port)
#> [1] "portfolio"
tan.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.0519
#> Portfolio standard deviation:  0.112
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3
#>   1.027  -0.326   0.299
```

The the set of efficient portfolios of risky assets can be computed as a convex combination of any two efficient portfolios. It is convenient to use the global minimum variance portfolio as one portfolio and an efficient portfolio with target expected return equal to the maximum expected return of the assets under consideration as the other portfolio. Call these portfolios $\mathbf{m}$ and $\mathbf{x}$, respectively. For any number α, another efficient portfolio can be computed as:

$$\mathbf{z} = \alpha\mathbf{x} + (1 - \alpha)\mathbf{m}$$

The function `efficient.frontier()` constructs the set of efficient portfolios using this method

for a collection of α values on an equally spaced grid between α_{min} and α_{max}. For example, to compute 20 efficient portfolios for values of α between -2 and 1.5 use:

```
ef = efficient_frontier(mu.vec, sigma.mat, alpha_min=-0.5,
                        alpha_max=2, nport=20)
attributes(ef)
#> $names
#> [1] "call"       "er_vec"     "sd_vec"     "weight_mat"
#>
#> $class
#> [1] "Markowitz"
ef
#> Call:
#> efficient_frontier(er_vec = mu.vec, cov_mat = sigma.mat, nport = 20,
#>     alpha_min = -0.5, alpha_max = 2)
#>
#> Frontier portfolios' expected returns and standard deviations
#>    port 1 port 2 port 3 port 4 port 5 port 6 port 7 port 8 port 9 port 10
#> ER 0.0160 0.0183 0.0207 0.0230 0.0254 0.0277 0.0300 0.0324 0.0347  0.0371
#> SD 0.0779 0.0755 0.0739 0.0729 0.0727 0.0732 0.0745 0.0764 0.0790  0.0821
#>    port 11 port 12 port 13 port 14 port 15 port 16 port 17 port 18 port 19
#> ER  0.0394  0.0418  0.0441  0.0464  0.0488  0.0511  0.0535  0.0558  0.0582
#> SD  0.0858  0.0899  0.0944  0.0993  0.1044  0.1098  0.1154  0.1212  0.1272
#>    port 20
#> ER  0.0605
#> SD  0.1333
```

Use the `summary()` method to show the weights of these portfolios. Use the `plot()` method to create a simple plot the efficient frontier:

```
plot(ef)
```

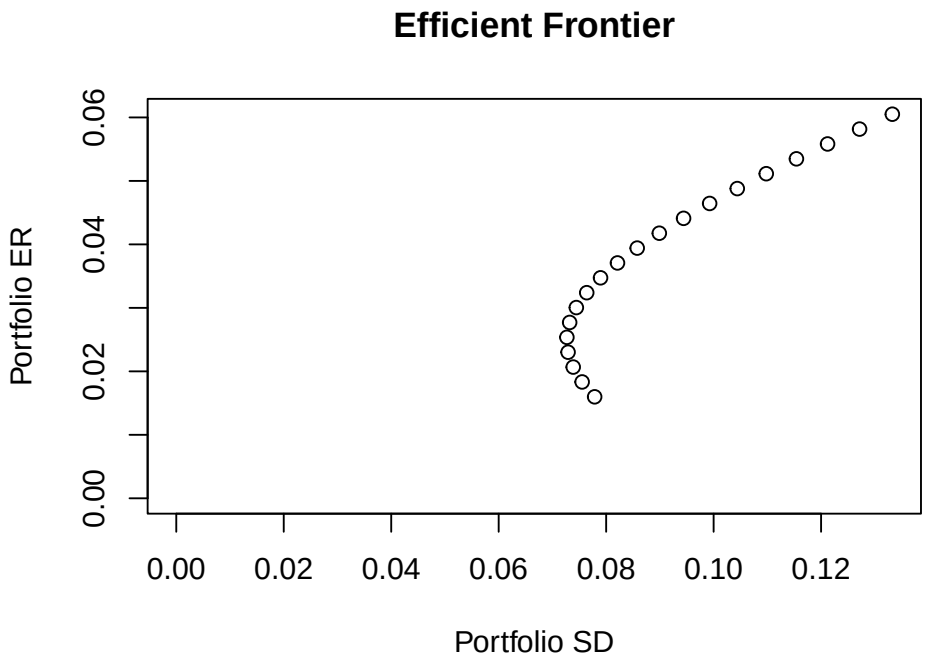


Figure 11.12: Plot method for objects of class Markowitz

The resulting plot is shown in Figure 11.12. To create a more elaborate plot of the efficient frontier showing the original assets and the tangency portfolio use:

```
plot(ef, plot.assets=T, col="blue", pch=16)
points(gmin.port$sd_port, gmin.port$er_port, col="green", pch=16, cex=2)
text(gmin.port$sd_port, gmin.port$er_port, labels = "Global min", pos = 4)
points(tan.port$sd_port, tan.port$er_port, col="red", pch=16, cex=2)
text(tan.port$sd_port, tan.port$er_port, labels = "Tangency", pos = 3)
sr.tan = (tan.port$er_port - r.f)/tan.port$sd_port
abline(a=r.f, b=sr.tan, col="green", lwd=2)
```

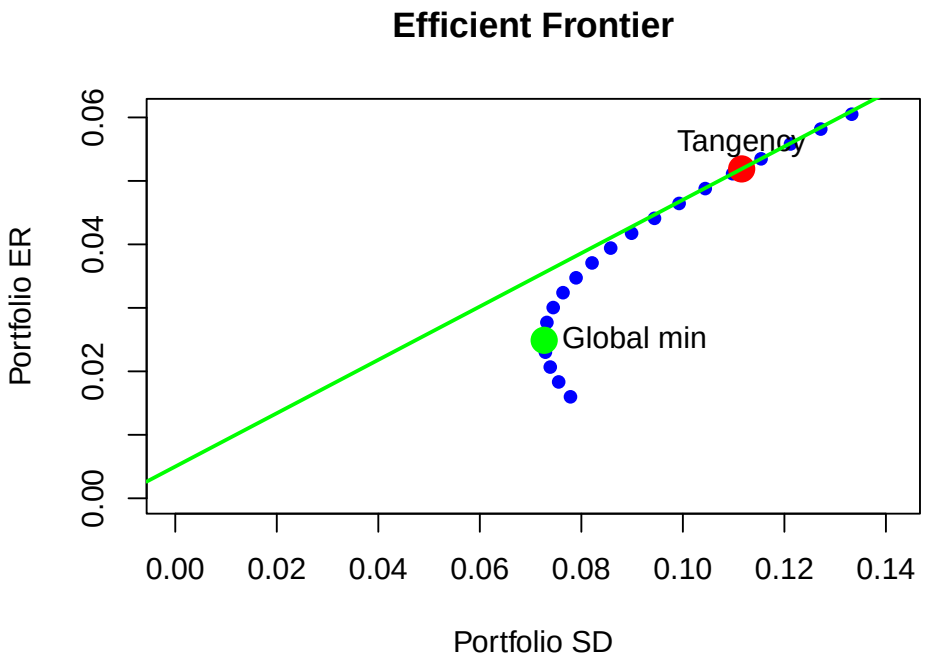


Figure 11.13: Efficient frontier plot.

The resulting plot is shown in Figure 11.13.

11.9 Application to Vanguard Mutual Fund

In this section, we use mean-variance portfolio theory to study asset allocation among a collection of Vanguard mutual Funds. To motivate this example, consider an investor who is deciding how to allocate savings in an employee-sponsored retirement plan.[7] Typical plans only allow participants to invest in a limited set of mutual funds. An important limitation of investing in mutual funds is that they cannot be sold short.

The **introcompfinr** data object `VanguardPrices` contains monthly closing prices on six diversified and passively managed Vanguard mutual funds over the period January, 1995 through December, 2014:

```
VanguardPrices = as.xts(VanguardPrices)
colnames(VanguardPrices)
#> [1] "vfinx" "veurx" "veiex" "vbltx" "vbisx" "vpacx"
```

[7] Common employee-sponsored retirement portfolios in the US are 401K plans (for employees of private companies) and 403B plans (for employees of government agencies).

```
range(index(VanguardPrices))
#> [1] "Jan 1995" "Dec 2014"
```

Table Table 11.3 gives a brief description of the six funds.[8]

Table 11.3: Vanguard mutual funds.

Fund Ticker	Fund Name	Description
vfinx	Vanguard 500 Index	Tracks the S&P 500 index
veurx	Vanguard European Stock Index	Tracks the MSCI Europe index
veiex	Vanguard Emerging Markets Index	Tracks the MSCI Emerging Markets index
vbltx	Vanguard Long-Term Bond Index	Tracks a US long-term bond index
vbisx	Vanguard Short-Term Bond Index	Tracks a US short-term bond index
vpacx	Vanguard Pacific Stock Index	Tracks the MSCI Pacific index

We construct mean-variance efficient portfolios using annualized GWN model estimates of the expected return vector and covariance matrix based on simple returns over the sub-period January, 2010 through December, 2014. These estimates are computed using:

```
VanguardRetS = na.omit(Return.calculate(VanguardPrices,
                                        method="simple"))
smpl = "2010-1::2014-12"
muhat = 12*colMeans(VanguardRetS[smpl])
covhat = 12*cov(VanguardRetS[smpl])
```

The estimated annualized expected returns and standard deviations are:

```
muhat
#>  vfinx  veurx  veiex  vbltx  vbisx  vpacx
#> 0.1513 0.0723 0.0356 0.0965 0.0196 0.0613
sqrt(diag(covhat))
#>  vfinx  veurx  veiex  vbltx  vbisx  vpacx
#> 0.1300 0.1947 0.1914 0.0864 0.0137 0.1467
```

The stock funds have higher average returns and volatilities than the bond funds. The annual risk-free rate at the end of this period is approximately 0.1%. Using this rate, the estimated Sharpe ratios on the six funds are:

[8] More detailed information on these funds is available at (finance.yahoo.com).

```
r.f = 0.001
(muhat - r.f)/sqrt(diag(covhat))
#> vfinx veurx veiex vbltx vbisx vpacx
#> 1.156 0.366 0.181 1.105 1.359 0.411
```

Interestingly, the short term US bond fund vbisx has the highest estimated annualized Sharpe ratio at 1.36 followed by vfinx (S&P 500 fund) and vbltx (long-term US bond fund). The Sharpe ratios for the other stock funds are much smaller than one.

The estimated return correlation matrix of the six funds is illustrated in Figure 11.14, created with the **corrplot** function `corrplot.mixed()`:

```
corrplot.mixed(cov2cor(covhat), upper = "ellipse")
```

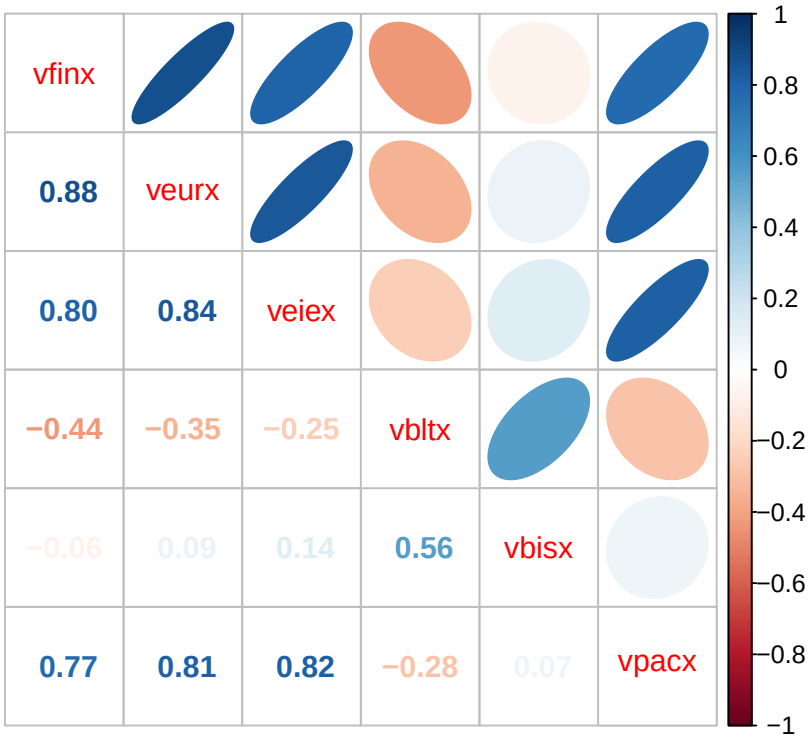


Figure 11.14: Estimated return correlations between six Vanguard mutual funds

The stock funds are all strongly positively correlated with each other, whereas the bond funds are either negatively correlated or very weakly positively correlated with the stock funds. Hence, combining the stock and bond funds in a portfolio should be beneficial in terms of risk diversification.

We use the **introcompfinr** portfolio functions to compute estimates of the global minimum variance portfolio, the tangency portfolio, and the efficient frontier of risky assets:

```
gmin.port = globalmin_portfolio(muhat, covhat)
tan.port = tangency_portfolio(muhat, covhat, r.f)
ef = efficient_frontier(muhat, covhat, alpha_min=-0.5,
                        alpha_max=1.5, nport=20)
```

The estimated global minimum variance portfolio is:

```
gmin.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.0197
#> Portfolio standard deviation:  0.0114
#> Portfolio weights:
#>    vfinx    veurx    veiex    vbltx    vbisx    vpacx
#>  0.0581 -0.0347 -0.0214 -0.0736  1.0647  0.0069
```

This portfolio is mostly concentrated in the short-term US bond fund vbisx (which has the smallest volatility) and has small short positions in veurx, veiex and vbltx. This portfolio is not feasible, however, because investors are not allowed to short sell mutual funds.

The estimated tangency portfolio is:

```
tan.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.0642
#> Portfolio standard deviation:  0.0209
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3 Asset_4 Asset_5 Asset_6
#>  0.3152 -0.0907 -0.0874  0.1159  0.7387  0.0083
```

This portfolio is also an infeasible long-short portfolio with long positions in the US funds vfinx, vbltx and vbisx and short positions in the non-US stock funds veurx, veiex and vpacx.

The mean-variance efficient frontiers are illustrated in Figure 11.15, created using:

```
plot(ef, plot.assets=TRUE, col="blue", pch=16)
text(gmin.port$sd_port, gmin.port$er_port,labels = "gmin", pos = 2)
text(tan.port$sd_port, tan.port$er_port,labels = "tan", pos = 3)
```

```
sr.tan = (tan.port$er_port - r.f)/tan.port$sd_port
text(0, r.f, labels=expression(r[f]), pos = 4)
abline(a=r.f, b=sr.tan, col="green", lwd=2)
abline(h=0, v=0)
```

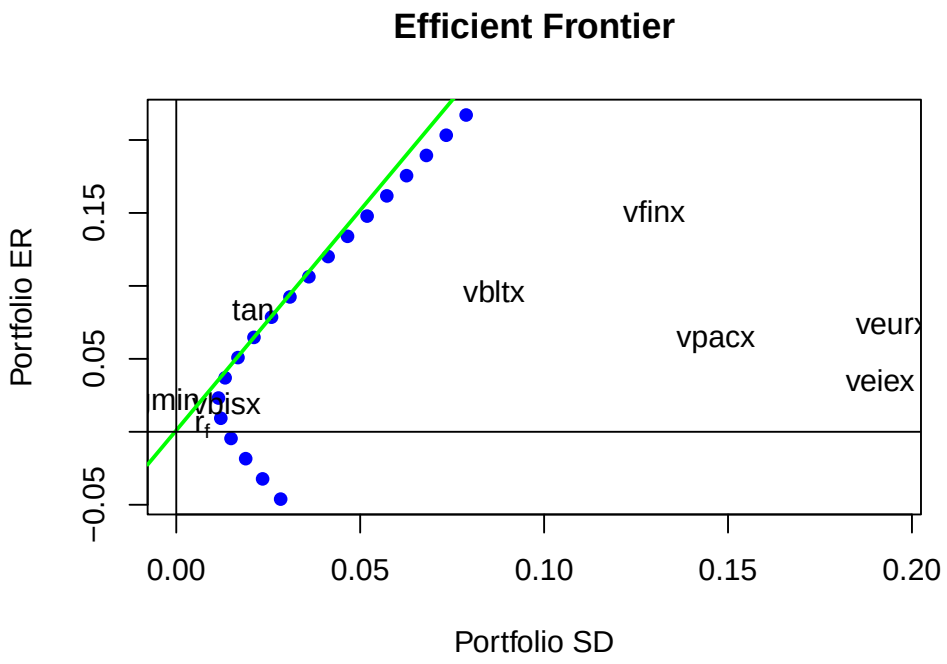


Figure 11.15: Mean variance efficient portfolios of the Vanguard mutual funds. The risky asset portfolio frontier is shown in blue. The efficient portfolios of the risk-free asset and the risky assets is shown in green. The global minimum variance portfolio is labeled gmin, and the tangency portfolio is labeled tan.

The portfolio frontier of risky assets is shown in blue. These portfolios are infeasible because they all contain short sales in one or more of the mutual funds:

```
ef$weight_mat
#>            vfinx    veurx    veiex     vbltx   vbisx   vpacx
#> port 1  -0.32213  0.04799  0.07623 -0.353776  1.5469 0.00480
#> port 2  -0.24208  0.03057  0.05568 -0.294783  1.4454 0.00524
#> port 3  -0.16203  0.01316  0.03512 -0.235789  1.3439 0.00568
#> port 4  -0.08198 -0.00426  0.01456 -0.176795  1.2423 0.00612
#> port 5  -0.00193 -0.02167 -0.00599 -0.117802  1.1408 0.00656
#> port 6   0.07813 -0.03909 -0.02655 -0.058808  1.0393 0.00700
```

```
#> port 7   0.15818 -0.05650 -0.04711  0.000186  0.9378 0.00744
#> port 8   0.23823 -0.07392 -0.06766  0.059179  0.8363 0.00788
#> port 9   0.31828 -0.09133 -0.08822  0.118173  0.7348 0.00832
#> port 10  0.39833 -0.10875 -0.10878  0.177167  0.6333 0.00876
#> port 11  0.47839 -0.12616 -0.12933  0.236160  0.5318 0.00920
#> port 12  0.55844 -0.14358 -0.14989  0.295154  0.4302 0.00964
#> port 13  0.63849 -0.16099 -0.17045  0.354148  0.3287 0.01008
#> port 14  0.71854 -0.17840 -0.19101  0.413141  0.2272 0.01052
#> port 15  0.79859 -0.19582 -0.21156  0.472135  0.1257 0.01096
#> port 16  0.87865 -0.21323 -0.23212  0.531128  0.0242 0.01140
#> port 17  0.95870 -0.23065 -0.25268  0.590122 -0.0773 0.01183
#> port 18  1.03875 -0.24806 -0.27323  0.649116 -0.1788 0.01227
#> port 19  1.11880 -0.26548 -0.29379  0.708109 -0.2804 0.01271
#> port 20  1.19885 -0.28289 -0.31435  0.767103 -0.3819 0.01315
```

The locations of vbisx and the global minimum variance portfolio are almost identical, which suggests that vbisx is a good feasible substitute for the infeasible global minimum variance portfolio. The remaining funds are well inside the risky asset efficient frontier. The set of efficient portfolios of risky assets plus the risk-free asset is shown in green. These are portfolios of the risk-free asset and the tangency portfolio. Since the tangency portfolio contains short sales, these portfolios are not feasible. However, the tangency portfolio is approximately 70% vbisx and 30% vfinx so a close substitute for the efficient portfolios are combinations of the risk-free asset and the approximate tangency portfolio that is 70% vbisx and 30% vfinx.

This example highlights some practical problems associated with applying mean-variance portfolio theory in a real-world asset allocation context. In particular, mean-variance efficient portfolios often contain short sales in some assets. When the underlying assets are mutual funds, investors are not allowed to directly short these assets and it may not be possible to exactly implement the theory. The next chapter shows how mean-variance portfolio theory can be modified to allow for additional constraints created by short sales restrictions.

11.10 Further Reading: Portfolio Theory with Matrix Algebra

Portfolio theory using matrix algebra is discussed in advanced textbooks on asset pricing and portfolio theory. Good treatments are given in (Ingersoll 1987), (Campbell, Lo, and MacKinlay 1997), and (Cochrane 2008). A nice review of portfolio theory with matrix algebra is given in (Constantinides and Malliaris 1995). In R, mean-variance optimized portfolios can be computed using the packages **fPortfolio**, **PARMA**, **PortfolioAnalytics**, and **stockPortfolio**.

11.11 Exercises

Exercise 11.1 (Solve for global minimum variance portfolio)**.** The global minimum variance portfolio $\mathbf{m} = (m_1, m_2, m_3)'$ for the 3 asset case solves the constrained minimization problem:

$$\begin{aligned}\min_{m_1,m_2,m_3} \quad \sigma^2_{p,m} &= m_1^2\sigma_1^2 + m_2^2\sigma_2^2 + m_2^2\sigma_2^2 \\ &\quad + 2m_1m_2\sigma_{12} + 2m_1m_3\sigma_{13} + 2m_2m_3\sigma_{23} \\ &\text{s.t. } m_1 + m_2 + m_3 = 1.\end{aligned}$$

The Lagrangian for this problem is:

$$\begin{aligned}L(m_1, m_2, m_3, \lambda) &= m_A^2\sigma_A^2 + m_B^2\sigma_B^2 + m_C^2\sigma_C^2 \\ &\quad + 2m_Am_B\sigma_{AB} + 2m_Am_C\sigma_{AC} + 2m_Bm_C\sigma_{BC} \\ &\quad + \lambda(m_A + m_B + m_C - 1)\end{aligned}$$

(a) Write out the first order conditions

Exercise 11.2 (Solve for global minimum variance portfolio)**.**

- Use 2 x 2 matrix inversion rule to explicitly solve for

Exercise 11.3 (Solve for tangency portfolio)**.**

- Give explicit formula for tangency portfolio in 2 x 2 case

12 Portfolio Theory with Short Sales Constraints

Updated: September 12, 2026

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The portfolio theory I described in Chapters Chapter 10 and Chapter 11 assumed that investors can freely short sell all assets. In many situation, however, investors may not be allowed to short sell some or all assets and/or there may be restrictions on when short sales may occur. In particular, certain types of assets (e.g., mutual funds) cannot be sold short. This has implications for asset allocation in individual retirement accounts. Some institutions, such as pension funds, are prevented from directly short selling assets. Government regulations may prevent certain assets from being sold short, or prevent short sales from occurring at certain times. For example, during the 2008 financial crisis the Securities and Exchange Commission temporarily blocked the short sale of financial stocks. Because of these practical limitations on short sales, it is important to be able to incorporate short sales restrictions into mean-variance portfolio theory in practice and in this chapter I describe how to do it.

This chapter is organized as follows. In Section 12.2 I describe the impact of short sales constraints, on the investment in risky assets, on efficient portfolios in the simplified two risky asset setting presented in Chapter 10. In Section 12.3 I discuss the computation of mean-variance efficient portfolios when there are short sales constraints in the general case of N risky assets, as described in Chapter 11, using quadratic programming. In Section 12.5 I give a real-world application of the theory to asset allocation among Vanguard mutual funds.

The examples I cover in this chapter use the **introcompfinr, PerformanceAnalytics**, and **quadprog** packages. Make sure these packages are installed and loaded in R before replicating the chapter examples.

12.1 Overview of Short Selling

Most investments involve purchases of assets that you believe will increase in value over time. When you buy an asset you establish a *long* position in the asset. You profit from the long position by selling the asset in the future at a price higher than the original purchase price. However, sometimes you may believe that an asset will decrease in value over time. To profit on an investment in which the asset price is lower in the future requires *short selling* the asset. In general, short selling involves

selling an asset you borrow from someone else today and repurchasing the asset in the future and returning it to its owner. You profit on the short sale if you repurchase the asset for a lower price than you initially sold it. Because short selling involves borrowing it is a type of leveraged investment and so risk increases with the size of the short position.

In practice, a short sale of an asset works as follows. Suppose you have an investment account at some brokerage firm and you are interested in short selling one share of an individual stock, such as Microsoft, that you do not currently own. Because the brokerage firm has so many clients, one of the clients will own Microsoft. The brokerage firm will allow you to borrow Microsoft, sell it and add the proceeds of the short sale to your investment account. Doing so establishes a *short position* in Microsoft which is a liability to return the borrowed share sometime in the future. You close the short position in the future by repurchasing one share of Microsoft through the brokerage firm who returns the share to the original owner. However, borrowing is not free. You will have to pay interest on the amount borrowed. You will also need sufficient capital in your account to cover any potential losses from the short sale. In addition, if the price of the short sold asset rises substantially during the short sale period then the brokerage firm may require you to add additional margin capital to your account. If you cannot do this then the brokerage firm will force you to close the short position immediately.

12.2 Portfolio Theory with Short Sales Constraints in a Simplified Setting

In this section, we illustrate the impact of short sales constraints on risky assets on portfolios in the simplified setting described in Chapter Chapter 10. We first look at simple portfolios of two risky assets, then consider portfolios of a risky asset plus a risk-free asset, and finish with portfolios of two risky assets plus a risk-free asset.

12.2.1 Two Risky Assets

Consider the two risky asset example data from Chapter Chapter 10:

```
mu.A = 0.175
sig.A = 0.258
sig2.A = sig.A^2
mu.B = 0.055
sig.B = 0.115
sig2.B = sig.B^2
rho.AB = -0.164
sig.AB = rho.AB*sig.A*sig.B
```

```
assetNames = c("Asset A", "Asset B")
mu.vals = c(mu.A, mu.B)
names(mu.vals) = assetNames
Sigma = matrix(c(sig2.A, sig.AB, sig.AB, sig2.B), 2, 2)
colnames(Sigma) = rownames(Sigma) = assetNames
```

Asset A is the high average return and high risk asset, and asset B is the low average return and low risk asset. Consider the following set of 19 portfolios:

```
x.A = seq(from=-0.4, to=1.4, by=0.1)
x.B = 1 - x.A
n.port = length(x.A)
names(x.A) = names(x.B) = paste("p", 1:n.port, sep=".")
mu.p = x.A*mu.A + x.B*mu.B
sig2.p = x.A^2 * sig2.A + x.B^2 * sig2.B + 2*x.A*x.B*sig.AB
sig.p = sqrt(sig2.p)
```

Figure Figure 12.1 shows the risky asset only portfolio frontier created using:

```
cex.val = 1.5
plot(sig.p, mu.p, type="b", pch=16, cex = cex.val,
ylim=c(0, max(mu.p)), xlim=c(0, max(sig.p)),
xlab=expression(sigma[p]), ylab=expression(mu[p]), cex.lab=cex.val,
col=c(rep("lightblue", 4), rep("black", 11), rep("lightblue", 4)))
text(x=sig.A, y=mu.A, labels="Asset A", pos=4, cex = cex.val)
text(x=sig.B, y=mu.B, labels="Asset B", pos=4, cex = cex.val)
```

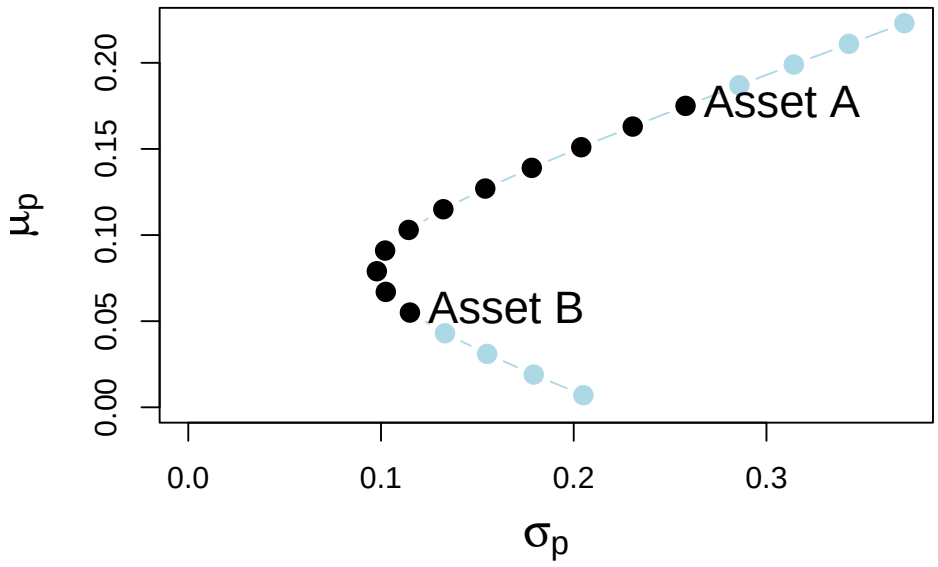


Figure 12.1: Two-risky asset portfolio frontier. Long-only portfolios are in black (between Asset B and Asset A), and long-short portfolios are in blue (below Asset B and above Asset A).

The long only (no short sale) portfolios, shown in black dots between the points labeled "Asset B" and "Asset A", are portfolios 5 to 15. The asset weights on these portfolios are:

```
longOnly = rbind(x.A[5:15], x.B[5:15])
rownames(longOnly) = assetNames
longOnly
#>         p.5 p.6 p.7 p.8 p.9 p.10 p.11 p.12 p.13 p.14 p.15
#> Asset A   0 0.1 0.2 0.3 0.4  0.5  0.6  0.7  0.8  0.9    1
#> Asset B   1 0.9 0.8 0.7 0.6  0.5  0.4  0.3  0.2  0.1    0
```

The long-short portfolios, shown in blue dots below the point labeled "Asset B" and above the point labeled "Asset A", are portfolios 1 to 4 and portfolios 16 to 19:

```
longShort = rbind(x.A[c(1:4, 16:19)], x.B[c(1:4, 16:19)])
rownames(longShort) = assetNames
longShort
#>          p.1  p.2  p.3  p.4 p.16 p.17 p.18 p.19
#> Asset A -0.4 -0.3 -0.2 -0.1  1.1  1.2  1.3  1.4
```

```
#> Asset B  1.4  1.3  1.2  1.1 -0.1 -0.2 -0.3 -0.4
```

Not allowing short sales for asset A eliminates the inefficient portfolios 1 to 4 (blue dots below the label "Asset B"), and not allowing short sales for asset B eliminates the efficient portfolios 16 to 19 (blue dots above the label "Asset A"). We make the following remarks:

- Not allowing short sales in both assets limits the set of feasible portfolios.
- It is not possible to invest in a portfolio with a higher expected return than asset A.
- It is not possible to invest in a portfolio with a lower expected return than asset B.

12.2.1.1 Short sales constraints and the global minimum variance portfolio

Recall, the two risky asset global minimum variance portfolio solves the unconstrained optimization problem:

$$\begin{aligned}&\min_{m_A, m_B} \sigma_p^2 = m_A^2\sigma_A^2 + m_B^2\sigma_B^2 + 2m_A m_B \sigma_{AB}\\ &s.t.\ m_A + m_B = 1.\end{aligned}$$

From Chapter Chapter 10, the two risky asset unconstrained global minimum variance portfolio weights are:

$$m_A = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}},\ m_B = 1 - m_A. \tag{12.1}$$

For the example data, the unconstrained global minimum variance portfolio weights are both positive:

```
gmin.port = globalmin_portfolio(mu.vals, Sigma)
gmin.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.0793
#> Portfolio standard deviation:  0.0978
#> Portfolio weights:
#> Asset A Asset B
#>   0.202   0.798
```

This means that if we disallow short-sales in both assets we still get the same global minimum variance portfolio weights. In this case, we say that the no short sales constraints are *not binding*

(have no effect) on the solution for the unconstrained optimization problem Equation 12.1. However, it is possible for one of the weights in the global minimum variance portfolio to be negative if correlation between the returns on assets A and B is sufficiently positive. To see this, the numerator in the formula for the global minimum variance weight for asset A can be re-written as:[1]

$$\sigma_B^2 - \sigma_{AB} = \sigma_B^2 - \rho_{AB}\sigma_A\sigma_B = \sigma_B^2\left(1 - \rho_{AB}\frac{\sigma_A}{\sigma_B}\right).$$

Hence, the weight in asset A in the global minimum variance portfolio will be negative if:

$$1 - \rho_{AB}\frac{\sigma_A}{\sigma_B} < 0 \Rightarrow \rho_{AB} > \frac{\sigma_B}{\sigma_A}.$$

For the example data, this cut-off value for the correlation is:

```
sig.B/sig.A
#> [1] 0.446
```

Figure Figure 12.2 shows the two risky asset portfolio frontier computed with $\rho_{AB} = 0.8$. Here, the global minimum variance portfolio is:

```
rho.AB = 0.8
sig.AB = rho.AB*sig.A*sig.B
Sigma = matrix(c(sig2.A, sig.AB, sig.AB, sig2.B), 2, 2)
gmin.port = globalmin_portfolio(mu.vals, Sigma)
gmin.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.016
#> Portfolio standard deviation:  0.099
#> Portfolio weights:
#> Asset A Asset B
#>  -0.325   1.325
```

and contains a negative weight in asset A. This portfolio is shown as the green dot in Figure Figure 12.2, and plots below the point labeled "Asset B" . Here, the no-short sales constraint on asset A is a *binding constraint*: it is not possible to invest in the global minimum variance portfolio without shorting asset A. Visually, we can see that the feasible long-only portfolio with

[1] By the Cauchy-Schwarz inequality the denominator for m_A in Equation 12.1 is always positive.

the smallest variance is 100% invested in asset B. This is the *no short sales constrained* global minimum variance portfolio. Notice that the variance of this portfolio is larger than the variance of the global minimum variance portfolio. This is the cost, in terms of variance, of imposing the no-short sales constraint.

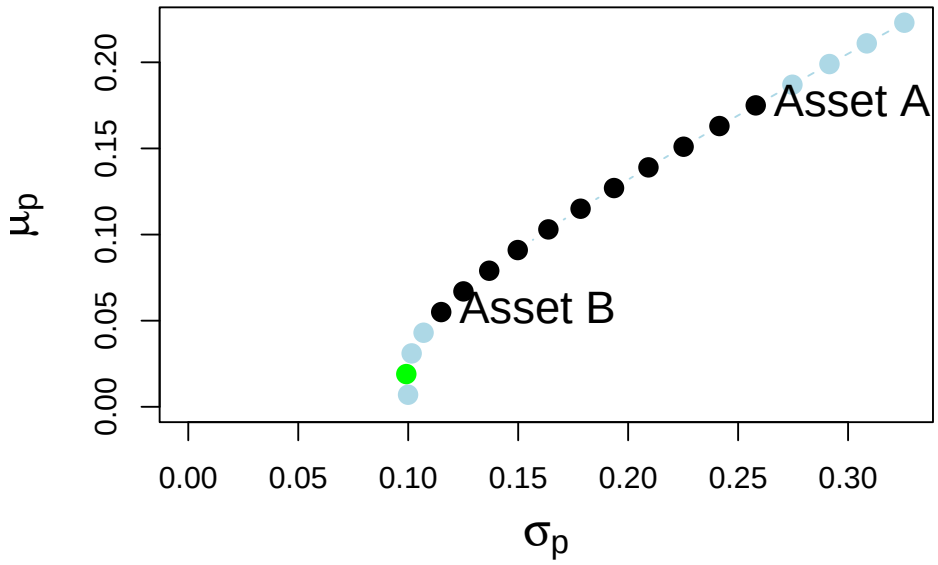


Figure 12.2: Two risky asset portfolio frontier with short sale in Asset A in the global minimum variance portfolio.

12.2.2 One risky asset and a risk-free asset

Now consider portfolios of a risky asset (e.g., asset A) and a risk-free asset (e.g., U.S. T-Bill) with risk-free rate $r_f > 0$. Let x denote the portfolio weight in the risky asset. Then $1 - x$ gives the weight in the risk-free asset. From Chapter Chapter 10, the expected return and volatility of portfolios of one risky asset and one risk-free asset are:

$$\begin{aligned} \mu_p &= r_f + x(\mu - r_f), \\ \sigma_p &= |x|\sigma, \end{aligned}$$

where μ and σ are the expected return and volatility of the risky asset, respectively. For example, Figure Figure 12.3 shows 29 portfolios of asset A and the risk-free asset with $r_f = 0.03$ created with:

```
r.f = 0.03
x.A = seq(from=-1.4, to=1.4, by=0.1)
mu.p.A = r.f + x.A*(mu.A - r.f)
sig.p.A = abs(x.A)*sig.A
plot(sig.p.A, mu.p.A, type="b", ylim=c(min(mu.p.A), max(mu.p.A)),
xlim=c(-0.01, max(sig.p.A)), pch=16, cex = cex.val,
xlab=expression(sigma[p]), ylab=expression(mu[p]), cex.lab = cex.val,
col=c(rep("lightblue", 14), rep("black", 11), rep("lightblue", 4)))
abline(h=0)
text(x=sig.A, y=mu.A, labels="Asset A", pos=4, cex = cex.val)
text(x=0, y=r.f, labels=expression(r[f]), pos=2, cex = cex.val)
```

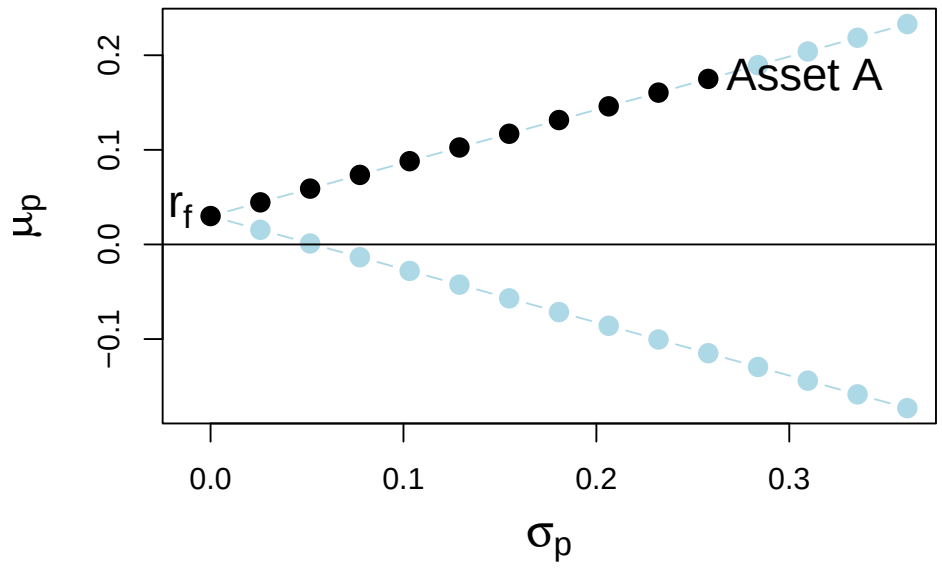


Figure 12.3: Return-risk characteristics of portfolios of asset A and a risk-free asset with $r_f = 0.03$. Long-only portfolios are in black (between r_f and Asset A), and long-short portfolios are in blue (below r_f and above Asset A).

The portfolios (blue dots) above the point labeled "Asset A" are short the risk-free asset and long more than 100% of asset A. The portfolios (blue dots) below $r_f = 0.03$ are short the risky asset and long more than 100% in the risk-free asset.

Here, we assume that the short sales constraint only applies to the risky asset and not the risk-free asset. That is, we assume that we can borrow and lend at the risk-free rate r_f without constraints.

In this situation, the no short sales constraint on the risky asset eliminates the inefficient portfolios (blue dots) below the risk-free rate r_f.

12.2.3 Two risky assets and risk-free asset

As discussed in Chapter Chapter 10, in the case of two risky assets (with no short sales restrictions) plus a risk-free asset the set of efficient portfolios is a combination of the tangency portfolio (i.e., the maximum Sharpe ratio portfolio of two risky assets) and the risk-free asset. For the example data with $\rho_{AB} = -0.164$ and $r_f = 0.03$, the tangency portfolio is:

```
rho.AB = -0.164
sig.AB = rho.AB*sig.A*sig.B
Sigma = matrix(c(sig2.A, sig.AB, sig.AB, sig2.B), 2, 2)
r.f = 0.03
tan.port = tangency_portfolio(mu.vals, Sigma, r.f)
summary(tan.port, r.f)
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.111
#> Portfolio standard deviation:  0.125
#> Portfolio Sharpe Ratio:        0.644
#> Portfolio weights:
#> Asset_1 Asset_2
#>   0.463   0.537
```

No asset is sold short in the unconstrained tangency portfolio, and so this is also the short-sales constrained tangency portfolio. The set of efficient portfolios is illustrated in Figure Figure 12.4, created with:

```
# risky asset only portfolios
x.A = seq(from=0, to=1, by=0.1)
x.B = 1 - x.A
mu.p = x.A*mu.A + x.B*mu.B
sig2.p = x.A^2 * sig2.A + x.B^2 * sig2.B + 2*x.A*x.B*sig.AB
sig.p = sqrt(sig2.p)
# T-bills plus tangency
x.tan = seq(from=0, to=2.4, by=0.1)
mu.p.tan.tbill = r.f + x.tan*(tan.port$er_port - r.f)
sig.p.tan.tbill = x.tan*tan.port$sd_port
```

```
# global minimum variance portfolio
gmin.port = globalmin_portfolio(mu.vals, Sigma)
# plot efficient portfolios
plot(sig.p, mu.p, type="b", pch=16, cex = cex.val,
     ylim=c(0, 0.20), xlim=c(-0.01, 0.35),
     xlab=expression(sigma[p]), ylab=expression(mu[p]), cex.lab = cex.val)
text(x=sig.A, y=mu.A, labels="Asset A", pos=4, cex = cex.val)
text(x=sig.B, y=mu.B, labels="Asset B", pos=4, cex = cex.val)
text(x=0, y=r.f, labels=expression(r[f]), pos=2, cex = cex.val)
text(x=tan.port$sd_port, y=tan.port$er_port, labels="Tangency", pos=2, cex = cex.val)
text(gmin.port$sd_port, gmin.port$er_port, labels="Global Min", pos = 4,
     cex = cex.val)
points(sig.p.tan.tbill, mu.p.tan.tbill, type="l", col="green",
       lwd=2, cex = cex.val)
```

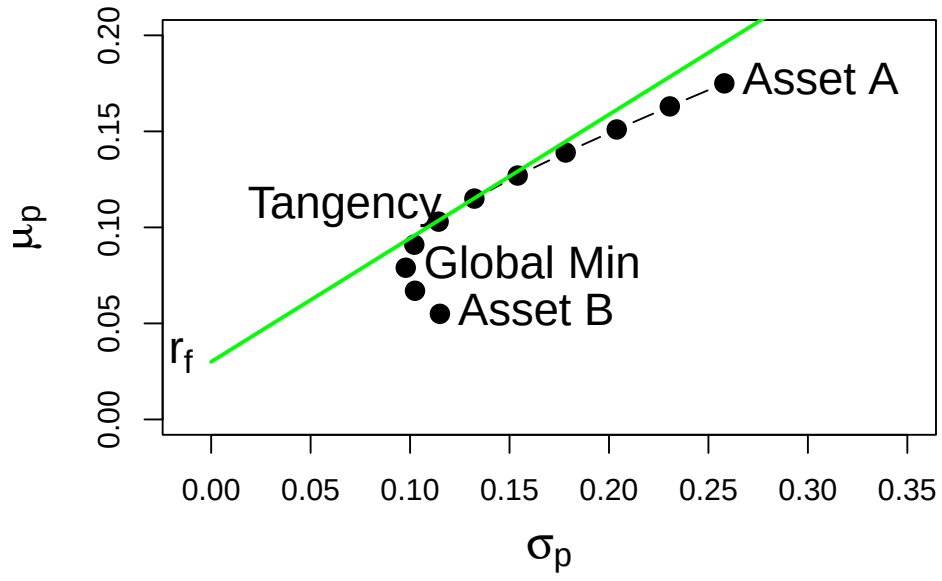


Figure 12.4: Efficient portfolios of two risky assets plus a risk-free asset: $\rho_{AB} = -0.164$ and $r_f = 0.03$. No short sales in unconstrained tangency portfolio.

Figure Figure 12.4 shows that when the tangency portfolio is located between the global minimum variance portfolio (tip of the Markowitz bullet) and asset A, there will be no short sales in the tangency portfolio. However, if the correlation between assets A and B is sufficiently positive (and

the risk-free rate is adjusted so that it is less than the expected return on the global minimum variance portfolio) then asset B can be sold short in the unconstrained tangency portfolio. For example, if $\rho_{AB} = 0.7$ and $r_f = 0.01$ the unconstrained tangency portfolio becomes:

```
rho.AB = 0.7
sig.AB = rho.AB*sig.A*sig.B
Sigma = matrix(c(sig2.A, sig.AB, sig.AB, sig2.B), 2, 2)
r.f = 0.01
tan.port = tangency_portfolio(mu.vals, Sigma, r.f)
summary(tan.port, r.f)
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.238
#> Portfolio standard deviation:  0.355
#> Portfolio Sharpe Ratio:        0.644
#> Portfolio weights:
#> Asset_1 Asset_2
#>   1.529  -0.529
```

Now, asset B is sold short in the tangency portfolio. The set of efficient portfolios in this case is illustrated in Figure Figure 12.5. Notice that the tangency portfolio is located on the set of risky asset only portfolios above the point labeled "Asset A", which indicates that asset B is sold short and more than 100% is invested in asset A.[2]

When shorts sales of risky assets are not allowed, the unconstrained tangency portfolio is infeasible. In this case, the constrained tangency portfolio is not the tangency portfolio but is 100% invested in asset A. This portfolio is located at the tangency point of a straight line drawn from the risk-free rate to the long-only risky asset frontier. Here, the cost of the no-short sales constraint is the reduction in the Sharpe ratio of the tangency portfolio. The Sharpe ratio of the constrained tangency portfolio is the Sharpe ratio of asset A:

```
(mu.A - r.f)/sig.A
#> [1] 0.64
```

Hence, the cost of the no short sales constraint in this example is very small.

[2] Also notice that when $\rho_{AB} = 0.7$ the expected return on the global minimum variance drops (frontier of risky assets is shifted down) below 0.03 so we need to reduce r_f to 0.01 to ensure that the tangency portfolio has a positive slope.

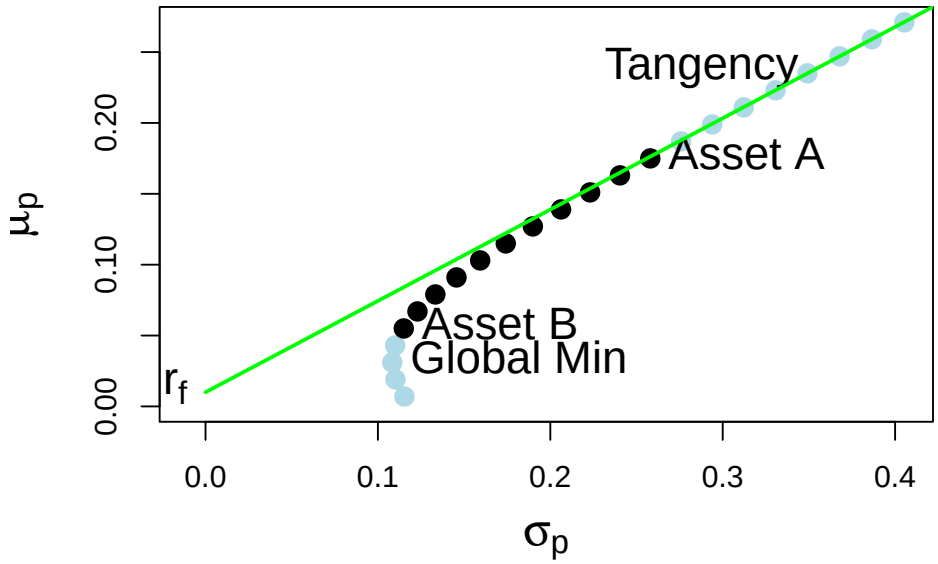


Figure 12.5: Efficient portfolios of two risky assets plus a risk free asset: $\rho_{AB} = 0.7$ and $r_f = 0.01$. Asset B sold short in unconstrained tangency portfolio. The short sale constrained tangency portfolio is the point labeled "Asset A".

12.3 Portfolio Theory with Short Sales Constraints in a General Setting

In this section we extend the analysis of the previous section to the more general setting of N risky assets plus a risk free asset.

12.3.1 Multiple risky assets

Example 12.1 (Random portfolios of Microsoft Nordstrom and Starbucks with no-short sales)**.** The example data on Microsoft, Nordstrom and Starbucks from Chapter Chapter 11 is used to illustrate the computation of mean-variance efficient portfolios with short sales constraints. These data are created using:

```
asset.names <- c("MSFT", "NORD", "SBUX")
mu.vec = c(0.0427, 0.0015, 0.0285)
names(mu.vec) = asset.names
sigma.mat = matrix(c(0.0100, 0.0018, 0.0011,
           0.0018, 0.0109, 0.0026,
           0.0011, 0.0026, 0.0199),
         nrow=3, ncol=3)
dimnames(sigma.mat) = list(asset.names, asset.names)
sd.vec = sqrt(diag(sigma.mat))
r.f=0.005
```

The impact of the no short sales restrictions on the risk-return characteristics of feasible portfolios is nicely illustrated by constructing random long-only portfolios of these three assets:

```
set.seed(123)
x.msft = runif(400, min=0, max=1)
x.nord = runif(400, min=0, max=1)
x.sbux = 1 - x.msft - x.nord
long.only = which(x.sbux > 0)
x.msft = x.msft[long.only]
x.nord = x.nord[long.only]
x.sbux = x.sbux[long.only]
length(long.only)
#> [1] 191
```

Here we first create 400 random weights for Microsoft and Nordstrom that lie between zero and one, respectively. Then, we determine the weight on Starbucks and throw out portfolios for which

the weight on Starbucks is negative. The remaining 191 portfolios are then long-only portfolios. To be sure, these weights are illustrated in Figure Figure 12.6 created using:

```
chart.StackedBar(cbind(x.msft, x.nord, x.sbux),
                 main="Random long-only portfolio weight vectors",
                 xlab = "portfolio", ylab = "weights",
                 xaxis.labels=as.character(1:length(long.only)))
```

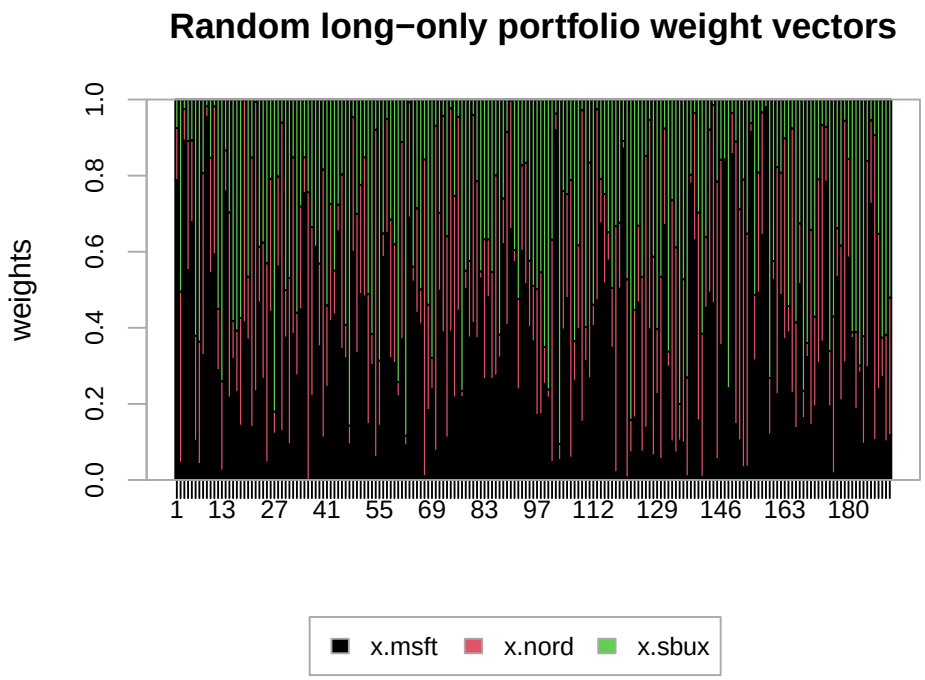


Figure 12.6: Weights on 191 random long-only portfolios of Microsoft, Nordstrom and Starbucks.

```
plot(sd.vec, mu.vec,  ylim=c(0, 0.05), xlim=c(0, 0.2),
     ylab=expression(mu[p]),
     xlab=expression(sigma[p]), type="n", cex.lab=1.5)
for (i in 1:length(long.only)) {
   z.vec = c(x.msft[i], x.nord[i], x.sbux[i])
   mu.p = crossprod(z.vec,mu.vec)
   sig.p = sqrt(t(z.vec)%*%sigma.mat%*%z.vec)
   points(sig.p, mu.p, pch=16, col="grey", cex=1.5)
}
points(sd.vec, mu.vec, pch=16, col="black", cex=2.5, cex.lab=1.75)
text(sd.vec, mu.vec, labels=asset.names, pos=4, cex = cex.val)
```

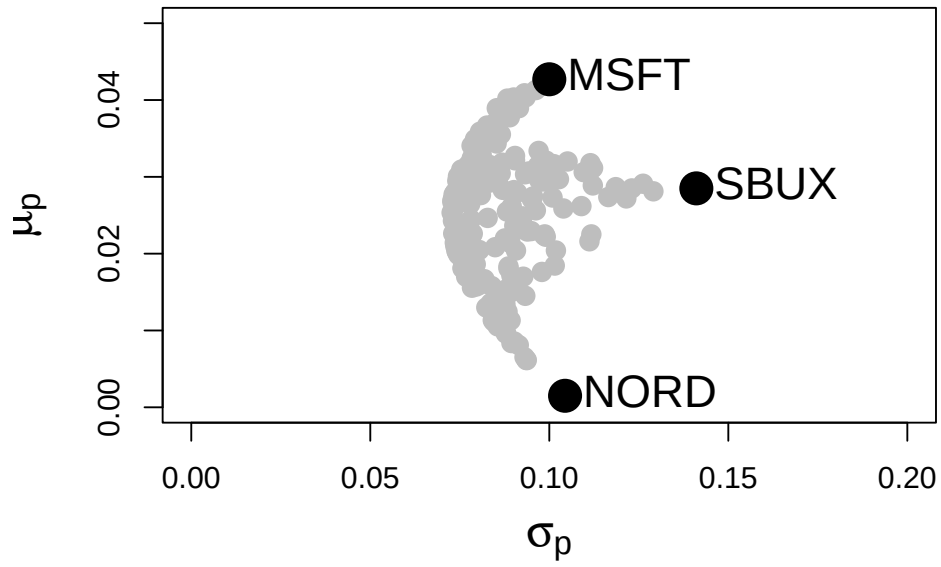


Figure 12.7: Risk-return characteristics of 191 random long-only portfolios of Microsoft, Nordstrom and Starbucks.

The shape of the risk-return scatter is revealing. No random long-only portfolio has an expected return higher than the expected return on Microsoft, which is the asset with the highest expected return. Similarly, no random portfolio has an expected return lower than the expected return on Nordstrom - which is the asset with the lowest expected return. Finally, no random portfolio has a volatility higher than the volatility on Starbucks, the asset with the highest volatility. The scatter of points has a convex (to the origin) parabolic-shaped outer boundary with endpoints at Microsoft and Nordstrom, respectively. The points inside the boundary taper outward from Starbucks towards the outer boundary. The entire risk-return scatter resembles an umbrella tipped on its side with spires extending from the outer boundary to the individual assets. ■

12.3.2 Portfolio optimization imposing no-short sales constraints

Consider an investment universe with N risky assets whose simple returns are described by the GWN model with expected return vector μ and covariance matrix Σ. The optimization problem to solve for the global minimum variance portfolio $\mathbf{m}$ with no-short sales constraints on all assets is a straightforward extension of the problem that allows short sales. We simply add the N no-short

sales inequality constraints $m_i \geq 0\,(i = 1, \ldots, N)$. The optimization problem is:

$$\begin{aligned} \min_{\mathbf{m}} \sigma^2_{p,m} &= \mathbf{m}'\Sigma\mathbf{m} \\ \text{s.t. } \mathbf{m}'\mathbf{1} &= 1, \\ m_i &\geq 0 \ (i = 1, \ldots, N). \end{aligned}$$

Here, there is one equality constraint, $\mathbf{m}'\mathbf{1} = 1$, and N inequality constraints, $m_i \geq 0\ (i = 1, \ldots, N)$.

Similarly, the optimization problem to solve for a minimum variance portfolio $\mathbf{x}$ with target expected return μ_p^0 and no short sales allowed for any asset becomes:

$$\begin{aligned} \min_{\mathbf{x}} \sigma^2_{p,x} &= \mathbf{x}'\Sigma\mathbf{x} \\ \text{s.t. } \mu_{p,x} &= \mathbf{x}'\mu = \mu_p^0, \\ \mathbf{x}'\mathbf{1} &= 1, \\ x_i &\geq 0 \ (i = 1, \ldots, N). \end{aligned} \tag{12.2}$$

Here there are two equality constraints, $\mathbf{x}'\mu = \mu_p^0$ and $\mathbf{x}'\mathbf{1} = 1$, and N inequality constraints, $x_i \geq 0\ (i = 1, \ldots, N)$

Remarks:

1. The optimization problems Equation 12.2 and Equation 12.2, unfortunately, cannot be analytically solved using the method of Lagrange multipliers due to the inequality constraints. They must be solved numerically using an optimization algorithm. Fortunately, the optimization problems Equation 12.2 and Equation 12.2 are special cases of a more general *quadratic programming* problem for which there is a specialized optimization algorithm that is fast and efficient and available in R in the package **quadprog**.[3]
2. There may not be a feasible solution to Equation 12.2. That is, there may not exist a no-short sale portfolio $\mathbf{x}$ that reaches the target return μ_p^0. For example, in Figure Figure 12.7 there is no efficient portfolio that has target expected return higher than the expected return on Microsoft. In general, with N assets there is no feasible solution to Equation 12.2 for target expected returns higher than the maximum expected return among the N assets. If you try to solve Equation 12.2 for an infeasible target expected return, the numerical optimizer will return an error message that indicates no solution is found.
3. The portfolio variances associated with the solutions to Equation 12.2 and Equation 12.2, respectively, must be at least as large as the variances associated with the solutions that do not impose the short sales restrictions. Intuitively this makes sense. Adding additional constraints to a minimization problem will lead to a higher minimum at the solution if the

[3] See the CRAN Optimization Task View for additional R packages that can be used to solve QP problems.

additional constraints are binding. If the constraints are not binding, then the solution with and without the additional constraints are the same.

4. The entire frontier of minimum variance portfolios without short sales exists only for target expected returns between the expected return on the global minimum variance portfolio with no short sales that solves Equation 12.2 and the target expected return equal to the maximum expected return among the N assets. For these target returns, the no-short sales frontier will lie inside and to the right of the frontier that allows short sales if the no-short sales restrictions are binding for some asset.

5. The entire frontier of minimum variance portfolios without short sales can no longer be constructed from any two frontier portfolios. It has to be computed by brute force for each portfolio solving Equation 12.2 with target expected return μ_p^0 above the expected return on the the global minimum variance portfolio that solves Equation 12.2 and below the target return equal to the maximum expected return among the N assets.

12.4 Quadratic Programming Problems

In this Section, we show that the inequality constrained portfolio optimization problems Equation 12.2 and Equation 12.2 are special cases of more general quadratic programming problems and we show how to use the function `solve.QP()` from the R package **quadprog** to numerically solve these problems.

Quadratic programming (QP) problems are of the form:

$$\min_{\mathbf{x}} \ \frac{1}{2}\mathbf{x}'\mathbf{D}\mathbf{x} - \mathbf{d}'\mathbf{x}, \text{ s.t.} \tag{12.3}$$

$$\mathbf{A}_{eq}'\mathbf{x} \geq \mathbf{b}_{eq} \text{ for } l \text{ equality constraints,} \tag{12.4}$$

$$\mathbf{A}_{neq}'\mathbf{x} = \mathbf{b}_{neq} \text{ for } m \text{ inequality constraints,} \tag{12.5}$$

where $\mathbf{D}$ is an $N \times N$ matrix, $\mathbf{x}$ and $\mathbf{d}$ are $N \times 1$ vectors, $\mathbf{A}_{neq}'$ is an $m \times N$ matrix, $\mathbf{b}_{neq}$ is an $m \times 1$ vector, $\mathbf{A}_{eq}'$ is an $l \times N$ matrix, and $\mathbf{b}_{eq}$ is an $l \times 1$ vector.

The `solve.QP()` function from the R package **quadprog** can be used to solve QP functions of the form Equation 12.3 - Equation 12.5. The function `solve.QP()` takes as input the matrices and vectors $\mathbf{D}$, $\mathbf{d}$, $\mathbf{A}_{eq}$, $\mathbf{b}_{eq}$, $\mathbf{A}_{neq}$ and $\mathbf{b}_{neq}$:

```
library(quadprog)
args(solve.QP)
#> function (Dmat, dvec, Amat, bvec, meq = 0, factorized = FALSE)
#> NULL
```

where the arguments `Dmat` and `dvec` correspond to the matrix $\mathbf{D}$ and the vector $\mathbf{d}$, respectively. The function `solve.QP()`, however, assumes that the equality and inequality matrices ($\mathbf{A}_{eq}$, $\mathbf{A}_{neq}$) and vectors ($\mathbf{b}_{eq}$, $\mathbf{b}_{neq}$) are combined into a single $(l+m) \times N$ matrix $\mathbf{A}'$ and a single $(l+m)\times 1$ vector $\mathbf{b}$ of the form:

$$\mathbf{A}' = \begin{bmatrix} \mathbf{A}'_{eq} \\ \mathbf{A}'_{neq} \end{bmatrix}, \mathbf{b} = \begin{pmatrix} \mathbf{b}_{eq} \\ \mathbf{b}_{neq} \end{pmatrix}.$$

In `solve.QP()`, the argument `Amat` represents the matrix $\mathbf{A}$ (not $\mathbf{A}'$) and the argument `bmat` represents the vector $\mathbf{b}$. The argument `meq` determines the number of linear equality constraints (i.e., the number of rows l of $\mathbf{A}'_{eq}$) so that $\mathbf{A}'$ can be separated into $\mathbf{A}'_{eq}$ and $\mathbf{A}'_{neq}$, respectively.

12.4.1 No short sales global minimum variance portfolio

The portfolio optimization problems Equation 12.2 and Equation 12.2 are special cases of the general QP problem Equation 12.3 - Equation 12.5. Consider first the problem to find the global minimum variance portfolio Equation 12.2. The objective function $\sigma^2_{p,m} = \mathbf{m}'\Sigma\mathbf{m}$ can be recovered from Equation 12.3 by setting $\mathbf{x} = \mathbf{m}$, $\mathbf{D} = 2 \times \Sigma$ and $\mathbf{d} = (0, \dots, 0)'$. Then,

$$\frac{1}{2}\mathbf{x}'\mathbf{D}\mathbf{x} - \mathbf{d}'\mathbf{x} = \mathbf{m}'\Sigma\mathbf{m} = \sigma^2_{p,m}.$$

Here, there is one equality constraint, $\mathbf{m}'\mathbf{1} = 1$, and N inequality constraints, $m_i \geq 0$ $(i = 1, \dots, N)$, which can be expressed compactly as $\mathbf{m} \geq \mathbf{0}$. These constraints can be expressed in the form Equation 12.4 and Equation 12.5 by defining the restriction matrices:

$$\underset{(1\times N)}{\mathbf{A}'_{eq}} = \mathbf{1}', \ \underset{(1\times 1)}{\mathbf{b}_{eq}} = 1$$
$$\underset{(N\times N)}{\mathbf{A}'_{neq}} = \mathbf{I}_N, \underset{(N\times 1)}{\mathbf{b}_{neq}} = (0, \dots, 0)' = \mathbf{0},$$

so that

$$\mathbf{A}' = \begin{bmatrix} \mathbf{1}' \\ \mathbf{I}_N \end{bmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ \mathbf{0} \end{pmatrix}.$$

Here,

$$\begin{aligned} \mathbf{A}'_{eq}\mathbf{m} &= \mathbf{1}'\mathbf{m} = 1 = b_{eq}, \\ \mathbf{A}'_{neq}\mathbf{m} &= \mathbf{I}_N\mathbf{m} = \mathbf{m} \geq \mathbf{0} = \mathbf{b}_{neq}. \end{aligned}$$

Example 12.2 (Global minimum variance portfolio with no short sales for three asset example data)**.** The unconstrained global minimum variance portfolio of Microsoft, Nordstrom and Starbucks is:

```
gmin.port = globalmin_portfolio(mu.vec, sigma.mat)
gmin.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.0249
#> Portfolio standard deviation:  0.0727
#> Portfolio weights:
#>  MSFT  NORD  SBUX
#> 0.441 0.366 0.193
```

This portfolio does not have any negative weights and so the no-short sales restriction is not binding. Hence, the short sales constrained global minimum variance is the same as the unconstrained global minimum variance portfolio.

The restriction matrices and vectors required by `solve.QP()` to compute the short sales constrained global minimum variance portfolio are:

```
D.mat = 2*sigma.mat
d.vec = rep(0, 3)
A.mat = cbind(rep(1,3), diag(3))
b.vec = c(1, rep(0,3))
D.mat
#>        MSFT   NORD   SBUX
#> MSFT 0.0200 0.0036 0.0022
#> NORD 0.0036 0.0218 0.0052
#> SBUX 0.0022 0.0052 0.0398
d.vec
#> [1] 0 0 0
t(A.mat)
#>      [,1] [,2] [,3]
#> [1,]    1    1    1
#> [2,]    1    0    0
#> [3,]    0    1    0
#> [4,]    0    0    1
b.vec
#> [1] 1 0 0 0
```

To find the short sales constrained global minimum variance portfolio, call `solve.QP()` with the above inputs and set `meq=1` to indicate one equality constraint:

```
qp.out = solve.QP(Dmat=D.mat, dvec=d.vec,
                  Amat=A.mat, bvec=b.vec, meq=1)
class(qp.out)
#> [1] "list"
names(qp.out)
#> [1] "solution"               "value"                  "unconstrained.solution"
#> [4] "iterations"             "Lagrangian"             "iact"
```

The returned object, `qp.out`, is a list with the following components:

```
qp.out
#> $solution
#> [1] 0.441 0.366 0.193
#>
#> $value
#> [1] 0.00528
#>
#> $unconstrained.solution
#> [1] 0 0 0
#>
#> $iterations
#> [1] 2 0
#>
#> $Lagrangian
#> [1] 0.0106 0.0000 0.0000 0.0000
#>
#> $iact
#> [1] 1
```

The portfolio weights are in the `solution` component, which match the global minimum variance weights allowing short sales, and satisfy the required constraints (weights sum to one and all weights are positive). The minimized value of the objective function (portfolio variance) is in the `value` component. See the help file for `solve.QP()` for explanations of the other components.

You can also use the **introcompfinr** function `globalmin_portfolio()` to compute the global minimum variance portfolio subject to short-sales constraints by specifying the optional argument `shorts=FALSE`:

```
gmin.port = globalmin_portfolio(mu.vec, sigma.mat,
                                shorts=FALSE)
```

```
gmin.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.0249
#> Portfolio standard deviation:  0.0727
#> Portfolio weights:
#>  MSFT  NORD  SBUX
#> 0.441 0.366 0.193
```

When `shorts=FALSE`, `globalmin_portfolio()` uses `solve.QP()` as described above to do the optimization. ■

12.4.2 No short sales minimum variance portfolio with target expected return

Next, consider the problem Equation 12.2 to find a minimum variance portfolio with a given target expected return. The objective function Equation 12.3 has $\mathbf{D} = 2 \times \Sigma$ and $\mathbf{d} = (0, \dots, 0)'$. The two linear equality constraints, $\mathbf{x}'\mu = \mu_p^0$ and $\mathbf{x}'\mathbf{1} = 1$, and N inequality constraints $\mathbf{x} \geq \mathbf{0}$ have restriction matrices and vectors:

$$\underset{(2\times N)}{\mathbf{A}'_{eq}} = \begin{pmatrix} \mu' \\ \mathbf{1}' \end{pmatrix}, \ \underset{(2\times 1)}{\mathbf{b}_{eq}} = \begin{pmatrix} \mu_p^0 \\ 1 \end{pmatrix}.$$
$$\underset{(N\times N)}{\mathbf{A}'_{neq}} = \mathbf{I}_N, \ \underset{(N\times 1)}{\mathbf{b}_{neq}} = (0, \dots, 0)'.$$

so that

$$\mathbf{A}' = \begin{pmatrix} \mu' \\ \mathbf{1}' \\ \mathbf{I}_N \end{pmatrix}, \ \mathbf{b} = \begin{pmatrix} \mu_p^0 \\ 1 \\ \mathbf{0} \end{pmatrix}.$$

Here,

$$\begin{aligned} \mathbf{A}'_{eq}\mathbf{x} &= \begin{pmatrix} \mu' \\ \mathbf{1}' \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mu'\mathbf{x} \\ \mathbf{1}'\mathbf{x} \end{pmatrix} = \begin{pmatrix} \mu_p^0 \\ 1 \end{pmatrix} = \mathbf{b}_{eq}, \\ \mathbf{A}'_{neq}\mathbf{x} &= \mathbf{I}_N\mathbf{x} = \mathbf{x} \geq 0. \end{aligned}$$

Example 12.3 (Minimum variance portfolio with target expected return and no short sales for three asset example data)**.** Now, we consider finding the minimum variance portfolio that has the same mean as Microsoft but does not allow short sales. First, we find the minimum variance portfolio allowing for short sales using the **introcompfinr** function `efficient_portfolio()`:

```
eMsft.port = efficient_portfolio(mu.vec, sigma.mat,
                                 target_return = mu.vec["MSFT"])
eMsft.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w)
#>
#> Portfolio expected return:     0.0427
#> Portfolio standard deviation:  0.0917
#> Portfolio weights:
#>    MSFT    NORD    SBUX
#>  0.8275 -0.0907  0.2633
```

This portfolio has a short sale (negative weight) in Nordstrom. Hence, when the short sales restriction is imposed it will be binding. Next, we set up the restriction matrices and vectors required by `solve.QP()` to compute the minimum variance portfolio with no-short sales:

```
D.mat = 2*sigma.mat
d.vec = rep(0, 3)
A.mat = cbind(mu.vec, rep(1,3), diag(3))
b.vec = c(mu.vec["MSFT"], 1, rep(0,3))
t(A.mat)
#>          MSFT   NORD   SBUX
#> mu.vec 0.0427 0.0015 0.0285
#>        1.0000 1.0000 1.0000
#>        1.0000 0.0000 0.0000
#>        0.0000 1.0000 0.0000
#>        0.0000 0.0000 1.0000
b.vec
#>   MSFT
#> 0.0427 1.0000 0.0000 0.0000 0.0000
```

Then we call `solve.QP()` with `meq=2` to indicate two equality constraints:

```
qp.out = solve.QP(Dmat=D.mat, dvec=d.vec,
                  Amat=A.mat, bvec=b.vec, meq=2)
names(qp.out$solution) = names(mu.vec)
round(qp.out$solution, digits=3)
#> MSFT NORD SBUX
#>    1    0    0
```

With short sales not allowed, the minimum variance portfolio with the same mean as Microsoft

is 100% invested in Microsoft. This is consistent with what we see in Figure Figure 12.7. The volatility of this portfolio is slightly higher than the volatility of the portfolio that allows short sales:

```
sqrt(qp.out$value)
#> [1] 0.1
eMsft.port$sd
#> [1] 0.0917
```

You can also use the **introcompfinr** function `efficient_portfolio()` to compute a minimum variance portfolio with target expected return subject to short-sales constraints by specifying the optional argument `shorts=FALSE`:

```
efficient_portfolio(mu.vec, sigma.mat,
                    target_return = mu.vec["MSFT"], shorts = FALSE)
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w)
#>
#> Portfolio expected return:     0.0427
#> Portfolio standard deviation:  0.1
#> Portfolio weights:
#> MSFT NORD SBUX
#>    1    0    0
```

Suppose you try to find a minimum variance portfolio with target return higher than the mean of Microsoft while imposing no short sales:

```
# efficient_portfolio(mu.vec, sigma.mat,
#                     target_return = mu.vec["MSFT"]+0.01,
#                     shorts = FALSE)
# uncomment this line to see the results
```

Here, `solve.QP()` reports the error "constraints are inconsistent, no solution!" This agrees with what we see in Figure Figure 12.7. ■

Example 12.4 (Compute efficient frontier portfolios from three asset example data)**.** In this example we compare the efficient frontier portfolios computed with and without short sales. This can be easily done using the **IntroCompFinR** function `efficient.frontier()`. First, we compute the efficient frontier allowing short sales for target returns between the mean of the global minimum variance portfolio and the mean of Microsoft:

```
ef = efficient_frontier(mu.vec, sigma.mat, alpha_min=0,
                        alpha_max=1, nport=10)
ef$weight_mat
#>           MSFT    NORD  SBUX
#> port 1   0.441  0.3656 0.193
#> port 2   0.484  0.3149 0.201
#> port 3   0.527  0.2642 0.209
#> port 4   0.570  0.2135 0.217
#> port 5   0.613  0.1628 0.224
#> port 6   0.656  0.1121 0.232
#> port 7   0.699  0.0614 0.240
#> port 8   0.742  0.0107 0.248
#> port 9   0.785 -0.0400 0.256
#> port 10 0.827 -0.0907 0.263
```

Here, portfolio 1 is the global minimum variance portfolio and portfolio 10 is the efficient portfolio with the same mean as Microsoft. Notice that portfolios 9 and 10 have negative weights in Nordstrom. Next, we compute the efficient frontier not allowing short sales for the same range of target returns:

```
ef.ns = efficient_frontier(mu.vec, sigma.mat, alpha_min=0,
                           alpha_max=1, nport=10, shorts=FALSE)
ef.ns$weight_mat
#>           MSFT   NORD  SBUX
#> port 1   0.441 0.3656 0.193
#> port 2   0.484 0.3149 0.201
#> port 3   0.527 0.2642 0.209
#> port 4   0.570 0.2135 0.217
#> port 5   0.613 0.1628 0.224
#> port 6   0.656 0.1121 0.232
#> port 7   0.699 0.0614 0.240
#> port 8   0.742 0.0107 0.248
#> port 9   0.861 0.0000 0.139
#> port 10 1.000 0.0000 0.000
```

Portfolios 1 - 8 have all positive weights and are the same as portfolios 1 - 8 when short sales are allowed. However, for portfolios 9 and 10 the short sales constraint is binding. For portfolio 9, the weight in Nordstrom is forced to zero and the weight in Starbucks is reduced. For portfolio 10, the weights on Nordstrom and Starbucks are forced to zero. The two frontiers are illustrated in Figure Figure 12.8. For portfolios 1-8, the two frontiers coincide. For portfolios 9 and 10, the no-shorts

frontier lies inside and to the right of the short-sales frontier. The cost of the short sales constraint is the increase in the volatility of minimum variance portfolios for target returns near the expected return on Microsoft.

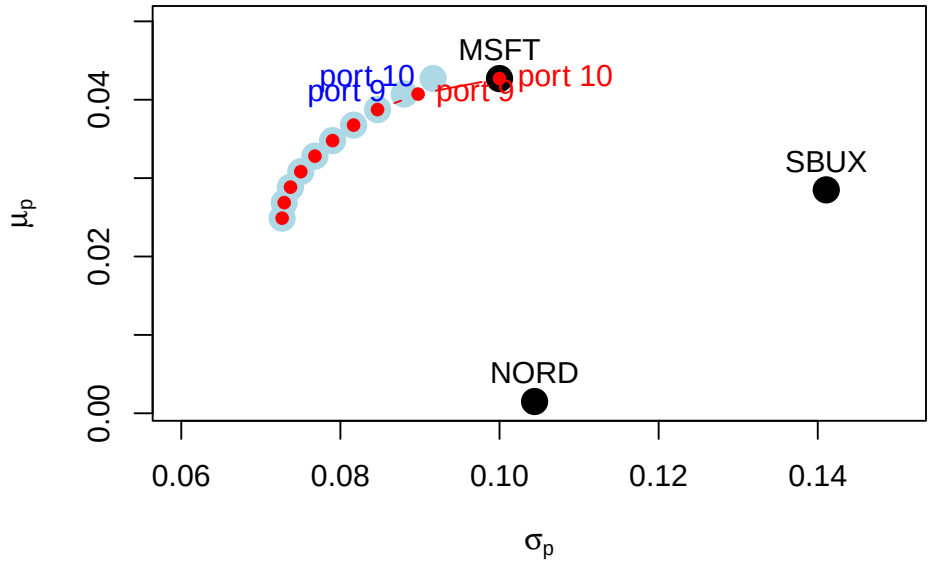


Figure 12.8: Efficient frontier portfolios with and without short sales. The unconstrained efficient frontier portfolios are in blue, and the short sales constrained efficient portfolios are in red. The unconstrained portfolios labeled port 9 and port 10 have short sales in Nordstrom. The constrained portfolio labeled port 9 has zero weight in Nordstrom, and the constrained portfolio labeled port 10 has zero weights in Nordstrom and Starbucks.

■

12.4.3 No short sales tangency portfolio

Consider an investment universe with N risky assets and a single risk-free asset. We assume that short sales constraints only apply to the N risky assets, and that investors can borrow and lend at the risk-free rate r_f. As shown in Chapter Chapter 11, with short sales allowed, mean-variance efficient portfolios are combinations of the risk-free asset and the tangency portfolio. The tangency portfolio is the maximum Sharpe ratio portfolio of risky assets and is the solution to the maximization problem:

$$\begin{aligned} \max_{\mathbf{t}} \frac{\mu_t - r_f}{\sigma_t} &= \frac{\mathbf{t}'\mu - r_f}{(\mathbf{t}'\Sigma\mathbf{t})^{1/2}} \; s.t. \\ \mathbf{t}'\mathbf{1} &= 1. \end{aligned}$$

Example 12.5 (Compute efficient portfolios allowing short sales from three asset example data when there is a risk-free asset)**.** Assuming a risk-free rate $r_f = 0.005$, we can compute the unconstrained tangency portfolio for the three asset example data using the **IntroCompFinR** function `tangency.portfolio()`:

```
r.f = 0.005
tan.port = tangency_portfolio(mu.vec, sigma.mat, r.f)
summary(tan.port, r.f)
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.0519
#> Portfolio standard deviation:  0.112
#> Portfolio Sharpe Ratio:        0.42
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3
#>   1.027  -0.326   0.299
```

Here, the unconstrained tangency portfolio has a negative weight in Nordstrom so that the short sales constraint on the risky assets will be binding. The unconstrained set of efficient portfolios, that are combinations of the risk-free asset and the unconstrained tangency portfolio, is illustrated in Figure Figure 12.9.

Here, we see that the unconstrained tangency portfolio is located on the frontier of risky asset portfolios above the point labeled "MSFT". The green line represents portfolios of the risk-free asset and the unconstrained tangency portfolio. ■

Under short sales constraints on the risky assets, the maximum Sharpe ratio portfolio solves:

$$\begin{aligned} \max_{\mathbf{t}} \frac{\mu_t - r_f}{\sigma_t} &= \frac{\mathbf{t}'\mu - r_f}{(\mathbf{t}'\Sigma\mathbf{t})^{1/2}} \, s.t. \\ \mathbf{t}'\mathbf{1} &= 1, \\ t_i &\geq 0, \, i = 1, \ldots, N. \end{aligned}$$

This optimization problem *cannot* be written as a QP optimization as expressed in Equation 12.3 - Equation 12.5. Hence, it looks like we cannot use the function `solve.QP()` to find the short sales constrained tangency portfolio. However, it turns out we can use `solve.QP()` to find the short sales constrained tangency portfolio. To do this we utilize the alternative derivation of the tangency portfolio presented in Chapter Chapter 11. The alternative way to compute the tangency portfolio is to first find a minimum variance portfolio of risky assets and a risk-free asset whose excess expected return, $\tilde{\mu}_{p,x} = \mu' \mathbf{x} - r_f \mathbf{1}$, achieves a target excess return $\tilde{\mu}_{p,0} = \mu_{p,0} - r_f > 0$. This

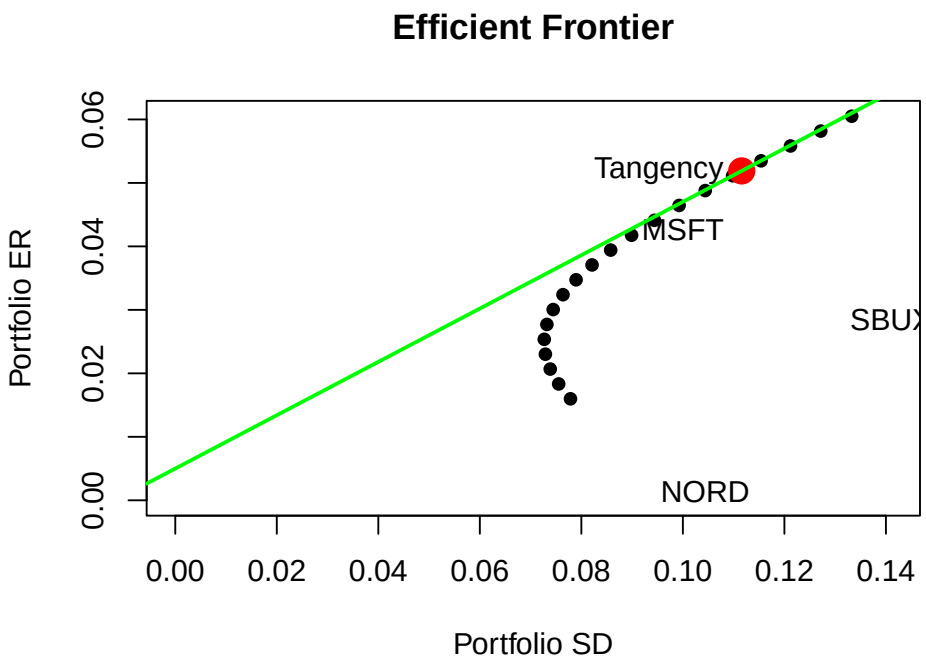


Figure 12.9: Efficient portfolios of three risky assets and a risk-free asset allowing short sales. Nordstrom is sold short in the unconstrained tangency portfolio.

portfolio solves the minimization problem:

$$\min_{\mathbf{x}} \ \sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x} \text{ s.t. } \tilde{\mu}_{p,x} = \tilde{\mu}_{p,0},$$

where the weight vector $\mathbf{x}$ is not constrained to satisfy $\mathbf{x}'\mathbf{1} = \mathbf{1}$. The tangency portfolio is then determined by normalizing the weight vector $\mathbf{x}$ so that its elements sum to one:[4]

$$\mathbf{t} = \frac{\mathbf{x}}{\mathbf{x}'\mathbf{1}} = \frac{\Sigma^{-1}(\mu - r_f \cdot \mathbf{1})}{\mathbf{1}'\Sigma^{-1}(\mu - r_f \cdot \mathbf{1})}.$$

An interesting feature of this result is that it does not depend on the value of the target excess return value $\tilde{\mu}_{p,0} = \mu_{p,0} - r_f > 0$. That is, every value of $\tilde{\mu}_{p,0} = \mu_{p,0} - r_f > 0$ gives the same value for the tangency portfolio.[5] We can utilize this alternative derivation of the tangency portfolio to find the tangency portfolio subject to short-sales restrictions on risky assets. First we solve the short-sales constrained minimization problem:

$$\min_{\mathbf{x}} \ \sigma^2_{p,x} = \mathbf{x}'\Sigma\mathbf{x} \text{ s.t.} \tag{12.6}$$

[4]This normalization ensures that the tangency portfolio is 100% invested in the risky assets.

[5]If $\tilde{\mu}_{p,0} = 0$ then the portfolio of risky assets and the risk-free asset has expected return equal to r_f. In this case, $\mathbf{x} = \mathbf{0}$ and $x_f = 1$.

$$\tilde{\mu}_{p,x} = \tilde{\mu}_{p,0}, \tag{12.7}$$

$$x_i \geq 0,\ i = 1, \ldots, N. \tag{12.8}$$

Here we can use any value for $\tilde{\mu}_{p,0}$ so for convenience we use $\tilde{\mu}_{p,0} = 1$. This is a QP problem with $\mathbf{D} = 2 \times \Sigma$ and $\mathbf{d} = (0, \ldots, 0)'$, one linear equality constraint, $\tilde{\mu}_{p,x} = \tilde{\mu}_{p,0}$, and N inequality constraints $\mathbf{x} \geq \mathbf{0}$. The restriction matrices and vectors are:

$$\underset{(1\times N)}{\mathbf{A}'_{eq}} = (\mu - r_f\mathbf{1})',\ \underset{(1\times 1)}{\mathbf{b}_{eq}} = 1,$$
$$\underset{(N\times N)}{\mathbf{A}'_{neq}} = \mathbf{I}_N,\ \underset{(N\times 1)}{\mathbf{b}_{neq}} = (0, \ldots, 0)'.$$

so that

$$\mathbf{A}' = \begin{pmatrix} (\mu - r_f\mathbf{1})' \\ \mathbf{I}_N \end{pmatrix},\ \mathbf{b} = \begin{pmatrix} 1 \\ \mathbf{0} \end{pmatrix}.$$

Here,

$$\begin{aligned} \mathbf{A}'_{eq}\mathbf{x} &= (\mu - r_f\mathbf{1})'\mathbf{x} = \tilde{}_{\mathbf{p,x}} = 1, \\ \mathbf{A}'_{neq}\mathbf{x} &= \mathbf{I}_N\mathbf{x} = \mathbf{x} \geq 0. \end{aligned}$$

After solving the QP problem for $\mathbf{x}$, we then determine the short-sales constrained tangency portfolio by normalizing the weight vector $\mathbf{x}$ so that its elements sum to one:

$$\mathbf{t} = \frac{\mathbf{x}}{\mathbf{x}'\mathbf{1}}.$$

Example 12.6 (Compute efficient portfolios not allowing short sales from three asset example data when there is a risk-free asset)**.** First, we use `solve.QP()` to find the short sales restricted tangency portfolio. The restriction matrices for the short sales constrained optimization Equation 12.6 - Equation 12.8 are:

```
D.mat = 2*sigma.mat
d.vec = rep(0, 3)
A.mat = cbind(mu.vec - rep(1, 3)*r.f, diag(3))
b.vec = c(1, rep(0,3))
```

The un-normalized portfolio $\mathbf{x}$ is found using:

```
qp.out = solve.QP(Dmat=D.mat, dvec=d.vec,
                  Amat=A.mat, bvec=b.vec, meq=1)
x.ns = qp.out$solution
names(x.ns) = names(mu.vec)
```

```
round(x.ns, digits=3)
#>  MSFT  NORD  SBUX
#> 22.74  0.00  6.08
```

The short sales constrained tangency portfolio is then:

```
t.ns = x.ns/sum(x.ns)
round(t.ns, digits=3)
#>  MSFT  NORD  SBUX
#> 0.789 0.000 0.211
```

In this portfolio, the allocation to Nordstrom, which was negative in the unconstrained tangency portfolio, is set to zero.

To verify that the derivation of the short sales constrained tangency portfolio does not depend on $\tilde{\mu}_{p,0} = \mu_{p,0} - r_f > 0$, we repeat the calculations with $\tilde{\mu}_{p,0} = 0.5$ instead of $\tilde{\mu}_{p,0} = 1$:

```
b.vec = c(0.5, rep(0,3))
qp.out = solve.QP(Dmat=D.mat, dvec=d.vec,
                  Amat=A.mat, bvec=b.vec, meq=1)
x.ns = qp.out$solution
names(x.ns) = names(mu.vec)
t.ns = x.ns/sum(x.ns)
round(t.ns, digits=3)
#>  MSFT  NORD  SBUX
#> 0.789 0.000 0.211
```

You can compute the short sales restricted tangency portfolio using the **introcompfinr** function `tangency_portfolio()` with the optional argument `shorts=FALSE`:

```
tan.port.ns = tangency_portfolio(mu.vec, sigma.mat, r.f, shorts=FALSE)
summary(tan.port.ns, r.f)
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.0397
#> Portfolio standard deviation:  0.0865
#> Portfolio Sharpe Ratio:        0.401
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3
#>   0.789   0.000   0.211
```

Notice that the Sharpe ratio on the short sales restricted tangency portfolio is slightly smaller than the Sharpe ratio on the unrestricted tangency portfolio.

The set of efficient portfolios are combinations of the risk-free asset and the short sales restricted tangency portfolio. These portfolios are illustrated in Figure Figure 12.10, created using:

```
ef.ns = efficient_frontier(mu.vec, sigma.mat, alpha_min=0,
                           alpha_max=1, nport=10, shorts=FALSE)
plot(ef.ns, plot.assets=TRUE, pch=16)
points(tan.port.ns$sd_port, tan.port.ns$er_port, col="red", pch=16, cex=2)
sr.tan.ns = (tan.port.ns$er_port - r.f)/tan.port.ns$sd_port
abline(a=r.f, b=sr.tan.ns, col="green", lwd=2)
text(tan.port$sd_port, tan.port$er_port, labels="Tangency", pos = 2)
```

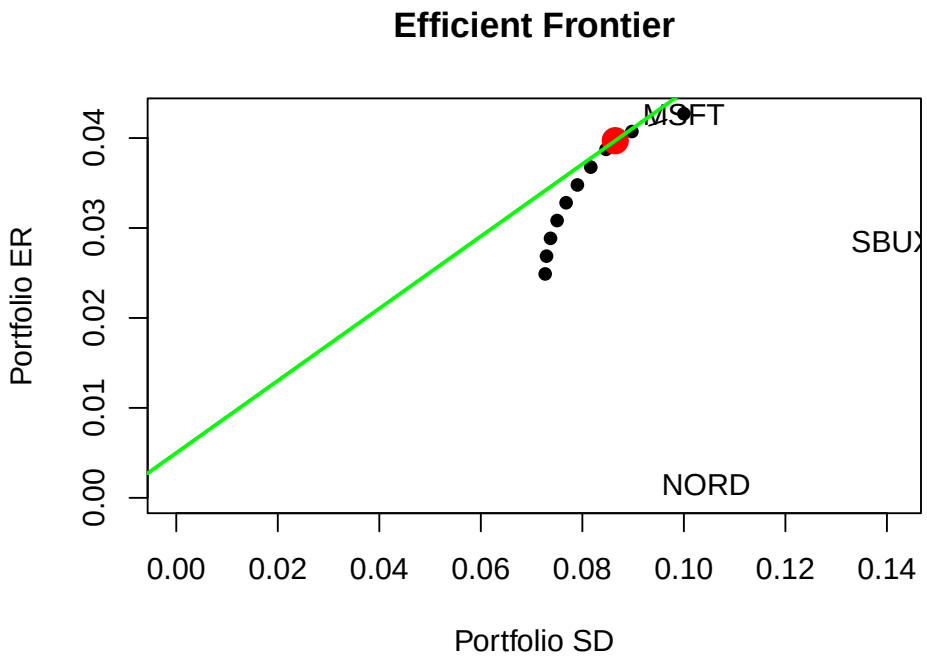


Figure 12.10: Efficient frontier portfolios of three risky assets and a risk-free asset not allowing short sales

■

12.5 Application to Vanguard Mutual Funds

In this section, we revisit the application of mean-variance portfolio theory to the problem of asset allocation among six Vanguard mutual funds that was discussed at the end of Chapter Chapter 11. For the sub-sample January, 2010 through December, 2014 it was shown that estimated mean-variance efficient portfolios had short sales in many of the Vanguard funds. Since it is not possible to short mutual funds, we re-estimate the efficient portfolios with the additional constraints that all portfolio weights must be positive. The monthly returns and annualized GWN model estimates for the Vanguard funds are created using:

```
VanguardPrices = as.xts(VanguardPrices)
VanguardRetS = na.omit(Return.calculate(VanguardPrices, method="simple"))
smpl = "2010-1::2014-12"
r.f = 0.001
muhat = 12*colMeans(VanguardRetS[smpl])
covhat = 12*cov(VanguardRetS[smpl])
```

We use the **introcompfinr** portfolio functions to compute unrestricted and short-sales restricted estimates of the global minimum variance portfolio, the tangency portfolio, and the efficient frontier of risky assets:

```
# unrestricted (short sales allowed) portfolios
gmin.port = globalmin_portfolio(muhat, covhat)
tan.port = tangency_portfolio(muhat, covhat, r.f)
ef = efficient_frontier(muhat, covhat, alpha_min=0,
                        alpha_max=1.5, nport=20)
# short sales restricted portfolios
gmin.port.ns = globalmin_portfolio(muhat, covhat, shorts=FALSE)
tan.port.ns = tangency_portfolio(muhat, covhat, r.f, shorts=FALSE)
ef.ns = efficient_frontier(muhat, covhat, alpha_min=0,
                           alpha_max=1, nport=20, shorts=FALSE)
```

The global minimum variance portfolios are:

```
# short sales allowed
gmin.port
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.0197
```

```
#> Portfolio standard deviation:  0.0114
#> Portfolio weights:
#>   vfinx   veurx   veiex   vbltx   vbisx   vpacx
#>  0.0581 -0.0347 -0.0214 -0.0736  1.0647  0.0069
# short sales restricted
gmin.port.ns
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_gmin)
#>
#> Portfolio expected return:     0.0218
#> Portfolio standard deviation:  0.0135
#> Portfolio weights:
#>  vfinx  veurx  veiex  vbltx  vbisx  vpacx
#> 0.0171 0.0000 0.0000 0.0000 0.9829 0.0000
```

In the short sales restricted global minimum variance portfolio there is 98% weight on vbisx (low volatility US short-term bond fund) and 2% weight on vfinx (S&P 500 fund). The weights on other funds are pushed to zero. The volatility of the short sales restricted portfolio is slightly higher than the volatility of the unrestricted portfolio and so the cost of the short sales restrictions is small.

The estimated tangency portfolios are:

```
# short sales allowed
summary(tan.port, r.f)
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.0642
#> Portfolio standard deviation:  0.0209
#> Portfolio Sharpe Ratio:        3.02
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3 Asset_4 Asset_5 Asset_6
#>  0.3152 -0.0907 -0.0874  0.1159  0.7387  0.0083
# short sales restricted
summary(tan.port.ns, r.f)
#> Call:
#> get_portfolio(er_vec = er_vec, cov_mat = cov_mat, weight_vec = w_t)
#>
#> Portfolio expected return:     0.0637
#> Portfolio standard deviation:  0.0287
```

```
#> Portfolio Sharpe Ratio:        2.18
#> Portfolio weights:
#> Asset_1 Asset_2 Asset_3 Asset_4 Asset_5 Asset_6
#>   0.192   0.000   0.000   0.245   0.563   0.000
```

The short sales restricted tangency portfolio has positive weights on vfinx, vbltx, and vbisx and zero weights for the remaining assets. The Sharpe ratio of the short sales restricted tangency portfolio, however, is substantially smaller than the Sharpe ratio of the unrestricted portfolio.

Figure Figure 12.11 shows the short sales restricted and unrestricted efficient portfolio frontiers, created using:

```
# unrestricted efficient frontiers
plot(ef, plot.assets=TRUE, col="blue", pch=16)
text(tan.port$sd_port, tan.port$er_port,labels = "tan", pos = 2)
sr.tan = (tan.port$er_port - r.f)/tan.port$sd_port
text(0, r.f, labels=expression(r[f]), pos = 4)
abline(a=r.f, b=sr.tan, col="green", lwd=2)
abline(h=0, v=0)
# short sales restricted frontiers
sr.tan.ns = (tan.port.ns$er_port - r.f)/tan.port.ns$sd_port
points(ef.ns$sd_vec, ef.ns$er_vec, type="b", pch=16, col="red")
abline(a=r.f, b=sr.tan.ns, col="green", lwd=2)
text(tan.port.ns$sd_port, tan.port.ns$er_port,labels = "tan (no shorts)", pos = 4)
```

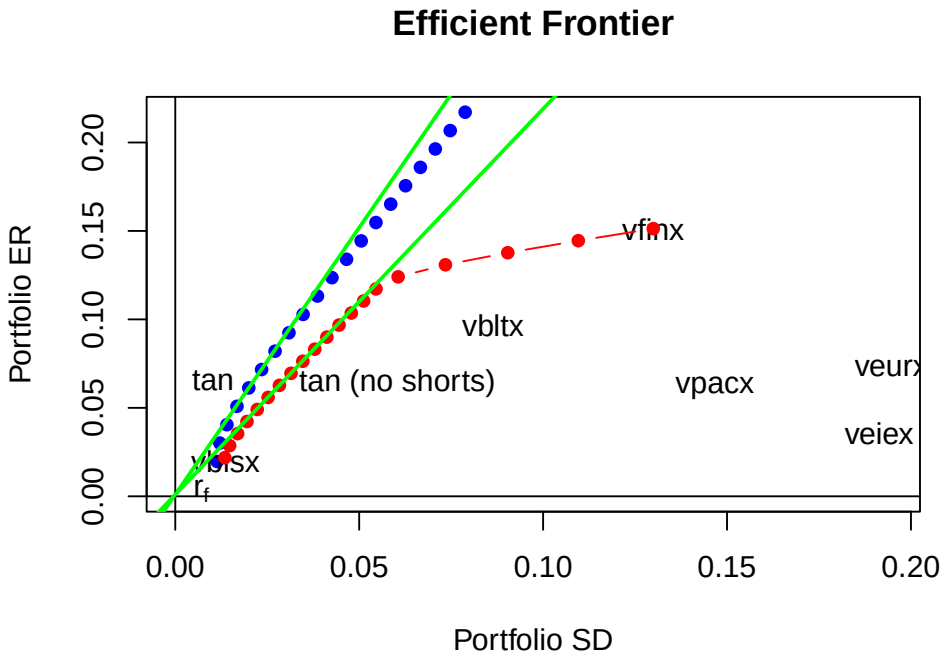


Figure 12.11: Unrestricted and short sales restricted efficient portfolio frontiers.

The short sales restricted risky asset only frontier, shown in red, lies underneath and to the right of the unrestricted risky asset frontier, shown in blue. The short sales restricted tangency portfolio has a similar expected return as the unrestricted tangency portfolio but has a higher volatility and a lower Sharpe ratio.

12.6 Further Reading: Portfolio Theory with Short Sales Constraints

- Mention other quadratic programming packages: ROI etc. See Doug's book
- Mention Pfaff's R book (Pfaff 2013) ??
- Mention Grilli's book etc.

To be completed...

12.7 Exercises

To be completed

{MartinSchererYollin} Martin, R.D., Scherer, B., and Yollin, G. (2016).

13 Portfolio Risk Budgeting

Updated: September 13, 2026

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The idea of portfolio risk budgeting is to decompose a measure of portfolio risk into risk contributions from the individual assets in the portfolio. In this way, an investor can see which assets are most responsible for portfolio risk and can then make informed decisions, if necessary, about re-balancing the portfolio to alter the risk. Portfolio risk budgeting reports, which summarize asset contributions to portfolio risk, are becoming standard in industry practice. In addition, it has become popular to consider portfolio construction using risk budgeting to create so-called "risk-parity" portfolios. In risk-parity portfolios, portfolio weights are constructed so that that each asset has equal risk contributions to portfolio risk.[1]

I first motivate risk budgeting using the portfolio return variance and standard deviation as measures of portfolio risk. I then show how Euler's theorem can be used to do risk budgeting for portfolio risk measures that are homogeneous of degree one in the portfolio weights. Such risk measures include portfolio return standard deviation (volatility) and Value-at-Risk for general return distributions.

- Need to finish introduction

The R packages I use in this chapter are **corrplot**, **introcompfinr** and **PerformanceAnalytics**. Make sure these packages are installed and loaded in R before replicating the chapter examples.

13.1 Risk Budgeting Using Portfolio Variance and Portfolio Standard Deviation

I motivate portfolio risk budgeting in the simple context of a two risky asset portfolio. To illustrate, consider forming a portfolio consisting of two risky assets (asset 1 and asset 2) with portfolio shares x_1 and x_2 such that $x_1 + x_2 = 1$. I assume that the GWN model holds for the simple returns on each asset. The portfolio return is given by $R_p = x_1 R_1 + x_2 R_2$, and the portfolio expected return and variance are given by $\mu_p = x_1\mu_1 + x_2\mu_2$ and $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\sigma_{12}$, respectively.

[1] Two hedge funds, Bridgewater and AQR capital management have helped to make risk parity investing popular.

The portfolio variance, σ_p^2, and standard deviation, σ_p, are natural measures of portfolio risk. The advantage of using σ_p is that it is in the same units as the portfolio return. Risk budgeting concerns the following question: what are the contributions of assets 1 and 2 to portfolio risk captured by σ_p^2 or σ_p?

13.1.1 Case 1: uncorrelated assets

To answer this question, I consider first the formula for portfolio variance. If the asset returns are uncorrelated ($\sigma_{12} = 0$) then

$$\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2$$

gives an *additive decomposition* of portfolio risk: $x_1^2\sigma_1^2$ is the variance contribution from asset 1, and $x_2^2\sigma_2^2$ is the variance contribution from asset 2. Clearly, asset 1 has a higher contribution than asset 2 if $x_1^2\sigma_1^2 > x_2^2\sigma_2^2$. For example, in an equally weighted portfolio asset 1's risk contribution is higher than asset 2's contribution if asset 1's return is more variable than asset 2's return. I can also define asset percent contributions to risk, which divide the asset contributions by portfolio variance, as $x_1^2\sigma_1^2/\sigma_p^2$ and $x_2^2\sigma_2^2/\sigma_p^2$, respectively. The percent contributions have the property that they add to 100%. Notice that an equally weighted portfolio may not have equal contributions to risk.

If I measure risk by portfolio volatility, $\sigma_p = \sqrt{x_1^2\sigma_1^2 + x_2^2\sigma_2^2}$, I may be tempted to define the contributions of assets 1 and 2 to risk as $x_1\sigma_1$ and $x_2\sigma_2$, respectively. However, this would not be correct because I would not get an additive decomposition

$$\sigma_p = \sqrt{x_1^2\sigma_1^2 + x_2^2\sigma_2^2} \neq x_1\sigma_1 + x_2\sigma_2.$$

To get an additive decomposition I have to define the risk contributions as $x_1^2\sigma_1^2/\sigma_p$ and $x_2^2\sigma_2^2/\sigma_p$, respectively. Then

$$\frac{x_1^2\sigma_1^2}{\sigma_p} + \frac{x_2^2\sigma_2^2}{\sigma_p} = \frac{\sigma_p^2}{\sigma_p} = \sigma_p.$$

Notice that the percent contributions to portfolio volatility are the same as the percent contributions to portfolio variance.

13.1.2 Case 2: correlated assets

If the asset returns are correlated ($\sigma_{12} \neq 0$) then the risk decomposition becomes a bit more complicated because I have to decide what to do with the covariance contribution to portfolio

variance. However, the formula for σ_p^2 suggests a simple additive decomposition:

$$\begin{aligned}\sigma_p^2 &= x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\sigma_{12} \\ &= (x_1^2\sigma_1^2 + x_1x_2\sigma_{12}) + (x_2^2\sigma_2^2 + x_1x_2\sigma_{12}) \\ &= (x_1^2\sigma_1^2 + x_1x_2\rho_{12}\sigma_1\sigma_2) + (x_2^2\sigma_2^2 + x_1x_2\rho_{12}\sigma_1\sigma_2)\end{aligned}$$

Now, $(x_1^2\sigma_1^2 + x_1x_2\sigma_{12})$ and $(x_2^2\sigma_2^2 + x_1x_2\sigma_{12})$ are portfolio variance contributions of assets 1 and 2, respectively. Here, I split the covariance contribution to portfolio variance, $2x_1x_2\sigma_{12}$, evenly between the two assets. Now, the correlation between the two asset returns also plays a role in the risk decomposition. The additive decomposition for σ_p is

$$\sigma_p = \frac{x_1^2\sigma_1^2 + x_1x_2\sigma_{12}}{\sigma_p} + \frac{x_2^2\sigma_2^2 + x_1x_2\sigma_{12}}{\sigma_p},$$

which gives $\left(\frac{x_1^2\sigma_1^2+x_1x_2\sigma_{12}}{\sigma_p}\right)$ and $\left(\frac{x_2^2\sigma_2^2+x_1x_2\sigma_{12}}{\sigma_p}\right)$ as the asset contributions to portfolio volatility, respectively.

13.1.3 The general case of N assets

The risk budgeting decompositions I presented above for two asset portfolios extend naturally to general N asset portfolios. In the case where all assets are mutually uncorrelated, asset i's contributions to portfolio variance and volatility are $x_i^2\sigma_i^2$ and $x_i^2\sigma_i^2/\sigma_p$, respectively. In the case of correlated asset returns, all pairwise covariances enter into asset i's contributions. The portfolio variance contribution is

$$x_i^2\sigma_i^2 + \sum_{j\neq i} x_ix_j\sigma_{ij}$$

and the portfolio volatility contribution is

$$\left(x_i^2\sigma_i^2 + \sum_{j\neq i} x_ix_j\sigma_{ij}\right)/\sigma_p$$

. As you might expect, these formulae can be simplified using matrix algebra.

13.2 Euler's Theorem and Risk Decompositions

When I used σ_p^2 or σ_p to measure portfolio risk, I was able to easily derive sensible risk decompositions in the two risky asset case. However, if I measure portfolio risk by value-at-risk or some other risk measure it is not so obvious how to define individual asset risk contributions. For portfolio risk measures that are *homogenous functions of degree one* in the portfolio weights, Euler's theorem provides a general method for decomposing risk into asset specific contributions.

13.2.1 Homogenous functions of degree one

First I define a homogenous function of degree one.

Definition 13.1 (homogenous function of degree one)**.** Let $f(x_1, x_2, \dots, x_N)$ be a continuous and differentiable function of the variables x_1, $x_2, \dots, x_N$. The function f is homogeneous of degree one if for any constant $c > 0$, $f(cx_1, cx_2, \dots, cx_N) = cf(x_1, x_2, \dots, x_N)$. In matrix notation I have $f(x_1, \dots, x_N) = f(\mathbf{x})$ where $\mathbf{x} = (x_1, \dots, x_N)'$. Then f is homogeneous of degree one if $f(c \cdot \mathbf{x}) = c \cdot f(\mathbf{x})$.

Example 13.1 (Homogenous functions of degree one)**.** Consider first the function $f(x_1, x_2) = x_1 + x_2$. Then

$$f(cx_1, cx_2) = cx_1 + cx_2 = c(x_1 + x_2) = cf(x_1, x_2)$$

so that $x_1 + x_2$ is homogenous of degree one. Next, let $f(x_1, x_2) = x_1^2 + x_2^2$. Then

$$f(cx_1, cx_2) = c^2x_1^2 + c^2x_2^2 = c^2(x_1^2 + x_2^2) \neq cf(x_1, x_2)$$

so that $x_1^2 + x_2^2$ is not homogenous of degree one. Finally, let $f(x_1, x_2) = \sqrt{x_1^2 + x_2^2}$ Then

$$f(cx_1, cx_2) = \sqrt{c^2x_1^2 + c^2x_2^2} = c\sqrt{(x_1^2 + x_2^2)} = cf(x_1, x_2)$$

so that $\sqrt{x_1^2 + x_2^2}$ is homogenous of degree one.

In matrix notation, define $\mathbf{x} = (x_1, x_2)'$ and $\mathbf{1} = (1, 1)'$. Let $f(x_1, x_2) = x_1 + x_2 = \mathbf{x}'\mathbf{1} = \mathbf{f}(\mathbf{x})$. Then

$$f(c \cdot \mathbf{x}) = (c \cdot \mathbf{x})' \mathbf{1} = c \cdot (\mathbf{x}'1) = c \cdot f(\mathbf{x}).$$

Let $f(x_1, x_2) = x_1^2 + x_2^2 = \mathbf{x}'\mathbf{x} = f(\mathbf{x})$. Then

$$f(c \cdot \mathbf{x}) = (c \cdot \mathbf{x})'(c \cdot \mathbf{x}) = c^2 \cdot \mathbf{x}'\mathbf{x} \neq c \cdot f(\mathbf{x}).$$

Let $f(x_1, x_2) = \sqrt{x_1^2 + x_2^2} = (\mathbf{x}'\mathbf{x})^{1/2} = f(\mathbf{x})$. Then

$$f(c \cdot \mathbf{x}) = ((c \cdot \mathbf{x})'(c \cdot \mathbf{x}))^{1/2} = c \cdot (\mathbf{x}'\mathbf{x})^{1/2} = c \cdot f(\mathbf{x}).$$

■

Consider a portfolio of N assets with weight vector $\mathbf{x}$, return vector $\mathbf{R}$, expected return vector μ and covariance matrix Σ. The portfolio return, expected return, variance, and volatility, are

functions of the portfolio weight vector $\mathbf{x}$:

$$\begin{aligned} R_p &= R_p(\mathbf{x}) = \mathbf{x}'\mathbf{R} \\ \mu_p &= \mu_p(\mathbf{x}) = \mathbf{x}'\mu, \\ \sigma_p^2 &= \sigma_p^2(\mathbf{x}) = \mathbf{x}'\Sigma\mathbf{x}, \\ \sigma_p &= \sigma_p(\mathbf{x}) = (\mathbf{x}'\Sigma\mathbf{x})^{1/2}. \end{aligned}$$

As a result, the portfolio return quantile and portfolio value-at-risk are also functions of the portfolio weight vector $\mathbf{x}$:

$$\begin{aligned} q_\alpha^{R_p}(x) &= \mu_p(\mathbf{x}) + \sigma_p(\mathbf{x})q_\alpha^Z, \\ \mathrm{VaR}_{p,\alpha}(\mathbf{x}) &= -q_\alpha^{R_p}(\mathbf{x})W_0. \end{aligned}$$

Proposition 13.1 (Portfolio quantites are homogenous of degree one in $\mathbf{x}$)**.** *The portfolio quantities* $R_p(\mathbf{x})$*,* $\mu_p(\mathbf{x})$*,* $\sigma_p(\mathbf{x})$*, the normal quantile* $q_\alpha^{R_p}(x)$*, and the normal value-at-risk* $\mathrm{VaR}_{p,\alpha}(\mathbf{x})$ *are homogenous functions of degree one in the portfolio weight vector* $\mathbf{x}$*.*

The result for $R_p(\mathbf{x})$ and $\mu_p(\mathbf{x})$ is trivial since they are linear functions of $\mathbf{x}$. For example, $R_p(c\mathbf{x}) = (c\mathbf{x})'\mathbf{R} = c\,(\mathbf{x}'\mathbf{R}) = cR_p(\mathbf{x})$. The result for $\sigma_p(\mathbf{x})$ is straightforward to show:

$$\begin{aligned} \sigma_p(c\cdot\mathbf{x}) &= ((c\cdot\mathbf{x})'\Sigma(c\cdot\mathbf{x}))^{1/2} \\ &= c\cdot(\mathbf{x}'\Sigma\mathbf{x})^{1/2} \\ &= c\cdot\sigma_p(\mathbf{x}). \end{aligned}$$

The result for $q_\alpha^{R_p}(\mathbf{x})$ follows because it is a linear function of $\mu_p(\mathbf{x})$ and $\sigma_p(\mathbf{x})$. The result for $\mathrm{VaR}_{p,\alpha}(\mathbf{x})$ follows from the linear homogeneity of the normal return quantile.

In the proposition, I stated that $q_\alpha^{R_p}(x)$ is homogenous of degree one in $\mathbf{x}$ when returns are normally distributed. It turns out that the homogeneity of $q_\alpha^{R_p}(x)$ holds generally for continuous distributions and even for the empirical quantile.

13.2.2 Euler's theorem

Euler's theorem gives an additive decomposition of a homogenous function of degree one.

Theorem 13.1 (Euler's theorem)**.** *Let* $f(x_1,\dots,x_N) = f(\mathbf{x})$ *be a continuous, differentiable and homogenous of degree one function of the variables* $\mathbf{x} = (x_1,\dots,x_N)'$*. Then,*

$$\begin{aligned} f(\mathbf{x}) &= x_1\cdot\frac{\partial f(\mathbf{x})}{\partial x_1} + x_2\cdot\frac{\partial f(\mathbf{x})}{\partial x_2} + \cdots + x_N\cdot\frac{\partial f(\mathbf{x})}{\partial x_N} \\ &= \mathbf{x}'\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}, \end{aligned}$$

where,

$$\underset{(N\times 1)}{\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}} = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_N} \end{pmatrix}.$$

Example 13.2 (Verifying Euler's theorem)**.** The function $f(x_1, x_2) = x_1 + x_2 = f(\mathbf{x}) = \mathbf{x}'\mathbf{1}$ is homogenous of degree one, and:

$$\begin{aligned} \frac{\partial f(\mathbf{x})}{\partial x_1} &= \frac{\partial f(\mathbf{x})}{\partial x_2} = 1, \\ \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} &= \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{1}. \end{aligned}$$

By Euler's theorem,

$$\begin{aligned} f(x) &= x_1 \cdot 1 + x_2 \cdot 1 = x_1 + x_2 \\ &= \mathbf{x}'\mathbf{1}. \end{aligned}$$

The function $f(x_1, x_2) = (x_1^2 + x_2^2)^{1/2} = f(\mathbf{x}) = (\mathbf{x}'\mathbf{x})^{1/2}$ is homogenous of degree one, and:

$$\begin{aligned} \frac{\partial f(\mathbf{x})}{\partial x_1} &= \frac{1}{2}\left(x_1^2 + x_2^2\right)^{-1/2} 2x_1 = x_1\left(x_1^2 + x_2^2\right)^{-1/2}, \\ \frac{\partial f(\mathbf{x})}{\partial x_2} &= \frac{1}{2}\left(x_1^2 + x_2^2\right)^{-1/2} 2x_2 = x_2\left(x_1^2 + x_2^2\right)^{-1/2}. \end{aligned}$$

By Euler's theorem,

$$\begin{aligned} f(\mathbf{x}) &= x_1 \cdot x_1\left(x_1^2 + x_1^2\right)^{-1/2} + x_2 \cdot x_2\left(x_1^2 + x_2^2\right)^{-1/2} \\ &= \left(x_1^2 + x_2^2\right)\left(x_1^2 + x_2^2\right)^{-1/2} \\ &= \left(x_1^2 + x_2^2\right)^{1/2}. \end{aligned}$$

Using matrix algebra, I have:

$$\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = \frac{\partial\left(\mathbf{x}'\mathbf{x}\right)^{1/2}}{\partial \mathbf{x}} = \frac{1}{2}\left(\mathbf{x}'\mathbf{x}\right)^{-1/2} 2\mathbf{x} = \left(\mathbf{x}'\mathbf{x}\right)^{-1/2}\mathbf{x} = \mathbf{x}\left(\mathbf{x}'\mathbf{x}\right)^{-1/2}.$$

Then by Euler's theorem:

$$f(\mathbf{x}) = \mathbf{x}'\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{x}'\mathbf{x}\left(\mathbf{x}'\mathbf{x}\right)^{-1/2} = \left(\mathbf{x}'\mathbf{x}\right)^{1/2}.$$

■

13.2.3 Risk decomposition using Euler's theorem

Proposition 13.2 (Additive risk decomposition for homogenous of degree one portfolio risk measures)**.** *Let* $\text{RM}_p(\mathbf{x})$ *denote a portfolio risk measure that is a homogenous function of degree one in the portfolio weight vector* $\mathbf{x}$*. For example,* $\text{RM}_p(\mathbf{x}) = \sigma_p(\mathbf{x})$ *or* $\text{RM}_p(\mathbf{x}) = \text{VaR}_{p,\alpha}(\mathbf{x})$*. Then, Euler's theorem gives the additive risk decomposition*

$$\begin{aligned}
\text{RM}_p(\mathbf{x}) &= x_1 \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_1} + x_2 \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_2} + \cdots + x_N \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_N} \\
&= \sum_{i=1}^{N} x_i \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_i} \\
&= \mathbf{x}' \frac{\partial \text{RM}_p(\mathbf{x})}{\partial \mathbf{x}}.
\end{aligned}$$

The partial derivatives in **?@eq-RiskBudgetRiskDecomposition** are called asset *marginal contributions to risk* (MCRs):

$$\text{MCR}_i^{RM} = \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_i} = \text{ marginal contribution of asset } i \tag{13.1}$$

The asset *contributions to risk* (CRs) are defined as the weighted marginal contributions:

$$\text{CR}_i^{RM} = x_i \cdot \text{MCR}_i^{RM} = \text{ contribution of asset } i \tag{13.2}$$

Then I can re-express the decomposition **?@eq-RiskBudgetRiskDecomposition** as

$$\begin{aligned}
\text{RM}_p(\mathbf{x}) &= x_1 \cdot \text{MCR}_1^{RM} + x_2 \cdot \text{MCR}_2^{RM} + \cdots + x_N \cdot \text{MCR}_N^{RM} \\
&= \text{CR}_1^{RM} + \text{CR}_2^{RM} + \cdots + \text{CR}_N^{RM}
\end{aligned}$$

If I divide both sides of Equation 13.3 by $\text{RM}_p(\mathbf{x})$ I get the asset *percent contributions to risk* (PCRs)

$$\begin{aligned}
1 &= \frac{\text{CR}_1^{RM}}{\text{RM}_p(\mathbf{x})} + \frac{\text{CR}_2^{RM}}{\text{RM}_p(\mathbf{x})} + \cdots + \frac{\text{CR}_N^{RM}}{\text{RM}_p(\mathbf{x})} \\
&= \text{PCR}_1^{RM} + \text{PCR}_2^{RM} + \cdots + \text{PCR}_N^{RM}
\end{aligned}$$

where

$$\text{PCR}_i^{RM} = \frac{\text{CR}_i^{RM}}{\text{RM}_p(\mathbf{x})} = \text{ percent contribution of asset } i \tag{13.3}$$

By construction the asset PCRs sum to one.

13.2.3.1 Risk decomposition using $\sigma_p(\mathbf{x})$

Let $\text{RM}_p(\mathbf{x}) = \sigma_p(\mathbf{x}) = (\mathbf{x}'\Sigma\mathbf{x})^{1/2}$. Because $\sigma_p(\mathbf{x})$ is homogenous of degree 1 in $\mathbf{x}$, by Euler's theorem

$$\begin{aligned}\sigma_p(\mathbf{x}) &= x_1\frac{\partial\sigma_p(\mathbf{x})}{\partial x_1} + x_2\frac{\partial\sigma_p(\mathbf{x})}{\partial x_2} + \cdots + x_n\frac{\partial\sigma_p(\mathbf{x})}{\partial x_n} \\ &= \mathbf{x}'\frac{\partial\sigma_p(\mathbf{x})}{\partial\mathbf{x}}.\end{aligned} \tag{13.4}$$

Now,

$$\begin{aligned}\frac{\partial\sigma_p(\mathbf{x})}{\partial\mathbf{x}} &= \frac{\partial(\mathbf{x}'\Sigma\mathbf{x})^{1/2}}{\partial\mathbf{x}} = \frac{1}{2}(\mathbf{x}'\Sigma\mathbf{x})^{-1/2}2\Sigma\mathbf{x} \\ &= \frac{\Sigma\mathbf{x}}{(\mathbf{x}'\Sigma\mathbf{x})^{1/2}} = \frac{\Sigma\mathbf{x}}{\sigma_p(\mathbf{x})}.\end{aligned}$$

Then,

$$\frac{\partial\sigma_p(\mathbf{x})}{\partial x_i} = \text{MCR}_i^\sigma = \text{i-th row of } \frac{\Sigma\mathbf{x}}{\sigma_p(\mathbf{x})} = \frac{(\Sigma\mathbf{x})_i}{\sigma_p(\mathbf{x})}, \tag{13.5}$$

and

$$\text{CR}_i^\sigma = x_i \times \frac{(\Sigma\mathbf{x})_i}{\sigma_p(\mathbf{x})}, \tag{13.6}$$

$$\text{PCR}_i^\sigma = x_i \times \frac{(\Sigma\mathbf{x})_i}{\sigma_p^2(\mathbf{x})}. \tag{13.7}$$

Notice that:

$$\sigma_p(\mathbf{x}) = \mathbf{x}'\frac{\partial\sigma_p(\mathbf{x})}{\partial\mathbf{x}} = \mathbf{x}'\frac{\Sigma\mathbf{x}}{\sigma_p(\mathbf{x})} = \frac{\sigma_p^2(\mathbf{x})}{\sigma_p(\mathbf{x})} = \sigma_p(\mathbf{x}).$$

Example 13.3 (Two asset example)**.** For a two asset portfolio you have:

$$\begin{aligned}\sigma_p(\mathbf{x}) = (\mathbf{x}'\Sigma\mathbf{x})^{1/2} &= \left(x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\sigma_{12}\right)^{1/2}, \\ \Sigma\mathbf{x} &= \begin{pmatrix}\sigma_1^2 & \sigma_{12}\\ \sigma_{12} & \sigma_2^2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} = \begin{pmatrix}x_1\sigma_1^2 + x_2\sigma_{12}\\ x_2\sigma_2^2 + x_1\sigma_{12}\end{pmatrix}, \\ \frac{\Sigma\mathbf{x}}{\sigma_p(\mathbf{x})} &= \begin{pmatrix}\left(x_1\sigma_1^2 + x_2\sigma_{12}\right)/\sigma_p(\mathbf{x})\\ \left(x_2\sigma_2^2 + x_1\sigma_{12}\right)/\sigma_p(\mathbf{x})\end{pmatrix},\end{aligned}$$

so that

$$\begin{aligned}\text{MCR}_1^\sigma &= \left(x_1\sigma_1^2 + x_2\sigma_{12}\right)/\sigma_p(\mathbf{x}), \\ \text{MCR}_2^\sigma &= \left(x_2\sigma_2^2 + x_1\sigma_{12}\right)/\sigma_p(\mathbf{x}).\end{aligned}$$

Then

$$\begin{aligned}
\mathrm{MCR}_1^{\sigma} &= \left(x_1\sigma_1^2 + x_2\sigma_{12}\right)/\sigma_p(\mathbf{x}),\\
\mathrm{MCR}_2^{\sigma} &= \left(x_2\sigma_2^2 + x_1\sigma_{12}\right)/\sigma_p(\mathbf{x}),\\
\mathrm{CR}_1^{\sigma} &= x_1 \times \left(x_1\sigma_1^2 + x_2\sigma_{12}\right)/\sigma_p(\mathbf{x}) = \left(x_1^2\sigma_1^2 + x_1x_2\sigma_{12}\right)/\sigma_p(\mathbf{x}),\\
\mathrm{CR}_2^{\sigma} &= x_2 \times \left(x_2\sigma_2^2 + x_2\sigma_2\right)/\sigma_p(\mathbf{x}) = \left(x_2^2\sigma_2^2 + x_1x_2\sigma_{12}\right)/\sigma_p(\mathbf{x}),
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{PCR}_1^{\sigma} &= \mathrm{CR}_1^{\sigma}/\sigma_p(\mathbf{x}) = \left(x_1^2\sigma_1^2 + x_1x_2\sigma_{12}\right)/\sigma_p^2(\mathbf{x}),\\
\mathrm{PCR}_2^{\sigma} &= \mathrm{CR}_2^{\sigma}/\sigma_p(\mathbf{x}) = \left(x_2^2\sigma_2^2 + x_1x_2\sigma_{12}\right)/\sigma_p^2(\mathbf{x}).
\end{aligned}$$

Notice that risk decomposition from Euler's theorem above is the same decomposition we motived at the beginning of the chapter. ■

13.2.3.2 Risk decomposition using $\mathrm{VaR}_{p,\alpha}(\mathbf{x})$

Let $\mathrm{RM}(\mathbf{x}) = \mathrm{VaR}_{p,\alpha}(\mathbf{x})$. Because $\mathrm{VaR}_{p,\alpha}(\mathbf{x})$ is homogenous of degree 1 in $\mathbf{x}$, by Euler's theorem

$$\begin{aligned}
\mathrm{VaR}_{p,\alpha}(\mathbf{x}) &= x_1\frac{\partial \mathrm{VaR}_{p,\alpha}(\mathbf{x})}{\partial x_1} + x_2\frac{\partial \mathrm{VaR}_{p,\alpha}(\mathbf{x})}{\partial x_2} + \cdots + x_n\frac{\partial \mathrm{VaR}_{p,\alpha}(\mathbf{x})}{\partial x_n}\\
&= \mathbf{x}'\frac{\partial \mathrm{VaR}_{p,\alpha}(\mathbf{x})}{\partial \mathbf{x}}.
\end{aligned} \tag{13.8}$$

Now,

$$\begin{aligned}
\mathrm{VaR}_{\alpha}(\mathbf{x}) &= W_0 \times q_{\alpha}^{R_p}(\mathbf{x}) = W_0 \times \left(\mu_p(\mathbf{x}) + \sigma_p(\mathbf{x}) \times q_{\alpha}^{Z}\right)\\
&= W_0 \times \left(\mathbf{x}'\mu + \left(\mathbf{x}'\Sigma\mathbf{x}\right)^{1/2} \times q_{\alpha}^{Z}\right),\\
\frac{\partial \mathrm{VaR}_{\alpha}(\mathbf{x})}{\partial \mathbf{x}} &= W_0 \times \frac{\partial}{\partial \mathbf{x}}\left(\mathbf{x}'\mu + \left(\mathbf{x}'\Sigma\mathbf{x}\right)^{1/2} \times q_{\alpha}^{Z}\right)\\
&= W_0 \times \left(\mu + \frac{\Sigma\mathbf{x}}{\sigma_p(\mathbf{x})} \times q_{\alpha}^{Z}\right).
\end{aligned}$$

Then

$$\mathrm{MCR}_i^{\mathrm{VaR}} = W_0 \times \left(\mu_i + \frac{(\Sigma\mathbf{x})_i}{\sigma_p(\mathbf{x})} \times q_{\alpha}^{Z}\right), \tag{13.9}$$

$$\mathrm{CR}_i^{\mathrm{VaR}} = x_i \times W_0 \times \left(\mu_i + \frac{(\Sigma\mathbf{x})_i}{\sigma_p(\mathbf{x})} \times q_{\alpha}^{Z}\right), \tag{13.10}$$

$$\mathrm{PCR}_i^{\mathrm{VaR}} = x_i \times W_0 \times \left(\mu_i + \frac{(\Sigma\mathbf{x})_i}{\sigma_p(\mathbf{x})} \times q_{\alpha}^{Z}\right) / \left(W_0 \times \left(\mu_p(\mathbf{x}) + \sigma_p(\mathbf{x}) \times q_{\alpha}^{Z}\right)\right). \tag{13.11}$$

It is often common practice to set $\mu = \mathbf{0}$ when computing $\text{VaR}_{p,\alpha}(\mathbf{x})$ especially for short time horizons such as a day. In this case,

$$\begin{aligned}
\text{MCR}_i^{\text{VaR}} &= W_0 \times \left(\frac{(\Sigma \mathbf{x})_i}{\sigma_p(\mathbf{x})} \times q_\alpha^Z \right) \propto \text{MCR}_i^\sigma, \\
\text{CR}_i^{\text{VaR}} &= x_i \times W_0 \times \left(\frac{(\Sigma \mathbf{x})_i}{\sigma_p(\mathbf{x})} \times q_\alpha^Z \right) \propto \text{CR}_i^\sigma, \\
\text{PCR}_i^{\text{VaR}} &= x_i \times \frac{(\Sigma \mathbf{x})_i}{\sigma_p^2(\mathbf{x})} = \text{PCR}_i^\sigma,
\end{aligned}$$

and we see that the portfolio normal VaR risk decompositions above give the same information as the portfolio volatility risk decompositions Equation 13.5 - Equation 13.7.

13.2.4 Interpreting marginal contributions to risk

The risk decomposition **?@eq-RiskBudgetRiskDecomposition** shows that any risk measure that is homogenous of degree one in the portfolio weights can be additively decomposed into a portfolio weighted average of marginal contributions to risk. The marginal contributions to risk Equation 13.1 are the partial derivatives of the risk meausre $\text{RM}_p(\mathbf{x})$ with respect to the portfolio weights, and so one may think that they can be interpreted as the change in the risk measure associated with a one unit change in a portfolio weight holding the other portfolio weights fixed:

$$\text{MCR}_i^\sigma = \frac{\partial \sigma_p(\mathbf{x})}{\partial x_i} \approx \frac{\Delta \sigma_p}{\Delta x_i} \Rightarrow \Delta \sigma_p \approx \text{MCR}_i^\sigma \cdot \Delta x_i.$$

If the portfolio weights were unconstrained this would be the correct interpretation. However, the portfolio weights are constrained to sum to one, $\sum_{i=1}^N x_i = 1$, so the increase in one weight implies an offsetting decrease in the other weights. Hence, the formula $\Delta \sigma_p \approx \text{MCR}_i^\sigma \cdot \Delta x_i$ ignores this re-allocation effect.

To properly interpret the marginal contributions to risk, consider the total derivative of $\text{RM}_p(\mathbf{x})$:

$$\begin{aligned}
d\text{RM}_p(\mathbf{x}) &= \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_1} dx_1 + \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_2} dx_2 + \cdots + \frac{\partial \text{RM}_p(\mathbf{x})}{\partial x_N} dx_N \\
&= \text{MCR}_1^{RM} dx_1 + \text{MCR}_2^{RM} dx_2 + \cdots + \text{MCR}_N^{RM} dx_N.
\end{aligned}$$

First, consider a small change in x_i offset by an equal change in x_j, $\Delta x_i = -\Delta x_j$. That is, the small increase in allocation to asset i is matched by a corresponding decrease in allocation to asset j. From Equation 13.12, the approximate change in the portfolio risk measure is

$$\Delta \text{RM}_p(\mathbf{x}) \approx \left(\text{MCR}_i^{RM} - \text{MCR}_j^{RM} \right) \Delta x_i. \tag{13.12}$$

When rebalancing a portfolio, the reallocation does not have to be limited to two assets. Suppose, for example, the reallocation is spread across all of the other assets $j \neq i$ so that

$$\Delta x_j = -\alpha_j \Delta x_i \text{ s.t. } \sum_{j \neq i} \alpha_j = 1.$$

Then

$$\sum_{j \neq i} \Delta x_j = -\sum_{j \neq i} \alpha_j \Delta x = -\Delta x_i \sum_{j \neq i} \alpha_j = -\Delta x_i,$$

and

$$\begin{aligned}\Delta \text{RM}(\mathbf{x}) &\approx \text{MCR}_i^{RM} \cdot \Delta x_i + \sum_{j \neq i} \text{MCR}_j^{RM} \cdot \Delta x_j \\ &= \left(\text{MCR}_i^{RM} - \sum_{j \neq i} \alpha_j \cdot \text{MCR}_j^{RM} \right) \Delta x_i.\end{aligned}$$

In matrix notation the result Equation 13.13 be written as

$$\Delta \text{RM}(\mathbf{w}) \approx \left((\mathbf{MCR}^{RM})' \alpha \right) \Delta x_i, \tag{13.13}$$

where

$$\begin{aligned}\mathbf{MCR}^{RM} &= (\text{MCR}_1^{RM}, \dots, \text{MCR}_n^{RM})', \\ \alpha &= (-\alpha_1, \dots, -\alpha_{i-1}, 1, -\alpha_{i+1}, \dots - \alpha_n)'.\end{aligned}$$

13.3 Portfolio risk reports

A *portfolio risk report* summarizes asset and portfolio risk measures as well as risk budgets. Table Table 13.1 shows a typical portfolio risk report. The individual asset information is in rows with the portfolio information at the bottom. The total dollar amount invested in the portfolio is W_0 and the dollar amounts invested in each asset are $d_i = x_i W_0$. The asset specific (standalone) risk measures are RM_i and the portfolio risk measure is $\text{RM}(\mathbf{x})$.

Table 13.1: Portfolio Risk Report

Asset	$\$\text{d}_i$	x_i	RM_i	MCR_i^{RM}	CR_i^{RM}	PCR_i^{RM}
Asset 1	$\$\text{d}_1$	x_1	RM_1	MCR_1^{RM}	CR_1^{RM}	PCR_1^{RM}
Asset 2	$\$\text{d}_2$	x_2	RM_2	MCR_2^{RM}	CR_2^{RM}	PCR_2^{RM}
⋮	⋮	⋮	⋮	⋮	⋮	⋮
Asset N	$\$\text{d}_N$	x_N	RM_N	MCR_N^{RM}	CR_N^{RM}	PCR_N^{RM}
Portfolio (Sum)	$\$W_0$	1			RM(x)	1

Example 13.4 (Portfolio volatility risk report)**.** Consider creating a portfolio volatility risk report from an equally weighted portfolio of Microsoft, Nordstrom, and Starbucks stock. The initial wealth invested in the portfolio is $100,000. The expected return vector and covariance matrix is based on sample statistics computed over the five-year period January, 1995 through January, 2000. The asset and portfolio expected return and volatility information is:

```
# asset information
asset.names <- c("MSFT", "NORD", "SBUX")
mu.vec = c(0.0427, 0.0015, 0.0285)
sigma.mat = matrix(c(0.0100, 0.0018, 0.0011,
                     0.0018, 0.0109, 0.0026,
                     0.0011, 0.0026, 0.0199),
                   nrow=3, ncol=3)
sig.vec = sqrt(diag(sigma.mat))
names(mu.vec) = names(sig.vec) = asset.names
dimnames(sigma.mat) = list(asset.names, asset.names)
# equally weighted portfolio information
W0 = 100000
x = rep(1/3, 3)
d = x*W0
names(x) = asset.names
mu.px = as.numeric(crossprod(x, mu.vec))
sig.px = as.numeric(sqrt(t(x)%*%sigma.mat%*%x))
```

The volatility risk budgeting calculations are:

```
MCR.vol.x = (sigma.mat%*%x)/sig.px
CR.vol.x = x*MCR.vol.x
PCR.vol.x = CR.vol.x/sig.px
```

The volatility risk report is computed using:

```
riskReportVol.px = cbind(d, x, sig.vec, MCR.vol.x,
                         CR.vol.x, PCR.vol.x)
PORT = c(W0, 1, NA, NA, sum(CR.vol.x), sum(PCR.vol.x))
riskReportVol.px = rbind(riskReportVol.px, PORT)
colnames(riskReportVol.px) = c("Dollar", "Weight", "Vol",
                               "MCR", "CR", "PCR")
riskReportVol.px
#>      Dollar Weight    Vol    MCR     CR   PCR
```

```
#> MSFT  33333  0.333 0.100 0.0567 0.0189 0.249
#> NORD  33333  0.333 0.104 0.0672 0.0224 0.295
#> SBUX  33333  0.333 0.141 0.1037 0.0346 0.456
#> PORT 100000  1.000    NA     NA 0.0759 1.000
```

In the equally weighted portfolio, the risk contributions are not equal across assets. Predictably, the ranking of the risk contributions follows the ranking of the individual asset volatilities σ_i, with Starbucks giving the highest contributions to portfolio volatility. So if the portfolio manager wants to reduce portfolio volatility the allocation to Starbucks should be reduced first.

To interpret the marginal contributions to risk, suppose that the portfolio manager wants to reduce portfolio volatility and chooses the rebalancing strategy: $\Delta x_{MSFT} = -\Delta x_{SBUX} = 0.1$ (i.e., x_{MSFT} increases to 0.433 and x_{SBUX} decreases to 0.233). Then, from Equation 13.12 the predicted change in portfolio standard deviation is:

```
delta.vol.px = (MCR.vol.x["MSFT",] - MCR.vol.x["SBUX",])*0.1
delta.vol.px
#>    MSFT
#> -0.0047
```

Hence, the predicted volatility after rebalancing is:

```
sig.px + delta.vol.px
#>   MSFT
#> 0.0712
```

The exact change in volatility from rebalancing is:

```
x1 = x + c(0.1, 0, -0.1)
sig.px1 = as.numeric(sqrt(t(x1)%*%sigma.mat%*%x1))
sig.px1
#> [1] 0.0729
```

The difference between the approximate change in portfolio volatility computed from the risk report and the actual change is:

```
sig.px1 - sig.px
#> [1] -0.00293
```

■

Example 13.5 (Portfolio normal VaR risk report)**.** Using the asset and portfolio data from the

previous example, consider creating a normal portfolio VaR report, where VaR is computed with 5% probability. The R calculations are:

```
# portfolio 5% normal VaR
alpha = 0.05
# risk budget calculations
VaR.px = abs(W0*(mu.px + sig.px*qnorm(alpha)))
MCR.VaR.x = abs(W0*(mu.vec + MCR.vol.x*qnorm(alpha)))
CR.VaR.x = x*MCR.VaR.x
PCR.VaR.x = CR.VaR.x/VaR.px
# risk report
VaR.vec = abs(W0*(mu.vec + sig.vec*qnorm(alpha)))
riskReportVaR.px = cbind(d, x, VaR.vec, MCR.VaR.x,
                         CR.VaR.x, PCR.VaR.x)
PORT = c(W0, 1, NA, NA, sum(CR.VaR.x), sum(PCR.VaR.x))
riskReportVaR.px = rbind(riskReportVaR.px, PORT)
colnames(riskReportVaR.px) = c("Dollar", "Weight", "VaR",
                               "MCR", "CR", "PCR")
riskReportVaR.px
#>      Dollar Weight   VaR   MCR    CR   PCR
#> MSFT  33333  0.333 12179  5053  1684 0.168
#> NORD  33333  0.333 17023 10907  3636 0.362
#> SBUX  33333  0.333 20354 14206  4735 0.471
#> PORT 100000  1.000    NA    NA 10055 1.000
```

The qualitative information in the portfolio normal VaR report is essentially the same as in the portfolio volatility report. Because the VaR calculation involve the asset expected returns, μ, the results are not exactly the same. If we set $\mu = \mathbf{0}$ in the VaR calculations then the two reports would give same risk information but on different scales (volatility is in return and VaR is in dollars).

To interpret the marginal contributions to risk, suppose that the portfolio manager wants to reduce portfolio VaR and chooses the rebalancing strategy: $\Delta x_{MSFT} = -\Delta x_{SBUX} = 0.1$ (i.e., x_{MSFT} increases to 0.433 and x_{SBUX} decreases to 0.233). Then, from Equation 13.12 the predicted change in portfolio VaR and new portfolio VaR are:

```
delta.VaR.px = (MCR.VaR.x["MSFT",] - MCR.VaR.x["SBUX",])*0.1
delta.VaR.px
#> MSFT
#> -915
VaR.px + delta.VaR.px
```

```
#> MSFT
#> 9140
```

The exact change in portfolio VaR from rebalancing is:

```
mu.px1 = as.numeric(crossprod(x1, mu.vec))
VaR.px1 = abs(W0*(mu.px1 + sig.px1*qnorm(alpha)))
VaR.px1 - VaR.px
#> [1] -624
VaR.px1
#> [1] 9431
```

Here, the predicted reduction in VaR is about \$300 more than the actual change in VaR. ■

13.4 Understanding Portfolio Volatility Risk Decompositions

Portfolio volatility risk reports described in the previous sub-section are commonly used in practice. As such, it is important to have a good understanding of the volatility risk decompositions and, in particular, the asset contributions to portfolio volatility. In the portfolio volatility risk decomposition Equation 13.4, the marginal contribution from asset i given by:

$$\text{MCR}_i^\sigma = \text{i-th row of } \frac{\Sigma \mathbf{x}}{\sigma_p(\mathbf{x})} = \frac{(\Sigma \mathbf{x})_i}{\sigma_p(\mathbf{x})}. \tag{13.14}$$

The interpretation of this formula, however, is not particularly intuitive. With a little bit of algebra, we can derive two alternative representations of Equation 13.14 that are easier to interpret and give better intuition about the asset contributions to portfolio volatility.

13.4.1 $\sigma - \rho$ decomposition for MCR_i^σ

Proposition 13.3 ($\sigma - \rho$ decomposition)**.** *The $\sigma - \rho$ decomposition for MCR_i^σ is*

$$\text{MCR}_i^\sigma = \sigma_i \rho_{i,p}(\mathbf{x}), \tag{13.15}$$

where $\rho_{i,p}(\mathbf{x}) = \text{corr}(R, R_p(\mathbf{x}))$ is the correlation between the return on asset i and the return on the portfolio.

Remarks:

1. Equation Equation 13.15 shows that an asset's marginal contribution to portfolio volatility depends on two components: (1) the asset's return volatility, σ_i (sometimes called standalone volatility); (2) the asset's correlation with the portfolio return, $\rho_{i,p}(\mathbf{x})$.
2. For a given standalone volatility, σ_i, the sign and magnitude of MCR_i^σ depends on $\rho_{i,p}(\mathbf{x})$. Because $-1 \leq \rho_{i,p}(\mathbf{x}) \leq 1$, it follows that $\text{MCR}_i^\sigma \leq \sigma_i$ and $\text{MCR}_i^\sigma = \sigma_i$ only if $\rho_{i,p}(\mathbf{x}) = 1$.
3. If an asset's return is uncorrelated with all of the returns in the portfolio then $\rho_{i,p}(\mathbf{x}) = (x_i\sigma_i)/\sigma_p$ and $\text{MCR}_i^\sigma = (x_i\sigma_i^2)/\sigma_p$.

Proposition 13.4 ($x - \sigma - \rho$ decomposition)**.** *Using Equation 13.15 the $x - \sigma - \rho$ decomposition for $\text{CR}_i^\sigma = x_i \times \text{MCR}_i^\sigma$ is:*

$$\begin{aligned} \text{CR}_i^\sigma &= x_i \times \sigma_i \times \rho_{i,p}(\mathbf{x}) \\ &= \textit{asset allocation} \times \textit{standalone volatility} \times \textit{correlation with portfolio} \end{aligned} \tag{13.16}$$

Remarks:

1. Equation Equation 13.16 shows that an asset's contribution to portfolio volatility depends on three components: (1) the asset's allocation weight x_i; (2) the asset's standalone return volatility, σ_i; (3) the asset's correlation with the portfolio return, $\rho_{i,p}(\mathbf{x})$.
2. $x_i \times \sigma_i$ = standalone contribution to portfolio volatility, which ignores correlation effects with other assets. In the typical situation, $\rho_{i,p} \neq 1$ which implies that $\text{CR}_i^\sigma < w_i \times \sigma_i$. Hence, an asset's contribution to portfolio volatility will almost always be less than its standalone contribution to portfolio volatility.
3. $\text{CR}_i^\sigma = x_i \times \sigma_i$ only when $\rho_{i,p}(\mathbf{x}) = 1$. That is, an asset's contribution to portfolio volatility is equal to its standalone contribution only when its return is perfectly correlated with the portfolio return.
4. If all assets are perfectly correlated (i.e., $\rho_{ij} = 1$ for all i and j) then all asset contributions to portfolio volatility are equal their standalone contributions, $x_i \times \sigma_i$, and $\sigma_p(\mathbf{x}) = x_1\sigma_1 + x_2\sigma_2 + \cdots + x_N\sigma_N$. This is the risk decomposition that would occur if there are no diversification effects.

The derivation of Equation 13.15 is straightforward. Recall, the vector of asset marginal contributions to portfolio volatility is given by

$$\frac{\partial \sigma_p(\mathbf{x})}{\partial \mathbf{x}} = \frac{\Sigma\mathbf{x}}{\sigma_p(\mathbf{x})}$$

Now,

$$\Sigma\mathbf{x} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1n} & \sigma_{n2} & \cdots & \sigma_n^2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Without loss of generality, consider the first row of $\Sigma\mathbf{x}$

$$x_1\sigma_1^2 + x_2\sigma_{12} + \cdots + x_n\sigma_{1n}$$

This expression is the covariance between the return on asset 1, R_1, and the return on the portfolio, $R_p(\mathbf{x})$:

$$\begin{aligned}
\text{cov}(R_1, R_p(\mathbf{x})) &= \text{cov}(R_1, x_1R_1 + x_2R_2 + \cdots + x_nR_n) \\
&= \text{cov}(R_1, x_1R_1) + \text{cov}(R_1, x_1R_1) + \cdots + \text{cov}(R_1, x_nR_n) \\
&= x_1\sigma_1^2 + x_2\sigma_{12} + \cdots + x_n\sigma_{1n} \\
&= \sigma_{1,p}(\mathbf{x})
\end{aligned}$$

Then we can re-write Equation 13.14 for $i = 1$ as

$$\text{MCR}_1^\sigma = \frac{(\Sigma\mathbf{x})_1}{\sigma_p(\mathbf{x})} = \frac{\sigma_{1,p}(\mathbf{x})}{\sigma_p(\mathbf{x})} \tag{13.17}$$

Define the correlation between the return on asset 1 and the return on the portfoio as

$$\rho_{1,p}(\mathbf{x}) = \text{corr}(R_1, R_p(\mathbf{x})) = \frac{\text{cov}(R_1, R_p(\mathbf{x}))}{\sigma_1\sigma_p(\mathbf{x})} = \frac{\sigma_{1,p}(\mathbf{x})}{\sigma_1\sigma_p(\mathbf{x})} \tag{13.18}$$

Then we can write

$$\sigma_{1,p}(\mathbf{x}) = \sigma_1\sigma_p(\mathbf{x})\rho_{1,p}(\mathbf{x}) \tag{13.19}$$

Substituting Equation 13.19 into Equation 13.17 gives the $\sigma - \rho$ decomposition for MCR_1^σ

$$\text{MCR}_1^\sigma = \frac{\sigma_1\sigma_p(\mathbf{x})\rho_{1,p}(\mathbf{x})}{\sigma_p(\mathbf{x})} = \sigma_1\rho_{1,p}(\mathbf{x})$$

Example 13.6 (Adding $\rho_{i,p}(\mathbf{x})$ to the portfolio volatility risk report)**.** Once you have the volatility risk report, you can easily calculate $\rho_{i,p}(\mathbf{x})$ from Equation 13.15 using

$$\rho_{i,p}(\mathbf{x}) = \frac{\text{MCR}_i^\sigma}{\sigma_i}$$

For example, to calculate $\rho_{i,p}(\mathbf{x})$ for the volatility risk report for the equally weighted portfolio of Microsoft, Nordstrom, and Starbucks use:

```
rho.x = MCR.vol.x/sig.vec
rho.x
#>        [,1]
#> MSFT 0.567
```

```
#> NORD 0.644
#> SBUX 0.735
```

Here we see that all assets are positively correlated with the portfolio, Microsoft has the smallest correlation (0.567), and Starbucks has the largest correlation (0.735). In this respect, Microsoft is most beneficial in terms of diversification and Starbucks is least beneficial. If all correlations were equal to one, then each asset MCR would be equal to its standalone volatility.

The expanded portfolio risk report is:

```
#>       Dollar Weight   Vol   Rho    MCR     CR   PCR
#> MSFT   33333  0.333 0.100 0.567 0.0567 0.0189 0.249
#> NORD   33333  0.333 0.104 0.644 0.0672 0.0224 0.295
#> SBUX   33333  0.333 0.141 0.735 0.1037 0.0346 0.456
#> PORT 100000  1.000    NA    NA     NA 0.0759 1.000
```

■

13.4.2 $\sigma - \beta$ decomposition for MCR_i^σ

Proposition 13.5 ($\sigma - \beta$ decomposition)**.** *The $\sigma - \beta$ decomposition for MCR_i^σ is*

$$\text{MCR}_i^\sigma = \beta_{i,p}(\mathbf{x})\sigma_p(\mathbf{x}) \tag{13.20}$$

where

$$\beta_{i,p}(\mathbf{x}) = \frac{\text{cov}(R_i, R_p(\mathbf{x}))}{\text{var}(R_p(\mathbf{x}))} = \frac{\text{cov}(R_i, R_p(\mathbf{x}))}{\sigma_p^2(\mathbf{x})} \tag{13.21}$$

Here, $\beta_{i,p}(\mathbf{x})$ is called asset i's *beta to the portfolio.*[2] This decomposition follows directly from the $\sigma - \rho$ decomposition Equation 13.15 since

$$\beta_{i,p}(\mathbf{x}) = \frac{\text{cov}(R_i, R_p(\mathbf{x}))}{\sigma_p^2(\mathbf{x})} = \frac{\rho_{i,p}(\mathbf{x})\sigma_i\sigma_p(\mathbf{x})}{\sigma_p^2(\mathbf{x})} \Rightarrow \rho_{i,p}(x) = \frac{\beta_{i,p}(\mathbf{x})\sigma_p(\mathbf{x})}{\sigma_i}.$$

It follows that:

$$\text{CR}_i^\sigma = x_i \times \beta_{i,p}(\mathbf{x}) \times \sigma_p(\mathbf{x}) \tag{13.22}$$

$$\text{PCR}_i^\sigma = x_i \times \beta_{i,p}(\mathbf{x}) \tag{13.23}$$

Remarks:

[2] It can be thought of as the population regression coefficient obtained by regressing the asset return, R_i,on the portfolio return, $R_p(\mathbf{x})$.

- By construction, the beta of the portfolio is 1

$$\beta_{p,p}(\mathbf{x}) = \frac{\text{cov}(R_p(\mathbf{x}), R_p(\mathbf{x}))}{\text{var}(R_p(\mathbf{x}))} = \frac{\text{var}(R_p(\mathbf{x}))}{\text{var}(R_p(\mathbf{x}))} = 1.$$

- When $\beta_{i,p}(\mathbf{x}) = 1$, $\text{MCR}_i^\sigma = \sigma_p(\mathbf{x})$, $\text{CR}_i^\sigma = x_i\sigma_p(\mathbf{x})$, and $\text{PCR}_i^\sigma = x_i$. In this case, an asset's marginal contribution to portfolio volatility is portfolio volatility and its percent contribution to portfolio volatility is its allocation weight. Intuitively, when $\beta_{i,p}(x) = 1$ the asset has the same risk, in terms of volatility contribution, as the portfolio.
- When $\beta_{i,p}(\mathbf{x}) > 1$, $\text{MCR}_i^\sigma > \sigma_p(\mathbf{x})$, $\text{CR}_i^\sigma > w_i\sigma_p(\mathbf{x})$, and $\text{PCR}_i^\sigma > x_i$. In this case, the asset's marginal contribution to portfolio volatility is more than portfolio volatility and its percent contribution to portfolio volatility is more than its allocation weight. Intuitively, when $\beta_{i,p}(\mathbf{x}) > 1$ the asset has more risk, in terms of volatility contribution, than the portfolio. That is, having the asset in the portfolio increases the portfolio volatility.
- When $\beta_{i,p}(\mathbf{x}) < 1$, $\text{MCR}_i^\sigma < \sigma_p(\mathbf{x})$, $\text{CR}_i^\sigma < x_i\sigma_p(\mathbf{x})$, and $\text{PCR}_i^\sigma < x_i$. In this case, the asset's marginal contribution to portfolio volatility is less than portfolio volatility and its percent contribution to portfolio volatility is less than its allocation weight. Intuitively, when $\beta_{i,p}(\mathbf{x}) < 1$ the asset has less risk, in terms of volatility contribution, than the portfolio. That is, having the asset in the portfolio decreases the portfolio volatility.

Example 13.7 (Adding $\beta_{i,p}(\mathbf{x})$ to the portfolio volatility risk report)**.** Once you have the volatility risk report, you can easily calculate $\beta_{i,p}(\mathbf{x})$ from Equation 13.20 using

$$\beta_{i,p}(\mathbf{x}) = \frac{\text{MCR}_i^\sigma}{\sigma_p(\mathbf{x})}$$

For example, to calculate $\beta_{i,p}(\mathbf{x})$ for the volatility risk report for the equally weighted portfolio of Microsoft, Nordstrom, and Starbucks use:

```
beta.x = MCR.vol.x/sig.px
beta.x
#>       [,1]
#> MSFT 0.747
#> NORD 0.886
#> SBUX 1.367
```

Here, both $\beta_{MSFT,p}(\mathbf{x})$ and $\beta_{NORD,p}(\mathbf{x})$ are less than one whereas $\beta_{SBUX,p}(\mathbf{x})$ is greater than one. We can say that Microsoft and Nordstrom are portfolio volatility reducers whereas Starbucks is a portfolio volatility enhancer.

The expanded risk report is:

```
#>      Dollar Weight   Vol   Rho  Beta    MCR     CR   PCR
```

```
#> MSFT   33333  0.333 0.100 0.567 0.747 0.0567 0.0189 0.249
#> NORD   33333  0.333 0.104 0.644 0.886 0.0672 0.0224 0.295
#> SBUX   33333  0.333 0.141 0.735 1.367 0.1037 0.0346 0.456
#> PORT 100000  1.000    NA    NA    NA     NA 0.0759 1.000
```

■

13.5 Risk Budgeting for Optimized Portfolios

To be completed

- For global minimum variance portfolio, FOCs show that $MCR_i = MCR_j$. Also plugging in solution for m shows that vector of MCR is the same.
- For an efficient portfolio with target return equal to the mean return on an asset in the portfolio, the portfolio beta is 1: $\beta_{i,p} = 1$ when $\mu_p = \mu_i$. Hence, $CR_i = x_i\sigma_p$
- For the tangency portfolio, FOCs (from re-written equivalent problem) show that

$$\frac{\mu_T - r_f}{\sigma_T^2} = \frac{\mu_i - r_f}{\sigma_{i,T}}$$

Volatility risk budgets for optimized portfolios have some interesting properties. For example, consider the global minimum variance portfolio allowing for short sales. In chapter Chapter 11 we showed that

$$\mathbf{m} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}.$$

Plugging this portfolio weight vector into Equation 13.5 gives

$$\frac{\partial\sigma_p(\mathbf{m})}{\partial\mathbf{m}} = \frac{\Sigma\mathbf{m}}{\sigma_p(\mathbf{m})} = \frac{\Sigma\Sigma^{-1}\mathbf{1}}{\sigma_p(\mathbf{m}) \times \mathbf{1}'\Sigma^{-1}\mathbf{1}} = \left(\frac{1}{\sigma_p(\mathbf{m}) \times \mathbf{1}'\Sigma^{-1}\mathbf{1}}\right) \times \mathbf{1} = c \times \mathbf{1}$$

which is a constant c times the one vector where

$$c = \frac{1}{\sigma_p(\mathbf{m}) \times \mathbf{1}'\Sigma^{-1}\mathbf{1}}.$$

As a result, for each asset $i = 1, \dots, n$ we have

$$\text{MCR}_i^\sigma = c.$$

That is, each asset in the global minimum variance portfolio has the same marginal contribution to portfolio volatility. This has to be the case, otherwise we could rebalance the portfolio and lower the volatility. To see this, suppose $\text{MCR}_i^\sigma > c$ for some asset i. Then we could lower portfolio volatility by reducing the allocation to asset i and increasing the allocation to any other asset in

the portfolio. Similarly, if $\text{MCR}_i^\sigma < c$ we could lower portfolio volatility by increasing the allocation to asset i and decreasing the allocation to any other asset in the portfolio.

Example 13.8 (Volatility risk budgeting for the global minimum variance portfolio)**.** to be completed

■

13.6 Risk Parity Portfolios

To be completed.

- Use stock-bond portfolio example like the WealthFront example: https://support.wealthfront.com/hc/en-us/articles/360000117963-How-does-Risk-Parity-work-
- Simple risk-parity portfolio: first equalize the risk contributions of the two asset classes and then apply leverage to achieve a target volatility
- Inputs required for MV portfolio theory: expected returns, variances and covariances (covariance matrix).
  - Problem: expected returns have a lot of estimation noise.
  - Exception is the global minimum variance portfolio
  - Long-only portfolios may be concentrated in only a few assets (not diverse)
- Risk parity only requires variances and covariances (ignores expected returns)
  - produces a long-only diverse portfolio (Q: what happens if an asset has a negative risk contribution?)
- Equal share weighted portfolio ignores expected returns.
  - Equal share weight does not mean equal risk contribution

13.6.1 Stock and Bond Portfolios

Consider forming portfolios consisting of two assets: (1) bond fund, and (2) stock fund. Let R_B and R_S denote the simple per-period returns on the bond and stock funds, respectively, and assume returns are characterized by the GWN model:

$$R_B \sim N(\mu_B, \sigma_B^2),\ R_S \sim N(\mu_S, \sigma_S^2),$$
$$\text{cov}(R_B, R_S) = \sigma_{BS},\ \text{cor}(R_B, R_S) = \rho_{BS}.$$

Let $\mathbf{R} = (R_B, R_S)'$. The GWN model in matrix form is:

$$\begin{aligned}\mathbf{R} &\sim N(\mu, \Sigma)\\ \mu &= \begin{pmatrix}\mu_B\\ \mu_S\end{pmatrix}, \ \begin{pmatrix}\sigma_B^2 & \sigma_{BS}\\ \sigma_{BS} & \sigma_S^2\end{pmatrix}.\end{aligned}$$

13.6.2 Defining Risk Parity Portfoliofolio

The idea behind risk parity is to use the risk budgeting information for a portfolio to create a long only portfolio with equal risk contributions. To illustrate, consider a two asset stock and bond portfolio with return $R_p = x_B R_B + X_S R_S$. Using Euler's Theorem we can decompose portfolio volatility $\sigma_p = \sqrt{\mathbf{x}'\Sigma\mathbf{x}}$ as:

$$\sigma_p = x_B \frac{\partial \sigma_p}{\partial x_B} + x_S \frac{\partial \sigma_p}{\partial x_B} = \text{CR}_B + \text{CR}_S.$$

In the risk parity portfolio we find $x_B > 0$ and $x_S > 0$ such that $\text{CR}_B = \text{CR}_S$ and $x_B + x_S = 1$. To do this we can utilize the $x - \sigma - \rho$ decomposition of CR_B and CR_S:

$$\begin{aligned}\text{CR}_B &= x_B \times \sigma_B \times \rho_{Bp},\\ \text{CR}_S &= x_S \times \sigma_S \times \rho_{Sp},\\ \rho_{Bp} &= \text{cor}(R_B, R_p) = \frac{\text{cov}(R_B, R_p)}{\sigma_B \sigma_p},\\ \rho_{Sp} &= \text{cor}(R_S, R_p) = \frac{\text{cov}(R_S, R_p)}{\sigma_S \sigma_p}.\end{aligned}$$

Now,

$$\begin{aligned}\text{cov}(R_B, R_p) &= \text{cov}(R_B, x_B R_B + x_S R_S) = \text{cov}(R_B, x_B R_B) + \text{cov}(R_B, x_S R_S)\\ &= x_B \text{cov}(R_B, R_B) + x_S \text{cov}(R_B, R_S)\\ &= x_B \sigma_B^2 + x_S \sigma_{BS},\\ \text{cov}(R_S, R_p) &= x_S \sigma_S^2 + x_B \sigma_{BS},\end{aligned}$$

so that

$$\begin{aligned}\rho_{Bp} &= \frac{x_B \sigma_B^2 + x_S \sigma_{BS}}{\sigma_B \sigma_p},\\ \rho_{Sp} &= \frac{x_S \sigma_S^2 + x_B \sigma_{BS}}{\sigma_S \sigma_p}.\end{aligned}$$

Setting $\text{CR}_B = \text{CR}_S$ then gives:

$$\begin{aligned} x_B \times \sigma_B \frac{x_B\sigma_B^2 + x_S\sigma_{BS}}{\sigma_B\sigma_p} &= x_S \times \sigma_S \frac{x_S\sigma_S^2 + x_B\sigma_{BS}}{\sigma_S\sigma_p} \Rightarrow \\ x_B^2\sigma_B^2 + x_Bx_S\sigma_{BS} &= x_S^2\sigma_S^2 + x_Sx_B\sigma_{BS} \Rightarrow \\ x_B^2\sigma_B^2 &= x_S^2\sigma_S^2 \Rightarrow \\ x_B\sigma_B &= x_S\sigma_S, \end{aligned}$$

where the last line uses the assumption that $x_B > 0$ and $x_S > 0$. Using $x_S = 1 - x_B$ we can solve for x_B:

$$\begin{aligned} x_B\sigma_B = (1 - x_B)\sigma_S = \sigma_S - x_B\sigma_S \Rightarrow \\ x_B(\sigma_B + \sigma_S) = \sigma_S \Rightarrow \\ x_B = \frac{\sigma_S}{\sigma_B + \sigma_S}. \end{aligned}$$

Then

$$x_S = 1 - x_B = \frac{\sigma_B + \sigma_S}{\sigma_B + \sigma_S} - \frac{\sigma_S}{\sigma_B + \sigma_S} = \frac{\sigma_B}{\sigma_B + \sigma_S}.$$

Remarks:

1. In the RP portfolio the amount invested in bonds is proportional to the volatility of stocks, and the amount invested in stocks is proportional to the volatility of bonds. Since stocks are generally more volatile that bonds, this means that the amount invested in bonds will be greater than the amount invested in stocks.
2. The solution of the RP portfolio does not depend on the correlation between bonds and stocks, ρ_{BS}. We will see in the next section that this result only holds for a two asset portfolio.
3. Using a little algebra, we can show that in the RP portfolio the amount invested in bonds (stocks) is proportional to the inverse of the volatility of bonds (stocks). To see this, write:

$$\begin{aligned} x_B &= \frac{\sigma_S}{\sigma_B + \sigma_S} = \frac{\frac{\sigma_S\sigma_B}{\sigma_B}}{\sigma_B + \sigma_S} \\ &= \frac{\frac{1}{\sigma_B}}{\frac{\sigma_B + \sigma_S}{\sigma_B\sigma_S}} = \frac{\frac{1}{\sigma_B}}{\frac{\sigma_B}{\sigma_B\sigma_S} + \frac{\sigma_S}{\sigma_B\sigma_S}} \\ &= \frac{\frac{1}{\sigma_B}}{\frac{1}{\sigma_S} + \frac{1}{\sigma_B}} = \frac{\sigma_B^{-1}}{\sigma_S^{-1} + \sigma_B^{-1}}, \\ x_S &= \frac{\frac{1}{\sigma_S}}{\frac{1}{\sigma_S} + \frac{1}{\sigma_B}} = \frac{\sigma_S^{-1}}{\sigma_S^{-1} + \sigma_B^{-1}}. \end{aligned}$$

4. With inverse volatility weighting, we see that an asset's weight is inversely related to its volatility. This means that highly volatile assets like stocks receive small weights, and mildly

volatile assets like bonds receive large weights. Notice that a RP portfolio becomes an equally weighted portfolio when $\sigma_B = \sigma_S$.

5. Inverse volatility weighting of assets (and then normalized so weights sum to 1) is a common way to form a RP portfolio as it is easy to do (just need estimates of volatility). However, for a portfolio of more than two assets inverse volatility weighting to form a RP portfolio works only when all assets are uncorrelated.

6. We can also determine the RP portfolio weights using the $x - \beta - \sigma$ decomposition of CR_B and CR_S:

$$\begin{aligned}
\text{CR}_B &= x_B \beta_{Bp} \sigma_p, \\
\text{CR}_S &= x_S \beta_{Sp} \sigma_p, \\
\beta_{Bp} &= \frac{\text{cov}(R_B, R_p)}{\text{var}(R_p)} = \frac{\sigma_{Bp}}{\sigma_p^2}, \\
\beta_{Bp} &= \frac{\text{cov}(R_S, R_p)}{\text{var}(R_p)} = \frac{\sigma_{Sp}}{\sigma_p^2}.
\end{aligned}$$

Then

$$\begin{aligned}
\text{CR}_B &= \text{CR}_S \Rightarrow \\
x_B \beta_{Bp} \sigma_p &= x_S \beta_{Sp} \sigma_p \Rightarrow \\
x_B \beta_{Bp} &= x_S \beta_{Sp} = (1 - x_B) \beta_{Sp} \Rightarrow \\
x_B &= \frac{\beta_{Sp}}{\beta_{Sp} + \beta_{Bp}} = \frac{\beta_{Bp}^{-1}}{\beta_{Sp}^{-1} + \beta_{Bp}^{-1}}, \\
x_S &= \frac{\beta_{Bp}}{\beta_{Sp} + \beta_{Bp}} = \frac{\beta_{Sp}^{-1}}{\beta_{Sp}^{-1} + \beta_{Bp}^{-1}}.
\end{aligned}$$

Here, we see that in a RP portfolio an asset's weights is inversely related to its beta to the portfolio. This result holds in general for portfolios with more than two assets. The higher an asset's beta to the portfolio is, the higher is its contribution to portfolio risk and the smaller is its weight in the RP portfolio. Notice that the RP portfolio is an equally weighted portfolio when both assets have the same beta to the portfolio.

7. For the RP portfolio, it can be shown that $\rho_{Bp} = \rho_{Sp}$. That is, both assets have the same correlation with the RP portfolio.

13.6.3 Example Data

The example data from WealthFront is:

Table 13.2: Example data for two asset portfolio

μ_{B}	μ_{S}	σ_{B}	σ_{S}	r_{f}	SR_{B}	$\mathrm{S
0.04	0.062	0.045	0.172	0.018	0.5	

```
r.f = 0.0175
mu.S = 0.0625
sig.S = 0.1725
sig2.S = sig.S^2
mu.B = 0.04
sig.B = 0.045
sig2.B = sig.B^2
rho.BS = 0.2
sig.BS = rho.BS*sig.B*sig.S
SR.B = (mu.B - r.f)/sig.B
SR.S = (mu.S - r.f)/sig.S
data.tbl = data.frame(t(c(mu.B, mu.S, sig.B, sig.S, r.f, SR.B, SR.S, sig.BS, rho.BS)))
col.names = c("$\\mu_{B}$","$\\mu_{S}$","$\\sigma_{B}$","$\\sigma_{S}$", "$r_{f}$",
              "$\\mathrm{SR}_{B}$","$\\mathrm{SR}_{S}$","$\\sigma_{BS}$","$\\rho_{BS}$")
kbl(data.tbl, col.names=col.names, caption = "Example data for two asset portfolio",
    label = "TableExampleData") %>%
   kable_styling(full_width=FALSE)
```

The matrix input are:

```
mu.vec = c(mu.B, mu.S)
cov.mat = matrix(c(sig2.B, sig.BS, sig.BS, sig2.S), 2,2, byrow=TRUE)
```

Portfolios of B and S

```
x.B = seq(from=-0.4, to=1.4, by=0.1)
x.S = 1 - x.B
mu.p = x.B*mu.B + x.S*mu.S
sig2.p = x.B^2 * sig2.B + x.S^2 * sig2.S + 2*x.B*x.S*sig.BS
sig.p = sqrt(sig2.p)
```

Compute equal weight, 60-40, global minimum variance, and tangency portfolios:

```
# equal weight
mu.ew = 0.5*mu.B + 0.5*mu.S
```

```
x.ew = c(0.5, 0.5)
sig2.ew = 0.5^2 * sig2.B + 0.5^2 * sig2.S + 2*0.5*0.5*sig.BS
sig.ew = sqrt(sig2.ew)
# 60-40
mu.64 = 0.4*mu.B + 0.6*mu.S
x.64 = c(0.4, 0.6)
sig2.64 = 0.4^2 * sig2.B + 0.6^2 * sig2.S + 2*0.4*0.6*sig.BS
sig.64 = sqrt(sig2.64)
# global minimum variance
xB.min = (sig2.S - sig.BS)/(sig2.B + sig2.S - 2*sig.BS)
xS.min = 1 - xB.min
x.min = c(xB.min, xS.min)
mu.min = xB.min*mu.B + xS.min*mu.S
sig2.min = xB.min^2 * sig2.B + xS.min^2 * sig2.S + 2*xB.min*xS.min*sig.BS
sig.min = sqrt(sig2.min)
# tangency
top = (mu.B - r.f)*sig2.S - (mu.S - r.f)*sig.BS
bot = (mu.B - r.f)*sig2.S + (mu.S - r.f)*sig2.B - (mu.B - r.f + mu.S - r.f)*sig.BS
x.B.tan = top/bot
x.S.tan = 1 - x.B.tan
x.tan = c(x.B.tan, x.S.tan)
mu.tan = x.B.tan*mu.B + x.S.tan*mu.S
sig2.tan = x.B.tan^2 * sig2.B + x.S.tan^2 * sig2.S + 2*x.B.tan*x.S.tan*sig.BS
sig.tan = sqrt(sig2.tan)
```

Compute risk parity portfolio:

```
xB.rp = sig.S/(sig.B + sig.S)
xS.rp = 1 - xB.rp
x.rp = c(xB.rp, xS.rp)
mu.rp = xB.rp*mu.B + xS.rp*mu.S
sig2.rp = xB.rp^2 * sig2.B + xS.rp^2 * sig2.S + 2*xB.rp*xS.rp*sig.BS
sig.rp = sqrt(sig2.rp)
```

Compute risk contributions for portfolios:

```
# risk parity portfolio
riskReport(mu.vec, cov.mat, x.rp)
#>         Weight   Vol Rho.p Beta.p    MCR     CR PCR
#> Asset B  0.793 0.045 0.775   0.63 0.0349 0.0276 0.5
```

```
#> Asset S  0.207 0.172 0.775   2.42 0.1336 0.0276 0.5
```

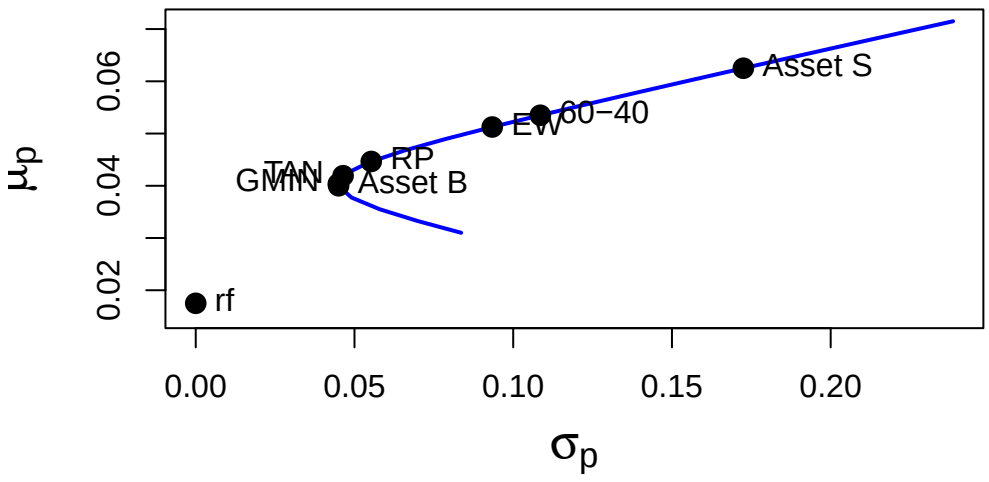
```
# equal weight portfolio
riskReport(mu.vec, cov.mat, x.ew)
#>         Weight   Vol Rho.p Beta.p    MCR      CR   PCR
#> Asset B    0.5 0.045 0.426  0.205 0.0192 0.00958 0.103
#> Asset S    0.5 0.172 0.972  1.795 0.1676 0.08381 0.897
```

```
# 60-40 portfolio
riskReport(mu.vec, cov.mat, x.64)
#>         Weight   Vol Rho.p Beta.p   MCR      CR    PCR
#> Asset B    0.4 0.045 0.357  0.148 0.016 0.00642 0.0591
#> Asset S    0.6 0.172 0.987  1.568 0.170 0.10212 0.9409
```

```
# global minimum variance portfolio
riskReport(mu.vec, cov.mat, x.min)
#>         Weight   Vol Rho.p Beta.p    MCR      CR    PCR
#> Asset B 0.9835 0.045 0.998      1 0.0449 0.04417 0.9835
#> Asset S 0.0165 0.172 0.260      1 0.0449 0.00074 0.0165
```

```
# tangency portfolio
riskReport(mu.vec, cov.mat, x.tan)
#>         Weight   Vol Rho.p Beta.p    MCR      CR   PCR
#> Asset B 0.9143 0.045 0.950  0.921 0.0428 0.03909 0.842
#> Asset S 0.0857 0.172 0.496  1.842 0.0855 0.00733 0.158
```

Plot of portfolios

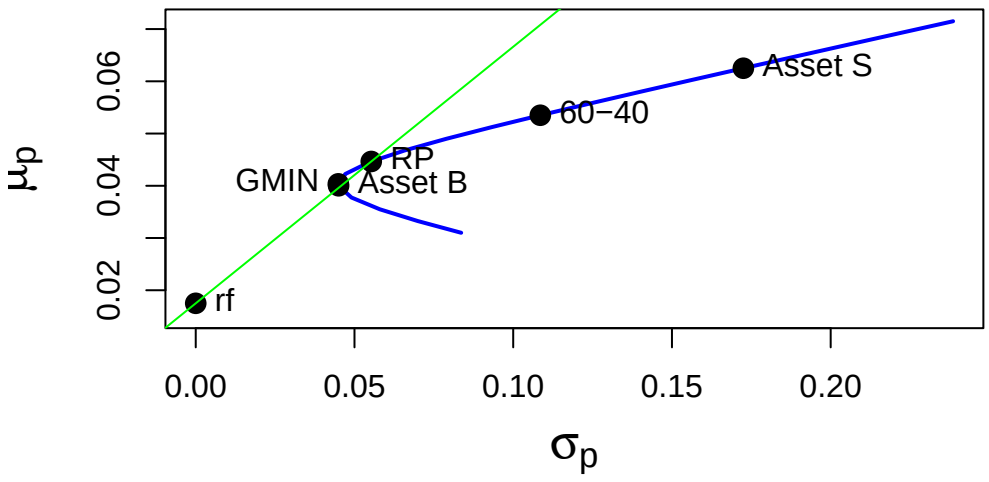


Leverage RP portfolio to have the same volatility as EW:

```
y = sig.ew/sig.rp
mu.y = r.f + y*(mu.rp - r.f)
sig.y = y*sig.rp
c(mu.y, sig.y)
#> [1] 0.0634 0.0934
```

Compute SR

```
SR.rp = (mu.rp - r.f)/sig.rp
SR.64 = (mu.64-r.f)/sig.64
c(SR.rp, SR.64)
#> [1] 0.491 0.332
```



13.6.4 Further Reading: Portfolio Risk Budgeting

To be completed

- Roncalli's book on risk budgeting and risk parity
- Qin's books on risk parity

13.7 Exercises

Example 13.9 (Risk report for equally weighted portfolio of Vanguard Mutual Funds)**.**

- To be completed

14 Statistical Analysis of Portfolios

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In Chapter 10 I introduced mean-variance portfolio theory in the context of simple portfolios with two risky assets and a safe asset. The theory presented assumed that investors know the true expected returns, variances, and correlations between returns (i.e., the parameters of the GWN model of returns). Of course, the real world implementation of mean-variance portfolio theory requires investors to estimate expected returns, variances, and correlations. In this chapter I explore the implications of estimation errors in the parameters of the GWM for asset returns for the application of mean-variance portfolio theory. The quantification of estimation error in mean-variance portfolio analysis is novel and under appreciated by many practitioners, and provides insights into some pitfalls in the practical application of mean-variance portfolio theory. Because mean-variance portfolios are functions of GWN model parameters, I use tools for the statistical analysis of functions of GWN model parameters presented in Chapter 8 throughout this chapter.

The R packages used in this chapter are **boot**, **bootstrap**, **car**, **ellipse**, **introcompfinr**, **mvtnorm**, and **PerformanceAnalytics**. Have these packages installed and loaded in R before replicating the chapter examples.

14.1 Portfolios of Two Risky Assets

Most of the issues involved with the statistical analysis of portfolios can be illustrated in the simple case of a two asset portfolio. Let R_A and R_B denote the simple returns on two risky assets and assume these returns are characterized by the GWN model:

$$\begin{pmatrix} R_A \\ R_B \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_A \\ \mu_B \end{pmatrix}, \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}\right). \tag{14.1}$$

In matrix notation:

$$\mathbf{R} \sim N(\mu, \Sigma),$$

where

$$\mathbf{R} = \begin{pmatrix} R_A \\ R_B \end{pmatrix}, \mu = \begin{pmatrix} \mu_A \\ \mu_B \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}.$$

A portfolio with weight vector $\mathbf{x} = (x_A, x_B)'$ has return $R_p = \mathbf{x}'\mathbf{R} = x_A R_A + x_B R_B$ which is also described by the GWN model:

$$\begin{aligned} R_p &\sim N(\mu_p, \sigma_p^2), \\ \mu_p &= \mathbf{x}'\mu = x_A\mu_A + x_B\mu_B, \\ \sigma_p^2 &= \mathbf{x}'\Sigma\mathbf{x} = x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B\sigma_{AB}. \end{aligned}$$

I assume that $\{\mathbf{r}_t\}_{t=1}^T$ is the observed sample of asset returns of size T, where $\mathbf{r}_t = (r_{At}, r_{Bt})'$, is generated from the GWN model Equation 14.1 from which I create the sample portfolio returns $r_{p,t} = \mathbf{x}'\mathbf{r}_t = x_A r_{At} + x_B r_{Bt}$. I assume that the portfolio weight vector $\mathbf{x}$ remains fixed over the observed sample and that the portfolio is rebalanced every period to maintain the fixed weights.

The true GWN model parameters are unknown in practice and must be estimated from the observed data. As described in Chapter 7, the GWN model parameters are estimated using the sample statistics:

$$\begin{aligned} \hat{\mu}_i &= \frac{1}{T}\sum_{t=1}^T r_{it},\ \hat{\sigma}_i^2 = \frac{1}{T-1}\sum_{t=1}^T (r_{it} - \hat{\mu}_i)^2, \\ \hat{\sigma}_i &= \sqrt{\hat{\sigma}_i^2},\ \hat{\sigma}_{ij}^2 = \frac{1}{T-1}\sum_{t=1}^T (r_{it} - \hat{\mu}_i)(r_{jt} - \hat{\mu}_j). \end{aligned}$$

These estimates are consistent, asymptotically normally distributed, and have estimation errors which can be quantified by standard error formulas:[1]

$$\text{se}(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{T}},\ \text{se}(\hat{\sigma}_i^2) \approx \frac{\sigma_i^2}{\sqrt{T/2}},\ \text{se}(\hat{\sigma}_i) \approx \frac{\sigma_i}{\sqrt{2T}},\ \text{se}(\hat{\sigma}_{ij}) \approx \frac{\sqrt{\sigma_i^2\sigma_j^2 + \sigma_{ij}^2}}{\sqrt{T}}. \tag{14.2}$$

14.1.1 Estimation of GWN Model Parameters for Portfolios

You can estimate the GWN model parameters μ_p, σ_p^2 and σ_p for the portfolio return, R_p, using two equivalent methods. In the first method, you use the sample portfolio returns $\{r_{p,t}\}_{t=1}^T$ and compute the sample statistics:

$$\hat{\mu}_p = \frac{1}{T}\sum_{t=1}^T r_{p,t},\ \hat{\sigma}_p^2 = \frac{1}{T-1}\sum_{t=1}^T (r_{p,t} - \hat{\mu}_p)^2,\ \hat{\sigma}_p = \sqrt{\hat{\sigma}_p^2}. \tag{14.3}$$

[1] $\hat{\sigma}_i$ is asymptotically unbiased and the bias for finite T is typically very small so that it is essentially unbiased for moderately sized T.

These estimates are unbiased and asymptotically normal with standard errors:

$$\text{se}(\hat{\mu}_p) = \frac{\sigma_p}{\sqrt{T}}, \text{se}(\hat{\sigma}_p^2) \approx \frac{\sigma_p^2}{\sqrt{T/2}}, \text{se}(\hat{\sigma}_p) \approx \frac{\sigma_p}{\sqrt{2T}}. \tag{14.4}$$

In the second method, I compute estimates of μ_p and σ_p^2 directly from the individual asset estimates in Equation 14.2:

$$\hat{\mu}_p = \mathbf{x}'\hat{\mu} = x_A\hat{\mu}_A + x_B\hat{\mu}_B, \tag{14.5}$$

$$\hat{\sigma}_p^2 = \mathbf{x}'\widehat{\Sigma}\mathbf{x} = x_A^2\hat{\sigma}_A^2 + x_B^2\hat{\sigma}_B^2 + 2x_Ax_B\hat{\sigma}_{AB}. \tag{14.6}$$

The estimates of μ_p and σ_p^2 in Equation 14.5 and Equation 14.6, respectively, are numerically identical to the estimates in Equation 14.3. To see this, consider the calculation of $\hat{\mu}_p$ in Equation 14.5:

$$\begin{aligned}
\hat{\mu}_p &= x_A\hat{\mu}_A + x_B\hat{\mu}_B = x_A\left(\frac{1}{T}\sum_{t=1}^{T} r_{At}\right) + x_B\left(\frac{1}{T}\sum_{t=1}^{T} r_{Bt}\right) \\
&= \frac{1}{T}\sum_{t=1}^{T}(x_Ar_{At} + x_Br_{Bt}) = \frac{1}{T}\sum_{t=1}^{T} r_{p,t}.
\end{aligned}$$

The proof of the equivalence of $\hat{\sigma}_p^2$ in Equation 14.6 with $\hat{\sigma}_p^2$ in Equation 14.3 is left as an end-of-chapter exercise.

The standard errors of $\hat{\mu}_p$ and $\hat{\sigma}_p^2$ can also be directly calculated from Equation 14.5 and Equation 14.6. Consider the calculation for $\text{se}(\hat{\mu}_p)$:

$$\text{var}(\hat{\mu}_p) = \text{var}(x_A\hat{\mu}_A + x_B\hat{\mu}_B) = x_A^2\text{var}(\hat{\mu}_A) + x_B^2\text{var}(\hat{\mu}_B) + 2x_Ax_B\text{cov}(\hat{\mu}_A, \hat{\mu}_B). \tag{14.7}$$

In chapter Chapter 7 it was shown that

$$\text{var}(\hat{\mu}_A) = \frac{\sigma_A^2}{T}, \text{var}(\hat{\mu}_B) = \frac{\sigma_B^2}{T}, \text{cov}(\hat{\mu}_A, \hat{\mu}_B) = \frac{\sigma_{AB}}{T}.$$

Then Equation 14.7 can be rewritten as:

$$\begin{aligned}
\text{var}(\hat{\mu}_p) &= x_A\frac{\sigma_A^2}{T} + x_B\frac{\sigma_B^2}{T} + 2x_Ax_B\frac{\sigma_{AB}}{T} \\
&= \frac{1}{T}\left(x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_Ax_B\sigma_{AB}\right) \\
&= \frac{\sigma_p^2}{T}.
\end{aligned}$$

Hence,

$$\text{se}(\hat{\mu}_p) = \sqrt{\text{var}(\hat{\mu}_p)} = \frac{\sigma_p}{\sqrt{T}},$$

which is the same formula given in Equation 14.4.

Unfortunately, the calculation of $\text{se}(\hat{\sigma}_p^2)$ is a bit horrific and I only sketch out some of the details. Now,

$$\begin{aligned}\text{var}(\hat{\sigma}_p^2) &= \text{var}(x_A^2\hat{\sigma}_A^2 + x_B^2\hat{\sigma}_B^2 + 2x_A x_B\hat{\sigma}_{AB}) \\ &= x_A^4\text{var}(\hat{\sigma}_A^2) + x_B^4\text{var}(\hat{\sigma}_B^2) + 4x_A^2x_B^2\text{var}(\hat{\sigma}_{AB}) \\ &+ 2x_A^2x_B^2\text{cov}(\hat{\sigma}_A^2, \hat{\sigma}_B^2) + 4x_A^3x_B\text{cov}(\hat{\sigma}_A^2, \hat{\sigma}_{AB}) \\ &+ 4x_Ax_B^3\text{cov}(\hat{\sigma}_B^2, \hat{\sigma}_{AB})\end{aligned}$$

Unfortunately, there are no easy formulas for $\text{var}(\hat{\sigma}_{AB})$, $\text{cov}(\hat{\sigma}_A^2, \hat{\sigma}_B^2)$, $\text{cov}(\hat{\sigma}_A^2, \hat{\sigma}_{AB})$ and $\text{cov}(\hat{\sigma}_B^2, \hat{\sigma}_{AB})$. With a CLT approximation and much tedious calculation it can be shown that Equation 14.8 is equal to the square of $\text{se}(\hat{\sigma}_p^2)$ given in Equation 14.4. Obviously, the method of computing standard errors for $\hat{\mu}_p$ and $\hat{\sigma}_p^2$ directly from the GWN model estimates computed from the sample portfolio returns is much easier than the indirect calculations based on the individual asset estimates, especially for $\hat{\sigma}_p^2$.

Example 14.1 (Example data for two risky asset portfolio)**.** To be completed... table example data

Table 10.1 from Chapter 10 gives annual return distribution parameters for two hypothetical assets A and B. I use this distribution but re-scale the values to represent a monthly return distribution. Asset A is the high risk asset with a monthly return of $\mu_A = 1.46\%$ and monthly standard deviation of $\sigma_A = 7.45\%$. Asset B is a lower risk asset with monthly return $\mu_B = 0.458\%$ and monthly standard deviation of $\sigma_B = 3.32\%$. The assets are assumed to be slightly negatively correlated with correlation coefficient $\rho_{AB} = -0.164$. Given the standard deviations and the correlation, the covariance can be determined from $\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B = -0.0004$. In R, these GWN model parameters are created using:

```
mu.A = 0.175/12
sig.A = 0.258/sqrt(12)
sig2.A = sig.A^2
mu.B = 0.055/12
sig.B = 0.115/sqrt(12)
sig2.B = sig.B^2
rho.AB = -0.164
sig.AB = rho.AB*sig.A*sig.B
sigma.mat = matrix(c(sig2.A, sig.AB, sig.AB, sig2.B),
                   2, 2, byrow=TRUE)
mu.vec = c(mu.A, mu.B)
sd.vec = c(sig.A, sig.B)
names(mu.vec) = names(sd.vec) = c("asset.A", "asset.B")
dimnames(sigma.mat) = list(names(mu.vec), names(mu.vec))
```

For portfolio analysis, I consider an equally weighted portfolio of assets A and B. The GWN model parameters for this portfolio are:

```
x1.A = 0.5
x1.B = 1 - x1.A
x1.vec = c(x1.A, x1.B)
names(x1.vec) = names(mu.vec)
mu.p1 = x1.A*mu.A + x1.B*mu.B
sig2.p1 = x1.A^2 * sig2.A + x1.B^2 * sig2.B + 2*x1.A*x1.B*sig.AB
sig.p1 = sqrt(sig2.p1)
cbind(mu.p1, sig2.p1, sig.p1)
#>        mu.p1 sig2.p1 sig.p1
#> [1,] 0.00958 0.00146 0.0382
```

I use the above GWN model parameters to simulate $T = 60$ hypothetical returns for assets A and B and the equally weighted portfolio:

```
n.obs = 60
set.seed(125)
returns.sim = rmvnorm(n.obs, mean=mu.vec, sigma=sigma.mat)
colnames(returns.sim) = names(mu.vec)
rp1.sim = returns.sim%*%x1.vec
```

The estimates of the GWN model parameters from the simulated returns for the individual assets are:

```
muhat.vals = colMeans(returns.sim)
sig2hat.vals = apply(returns.sim, 2, var)
sigmahat.vals = apply(returns.sim, 2, sd)
covmat.hat = cov(returns.sim)
covhat = covmat.hat[1,2]
rbind(muhat.vals, sig2hat.vals, sigmahat.vals)
#>               asset.A asset.B
#> muhat.vals    0.03077 0.00212
#> sig2hat.vals  0.00539 0.00127
#> sigmahat.vals 0.07344 0.03560
covhat
#> [1] -0.000848
```

The estimated standard errors and 95% confidence intervals for $\hat{\mu}_i$ and $\hat{\sigma}_i$ $(i = A, B)$ are:

```
# estimated standard errors
se.muhat = sigmahat.vals/sqrt(n.obs)
se.sigmahat = sigmahat.vals/sqrt(2*n.obs)
rbind(se.muhat, se.sigmahat)
#>             asset.A asset.B
#> se.muhat    0.00948 0.00460
#> se.sigmahat 0.00670 0.00325
# 95% confidence intervals
lower.mu = muhat.vals - 2*se.muhat
upper.mu = muhat.vals + 2*se.muhat
lower.sigma = sigmahat.vals - 2*se.sigmahat
upper.sigma = sigmahat.vals + 2*se.sigmahat
cbind(lower.mu, upper.mu)
#>         lower.mu upper.mu
#> asset.A  0.01181   0.0497
#> asset.B -0.00707   0.0113
cbind(lower.sigma, upper.sigma)
#>         lower.sigma upper.sigma
#> asset.A      0.0600      0.0869
#> asset.B      0.0291      0.0421
```

Here, you see that the means are not estimated as precisely as the volatilities and that all of the true values are contained in the 95% confidence intervals.

For the equally weighted portfolio, the GWN model estimates computed directly from the simulated portfolio returns using Equation 14.3 are:

```
muhat.p1 = mean(rp1.sim)
sig2hat.p1 = as.numeric(var(rp1.sim))
sigmahat.p1 = sd(rp1.sim)
cbind(muhat.p1, sig2hat.p1, sigmahat.p1)
#>      muhat.p1 sig2hat.p1 sigmahat.p1
#> [1,]   0.0164    0.00124      0.0352
```

The portfolio estimates computed from the asset estimates Equation 14.5 and Equation 14.6 are numerically equivalent:

```
muhat.p1 = as.numeric(x1.A*muhat.vals[1] + x1.B*muhat.vals[2])
sig2hat.p1 = as.numeric(x1.A^2 * sig2hat.vals[1] +
                        x1.B^2 * sig2hat.vals[2] +
```

```
                        2*x1.A*x1.B*covhat)
sighat.p1 = sqrt(sig2hat.p1)
cbind(muhat.p1, sig2hat.p1, sigmahat.p1)
#>      muhat.p1 sig2hat.p1 sigmahat.p1
#> [1,]   0.0164    0.00124      0.0352
```

The estimated standard errors for $\hat{\mu}_p$ and $\hat{\sigma}_p$, and 95% confidence intervals for μ_p and σ_p computed using Equation 14.4 are:

```
# estimated standard errors
se.muhat.p1 = sigmahat.p1/sqrt(n.obs)
se.sigmahat.p1 = sigmahat.p1/sqrt(2*n.obs)
cbind(se.muhat.p1, se.sigmahat.p1)
#>      se.muhat.p1 se.sigmahat.p1
#> [1,]     0.00455        0.00322
# 95% confidence intervals
lower.mu = muhat.p1 - 2*se.muhat.p1
upper.mu = muhat.p1 + 2*se.muhat.p1
lower.sigma = sigmahat.p1 - 2*se.sigmahat.p1
upper.sigma = sigmahat.p1 + 2*se.sigmahat.p1
ans = rbind(c(lower.mu, upper.mu),
c(lower.sigma, upper.sigma))
colnames(ans) = c("lower", "upper")
rownames(ans) = c("mu.p", "sigma.p")
ans
#>           lower  upper
#> mu.p    0.00735 0.0255
#> sigma.p 0.02880 0.0417
```

As with the individual assets, the portfolio mean is estimated less precisely than the portfolio volatility and the true values are in the 95% confidence intervals. ■

14.1.2 Estimation error in the risk-return diagram

In the analysis of portfolios in Chapter 10, the risk return trade-off between portfolios was visualized by plotting the volatility-expected return pairs $(\hat{\sigma}_p, \hat{\mu}_p)$ of different portfolios. This was done without considering estimation error in $\hat{\sigma}_p$ and $\hat{\mu}_p$. Estimation error in $\hat{\sigma}_p$ and $\hat{\mu}_p$ creates uncertainty about the location of the true values (σ_p, μ_p) in the risk-return diagram.

Example 14.2 (True and estimated risk return diagram for example data)**.** Figure 14.1 shows the true values and sample estimates of the pairs (σ_A, μ_A), (σ_B, μ_B), and (σ_p, μ_p) for the equally weighted portfolio created using:

```
# true values
plot(sd.vec, mu.vec, pch=16, col="black",
     ylim=c(-0.01, 0.04), xlim=c(0, 0.1),
     xlab=expression(sigma[p]), ylab=expression(mu[p]),
     cex=2)
abline(h=0)
points(sig.p1, mu.p1, pch=16, col="black", cex=2)
text(x=sig.A, y=mu.A, labels="A", pos=4, cex = 1.5)
text(x=sig.B, y=mu.B, labels="B", pos=4, cex = 1.5)
text(x=sig.p1, y=mu.p1, labels="P", pos=4, cex = 1.5)
# estimates
points(sigmahat.vals, muhat.vals, pch=16, col="blue", cex=2)
points(sigmahat.p1, muhat.p1, pch=16, col="blue", cex=2)
text(x=sigmahat.vals[1], y=muhat.vals[1], labels=expression(hat(A)),
     col="blue", pos=4, cex = 1.5)
text(x=sigmahat.vals[2], y=muhat.vals[2], labels=expression(hat(B)),
     col="blue", pos=4, cex = 1.5)
text(x=sigmahat.p1, y=muhat.p1, labels=expression(hat(P)),
     col="blue", pos=4, cex = 1.5)
```

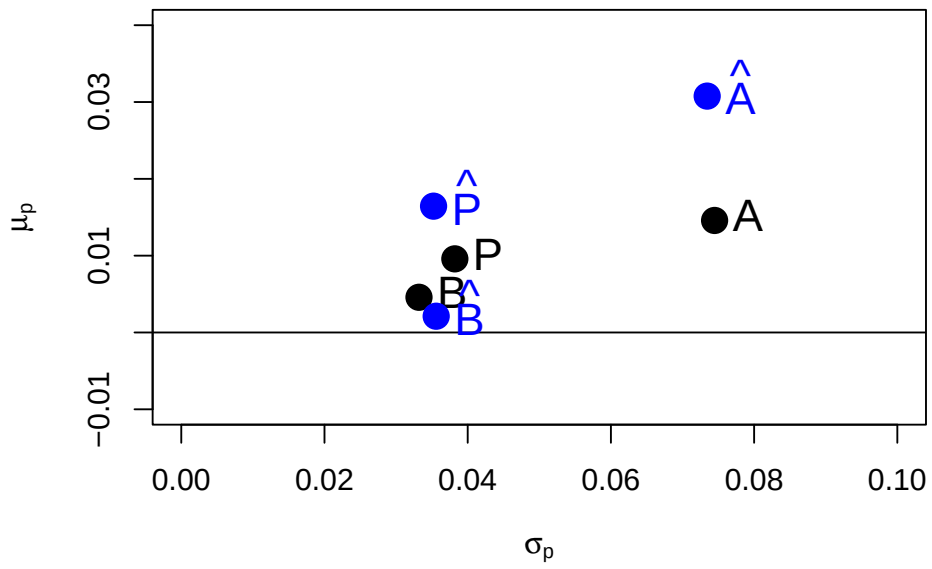


Figure 14.1: Risk-return diagram for example data. A $= (\sigma_A, \mu_A)$, B $= (\sigma_B, \mu_B)$, $\hat{\text{A}} = (\hat{\sigma}_A,\ \hat{\mu}_A)$, $\hat{\text{B}} = (\hat{\sigma}_B,\ \hat{\mu}_B)$.

You see that $(\hat{\sigma}_B, \hat{\mu}_B)$ is fairly close to (σ_B, μ_B) but that $(\hat{\sigma}_A, \hat{\mu}_A)$ is quite far above (σ_A, μ_A) and $(\hat{\sigma}_P, \hat{\mu}_P)$ is moderately far above (σ_P, μ_P). The large positive estimation errors in $\hat{\mu}_P$ and $\hat{\mu}_A$ greatly overstate the risk-return characteristics of the equally weighted portfolio and asset A. ■

To illustrate estimation error in the risk return diagram, the individual 95% confidence intervals for μ_i and σ_i could be superimposed on the plot as rectangles centered at the pairs $(\hat{\sigma}_i, \hat{\mu}_i)$. However, the probability that these rectangles contain the true pairs (σ_i, μ_i) is not equal to 95% because the confidence intervals for μ_i are created independently from the confidence intervals for σ_i. To create a joint confidence set that cover the pair (σ_i, μ_i) with probability 95% requires knowing the joint probability distribution of $(\hat{\sigma}_i, \hat{\mu}_i)$.

In Chapter 7, it was shown that in the GWN model:

$$\begin{pmatrix} \hat{\sigma}_i \\ \hat{\mu}_i \end{pmatrix} \sim N\left(\begin{pmatrix} \sigma_i \\ \mu_i \end{pmatrix}, \begin{pmatrix} \text{se}(\hat{\sigma}_i)^2 & 0 \\ 0 & \text{se}(\hat{\mu}_i)^2 \end{pmatrix} \right), \tag{14.8}$$

for large enough T. Hence, $\hat{\sigma}_i$ and $\hat{\mu}_i$ are (asymptotically) jointly normally distributed and $\text{cov}(\hat{\sigma}_i, \hat{\mu}_i) = 0$ which implies that they are also independent. Let $\theta = (\sigma_i, \mu_i)', \hat{\theta} = (\hat{\sigma}_i, \hat{\mu}_i)'$ and $\mathbf{V} = \text{diag}(\text{se}(\hat{\sigma}_i)^2, \text{se}(\hat{\mu}_i)^2)$. Then Equation 14.8 can be expressed as $\hat{\theta} \sim N(\theta, \mathbf{V})$. It follows

that the quadratic form $\left(\hat{\theta}-\theta\right)' \mathbf{V}^{-1} \left(\hat{\theta}-\theta\right) \sim \chi^2(2)$, and so

$$Pr\left\{\left(\hat{\theta}-\theta\right)' \mathbf{V}^{-1} \left(\hat{\theta}-\theta\right) \leq q_{.95}^{\chi^2(2)}\right\} = 0.95 \tag{14.9}$$

where $q_{.95}^{\chi^2(2)}$ is the 95% quantile of the $\chi^2(2)$ distribution. The equation $\left(\hat{\theta}-\theta\right)' \mathbf{V}^{-1} \left(\hat{\theta}-\theta\right) = q_{.95}^{\chi^2(2)}$ defines an un-tilted ellipse in $\sigma_i - \mu_i$ space centered at $(\hat{\sigma}_i, \hat{\mu}_i)$ with axes proportional to $\text{se}(\hat{\sigma}_i)$ and $\text{se}(\hat{\mu}_i)$, respectively. This ellipse is the joint 95% confidence set for (σ_i, μ_i).

Example 14.3 (True and estimated risk return diagram with confidence ellipses for example data)**.** Figure 14.2 repeats Figure 14.1 with the addition of the joint confidence sets for (σ_A, μ_A), (σ_B, μ_B), and (σ_p, μ_p) , created using

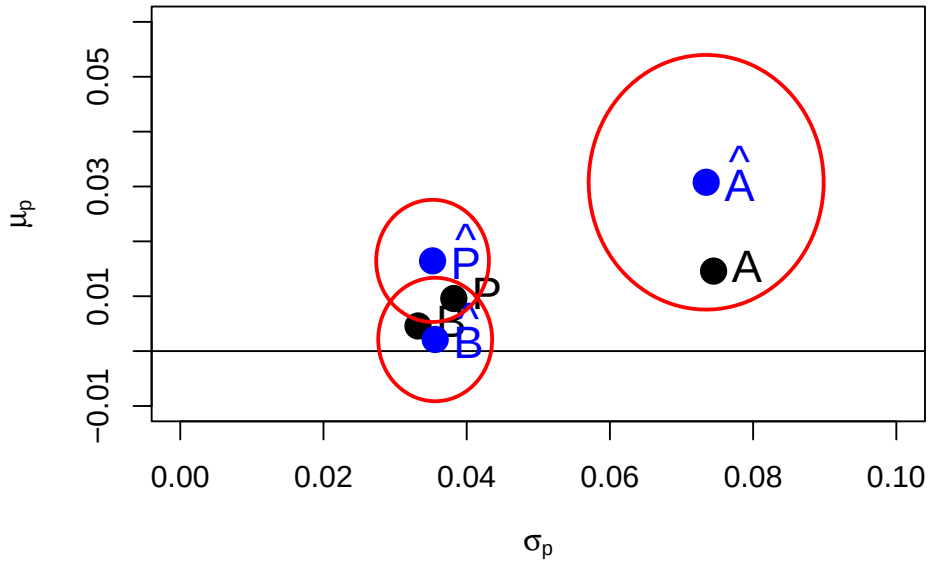


Figure 14.2: Risk return tradeoff with 95% confidence ellipses

The `ellipse()` function from the R package **ellipse** is used to draw the 95% confidence sets. The confidence ellipses are longer in the μ_i direction because $\text{se}(\hat{\mu}_i)$ is larger than $\text{se}(\hat{\sigma}_i)$ and indicate more uncertainty about the true values of μ_i than the true values of σ_i. The ellipse for (σ_A, μ_A) is much bigger than the ellipses for (σ_B, μ_B) and (σ_p, μ_p) and indicates a wide range of possible values for (σ_A, μ_A) . For all pairs, the true values are inside the 95% confidence ellipses.[2] ■

[2]Notice, however, that the pair (σ_p, μ_p) is inside the confidence ellipse for (σ_B, μ_B) but that (σ_B, μ_B) is not in the confidence ellipse for (σ_p, μ_p).

The estimation errors in the points $(\hat{\sigma}_i, \hat{\mu}_i)$ can also be illustrated using the bootstrap. You simply sample with replacement from the observed returns B times and compute estimates of (σ_i, μ_i) from each bootstrap sample. You can then plot the B bootstrap pairs $(\hat{\sigma}_i^*, \hat{\mu}_i^*)$ on the risk return diagram instead of the confidence ellipses.

Example 14.4 (True and estimated risk return diagram with bootstrap estimates for example data)**.** The following R code creates $B = 500$ bootstrap estimates of (σ_i, μ_i) for $i = A, B, P$:

```
n.boot = 500
mu.boot = matrix(0, n.boot, 3)
sd.boot = matrix(0, n.boot, 3)
colnames(mu.boot) = colnames(sd.boot) = c("A", "B", "P")
set.seed(123)
for (i in 1:n.boot) {
   boot.idx = sample(n.obs, replace=TRUE)
   ret.boot = returns.sim[boot.idx, ]
   rp1.boot = ret.boot%*%x1.vec
   ret.boot = cbind(ret.boot, rp1.boot)
   mu.boot[i, ] = colMeans(ret.boot)
   sd.boot[i, ] = apply(ret.boot, 2, sd)
}
```

Figure 14.3 repeats Figure 14.2 and adds the bootstrap estimates of (σ_i, μ_i) for $i = A, B, P$. Notice that the bootstrap estimates for each asset produce a scatter that fills the 95% confidence ellipses with just a few estimates lying outside the ellipses. Hence, the bootstrap is an easy and effective way to illustrate estimation error in the risk-return diagram. ■

14.1.3 Estimation error in the portfolio frontier

In the case of two risky assets, the portfolio frontier is a plot of the risk-return characteristics of all feasible portfolios. From this frontier, you can readily determine the set of efficient portfolios which are those portfolios that have the highest expected return for a given risk level. If you construct the portfolio frontier from portfolio risk and return estimates from the GWN model, then the estimation error in these estimates leads to estimation error in the entire portfolio frontier.

Example 14.5 (True and estimated portfolio frontier for example data)**.** Using the example data, you can construct the true unobservable portfolio frontier as well as an estimate of this frontier from the simulated returns. For example, consider creating the true and estimated portfolio frontiers for the following portfolios:

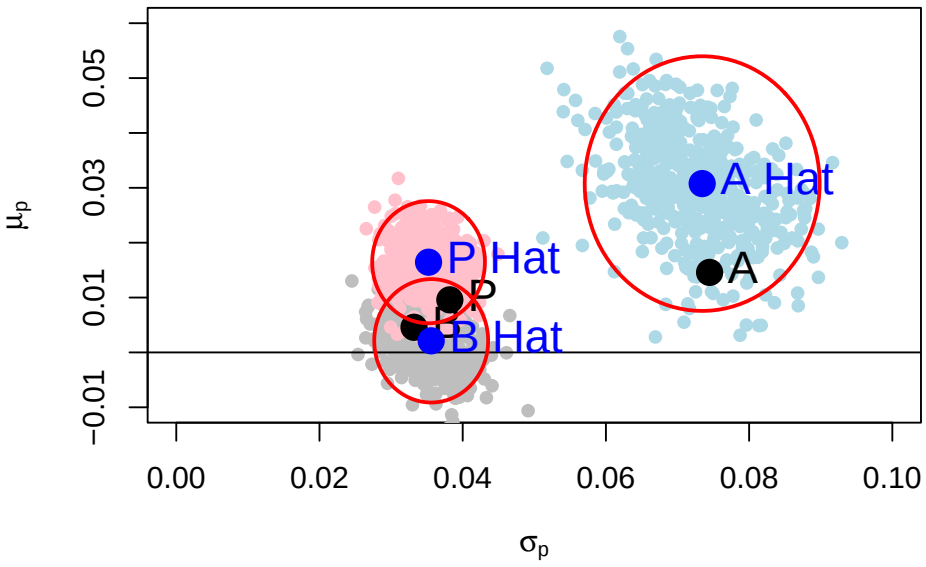


Figure 14.3: Risk return trade off with bootstrap estimates.

```
x.A = seq(from=-0.4, to=1.4, by=0.1)
x.B = 1 - x.A
```

The risk-return characteristics of the portfolios on the true portfolio frontier are:

```
mu.p = x.A*mu.A + x.B*mu.B
sig2.p = x.A^2 * sig2.A + x.B^2 * sig2.B + 2*x.A*x.B*sig.AB
sig.p = sqrt(sig2.p)
```

The risk-return characteristics of the portfolios on the estimated frontier are:

```
muhat.p = x.A*muhat.vals["asset.A"] + x.B*muhat.vals["asset.B"]
sig2hat.p = x.A^2 * sig2hat.vals["asset.A"] +
   x.B^2 * sig2hat.vals["asset.B"] + 2*x.A*x.B*covhat
sighat.p = sqrt(sig2hat.p)
```

The true and estimated portfolios are illustrated in Figure 14.4, created with

```
plot(sig.p, mu.p, pch=16, type="b", col="black",
     ylim=c(-0.01, 0.041), xlim=c(0, 0.125),
```

```
     xlab=expression(sigma[p]), ylab=expression(mu[p]), cex=1)
abline(h=0)
points(sd.vec, mu.vec, pch=16, col="black", cex=2)
text(x=sig.A, y=mu.A, labels="A", pos=3, cex = 1.5)
text(x=sig.B, y=mu.B, labels="B", pos=3, cex = 1.5)
# estimated assets and frontier
points(sighat.p, muhat.p, type="b", pch=16, col="blue", cex=1)
points(sigmahat.vals, muhat.vals, pch=16, col="blue", cex=2)
text(x=sigmahat.vals[1], y=muhat.vals[1], labels="A Hat",
     col="blue", pos=3, cex = 1.5)
text(x=sigmahat.vals[2], y=muhat.vals[2], labels="B Hat",
     col="blue", pos=2, cex = 1.5)
```

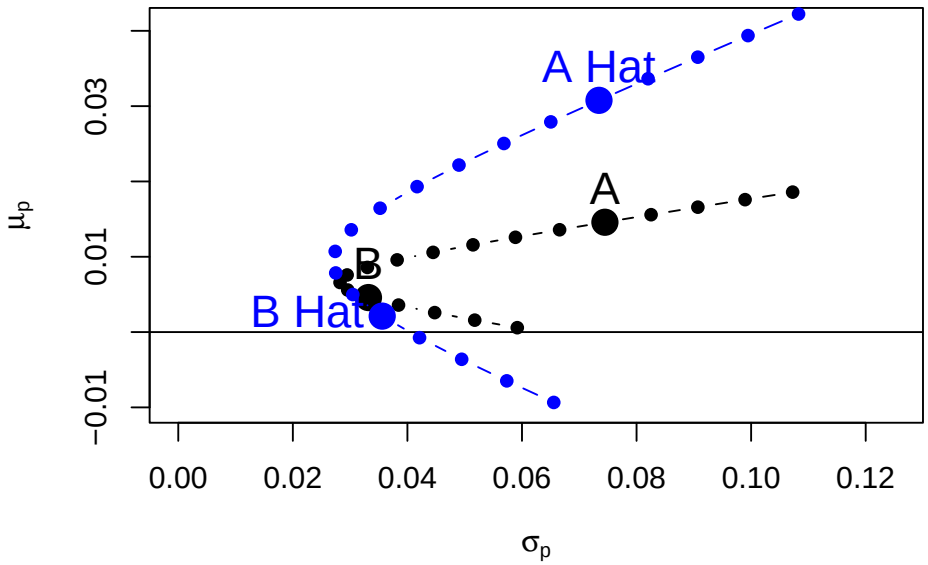


Figure 14.4: True and estimated portfolio frontiers

Due to the large estimation error in $\hat{\mu}_A$, the estimated frontier is considerably higher than the true frontier. As a result, portfolios on the estimated frontier appear to have higher reward-to-risk properties than they actually do. ■

Sampling uncertainty about the estimated frontier can be easily computed using the bootstrap. The process is the same as in the case of a single asset. For each bootstrap sample, estimate the

expected return and volatility of each portfolio on the frontier and then plot these bootstrap pairs on the plot showing the estimated frontier from the sample returns.

Example 14.6 (True and estimated risk portfolio frontier with bootstrap estimates for example data)**.** The R code to create $B = 1000$ bootstrap estimates of risk and return pairs for the frontier portfolios is

```
# initialize matrices
n.boot = 1000
mu.boot = matrix(0, n.boot, length(x.A))
sd.boot = matrix(0, n.boot, length(x.A))
colnames(mu.boot) = colnames(sd.boot) =
      paste("P", 1:length(x.A), sep=".")
# bootstrap loop
set.seed(123)
for (i in 1:n.boot) {
   boot.idx = sample(n.obs, replace=TRUE)
   ret.boot = returns.sim[boot.idx, ]
# GWN model estimates
   muhat.boot = colMeans(ret.boot)
   sig2hat.boot = apply(ret.boot, 2, var)
   sigmahat.boot = sqrt(sig2hat.boot)
   covhat.boot = cov(ret.boot)[1,2]
# portfolio risk return estimates
   mu.boot[i, ] = x.A*muhat.boot[1] + x.B*muhat.boot[2]
   sig2.boot = x.A^2 * sig2hat.boot[1] +
      x.B^2 * sig2hat.boot[2] + 2*x.A*x.B*covhat.boot
   sd.boot[i, ] = sqrt(sig2.boot)
}
```

The true and estimated frontier together with the bootstrap risk-return pairs for each portfolio on the estimated frontier is illustrated in Figure 14.5, create using

```
# set up plot area
plot(sig.p, mu.p, type="n", ylim=c(-0.02, 0.07), xlim=c(0, 0.13),
     xlab=expression(sigma[p]), ylab=expression(mu[p]))
# plot bootstrap estimates
for (i in 1:length(x.A)) {
   points(sd.boot[, i], mu.boot[, i], pch=16, col="lightblue")
}
```

```
# plot true frontier
points(sig.p, mu.p, pch=16, type="b", col="black", cex=1)
abline(h=0)
# plot true assets
points(sd.vec, mu.vec, pch=16, col="black", cex=2)
text(x=sig.A, y=mu.A, labels="A", pos=3, cex = 1.5)
text(x=sig.B, y=mu.B, labels="B", pos=3, cex = 1.5)
# plot estimated frontier
points(sighat.p, muhat.p, type="b", pch=16, col="blue", cex=1)
# plot estimated assets
points(sigmahat.vals, muhat.vals, pch=16, col="blue", cex=2)
text(x=sigmahat.vals[1], y=muhat.vals[1], labels="A Hat",
     col="blue", pos=3, cex = 1.5)
text(x=sigmahat.vals[2], y=muhat.vals[2], labels="B Hat",
     col="blue", pos=2, cex = 1.5)
```

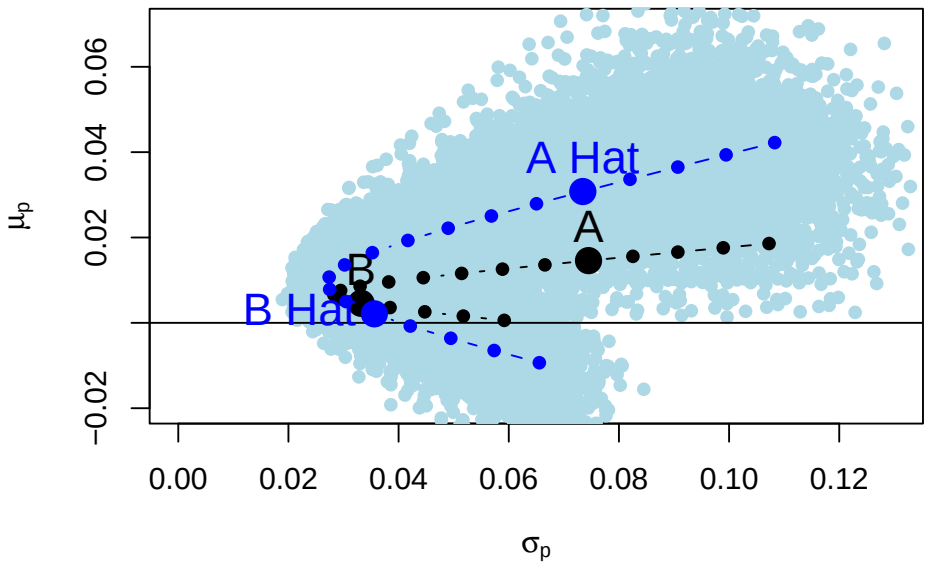


Figure 14.5: Estimated frontier with bootstrap estimates.

The true frontier portfolios are the black dots, the estimated frontier portfolios are the dark blue dots and the bootstrap risk-return pairs are the light blue dots. The bootstrap estimates form a light blue cloud around the estimated frontier. The bootstrap cloud can be interpreted as approximating

a confidence interval around estimated frontier. From Equation 14.9, this confidence interval can be thought of the union of all of the confidence ellipses about the portfolios on the frontier. The true frontier of portfolios (black dots) is within the blue cloud. ■

14.1.4 Statistical properties of the global minimum variance portfolio

For portfolios of two risky assets, the global minimum variance portfolio weights satisfy

$$x_A^{\min} = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}}, \; x_B^{\min} = 1 - x_A^{\min} = \frac{\sigma_A^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}}. \tag{14.10}$$

The expected return and variance of the global minimum variance portfolio are

$$\begin{aligned} \mu_{p,\min} &= x_A^{\min}\mu_A + x_B^{\min}\mu_B, \\ \sigma_{p,\min}^2 &= \left(x_A^{\min}\right)^2 \sigma_A^2 + \left(x_B^{\min}\right)^2 \sigma_B^2 + 2x_A^{\min}x_B^{\min}\sigma_{AB}. \end{aligned}$$

The estimated global minimum variance portfolio weights are then

$$\hat{x}_A^{\min} = \frac{\hat{\sigma}_B^2 - \hat{\sigma}_{AB}}{\hat{\sigma}_A^2 + \hat{\sigma}_B^2 - 2\hat{\sigma}_{AB}}, \; x_B^{\min} = 1 - \hat{x}_A^{\min} = \frac{\hat{\sigma}_A^2 - \hat{\sigma}_{AB}}{\hat{\sigma}_A^2 + \hat{\sigma}_B^2 - 2\hat{\sigma}_{AB}}. \tag{14.11}$$

From Equation 14.11 you see that estimation error in the global minimum variance weights is related to estimation error in the asset variances and the covariance in a complicated nonlinear way. However, the estimation error does not depend on estimation error in the expected returns. The estimated expected returns and variance of the global minimum variance portfolio are

$$\hat{\mu}_{p,\min} = \hat{x}_A^{\min}\hat{\mu}_A + \hat{x}_B^{\min}\hat{\mu}_B, \tag{14.12}$$

$$\hat{\sigma}_{p,\min}^2 = \left(\hat{x}_A^{\min}\right)^2 \hat{\sigma}_A^2 + \left(\hat{x}_B^{\min}\right)^2 \hat{\sigma}_B^2 + 2\hat{x}_A^{\min}\hat{x}_B^{\min}\hat{\sigma}_{AB} \tag{14.13}$$

Here, estimation error in $\hat{\mu}_{p,\min}$ depends on two sources: estimation error in the global minimum variance weights (which depend on estimation error in the asset variances and the covariance), and estimation error in the asset expected returns. However, estimation error in $\hat{\sigma}_{p,\min}^2$ only depends on estimation error in the asset variances and the covariance. Because there is more estimation error in the asset expected returns than the asset variances $\hat{\mu}_{p,\min}$, will be estimated more imprecisely than $\hat{\sigma}_{p,\min}^2$.

Example 14.7 (True and estimated global minimum variance portfolio for example data)**.** For the example data, the true and estimated global minimum variance portfolio are:

```
# True global min variance portfolio weights
xA.min = (sig2.B - sig.AB)/(sig2.A + sig2.B - 2*sig.AB)
xB.min = 1 - xA.min
c(xA.min, xB.min)
#> [1] 0.202 0.798
# Estimated min variance portfolio weights
xA.min.hat = (sig2hat.vals[2] - covhat)/
             (sig2hat.vals[1] + sig2hat.vals[2] - 2*covhat)
xB.min.hat = 1 - xA.min.hat
c(xA.min.hat, xB.min.hat)
#> asset.B asset.B
#>   0.253   0.747
```

The estimated weights are close to the true weights. The true and estimated expected return and volatility of the global minimum variance portfolio are:

```
# Expected return and volatility of true global minimum variance portfolio
mu.p.min = xA.min*mu.A + xB.min*mu.B
sig2.p.min = xA.min^2 * sig2.A + xB.min^2 * sig2.B +
               2*xA.min*xB.min*sig.AB
sig.p.min = sqrt(sig2.p.min)
c(mu.p.min, sig.p.min)
#> [1] 0.0066 0.0282
# Expected return and volatility of estimated global
# minimum variance portfolio
mu.p.min.hat = as.numeric(xA.min.hat*muhat.vals[1] +
                            xB.min.hat*muhat.vals[2])
sig2.p.min.hat = xA.min.hat^2 * sig2hat.vals[1] +
                 xB.min.hat^2 * sig2hat.vals[2] +
                 2*xA.min.hat*xB.min.hat*covhat
sig.p.min.hat = as.numeric(sqrt(sig2.p.min.hat))
c(mu.p.min.hat, sig.p.min.hat)
#> [1] 0.00937 0.02706
```

The estimated volatility of the global minimum variance portfolio is close to the true volatility but the estimated expected return is much larger than the true expected return. These portfolios are illustrated in Figure 14.6. ■

The statistical properties of Equation 14.11, Equation 14.12 and Equation 14.13 are difficult to derive analytically. For example, suppose you would like to evaluate the bias in the global minimum

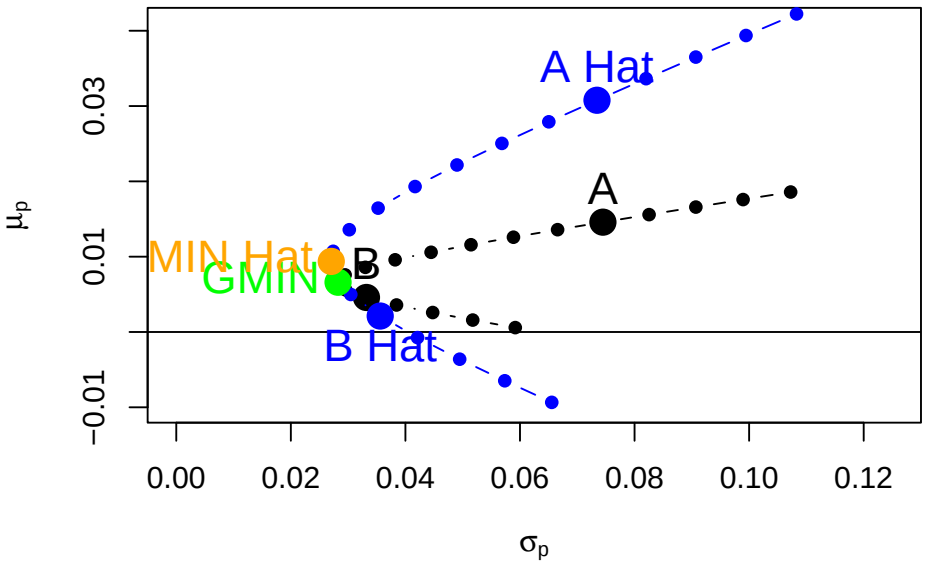


Figure 14.6: True and estimated global minimum variance portfolios.

variance weights. You would need to evaluate

$$E\left[\hat{x}_A^{\min}\right] = E\left[\frac{\hat{\sigma}_B^2 - \hat{\sigma}_{AB}}{\hat{\sigma}_A^2 + \hat{\sigma}_B^2 - 2\hat{\sigma}_{AB}}\right]. \tag{14.14}$$

Now,

$$E\left[\frac{\hat{\sigma}_B^2 - \hat{\sigma}_{AB}}{\hat{\sigma}_A^2 + \hat{\sigma}_B^2 - 2\hat{\sigma}_{AB}}\right] \neq \frac{E\left[\hat{\sigma}_B^2\right] - E\left[\hat{\sigma}_{AB}\right]}{E\left[\hat{\sigma}_A^2\right] + E\left[\hat{\sigma}_B^2\right] - 2E\left[\hat{\sigma}_{AB}\right]}$$

because $E[g(X)] \neq g(E[X])$ for nonlinear functions $g(\cdot)$. Hence, Equation 14.14 is extremely difficult to evaluate analytically. Similarly, suppose you would like to compute $\text{se}(\hat{x}_A^{\min})$. You would need to compute

$$\text{var}\left(\hat{x}_A^{\min}\right) = \text{var}\left(\frac{\hat{\sigma}_B^2 - \hat{\sigma}_{AB}}{\hat{\sigma}_A^2 + \hat{\sigma}_B^2 - 2\hat{\sigma}_{AB}}\right),$$

which is a difficult and tedious calculation and can only be approximated based on the CLT.

The computations are even more difficult for evaluating bias and computing standard errors for Equation 14.12 and Equation 14.13. For example, to evaluate the bias of $\hat{\mu}_{p,\min}$ you would need to calculate

$$E\left[\hat{\mu}_{p,\min}\right] = E\left[\hat{x}_A^{\min}\hat{\mu}_A + \hat{x}_B^{\min}\hat{\mu}_B\right] = E\left[\hat{x}_A^{\min}\hat{\mu}_A\right] + E\left[\hat{x}_B^{\min}\hat{\mu}_B\right]. \tag{14.15}$$

Without knowing more about the joint distributions of the asset means and weights you cannot

simplify Equation 14.15. To compute the standard error of $\hat{\mu}_{p,\min}$ you need to calculate

$$\begin{aligned}\text{var}\left(\hat{\mu}_{p,\min}\right) &= \text{var}\left(\hat{x}_A^{\min}\hat{\mu}_A + \hat{x}_B^{\min}\hat{\mu}_B\right) \\ &= \text{var}(\hat{x}_A^{\min}\hat{\mu}_A) + \text{var}(\hat{x}_B^{\min}\hat{\mu}_B) + 2\text{cov}\left(\hat{x}_A^{\min}\hat{\mu}_A,\, \hat{x}_B^{\min}\hat{\mu}_B\right).\end{aligned}$$

Again, without knowing more about the joint distributions of the asset means and estimated weights you cannot simplify Equation 14.15.

Fortunately, statistical properties of Equation 14.11, Equation 14.12 and Equation 14.13 can be easily quantified using the bootstrap. For each bootstrap sample you calculate these statistics and from the bootstrap distributions you can then evaluate bias and compute standard errors and confidence intervals.

Example 14.8 (True and estimated global minimum variance portfolio with bootstrap estimates for example data)**.** To compute $B = 1000$ bootstrap estimates of Equation 14.11, Equation 14.12 and Equation 14.13 use:

```
# initialize matrices
n.boot = 1000
weights.boot = matrix(0, n.boot, 2)
stats.boot = matrix(0, n.boot, 2)
colnames(weights.boot) = names(mu.vec)
colnames(stats.boot) = c("mu", "sigma")
# bootstrap loop
set.seed(123)
for (i in 1:n.boot) {
   boot.idx = sample(n.obs, replace=TRUE)
   ret.boot = returns.sim[boot.idx, ]
# GWN model estimates
   muhat.boot = colMeans(ret.boot)
   sig2hat.boot = apply(ret.boot, 2, var)
   sigmahat.boot = sqrt(sig2hat.boot)
   covhat.boot = cov(ret.boot)[1,2]
# global minimum variance portfolio weights
   weights.boot[i, 1] = (sig2hat.boot[2] - covhat.boot)/
                        (sig2hat.boot[1] + sig2hat.boot[2] -
                            2*covhat.boot)
   weights.boot[i, 2] = 1 - weights.boot[i, 1]
# portfolio risk return estimates
   stats.boot[i, "mu"] = weights.boot[i, 1]*muhat.boot[1] +
```

```
                        weights.boot[i, 2]*muhat.boot[2]
  sig2.boot = (weights.boot[i, 1]^2 * sig2hat.boot[1] +
               weights.boot[i, 2]^2 * sig2hat.boot[2]
               + 2*weights.boot[i, 1]*weights.boot[i, 2]*covhat.boot)
  stats.boot[i, "sigma"] = sqrt(sig2.boot)
}
```

The bootstrap bias estimates for the global minimum variance portfolio weights are:

```
colMeans(weights.boot) - c(xA.min.hat, xB.min.hat)
#>   asset.A   asset.B
#>  0.000886 -0.000886
```

These values are close to zero suggesting that the estimated weights are unbiased. Notice that the estimates are identical but opposite in sign. This arises because the bootstrap estimates are perfectly negatively correlated:

```
cor(weights.boot)[1,2]
#> [1] -1
```

The bootstrap standard error estimates for the weights are:

```
apply(weights.boot, 2, sd)
#> asset.A asset.B
#>  0.0454  0.0454
```

The standard errors estimates are identical and close to 0.05 which is not too large. Figure Figure 14.7 shows histograms and normal QQ-plots for the bootstrap estimates of the weights. These distributions are centered at the sample estimates (white vertical lines) and look slightly asymmetric.

The bootstrap estimates of bias for the estimated expected return and volatility of the global minimum variance portfolio are:

```
colMeans(stats.boot) - c(mu.p.min.hat, sig.p.min.hat)
#>        mu     sigma
#>  0.000382 -0.000644
```

These small values indicate that the estimates are roughly unbiased. The bootstrap standard error estimates are:

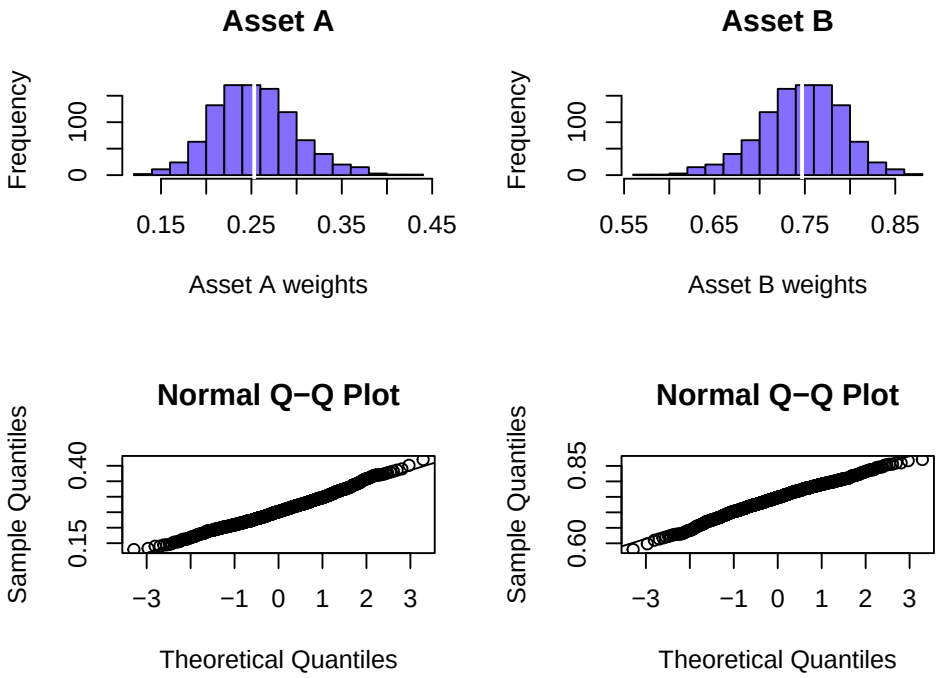


Figure 14.7: Empirical distribution of bootstrap estimates of global minimum variance portfolio weights.

```
apply(stats.boot, 2, sd)
#>      mu   sigma
#> 0.00345 0.00234
```

Here, the bootstrap standard error estimate for $\hat{\mu}_{p,\min}$ is larger than the estimate for $\hat{\sigma}_{p,\min}$ indicating more uncertainty about the mean of the global minimum variance portfolio than the volatility of the portfolio. Interestingly, the bootstrap standard error estimates are close to the naive analytical standard error formulas based on the CLT that ignore estimation error in the weights:

```
sig.p.min.hat/sqrt(n.obs)
#> [1] 0.00349
sig.p.min.hat/sqrt(2*n.obs)
#> [1] 0.00247
```

That is,

$$\text{se}_{\text{boot}}(\hat{\mu}_{p,\min}) \approx \frac{\hat{\sigma}_{p,\min}}{\sqrt{T}}, \; \text{se}_{\text{boot}}\left(\hat{\sigma}_{p,\min}\right) \approx \frac{\hat{\sigma}_{p,\min}}{\sqrt{2 \cdot T}}.$$

The histograms and normal QQ-plots of the bootstrap estimates of $\mu_{p,min}$ and $\sigma_{p,min}$ are illustrated in Figure 14.8, and look like normal distributions.

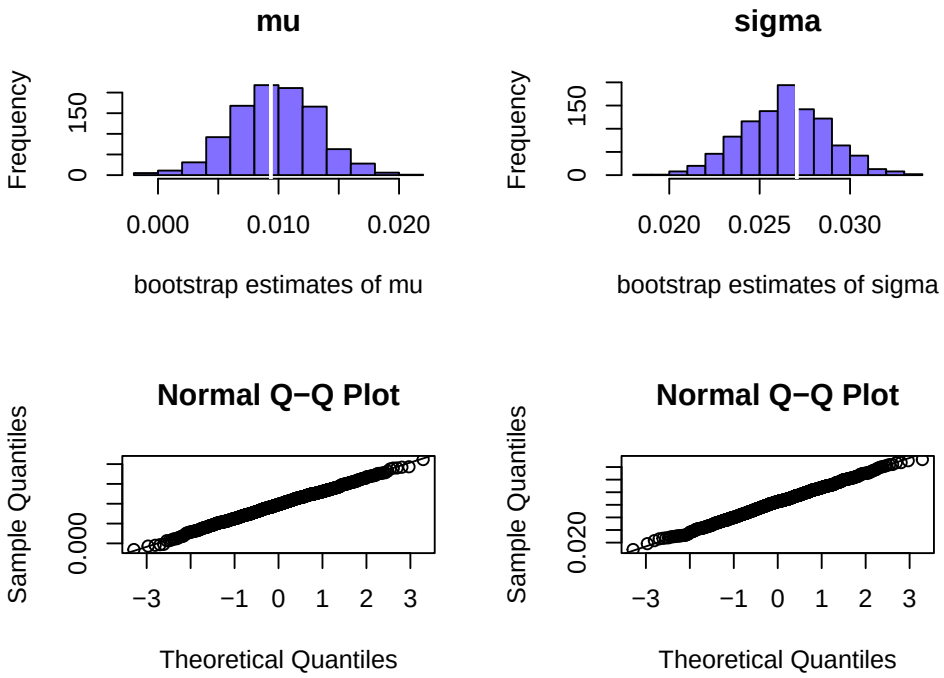


Figure 14.8: Empirical distribution of bootstrap estimates of $\mu_{p,min}$ and $\sigma_{p,min}$.

The sampling uncertainty in $\hat{\mu}_{p,\text{min}}$ and $\hat{\sigma}_{p,\text{min}}$ can be visualized by plotting the bootstrap estimates of $\mu_{p,min}$ and $\sigma_{p,min}$ on the risk return diagram, as shown in Figure 14.9. Clearly there is much more uncertainty about the location of $\mu_{p,min}$ than the location of $\sigma_{p,min}$. ■

To sum up, for the global minimum variance portfolio you have roughly unbiased estimates of the weights, expected return and variance. You have a fairly precise estimate of volatility but an imprecise estimate of expected return.

14.1.5 Statistical properties of the Sharpe ratio

Let r_f denote the rate of return on a risk free asset with maturity equal to the investment horizon of the return R_i on risky asset i. Recall from Chapter 10, the Shapre ratio of asset i is defined as

$$\text{SR}_i = \frac{\mu_i - r_f}{\sigma_i},$$

and is a measure of risk-adjusted expected return. Graphically in the risk-return diagram, SR_i is the slope of a straight line from the risk-free rate that passes through the point (σ_i, μ_i) and represents the risk-return tradeoff of portfolios of the risk-free asset and the risky asset. The estimated Sharpe ratio is

$$\widehat{\text{SR}}_i = \frac{\hat{\mu}_i - r_f}{\hat{\sigma}_i},$$

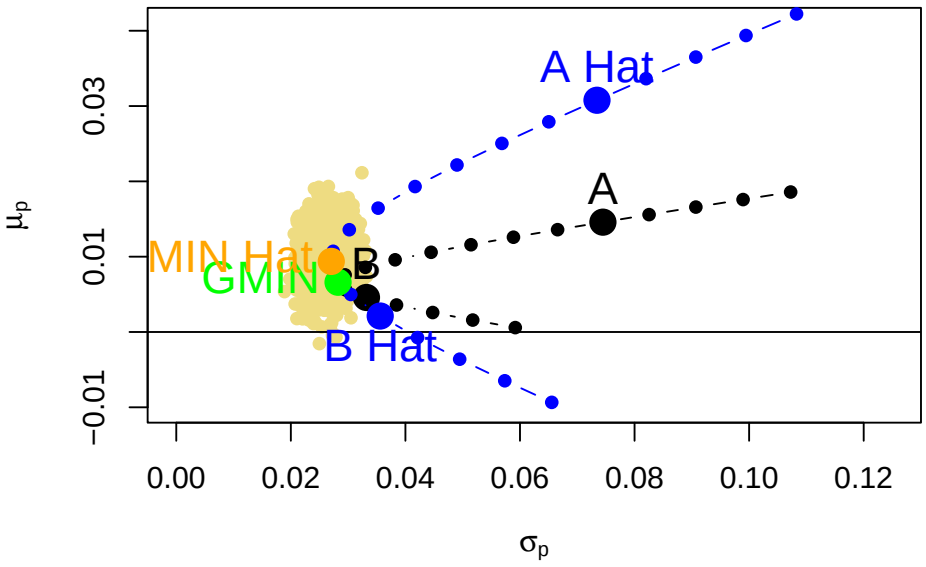


Figure 14.9: Estimation error in estimates of the global minimum variance portfolio.

which inherits estimation error from $\hat{\mu}_i$ and $\hat{\sigma}_i$ in a nonlinear way. As a result, analytically computing the bias and standard error for $\widehat{\text{SR}}_i$ is complicated and typically involves approximations based on the CLT. However, computing numerical estimates of the bias and the standard error for $\widehat{\text{SR}}_i$ using the bootstrap is simple.

Example 14.9 (True and estimated Sharpe ratios for example data)**.** For the example data, assume a monthly risk-free rate of $r_f = 0.03/12 = 0.0025$. The true Sharpe ratios are:

```
r.f = 0.03/12
SR.true = (mu.vec - r.f)/sd.vec
SR.true
#> asset.A asset.B
#>  0.1622  0.0628
```

Here, asset A has a much higher Sharpe ratio than asset B. For the simulated data, the estimated Sharpe ratios are:

```
SR.hat = (muhat.vals - r.f)/sigmahat.vals
SR.hat
#> asset.A asset.B
```

```
#>  0.3849 -0.0107
```

For both assets the estimated Sharpe ratios are quite different from the true Sharpe ratios and are highly misleading. The estimated Sharpe ratio for asset A is more than two times larger than the true Sharpe ratio and indicates a much higher risk adjusted preformance than is actually available. For asset B, the estimated Sharpe ratio is slightly negative when, in fact, the true Sharpe ratio is slightly positive. The true and estimated Sharpe ratios are illustrated in Figure Figure 14.10. The slopes of the black lines are the true Sharpe ratios, and the slopes of the blues lines are the estimated Sharpe ratios. ■

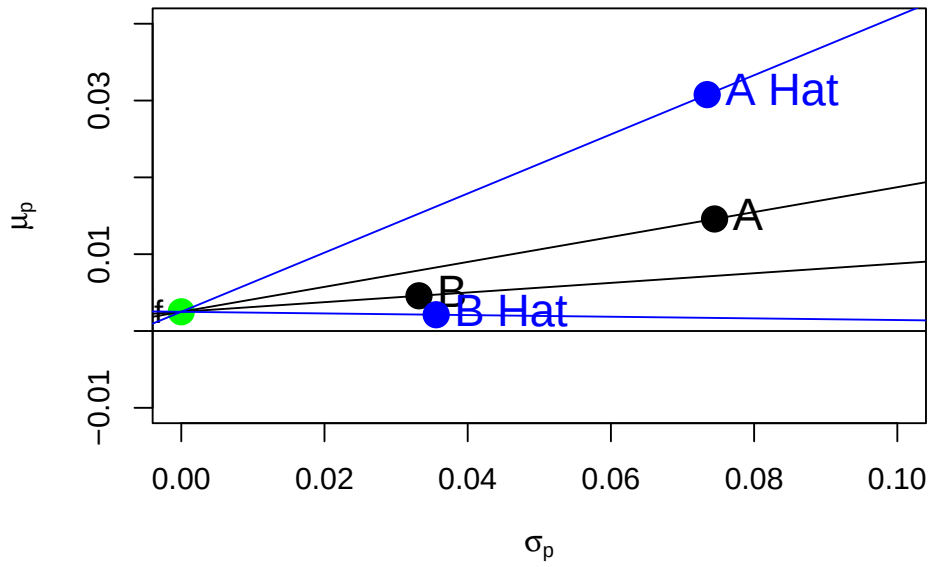


Figure 14.10: True and estimated Sharpe ratios.

Example 14.10 (Delta method standard errors for estimated Sharpe ratio)**.** Let $\theta = (\mu, \sigma)'$. Then the Sharpe ratio can be considered as a function of θ:

$$\text{SR}(\theta) = \frac{\mu - r_f}{\sigma}$$

Since $\hat{\theta}$ is asymptotically normally distributed with covariance given by (), the delta method can be used to obtain an analytic standard error for $\widehat{\text{SR}} = \text{SR}(\hat{\theta})$. It is straightforward to show (see end-of-chapter exercises) that the gradient function is:

$$\mathbf{g}(\theta) = \frac{\partial \text{SR}(\theta)}{\partial \theta} = \begin{pmatrix} \frac{-1}{\sigma} \\ -(\sigma)^{-2}(\mu - r_f) \end{pmatrix}.$$

The delta method variance is then:

$$\mathbf{g}(\theta)' \text{var}(\hat{\theta}) \mathbf{g}(\theta) = \frac{1}{T}\left(1 + \frac{1}{2}\text{SR}^2\right) = \text{se}(\widehat{\text{SR}})^2,$$

The corresponding standard error is:

$$\text{se}(\widehat{\text{SR}}) = \sqrt{\frac{1}{T}\left(1 + \frac{1}{2}\text{SR}^2\right)} \tag{14.16}$$

Notice that $\text{se}(\widehat{\text{SR}})$ increases with SR^2 and decreases with T. The practically useful estimated standard error replace the unknown value of SR with its estimated value $\widehat{\text{SR}}$ and is given by

$$\hat{\text{se}}(\widehat{\text{SR}}) = \sqrt{\frac{1}{T}\left(1 + \frac{1}{2}\widehat{\text{SR}}^2\right)} \tag{14.17}$$

■

Example 14.11 (Applying the analytic delta method to the Sharpe Ratio)**.** The estimated monthly SRs for assets A and B (using $r_f = 0.03/12$), their standard errors, and 95% confidence intervals are computed using:

```
seSR.hat = sqrt((1/n.obs)*(1 + 0.5*(SR.hat^2)))
lower.SR = SR.hat - 2*seSR.hat
upper.SR = SR.hat + 2*seSR.hat
width.SR = upper.SR - lower.SR
ans = rbind(SR.hat, seSR.hat, lower.SR, upper.SR, width.SR)
colnames(ans) = c("Asset A", "Asset B")
rownames(ans) = c("Estimate", "Std Error", "Lower", "Upper", "Width")
ans
#>           Asset A Asset B
#> Estimate    0.385 -0.0107
#> Std Error   0.134  0.1291
#> Lower       0.117 -0.2689
#> Upper       0.653  0.2475
#> Width       0.535  0.5164
```

The estimated monthly SRs for assets A and B are close to zero, with a large estimated standard errors and a 95% confidence interval containing both negative and positive values. Clearly, the SR is not estimated very well. ■

Example 14.12 (Applying the numerical delta method to the Sharpe Ratio)**.** I first create the named inputs $\hat{\theta}$ and $\widehat{\text{var}}(\hat{\theta})$ from the GWN model for simple returns:

```
thetahat = c(muhat.vals[1], sigmahat.vals[1])
names(thetahat) = c("mu", "sigma")
var.thetahat = matrix(c(se.muhat[1]^2,0,0,se.sigmahat[1]^2), 2, 2, byrow=TRUE)
rownames(var.thetahat) = colnames(var.thetahat) = names(thetahat)
thetahat
#>     mu  sigma
#> 0.0308 0.0734
var.thetahat
#>              mu    sigma
#> mu    8.99e-05 0.00e+00
#> sigma 0.00e+00 4.49e-05
```

I use the **car** function `deltaMethod()` to compute the numerical delta method for the estimated Sharpe ratio $\text{SR}(\hat{\theta})$ for asset A as follows:

```
dmSR.A = deltaMethod(thetahat, "(mu-r.f)/sigma", var.thetahat)
dmSR.A
#>                   Estimate    SE 2.5 % 97.5 %
#> (mu - r.f)/sigma    0.385 0.134 0.123   0.65
```

The numerical delta method estimated standard error exactly matches the analytic standard error. The 95% confidence interval returned by the function `deltaMethod()` uses the formula $\hat{\theta} \pm 1.96 \times \widehat{\text{se}}(\hat{\theta})$ as so is slightly different than the analytic result given above. Using similar computations, The numerical delta method for asset B is

```
#>                   Estimate      SE   2.5 % 97.5 %
#> (mu - r.f)/sigma  -0.0107  0.1291 -0.2638   0.24
```

■

Example 14.13 (Delta method for joint distribution of two Shapre ratios)**.** Let $\theta = (\mu_1, \mu_2, \sigma_1, \sigma_2)'$ and define the 2×1 vector of Sharpe ratios:

$$\eta = f(\theta) = \begin{pmatrix} \frac{\mu_A - r_f}{\sigma_A} \\ \frac{\mu_B - r_f}{\sigma_B} \end{pmatrix} = \begin{pmatrix} \text{SR}_A(\theta) \\ \text{SR}_B(\theta) \end{pmatrix},$$

so that $\eta_A = \text{SR}_A(\theta)$ and $\eta_B = \text{SR}_B(\theta)$ where r_f denotes the constant risk-free rate. Consider the $\theta = (\mu_A, \mu_B, \sigma_A, \sigma_B)'$ and $\eta = \text{SR}(\theta) = (\text{SR}_A(\theta), \text{SR}_B(\theta))'$ where

$$\text{SR}_A(\theta) = \frac{\mu_A - r_f}{\sigma_A},\ \text{SR}_B(\theta) = \frac{\mu_B - r_f}{\sigma_B}.$$

Here, I want to use the delta method to get the joint distribution of η. Now, using the chain-rule:

$$\begin{aligned}
&\frac{\partial \text{SR}_A(\theta)}{\partial \mu_A} = \sigma_A^{-1},\ \frac{\partial \text{SR}_A(\theta)}{\partial \mu_B} = 0, \qquad \frac{\partial \text{SR}_A(\theta)}{\partial \sigma_1} = -\sigma_1^{-2}(\mu_1 - r_f),\ \frac{\partial \text{SR}_A(\theta)}{\partial \sigma_2} = 0, \\
&\frac{\partial \text{SR}_B(\theta)}{\partial \mu_1} = 0,\ \frac{\partial \text{SR}_B(\theta)}{\partial \mu_2} = \sigma_2^{-1}, \qquad \frac{\partial \text{SR}_B(\theta)}{\partial \sigma_1} = 0,\ \frac{\partial \text{SR}_B(\theta)}{\partial \sigma_2} = -\sigma_1^{-2}(\mu_1 - r_f).
\end{aligned}$$

Then, using $\text{SR}_i = (\mu_i - r_f)/\sigma_i$ $(i = A, B)$, I have:

$$\mathbf{G}(\theta)' = \begin{pmatrix} \sigma_A^{-1} & 0 & -\sigma_A^{-1}\text{SR}_A & 0 \\ 0 & \sigma_B^{-1} & 0 & -\sigma_B^{-1}\text{SR}_B \quad 0 \end{pmatrix}$$

Using Proposition 7.10 and the results from the previous example I have:

$$\text{var}(\hat{\theta}) = \frac{1}{T}\begin{pmatrix} \sigma_A^2 & \sigma_{AB} & 0 & 0 \\ \sigma_{AB} & \sigma_B^2 & 0 & 0 \\ 0 & 0 & \frac{1}{2}\sigma_A^2 & \frac{1}{2}\sigma_{AB}^2\sigma_A^{-1}\sigma_B^{-1} \\ 0 & 0 & \frac{1}{2}\sigma_{AB}^2\sigma_A^{-1}\sigma_B^{-1} & \frac{1}{2}\sigma_B^2 \end{pmatrix}.$$

Then, after some straightforward matrix algebra calculations (see end-of-chapter exercises) I get:

$$\begin{aligned}
\widehat{\text{var}}(\hat{\eta}) &= \mathbf{G}(\hat{\theta})'\widehat{\text{var}}(\hat{\theta})\mathbf{G}(\hat{\theta}) \\
&= \begin{pmatrix} \widehat{\text{var}}(\widehat{\text{SR}}_A) & \widehat{\text{cov}}(\widehat{\text{SR}}_A, \widehat{\text{SR}}_B) \\ \widehat{\text{cov}}(\widehat{\text{SR}}_A, \widehat{\text{SR}}_B) & \widehat{\text{var}}(\widehat{\text{SR}}_B) \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{T}(1 + \frac{1}{2}\widehat{\text{SR}}_A^2) & \frac{\hat{\rho}_{AB}}{T}(1 + \frac{1}{2}\hat{\rho}_{AB}\widehat{\text{SR}}_A\widehat{\text{SR}}_B) \\ \frac{\hat{\rho}_{AB}}{T}(1 + \frac{1}{2}\hat{\rho}_{AB}\widehat{\text{SR}}_A\widehat{\text{SR}}_B) & \frac{1}{T}(1 + \frac{1}{2}\widehat{\text{SR}}_B^2) \end{pmatrix}.
\end{aligned}$$

The diagonal elements match the delta method variances for the estimated Sharpe ratio derived earlier. The off-diagonal covariance term shows that the two estimated Sharpe ratios are correlated. You explore this correlation in more depth in the end-of-chapter exercises. ■

Example 14.14 (Jackknife standard error for estimated Sharpe ratio)**.** I use the **bootstrap** function `jackknife()` to compute the jacknife estimated standard for the Sharpe ratio for assets A and B:

```
theta.SR = function(x, xdata, r.f) {
  mu.hat = mean(xdata[x, ])
  sigma.hat = sd(xdata[x, ])
  fun.hat = (mu.hat-r.f)/sigma.hat
  fun.hat
}
SR.jack = jackknife(1:n.obs, theta=theta.SR,
                    returns.sim[,"asset.A",drop=FALSE], r.f)
ans = cbind(SR.jack$jack.se, dmSR.A$SE)
colnames(ans) = c("Jackknife", "Delta Method")
rownames(ans) = "SR.A"
ans
#>      Jackknife Delta Method
#> SR.A      0.15        0.134
```

```
#>      Jackknife Delta Method
#> SR.B     0.132        0.129
```

■

Example 14.15 (Bootstraped Sharpe ratios for example data)**.** To illustrate the magnitude of the estimation errors in the estimated Sharpe ratios, Figure 14.11 shows 500 bootstrap estimates of the risk-return pairs (σ_A, μ_A) and (σ_B, μ_B) along with the maximum and minimum bootstrap Sharpe ratio estimates for each asset. The bootstrap Sharpe ratio estimates are computed using:

```
n.boot = 500
mu.boot = matrix(0, n.boot, 2)
sd.boot = matrix(0, n.boot, 2)
SR.boot = matrix(0, n.boot, 2)
colnames(mu.boot) = colnames(sd.boot) = colnames(SR.boot) = c("Asset.A", "Asset.B")
set.seed(123)
for (i in 1:n.boot) {
   boot.idx = sample(n.obs, replace=TRUE)
   ret.boot = returns.sim[boot.idx, ]
   mu.boot[i, ] = colMeans(ret.boot)
   sd.boot[i, ] = apply(ret.boot, 2, sd)
```

```
    SR.boot[i, ] = (mu.boot[i, ] - r.f)/sd.boot[i, ]
}
# find index of max and min bootstrap Sharpe ratios
maxSRidx.A = which(SR.boot[, 1] == max(SR.boot[, 1]))
minSRidx.A = which(SR.boot[, 1] == min(SR.boot[, 1]))
maxSRidx.B = which(SR.boot[, 2] == max(SR.boot[, 2]))
minSRidx.B = which(SR.boot[, 2] == min(SR.boot[, 2]))
```

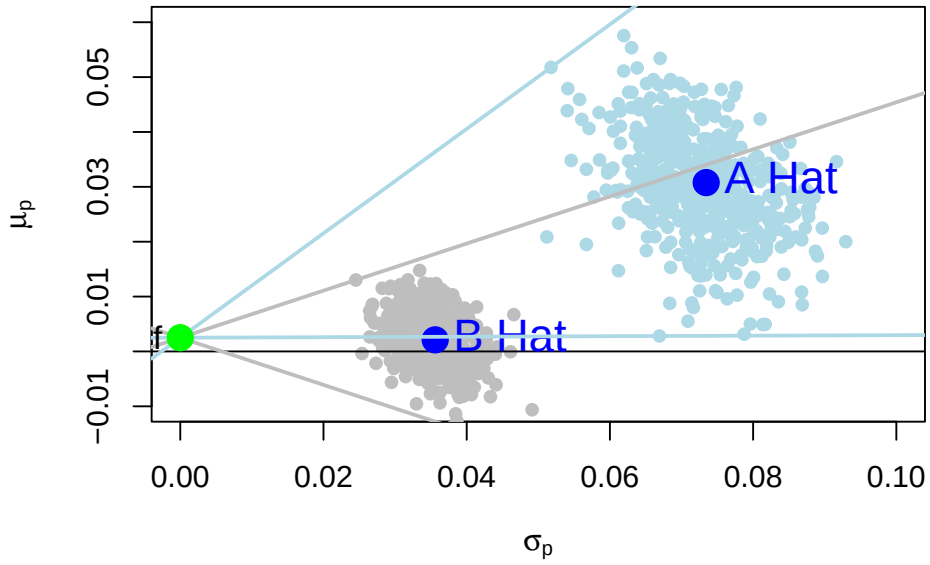


Figure 14.11: Bootstrap risk-return estimates with maximum and minimum Sharpe ratios.

The maximum and minimum bootstrap Sharpe ratio estimates for asset B are the two grey straight lines from the risk-free rate that have the highest and lowest slopes, respectively. The maximum and minimum bootstrap Sharpe ratio estimates for asset A are the two light blue lines with the highest and lowest slopes, respectively. These maximum and minimum estimates are:

```
apply(SR.boot, 2, max)
#> Asset.A Asset.B
#>   0.952   0.429
apply(SR.boot, 2, min)
#>  Asset.A  Asset.B
#>  0.00467 -0.42946
```

For each asset the difference in the maximum and minimum slopes is substantial and indicates a

large about of estimation error in the estimated Sharpe ratios.

The statistical properties of the estimated Sharpe ratios can be estimated from the bootstrap samples. The bootstrap estimates of bias are:

```
# bias
biasSR.boot = colMeans(SR.boot) - c(SR.true)
biasSR.boot
#> Asset.A Asset.B
#>  0.2316 -0.0642
# bias relative to true values
biasSR.boot/SR.true
#> Asset.A Asset.B
#>    1.43   -1.02
```

Here you see there is a substantial upward bias in $\widehat{\text{SR}}_A$ (bias relative to true Sharpe ratio is 141%) and a substantial downward bias in $\widehat{\text{SR}}_B$ (bias relative to true Sharpe ratio is -105%). A crude bias adjusted Sharpe ratio estimate, which substracts the estimated bias from the sample estimate, gives results much closer to the true Sharpe ratios:

```
SR.hat.biasAdjust = SR.hat - biasSR.boot
SR.hat.biasAdjust
#> asset.A asset.B
#>  0.1533  0.0534
SR.true
#> asset.A asset.B
#>  0.1622  0.0628
```

The bootstrap estimated standard errors for both assets are big and indicate that the Sharpe ratios are not estimated well:

```
apply(SR.boot, 2, sd)
#> Asset.A Asset.B
#>   0.154   0.132
```

Figure 14.12 shows the histograms and normal QQ-plots of the bootstrap distributions for the estimated Sharpe ratios. The histograms for both assets show the wide dispersion of the bootstrap Sharpe ratio estimates that was illustrated in Figure 14.11. The histogram and QQ-plot for asset A shows a positive skewness whereas the histogram and QQ-plot for asset B show a more symmetric distribution. The quantile-based 95% confidence intervals for the true Sharpe ratios are:

```r
# 95% CI for SR.A
quantile(SR.boot[, 1], probs=c(0.025, 0.975))
#>   2.5%  97.5%
#> 0.0964 0.7093
# 95% CI for SR.B
quantile(SR.boot[, 2], probs=c(0.025, 0.975))
#>   2.5%  97.5%
#> -0.261  0.263
```

These intervals are quite large and indicate much uncertainty about the true Sharpe ratio values.
■

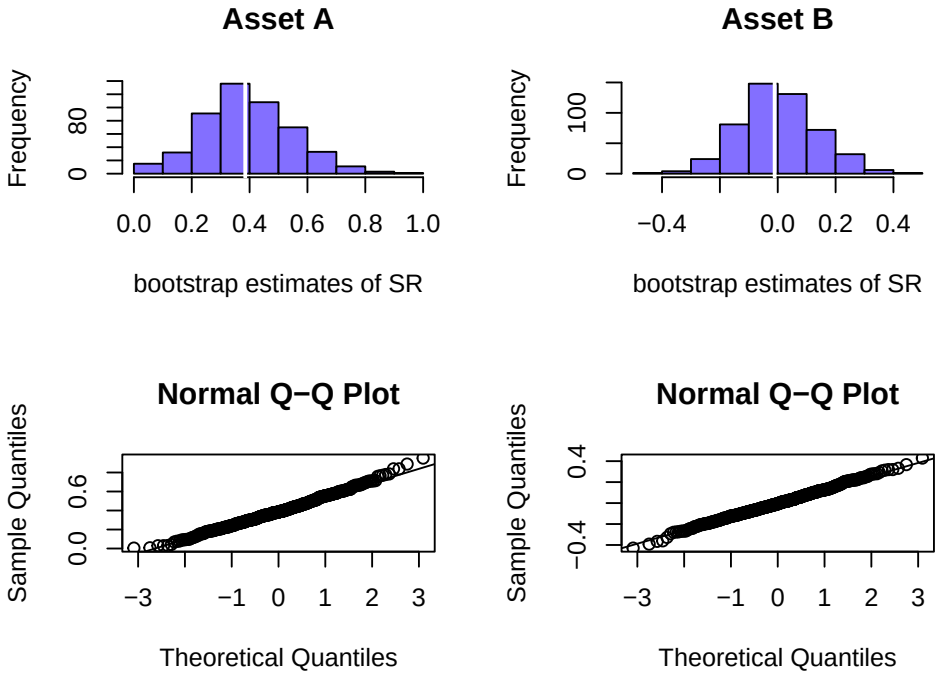


Figure 14.12: Bootstrap distributions of estimated Sharpe ratios.

14.2 Hypothesis Testing for Sharpe Ratio

To be completed

14.3 Rolling Analysis of Portfolios

To be completed...

14.4 Further Reading: Statistical Analysis of Portfolios

- Mention other quadratic programming packages: ROI etc. See Doug's book
- Mention Pfaff's R book
- Mention Grilli's book etc.
- Broadie, M. (1993). "Computing efficient frontiers using estimated parameters", Annals of Operations Research 45, 21-58. Describes the logic behind actual frontiers
- Jorion, P. (1992). "Portfolio Optimization in Practice. Financial Analysts Journal", 48(1), 68–74.
- Michaud (2008), *Efficient Asset Management: A Practical Guide to Stock Portfolio Optimization and Asset Allocation*, Second Edition. To be completed...

14.5 Exercises

Exercise 14.1. Show that $\hat{\sigma}_p^2$ in Equation 14.6 with $\hat{\sigma}_p^2$ in Equation 14.3 are the same when the sample covariance matrix is used in Equation 14.6.

{Ruppert} Ruppert, D.A.. *Statistics and Data Analysis for Financial Engineering*, Springer-Verlag.

{MartinSchererYollin}Martin, R.D., Scherer, B., and Yollin, G. (2016).

15 Single Index Model

Updated: September 18, 2026

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15.1 Introduction

To be completed

Notes:

1. For consistency follow the chapter format of the GWN model
2. Give the model in excess return form - stress the interpretation and linkage later on to the CAPM and general factor models

Outline

1. Monte Carlo Simulation
2. Single Index Model and Portfolio Theory
    a. portfolio optimization using SI covariance matrix
        i. always pd, reduces number of estimated components in matrix.
        ii. Compare results with sample covariance matrix - in estimation chapter
    b. Risk analysis of asset and portfolios: factor risk reports

The R packages used in this chapter are **corrplot**, **introcompfinr**, **xts**, and **PerformanceAnalytics**.

15.2 Motivation

The GWN model for monthly asset returns assumes that returns are jointly normally distributed with a constant mean vector and covariance matrix. This model captures many of the stylized facts for monthly asset returns. While the GWN model captures the covariances and correlations between different asset returns it does not explain where these covariances and correlations come

from. The single index model is an extension of the GWN model that explains the covariance and correlation structure among asset returns as resulting from common exposures to an underlying market index. The intuition behind the single index model can be illustrated by examining some example monthly returns.

Example 15.1 (Monthly returns for single index model examples)**.** For the single index model examples in this chapter, consider the monthly adjusted closing prices on four Northwest stocks (with tickers in parentheses): Boeing (BA), Nordstrom (JWN), Microsoft (MSFT), Starbucks (SBUX). I will also use the monthly adjusted closing prices on the S&P 500 index as the proxy for the market index. These prices are extracted from the **introcompfinr** package as follows:

```
baPrices = to.monthly(baDailyPrices, OHLC=FALSE)
jwnPrices = to.monthly(jwnDailyPrices, OHLC=FALSE)
msftPrices = to.monthly(msftDailyPrices, OHLC=FALSE)
sbuxPrices = to.monthly(sbuxDailyPrices, OHLC=FALSE)
sp500Prices = to.monthly(sp500DailyPrices, OHLC=FALSE)
siPrices = merge(baPrices, jwnPrices, msftPrices,
                 sbuxPrices, sp500Prices)
```

The data sample is from January, 1998 through May, 2012 and returns are simple returns:

```
smpl = "1998-01::2012-05"
siPrices = siPrices[smpl]
siRetS = na.omit(Return.calculate(siPrices, method="simple"))
```

Prices are shown in Figure 15.1. The time plots of prices show some common movements among the stocks that are similar to movements of the S&P 500 index . During the dot-com boom-bust at the beginning of the sample, except for Starbucks, prices rise during the boom and fall during the bust. For all assets, prices rise during the five year boom period prior to the financial crisis, fall sharply after 2008 during the bust, and then rise afterward.

Figure 15.2 shows time plots of the monthly returns on each stock together with the return on the S&P 500 index.[1]

Figure 15.2 shows that the individual stock returns are more volatile than the S&P 500 returns, and that the movement in stock returns (orange lines) tends to follow the movements in the S&P 500 returns indicating positive covariances and correlations.

The sample return covariance and correlation matrices are computed using:

[1] The `plot()` method for **xts** objects creates a plot object. Calling `lines()` adds additional series to the plot. You only want to print the final plot object.

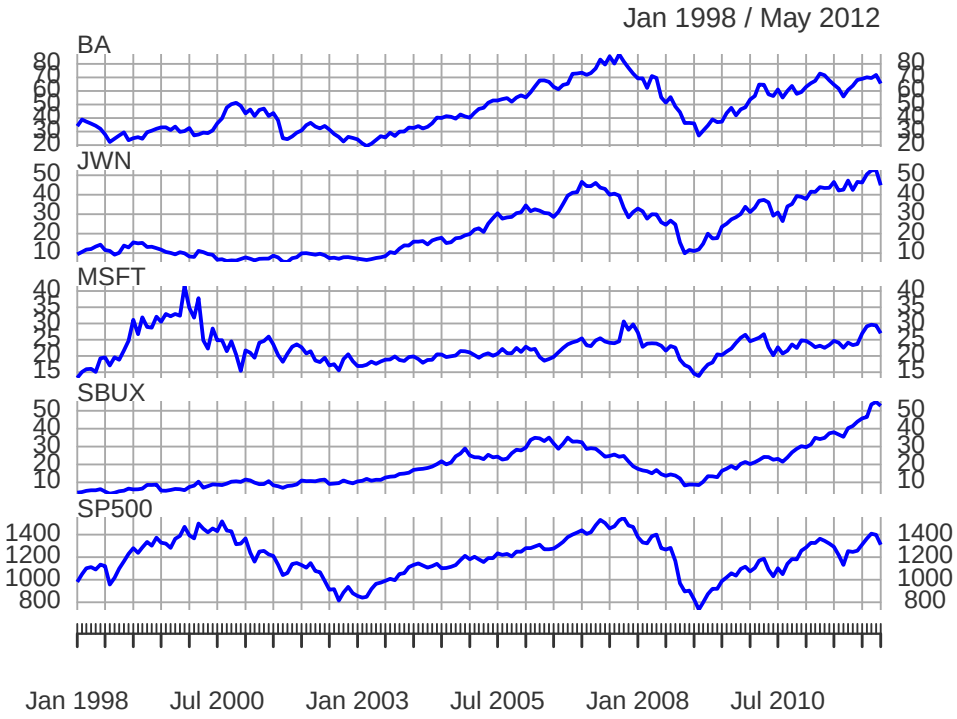


Figure 15.1: Monthly closing prices on four Northwest stocks and the S&P 500 index.

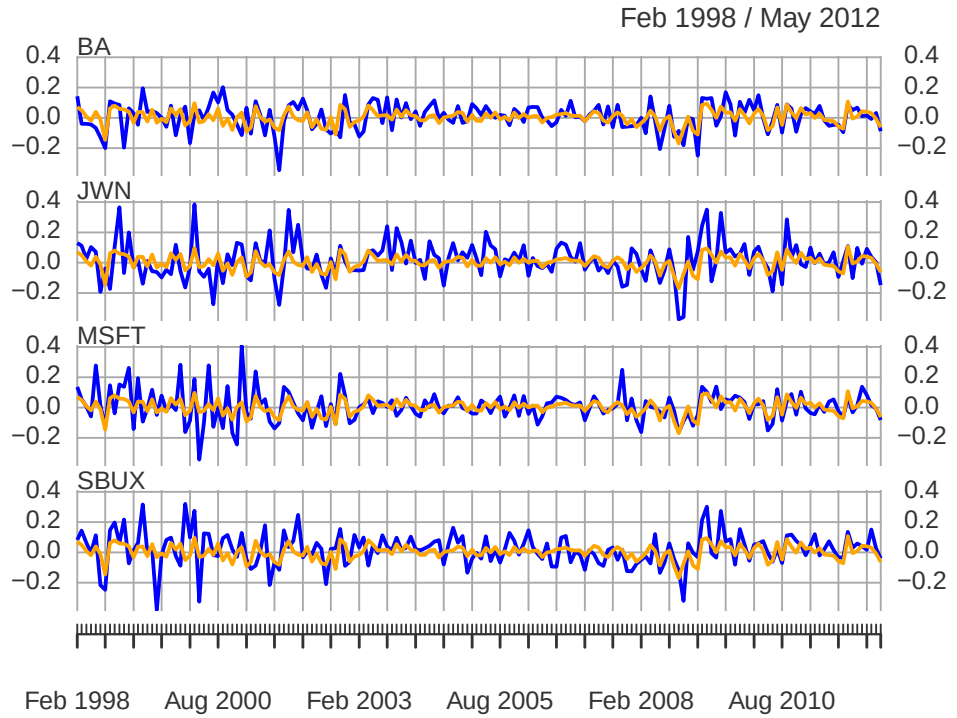


Figure 15.2: Monthly returns on four Northwest stocks. The orange line in each panel is the monthly return on the stock, and the blue line is the monthly return on the S&P 500 index.

```
covmatHat = cov(siRetS)
cormatHat = cov2cor(covmatHat)
```

The sample correlation matrix is visualized in Figure 15.3 using the **corrplot** function `corrplot.mixed()`:

```
corrplot.mixed(cormatHat, upper="ellipse")
```

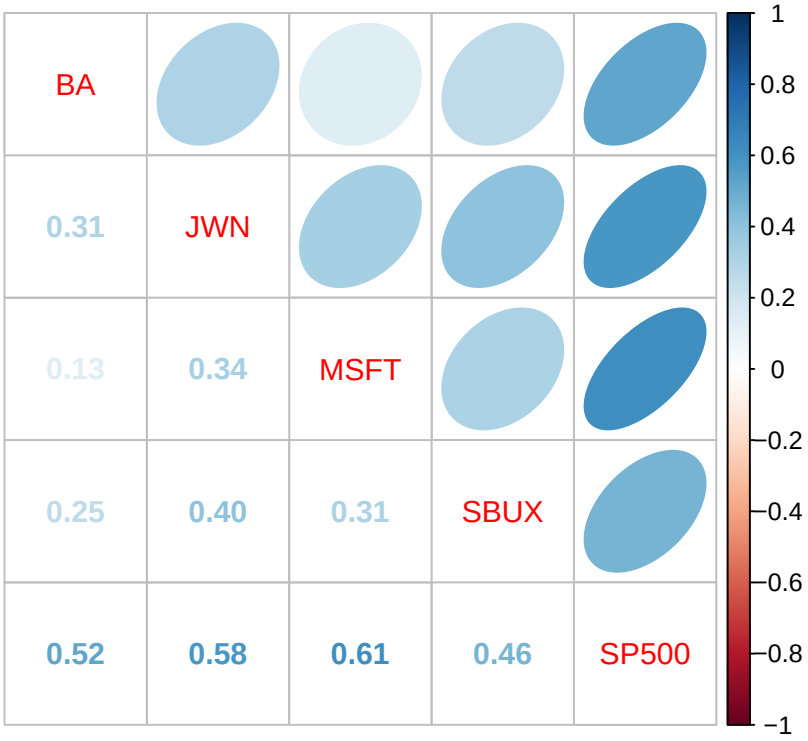


Figure 15.3: Sample correlation matrix of monthly returns.

All returns are positively correlated and each stock return has the highest positive correlation with the S&P 500 index.

The positive covariance and correlation of each stock return with the market return can also be visualized with scatterplots as illustrated in Figure 15.4:

The scatterplots show that as the market index return increases, the returns on each stock increase in a linear way. ■

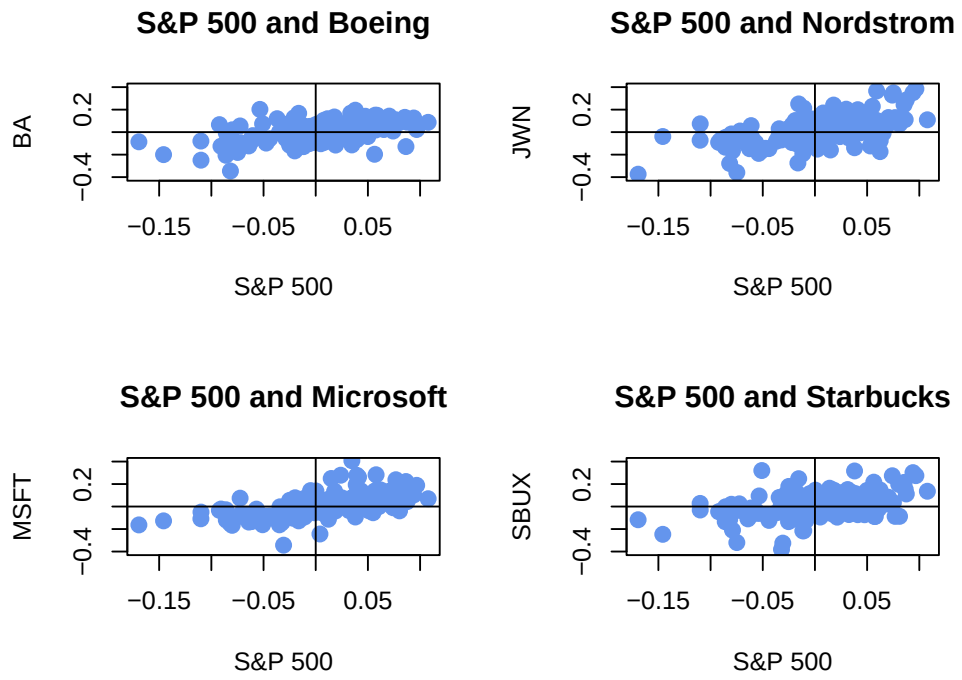


Figure 15.4: Monthly return scatterplots of each stock vs. the S&P 500 index.

15.3 William Sharpe's SI Model

- What is SI model used for? Extension of the GWN model to capture the stylized fact of a common component in the returns of many assets.
  - Provides an explanation as to why assets are correlated with each other: share an exposure to a common source.
  - Provides a simplification of the covariance matrix of a large number of asset returns.
  - Provides additional intuition about risk reduction through diversification.
- Example of a factor model for asset returns. Factor models are used heavily in academic theories of asset pricing and in industry for explaining asset returns, portfolio construction and risk analysis. Sharpe's SI model is the most widely used.
- Motivate by showing positive correlation between individual asset returns and market index (sp500)
- Mostly used for simple returns; but can be used for cc returns for risk analysis.

Let R_{it} denote the simple return on asset i over the investment horizon between times $t-1$ and t. Let R_{Mt} denote the simple return on a well diversified market index portfolio, such as the S&P 500 index. The *single index (SI) model* for R_{it} has the form:[2]

[2]The single index model is also called the* market model* or the *single factor model*.

$$R_{it} = \beta_{i,0} + \beta_{i,M} R_{Mt} + \varepsilon_{it}, \tag{15.1}$$

$$R_{Mt} \sim iid\, N(\mu_M, \sigma_M^2), \tag{15.2}$$

$$\varepsilon_{it} \sim \text{GWN}\,(0, \sigma_{\varepsilon,i}^2), \tag{15.3}$$

$$\text{cov}(R_{Mt}, \varepsilon_{is}) = 0, \ \textit{for all } t \textit{ and } s, \tag{15.4}$$

$$\text{cov}(\varepsilon_{it}, \varepsilon_{js}) = 0, \text{ for all } i \neq j \text{ and all } t \text{ and } s. \tag{15.5}$$

The SI model Equation 15.1 - Equation 15.5 assumes that all individual asset returns, R_{it}, are: (1) iid, (2) normally distributed, (3) a linear function of the market index return, R_{Mt}, and an asset specific error term, ε_{it}. The market index, R_{Mt} is assumed to be independent of the asset specific error term ε_{it}. In contrast to the GWN model, in the SI model the asset specific error terms are assumed to be independent across assets. That is, $\text{cov}(\varepsilon_{it}, \varepsilon_{js}) = 0$ for all $i \neq j$.

The SI model is an important extension of the regression form of the GWN model. In the SI model individual asset returns are explained by two distinct sources: (1) a common (to all assets) market-wide source R_{Mt}, and (2) an asset specific source ε_{it}. R_{Mt} represents common news that impacts all assets and ε_{it} represents news that is only specific to asset i. In the SI model, different assets are contemporaneously correlated because they share an exposure to a common market index.

15.3.1 Interpretation of single index model

Interpretation of $\beta_{i,M}$

First, consider the interpretation of $\beta_{i,M}$ in Equation 15.1. The coefficient $\beta_{i,M}$ is called the asset's *market exposure* or *market "beta"* . It represents the slope coefficient in the linear relationship between R_{it} and R_{Mt}. Because it is assumed that R_{Mt} and ε_{it} are independent, $\beta_{i,M}$ can be interpreted as the partial derivative

$$\frac{\partial R_{it}}{\partial R_{Mt}} = \frac{\partial}{\partial R_{Mt}}(\beta_{i,0} + \beta_{i,M} R_{Mt} + \varepsilon_{it}) = \beta_{i,M}.$$

Then, for small changes in R_{Mt} and holding ε_{it} fixed, I have the approximation

$$\frac{\Delta R_{it}}{\Delta R_{Mt}} \approx \beta_{i,M} \Rightarrow \Delta R_{it} \approx \beta_{i,M} \times \Delta R_{Mt},$$

Hence, for a given change in the market return $\beta_{i,M}$ determines the magnitude of the response of asset $i's$ return. The larger (smaller) is $\beta_{i,M}$ the larger (smaller) is the response of asset i to the movement in the market return.

The coefficient $\beta_{i,M}$ in Equation 15.1 has another interpretation that is directly related to portfolio risk budgeting. To see this, assume that the market index is a diversified portfolio of N assets

$$R_{Mt} = x_1 R_1 + \cdots + x_N R_N.$$

In Chapter 13 I derived the following results using the $\sigma - \beta$ decomposition:

$$\beta_{i,M} = \frac{\text{cov}(R_{it}, R_{Mt})}{\text{var}(R_{Mt})} = \frac{\sigma_{iM}}{\sigma_M^2}, \tag{15.6}$$

$$\text{MCR}_i^{\sigma_M} = \beta_{i,M}\sigma_M, \tag{15.7}$$

$$\text{CR}_i^{\sigma_M} = x_i\beta_{i,M}\sigma_M \tag{15.8}$$

$$\text{PCR}_i^{\sigma_M} = x_i\beta_{i,M} \tag{15.9}$$

Here, you see that $\beta_{i,M}$ is the "beta" of asset i with respect to the market index portfolio and is proportional to the marginal contribution of asset i to the volatility of the market index portfolio. Therefore, assets with large (small) values of $\beta_{i,M}$ have large (small) contributions to the volatility of the market index portfolio. In this regard, $\beta_{i,M}$ can be thought of as a measure of an asset's portfolio risk. Increasing allocations to assets with high (low) $\beta_{i,M}$ values will increase (decrease) portfolio risk (as measured by portfolio volatility). In particular, when $\beta_{i,M} = 1$ asset $i's$ percent contribution to the volatility of the market portfolio is equal to its allocation weight. When $\beta_{i,M} > 1$ asset $i's$ percent contribution to the volatility of the market portfolio is greater than its allocation weight, and when $\beta_{i,M} < 1$ asset $i's$ percent contribution to the volatility of the market portfolio is less than its allocation weight.

The derivation of Equation 15.6 using Equation 15.1 - Equation 15.5 is straightforward. Using Equation 15.1 I can write

$$\begin{aligned}
\text{cov}(R_{it}, R_{Mt}) &= \text{cov}(\beta_{i,0} + \beta_{i,M}R_{Mt} + \varepsilon_{it}, R_{Mt}) \\
&= \text{cov}(\beta_{i,M}R_{Mt}, R_{Mt}) + \text{cov}(\varepsilon_{it}, R_{Mt}) \\
&= \beta_{i,M}\text{var}(R_{Mt}) \quad (\text{since } \text{cov}(\varepsilon_{it}, R_{Mt}) = 0) \\
\Rightarrow \beta_{i,M} &= \frac{\text{cov}(R_{it}, R_{Mt})}{\text{var}(R_{Mt})} = \frac{\sigma_{iM}}{\sigma_M^2}.
\end{aligned}$$

Interpretation of $\beta_{i,0}$

The parameter $\beta_{i,0}$ is the intercept in the linear SI model relationship between R_{it} and R_{Mt}. To interpret $\beta_{i,0}$, use Equation 15.1 and take expectations:

$$\begin{aligned}
E[R_{it}] = \mu_i &= \beta_{i,0} + \beta_{i,M}E[R_{Mt}] + E[\varepsilon_{it}] \\
&= \beta_{i,0} + \beta_{i,M}\mu_M
\end{aligned}$$

Solving for $\beta_{i,0}$ gives

$$\beta_{i,0} = \mu_i - \beta_{i,M}\mu_M. \tag{15.10}$$

Hence, the intercept term $\beta_{i,0}$ can be interpreted as the average return on asset i that is in excess of the average return due to the market.

Using Equation 15.10, the SI model model Equation 15.1 can be re-expressed as

$$\begin{aligned} R_{it} &= \beta_{i,0} + \beta_{i,M}R_{Mt} + \varepsilon_{it} \\ &= \mu_i - \beta_{i,M}\mu_M + \varepsilon_{it} \\ &= \mu_i + \beta_{i,M}(R_{Mt} - \mu_M) + \varepsilon_{it}. \end{aligned} \tag{15.11}$$

Interpretation of R_{Mt} and ε_{it}

To aid in the interpretation of R_{Mt} and ε_{it} in Equation 15.1, consider Equation 15.11 expressed in terms of asset $i's$ suprise return $R_{it} - \mu_i$:

$$R_{it} - \mu_i = \beta_{i,M}(R_{Mt} - \mu_M) + \varepsilon_{it}. \tag{15.12}$$

In Equation 15.12, you see that asset $i's$ suprise return $R_{it} - \mu_i$ is equal to a component proportional to the surprise return on the market index, $\beta_{i,M}(R_{Mt} - \mu_M)$, and an asset specific surprise component, ε_{it}. You can think of $R_{Mt} - \mu_M$ as capturing unexpected "market-wide" news at time t that is common to all assets, and $\beta_{i,M}$ captures the sensitivity or exposure of asset i to this unexpected market-wide news. Equation 15.12 shows that an asset's surprise return is zero if the market's surprise return is zero and the asset specific surprise return is zero (or the two surprise returns exactly counter-act each other). An example of unexpected market-wide news is the release of information by the government about the national unemployment rate. If the news is good then you might expect $R_{Mt} - \mu_M$ to increase because general business conditions are good. Different assets will respond differently to this news and this differential impact is captured by $\beta_{i,M}$. Assets with positive values of $\beta_{i,M}$ will see their returns increase because of good unexpected market-wide news, and assets with negative values of $\beta_{i,M}$ will see their returns decrease because of this good news. Hence, the magnitude and direction of correlation between asset returns can be partially explained by their exposures to common unexpected market-wide news. Because ε_{it} is assumed to be independent of R_{Mt} and of ε_{jt}, you can think of ε_{it} as capturing unexpected specific news to asset i that is unrelated to market news or to unexpected specific news to any other asset j. Examples of unexpected asset-specific news are company earnings reports and reports about corporate restructuring (e.g., CEO resigning). Hence, unexpected specific news for asset i only effects the return on asset i and not the return on any other asset j.

The GWN model

$$R_{it} = \mu_i + \varepsilon_{it}$$

does not distinguish between overall market and asset specific news and so allows the unexpected news shocks ε_{it} to be correlated across assets. Some news is common to all assets but some is specific to a given asset and the GWN model error includes both types of news. In this regard, the GWN model for an asset's return is not a special case of the SI model when $\beta_{i,M} = 0$ because the SI model assumes that ε_{it} is uncorrelated across assets.

15.3.2 Statistical properties of returns in the SI model

In this sub-section I present and derive the statistical properties of returns in the SI model Equation 15.1 - Equation 15.5. I will derive two types of statistical properties: unconditional and conditional. The unconditional properties are based on unconditional or marginal distribution of returns. The conditional properties are based on the distribution of returns conditional on the value of the market return.

15.3.2.1 Unconditional properties

The unconditional properties of returns in the SI model Equation 15.1 - Equation 15.5 are summarized in the following Proposition:

Proposition 15.1 (Unconditional properties of returns in the SI model)**.** *Let returns be described by the SI model Equation 15.1 - Equation 15.5. Then*

$$E[R_{it}] = \mu_i = \beta_{i,0} + \beta_{i,M}\mu_M, \tag{15.13}$$

$$\text{var}(R_{it}) = \sigma_i^2 = \beta_{i,M}^2\sigma_M^2 + \sigma_{\varepsilon,i}^2, \tag{15.14}$$

$$\text{cov}(R_{it}, R_{jt}) = \sigma_{ij} = \beta_{i,M}\beta_j\sigma_M^2, \tag{15.15}$$

$$\text{cor}(R_{it}, R_{jt}) = \rho_{ij} = \frac{\beta_{i,M}\beta_j\sigma_M^2}{\sqrt{(\beta_{i,M}^2\sigma_M^2 + \sigma_{\varepsilon,i}^2)(\beta_j^2\sigma_M^2 + \sigma_{\varepsilon,j}^2)}}, \tag{15.16}$$

$$R_{it} \sim iid\, N(\mu_i, \sigma_i^2) = N(\beta_{i,0} + \beta_{i,M}\mu_M, \beta_{i,M}^2\sigma_M^2 + \sigma_{\varepsilon,i}^2) \tag{15.17}$$

The derivations of these properties are straightforward and are left as end-of-chapter exercises.

From property Equation 15.14, an asset's return variance is additively decomposed into two independent components:

$$\begin{aligned} \text{var}(R_{it}) = & \quad \sigma_i^2 = \beta_{i,M}^2\sigma_M^2 + \sigma_{\varepsilon,i}^2 \\ (\text{total asset } i \text{ variance} = & \quad \text{market variance} + \text{asset specific variance}) \end{aligned} \tag{15.18}$$

Here, $\beta_{i,M}^2\sigma_M^2$ is the contribution of the market index return R_{Mt} to the total variance of asset i, and $\sigma_{\varepsilon,i}^2$ is the contribution of the asset specific component ε_{it} to the total return variance, respectively. If you divide both sides of Equation 15.18 by σ_i^2 you get

$$\begin{aligned} 1 &= \frac{\beta_{i,M}^2\sigma_M^2}{\sigma_i^2} + \frac{\sigma_{\varepsilon,i}^2}{\sigma_i^2} \\ &= R^2 + (1 - R^2), \end{aligned}$$

where

$$R^2 = \frac{\beta_{i,M}^2\sigma_M^2}{\sigma_i^2} \tag{15.19}$$

is the proportion of asset $i's$ variance that is explained by the variability of the market return R_{Mt} and

$$1 - R^2 = \frac{\sigma_{\varepsilon,i}^2}{\sigma_i^2} \tag{15.20}$$

is the proportion of asset $i's$ variance that is explained by the variability of the asset specific component ε_{it}.

- Sharpe's rule of thumb. A typical stock has $R^2 = 0.30$. That is, 30% of an asset's variability is explained by the market movements. - R^2 can be interpreted as the fraction of risk that is non-diversifiable, $1 - R^2$ gives the fraction of risk that is diversifiable. Come back to this point after discussing the SI model and portfolios below.
- This is an example of factor risk budgeting

Properties Equation 15.15 and Equation 15.16 show how assets are correlated in the SI model. In particular,

- $\sigma_{ij} = 0$ if $\beta_{i,M} = 0$ or $\beta_j = 0$ or both. Assets i and j are uncorrelated if asset i or asset j or both do not respond to market news.
- $\sigma_{ij} > 0$ if $\beta_{i,M}, \beta_j > 0$ or $\beta_{i,M}, \beta_j < 0$. Assets i and j are positively correlated if both assets respond to market news in the same direction.
- $\sigma_{ij} < 0$ if $\beta_{i,M} > 0$ and $\beta_j < 0$ or if $\beta_{i,M} < 0$ and $\beta_j > 0$. Assets i and j are negatively correlated if they respond to market news in opposite directions.

From Equation 15.16, assets i and j are perfectly correlated ($\rho_{ij} = \pm 1$) only if $\sigma_{\varepsilon,i} = \sigma_{\varepsilon,j} = 0$.

Property Equation 15.17 shows that the distribution of asset returns in the SI model is normal with mean and variance given by Equation 15.13 and Equation 15.14, respectively.

In summary, the unconditional properties of returns in the SI model are similar to the properties of returns in the GWN model: Returns are covariance stationary with constant means, variances, and covariances. Returns on different assets can be contemporaneously correlated and all asset returns uncorrelated over time. The SI model puts more structure on the expected returns, variances and

covariances than the GWN model and this allows for a deeper understanding of the behavior of asset returns.

15.3.2.2 Conditional properties

The properties of returns in the SI model Equation 15.1 - Equation 15.5 conditional on $R_{Mt} = r_{Mt}$ are:

$$\begin{aligned}
E[R_{it}|R_{Mt} &= r_{Mt}] = \beta_{i,0} + \beta_{i,M} r_{Mt}, && (15.21)\\
\text{var}(R_{it}|R_{Mt} &= r_{Mt}] = \sigma^2_{\varepsilon,i}, && (15.22)\\
\text{cov}(R_{it}, R_{jt}|R_{Mt} &= r_{Mt}] = 0, && (15.23)\\
\text{cor}(R_{it}, R_{jt}|R_{Mt} &= r_{Mt}] = 0, && (15.24)\\
R_{it}|R_{Mt} &\sim iid\, N(\beta_{i,0} + \beta_{i,M} r_{Mt}, \sigma^2_{\varepsilon,i}). && (15.25)
\end{aligned}$$

Recall, conditioning on a random variable means you observe its value. In the SI model, once you observe the market return two important things happen: (1) an asset's return variance reduces to its asset specific variance; and (2) asset returns become uncorrelated.

15.3.3 SI model and portfolios

A nice feature of the SI model for asset returns is that it also holds for a portfolio of asset returns. This property follows because asset returns are a linear function of the market return. To illustrate, consider a two asset portfolio with investment weights $x_1$and x_2 where each asset return is explained by the SI model:

$$\begin{aligned}
R_{1t} &= \beta_{1,0} + \beta_{1,M} R_{Mt} + \varepsilon_{1t},\\
R_{2t} &= \beta_{2,0} + \beta_{2,M} R_{Mt} + \varepsilon_{2t}.
\end{aligned}$$

Then the portfolio return is

$$\begin{aligned}
R_{p,t} &= x_1 R_{1t} + x_2 R_{2t}\\
&= x_1(\beta_{1,0} + \beta_{1,M} R_{Mt} + \varepsilon_{1t}) + x_2(\beta_{2,0} + \beta_{2,M} R_{Mt} + \varepsilon_{2t})\\
&= \left(x_1\beta_{1,0} + x_2\beta_{2,0}\right) + \left(x_1\beta_{1,M} + x_2\beta_{2,M}\right) R_{Mt} + (x_1\varepsilon_{1t} + x_2\varepsilon_{2t})\\
&= \beta_{p,0} + \beta_{p,M} R_{Mt} + \varepsilon_{p,t}
\end{aligned}$$

where $\beta_{p,0} = x_1\beta_{1,0} + x_2\beta_{2,0}$, $\beta_{p,M} = x_1\beta_{1,M} + x_2\beta_{2,M}$, and $\varepsilon_{p,t} = x_1\varepsilon_{1t} + x_2\varepsilon_{2t}$.

This result easily extends to portfolios with N assets. Let $R_{it} = \beta_{i,0} + \beta_{i,M} R_{Mt} + \varepsilon_{it}$ for $i = 1, \ldots, N$, and let $R_{p,t} = \sum_{i=1}^{N} x_i R_{it}$. Then

$$R_{p,t} = \beta_{p,0} + \beta_{p,M} R_{Mt} + \varepsilon_{p,t} \tag{15.26}$$

where

$$\beta_{p,0} = x_1\beta_{1,0} + \cdots + x_N\beta_{N,0}$$
$$\beta_{p,M} = x_1\beta_{1,M} + \cdots + x_N\beta_{N,M}$$

15.3.3.1 SI model and large portfolios

Consider an equally weighted portfolio of N assets, where N is a large number (e.g. $N = 500$) whose returns are described by the SI model. Here, $x_i = 1/N$ for $i = 1, \ldots, N$. Then the portfolio return is

$$\begin{aligned}
R_{p,t} &= \sum_{i=1}^{N} x_i R_{it} \\
&= \sum_{i=1}^{N} x_i \left(\beta_{i,0} + \beta_{i,M} R_{Mt} + \varepsilon_{it}\right) \\
&= \sum_{i=1}^{N} x_i \beta_{i,0} + \left(\sum_{i=1}^{N} x_i \beta_{i,M}\right) R_{Mt} + \sum_{i=1}^{N} x_i \varepsilon_{it} \\
&= \frac{1}{N}\sum_{i=1}^{N} \beta_{i,0} + \left(\frac{1}{N}\sum_{i=1}^{N} \beta_{i,M}\right) R_{Mt} + \frac{1}{N}\sum_{i=1}^{N} \varepsilon_{it} \\
&= \bar{\beta}_0 + \bar{\beta} R_{Mt} + \bar{\varepsilon}_t,
\end{aligned}$$

where $\bar{\beta}_{i,0} = \frac{1}{N}\sum_{i=1}^{N} \beta_{i,0}$, $\bar{\beta}_{i,M} = \frac{1}{N}\sum_{i=1}^{N} \beta_{i,M}$ and $\bar{\varepsilon}_t = \frac{1}{N}\sum_{i=1}^{N} \varepsilon_{it}$. Now,

$$\text{var}(\bar{\varepsilon}_t) = \text{var}\left(\frac{1}{N}\sum_{i=1}^{N} \varepsilon_{it}\right) = \frac{1}{N^2}\sum_{i=1}^{N} \text{var}(\varepsilon_{it}) = \frac{1}{N}\left(\frac{1}{N}\sum_{i=1}^{N} \sigma^2_{\varepsilon,i}\right) = \frac{1}{N}\bar{\sigma}^2$$

where $\bar{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N} \sigma^2_{\varepsilon,i}$ is the average of the asset specific variances. For large N, $\frac{1}{N}\bar{\sigma}^2 \approx 0$ and you have the Law of Large Numbers result

$$\bar{\varepsilon}_t = \frac{1}{N}\sum_{i=1}^{N} \varepsilon_{it} \approx E[\varepsilon_{it}] = 0.$$

As a result, in a large equally weighted portfolio you have the following:

- $R_{p,t} \approx \bar{\beta}_{i,0} + \bar{\beta}_{i,M} R_{Mt}$: all non-market asset-specific risk is diversified away and only market risk remains.
- $\text{var}(R_{p,t}) = \bar{\beta}^2 \text{var}(R_{Mt}) \Rightarrow \text{SD}(R_{p,t}) = |\bar{\beta}| \times \text{SD}(R_{Mt})$: portfolio volatility is proportional to market volatility where the factor of proportionality is the absolute value of portfolio beta.
- $R^2 \approx 1$: Approximately 100% of portfolio variance is due to market variance.

- $\bar{\beta} \approx 1$. A large equally weighted portfolio resembles the market portfolio (e.g., as proxied by the S&P 500 index) and so the beta of a well diversified portfolio will be close to the beta of the market portfolio which is one by definition.[3]

These results help us to understand the type of risk that gets diversified away and the type of risk that remains when you form diversified portfolios. Asset specific risk, which is uncorrelated across assets, gets diversified away whereas market risk, which is common to all assets, does not get diversified away.

- (Relate to average covariance calculation from portfolio theory chapter).
- Related to asset R2 discussed earlier. R2 of an asset shows the portion of risk that cannot be diversified away when forming portfolios.

15.4 The SI model in matrix notation

- Need to emphasize that the SI model covariance matrix is always positive definite. This is an important result because it allows for the mean-variance analysis of very large portfolios.

For $i = 1, \dots, N$ assets, stacking Equation 15.1 gives the SI model in matrix notation

$$\begin{pmatrix} R_{1t} \\ \vdots \\ R_{Nt} \end{pmatrix} = \begin{pmatrix} \beta_{1,0} \\ \vdots \\ \beta_{N,0} \end{pmatrix} + \begin{pmatrix} \beta_{1,M} \\ \vdots \\ \beta_{N,M} \end{pmatrix} R_{Mt} + \begin{pmatrix} \varepsilon_{1t} \\ \vdots \\ \varepsilon_{Nt} \end{pmatrix},$$

or

$$\mathbf{R}_t = \beta_0 + \beta_M R_{Mt} + \varepsilon_t. \tag{15.27}$$

The unconditional statistical properties of returns Equation 15.13, Equation 15.14, Equation 15.15 and Equation 15.17 can be re-expressed using matrix notation as follows:

$$E[\mathbf{R}_t] = \mu = \beta_0 + \beta_M \mu_M, \tag{15.28}$$

$$\text{var}(\mathbf{R}_t) = \Sigma = \sigma_M^2 \beta_M \beta_M' + \mathbf{D}, \tag{15.29}$$

$$\mathbf{R}_t \sim iid\, N(\mu, \Sigma) = N(\beta_0 + \beta_M \mu_M, \sigma_M^2 \beta_M \beta_M' + \mathbf{D}), \tag{15.30}$$

[3]The S&P 500 index is a value weighted index and so is not equal to an equally weighted portfolio of the stocks in the S&P 500 index. However, since most stocks in the S&P 500 index have large market capitalizations the values weights are not too different from equal weights.

where

$$\begin{aligned}\mathbf{D} &= \text{var}(\varepsilon_t) = \begin{pmatrix} \sigma^2_{\varepsilon,1} & 0 & \cdots & 0 \\ 0 & \sigma^2_{\varepsilon,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2_{\varepsilon,N} \end{pmatrix} \\ &= \text{diag}(\sigma^2_{\varepsilon,1}, \sigma^2_{\varepsilon,1}, ..., \sigma^2_{\varepsilon,N}).\end{aligned} \tag{15.31}$$

The derivation of the SI model covariance matrix Equation 15.29 is

$$\begin{aligned}\text{var}(\mathbf{R}_t) &= \Sigma = \beta_M \text{var}(R_{Mt})\beta'_M + \text{var}(\varepsilon_t) \\ &= \sigma^2_M \beta_M \beta'_M + \mathbf{D},\end{aligned}$$

which uses the assumption that the market return R_{Mt} is uncorrelated will all asset specific error terms in ε_t.

Example 15.2 (SI covariance matrix for three asset portfolio.)**.** It is useful to examine the SI covariance matrix Equation 15.29 for a three asset portfolio. In this case, you have

$$\begin{aligned}R_{it} &= \beta_{i,0} + \beta_{i,M} R_{Mt} + \varepsilon_{it}, \ i = 1, 2, 3 \\ \sigma^2_i &= \text{var}(R_{it}) = \beta^2_{i,M}\sigma^2_M + \sigma^2_{\varepsilon,i} \\ \sigma_{ij} &= \text{cov}(R_{it}, R_{jt}) = \sigma^2_M \beta_{i,M}\beta_{j,M}\end{aligned}$$

The 3×3 covariance matrix is

$$\begin{aligned}\Sigma &= \begin{pmatrix} \sigma^2_1 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma^2_2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma^2_3 \end{pmatrix} \\ &= \begin{pmatrix} \beta^2_{1,M}\sigma^2_M + \sigma^2_{\varepsilon,1} & \sigma^2_M \beta_{1,M}\beta_{2,M} & \sigma^2_M \beta_{1,M}\beta_{3,M} \\ \sigma^2_M \beta_{1,M}\beta_{2,M} & \beta^2_{2,M}\sigma^2_M + \sigma^2_{\varepsilon,2} & \sigma^2_M \beta_{2,M}\beta_{3,M} \\ \sigma^2_M \beta_{1,M}\beta_{3,M} & \sigma^2_M \beta_{2,M}\beta_{3,M} & \beta^2_{3,M}\sigma^2_M + \sigma^2_{\varepsilon,3} \end{pmatrix} \\ &= \sigma^2_M \begin{pmatrix} \beta^2_{1,M} & \beta_{1,M}\beta_{2,M} & \beta_{1,M}\beta_{3,M} \\ \beta_{1,M}\beta_{2,M} & \beta^2_{2,M} & \beta_{2,M}\beta_{3,M} \\ \beta_{1,M}\beta_{3,M} & \beta_{2,M}\beta_{3,M} & \beta^2_{3,M} \end{pmatrix} + \begin{pmatrix} \sigma^2_{\varepsilon,1} & 0 & 0 \\ 0 & \sigma^2_{\varepsilon,2} & 0 \\ 0 & 0 & \sigma^2_{\varepsilon,3} \end{pmatrix}.\end{aligned}$$

The first matrix shows the return variance and covariance contributions due to the market returns, and the second matrix shows the contributions due to the asset specific errors. Define $\beta_M =$

$(\beta_{1,M}, \beta_{2,M}, \beta_{3,M})'$. Then

$$\sigma_M^2 \beta_M \beta_M' = \sigma_M^2 \begin{pmatrix} \beta_{1,M} \\ \beta_{2,M} \\ \beta_{3,M} \end{pmatrix} \begin{pmatrix} \beta_{1,M} & \beta_{2,M} & \beta_{3,M} \end{pmatrix} = \sigma_M^2 \begin{pmatrix} \beta_{1,M}^2 & \beta_{1,M}\beta_{2,M} & \beta_{1,M}\beta_{3,M} \\ \beta_{1,M}\beta_{2,M} & \beta_{2,M}^2 & \beta_{2,M}\beta_{3,M} \\ \beta_{1,M}\beta_{3,M} & \beta_{2,M}\beta_{3,M} & \beta_{3,M}^2 \end{pmatrix},$$

$$\mathbf{D} = \text{diag}(\sigma_{\varepsilon,1}^2, \sigma_{\varepsilon,2}^2, \sigma_{\varepsilon,3}^2) = \begin{pmatrix} \sigma_{\varepsilon,1}^2 & 0 & 0 \\ 0 & \sigma_{\varepsilon,2}^2 & 0 \\ 0 & 0 & \sigma_{\varepsilon,3}^2 \end{pmatrix},$$

and so

$$\Sigma = \sigma_M^2 \beta_M \beta_M' + \mathbf{D}.$$

The matrix form of the SI model Equation 15.28 - Equation 15.30 is useful for portfolio analysis. For example, consider a portfolio with $N \times 1$ weight vector $\mathbf{x} = (x_1, \ldots, x_N)'$. Using Equation 15.27, the SI model for the portfolio return $R_{p,t} = \mathbf{x}'\mathbf{R}_t$ is

$$\begin{aligned} R_{p,t} &= \mathbf{x}'(\beta_0 + \beta_M R_{Mt} + \varepsilon_t) \\ &= \mathbf{x}'\beta_0 + \mathbf{x}'\beta_M R_{Mt} + \mathbf{x}'\varepsilon_t \\ &= \beta_{p,0} + \beta_{p,M} R_{Mt} + \varepsilon_{p,t}, \end{aligned}$$

where $\beta_{p,0} = \mathbf{x}'\beta_0$, $\beta_p = \mathbf{x}'\beta_M$ and $\varepsilon_{p,t} = \mathbf{x}'\varepsilon_t$. ■

15.5 Single index model portfolio theory

1. Reduces the number of free parameters to estimate
2. Give conditions for which the covariance matrix is positive definite and full rank
3. solves the problem of less than full return covariance matrix.
4. Does not provide an alternative way to estimate expected returns

15.6 Monte Carlo Simulation of the SI Model

To give a first step reality check for the SI model, consider simulating data from the SI model for a single asset. The steps to create a Monte Carlo simulation from the SI model are:

1. Fix values for the SI model parameters $\beta_{i,0}$, $\beta_{i,M}$, $\sigma_{\varepsilon,i}$, μ_M and σ_M
2. Determine the number of simulated returns, T, to create.
3. Use a computer random number generator to simulate T iid values of $R_{Mt} \sim N(\mu_M, \sigma_M^2)$ and $\varepsilon_{i,t} \sim N(0, \sigma_{\varepsilon,i}^2)$, where R_{Mt} is simulated independently from $\varepsilon_{i,t}$.
4. Create the simulated asset returns $\tilde{R}_{it} = \beta_{i,0} + \beta_{i,M}\tilde{R}_{Mt} + \tilde{\varepsilon}_{it}$ for $t = 1, \ldots, T$

The following example illustrates simulating SI model returns for Boeing.

Example 15.3 (Simulating monthly returns from the SI model for Boeing)**.** Consider simulating monthly returns on Boeing from the SI model where the S&P 500 index is used as the market index. The SI model parameters are calibrated from the actual monthly returns on Boeing and the S&P 500 index using sample statistics, as discussed in the next section, and are given by:

$$\beta_{BA,0} = 0.005,\ \beta_{BA,M} = 0.98,\ \sigma_{\varepsilon,BA} = 0.08,\ \mu_M = 0.003,\ \sigma_M = 0.048.$$

These values are set in R using:

```
alpha.BA = 0.005
beta.BA = 0.98
sd.e.BA = 0.08
mu.sp500 = 0.003
sd.sp500 = 0.048
```

The simulated returns for the S&P 500 index and Boeing are created using:

```
n.sim = nrow(siRetS)
set.seed(123)
sp500.sim = rnorm(n.sim, mu.sp500, sd.sp500)
e.BA.sim = rnorm(n.sim, 0, sd.e.BA)
BA.sim = alpha.BA + beta.BA*sp500.sim + e.BA.sim
BA.sim = xts(BA.sim, index(siRetS))
sp500.sim = xts(sp500.sim, index(siRetS))
colnames(BA.sim) = "BA.sim"
colnames(sp500.sim) = "SP500.sim"
```

The simulate returns are illustrated in Figure 15.5 and Figure 15.6. The time plot and scatterplot of the simulated returns on the S&P 500 index look much like the corresponding plots of the actual returns shows in the top left panels of Figure 15.2 and Figure 15.4, respectively. Hence, the SI model for the S&P 500 index and Boeing passes the first step reality check. In Figure 15.6, the line $\beta_{i,0} + \beta_{i,M}R_{Mt} = 0.005 + 0.98R_{Mt}$ is shown in orange. This line represents the predictions of Boeing's return given the S&P 500 return. For example, if $R_{Mt} = 0.05$ then the predicted return for Boeing is $R_{BA} = 0.005 + 0.98 \times (0.05) = 0.054$. The differences between the observed Boeing returns and (blue dots) and the predicted returns (orange line) are the asset specific error terms. These are the random asset specific news components. The standard deviation of these components, $\sigma_{\varepsilon,BA} = 0.08$, represents the typical magnitude of these components. Here, you would not be too surprised if the observed Boeing return is 0.08 above or below the orange line. How close the Boeing returns are to the predicted returns is determined by the R^2 value given by

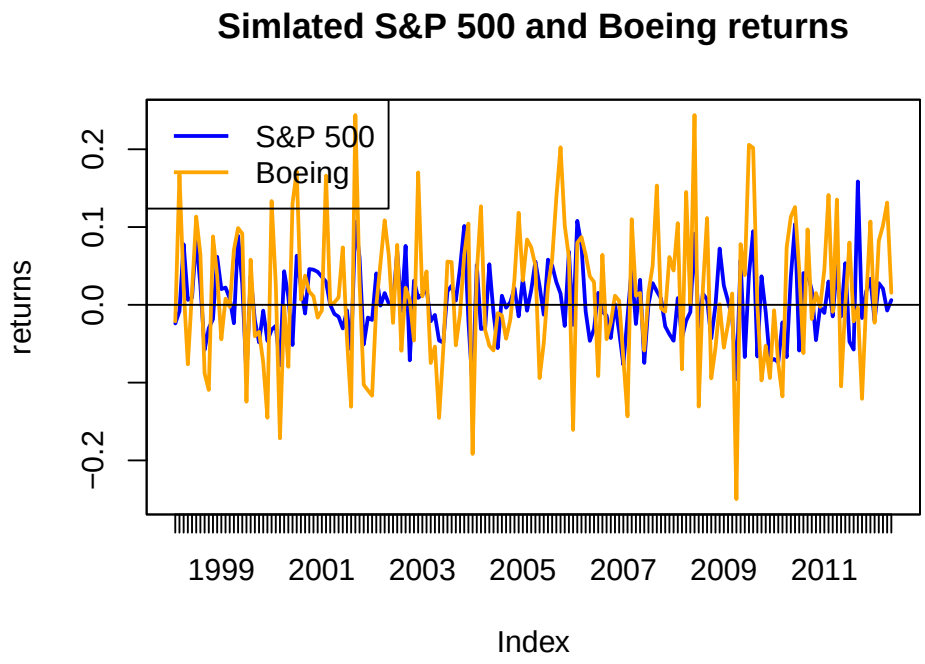


Figure 15.5: Simulated SI model monthly returns on the S&P 500 index and Boeing.

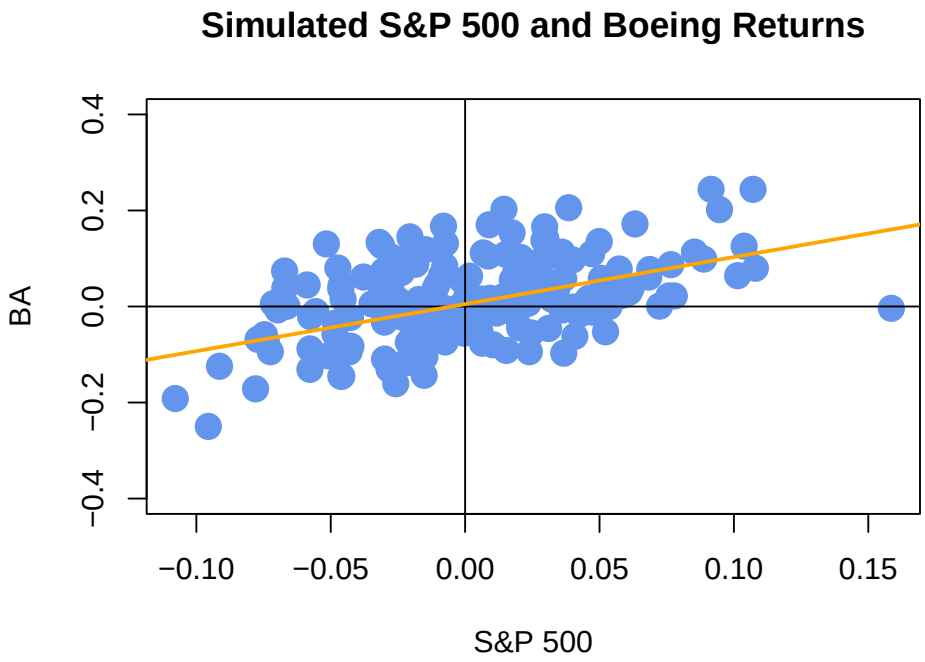


Figure 15.6: Simulated SI model monthly returns on the S&P 500 index and Boeing. The orange line is the equation $R_{BA,t} = \alpha_{BA} + \beta_{BA} R_{Mt} = 0.005 + 0.98 \times R_{Mt}$.

Equation 15.19. Here, $\sigma^2_{BA} = \beta^2_{BA,M}\sigma^2_M + \sigma^2_{\varepsilon,BA} = (0.98)^2(0.048)^2 + (0.08)^2 = 0.00861$ and so $R^2 = (0.98)^2(0.048)^2/0.00861 = 0.257$. Hence, 25.7% of the variability (risk) of Boeing returns are explained by the variability of the S&P 500 returns. As a result, 74.3% $(1 - R^2 = 0.743)$ of the variability of Boeing returns are due to the random asset specific terms. ■

15.7 Further Reading: Single Index Model

To be completed ...

15.8 Exercicses

Exercise 15.1 (Unconditional properties of returns in the SI model.)**.** Derive equations Equation 15.13 - Equation 15.16.

{Ruppert} Ruppert, D.A.. *Statistics and Data Analysis for Financial Engineering*, Springer-Verlag.

{MartinSchererYollin}Martin, R.D., Scherer, B., and Yollin, G. (2016).

16 Estimation of the Single Index Model

Updated: September 19, 2026

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16.1 Introduction

To be completed

Notes:

1. For consistency follow the chapter format of the GWN model
2. Covers estimation and hypothesis testing

Outline

1. Estimation using Method of Moments
2. Estimation using Least Squares
3. Statistical properties of method of moments/LS estimates
4. Linear regression in R
5. Examples, examples, examples.
6. Hypothesis Testing in the Single Index Model
    a. Tests for coefficients
    b. Tests for model assumptions: check to see if residual correlation matrix is diagonal!
    c. Diagnostics for covariance stationarity
    d. Rolling estimates from the single index model
7. Single Index Model and Portfolio Theory
    a. portfolio optimization using SI covariance matrix
        i. always pd, reduces number of estimated components in matrix.
        ii. Compare results with sample covariance matrix - in estimation chapter
    b. Risk analysis of asset and portfolios: factor risk reports

The R packages used in this chapter are **corrplot**, **introcompfinr**, **xts**, and **PerformanceAnalytics**.

16.2 Data for examples

I use the example data from @ref() for the examples in this chapter. The data are the monthly adjusted closing prices on four Northwest stocks (with tickers in parentheses): Boeing (BA), Nordstrom (JWN), Microsoft (MSFT), Starbucks (SBUX), and the S&P 500 index (SPY, proxy for the market index). These prices are extracted from the **introcompfinr** package as follows:

```
baPrices = to.monthly(baDailyPrices, OHLC=FALSE)
jwnPrices = to.monthly(jwnDailyPrices, OHLC=FALSE)
msftPrices = to.monthly(msftDailyPrices, OHLC=FALSE)
sbuxPrices = to.monthly(sbuxDailyPrices, OHLC=FALSE)
sp500Prices = to.monthly(sp500DailyPrices, OHLC=FALSE)
siPrices = merge(baPrices, jwnPrices, msftPrices,
                 sbuxPrices, sp500Prices)
```

The data sample is from January, 1998 through May, 2012 and returns are simple returns:

```
smpl = "1998-01::2012-05"
siPrices = siPrices[smpl]
siRetS = na.omit(Return.calculate(siPrices, method="simple"))
```

16.3 Estimation of SI Model

Consider the SI model Equation 15.1 - Equation 15.5. The asset specific parameters to be estimated are $\beta_{i,0}$, $\beta_{i,M}$ and $\sigma^2_{\varepsilon,i}$, $(i = 1, \dots, N)$, and the market parameters to be estimated are μ_M and σ^2_M. These parameters can be estimated using the plug-in principle, linear regression, and maximum likelihood. All methods give essentially the same estimators for the SI model parameters.

16.3.1 Plug-in principle estimates

Let $\{(r_{it}, r_{Mt})\}_{t=1}^T$ denote a sample of size T of observed returns on asset i and the market return which are assumed to be generated from the SI model. Equation 15.1 - Equation 15.17. Recall, the plug-in principle says to estimate population model parameters using appropriate sample statistics. For the market parameters, the plug-in principle estimates are the same as the GWN model

estimates

$$\begin{aligned}\hat{\mu}_M &= \frac{1}{T}\sum_{t=1}^{T} r_{Mt},\\ \hat{\sigma}_M^2 &= \frac{1}{T-1}\sum_{t=1}^{T}(r_{Mt}-\hat{\mu}_M)^2.\end{aligned}$$

From Equation 15.10 - Equation 15.6 we see that $\beta_{i,0}$ and $\beta_{i,M}$ are functions of population parameters

$$\begin{aligned}\beta_{i,0} &= \mu_i - \beta_{i,M}\mu_M,\\ \beta_{i,M} &= \frac{\text{cov}(R_{it}, R_{Mt})}{\text{var}(R_{Mt})} = \frac{\sigma_{iM}}{\sigma_M^2}.\end{aligned}$$

The corresponding plug-in principle estimates are then:

$$\hat{\beta}_{i,0} = \hat{\mu}_i - \hat{\beta}_{i,M}\hat{\mu}_M, \tag{16.1}$$

$$\hat{\beta}_{i,M} = \frac{\hat{\sigma}_{iM}}{\hat{\sigma}_M^2}, \tag{16.2}$$

where

$$\begin{aligned}\hat{\mu}_i &= \frac{1}{T}\sum_{t=1}^{T} r_{it},\\ \hat{\sigma}_{iM} &= \frac{1}{T-1}\sum_{t=1}^{T}(r_{it}-\hat{\mu}_i)(r_{Mt}-\hat{\mu}_M).\end{aligned}$$

Given the plug-in principle estimates $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$, the plug-in principle estimate of ε_{it} is

$$\hat{\varepsilon}_{it} = r_{it} - \hat{\beta}_{i,0} - \hat{\beta}_{i,M} r_{Mt},\ t = 1, \ldots, T. \tag{16.3}$$

Using Equation 16.3, the plug-in principle estimate of $\sigma_{\varepsilon,i}^2$ is the sample variance of $\{\hat{\varepsilon}_{it}\}_{t=1}^{T}$ (adjusted for the number of degrees of freedom):

$$\hat{\sigma}_{\varepsilon,i}^2 = \frac{1}{T-2}\sum_{t=1}^{T}\hat{\varepsilon}_t^2 = \frac{1}{T-2}\sum_{t=1}^{T}\left(r_{it} - \hat{\beta}_{i,0} - \hat{\beta}_{i,M} r_{Mt}\right)^2. \tag{16.4}$$

Plug-in principle estimates of R^2 based on Equation 15.19 can be computed using

$$\hat{R}^2 = \frac{\hat{\beta}_i^2\hat{\sigma}_M^2}{\hat{\sigma}_i^2} = 1 - \frac{\hat{\sigma}_{\varepsilon,i}^2}{\hat{\sigma}_i^2}. \tag{16.5}$$

Example 16.1 (Computing plug-in principle estimators for SI model parameters)**.** Consider com-

puting the plug-in principle estimates for $\beta_{i,0}$, $\beta_{i,M}$ and $\sigma^2_{\varepsilon,i}$ from the example data using the formulas Equation 16.1, Equation 16.2 and Equation 16.4, respectively. First, extract the sample statistics $\hat{\mu}_i$, $\hat{\sigma}_{iM}$, $\hat{\mu}_M$, and $\hat{\sigma}^2_M$:

```
assetNames = colnames(siRetS)[1:4]
muhat = colMeans(siRetS)
Sigmahat = var(siRetS)
sig2hat = diag(Sigmahat)
covAssetsSp500 = Sigmahat[assetNames, "SP500"]
```

Next, estimate $\hat{\beta}_{i,M}$ using

```
betaMHat = covAssetsSp500/sig2hat["SP500"]
betaMHat
#>    BA   JWN  MSFT  SBUX
#> 0.978 1.485 1.303 1.057
```

Here, we see that $\hat{\beta}_{BA,M}$ and $\hat{\beta}_{SBUX,M}$ are very close to one and that $\hat{\beta}_{JWN,M}$ and $\hat{\beta}_{MSFT,M}$ are slightly bigger than one. Using the estimates of $\hat{\beta}_{i,M}$ and the sample statistics $\hat{\mu}_i$ and $\hat{\mu}_M$ the estimates for $\hat{\beta}_{i,0}$ are

```
beta0Hat = muhat[assetNames] - betaMHat*muhat["SP500"]
beta0Hat
#>      BA     JWN    MSFT    SBUX
#> 0.00516 0.01231 0.00544 0.01785
```

All of the estimates of $\hat{\beta}_{i,0}$ are close to zero.

The estimates of $\sigma^2_{\varepsilon,i}$ and $\sigma_{\varepsilon,i}$ can be computed using:

```
sig2eHat = rep(0, length(assetNames))
names(sig2eHat) = assetNames
for (aName in assetNames) {
     eHat = siRetS[, aName] - beta0Hat[aName] - betaMHat[aName]*siRetS[, "SP500"]
     sig2eHat[aName] = crossprod(eHat)/(length(eHat) - 2)
}
sigeHat = sqrt(sig2eHat)
ans = rbind(sig2eHat,sigeHat)
ans
#>                BA     JWN    MSFT    SBUX
#> sig2eHat 0.00581 0.00994 0.00646 0.00941
```

```
#> sigeHat  0.07624 0.09971 0.08036 0.09703
```

Here, Nordstrom (JWN) has the largest asset specific volatility and Boeing (BA) has the lowest.

Lastly, the estimates of R^2 can be computed using

```
R2 = 1 - sig2eHat/sig2hat[assetNames]
R2
#>    BA   JWN  MSFT  SBUX
#> 0.270 0.334 0.373 0.210
```

Microsoft has the largest fraction of returns explained by the market index and Starbucks has the lowest.

The fitted SI model is visualized in the figure below created using:

■

16.3.2 Least squares estimates

The SI model representation Equation 15.1 shows that returns are a linear function of the market return and an asset specific error term

$$R_{it} = \beta_{i,0} + \beta_{i,M} R_{Mt} + \varepsilon_{it},$$

here $\beta_{i,0}$ is the intercept and $\beta_{i,M}$ is the slope. *Least squares regression* is a method for estimating $\beta_{i,0}$ and $\beta_{i,M}$ by finding the "best fitting" line to the scatterplot of returns where R_{it} is on the vertical axis and R_{Mt} is on the horizontal axis.

To see how the method of least squares determines the "best fitting" line, consider the scatterplot of the sample returns on Boeing and the S&P 500 index illustrated in Figure xxx. In the figure, the black line is a fitted line with initial guess $\hat{\beta}_{BA,0} = 0$ and $\hat{\beta}_{BA,M} = 0.5$. The differences between the observed returns (blue dots) and the values on the fitted line are the estimated errors $\hat{\varepsilon}_{BA,t} = r_{BA,t} - \hat{\beta}_{BA,0} - \hat{\beta}_{BA,M} r_{Mt} = r_{BA,t} - 0 - 0.5 \times R_{Mt}$. Some estimated errors are big and some are small. The overall fit of the line can be measured using a statistic based on all $t = 1, \ldots, T$ of the estimated errors. A natural choice is the sum of the errors $\sum_{t=1}^{T} \hat{\varepsilon}_t$. However, this choice can be misleading due to the canceling out of large positive and negative errors. To avoid this problem, it is better to measure the overall fit using $\sum_{t=1}^{T} |\hat{\varepsilon}_t|$ or $\sum_{t=1}^{T} \hat{\varepsilon}_t^2$. Then the best fitting line can be determined by finding the intercept and slope values that minimize $\sum_{t=1}^{T} |\hat{\varepsilon}_t|$ or $\sum_{t=1}^{T} \hat{\varepsilon}_t^2$.

The method of least squares regression defines the "best fitting" line by finding the intercept and slope values that minimize the sum of squared errors

$$\text{SSE}(\hat{\beta}_{i,0}, \hat{\beta}_{i,M}) = \sum_{t=1}^{T} \hat{\varepsilon}_{it}^2 = \sum_{t=1}^{T} \left(r_{it} - \hat{\beta}_{i,0} - \hat{\beta}_{i,M} r_{Mt} \right)^2. \tag{16.6}$$

Because $\text{SSE}(\hat{\beta}_{i,0}, \hat{\beta}_{i,M})$ is a continuous and differential function of $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$, the minimizing values of $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ can be determined using simple calculus. The first order conditions for a minimum are:

$$0 = \frac{\partial \text{SSE}(\hat{\beta}_{i,0}, \hat{\beta}_{i,M})}{\partial \hat{\beta}_{i,0}} = -2 \sum_{t=1}^{T} (r_{it} - \hat{\beta}_{i,0} - \hat{\beta}_{i,M} r_{Mt}) = -2 \sum_{t=1}^{T} \hat{\varepsilon}_{it}, \tag{16.7}$$

$$0 = \frac{\partial \text{SSE}(\hat{\beta}_{i,0}, \hat{\beta}_{i,M})}{\partial \hat{\beta}_{i,M}} = -2 \sum_{t=1}^{T} (r_{it} - \hat{\beta}_{i,0} - \hat{\beta}_{i,M} r_{Mt}) r_{Mt} = -2 \sum_{t=1}^{T} \hat{\varepsilon}_{it} r_{Mt}. \tag{16.8}$$

These are two linear equations in two unknowns which can be re-expressed as

$$\begin{aligned} \hat{\beta}_{i,0} T + \hat{\beta}_{i,M} \sum_{t=1}^{T} r_{Mt} &= \sum_{t=1}^{T} r_{it}, \\ \hat{\beta}_{i,0} \sum_{t=1}^{T} r_{Mt} + \hat{\beta}_{i,M} \sum_{t=1}^{T} r_{Mt}^2 &= \sum_{t=1}^{T} r_{it} r_{Mt}. \end{aligned}$$

Using matrix algebra, we can write these equations as:

$$\begin{pmatrix} T & \sum_{t=1}^{T} r_{Mt} \\ \sum_{t=1}^{T} r_{Mt} & \sum_{t=1}^{T} r_{Mt}^2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_{i,0} \\ \hat{\beta}_{i,M} \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^{T} r_{it} \\ \sum_{t=1}^{T} r_{it} r_{Mt} \end{pmatrix}, \tag{16.9}$$

which is of the form $\mathbf{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbf{A} = \begin{pmatrix} T & \sum_{t=1}^{T} r_{Mt} \\ \sum_{t=1}^{T} r_{Mt} & \sum_{t=1}^{T} r_{Mt}^2 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} \hat{\beta}_{i,0} \\ \hat{\beta}_{i,M} \end{pmatrix}, \mathbf{b} = \begin{pmatrix} \sum_{t=1}^{T} r_{it} \\ \sum_{t=1}^{T} r_{it} r_{Mt} \end{pmatrix}.$$

Hence, we can determine $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ by solving $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. Now,[1]

$$\begin{aligned} \mathbf{A}^{-1} &= \frac{1}{\det(\mathbf{A})} \begin{pmatrix} \sum_{t=1}^{T} r_{Mt}^2 & -\sum_{t=1}^{T} r_{Mt} \\ -\sum_{t=1}^{T} r_{Mt} & T \end{pmatrix}, \\ \det(\mathbf{A}) &= T \sum_{t=1}^{T} r_{Mt}^2 - \left(\sum_{t=1}^{T} r_{Mt} \right)^2 = T \sum_{t=1}^{T} (r_{Mt} - \hat{\mu}_M)^2, \\ \hat{\mu}_M &= \frac{1}{T} \sum_{t=1}^{T} r_{Mt}. \end{aligned}$$

[1]The matrix $\mathbf{A}$ is invertible provided $\det(A) \neq 0$. This requires the sample variance of R_{Mt} to be non-zero.

Consequently,

$$\begin{pmatrix} \hat{\beta}_{i,0} \\ \hat{\beta}_{i,M} \end{pmatrix} = \frac{1}{T\sum_{t=1}^{T}\left(r_{Mt}-\hat{\mu}_M\right)^2} \begin{pmatrix} \sum_{t=1}^{T} r_{Mt}^2 & -\sum_{t=1}^{T} r_{Mt} \\ -\sum_{t=1}^{T} r_{Mt} & T \end{pmatrix} \begin{pmatrix} \sum_{t=1}^{T} r_{it} \\ \sum_{t=1}^{T} r_{it} r_{Mt} \end{pmatrix} \tag{16.10}$$

and so

$$\hat{\beta}_{i,0} = \frac{\sum_{t=1}^{T} r_{Mt}^2 \sum_{t=1}^{T} r_{it} - \sum_{t=1}^{T} r_{Mt}^2 \sum_{t=1}^{T} r_{it} r_{Mt}}{T\sum_{t=1}^{T}\left(r_{Mt}-\hat{\mu}_M\right)^2}, \tag{16.11}$$

$$\hat{\beta}_{i,M} = \frac{T\sum_{t=1}^{T} r_{it} r_{Mt} - \sum_{t=1}^{T} r_{Mt} \sum_{t=1}^{T} r_{it}}{T\sum_{t=1}^{T}\left(r_{Mt}-\hat{\mu}_M\right)^2}. \tag{16.12}$$

After a little bit of algebra (see end-of-chapter exercises) it can be shown that

$$\begin{aligned} \hat{\beta}_{i,0} &= \hat{\mu}_i - \hat{\beta}_i \hat{\mu}_M, \\ \hat{\beta}_{i,M} &= \frac{\hat{\sigma}_{iM}}{\hat{\sigma}_M^2}, \end{aligned}$$

which are plug-in estimates for $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ determined earlier. Hence, the least squares estimates of $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ are identical to the plug-in estimates.

The solution for the least squares estimates in Equation 16.10 has an elegant representation using matrix algebra. To see this, define the $T \times 1$ vectors $\mathbf{r}_i = (r_{i1}, \ldots, r_{iT})'$, $\mathbf{r}_M = (r_{M1}, \ldots, r_{MT})'$ and $\mathbf{1} = (1, \ldots, 1)'$. Then we can re-write Equation 16.9 as

$$\begin{pmatrix} \mathbf{1}'\mathbf{1} & \mathbf{1}'\mathbf{r}_M \\ \mathbf{1}'\mathbf{r}_M & \mathbf{r}_M'\mathbf{r}_M \end{pmatrix} \begin{pmatrix} \hat{\beta}_{i,0} \\ \hat{\beta}_{i,M} \end{pmatrix} = \begin{pmatrix} \mathbf{1}'\mathbf{r}_i \\ \mathbf{r}_M'\mathbf{r}_i \end{pmatrix}$$

or

$$\mathbf{X}'\mathbf{X}\hat{\beta}_i = \mathbf{X}'\mathbf{r}_i \tag{16.13}$$

where $\mathbf{X} = (\ \mathbf{1} \quad \mathbf{r}_M)$ is a $T \times 2$ matrix and $\hat{\beta}_i = (\hat{\beta}_{i,0}, \hat{\beta}_{i,M})'$. Provided $\mathbf{X}'\mathbf{X}$ is invertible, solving Equation 16.13 for $\hat{\beta}_i$ gives the least squares estimates in matrix form:

$$\hat{\beta}_i = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{r}_i. \tag{16.14}$$

The matrix form solution Equation 16.14 is especially convenient for computation in R.

The least squares estimates of ε_t, $\sigma_{\varepsilon,i}^2$ and R^2 are the same as the plug-in estimators Equation 16.3, Equation 16.4 and Equation 16.5, respectively. In the context of least squares estimation, the estimate $\hat{\sigma}_{\varepsilon,i} = \sqrt{\hat{\sigma}_{\varepsilon,i}^2}$ is called the *standard error of the regression* and measures the typical magnitude of $\hat{\varepsilon}_t$ (difference between observed return and fitted regression line).

16.3.3 Simple linear regression in R

- don't do regression examples until statistical theory is discussed
- computing least squares estimates using matrix algebra formulas
- computing least squares estimates using `lm()`
  - See discussion from my regression chapter in MFTSR
  - describe structure of `lm()` function, extractor and method functions
- Do analysis of example data To be completed...

16.4 Statistical Properties of SI Model Estimates

To determine the statistical properties of the plug-in principle/least squares/ML estimators $\hat{\beta}_{i,0}$, $\hat{\beta}_{i,M}$ and $\hat{\sigma}^2_{\varepsilon,i}$ in the SI model, we treat them as functions of the random variables $\{(R_{i,t}, R_{Mt})\}_{t=1}^T$ where R_t and R_{Mt} are assumed to be generated by the SI model Equation 15.1 - Equation 15.5.

16.4.1 Bias

In the SI model, the estimators $\hat{\beta}_{i,0}$, $\hat{\beta}_{i,M}$ and $\hat{\sigma}^2_{\varepsilon,i}$ (with degrees-of-freedom adjustment) are unbiased:

$$\begin{aligned} E[\hat{\beta}_{i,0}] &= \beta_{i,0}, \\ E[\hat{\beta}_{i,M}] &= \beta_{i,M}, \\ E[\hat{\sigma}^2_{\varepsilon,i}] &= \sigma^2_{\varepsilon,i}. \end{aligned}$$

To shows that $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ are unbiased, it is useful to consider the SI model for asset i in matrix form for $t = 1, \ldots, T$:

$$\mathbf{R}_i = \beta_{i,0}\mathbf{1} + \beta_{i,M}\mathbf{R}_M + \varepsilon_i = \mathbf{X}\beta_i + \varepsilon_i,$$

where $\mathbf{R}_i = (R_{i1}, \ldots, R_{iT})'$, $\mathbf{R}_M = (R_{M1}, \ldots, R_{MT})'$, $\varepsilon_i = (\varepsilon_{i1}, \ldots, \varepsilon_{iT})'$, $\mathbf{1} = (1, \ldots, 1)'$, $\mathbf{X} = \begin{pmatrix} \mathbf{1} & \mathbf{R}_M \end{pmatrix}$, and $\beta_i = (\beta_{i,0}, \beta_{i,M})'$. The estimator for β_i is

$$\hat{\beta}_i = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{R}_i.$$

Pluging in $\mathbf{R}_i = \mathbf{X}\beta_i + \varepsilon_i$ gives

$$\begin{aligned} \hat{\beta}_i &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\gamma_i + \varepsilon_i) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\gamma_i + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon_i \\ &= \beta_i + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon_i. \end{aligned}$$

Then

$$\begin{aligned} E[\hat{\beta}_i] &= \beta_i + E\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon_i\right] \\ &= \beta_i + E\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right] E[\varepsilon_i] \,(\text{because } \varepsilon_{it} \text{ is independent of } R_{Mt}) \\ &= \beta_i \,(\text{because } E[\varepsilon_i] = \mathbf{0}). \end{aligned}$$

The derivation of $E[\hat{\sigma}^2_{\varepsilon,i}] = \sigma^2_{\varepsilon,i}$ is beyond the scope of this book and can be found in graduate econometrics textbooks such as Hayashi (1980).

16.4.2 Precision

Under the assumptions of the SI model, analytic formulas for estimates of the standard errors for $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ are given by:

$$\widehat{\text{se}}(\hat{\beta}_{i,0}) \approx \frac{\hat{\sigma}_{\varepsilon,i}}{\sqrt{T \cdot \hat{\sigma}^2_M}} \cdot \sqrt{\frac{1}{T}\sum_{t=1}^{T} r^2_{Mt}}, \tag{16.15}$$

$$\widehat{\text{se}}(\hat{\beta}_{i,M}) \approx \frac{\hat{\sigma}_{\varepsilon,i}}{\sqrt{T \cdot \hat{\sigma}^2_M}}, \tag{16.16}$$

where "$\approx$" denotes an approximation based on the CLT that gets more accurate the larger the sample size.

Remarks:

- $\widehat{\text{se}}(\hat{\beta}_{i,0})$ and $\widehat{\text{se}}(\hat{\beta}_{i,M})$ are smaller the smaller is $\hat{\sigma}_{\varepsilon,i}$. That is, the closer are returns to the fitted regression line the smaller are the estimation errors in $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$.
- $\widehat{\text{se}}(\hat{\beta}_{i,M})$ is smaller the larger is $\hat{\sigma}^2_M$. That is, the greater the variability in the market return R_{Mt} the smaller is the estimation error in the estimated slope coefficient $\hat{\beta}_{i,M}$. This is illustrated in Figure xxx. The left panel shows a data sample from the SI model with a small value of $\hat{\sigma}^2_M$ and the right panel shows a sample with a large value of $\hat{\sigma}^2_M$. The right panel shows that the high variability in R_{Mt} makes it easier to identify the slope of the line.
- Both $\widehat{\text{se}}(\hat{\beta}_{i,0})$ and $\widehat{\text{se}}(\hat{\beta}_{i,M})$ go to zero as the sample size, T, gets large. Since $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ are unbiased estimators, this implies that they are also consistent estimators. That is, they converge to the true values $\beta_{i,0}$ and $\beta_{i,M}$, respectively, as $T \to \infty$.
- In R, the standard error values Equation 16.15 and Equation 16.16 are computed using the `summary()` function on an "`lm`" object.

There are no easy formulas for the estimated standard errors for $\hat{\sigma}^2_{\varepsilon,i}$, $\hat{\sigma}_{\varepsilon,i}$ and $\hat{R}^2$. Estimated standard errors for these estimators, however, can be easily computed using the bootstrap.

Example 16.2 (Computing estimated standard errors for $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ in R)**.** to be completed

■

16.4.3 Sampling distribution and confidence intervals.

Using arguments based on the CLT, it can be shown that for large enough T the estimators $\hat{\beta}_{i,0}$ and $\hat{\beta}_{i,M}$ are approximately normally distributed:

$$\begin{aligned} \hat{\beta}_{i,0} &\sim N(\beta_{i,0}, \widehat{\text{se}}(\hat{\beta}_{i,0})^2), \\ \hat{\beta}_{i,M} &\sim N(\beta_{i,M}, \widehat{\text{se}}(\hat{\beta}_{i,M})^2), \end{aligned}$$

where $\widehat{\text{se}}(\hat{\beta}_{i,0})$ and $\widehat{\text{se}}(\hat{\beta}_{i,M})$ are given by Equation 16.15 and Equation 16.16, respectively.

16.5 Further Reading: Single Index Model

To be completed ...

16.6 Exercicses

To be completed ...

{Ruppert} Ruppert, D.A.. *Statistics and Data Analysis for Financial Engineering*, Springer-Verlag.

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